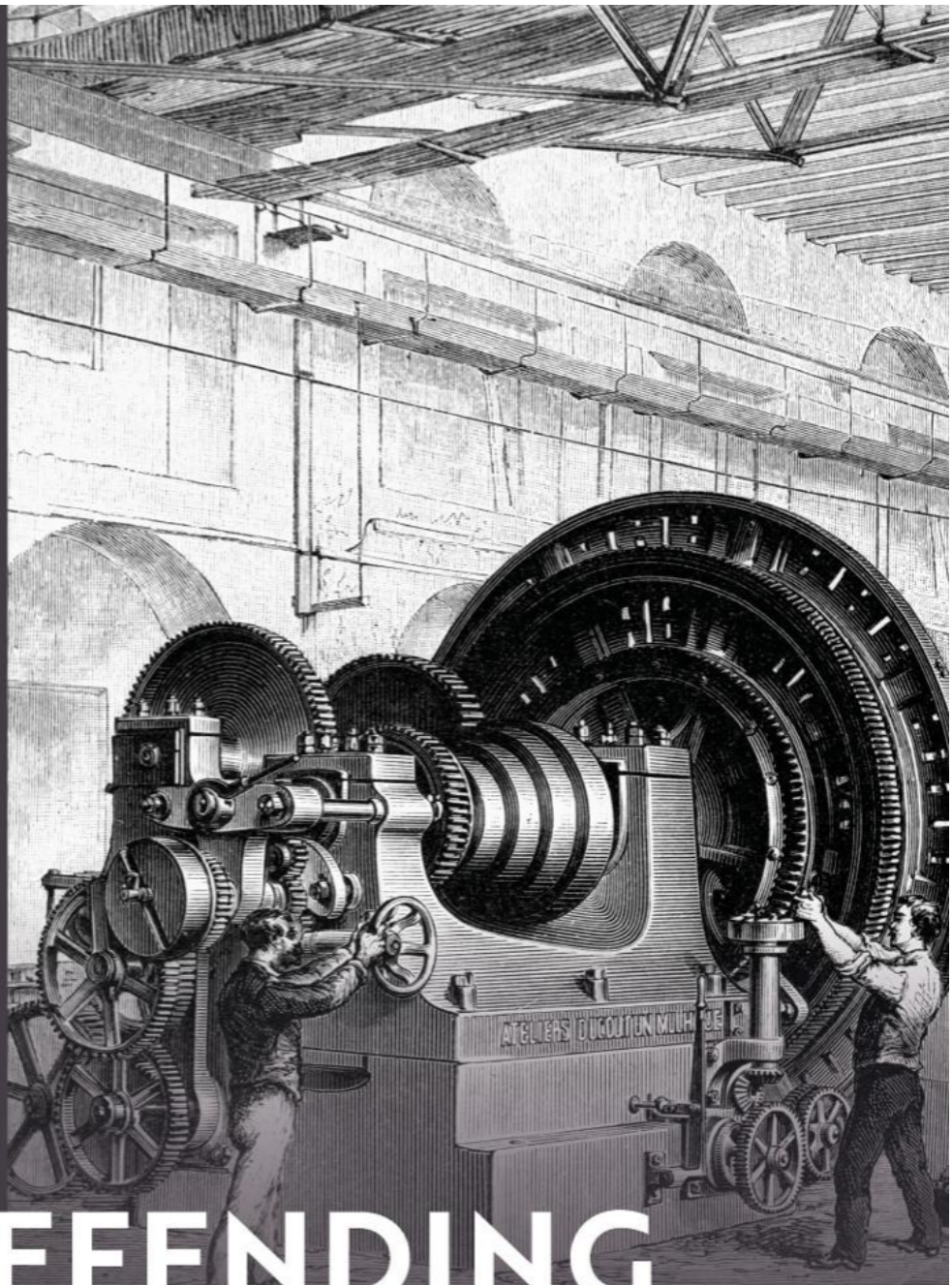


DEFENDING MATERIALISM

The Uneasy History of the Atom in Science and Philosophy

KATARINA KOLOZOVA,
WILLIAM PAUL COCKSHOTT
& GREG MICHAELSON

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Katarina Kolozova, William Paul Cockshott, Greg Michaelson

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第 1 章引言

Chapter 1. INTRODUCTION

概述

Overview

p. 1

我们走出教堂后，站着谈了一会儿贝克莱主教的巧妙诡辩：他试图证明物质并不存在，宇宙中的一切都只是观念。我说，尽管我们确信他的学说不正确，却无法驳倒它。我永远不会忘记约翰逊那敏捷的回答：他用脚猛踢一块大石头，踢得自己都弹了回来，说："我就这样反驳它。"——詹姆斯·博斯韦尔[1]

After we came out of the church, we stood talking for some time together of Bishop Berkeley's ingenious sophistry to prove the nonexistence of matter, and that every thing in the universe is merely ideal. I observed, that though we are satisfied his doctrine is not true, it is impossible to refute it. I never shall forget the alacrity with which Johnson answered, striking his foot with mighty force against a large stone, till he rebounded from it – 'I refute it thus'. James Boswell[1]

唯物主义与唯心主义是一切哲学的两极，二者的争论已持续了 2500 年。约 250 年前，詹姆斯·博斯韦尔与塞缪尔·约翰逊的上述对话，简明地概括了这种对立在哲学和社会层面的复杂性。

Materialism and idealism, the twin poles of all philosophies, have contended for 2,500 years. The complexity of their opposition, both philosophical and social, is neatly encapsulated by the above exchange between James Boswell and Samuel Johnson some 250 years ago.

这场争论发生在 18 世纪末，当时英国工业革命正在加速，其基础是迅速发展的数学、物理、化学和工程原理。这些原理反过来威胁到宗教意识形态的古老根基，而宗教意识形态曾是英国统治阶级的基石。

This argument took place in the late eighteenth century, as the British Industrial Revolution was accelerating, founded on burgeoning mathematical, physical, chemical and engineering principles. In turn, these principles threatened the age-old bases of the religious ideologies that had been the bedrock of the British ruling classes.

乔治·贝克莱（1685—1753）是一位爱尔兰圣公会主教，他在 18 世纪初试图用哲学而非《圣经》权威来支撑自然神论。如今他被视为绝对唯心主义的奠基人，该学说否认一切物质性。

George Berkeley (1685 – 1753) was an Irish Anglican prelate who, at the start of the eighteenth century, sought to use philosophy rather than biblical authority to buttress deism. He is now seen as a founder of absolute idealism, which denies all materiality.

詹姆斯·博斯韦尔（1740—1795）是苏格兰人，在启蒙运动时期成年。他的信仰立场多变，从小受加尔文主义熏陶，也曾动过皈依天主教的念头。在爱丁堡大学，亚当·斯密曾给他授课；斯密的经济学对制造业的发展和马克思都有重大影响。博斯韦尔一向交游广泛，甚至曾与临终的伟大怀疑论哲学家大卫·休谟讨论来世是否可能存在，而休谟坚决否认这一观念。博斯韦尔有私人财产维持生活，尽管自我厌弃，仍过着放纵的生活；他最为人所知的身份是约翰逊的传记作者。

James Boswell (1740–95) was a Scot who came of age during the Enlightenment. Of fluid persuasion, he was raised a Calvinist and toyed with conversion to Catholicism. At the University of Edinburgh, he was taught, in part, by Adam Smith, whose economics greatly influenced both the development of manufacture and Marx. Always well connected, Boswell even discussed the possibility of an afterlife with the dying David Hume, the great sceptical philosopher, who firmly repudiated the notion. With private means, Boswell led a licentious, if self-loathing, life and is best known as Johnson's biographer.

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塞缪尔·约翰逊（1709—1784）是一位英国文人，以其权威的英语词典闻名。他言语直率，曾在牛津大学受教育，博览群书，是托利党人，也是圣公会信徒。因此，约翰逊的常识唯物主义竟能穿透那些令博斯韦尔如此困惑的诡辩，实在引人注目。

Samuel Johnson (1709 – 84) was an English man of letters, who is remembered for his authoritative dictionary of the English language. Plain-speaking, Oxford-educated and widely read, he was a Tory and an Anglican. Thus, it is striking that Johnson's common-sense materialism cut through the sophistries that so perplexed Boswell.

p. 2

我们的唯物主义同样以同外部现实的互动为基础，但它当然不像约翰逊的唯物主义那样迁就唯心主义。事实上，本书旨在捍卫原子论唯物主义和机械唯物主义，反对唯心主义的解释。

Our materialism is equally grounded in interaction with external reality, though certainly not as accommodating of idealism as Johnson's. Indeed, the intention

我们的论述采取历史视角，从古希腊原子论者一直到当代，将其视为贯穿科学与哲学学科的哲学分化。我们关心的是这种分化所带来的政治哲学后果。

我们的批判立场来自卡尔·马克思的唯物主义学说，而不是马克思主义的全部遗产——后者并非铁板一块。因此，我们选择推演并追随马克思本人的思路；我们认为，这种思路已经在他关于希腊原子论的博士论文中成形。因此，我们的诠释学是对马克思本人方法的一种模仿。我们拆解辩证法与唯物主义的统一体，试图把二者作为独立但常常结合或交叉的思想线索来追踪，并根据所讨论的作者和学科，分析这两个概念本身的辩证关系。

本书追踪若干贯穿时间的平行线索：

• 原子观念在数世纪中如何既变化又保持同一。我们把它视为真正唯物主义与带有唯心主义色彩的唯物主义——或隐蔽唯心主义的唯物主义——之间的分界点，这在哲学以及数学和计算等科学中都存在；

• 逻辑的发展及其在多位思想家著作中与“辩证法”的关系。我们在这里探究其政治影响，以及意识形态如何影响科学沿唯物主义与唯心主义这一分界作出的认识论选择。

• “无中不能生有”这一原则反复出现。它源于古希腊哲学，并在守恒定律的发展中不断被重述和激活，最终被理解为这些定律都源于对称性。这一形而上学困境隐秘地存在于科学、政治哲学和经济学中。作为欧洲观念史组织结构的支柱之一，这种无意的、常常不经意间存在的返祖现象，塑造了我们今天所知的当代科学、技术和社会经济理论的很大部分；

• 物理学、生物学以及对人类社会的反思中的时间性质；后一类反思主要见于哲学，也见于社会科学。对贯穿这些学科的时间观念作诠释学考察，能够为历史唯物主义提供一种新的理解，使之摆脱 diamat 的教条，即苏联所确立的辩证唯物主义的教条。

我们的焦点是列宁和爱因斯坦批判的显性唯心主义，以及影响苏联 diamat 的隐性黑格尔式唯心主义。从古希腊思想中唯物主义的起源开始，我们探讨它

of this book is to defend atomistic and mechanical materialism against idealist accounts.

Our extended presentation is from a historical perspective, from the Greek atomists to the current day, as a matter of philosophical differentiation running across scientific and philosophical disciplines. We are interested in exploring the political-philosophical consequences of such differentiation.

Our critical stance is informed by the materialist doctrine of Karl Marx rather than the entire legacy of Marxism, which is not a monolithic phenomenon. Hence, we chose to extrapolate and follow Marx’s original line of thinking, which, we argue, already takes shape in his doctoral dissertation on Greek atomism. Our hermeneutics is, therefore, a type of mimesis of Marx’s own approach. While we dismantle the unity of dialectics and materialism, we try to follow their developments as independent yet often united or intersecting strands of thought, and analytically examine the dialectics of the two notions themselves, depending on the authors and disciplines discussed.

The book covers several parallel threads which are traced through time:

• *How the idea of atoms has both changed and remained the same across the centuries.* We see this as the point of delineation between materialism proper and materialism with a touch of idealism, or materialism as covert idealism, both in philosophy and the sciences, which include mathematics and computing;

• *The development of logic and its relationship to ‘dialectics’ in the work of multiple thinkers.* Here we explore the political repercussions as well as the influence of ideology on the epistemological choices made in sciences along the materialism/idealism divide;

• *The repeated appearance of the principle that nothing comes from nothing.* This originates in ancient Greek philosophy and is reprised and animated in the development of conservation laws, culminating in the understanding that these all derive from the properties of symmetry. This underpinning metaphysical dilemma is insidiously present in the sciences, political philosophy and economics. As one of the pillars of the organization of the European history of ideas, this unintended, often inadvertently present, atavism has moulded much of what we now know as contemporary science, technology and socio-economic theory.

• *The nature of time in physics, biology and reflections on human society, mainly in philosophy but also in the social sciences.* A hermeneutics of the ideas of time running through these disciplines offers a fresh view on the idea of historical materialism, disentangled from the dogmas of the diamat, that is, dialectical materialism as set in the USSR.

Our focus is the explicit idealism criticized by Lenin and Einstein and the implicit Hegelian idealism that influenced Soviet diamat. Starting with the origins of ma-

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在牛顿物理学和牛顿世界观中的复兴。我们说明马克思如何借鉴原子论者，为他批判黑格尔唯心主义提供基础。然后考察历史唯物主义的兴起和生物学在达尔文主义下的转型。我们进一步探讨达尔文主义更广泛的认识论影响，这种影响可见于马克思主义以及若干科学中的其他哲学流派。

terialism in ancient Greek thinkers, we explore its revival in the physics of Newton and the Newtonian world view. We show how Marx drew on the atomists to provide the basic foundation for his critique of Hegelian idealism. We then look at the rise of historical materialism and the Darwinian transformation of biology. We further explore the wider epistemological effect of Darwinism, which can be traced in Marxism and other philosophical strands and schools of thought in a number of sciences.

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19 世纪末 20 世纪初，围绕原子爆发了激烈的哲学争论，如今这常被遗忘。植根于科学解释模型中的唯心主义认识论，阻止人们把原子论接受为超出边缘推测的东西。diamat 教条及其核心的黑格尔权威，对苏联科学尤其是数学和计算的发展产生了抑制作用。在自称唯物主义认识论的最深层信条中，唯心主义内核影响了 20 世纪社会主义世界技术发展的步伐。我们以此为例，说明科学作为隐蔽哲学的影响：正如马克思所论证的，这种隐蔽哲学会不可避免地产生一种唯心主义，并影响技术以及更广泛科学发展的路径。这是科学受自身形而上学选择之苦的症状，即使它明确把自己定位为反形而上学。矛盾的是，我们在苏联社会主义科学传统中比在 20 世纪至 21 世纪西方科学发展中更注意到这种影响。

The late nineteenth and early twentieth centuries saw intense philosophical controversy over atoms, something which tends to be forgotten now. Idealist epistemologies, embedded in scientific models of interpretation, prevented the acceptance of atomism as anything more than a marginal speculation. The dogma of diamat, and the authority of Hegel at its centre, had an inhibiting effect on the development of sciences, particularly mathematics and computing, in the Soviet Union. The idealist core in the deepest tenets of what purported to be a materialist epistemology affected the pace of technological development in the socialist world of the twentieth century. We explore this as an illustration of the effect of science as covert philosophy, the latter unavoidably yielding a form of idealism, as argued by Marx, on the pathways of technological and broader scientific development. This is a symptom of science suffering from its own metaphysical choices, even when it positions itself decisively as anti-metaphysical. Paradoxically, we note such an effect more in the tradition of Soviet socialist science than in Western twentieth to twenty-first century scientific development.

原子能开发出来以后，已没有人怀疑原子的真实存在，但在 1900 年前后，原子的实在性还广受质疑。我们追溯这场哲学争论的政治和理论背景。我们考察人们对于原子论的“机械论”世界观中所蕴含的“粗糙的物质性”（马克思，《资本论》第 1 卷第 1 章）的反感；这种物质性与一切存在所依据的任何“空灵本原”（ἀρχή）相对。随后我们论证，正是这种反感，使那些公认属于唯物主义的哲学和科学仍然留着一扇后门，其中也包括那些甚至不愿在唯心主义与唯物主义之间作出形而上学选择的学科。“空灵本原”这一隐喻指向从古希腊到当代数学和计算机科学的一系列观念。这既可能表现为一个悬而未决的问题，也可能表现为干脆坚持“一切都是语言，外部现实并不存在”——至少不是以一种应当被认真对待的方式存在；后结构主义反复念诵的口号便是一例。

After the development of atomic energy, nobody doubted that atoms were real, but back in 1900 their reality was widely questioned. We trace the political and theoretical background to this philosophical controversy. We identify the aversion towards ‘coarse materiality’ (Marx, Cap. V. 1, Ch. 1) nesting in the ‘mechanistic’ world view of atomism, as opposed to any ‘ethereal’ principle (ἀρχή) of all existence. We then argue that this aversion is why the admittedly materialist philosophy and sciences kept an open back door, including those which did not even bother to make the metaphysical choice of idealism versus materialism. The metaphor of ‘ethereal principle’ refers to a plethora of notions dating from Greek antiquity to contemporary mathematics and computer sciences. This can range from an open question to simply maintaining the position that ‘all is language and the outside reality does not exist’, at least not in a way that should count, as in the post-structuralist mantra, for example.

方法

Approach

我们对唯物主义的论述尽可能以一手资料为依据；原始材料不是英语时，则使用译本。我们基本上避免与那些阐释我们所引哲学家和科学家的人展开讨论。我们也基本上避免采用范围更广的历史叙述，

Our account of materialism is drawn, as far as possible, from primary sources, using translations where original material is not in English. We have mostly avoided engaging with interpreters of the philosophers and scien-

除非找不到一手资料，或这些叙述提供了没有争议的概述。

tists we cite. We have also mostly avoided wider historical accounts, except where we could not locate primary sources or where they offered uncontroversial summaries.

我们承认自己的“西方”哲学取向，也承认未介入当代科学史领域的争论。我们希望为理解唯物主义带来一种新视角，引导读者像我们努力做到的那样，直接接触原始论证。如果有人指责我们无知或修正主义，那就由他们去吧。当然，我们乐于接受纠正，并进一步讨论自己的立场，所秉持的是达成共同理解的精神，而不是争胜计分的经院式论辩。

We acknowledge our ‘Western’ philosophical orientation and our lack of engagement with contemporary debates in the history of science. We wish to bring a fresh perspective to the understanding of materialism, which leads our readers to engage directly with original arguments as we have tried to do. If we are accused of ignorance or of revisionism, then so be it. We are happy, of course, to be corrected and to further debate our positions in the spirit of coming to a shared understanding rather than point-scoring scholasticism.

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为了始终聚焦唯物主义在哲学和自然哲学中的发展，我们有意作出了一些省略。尤其是，我们不讨论那些“重大”的哲学问题：我们从哪里来？为什么在这里？为什么是这样？又将走向何方？

To maintain the core focus on the development of materialism in philosophy and natural philosophy, we have made a number of intentional omissions. In particular, we do not address the ‘big’ philosophical questions: where do we come from; why are we here; why are we like this; where are we going?

除非其他世界观曾直接促进或阻碍唯物主义，我们不予讨论。尤其是，总体而言，我们很少谈及信仰或宗教。最后，我们也不讨论这里阐述的唯物主义所具有的道德、伦理、社会或直接政治后果。也许最重要的一项省略，是没有探讨意识的性质。我们对唯心主义的拒斥，必然使我们把意识视为物质系统的一种属性，无论这种系统是否属于人类。

We do not consider other world views, other than where they directly contributed to or hindered materialism. In particular, overall we say little about belief or religion. Finally, we do not consider the moral, ethical, social or immediate political consequences of the materialism we elaborate. Perhaps the most significant omission is that of the nature of consciousness. Our rejection of idealism necessarily leads us to situate consciousness as a property of material systems, whether or not human.

请注意，引文均照原文逐字引用，不对其中诸如带有性别色彩的语言作出评论。斜体沿用原文。人物的生卒年往往直接取自维基百科。

Note that quotations are verbatim, without comment on, say, gendered language. Italics are as in the original. Dates of birth and death are often purloined from Wikipedia.

各章

Chapters

下面逐章简要概述本书内容。

We next briefly survey the book’s contents, chapter by chapter.

第 2 章对古希腊所提供的概念材料及其开创的跨千年交流的语文学与哲学考察

Chapter 2. Philologico-philosophical examination of the conceptual material proffered by Greek Antiquity and the trans-millennial exchanges it has foregrounded

• 我们指出希腊哲学中唯心主义与唯物主义之间张力的根基：这种张力历经数世纪仍然存在，其基础前提仍与古代确立的前提相同。亚里士多德和原子论者既试图使变化成为可思考的对象，又仍困在必须遵守埃利亚学派“无中不能生有”原则的要求之中。原子论者成功地表明，“虚空”或“虚无”是物质的一个方面，一切存在的本原都是物质性的——即不可分割的物质量，atomoi，也就是“不可切分者”；伊壁鸠鲁派的卢克莱修 • 卡鲁斯的著作尤其如此。

• We identify the foundations of the tension between idealism and materialism in Greek philosophy: a tension that has persevered throughout the centuries while retaining the same grounding premises identified in Antiquity. Both Aristotle and the atomists remain trapped in observing the Eleatic Principle, ‘nothing comes from nothing’, while trying to enable a thought of change. The atomists succeed, most notably in the work of the Epicurean Lucretius Carus, in demonstrating that ‘the void’ or ‘nothingness’ is an aspect of matter and that

the principle of all existing is material –the indivisible quantity of matter, the atomoi or the uncuttables.

第 3 章古典原子论

Chapter 3. Classical atomism

p. 4–5

• 我们考察从原子论者到 18 世纪法国机械学派完成牛顿世界观的历程。阿基米德式机械唯物主义对牛顿影响巨大，牛顿由此提出自然哲学的方法。法国学派通过能量守恒和最小作用量原理扩展了牛顿力学。

• We explore the atomists to the completion of the Newtonian world view by the eighteenth-century French mechanical school. Archimedean mechanical materialism was enormously influential on Newton, who enunciated a methodology of natural philosophy. The French school elaborated on Newton’s mechanics by postulating the conservation of energy and the principle of least action.

第 4 章辩证法、唯物主义与变化：从伊壁鸠鲁经亚里士多德到马克思

Chapter 4. Dialectics, materialism, change from Epicurus to Marx via Aristotle

p. 5

• 我们重建辩证法作为一种本体论范畴在欧洲哲学中出现的过程，并将其与古代的起点区分开来：在古代，尤其在亚里士多德关于我们今天称为逻辑学的论著中，辩证法原本是一种方法或本体论范畴。我们在共产主义的 diamat 学说中所见的唯物辩证法，其起源可以追溯到黑格尔，却不能再上溯到更早的时代。马克思对这一术语的使用更接近亚里士多德，而非受黑格尔启发的 diamat。我们把辩证法和唯物主义视为马克思思想中的两个不同范畴，也在马克思的思想来源——原子论和黑格尔——中将二者区分开来。

• We reconstruct the appearance of dialectics as an ontological category in European philosophy, as opposed to its origin as a category of method or ontology found in Antiquity, and in particular in Aristotle’s writings on what we now call logic. The origins of materialist dialectics, as we find it in the communist doctrine of diamat, can be traced back to Hegel but not further in the past. Marx’s use of the term is closer to that of Aristotle’s than to the Hegel-inspired diamat. We look at dialectics and materialism as two distinct categories in Marx, but also in what served as Marx’s inspiration, namely, atomism and Hegel.

第 5 章历史唯物主义与机械唯物主义

Chapter 5. Historical and mechanical materialism

• 我们介绍应用于社会和生物变化的唯物主义。历史唯物主义被视为斯密与恩格斯各自独立作出的共同发现。马克思的《资本论》受到牛顿关于守恒定律，以及作用力与反作用力大小相等、方向相反的观念的影响。恩格斯将历史与生物学中目的论的消解归功于马克思和达尔文二人各自独立的贡献。我们把达尔文之前以人类为中心的目的论生物学解释，与达尔文提出的因果模式相对照。我们考察达尔文主义如何取消人与动物之间的截然区分，并考察人类的情感和行为特征如何像其生理特征一样，有着深厚的哺乳动物根源。原子论在 19 世纪初复兴。道尔顿等人的化学进展，使之成为化学研究的实际基础。这又为理解热、压力等现象奠定了基础；蒸汽革命正建立在对这些现象的理解之上，而这也促成了热力学关键概念的发展。

• We introduce materialism as applied to social and biological change. Historical materialism is treated as the independent co-discovery of Smith and Engels. Marx’s *Capital* was influenced by Newtonian ideas of conservation laws and of equal and opposite actions and reactions. Engels credited Marx and Darwin with jointly and independently displaced teleology from history and biology. Pre-Darwinian anthropocentric teleological accounts of biology are contrasted with the mode of causality proposed by Darwin. We consider Darwinism’s abolition of the human as distinct from the animal, and how human traits of emotion and behaviour, as much as physiology, have deep mammalian roots. Atomism revived at the start of the nineteenth century. Advances in chemistry by Dalton and others established it as the working basis of chemistry. This was the basis for understanding heat, pressure, etc., on which the steam revolution was based, and led to the development of the key concepts of thermodynamics.

第 6 章唯心主义的复兴与回应

Chapter 6. Idealist reprise and responses

• 我们讨论 20 世纪物理学中唯物主义与唯心主义争论的关键分界。原子论复兴恰逢社会主义兴起，一些国家担心新的原子论世界观会鼓励无神论唯物主义和社会主义。杰出原子论者玻尔兹曼遭到以马赫为首的实证主义/工具主义学派强烈反对。1905 年，爱因斯坦发表三篇论文，共同彻底改变我们对自然界的理解，并决定性地驳斥马赫主义。这些进展导致量子力学，并通过海森堡学派导致一种对新物理学的唯心主义解释，该解释大量借鉴马赫主义。

• We cover the key dividing lines in the debate between materialistic and idealistic schools of twentieth-century physics. The revival of atomism coincided with the growth of socialism, and there was, in some countries, a fear that atheistic materialism and hence socialism would be encouraged by the new atomistic account of the world. The prominent atomist Boltzmann experienced strong opposition from the positivist/instrumentalist school of philosophy and natural philosophy headed by Mach. In 1905, Einstein published three papers that together revolutionized our understanding of the natural world and decisively repudiated Machism. These advances though led to quantum mechanics and, via the Heisenberg school, an idealist interpretation of the new physics that drew heavily on Machism.

第 7 章逻辑与唯物主义

Chapter 7. Logic and materialism

• 亚里士多德认识论中三段论逻辑与辩证法的统一，在宗教改革前深刻影响欧洲思想以及天主教框架下的教育。此后，英国传统拒绝逻辑和辩证法，转向归纳、经验主义和唯物主义；随后逻辑被恢复并置于形式化基础上。数学的逻辑基础似乎触手可及。

• The unity of syllogistic logic and dialectics in Aristotle’s epistemology heavily influenced European thought and hence education, framed by Catholicism, until the Reformation. Thereafter, following the rejection of both logic and dialectics in favour of induction, empiricism and materialism in the British tradition, logic was rehabilitated and placed on a formal footing. It seemed that the logical foundations of all mathematics were within reach.

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第 8 章逻辑与辩证唯物主义

Chapter 8. Logic and dialectical materialism

• 黑格尔对亚里士多德思想中辩证成分的强调，深刻塑造了 19 世纪及 20 世纪初马克思主义者对逻辑的态度。1925 年恩格斯的《自然辩证法》出版后，苏联围绕辩证唯物主义内部形式主义（此时已与逻辑相联系）与辩证法之间的平衡，展开了激烈斗争。这导致一类被混同归并为“孟什维克化的唯心主义”的思想受到严厉谴责，也导致公式化的“diamat”的形成；直至斯大林去世，这种 diamat 一直规定着官方共产主义哲学的面貌。

• Hegel’s emphasis on the dialectical component of Aristotelian thought strongly framed the attitude of nineteenth- and early twentieth-century Marxists towards logic. Following the publication of Engels’ *Dialectics of Nature* in 1925, there was intense struggle in the Soviet Union over the balance in dialectical materialism between formalism, now associated with logic, and dialectics. This led to the anathematizing of a conflated Menshevizing Idealism and the formulaic ‘diamat’ that defined official communist philosophy until the death of Stalin.

第 9 章逻辑危机与反形式主义的顶峰

Chapter 9. The crisis in logic and the apotheosis of anti-formalism

• 20 世纪初，逻辑学的进展使其能够将自身形式化，由此发现了其核心深处的悖论。与此同时，这些进展也为计算机的发展奠定了基础，并对我们如何将现实刻画为彼此建模的反映系统，产生了深远影响。

• In the early twentieth-century, advances in logic enabling it to formalize itself led to the identification of profound paradoxes at its heart. At the same time, these advances underpinned the development of computers,

with profound implications for how reality may be characterized as reflective systems that model each other.

第 10 章语言、自动机与意义

Chapter 10. Language, automata and meaning

• 我们把可计算性作为唯物主义的统一框架，整合模型论、语言游戏、自动机理论和形式语言。马尔可夫的构造主义被视为对直觉主义的唯物主义重构，为现实的有限主义解释提供基础。

• We present computability as a unifying framework for materialism, integrating model theory, language games, automata theory and formal languages. Markov’s constructivism is considered as a materialist reformulation of intuitionism, underpinning a finitist interpretation of reality.

第 11 章辩证唯物主义与随机唯物主义

Chapter 11. Dialectical and stochastic materialism

• 我们阐述信息论的含义：信息是客观的、物理的，完全独立于人类意识。我们探讨信息与随机性的关系，表明表面偶然性受规则支配。最后，我们探讨量、质和相变概念。

• We elaborate on the implication of information theory, which is that information is something objective and physical, quite independent of human consciousness. The relationship between information and randomness is explored, demonstrating that apparent chance is governed by rules. Finally, we explore the notions of quantity, quality and phase changes.

第 1 章原注 / Chapter 1 Notes

↑1. J. 博斯韦尔：《塞缪尔·约翰逊传》（The Life of Samuel Johnson），第 1 卷（Everyman’s Libray，J,. M. Dent & Sons Ltd，1906），第 292 页。

〔校注：底本中的 Libray 与 J,. 拼写、标点照录；未独立核验该引文所据版本。〕

1 J. Boswell, *The Life of Samuel Johnson*, vol. I (Everyman’s Libray, J,. M. Dent & Sons Ltd, 1906), 292.

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第 2 章探索古希腊所提供的概念材料及其影响的跨千年交流

Chapter 2. EXPLORING THE CONCEPTUAL MATERIAL PRESENTED BY GREEK ANTIQUITY AND THE TRANS-MILLENNIAL EXCHANGES IT HAS INFLUENCED

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细读巴门尼德——本体论之父，因而也是我们数世纪以来所知的（欧洲）[1]哲学之父——提出的公设，便会发现：西方理性思想诞生之初提出的谜题从未真正解决。芝诺悖论至今仍是悖论——对哲学而言更是如此，而非对科学——而“无中不能生有”在今日仍是物理—形而上学定律，一如它在公元前六、五世纪的欧洲文明中那样。前苏格拉底哲学家接受这一原则；甚至那些主张变化与运动可能的人，如原子论者，也必须设法绕过它，而不是挑战它。在埃利亚学派体系中——直到亚里士多德都未受挑战——“无中不能生有”意味着：变化不可能，虚空不存在，万物始终以坚实不变的形式存在。“虚空”在本体论意义上意味着虚无；而运动则意味着本体论意义上的“生成”，这一论述诉诸的是本体论，而不是自然哲学。运动被理解为成为一个“ousia”（本质或实体），这与“本质具有不可毁灭的永恒性质——因此始终已经存在”这一原则相矛盾。变化的观念违反了埃利亚学派的基本前提，即其公理中的公理：无不存在，虚无不存在，而“成为”将意味着占据先前“空”的（虚无的）空间。在巴门尼德那里，绝对意义上的虚无、非存在与虚空被等同起来，并在整个古代一直如此；不过也有一些例外，即直接与这种逻辑对抗的例子，我们将在本书后面讨论。至于原子论者，他们找到了一种弯曲巴门尼德规则的方法，而不是挑战或反驳它。

A close reading of the postulates put forward by Parmenides –the father of ontology and thus of (European)[1] philosophy as we have known it for centuries –discloses the fact that the riddles posed at the birth of Western rational thought have never been resolved. Zeno’s paradox is still a paradox –more so to philosophy than to the sciences –and ‘nothing comes from nothing’ is as much of a physical-metaphysical law nowadays as it was in sixth and fifth century BC of the European civilization. The principle was accepted by the pre-Socratics, and even those who advocated that change and movement were possible, such as the atomists, had to find a way around it instead of challenging it. In the Eleatic system, unchallenged till Aristotle, ‘nothing comes from nothing’ meant: change was impossible, empty space didn’t exist and everything had always existed in solid and unchangeable form. ‘Empty space’ meant nothingness in the ontological sense, whereas movement implied coming into being in the ontological sense argued in terms of ontology instead of physical philosophy. Movement was taken to be becoming an ousia, an essence or substance, which contradicts the principle that essence is of indestructible everlasting nature –and has, thus, always already existed. The idea of change violates the principle premise, an axiom of the axiom of the Eleatics – nothing is not, nothingness does not exist and becoming would mean *and thus taking up previously ‘empty space’ (nothingness)*. Nothingness, non-being and empty space (or void) in the absolute sense are equated in Parmenides and continue to be so throughout antiquity, with some exceptional examples of direct confrontation with this logic that we will discuss further in this book. As for the atomist approach, it found a way to bend the Parmenidean rule rather than challenge or refute it.

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因此，后来几个世纪的自然哲学和科学所处理的空间范畴，既缺席于巴门尼德的推理，也缺席于古希腊那些反对他但从未挑战该原则及其等式的人——即“虚空等于本体论意义上的虚无”（意译）。最早挑战这一公设的，且仍是罕见例子的，是亚里士多德；他基于该公设本身的性质或本质提出质疑：每个谓词都与某种“某物”相关（希腊语 ousia 即 to ti，意为“某物”或“什么”）。此外，ousia 概念植根于多样性；按照亚里士多德的范畴体系，存在（to on）并非只有一种 ousia。

Thus, the category of space as treated by the natural philosophy and sciences in the centuries to follow was absent from the Parmenidean reasoning, as well as from the reasoning of those who opposed it in Greek Antiquity but never challenged the principle and the equation at issue, namely that ‘empty space equals nothingness in the ontological sense’ (a paraphrase). The first to challenge this postulate, remaining a rare example, was Aristotle, and he did so on grounds of the quality or nature of the postulation itself – every predicate relates to a certain ‘something’ (*ousia* as *tò tí* in Greek, a ‘something’ or a ‘what’). Also, the notion of *ousia* is grounded in multiplicity; according to Aristotle’s categorical system, there is not only one *ousia* of Being (*tò ōn*).

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并不存在超越事物本身的运动。变化总是相对于实体、数量、质量或位置而发生的。（亚里士多德，《物理学》200b33–201a3）[23]

There is no such thing as motion over and above the things. It is always with respect to substance or to quantity or to quality or to place that what changes. (Aristotle, *Phys.* 200b33–201a3)[23]

亚里士多德在“纯粹理性”与“物自体”之间进行了某种准康德式划分——推理的界限和条件从语言的可能性本身中寻找——修辞学、逻辑学、分析科学以及对三段论的理解，都是日常语言进一步精确化和形式化的结果。（术语在某种程度上已经形式化，且与其日常含义难以区分——例如“虚空”“非存在”等——这种现象自前苏格拉底时期就已存在。）话虽如此，亚里士多德在其全部著作中似乎都在论证：（人类）知识的主体已经完善了用来解释外部现实的概念工具。尽管如此，二者仍受各自内在法则支配，是截然不同的事物。正因如此，ousia 是关于“某物”的（希腊语 to ti，即某种“什么”），并且总是“被用来述说许多事物”[4]——这再次挑战了巴门尼德把 Being、to on 和 ousia 视为一体的观点——但它并不等同于那个“某物”。正因如此，我们把亚里士多德对哲学和科学思想发展的贡献比作康德的贡献。此外，我们认为，这同样是一种借助形式推理，并援引经验证明和实验权威，从而摆脱形而上学（本体论）权威的“前科学”突破。

Aristotle carries out a quasi-Kantian division between ‘pure reason’ and the *noumena* –the limits and conditions of reasoning are sought for within the very possibility of language –rhetoric, logic, the sciences of analysis and the understanding of syllogism are the result of further precision and formalization of everyday language. (A certain degree of formalization of terminology indistinguishable from its meaning in daily life –such as ‘the empty’, ‘that which is not’ etc. –has existed since the pre-Socratics.) That said, Aristotle seems to argue throughout his opus that the subject of (human) knowledge has perfected its conceptual tools serving to explain the outside reality. Nonetheless, the two are radically different things governed by their own immanent laws. That is why an *ousia* is of ‘something’ (a certain ‘what’ – *to ti* in Greek) and is always ‘spoken of many’[4] –another provocation to Parmenides’ Being, *to on* and *ousia* as one –but it is not the same as that something. It is for this reason that we liken Aristotle’s contribution to the development of philosophical and scientific thought to that of Kant. Furthermore, we would argue that it is also a proto-scientific breaking away from the authority of metaphysics (ontology) by virtue of employing formal reasoning, and also invoking the authority of empirical proof and experimentation.

虚空作为运动的先决条件

Empty space as a precondition of movement

从巴门尼德到晚期原子论者——以及在与教条有显著偏离时包括亚里士多德——乃至整个古代，要产生运动，就必须存在未被任何程度实体填充或占据的虚空。无论是“（某）物”（to ti）这一物理概念，还是诸如 toon——即存在，或任何形式的实体/本质/同一性（ousia）——这样的本体论概念，对古代思想家而言，运动不过是某物或某实体出现在此前空无一物的场所的一种表象：“因为虚空就是无”[原文：τὸ γὰρ κενεὸν οὐδέν ἐστιν]，梅利索斯说（DK 30B7.7）。[5]即使空间只在极短瞬间被（绝对地）“清空”，虚无的可能性——进而虚空的可能性——也被允许。这被视为违反巴门尼德法则“无中不能生有”。这种可能性会与最高真理、最纯粹的同一反复相矛盾——“无中不能生有”，或者说“存在即存在，不能是非存在”——这似乎在古代哲学中被普遍接受，即使像亚里士多德那样对其提出质疑。变化也是运动，因为它涉及成为、生成和产生，因此是某种本质上显然不在的东西的产生。对变化——进而对成为或生成——的严格巴门尼德式反对是：如果“无中不能生有”，那么存在（to on）或 ousia 已经以其绝对形式存在，因此成为被排除。于是万物始终如其所是地存在，不变、不动、坚实——不允许任何虚空或虚无的裂缝。此外，万物不朽——某物不能转化为无，因为无不能成为某物。变化只是幻象；可见世界是幻象、梦境、幻觉，正如苏格拉底在临终时所解释的（柏拉图，66e–67a）。[6]这种观点一直作为主导观点延续到新柏拉图主义、中世纪哲学，以及更近期的“新理性主义”对理性——即科学思维——的迷恋，与威尔弗里德·塞拉斯所说的“现实的显

From Parmenides to the late atomists –and, with a significant aberration from the dogma, through Aristotle – as well as throughout antiquity, for there to be movement, there has to be empty space not filled with or occupied by a body of any sort of degree. Be it a physical notion of a ‘(some)thing’ (tò tí) or an ontological concept such as *tò ōn* –that is, the Being or substance/essence/identity of any sort (*ousia*) –to the Ancient thinkers, movement is simply an appearance of a thing or an entity at a place where there was nothing prior to it: ‘for the empty is nothing’ [in the original: τὸ γὰρ κενεὸν οὐδέν ἐστιν], argues Melissus (DK 30B7.7).[5] Even if the space is ‘vacated’ (in an absolute way) for only an infinitesimally short moment of time, the possibility of nothingness, and thus empty space, is allowed. This is considered a breach of the Parmenidean law ‘nothing comes from nothing’. Such a possibility would contradict the utmost truth, the purest tautology of them all –nothing comes from nothing, or, being is being and cannot be not-being –that seems to be universally accepted in ancient philosophy even when problematized, for example, by Aristotle. Change is movement too, *as it is a matter of becoming, generation and generation*, and thus coming to being of something which is apparently not in its essence. A strictly Parmenidean objection to change, and thus to becoming or generation, would be that if nothing comes from nothing, the being (*tò ōn*) or the *ousia* is already there in its absolute form, and thus becoming is precluded. Everything, therefore, has al-

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现图景”相对。（我们这里指对塞拉斯的一种特定接受[7]，而不是他本人承认“民间推理”与科学思维连续性的著作。[8]）哲学的诞生、存在概念的诞生以及完全理性思维（同一反复）的诞生，恰与一种世界观同时诞生：它宣称直接经验的现实是幻象。它也恰与“身体恐惧症”的诞生同时发生，因为真理不可能也不存在于腐朽肉体的可见世界中，而存在于完美、不可见、没有任何身体或物质的世界中（柏拉图，《斐多篇》，64c–65c）。让我们考察巴门尼德论证的诸范畴，它们标记了数世纪乃至数千年的唯心主义和反唯物主义，并看看它们如何形成一个界定明确、不可穿透的立场。

ways existed as it ever was, unchanged, unmoving, solid – no crack of empty space or nothingness is allowed. Also, everything is undying –something cannot be transformed into nothing since nothing cannot become something. Change is but an illusion; the visible world is an illusion, a dream, a hallucination, as Socrates explains at his deathbed (Plato, 66e – 67a).[6] This view has persisted as the dominant one well into Neoplatonism, medieval philosophy and, more recently, in the neorationalist fetishizing of rationality, that is, of scientific thought as opposed to ‘the manifest image of reality’ as per Wilfrid Sellars. (We are referring here to a particular reception of Sellars[7] rather than his own writings that admit continuity between ‘folk reasoning’ and scientific thought.[8]) The birth of philosophy, of the notion of being and of perfectly rational thought (tautology) coincides with the birth of a world view that declares the immediately experienced reality an illusion. It coincides with the birth of somatophobia too, as the truth cannot and does not lie in the visible world of rotting flesh but in the perfect, invisible world without bodies or matter of any sort (Plato, *Phaedo*, 64c – 65c). Let us consider the categories of the Parmenidean arguments that have marked centuries and millennia of idealism and anti-materialism and how they form a perfectly circumscribed and impenetrable position.

“无中不能生有”这一原则至今有效：其科学与哲学本体论阐释

Nothing comes from nothing, a principle that is valid to this date: Scientific and philosophical-ontological accounts of it

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迄今为止，从牛顿到霍金，这一原则——“无中不能生有”——似乎在科学中未受质疑。哲学似乎允许更多可能：随着启蒙运动后的思想变革，哲学越来越成为独立于科学的沉思散文，也就越来越允许自相矛盾的立场和与科学直接矛盾的观点。因此，“无中不能生有”在现代哲学中并不必然是普遍共识。虚无本身似乎也是一种存在；根据某些哲学家的说法，它在本体论上存在，尤其在现代性开始之后；试看存在主义、虚无主义以及黑格尔的非存在概念。[9]

〔校注：此处底本拼作 Hawkins，原书第 10 页同一论述使用 Hawking；中文沿用所供译稿的“霍金”，英文保留底本拼写。〕

To this date, from Newton to Hawkins, this principle –‘nothing comes from nothing’ –seems undisputed by the sciences. It seems that philosophy has allowed for more possibilities: the more it has become meditational prose, independent of the sciences following the intellectual overhaul of the Enlightenment, the more paradoxical positions and those in direct contradiction with science have been permitted. Thus, ‘nothing comes from nothing’ is not necessarily a matter of universal consensus for modern philosophy. Nothingness in its own right is also being, it seems,and it is ontologically existent according to some philosophers, in particular with the beginning of modernity; consider existentialism, nihilism and Hegel’s notion of non-being.[9]

这种对科学讨论的漠视和对经验材料的蔑视，表明本章导言中所述的定义希腊哲学的“可见”与“不可见”、有形与无形、物质与观念之间的原始二分仍在延续。柏拉图认为可见世界是幻象，经验记录不足以对存在作出合理说明，至少不能被视为绝对。

This absence of heed for the scientific discussions and contempt for the empirical point to the continuation of the originary division between the ‘visible’ and ‘invisible’, bodily and incorporeal and physical and ideal defining Greek philosophy, as noted in the introduction to this chapter. Plato argued that the visible world is an illusion and that the empirical record does not amount to a viable account of making sense of being, at least not one seen as an absolute.

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哲学从古至今一直以人类为中心：马克思在《对黑格尔哲学的总体批判》（收录于《1844 年经济哲学手稿》）中阐明，所谓的绝对精神不过是自哲学诞

Philosophy has always been anthropocentric, from antiquity to this day: in his ‘Critique of Hegel’s Philosophy in General’ (part of the *Economic and Philo-*

生以来一切哲学中人类中心主义狂妄的顶点，实质上是普遍利己主义者的概念。[10]人类中心主义也根植于费尔巴哈的唯物主义中，而正是这一点使它成为唯心主义——事实上，一切以主体性为中心的思想，即以人类主体性为中心的思想，归根结底都是唯心主义。[11]正因如此，马克思认为哲学历来都是唯心主义的，因此他提议通过采取第三方视角来摆脱哲学，把哲学转变为科学。因此，在马克思看来，科学的前提是放弃以主体性为中心的思想，并模仿第三方的（想象中的）视角。[12]当代哲学——与“分析哲学”（据称占据真正哲学的霸权地位）相对的“大陆哲学”——尤其是批判理论，鄙视经验主义。它对技术感兴趣，只是为了理论化其社会效果[13]；对理论数学好奇（例如前述新理性主义者），但可量化性、通过经验归纳进行的证明，却遭到隐晦蔑视。正是这种对物质（证明）的蔑视，使它至今仍是哲学，与巴门尼德和柏拉图式的“身体恐惧”叙事并无二致，即使它选择以其他名称自称（理论、后哲学等）。

sophical Manuscripts of 1844), Marx explains that the Absolute Spirit is simply the apex of the anthropocentric pretensions of all philosophy since its inception, amounting to the notion of the universal egoist.[10] Anthropocentrism is embedded in Feuerbach’s materialism too, and that is what renders it idealism –in fact, all subjectivity-centred thought –amounting to human-subjectivity-centred thought–comes down to idealism.[11] It is for this reason that, according to Marx, philosophy has always been idealist, and he thus proposes an exit from it by virtue of assuming the third party’s perspective, transforming philosophy into science. Therefore, in Marx, science is premised on the abandonment of subjectivity-centred thought and miming the third party’s (imagined) perspective.[12] Contemporary philosophy, the kind one calls ‘continental philosophy’ as opposed to ‘analytic philosophy’ (supposedly occupying the hegemonic position of philosophy proper), and critical theory in particular, despises empiricism. It displays interest in technology to simply theorize its social effects[13] and curiosity about theoretical mathematics (such as among the aforementioned neorationalists), but quantifiability, proof executed through empirical induction, is subject to tacit contempt. The contempt for the material (proof) is what makes it philosophy to this day, not much unlike the Parmenidean and Platonic somatophobic narratives, even when it chooses to call itself differently (theory, post-philosophy etc.).

进入 21 世纪后，那种源自笛卡尔和柏拉图主义对物理事物的蔑视，仍在思辨现实主义启发的新理性主义、加速主义以及后人类主义网络理论中隐含或明示；这些理论往往不过是变相的超人类主义，也包括异质女性主义（xenofeminism）。身体和物理事物也属于运动、变化和衰败的范畴，被视为应予废除的幻象，以便纯粹理智——黑格尔的精神——以人工智能和人机奇点形式统治。自前苏格拉底哲学以来，讨论变得更精致复杂，但其根基似乎仍是同样的关切，因为若移除这些奠基前提，整个哲学大厦就会崩塌：存在是核心概念，本体论是哲学的归宿，形式理性与本体论一致，从而产生形而上学；而对物质世界及其应用的研究则被视为次要，正如“可见世界”一样。

Well into the twenty-first century, the old Cartesian and Platonist contempt for the physical is also implied or declared, as one can note in the speculative-realism-inspired neorationalism, accelerationism and posthumanist cyber theory that more often than not amounts to simply transhumanist but also xenofeminism Body, and the physical, is also of the register of movement, change and decay, and it is an illusion to be abolished so that pure Intellect –Hegel’s Spirit –would rule in the form of AI and machine-human singularity. Discussions have become more elaborate and more complex since the times of pre-Socratic philosophy, but they seem to be underpinned by the same concerns since the founding premises cannot be removed without the whole edifice falling apart: Being is the central notion, ontology is what philosophy comes down to, formal reason is aligned with ontology and thus produces metaphysics and the study of the material world and its application is of inferior importance, as is the ‘visible world’. p. 10

“无中不能生有”——或存在来自存在——这一原则在科学探索中有意义，尤其在物理学（牛顿、霍金等）和化学中；这无需任何曲折的悖论和矛盾论证，不像埃利亚学派主张导致的困境。亚里士多德用其范畴论和逻辑揭示了埃利亚学派主张的不一致，从而证明将人类推理（范畴应用）与存在本身混同是一种谬误。通过把 ousia 与质量、数量、位置和时间区分开，亚里士多德证明谓述总是关于某种特定事物；由于可能的谓述实际上无穷无尽，因此也存在多种 ousia：“简言之，存在总是以复数形式被述说”[τὸ ὂν τὸ ἁπλῶς λεγόμενον λέγεται πολλαχῶς]（亚里士多德，《形而上学》E.2 1026a33–34）。作为这

Nothing comes from nothing –or being becomes out of being –is a principle that makes sense in scientific explorations, in particular in physics (Newton, Hawking etc.) and in chemistry, and this is possible without any convoluted affirmations of paradoxes and contradictions, unlike the *aporia* to which the Eleatic proposal led. Aristotle divulges the inconsistencies of the Eleatic proposal in terms of his categories and logic and thus demonstrates that the collapse of human reasoning (application of categories) and being itself is a fallacy. By distinguishing *ousia* from quality, quantity, location and time, Aristotle demonstrates that predication is always about a certain p. 10–11

一论证的一部分，亚里士多德考察了人类思维运作、言语方式以及唯一可能可信且合乎逻辑的言语模式中可能存在的矛盾、不一致和悖论——而不是存在本身中的矛盾。这是与整个巴门尼德之后传统的革命性决裂：它从言语的逻辑一致性出发，即人类能够控制和完善的部分，而不是从存在本身出发。人类语言反映现实和外部世界的一致性，但后者本身似乎是封闭的，只有我们的范畴体系和逻辑发展到一定程度时才可理解。如果撞上矛盾之墙，那是人类推理的问题，而不是存在、外部世界或本质超出人类思维的现实的问题。在《物理学》中，亚里士多德写道：

something, and as there is a practically endless or unbound multiplicity of possible predications, there is thus a multiplicity of *ousiai*: ‘the being, put simply, is always said in plural’ [τὸ ὂν τὸ ἁπλῶς λεγόμενον λέγεται πολλαχῶς] (Aristotle, Meta. E.2 1026a33–34). As part of the elaboration of this argument, Aristotle examines possible contradictions, inconsistencies and paradoxes in how the human mind might work, or in the way one speaks and in the only possibly credible modes of speech that make sense – not in Being itself. This is a revolutionary break from the entire post-Parmenidean tradition –one which departs from the logical consistency of speech, that which is within the reach of human control and perfection, not from ‘Being itself’. Human language reflects the consistency of reality and the outside world, but the latter seems to be in and of itself foreclosed and accessible only to the degree to which our categorical system and logic are developed. If one runs against the wall of contradiction, it is due to a problem in human reasoning, not a problem of Being, of the outside world, of the real whose essence evades human mind. In *Physics, Aristotle writes:*

p. 11

人们不容易把运动和变化归入另一个属（genus）——只要看看有些人把它归入何处就清楚了：他们把运动等同于差异、不等或非存在；但这些东西并不必然运动，无论它们不同、不等还是不存在。（201b19–201b23）[14]

One could not easily put motion and change in another genus —this is plain if we consider where some people put it: they identify motion with difference or inequality or not being; but such things are not necessarily moved, whether they are different or unequal or non-existent. (201b19–201b23)[14]

此外，他论证道：

Furthermore, he argues:

因此，有人说虚空是物体的质料（他们也把位置等同于它），这种说法不正确；因为质料并不与事物分离，而他们却把虚空当作某种分离的东西来探究。（214a12–214a16）[15]

So some say that the void is the matter of the body (they identify the place, too, with this), and in this they speak incorrectly; for the matter is not separable from the things, but they are inquiring about the void as about something separable. (214a12–214a16)[15]

除了作为抽象范畴，人们不能把一般物质说成一个独立的存在者，正如也不能这样谈论虚空。因此，亚里士多德以“人们这样说是错误的”为理由驳斥了埃利亚学派的不可能性。亚里士多德考察了“organon”（理性推理及其规则的工具）及其在当时的可能性和局限，以便使对存在者（to on）以及非存在者的说明成为可能。所谓“非存在者”不是对存在的否定，而是对某种特定 ousia、对“某物”的否定。

Except as an abstract category, one cannot speak of matter in general as a distinct ontic entity, just as one cannot speak of the void in such fashion either. Thus, the Eleatic impossibility is refuted by Aristotle on the grounds of ‘one [thus] speaks incorrectly’. Aristotle examines the ‘organon’ (the instrumentarium of rational reasoning and its rules) and its possibilities and its shortcomings of the era in order to make the account of *that which is* (tò ōn), as well as of *that which is* not, possible. That ‘which is not’ is not a negation of Being but rather of a particular *ousia* , of a ‘some-thing’.

p. 11–12

亚里士多德《物理学》的上述引文带有鲜明的唯物主义色彩：物质与“存在之物”——某物，或 tò tí，或 ousia——不可分离。因此，物体并不是能够“承载”或容纳虚空或空的空间的东西，不能像留基伯和德谟克利特所尝试的那样，借此既使运动成为可能，又避免反驳埃利亚学派。恰恰相反，亚里士多德确实进行了这种反驳。按照亚里士多德的说法，早期原子论者所提出的妥协是：“他们说元素是充实与虚空，并分别把它们称为存在与非存在”［στοιχεῖα μὲν τὸ πλῆρες καὶ τὸ κενὸν εἶναί φασι, λέγοντες τὸ μὲν ὂν τὸ δὲ μὴ ὄν］（亚里士多德，《形而上学》A.4 985b5–6）。[16]这是一种谬误，一种推理错误，而且极可能是一种本体论上的不可能性。我们在这里使用“极可能”，因为亚里士多德似乎并不认为：仅仅由于我们

The above quote from Aristotle’s *Physics* resonates as staunchly materialistic: matter and ‘what is’ –the some-thing, or *tò tí* or *ousia* –are inalienable; therefore, bodies are not something that can ‘carry’ or contain the void or empty space, so that the possibility of movement is foregrounded while avoiding refutation of the Eleatic, as Leucippus and Democritus have attempted to do. Quite to the contrary, Aristotle engages in such refutation. The compromise the early atomists offered was, according to Aristotle, that ‘the elements are said to be full and empty, calling them either being or non-being’ [στοιχεῖα μὲν τὸ πλῆρες καὶ τὸ κενὸν εἶναί φασι, λέγοντες τὸ μὲν ὂν τὸ δὲ μὴ ὄν] (Aristotle, *Metaphysics*, A.4 985b5-6).[16] This is a fallacy, an error in reasoning

所能依靠的是认知工具（organon）的完善，外部世界就不可通达。非存在是相对于某种 ousia 而言的偶发、巧合，或更确切地说，偶然性（sumbebekos）；它可以是一种虚假，最后，也可以是一种潜能（即作为存在者现象的生成）。[17]正如质料依存于某种 ousia，存在因而是正确谓述与现实性的范畴。正确使用范畴，能够导向对实在或存在的正确说明。亚里士多德的方法可以与今天所谓的科学推理相比：证明的工具（organon）的完善，能够使我们对外部世界、实在或存在的说明更加准确。

and most probably an ontological impossibility. We are using ‘most probably’ here as Aristotle does not seem to think that the outside is impenetrable just because what we can rely on is the perfection of our cognitive instruments, the *organon*. Non-being is a matter of accident or coincidence or rather contingency (*sumbebekos*) with regard to an *ousia*; it can be a falsity and, finally, a potentiality (of becoming, as an ontic phenomenon).[17] Just as matter is contingent on an *ousia*, Being is therefore a category of correct predication and actuality. Correct use of categories can lead to a correct explanation of the real or of Being. Aristotle’s approach is comparable to what one might today call scientific reasoning – the perfection of the *organon* of demonstration leads to accuracy in one’s account of the outside world, of the real or Being.

尽管如此，亚里士多德仍是哲学家——存在一种最高 ousia 的终极真理，凌驾于众多 ousiai 之上；在此意义上，实在与真理建立统一，因为实在通过真理或纯粹沉思超越自身——作为包括可感世界在内的一切存在者的“第一推动者”，是 noesis（理性/思想）。（亚里士多德，《形而上学》1072a30–32）[18]又见：“被欲求者和思想的对象推动运动而自身不动。它们是第一的，也是同一的。”（《形而上学》1072a26–27）尽管如此，亚里士多德关于第一推动者的观念，是他整个著作尤其是《形而上学》和《物理学》中对埃利亚学派论证最直接的反驳。基于这一立场，亚里士多德既可被视为唯物主义者，也可被视为推动唯物主义立场发展的人：运动是一切存在者的原因，因此一切存在都是运动。

Aristotle remains a philosopher nonetheless – there is an ultimate truth of the ultimate *ousia*, overarching the plurality of *ousiae*, whereby real and truth establish unity insofar as the real transcends itself through truth or pure contemplation – the ‘prime mover’ of all existing, including the sensible world, is *noesis* (reason/ing). (Aristotle, Meta. 1072a30-32)[18] Also: ‘the desired and that which is subject of thought move while remaining unmoved. They are the first [primary] and the same’ (Meta. 1072a26-27) In spite of this, Aristotle’s notion of the prime mover is the most direct opposition to the Eleatic argument to be elaborated throughout his oeuvre, and primarily in *Metaphysics* and *Physics*. By virtue of that stance, Aristotle can be considered either a materialist or working in favour of a development of a materialist position: movement is the cause of all existent, consequently, all existence is movement.

p. 12

亚里士多德的哲学，或其唯物主义变体，站在前苏格拉底哲学家尤其是原子论者的肩上。

Aristotle’s philosophy, or its materialist variant, stands on the shoulders of the pre-Socratics, in particular the atomists.

原子论者

Atomists

如前所述，巴门尼德禁止对虚无、非存在进行哲学说明，意味着不可能说明运动和变化。许多前苏格拉底哲学家试图在遵守巴门尼德“禁止非存在”的前提下，构想一种能够说明变化和运动的认识论可能性。原子论者德谟克利特和留基伯遵守巴门尼德公理：“存在来自存在”，“非存在来自非存在”，因此本体论上不存在非存在。尽管如此，他们找到一种绕过这一哲学困境的方法，以构建一种允许变化和运动的物理世界说明。显然，感官世界中可见的生长与衰败、出现与消失、空间位移中的运动，促使原子论者揭示现象背后的隐藏原理，并给出哲学解释。他们的解释显然是唯物主义的——它以物理学（physike）即自然哲学的形式表述。在早期原子论者时代以及整个古希腊，哲学一词涵盖科学或对周围现实的科学解释方法：经验观察被形式化，自然和存在的抽象与规律既从经验洞见中提炼，也被经验洞见证实。

Parmenides’ prohibition on a philosophical account of nothingness, non-being, as already noted, implied the impossibility of offering an account of movement and change. Many of the pre-Socratics have sought to conceptualize an epistemic possibility for giving an account of change and movement while observing Parmenides’ ‘ban on non-being’. The atomists, Democritus and Leucippus, adhere to the Parmenidean axiom, according to which ‘being comes from being’, whereas ‘non-being comes from non-being’, and there is therefore no such thing as non-being, ontologically speaking. Nonetheless, they have found a way around this philosophical *aporia* in order to create an account of the physical world that allows for change and movement. Obviously, what was visible in the sensible world, such as growth and decay, emergence and disappearance and movement through space displacement, urged the atomists to unravel the hidden principles of the phenomena

and offer a philosophical explication of it. Their explication was, apparently, materialist –it was formulated in terms of physics, or *physikē*, a *philosophy* of nature. In the time of the early atomists and throughout the period of Greek Antiquity, the word *philosophy* encompassed sciences or the scientific approach to the explication of the surrounding reality: empirical observations were formalized, and abstractions and laws of nature and Being were extracted from such empirical insights as well as corroborated by them.

有限性——存在之坚实与完满不可或缺的条件

Finitude –the indispensable condition for the solidity and fullness of being

p. 13

存在确实不可分割，芝诺悖论被颠倒过来：遵循埃利亚原则，我们只能断言，必然存在某种存在实例，其分割不可能，如果我们遵守“无中不能生有”原则。也就是说，存在一个 a-tomos 点，一个确实坚实、不变、永恒的自然元素——原子，即“不可分割者”（DK 67A7，DK 68A37）。在《论生成与消灭》（325b12–325b33）中，亚里士多德论证，原子论者设定不可分割物体的动机，是为埃利亚学派提出的运动、变化和多样性（不）可能性悖论提供答案。[19]

Being is indeed indivisible, and Zeno’s paradox is reversed: by observing the Eleatic principle, we can only claim that there must be an instance of Being where its division becomes impossible if we are to observe the principle of ‘nothing comes from nothing’. Namely, there is a point of a-tomos, an element of nature that is indeed solid, unchangeable and permanent –the *atom*, the ‘indivisible’ (DK 67A7, DK 68A37). In *On Generation and Corruption* (325b12–325b33), Aristotle argues that the motivation for the atomists’ postulation of indivisible bodies was to proffer an answer to the *aporia* brought forth by the Eleatics, namely that of the (im)possibility of movement, change and multiplicity.[19]

因此，正是通过遵守埃利亚原则，原子论者才达到一个允许设想多样性和运动的实例。这就是不可分割（“不可切割”）、现实化的物质之点——atomos。（亚里士多德，《物理学》206a14–206a18）如果空间确实被视为无限可分，那么存在之坚实、稳定和固定的论证就会被削弱甚至完全矛盾。因此，埃利亚立场在术语上就是矛盾的——它试图证明万物始终如其所是、不变且不可变，却否定了变化可能，从而否定了运动。这隐含着无限。

Thus, by observing precisely the Eleatic principle, the atomists have reached an instance that allows for the conceiving of multiplicity and movement. That is the point of an indivisible (‘uncuttable’), actualized matter –*atomos*. (Aristotle, *Phys*. 206a14–206a18) If space should indeed be considered subject to limitless divisibility, the argument for solidity, stability and fixity of Being would be undermined if not contradicted entirely. Therefore, the Eleatic position is one of contradiction in terms – by seeking to demonstrate that everything has ever existed as it is, unchanging and unchangeable, it rejects the possibility of change and thus movement. Infinity is implied.

所有这一切都以“无中不能生有”原则为依据。然而，正因接受“无中不能生有”，原子论者才证明变化不仅可能而且必然，因为有限性实际上是其前提——无限意味着可分，无限可分意味着任何形式的坚实和完满存在都不可能。古典埃利亚论证是本体论的，因为它依赖把存在物化为抽象的形式论证。然而，原子论者的论证属于自然哲学：对自然（physis）的一种以经验为基础的说明——物质的无限分割或切割意味着，在 physis 的核心，否定或消灭行为愈发在场且压倒性。引入 atomos 点，即不可切割、不可分割者，就是确认一种超越毁灭的物质现实实例和形式。因此，有限性保证了“无中不能生有”原则得到遵守。这种立场由原子论者阐述，但毕达哥拉斯学派似乎在原子论者之前就持有类似观点。

All of this is justified by the principle of ‘nothing comes from nothing’. However, by endorsing the principle of ‘nothing comes from nothing’, the atomists have proven that change is not only possible but necessary, as finitude is actually its prerequisite – infinity implies divisibility, and infinite divisibility implies the impossibility of solidity and fullness of being of any sort. The classical Eleatic argument is ontological, as it relies on a formal argument of the being as reified abstraction. The argument of the atomists, however, is that of the natural philosophy: an empirically grounded account of *physis* – infinite division or cutting of matter implies an ever more present and overwhelming presence of the act of introducing negation or annihilation at the heart of *physis*. Introducing a point of *atomos*, or of the uncuttable, of the indivisible, is the identification of an instance and form of material reality that is beyond destruction. Thus, finitude is what vouches for the observance of the principle

‘nothing comes from nothing’. Such a position is elaborated by the atomists, but it seems that the Pythagoreans held a similar stance prior to the atomists.

亚里士多德进一步为原子论立场提供佐证，他论证道：

Aristotle offers further corroboration of the atomist stance by arguing:

事物既被认为潜能地存在，也被认为现实地存在。此外，事物要么通过增加，要么通过分割而无限。正如我们所见，大小并非现实无限。但通过分割，它是无限的。（驳斥不可分割线的理论并不困难。）那么剩下的选择是：无限具有潜在存在。但我们不能像说“这可能是一尊雕像”那样理解潜在存在——这会是一尊雕像，但无限不会现实存在。存在以多种方式被述说，我们说无限存在，其意义如同我们说“现在是白天”或“现在是竞技会”，因为事物总是一个接一个地产生。（《物理学》206a14–206a25）

Now things are said to exist both potentially and in fulfilment. Further, a thing is infinite either by addition or by division. Now, as we have seen, magnitude is not actually infinite. But by division it is infinite. (There is no difficulty in refuting the theory of indivisible lines.) The alternative then remains that the infinite has a potential existence. But we must not construe potential existence in the way we do when we say that it is possible for this to be a statue—this will be a statue, but something infinite will not be in actuality. Being is spoken of in many ways, and we say that the infinite is in the sense in which we say it is day or it is the games, because one thing after another is always coming into existence. (*Phys*. 206a14–206a25)

p. 13–14

由否定性或“空无物质”促成的运动与变化

Movement and change enabled by negative or ‘empty matter’

亚里士多德在说明早期原子论者时写道：“元素或者是充实的，或者是空的，也可以分别称为存在或非存在”；原文为：“στοιχεῖα μὲν τὸ πλῆρες καὶ τὸ κενὸν εἶναί φασι, λέγοντες τὸ μὲν ὂν τὸ δὲ μὴ ὄν.”（《形而上学》A.4 985b5–6）这样，“非存在”就参与存在，因为它的在场是对构成物质的原子的掏空——kenosis。因此，虚无参与存在，却不是存在的起源；它毋宁嵌在主要被视为物质或自然（physis）的存在的核心之中。原子论者必须提出两个论点：存在是多重的，而空或虚空——空间的缺失，或任何东西的缺失——是物质现实的一个方面。这是在试图克服一种困境：遵守埃利亚法则，阻碍了对运动、变化、生成和衰亡问题的科学与哲学考察。这一尝试类似赫拉克利特的解决办法：“一”实际上是多；即使它几乎熄灭，也不会彻底熄灭，而会“按照一定的尺度”重新燃起，也就是遵循自然与存在的内在法则，亦即科学与本体论的法则。（DK22B30）原文为：κόσμον τόνδε, τὸν αὐτὸν ἁπάντων, οὔτε τις θεῶν οὔτε ἀνθρώπων ἐποίησεν, ἀλλ’ ἦν ἀεὶ καὶ ἔστιν καὶ ἔσται πῦρ ἀείζωον, ἁπτόμενον μέτρα καὶ ἀποσβεννύμενον μέτρα（DK22B30）。赫拉克利特假定存在几乎完全熄灭；与之不同，原子论者谈论的是原子内部的绝对虚空。

‘The elements are either full or empty, or they could be called either being or non-being’, explains Aristotle in his account of the early atomists; or, in the original: ‘στοιχεῖα μὲν τὸ πλῆρες καὶ τὸ κενὸν εἶναί φασι, λέγοντες τὸ μὲν ὂν τὸ δὲ μὴ ὄν.’ (*Meta*. A.4 985b5-6) In this way, ‘non-being’ participates in Being as its presence is a hollowing out –*kenosis* –of the atoms that constitute matter. Therefore, nothingness participates in Being without being its origin and is rather nested at the heart of Being seen primarily as matter or *physis*. The atomists had to put forward the thesis that Being is multiple as well as the thesis that emptiness or void (the absence of space or of anything for that matter) is an aspect of material reality. This attempt to overcome the frustration of scientific and philosophical examination of the question of movement, change, generation and degeneration, created by the observance of the Eleatic law, resembles Heraclites’ s solution: the One is in fact multiple, and even though it might go virtually extinguished, it never fully goes out, instead rekindling ‘according to certain measures’, that is, internal laws of nature and Being, scientific and ontological. (DK22B30) Or, in the original: κόσμον τόνδε, τὸν αὐτὸν ἁπάντων, οὔτε τις θεῶν οὔτε ἀνθρώπων ἐποίησεν, ἀλλ’ ἦν ἀεὶ καὶ ἔστιν καὶ ἔσται πῦρ ἀείζωον, ἁπτόμενον μέτρα καὶ ἀποσβεννύμενον μέτρα (DK22B30). Unlike Heraclitus, who assumes a virtual extinction of existence, the atomists speak of an absolute void inside of an atom.

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希腊哲学中理解运动的困难——并不只见于前苏格拉底思想家——在于它对运动的本体论定义：运动是存在与生成的问题，是从一种 ousia 转变为另一种 ousia。从纯粹本体论上说，这种运动似乎不可能，而且确实涉及存在从非存在中产生这一前提。存在过去存在、现在存在，而且将永远存在；生成是不可能的。运动很少被同空间、自然哲学，或关于物

The trouble with conceiving movement in Greek philosophy, and not only among the pre-Socratics, lies in its ontological definition –it is a matter of Being and becoming, of movement from one *ousia* into another. Speaking purely ontologically, such movement seems impossible and does involve the premise of the emergence of Being from non-being. Being has been, is and will al-

质行为的精确推理联系起来。即使前苏格拉底思想家，尤其是原子论者，确实建立了这种联系，空的空间也成为非存在的一种可能性，存在的充实与无限在场因而受到损害。梅利索斯认为，“空”——显然被理解为一个空间范畴——就是无：“因为虚空就是无”，原文为 τὸ γὰρ κενεὸν οὐδέν ἐστιν（DK 30B7.7）。为了使某物生成，就必须假定此前没有这种 ousia；为了使一种 ousia“位移”，便必须设想空间被腾空，它又在一个此前绝对空无的空间中重新出现；而那个空间被理解为非存在。亚里士多德在《物理学》中记述了原子论者对运动的这种理解：“这些人认为，虚空是一道间隙，非存在以可感物体的方式在其中存在。”（《物理学》IV.6 213a27-9）希腊文底本为：οἱ δὲ ἄνθρωποι βούλονται κενὸν εἶναι διάστημα ἐν ᾧ μηδέν ἐστι σῶμα ἰσθητόν。

〔校注：这里的中文引句依本书英文释义译出。书中所附希腊文的句意为“人们认为虚空是其中没有可感物体的间隔”，与其英文释义不完全相合；底本希腊词形 ἰσθητόν 照录，不静默改写。〕

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亚里士多德最早把空间，从而把运动，归入可以作为谓词一部分但不参与“本质”或“实体”即 ousia 的范畴。这样，亚里士多德既在本体论上，也在自然哲学上提供了解决方案，允许对运动这一物质现象进行精确和数学的探索。

亚里士多德的介入以及随后数世纪的科学探究，使伊壁鸠鲁在提供令人信服的原子论说明时，任务变得更容易，也不再那么受埃利亚格言束缚。至少，卢克莱修·卡鲁斯《物性论》是这样读的：

nec porro augendis rebus spatio foret usus seminis ad coitum, si e nilo crescere possent. nam fierent iuuenes subito ex infantibus paruis e terraque exorta repente arbusta salirent. (Liber I, 184-7)[20]

按照古典语文学的规范，这里使用原典的页边编号，以便在不同版本与译本之间查找对应位置。

或者，依英译：“假如生命能从无中增长，万物的成长还需要空间吗？那样，小小的婴儿顷刻便会成为能行走的成人，枝繁叶茂的树也会突然从地上跃起——这些都是闻所未闻的奇迹；因为依自然之道，每一种东西都从合乎其规律的种子缓慢生长，并在这生长中保持自身的种类。”[21]伊壁鸠鲁派无需假设“原子包含虚空”便能解释运动与变化：原子是“坚实的”物质，而生长与衰亡遵循物质保存的法则：

sunt igitur solida ac sine inani corpora prima. praeterea quoniam genitis in rebus inanest, materiem circum solidam constare necessest. nec res ulla potest uera ratione probari corpore inane suo celare atque intus habere, si non, quod cohibet, solidum constare relinquas. id porro nihil esse potest nisi materiai concilium, quod inane queat rerum cohibere. (Titus Lucretius Carus, De Rerum Natura , Liber I, 510-15)

ways be – becoming is an impossibility. It is rarely linked with space and natural philosophy or exact reasoning concerning the behaviour of matter. With the pre-Socratics, and the atomists more specifically, even when it is, empty space becomes a possibility for non-being, and the fullness and limitless being-there of Being is compromised. Melissus argues that ‘the empty’, apparently conceived as a spatial category, is nothing: ‘as the empty is nothing’, or, in the original, τὸ γὰρ κενεὸν οὐδέν ἐστιν (DK 30B7.7). In order for something to become, the previous absence of *ousia* is assumed; in order for an *ousia* to ‘displace’ itself, one ought to imagine an evacuation of space and re-emergence in an absolutely empty prior space which is understood as non-being. This understanding of movement among the atomists is noted by Aristotle in his *Physics*: ‘these people reasoned that the empty is a fissure through which non-being is in the way of a sensible body’ (*Physics* IV.6 213a27-9), or, in the Greek original: οἱ δὲ ἄνθρωποι βούλονται κενὸν εἶναι διάστημα ἐν ᾧ μηδέν ἐστι σῶμα ἰσθητόν.

Aristotle was the first to place space, and therefore movement, among the categories that can become a part of a predicate but do not participate in the ‘essences’ or ‘substances’, that is, *ousiai*. In this way, Aristotle offers a solution both ontological as well as in terms of natural philosophy, allowing for exact and mathematical explorations of the material phenomenon of movement.

Aristotle’s intervention and the centuries of scientific enquiry that followed render Epicurus’ task of offering a convincing account of atomism easier or less contrived by the Eleatic dictum. At least, that is how it reads in Lucretius Carus’ *De Rerum Natura*:

nec porro augendis rebus spatio foret usus seminis ad coitum, si e nilo crescere possent. nam fierent iuuenes subito ex infantibus paruis e terraque exorta repente arbusta salirent. (Liber I, 184-7)[20]

Marginal pagination is used as per the standards of classical philology in order to enable easier navigation through different editions and translations.

Or in English translation: ‘Would space be needed for the growth of things; Were life an increment of nothing: then; The tiny babe forthwith would walk a man, And from the turf would leap a branching tree —Wonders unheard of; for, by Nature, each; Slowly increases from its lawful seed, And through that increase shall conserve its kind’.[21] The Epicureans did not need the conjecture of ‘atoms containing void’ in order to proffer an account of movement and change –atoms are ‘solid’ matter, and growth and decay observe the laws of its preservation:

sunt igitur solida ac sine inani corpora prima. praeterea quoniam genitis in rebus inanest, materiem circum solidam constare necessest. nec res ulla potest uera ratione probari corpore inane suo celare atque intus habere, si non, quod cohibet, solidum constare relinquas. id porro nihil esse potest nisi materiai concilium, quod inane queat rerum cohibere. (Titus Lucretius Carus, *De Rerum Natura* , Liber I, 510-15)

或者，依英译：“因此，原初物体是坚实的，内部没有虚空。但既然一切生成之物中都有虚空，坚实的物质就必然环绕着它。除非你承认那容纳虚空的东西本身是坚实的，否则便无法凭正确的推理证明：某物在自己的躯体中隐藏着、容纳着虚空。须知，能够将虚空容纳在事物之中的，只能是结合在一起的物质。”[22]为了使物质在其有限或受限却无界的存在中不断运动与生长[23]——因为存在之外别无所有，那里是无，因而无法在物质现实领域之外找到它——就需要空间，也需要物质宇宙内部所包含的虚无：“如此广袤的空间向万物四周敞开——各个方向都有余地，没有尽头。最后，我们亲眼可见，事物界定着事物：空气隔开山丘，山峦又围住空气；陆地终结海洋，海洋反过来又界定所有陆地；然而，对于全体而言，确实没有任何东西能从外部为它划定边界。”[24]

Or in English translation: ‘Thus primal bodies are solid, without a void.; But since there’s void in all begotten things; All solid matter must be round the same; Nor, by true reason canst thou prove aught hides; And holds a void within its body, unless; Thou grant what holds it be a solid. Know; That which can hold a void of things within; Can be naught else than matter in union knit’.[22] For the matter to constantly move and grow within its finite or limited yet unbound existence[23] – ‘as there is nothing beyond existence, it is naught and therefore it is not to be found outside the realm of material reality –space is required and nothingness contained within the material universe: Such huge abundance spreads for things around—/ Room off to every quarter, without end. Lastly, before our very eyes is seen / Thing to bound thing: air hedges hill from hill, / And mountain walls hedge air; land ends the sea, / And sea in turn all lands / but for the All / Truly is nothing which outside may bound’.[24]

p. 15–16

摆脱抽象理想性——无论是概念或数学理想、ousia，还是柏拉图的 eidoi（理念）——是自然存在的必要条件，因为自然是原子碰撞的产物，是“斗争”（参见卢克莱修《物性论》第二卷）或对抗，类似于赫拉克利特的 polemos。在《物性论》第二卷中，我们读到，原子从理想的“垂直向下运动”中发生极小（paulum）偏转是必要的（II, 214–20），以便碰撞发生，从而产生原子新组合，进而产生自然新形式：

A move away from abstract ideality, be it a conceptual or mathematical ideal, *ousia*, or Plato’s *eidoi* (forms), is the indispensable condition for the existence of nature as it is the product of collision between atoms, of ‘strife’ (cf. Liber II of *De Rerum Natura* by Lucretius Carus) or confrontation, not unlike Heraclitus’ *polemos*. In the second book of *De Rerum Natura*, we read that a very small (paulum) swerve from the ideal ‘downright movement’ of atoms is indispensable (II, 214–20) in order for the collision to take place, which gives birth to new combinations of atoms and thus new forms of nature:

p. 16

它们或者由自身的重量推动，或者偶然受到外来的撞击而运动。它们在不停的运动中频繁相遇、碰撞，便会猛然彼此弹开、相向而去；这并不奇怪——因为它们极其坚硬，具有坚实的重量，而且身后没有什么阻碍它们的运动。（卢克莱修 • 卡鲁斯，《物性论》第 2 卷，83–88）[25]

Be borne along, either by weight their own, Or haply by another’s blow without. For, when, in their incessancy so oft They meet and clash, it comes to pass amain; They leap asunder, face to face: not strange—Being most hard, and solid in their weights, And naught opposing motion, from behind. (Lucretius Carus, *On the Nature of Things: Liber II*, 83–88)[25]

必须注意，偶然——意外或不可预测的偶然性——是创造性碰撞的条件。偏转出现在不确定的时间和地点——原文“incerto tempore ferme incertisque locis spatio deflectere paulum”（卢克莱修 • 卡鲁斯，II, 219–20）——进而导致碰撞、分散和原子连接的重组，一种结构瓦解，新结构出现。运动持续不断，因为虚空的绝对性不允许原子静止——在绝对空虚的虚空中，它们无法占据任何位置，因为虚空不允许位置，它不产生任何对立、阻力或物质来构成这种假想位置。

It is important to note that *chance* – accident or unpredictable contingency –is the condition of the creative collision. The swerve appears at an uncertain time and location –in the original, ‘incerto tempore ferme incertisque locis spatio deflectere paulum’ (Lucretius Carus, Liber II, 219 – 20) – which then causes collision, dispersion and reorganization in the atomic linkages, one structure falling apart and a new one emerging. Movement is constant, as the absoluteness of the void does not allow for the atoms to stand still –there is no location that they could occupy in the void that is empty in an absolute way, disallowing position, as it does not create any opposition, resistance or matter that would constitute such supposed location.

也没有任何地方，当物体抵达那里时，能在虚空中失去重力而静止；虚空也不能为任何事物提供支撑——不，它必须按其本性，继续让路。因此，事物绝不可能像被征服一样，因渴求中心而保持结合。（卢克莱修 • 卡鲁斯，I, 1078–83）

Nor is there any place, where, when they’ve come, Bodies can be at standstill in the void, Deprived of force of weight; nor yet may void; Furnish support to any,—nay, it must, True to its bent of nature, still give way. Thus in such manner not at all can things; Be held in union, as if overcome; By craving for a centre. (Lucretius Carus. Liber I, 1078–83)

或依原文：

or in the original:

nec quisquam locus est quo corpora com venerunt ponderis amissa vi possint stare in inani, nec quod inane autem est, ulli subsistere debet, quin, sua quod natura petit, concedere pergat. haud igitur possunt tali ratione teneri res in concilium medii cuppedine victae. (1077–1082)[26]

p. 16–17

虚空实际上是虚无，是绝对的无，因此它只在概念上存在，而非本体论上——毋宁说，面对虚空的毁灭力量，生命以及只能以物理形式存在的存在不断诞生并重生。空间和位置没有物理表征，它们是无界且纯粹延展："空间无边界也无尺度，向四面八方无量延伸。"（卢克莱修·卡鲁斯，《物性论》II, 91–2）原文："quoniamspatium sine fine modoquest, inmensumque patere in cunctas undique partis"。空间无法度量，表明它被理解为虚空的一个方面，以及那种使变化成为可能、却不产生任何东西的虚无——"无中不能生有"是原子论原则。然而，它是一个能够让我们把握变化和运动的范畴。毁灭潜能是物质本身固有的，是原子行为的一个方面，而正是这种否定性维持生命。但它本身并不构成本体论现实——无即不存在之物，它只存在于原子碰撞的缝隙中，存在于除此以外皆为坚实物质的原子之间。卢克莱修·卡鲁斯用以下比喻解释空间（就虚空而言）在物质（物理）过程发生中的作用：

nec quisquam locus est quo corpora com venerunt ponderis amissa vi possint stare in inani, nec quod inane autem est, ulli subsistere debet, quin, sua quod natura petit, concedere pergat. haud igitur possunt tali ratione teneri res in concilium medii cuppedine victae. (1077–1082)[26]

The void is in fact nothingness, an absolute naught, and it therefore exists only conceptually, not ontologically – it is rather that against the annihilating power of the void life and thus existence that could be found only in a physical form are born and perpetually reborn. Space and location do not have a physical representation, they are unbound and pure extension: 'Space has no bound nor measure, and extends; Unmetered forth in all directions round. (Titus Lucretius Carus, *On the Nature of Things, Liber II*, 91–2) Or in the original: 'quoniam spatium sine fine modoquest, inmensumque patere in cunctas undique partis'. The fact that space has no measure points to its understanding as an aspect of the void and of that enabling nothingness that does not engender anything – 'nothing comes from nothing' is the atomist principle. Yet it is a category that enables change and movement to be grasped. The potentiality of annihilation is intrinsic to matter itself, and an aspect of the behaviour of the atoms and that very negativity is what sustains life. However, it does not constitute an ontological reality in and of itself –nothing is what is not, and it exists only in the fissures of collisions of atoms, amidst the otherwise solid matter the atoms are. Lucretius Carus uses the following metaphors to explain the role of space (with regard to void) in the occurrence of material (physical) processes:

p. 17

在光线所照的虚空中，它们不断战斗，仿佛永恒争斗；成队交锋，毫不停歇，在相遇与分离中上下奔忙。由此你便可推想，原初种子在更大的虚空中不停颠簸，究竟是怎样的情形——至少在这限度之内。（卢克莱修·卡鲁斯，第2卷，118–123）

Amid a void in the very light of the rays, And battling on, as in eternal strife, And in battalions contending without halt, In meetings, partings, harried up and down. From this thou mayest conjecture of what sort; The ceaseless tossing of primordial seeds; Amid the mightier void – at least so far. (Lucretius Carus, Liber II, 118–23)

或依原文：

Or in the original:

multa minuta modis multis per inane videbis corpora misceri radiorum lumine in ipso, et velut aeterno certamine proelia pugnas edere turmatim certantia nec dare pausam, conciliis et discidiis exercita crebris, 120 conicere ut possis ex hoc, primordia rerum quale sit in magno iactari semper inani—dumtaxat rerum magnarum parva potest res exemplare dare et vestigia notitiai. (Liber II 116–24)

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p. 17–18

虚无是一个本体论概念，但不是一种作为存在者的实在。然而，它引发曲线、偏转或 clinamen，因此不只是抽象。就其作为原子碰撞导致的衰败和毁灭过程的效果而言，它是真实的。然而，从因果上看，偏转先于碰撞，是虚空本身的属性。再者，虚空在空间上不存在，也没有时间固定点。人们或许会猜想，它是原子运动的属性，不构成时间因果性，而后者本身并非线性。虚空存在，是因为原子必须碰撞；它们碰撞，是因为不存在有中心、能保证静态位置的空间虚空。在这个意义上，虚空缺席于物质和原子的构成，却又作为"运算子"出现在永不停息的原子组合与分解活动中。人们可以把它比作今天所说的反粒子，或空间测量的抽象范畴，因为它

Nothingness is an ontological concept but not an ontic reality. Yet it occasions the curve, or the swerve or *clinamen*, and thus it is not only an abstraction. It is real insofar as it is an effect of processes of decay and destruction that occur due to the atoms' collisions. However, causally, the swerve precedes the collisions and is a property of the void itself. Yet again, the void does not exist spatially, nor does it have temporal fixed point. One might conjure that it is the property of atoms' movement and does not constitute temporal causality whereas the latter is not linear. The void is there because atoms must collide, and they do so as there is no spatial void with a centre that can ensure a static position. In that

既能引发产生存在的偏转和碰撞，又只是概念，没有事实上的、时间的或空间的位置，也没有物质实体，尽管它对原子现实产生物质效果。

〔校注：底本末句使用 special position；按上下文译为空间位置（spatial position），英文保留 special。关于虚空与反粒子的类比是作者的论述，未替作者作科学事实上的更正。〕

sense, the void is absent from matter and the constitution of atoms and yet again makes itself present as an ‘operator’ in the never-ceasing atomic actions of composition and decomposition. One might liken it to what is nowadays known as anti-particles or the abstract category of measurement of space, since it has the capacity to cause the swerve and the collision that engenders existence while being mere concept without factual, temporal and special position or body of matter, albeit with a material effect on the atomic reality.

第 2 章原注 / Chapter 2 Notes

p. 235 ↑1. “西方”通常与“欧洲”同义使用，并意在涵盖欧洲的跨大西洋延伸部分。我们选择使用“欧洲”，因为它对欧洲东南部和东部的排斥性较小，尤其是因为这一地理范围发挥作用的时间，远远长于其西方化的变体。伊斯兰哲学与数学对于其历史延续和科学进步都有关键影响。此外，我们在这里将哲学与欧洲哲学等同，因为我们赞同弗朗索瓦·拉吕埃勒，认为哲学是一种欧洲特有的现象；这并不否认阿拉伯文化对其保存和发展的贡献，事实上还将这种贡献包含在内。

1 West is usually used as synonymous to European and is meant to cover its transatlantic extension. We choose to operate with ‘European’ as it is less exclusive towards the Southeast and East of Europe, in particular because its location has been relevant for far longer than its Westernized variant. Islamic philosophy and mathematics is of critical influence for its historical continuity and also scientific advancement. Also, we are equating here philosophy with European philosophy because we argue, concurring with François Laruelle, philosophy is a European phenomenon proper, notwithstanding the Arabic contribution to its preservation and development –in fact, including it.

p. 235 ↑2. 亚里士多德：《物理学》，载《亚里士多德全集》，R. P. Hardie、R. K. Gaye 译，第 2 卷（普林斯顿大学出版社，1984），第 315–446 页；参见 WilliamDavid Ross 编，Aristotelis Physica，Clarendon Press（牛津大学出版社，1950），200b33–201a3。

〔校注：书目信息、卷行号及原文拼写依底本保留；未独立查证所有引文版本。〕

2 Aristotle, ‘Physics’, in *The Complete Works of Aristotle*, trans. R. P. Hardie and R. K. Gaye, vol. 2 (Princeton University Press, 1984), 315–446 cf WilliamDavid Ross, ed., *Aristotelis Physica Clarendon Press* (Oxford University Press, 1950) , 200b33–201a3.

p. 235 ↑3. 请注意，对于古典文本，我们使用原典的页边编号，因为这种编号可以跨越版本和语言得到普遍辨识。

3 Please note that for the classical texts, we are using marginal pagination as it is universally legible regardless of the edition and language.

p. 235 ↑4. Joan Kung：“亚里士多德论‘存在以多种方式被述说’”（Aristotle on “Being Is Said in Many Ways”），History of Philosophy Quarterly，第 3 卷第 1 期（1986），第 3–18 页。

4 Joan Kung, ‘Aristotle on “Being Is Said in Many Ways”’, *History of Philosophy Quarterly* 3, no. 1 (1986): 3–18.

p. 235 ↑5. Walther Kranz、Hermann Diels：《前苏格拉底哲学家残篇》（Die Fragmente der Vorsokratiker），第 3 卷（Weidmann，1952）。本文行内引用将其缩写为 DK，使用原出版物（以及后续所有版本）所采用的页边编号。

5 Walther Kranz and Hermann Diels, *Die Fragmente der Vorsokratiker*, vol. 3 (Weidmann, 1952), which in the inline citations here is abbreviated as DK and the marginal pagination used in the original publication (and in all future editions).

p. 235 ↑6. 柏拉图：《斐多篇》（Phaedo），C. J. Rowe 编订并作评注，Christopher J. Rowe 编（剑桥大学出版社，1992）。行内引文采用国际通行的柏拉图原典页码编号。

〔校注：书目信息、卷行号及原文拼写依底本保留；未独立查证所有引文版本。〕

6 Plato, *Phaedo*, edited and with commentaries by C.J. Rowe, ed. Christopher J. Rowe (Cambridge University Press, 1992), The inline reference follows the internationally accepted pagination of Plato’s originals.

p. 235 ↑7. Peter Wolfendale：《面向对象的哲学：物自身的新衣》（Object-oriented Philosophy: The Noumenon’s New Clothes），第 1 卷（MIT Press，2019）。

7 Peter Wolfendale, *Object-oriented Philosophy: The Noumenon’s New Clothes*, vol. 1 (MIT Press, 2019).

p. 235 ↑8. Wilfrid Sellars：《科学、知觉与实在》（Science, Perception, and Reality）（Routledge & Kegan Paul，1963），第 1–40 页。

8 Wilfrid Sellars, *Science, Perception, and Reality* (Routledge & Kegan Paul, 1963), 1–40.

p. 235 ↑9. Georg Wilhelm Fredrich Hegel，Georg Wilhelm Friedrich Hegel: The Science of Logic，A. V. Miller 译（牛津大学出版社，1977），第 82 页。

〔校注：书目信息、卷行号及原文拼写依底本保留；未独立查证所有引文版本。〕

9 Georg Wilhelm Fredrich Hegel, *Georg Wilhelm Friedrich Hegel: The Science of Logic*, trans. A. V. Miller (Oxford University Press, 1977), 82.

p. 235 ↑10. “这种自我抽象、被固定为自为的实体，就是作为抽象利己主义者的人——即在其纯粹抽象中被提升到思想层面的利己主义。”（Karl Marx、Friedrich Engels：《1844 年经济学哲学手稿与共产党宣言》，The

10 The self-abstracted entity, fixed for itself, is man as abstract egoist—egoism raised in its pure abstraction to the level of thought’ (Karl Marx and Friedrich Engels, *The Economic and Philosophic Manuscripts of*

Economic and Philosophic Manuscripts of 1844 and the Communist Manifesto，Prometheus Books，2009，“对黑格尔哲学的一般批判”。）

↑11. Karl Marx、Friedrich Engels、W. Lough，Theses On Feuerbach Ludwig Feuerbach and the End of Classical German Philosophy（1845），无页码：“以往一切唯物主义——包括费尔巴哈的唯物主义——的主要缺点是：对事物、现实、感性，只从客体或直观的形式去理解，而不是把它们理解为人的感性活动、实践，不从主体方面去理解。因此，与唯物主义相反，能动的方面被唯心主义抽象地发展了；当然，唯心主义并不知道现实的、感性的活动本身。”

〔校注：书目信息、卷行号及原文拼写依底本保留；未独立查证所有引文版本。〕

1844 and the Communist Manifesto (Prometheus Books, 2009) ‘Critique of Hegel’s Philosophy in General’).

11 Karl Marx, Friedrich Engels, and W. Lough, *Theses On Feuerbach Ludwig Feuerbach and the End of Classical German Philosophy* (1845) s.p.: ‘The chief defect of all hitherto existing materialism —that of Feuerbach included —is that the thing, reality, sensuousness, is conceived only in the form of the object or of contemplation, but not as a sensuous human activity, practice, not subjectively. Hence, in contradistinction to materialism, the active side was developed abstractly by idealism —which, of course, does not know real, sensuous activity as such.’ p. 235–236

↑12. 马克思、恩格斯：《1844 年经济学哲学手稿与共产党宣言》（The Economic and Philosophic Manuscripts of 1844 and the Communist Manifesto）：“对黑格尔哲学的一般批判”。

12 Marx and Engels, *The Economic and Philosophic Manuscripts of 1844 and the Communist Manifesto* : ‘Critique of Hegel’s Philosophy in General’. p. 236

↑13. Rosi Braidotti、Maria Hlavajova：《后人类术语表》（Posthuman Glossary）（Bloomsbury Publishing，2018）。

13 Rosi Braidotti and Maria Hlavajova, *Posthuman Glossary* (Bloomsbury Publishing, 2018). p. 236

↑14. 亚里士多德：《物理学》。

14 Aristotle, ‘Physics’. p. 236

↑15. 亚里士多德。

15 Aristotle. p. 236

↑16. 亚里士多德，Aristotelis Opera，Immanuelis Bekkeri、Olof Gigon、Hermann Bonitz 编（Walter de Gruyter，1970）。

16 Aristotle, *Aristotelis Opera*, ed. Immanuelis Bekkeri, Olof Gigon, and Hermann Bonitz (Walter de Gruyter, 1970). p. 236

↑17.《形而上学》ν.2，1089a15–19：ἔπειτα ἐκ ποίου μὴ ὄντος καὶ ὄντος τὰ ὄντα; πολλαχῶς γὰρ καὶ τὸ μὴ ὄν, ἐπειδὴ καὶ τὸ ὄν· καὶ τὸ μὲν μὴ ἄνθρωπον σημαίνει τὸ μὴ εἶναι τοδί, τὸ δὲ μὴ εὐθὺ τὸ μὴ εἶναι τοιονδί, τὸ δὲ μὴ τρίπηχυ τὸ μὴ εἶναι τοσονδί.〔中译：再者，存在者来自哪一种非存在与存在？非存在也有多种说法，因为存在亦然：“非人”表示不是这个，“非直”表示不是如此这般，“非三肘长”表示不是这样大的量。〕

〔校注：书目信息、卷行号及原文拼写依底本保留；未独立查证所有引文版本。〕

17 Meta. ν.2 1089a15–19, ἔπειτα ἐκ ποίου μὴ ὄντος καὶ ὄντος τὰ ὄντα; πολλαχῶς γὰρ καὶ τὸ μὴ ὄν, ἐπειδὴ καὶ τὸ ὄν· καὶ τὸ μὲν μὴ ἄνθρωπον σημαίνει τὸ μὴ εἶναι τοδί, τὸ δὲ μὴ εὐθὺ τὸ μὴ εἶναι τοιονδί, τὸ δὲ μὴ τρίπηχυ τὸ μὴ εἶναι τοσονδί. p. 236

↑18. 此处以及下文对原典的翻译均出自我本人。

18 This and the following translation of the original are my own. p. 236

↑19. 亚里士多德：“论生成与消灭”（On Generation and Corruption），载《亚里士多德全集》，Harold H. Joachim 译（普林斯顿大学出版社，2014）。

19 Aristotle, ‘On Generation and Corruption’, in *Complete Works of Aristotle*, trans. Harold H. Joachim (Princeton University Press, 2014). p. 236

↑20. Titus Lucretius Carus，De rerum natura Volume primo（Libri I–III），Marcus Deufert 编（de Gruyter GmbH，2019）。

20 Titus Lucretius Carus, *De rerum natura Volume primo (Libri I-III)*, ed. Marcus Deufert (de Gruyter GmbH, 2019). p. 236

↑21. Titus Lucretius Carus：《物性论》（On the Nature of Things），William Ellery Leonard 译，取自古腾堡计划，发布日期 2008 年 7 月 31 日［电子书第 785 号］，最后更新于 2013 年 2 月 4 日，访问于 2021 年 4 月。本书对该书的所有引文均取自这一译本。引用时不使用译本页码，而遵循古典文献规范，使用原典页边编号，以确保准确定位原文并便于与其他译本比较。

21 Titus Lucretius Carus, *On the Nature of Things*, trans. By William Ellery Leonard, retrieved from Project Gutenberg Release Date: 31 July 2008 [EBook #785] Last Updated: 4 February 2013, accessed in April 2021. All citations of this title are taken from this translation. Pages are not cited but rather the original marginal pagination, in line with the classical canon, in order to ensure accurate location of the original as well as be able to compare other translations p. 236

↑22. 如前所述，译文取自 Titus Lucretius Carus：《物性论》（On the Nature of Things），William Ellery

22 As already noted, the translations are taken from Titus Lucretius Carus, *On the Nature of Things*, trans. p. 236

Leonard 译，古腾堡计划，发布日期 2008 年 7 月 31 日［电子书第 785 号］，最后更新于 2013 年 2 月 4 日，访问于 2021 年 4 月。本书对该书的所有引文均取自这一译本。引用时不使用译本页码，而遵循古典文献规范，使用原典页边编号，以确保准确定位原文并便于与其他译本比较。

By William Ellery Leonard, retrieved from Project Gutenberg Release Date: 31 July 2008 [EBook #785] Last Updated: 4 February 2013, accessed in April 2021. All citations of this title are taken from this translation. Pages are not cited but rather the original marginal pagination, in line with the classical canon, in order to ensure accurate location of the original as well as be able to compare other translations,

p. 236–237

↑23. 拉丁文原文：Praeterea si iam finitum constituatur; omne quod est spatium, si quis procurrat ad oras; ultimus extremas iaciatque uolatile telum; (I, 970) id ualidis utrum contortum uribus ire; quo fuerit missum mauis longeque uolare; an prohibere aliquid censes obstareque posse?; alterutrum fatearis enim sumasque necessest.; quorum utrumque tibi effugium praecludit et omne; cogit ut exempta concedas fine patere; nam siue est aliquid quod probeat officiatque; quominus quo missum est ueniat finique locet se; siue foras fertur, non est a fine profectum. ; (I, 980) hoc pacto sequar atque, oras ubicumque locaris; extremas, quaeram: quid telo denique fiet?; fiet uti nusquam possit consistere finis; effugiumque fugae prolatet copia semper; Praeterea spatium summai totius omne; (I 984) undique si inclusum certis consisteret oris; finitumque foret, iam copia materiai; undique ponderibus solidis confluxet ad imum; nec res ulla geri sub caeli tegmine posset; nec foret omnino caelum neque lumina solis; quippe ubi materies omnis cumulata iaceret (I, 990); ex infinito iam tempore subsidendo. 991(Liber I, 970–90)。至于英文，这里破例使用 Martin Ferguson Smith 的译文，因为它更便于同拉丁原文逐行对照：“再假定一切现存的空间是有限的，有人跑到其最远边缘，将一支矛掷出去：你认为这支用强劲力量掷出的矛会远飞并抵达目标，还是认为会有什么东西挡住它的去路，使它停下？你必须认可并选择这两种假设中的一种，但二者都会堵住你的退路，迫使你承认宇宙的广延是无限的。因为，无论有什么东西挡住长矛，使它无法击中并留在目标上，还是它继续飞行，它都不是从宇宙的尽头出发的。我就这样紧追着你：无论你把最远边界设在哪里，我都会追问长矛最终怎样了。我们的结论是，任何地方都不能固定一条边界；面对飞行的无限可能性，永远无从逃避。”（第 1 卷，968–991）引自 Lucretius：《物性论》（On the Nature of Things），Martin Ferguson Smith 译并作导论与注释（Hackett Publishing Company，1969）。

〔校注：书目信息、卷行号及原文拼写依底本保留；未独立查证所有引文版本。〕

23 In the Latin original: Praeterea si iam finitum constituatur; omne quod est spatium, si quis procurrat ad oras; ultimus extremas iaciatque uolatile telum; (I, 970) id ualidis utrum contortum uribus ire; quo fuerit missum mauis longeque uolare; an prohibere aliquid censes obstareque posse?; alterutrum fatearis enim sumasque necessest.; quorum utrumque tibi effugium praecludit et omne; cogit ut exempta concedas fine patere; nam siue est aliquid quod probeat officiatque; quominus quo missum est ueniat finique locet se; siue foras fertur, non est a fine profectum. ; (I, 980) hoc pacto sequar atque, oras ubicumque locaris; extremas, quaeram: quid telo denique fiet?; fiet uti nusquam possit consistere finis; effugiumque fugae prolatet copia semper; Praeterea spatium summai totius omne; (I 984) undique si inclusum certis consisteret oris; finitumque foret, iam copia materiai; undique ponderibus solidis confluxet ad imum; nec res ulla geri sub caeli tegmine posset; nec foret omnino caelum neque lumina solis; quippe ubi materies omnis cumulata iaceret (I, 990); ex infinito iam tempore subsidendo. 991(Liber I, 970–90), or in English, let’s, as an exception, use Martin Ferguson Smith’s translation as it’s easier to follow the alignment with the Latin original: ‘Then again, just suppose that all the existing space were finite, and that someone ran forward to the edge of its farthest border and launched a spear into flight: do you favour the view that the spear, cast with virile vigor, would fly far and reach its target, or do you suppose that something could check it by obstructing its course? You must grant and adopt one or the other of these hypotheses, and yet both deny you a subterfuge and compel you to acknowledge that the expanse of the universe is infinite. For whether there is something to check the spear and prevent it from hitting its mark and lodging in its target, or whether it flies on, it did not start from the end of the universe. In this way I will dog you: wherever you locate the farthest border, I will ask about the ultimate fate of the spear. Our conclusion will be that nowhere can a boundary be fixed: no escape will ever be-found from the limitless possibility of flight (Liber I, 968–91), from Lucretius, *On the Nature of Things*, translated with Introduction and Notes by Martin Ferguson Smith (Hackett Publishing Company, 1969).

p. 237

↑24. 译文取自 Titus Lucretius Carus：《物性论》（On the Nature of Things），William Ellery Leonard 译，古腾堡计划，发布日期 2008 年 7 月 31 日［电子书第 785 号］，最后更新于 2013 年 2 月 4 日，访问于 2021 年 4 月。本书对该书的所有引文均取自这一译本。引用时不使用译本页码，而遵循古典文献规范，使用原典页边编号，以确保准确定位原文并便于与其他译本比较。

24 Translation is taken from Titus Lucretius Carus, *On the Nature of Things*, trans. By William Ellery Leonard, retrieved from Project Gutenberg Release Date: 31 July 2008 [EBook #785] Last Updated: 4 February 2013, accessed in April 2021. All citations of this title are taken from this translation. Pages are not cited but rather the original marginal pagination, in line with the classical

canon, in order to ensure accurate location of the original as well as be able to compare other translations.

↑25. 与注 24 相同，我们使用 William Ellery Leonard 的译文，并按原典页边编号引用。

25 Same as in note 24, we use William Ellery Leonard and cite the original marginal pagination. p. 237

↑26. 原始来源与以上引文相同。

26 The same original source as in the citations above. p. 237

第 3 章古典原子论

Chapter 3. CLASSICAL ATOMISM

p. 19

在蒸汽机发明之前，力学和机械论世界观建立在阿基米德的简单机械之上。这些机械可以通过几何学进行数学分析。阿基米德的著作提供了这种分析，以杠杆为核心范例，并把其他机械视为杠杆的推广。

Before the steam engine’s invention, mechanics and the mechanical world view were based on Archimedes’ simple machines. These machines were amenable to mathematical analysis through geometry. Archimedes’ works provided this analysis, with levers as the central example, viewing other machines as lever generalizations.

原子论唯物主义认为，现实的基本构成单元是物质粒子，被称为原子、corpora 或其他名称。机械唯物主义在此基础上进一步主张，物质遵循严格的守恒定律。卢克莱修阐述了“无中不能生有”原则；尽管在他那里这属于哲学语境，但在机械唯物主义中，该原则通过守恒定律和对称性获得了精确表达。

Atomistic materialism posits that the fundamental constituents of reality are particles of matter, referred to as atoms, corpora or other terms. Mechanical materialism builds on this by stating that matter adheres to strict conservation laws. Lucretius elaborated the principle ‘nothing comes from nothing’, which, although philosophical in his context, is given precise expression in mechanical materialism through conservation laws and symmetries.

阿基米德

Archimedes

古代思想家中，现存著作最清晰地表达这种机械唯物主义的是阿基米德[1]。他在《论平面的平衡》开篇写道：

The ancient thinker whose surviving works expressed this most clearly is Archimedes,[1] who opened his book *On the Equilibrium of Planes* as follows:

我提出以下公设：

1. 等距的等重物处于平衡；不等距的等重物不平衡，而是向距离较远的重物倾斜。2. 如果当位于特定距离的重物处于平衡时，向其中一个重物添加某物，它们便不再平衡，而是向被添加重物的一侧倾斜。3. 同样，如果从其中一个重物取走某物，它们便不再平衡，而是向未被取走重物的一侧倾斜。

I POSTULATE the following:

1. Equal weights at equal distances are in equilibrium, and equal weights at unequal distances are not in equilibrium but incline towards the weight which is at the greater distance. 2. If, when weights at certain distances are in equilibrium, something be added to one of the weights, they are not in equilibrium but incline towards the weight to which the addition was made. 3. Similarly, if anything be taken away from one of the weights, they are not in equilibrium but incline towards the weight from which nothing was taken.

原理（1）陈述了一条守恒定律：支点两侧的距离与重量之积必须守恒。他后来更明确地表述为：

Principle (1) states a conservation law that the distance times weight on each side of the fulcrum must be conserved. He later states this more explicitly as:

两个量，无论是否可通约，当它们与支点的距离与这两个量成反比时，便处于平衡。

Two magnitudes, whether commensurable or incommensurable, balance at distances reciprocally proportional to the magnitudes.

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原理（2）和（3）在这一语境中表达了 ex nihilo nihil fit（无中不能生有）原则：若不向处于平衡状态的物体添加物质，或从中移除物质，就不会产生运动。

Principles (2) and (3) express the *ex nihilo nihil fit* principle in the context that without the addition (or subtraction) of matter from an equilibrium position, there will be no motion.

我们略过了他详细分析各种几何形状重心所需的其他命题。他运用古典几何技术证明了一些命题，例如：“在任何三角形中，重心位于连接任意一个角与对边中点的直线上。”然而，在著作的其他部分，他使用了看似极为现代的技术：递归分解以及实质上属于积分的方法。过去两千年来孩子们不得不背诵的一些公式——圆的面积 πr^2 和球体面积 $4\pi r^2$ ——最初正是由他使用这些技术发现的。他最初把圆面积公式表述为：

We have missed other propositions, which he needed in order to make detailed analyses of the centres of gravity of various geometrical shapes. He proves some propositions, like: ‘In any triangle the centre of gravity lies on the straight line joining any angle to the middle point of the opposite side’ by classical geometrical techniques. In other parts of his work, however, he uses what seem to be remarkably modern techniques: recursive decomposition and what is effectively integration.

Some of the formulae that children have had to learn by heart these last two millennia –πr^2 for a circle's area and $4\pi r^2$ for a sphere's –were first discovered by him using these techniques. The first formula he stated in the form:

任何圆的面积等于一个直角三角形，其中直角旁的一条边等于半径，另一条边等于圆周长。

The area of any circle is equal to a right-angled triangle in which one of the sides about the right angle is equal to the radius, and the other to the circumference.

稍加思考便可知，这样描述的面积在我们的记号中就是 πr^2 。随后，他通过在圆内接和外切多边形来证明它，起初用正方形，然后正八边形，再然后十六边形等。

A moment's thought suffices to convince one that the area so described must be πr^2 in our notation. He then goes on to prove it by inscribing and circumscribing the circle, initially with squares, then regular octagons and then sixteen-sided polygons etc.

内接和外切于圆的多边形为圆面积提供上下界。利用递归细分，他通过构造收敛级数证明了一个命题，这些级数收敛于一个三角形的面积，该三角形的底由圆周长确定，高由半径确定。他还通过显式计算 96 边形的面积，推导出 π 的上下界：

Polygons inscribed in and circumscribed around a circle provide upper and lower bounds for its area. Using recursive subdivision, he proved a proposition by constructing convergent series for the bounds which converge on the area of a triangle with base set by the circle's circumference and height set by its radius. He also deduced, by explicitly calculating the areas of 96-sided polygons, the upper and lower bounds for π:

$$3\tfrac{10}{71} < \pi < 3\tfrac{1}{7}$$

用现代十进制表示，这相当于：

In modern decimal notation, this is equivalent to deducing

$$3.1408 < \pi < 3.1428$$

〔校注：小数界限依原书照录；其中上界 3.1428 是截短值，不等于分数上界 22/7 的精确小数展开。〕

现代计算器给出 $\pi = 3.141592654$ 。

A modern calculator gives it as 3.141592654.

关于阿基米德证明球体面积是其大圆面积四倍，维多利亚时期数学家、剑桥版《阿基米德全集》编者托马斯·利特尔·希思说，该推导本质上等同于“真正的积分”。阿基米德使用的方法，比牛顿早近两千年就已经等同于微积分。

Concerning *Archimedes*' proof that the area of a sphere is four times the area of its great circle, the Victorian mathematician and editor of the Cambridge edition of *Archimedes*, Thomas Little Heath, says that the derivation is essentially identical to 'true integration'. *Archimedes* was using methods equivalent to Newton's calculus almost two thousand years before the English philosopher.

阿基米德的方法显示了几何学与力学之间紧密且相互支持的关系。例如，他推导球体体积 $\frac{4}{3}\pi r^3$ 时明确使用了力学论证。第一步是证明：一个单位圆柱体的体积等于一个单位半球体与一个单位圆锥体体积之和。

Archimedes' approach shows a close and mutually supportive relationship between geometry and mechanics. For example, his derivation of the volume of a sphere as $\frac{4}{3}\pi r^3$ uses explicitly mechanical arguments. His first step is to show that a unit cylinder has the same volume as the sum of the volumes of a unit hemisphere plus a unit cone.

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单位圆柱指高和半径均为 1 的圆柱。单位圆锥指高和半径均为 1 的圆锥。单位半球是半径为 1 的半球。

A unit cylinder is one whose height and radius are each 1. A unit cone is one whose height and radius are 1. Of course, a unit hemisphere is the hemisphere with a radius of 1.

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他的证明使用了他所谓的“机械法”。假设圆锥、半球和圆柱是由相同材料制成的重物。他断言，如果把它们以等距悬挂在支点两侧，如图 3.1 所示，使圆锥和半球在一侧，圆柱在另一侧，那么它们会完全平衡。

His proof uses what he called his 'mechanical method'. Assume that the cone, hemisphere and cylinder are weights made of the same material. He asserts that if we attach them at equal distances from a fulcrum, as shown in Figure 3.1, with the cone and hemisphere on one side and the cylinder on the other, then they will exactly balance.

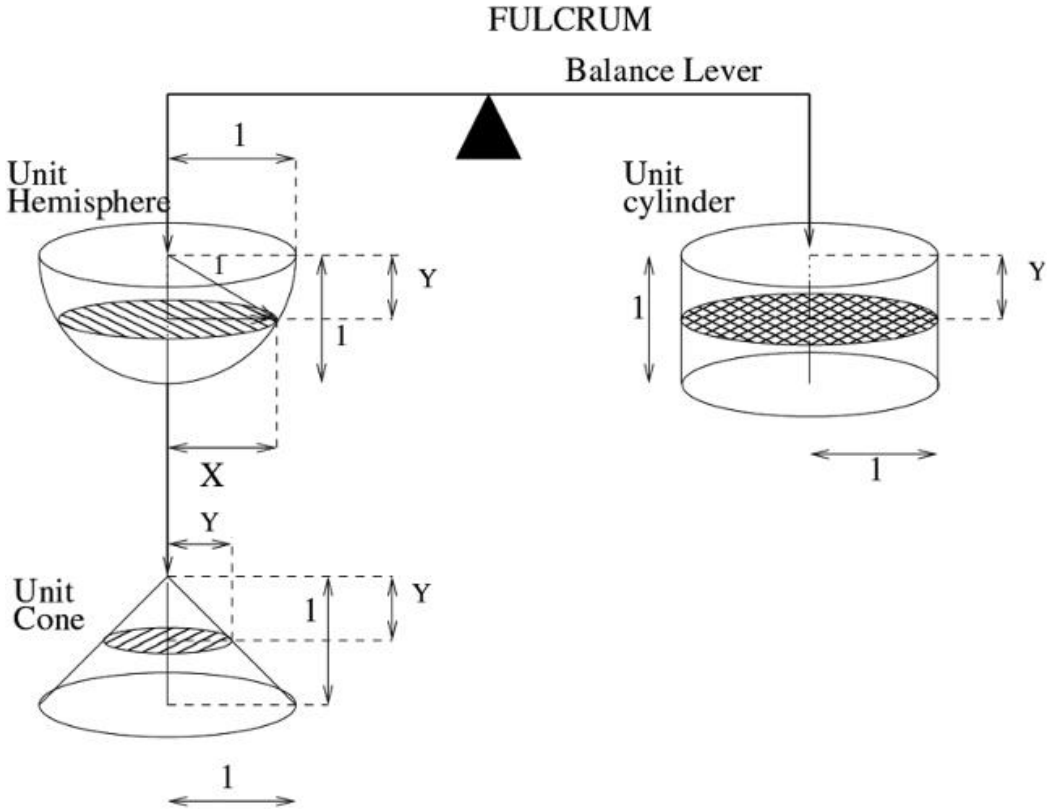


图 3.1 阿基米德证明，单位圆柱体的体积等于单位圆锥体与单位半球体的体积之和。作者绘图。

Figure 3.1 Archimedes proved that a unit cylinder has the same volume as a unit cone and a unit hemisphere. Figure by authors. FULCRUM：支点；Balance Lever：平衡杠杆；Unit Hemisphere：单位半球体；Unit cylinder：单位圆柱体；Unit Cone：单位圆锥体。X、Y、r 及数字为原图符号。

证明如下：从三个立体的顶部开始，逐层向下取无限小的圆形切片；在每一层，圆锥和半球的切片面积之和等于圆柱的切片面积。

The proof is that if you take infinitesimally small circular slices through all three solids, moving down in step from the top of each, then at each level the slices of the cone and hemisphere will be the same area as the slices of the cylinder.

1. 圆柱切片的面积很容易计算。它们是单位圆，所以面积为 $\pi = \pi 1^2$。2. 假设我们处于距每个立体顶端为 Y 的位置。由于圆锥的高与其底面半径相等，在 Y 处圆锥的半径也为 Y，因此切片面积为 πY^2。

1. The area of the cylinder slices is easy. Since they are unit circles, they have an area of $\pi = \pi 1^2$. 2. Suppose we are at distance Y from the top of each solid. The radius of the cone at Y will also be Y, since it is a cone whose height is the same as its radius. So the area will be πY^2.

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3. 根据毕达哥拉斯定理，穿过半球的切片的半径 X 必须为 $\pi(1^2 - Y^2) = \pi(1 - Y^2)$。

〔校注：原书称下式为半径 X，但式子实际表示截面面积。按图 3.1，半径应为 $X=\sqrt{(1-Y^2)}$，面积为 $\pi(1-Y^2)$。所供旧译已改为 $X^2=1^2-Y^2$，现将作者原文与校正说明分开。〕

3. By Pythagoras' theorem, the radius X of the slice through the hemisphere must be $\pi(1^2 - Y^2) = \pi(1 - Y^2)$

4. 因此，半球与圆锥的圆形截面面积之和必定为 $\pi(1 - Y^2) + \pi Y^2 = \pi$，证毕。

4. Thus the sum of the circles in the hemisphere and cone must be $\pi(1 - Y^2) + \pi Y^2 = \pi$, Q.E.D.

阿基米德还独立于这一论证，证明了单位圆柱体的体积是单位圆锥体的三倍。因此，半球的体积必为 $\frac{2}{3}\pi$，单位球的体积必为 $\frac{4}{3}\pi$。将其按比例放大到任意半径 r，就得到熟悉的 $\frac{4}{3}\pi r^3$。数学抽象与实际力学在这里存在密切联系。先前公理化力学中关于天平的公设，被用作出发点，推导出一种实质上属于积分学的证明。

Archimedes has independently of this argument proven that the volume of a unit cylinder is three times the volume of the unit cone, so the volume of the hemisphere must be $\frac{2}{3}\pi$, and that of the unit sphere must be $\frac{4}{3}\pi$. Scaling for a sphere of arbitrary radius r we get the familiar $\frac{4}{3}\pi r^3$. There is an intimate relationship here between the mathematical abstraction and the real mechanics. Prior axiomatic mechanics postulates about balances are used as a stepping stone to derive what amounts to an integral calculus proof.

他并不满足于圆锥、球体和圆柱。他进而求抛物线下的面积，为此构想了更复杂的天平装置，把三角形和抛物线下的区域悬挂在支点不同距离处。但为什么他关注抛物线等曲线？

He did not content himself with just cones, spheres and cylinders. He moved on to integrate the areas under parabolas, for which proof he imagines more complex arrangements of the balance, with triangles and areas under a parabola suspended at different distances from the fulcrum. But why his concern with curves like parabolas?

鲁索[2]认为，阿基米德的研究，尤其是他开创静水力学这一学科，与晚期希腊化文明的经济和政治秩序密切相关。鲁索指出，阿基米德所生活的锡拉库萨的经济和军事目标可以解释这一点。锡拉库萨是一支主要海军力量，致力于建造越来越大的船只。正因如此，阿基米德在静水力学分析中才关注推导浮体的稳定静止角，包括对以抛物线描述的物体如何漂浮的分析。抛物线曲线是可能的船体形状。

Russo[2] argues that Archimedes' research, particularly his invention of hydrostatics as a discipline, was closely connected to the economic and political order of late Hellenistic civilization. Russo points to the economic and military objectives of Syracuse, the state in which Archimedes lived, as an explanation. Syracuse was a major naval power, striving to build ever larger ships. That is why Archimedes was so concerned in his hydrostatic analysis to derive the angles of stable repose of floating bodies. This included an analysis of how bodies described in terms of parabolas would float. Parabolic curves are possible hull forms.

抛物线形状已被阿基米德的前辈阿波罗尼乌斯分析为圆锥的截面。绘制和构造此类曲线的方法已为人知。根据这些规则，造船师可以切割出船体的成型肋骨。

The parabolic shape had already been analysed by one of Archimedes' predecessors, Apollonius, as segments of cones. The method of constructing and drawing such curves was understood. From these rules, shipwrights could cut out the forming ribs of a ship's hull.

在希腊化时期，相互竞争的商贸国家出于经济和政治原因资助科学研究。阿基米德开创的静水力学新科学由经济和军事需求驱动，他利用它推导船体所能承受的最大稳定载荷。然而，罗马征服终结了科学资助，阿基米德也被杀。一个统一国家在地中海地区的建立，切断了科学研究的资金来源。17 世纪，法国科学院和英国皇家学会在相互竞争的海上贸易经济推动下，恢复了对科学的国家资助。

During the Hellenistic period, competing merchant states funded scientific research for economic and political reasons. Archimedes' new science of hydrostatics was driven by economic and military necessity, and he used it to deduce the maximum stable load that hulls could bear. However, the Roman conquest ended funding for science, and Archimedes was famously killed. The establishment of a single state over the inland sea removed the sources of funding for science. The French Academy and British Royal Society resumed state funding of science in the seventeenth century, driven again by the existence of mercantile sea-borne trading economies in mutual competition.

在上述阿基米德的证明中，他利用等价性推导球体性质。但数学中的这种等价性论证依赖关于力学的先前的公理。这些公理又假设力学定律本身由等价关系定义。

In Archimedes' proof above, he is using equivalence to deduce the properties of the sphere. But this equivalence argument in maths is relying on prior axioms about mechanics. These in turn assume that mechanical laws themselves are defined in terms of equivalence relations.

这也适用于他在流体静力学方面那些更为人熟知的发现。这里的关键定律是：漂浮物体排开与自身等重的液体；浸没物体则排开与自身等体积的液体，这比它自身的重量要小。我们在这里再次遇到体积的相等与重量的相等，这正是上述证明的主题。

〔校注：底本“this being less than its own weight”指代不清，字面上混用了体积与重量。若指下沉物体所排液体的重量小于物体重量，则需限定密度条件；正文保留其原有表述。〕

The same applies to his much better-known discoveries in hydrostatics. The key laws here are that a floating body displaces its own weight of liquid, whereas one that is submerged displaces its own volume of liquid, this being less than its own weight. Here again we have equivalences of volumes and equivalences of weights, the themes in the proof we have run through.

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阿基米德用于确定质心和浮力的几何证明方法需要极高巧思和深刻理解，今天很少有人能掌握。我们改用有限元方法，本质上是数值积分，作为替代。然而，没有电子计算工具，几何学仍是解决这些问题的少数工具之一。在古典几何中，一方面有机械性的物质过程：绘制直线和相交圆；另一方面有用

Archimedes' geometrical proof methods for determining centres of mass and buoyancy demand ingenuity and depth of understanding that few can master today. Instead, we use finite element methods, essentially numerical integration, as a substitute. However, without electronic computational aids, geometry remains one of the only tools to solve these problems. With classical ge-

有理数进行的实际计算，阿基米德为 π 设定的明确数值界限就是例证。让我们稍作思考。

ometry, there is on the one hand a mechanical material process: the drawing of straight lines and intersecting circles; on the other hand, there is actual calculation in terms of rational numbers, as exemplified by the explicit numerical bounds Archimedes set for π. Let us reflect for a bit on what is involved here.

假设你是一位建造神庙的建筑师。你用木炭绘制图纸，也许在纸莎草纸上，也许在白色大理石板上。这些图纸构成未来神庙的物质模型。[3]图纸用直尺和圆规等机械工具绘制。但假设图纸只使用这些工具绘制，长度由例如“正方形面积加倍法”或黄金比例矩形确定。那么你也知道最终神庙的大致线性尺寸。这些尺寸也以比例形式存在，最终可从毕达哥拉斯定理推导，这些比例必然存在于设计元素的长度之间。理论几何学确立了缩尺图纸上必然近似的测量值与最终建筑更精确的相对尺寸之间的相似性。根据所需精度，可以一边放大比例一边重复相同构造，仍使用直线和圆。直线可以通过立柱的视觉对齐标记，大圆可以用锚定在固定点的绷紧链条描绘。

Suppose you were an architect constructing a temple. You draw plans in charcoal, perhaps on papyrus or perhaps on a white marble slab. These plans constitute a material model of the temple to come.[3] The plans are drawn with mechanical aids, ruler and compass. But suppose the drawing is one made using just these tools, with lengths set by, for example, the ‘square doubling technique’ or the drawing of a rectangle in the Golden Ratio. In that case, you also know the approximate linear dimensions of the final temple. These will also be in ratios, ultimately derivable from Pythagoras’ theorem, that must exist between the lengths of the elements of the design. The theoretical geometry establishes a similarity between the necessarily approximate measurements on the scale drawing and the much more precise relative dimensions of the final building. Depending on the accuracy required, the same constructions can be done while increasing scales, again using straight lines and circles. Straight lines can be marked by visual alignment of posts and large circles traced out with taut chains anchored to a fixed point.

机械唯物主义使一种物质系统——平面图——能够作为另一种物质系统——神庙——的模型。二者都是物质。平面图与建筑之间的相似性，通过在放大比例上重复相同构造步骤得到保证。所构建的图像既是客观知识，也是建筑的目的（telos）。物理建模过程使众多工匠能够协调合作，每人建造子组件、立柱、基座等，其各自尺寸由黄金比例协调。

Mechanical materialism allows one material system, the plan, to act as a model for another, the temple. Both are material. The similarity between plan and building is assured by the repetition of the same constructive steps on an enlarged scale. The constructed image is both objective knowledge and *telos* of the building. The physical modelling process allows the coordination of the multiple craft workers, each constructing subcomponents, pillars, plinths etc., whose individual dimensions are coordinated by golden ratios.

古代唯物主义者对原子论的批评

Criticism of atomism by ancient materialists

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古代存在不认同原子论的唯物主义者。原子论虽然根本有效，但在古典时期失去了公信力。这是因为它试图解释一些对当时粗浅原子论来说过于复杂的现象。伊壁鸠鲁知道静电和磁现象。他知道摩擦过的琥珀棒会吸引灰尘和谷壳，磁石会吸引铁。

In ancient times, there existed materialists who did not subscribe to atomism. The atomic theory, which is fundamentally valid, suffered a loss of credibility during the classical era. This was due to its attempts to account for phenomena that were far too complex for the rudimentary form of atomism available at the time. Epicurus knew of electrostatic and magnetic phenomena. He knew that a rubbed amber rod would attract dust and chaff. He knew that lodestone attracted iron.

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他对此的解释是：起吸引作用的物体（琥珀或磁石）释放出微小粒子，其大小比其他物质的原子小一万倍。这些更小的原子形状特殊，会与被吸引物体缠绕，把二者结合。

He interpreted this in terms of the attracting object (the amber or lodestone) emitting tiny particles, ten thousand times smaller than the atoms of other things. The shapes of these tinier atoms were such that they became entangled with the object being attracted and bound the two together.

我们现在知道，静电吸引确实涉及更小粒子：琥珀摩擦时发生电子转移。电子确实比构成琥珀的原子轻数万倍。但在没有正负电荷概念和超距作用力概念时，伊壁鸠鲁学派的解释似乎不可信。应用于磁学时更不可思议。

We now know that electrostatic attraction does involve tinier particles: electrons being transferred when amber is rubbed. An electron is indeed of the order of tens of thousands of times lighter than the atoms making up the amber. But without the notion of positive and negative charge and without the notion of action-at-a-distance forces, the Epicurean explanation seemed implausible. Applied to magnetism, it was still more incredible.

原子论是古典罗马医学中的重要哲学流派，主要由阿斯克勒皮亚德斯倡导。这一观点认为，疾病源于允许分子循环的微小体孔收缩或扩张。如果体孔收缩，疾病被归类为“停滞”；如果扩张，则被归类为“流溢”。

Atomism was a significant philosophical approach in classical Roman medicine, chiefly advocated by Asclepiades. This perspective suggested that diseases resulted from the constriction or dilation of tiny body pores that allowed molecules to circulate. Diseases were classified as ‘stasis’ if the pores were constricted and as ‘flux’ if they were dilated.

该假说并非完全荒谬，因为现代医学仍承认血管收缩和血管扩张可能是病理现象。但应用于解剖细节时，它可能变得荒谬。阿斯克勒皮亚德斯认为，水通过膀胱壁上不可见的微小通道从血液进入膀胱。

The hypothesis was not totally absurd, since modern medicine still recognizes vaso-constriction and vaso-dilation as possible pathologies. But when applied to the details of anatomy, it could become absurd. Asclepiades held that water passed from the blood into the bladder via invisible tiny channels in the wall of the bladder.

另一位医学导师盖伦是解剖学专家，他轻而易举地驳斥了这些关于膀胱的说法。他描述了活体解剖实验，证明膀胱通过输尿管流入尿液而充盈：

A rival medical teacher, Galen, who was an expert on anatomy, had no difficulty in refuting these claims about the bladder. He describes vivisection experiments which demonstrated that the bladder was filled via a flow of urine through the ureters.

现在，证明方法如下。必须切开输尿管前方的腹膜，然后用结扎线固定输尿管，接着给动物包扎好，再放它走（因为它不会继续排尿）。此后，松开外部绷带，显示膀胱是空的，而输尿管完全充盈并膨胀——事实上几乎要破裂；解开输尿管结扎线后，便可清楚看到膀胱充盈尿液。[4]

Now the method of demonstration is as follows. One has to divide the peritoneum in front of the ureters, then secure these with ligatures, and next, having bandaged up the animal, let him go (for he will not continue to urinate). After this one loosens the external bandages and shows the bladder empty and the ureters quite full and distended –in fact almost on the point of rupturing; on removing the ligature from them, one then plainly sees the bladder becoming filled with urine.[4]

在从解剖学上批判伊壁鸠鲁学派后，盖伦接着质疑他们对静电和磁吸引的解释。他说，伊壁鸠鲁认为铁被磁石吸引，谷壳被琥珀吸引。他解释说，从磁石流出的原子容易与从铁流出的原子相互交织，导致缠结和吸引。尽管假说缺乏说服力，伊壁鸠鲁承认吸引存在。但很难相信磁石粒子能吸引铁。即使接受这一点，仍无法解释为什么铁会粘在另一块铁上。这是如何发生的？

Having criticized the Epicureans on anatomical grounds, Galen then goes on to cast doubt on their explanations of electrostatic and magnetic attraction. Epicurus, he said, believes that iron is attracted to the lodestone and chaff to amber. He explains that the atoms flowing from the stone easily interlock with those flowing from the iron, causing entanglement and attraction. Despite his unconvincing hypotheses, Epicurus acknowledges the existence of attraction. But it’s hard to believe that particles from a lodestone can attract iron. Even if we accept this, it still doesn’t explain why iron sticks to other iron. How does this happen?

粒子与铁碰撞后反弹，使铁悬浮吗？还是它们穿透铁块，与另一块铁碰撞而无法穿透？然后它们回到第一块铁，像之前那样造成缠结吗？

Do the particles collide with the iron and bounce back, suspending it? Or do they penetrate the iron and collide with another piece without being able to pass through? Then do they return to the first piece and cause entanglements like before?

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这一假说因其荒谬而被明确驳斥：事实上，我曾见过五根铁制书写针成一排相互附着，只有第一根与磁石接触，而磁力通过它传递给其余几根。此外，不能说如果把第二根铁笔与第一根下端接触，它就被固定、附着并悬停；而如果接触针身侧面任何其他部位，它却不会附着。因为磁

The hypothesis here becomes clearly refuted by its absurdity. As a matter of fact, I have seen five writing-stylets of iron attached to one another in a line, only the first one being in contact with the lodestone, and the power being transmitted through it to the others. Moreover, it cannot be said that

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石的力量向四面八方扩散；它只需在任何一点与第一根针接触；力量便从这根针出发，快如思想般流经第二根，再从第二根流向第三根。现在，试想一块小磁石悬挂在屋内，其四周接触大量铁片，这些铁片又连接其他铁片，这些铁片再连接其他铁片，如此类推——所有这些铁片必然充满从磁石散发的微粒；因此，这块小磁石很可能因分解为这些散发物而消散。[5]

if you bring a second stylet into contact with the lower end of the first, it becomes held, attached, and suspended, whereas, if you apply it to any other part of the side it does not become attached. For the power of the lodestone is distributed in all directions; it merely needs to be in contact with the first stylet at any point; from this stylet again the power flows, as quick as a thought, all through the second, and from that again to the third. Now, if you imagine a small lodestone hanging in a house, and in contact with it all round a large number of pieces of iron, from them again others, from these others, and so on, —all these pieces of iron must surely become filled with the corpuscles which emanate from the stone; therefore, this first little stone is likely to become dissipated by disintegrating into these emanations.[5]

直到 20 世纪中叶，人们才真正理解铁的磁性。正确理解磁性需要一系列进一步进展：超距作用力、法拉第对电与磁现象的统一，以及电子和电子自旋的发现。

The magnetism of iron was not understood until the mid-twentieth century. A proper understanding of magnetism required a whole set of further advances: action-at-a-distance forces, Faraday’s unification of electric and magnetic phenomena and the discovery of electrons and electron spin.

盖伦致力于推广一种与之竞争的唯物主义生理学理论，允许物质相互转化。他认为食物通过肝门静脉的一种能力转化为血液。他嘲笑“面包中包含肉和骨的元素”这一说法，主张身体具有转化物质的能力。

Galen was concerned to promote a rival materialist theory of physiology which allowed transformation of substances, one into another. He believed that food was converted into blood via a faculty of the hepatic portal vein. He ridiculed the notion that bread contained within it the elements of flesh and bone, arguing instead that it was a faculty of the body to transmute substances.

盖伦认为，神经以气动方式工作。这一理论最早由埃拉西斯特拉图斯提出，他认为神经中有空腔，精神之气（psychic pneuma）[6]从中通过。气压沿神经传导，使肌肉膨胀并由此收缩。盖伦与埃拉西斯特拉图斯在生理学上都相信“气”（pneuma）的作用。吸入的空气在肺中转化为生命之气（vital pneuma），与血液混合并到达脑部。在那里，它被转化为精神之气，即“动物精气”（spiritus animalis），调节脑与神经的功能。[7]

Galen believed that the nerves worked pneumatically. This was a theory initially advanced by Erasistratus, who held that nerves contain a cavity through which *psychic pneuma*[6] passed. That air pressure was transmitted down the nerves and caused muscles to expand and thus contract. Galen and Erasistratus believed in ‘pneuma’ in physiology. Inhaling air transforms into vital pneuma in the lungs that mixes with blood and reaches the brain. It’s converted into *psychic pneuma*, ‘spiritus animalis’, which regulates brain and nerve function.[7]

生命依赖空气这一点显而易见。如果动物鼻孔和嘴巴被堵住，它就会窒息。他的解剖实验证明，如果切断神经，被解剖动物在该神经支配区域就失去运动能力。此外，古人懂得制造气动装置，甚至简单气动自动机。气动技术是那个时代的先进技术。[8]

That life depended on air was clear. If an animal had its nose and mouth stopped, it suffocated. His dissection experiments had demonstrated that if he cut a nerve, the animal he was dissecting lost the power of movement in the area supplied by the nerve. Moreover, the ancients knew how to build pneumatically powered machines, or even simple pneumatic automata. Pneumatics was the advanced technology of the age.[8]

盖伦的唯物主义为天主教话语中的唯心主义主题奠定基础。他否认不朽灵魂存在，认为意识取决于大脑。在随后的基督教时期，一种经过修改的理论出现：不朽精神栖息于大脑脑室，赋予身体生命。这把原本的唯物主义理论转变为其对立面。

Galen’s materialism formed the basis for idealistic themes in Catholic discourse. He denied the existence of an immortal soul and believed that consciousness depended on the brain. In the Christian period that followed, a modified theory emerged, stating that the immortal spirit resided in the brain’s ventricles, animating the body. This transformed the original materialist theory into its opposite.

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不再认为动物精气由身体从吸入肺部的物质空气（pneuma）制造，而是《天主教教理》声称：

Instead of the animal spirit being manufactured by the body from the material air (pneuma) inspired into the lungs, we get the Catholic Catechism stating:

基督教关于人的观点——人按上帝形象被造，既有灵性灵魂又有肉体——具有核心重要性。灵魂是我们内心承载的永恒之种，不能仅归结为物质，其起源只能来自上帝。

……

人体之所以是人，恰恰因为它被灵性灵魂赋予生命。

基督教教义是过去实验科学被扭曲的记忆或被误解的遗迹。spiritus（盖伦 pneuma 的拉丁译名）原本的气力机械含义完全丧失。这个词脱离原义，变成完全抽象的概念，神学家便可随意赋予它想象属性。

The Christian vision of the human person made in the image of God with a spiritual soul as well as a body is of central importance. The soul, the seed of eternity we bear in ourselves, irreducible to the merely material, can have its origin only in God.

. . .

The human body is human precisely because it is animated by a spiritual soul.

The Christian doctrine is the distorted memory or misunderstood relic of a past experimental science. The actual pneumatic mechanical meaning of spiritus (Latin translation of Galen's pneuma) is totally lost. The word, detached from its original real meaning, became something entirely abstract, to which the theologians became free to attach any imaginary properties.

牛顿

Newton

在牛顿时代，若有人提及古典思想家，指的是希腊人和罗马人。如今，随时间流逝，牛顿被视为机械唯物主义的典范哲学家，以至于“经典力学”和“牛顿力学”实际上成为同义词。

通过卢克莱修，宇宙由不断运动的粒子构成的观念在古典文明衰落后存续。通过阿基米德，后来被称为静力学的理论也存续。我们现在知道，阿基米德还发展了非常接近微积分的数学技术。但让我们了解这些内容的手稿直到 19 世纪末才印刷出版。鲁索[9]推测，也许牛顿接触过这部及其他后来因霉变和腐朽失传的古典文献。无论推测是否属实，微分学正是通过牛顿和莱布尼茨传播的。牛顿把力学从静止物体科学转变为运动物体科学。

In Newton's day, if someone referred to the classical thinkers, they meant the Greeks and the Romans. Now, with the passage of time, Newton is seen as the archetypal philosopher of mechanical materialism, so much so that the terms *classical mechanics* and *Newtonian mechanics* have effectively become synonymous.

Via Lucretius, the idea of a universe made up of constantly moving particles had survived the fall of classical civilization. Via Archimedes, what later came to be known as static mechanics had also survived. We now know that Archimedes had also developed mathematical techniques very close to calculus. But the manuscript by which we know these things was not published in print till the late nineteenth century. Russo[9] speculated that perhaps Newton had access to this and other classical texts, which, through mould and decay, have subsequently been lost to us. Whatever the truth of these speculations, it is via Newton and Leibniz that the differential calculus was subsequently promulgated. Newton transformed mechanics from a science of static bodies to one of moving bodies.

哲学的任务

The task of philosophy

牛顿把哲学任务阐述如下：

我们提供这部著作作为哲学的数学原理；因为哲学的全部困难似乎在于：从运动现象出发探究自然力，再从这些力出发证明其他现象。

从运动观察出发，从中推断力和运动定律。这使你能够审视新情况并预测将发生什么。每一艘抵达火星或木星的太空探测器都证实这一方法。

Newton sets out the task of philosophy as follows:

We offer this work as the mathematical principles of philosophy; for all the difficulty of philosophy seems to consist in this –from the phænomena of motions to investigate the forces of nature, and then from these forces to demonstrate the other phænomena;

Start with observations of motion. Infer from these the forces and laws of motion. This allows you to look at new situations and predict what will occur. Every space probe arriving at Mars or Jupiter confirms this approach.

p. 27

我们注意到，卢克莱修的原子论也基于对经验可观察现象的演绎。作为诗人，这些推论自然流露于叙述；作为哲学家，牛顿将其系统化为四条明确推理规则：

We noted that Lucretius also based his atomism on deductions from what was empirically observable. As a poet, these simply come out in the telling. As a philosopher, Newton systematized this in a set of four explicit rules of reasoning, which we quote below:

规则 I

RULE I

对于自然事物，我们承认的原因不应多于既能解释其现象又真实且充分的原因。

〔校注：本书英文底本此处作 time，语义不通；中文按上下文采用“真实”，英文仍照录 time，未将校正混入底本文字。〕

We are to admit no more causes of natural things than such as are both time and sufficient to explain their appearances.

为此，哲学家们说自然不做无用之事，当较少原因足够时，更多原因就是无用；因为自然喜欢简单，不追求多余原因的浮华。[10]

To this purpose, the philosophers say that Nature does nothing in vain, and more is in vain when less will serve, for Nature is pleased with simplicity and affects not the pomp of superfluous causes.[10]

这就是俗称的 KISS 原则：Keep It Simple, Stupid（笨蛋，保持简单）。在此之前，它被称为奥卡姆剃刀：不应在必要原因之外增加实体。牛顿只是重申前辈哲学家的共识：简单解释优于复杂解释。

This is what is colloquially called the KISS principle: Keep It Simple, Stupid. Before that, it was known as Occam’s razor, which holds that entities should not be multiplied beyond cause. Here Newton is simply repeating, as he says, the consensus of previous philosophers that simple explanations should always be preferred over complex ones.

在数学上，这意味着探索自然法则时应优先选择最简单公式，而非更复杂公式。牛顿的同时代人莱布尼茨也持相同观点：

In terms of mathematics, this means that we should always prefer the simplest mathematical formula over the more complex when trying to discover a natural law. Leibniz, Newton’s contemporary, argues the same point

因此，让我们假设，例如，有人像那些从事荒谬地占术（geomancy）的人一样，在纸上随意标出若干点。我认为，完全可能找到一条几何曲线，其运动恒定且均匀，遵循某种规则，使这条曲线按照手写下这些点的顺序依次穿过所有点。

Thus, let us assume, for example, that someone jots down a number of points at random on a piece of paper, as do those who practice the ridiculous art of geomancy. I maintain that it is possible to find a geometric line whose motion is constant and uniform, following a certain rule, such that this line passes through all the points in the same order in which the hand jotted them down.

但当规则极其复杂时，符合该规则的事物便被视为不规则。因此可以说，无论上帝以何种方式创造世界，它始终有规律并符合某种普遍秩序。然而上帝选择了最完美的世界，即在假设上最简单、在现象上最丰富的那一个，正如几何学中一条构造简单但性质与效果极其显著且影响广泛的曲线。[11]

But when a rule is extremely complex, what is in conformity with it passes for irregular. Thus, one can say in whatever manner God might have created the world, it would always have been regular and in accordance with a certain general order. But God has chosen the most perfect world, that is, the one which is at the same time the simplest in hypotheses and the richest in phenomena, as might be a line in geometry whose construction is easy and whose properties and effects are extremely remarkable and widespread.[11]

莱布尼茨打算用何种具体方法，构造一条穿过所有随机点的光滑曲线，并未说明。在 18 世纪，或许会用如下多项式来拟合这些点：

The precise means by which Leibniz intended to construct a smooth curve through all the random points is not specified. In the eighteenth century, perhaps the means would have been to fit a polynomial like

$$a + bx + cx^2 + dx^3 + ex^4 + \cdots$$

to the points. Fourier would later discover a technique by which you can arbitrarily fit a curve through points using sine and cosine functions. We will return to this.

p. 28

傅里叶后来发现一种技术，可以用正弦和余弦函数把曲线任意拟合到点上。我们稍后回到这一点。

牛顿能举出的最明显的例子，是他的太阳系解释比托勒密简单得多。

The most obvious example that Newton could invoke in terms of simplicity was how much simpler his expla-

nation of the solar system was than that advocated by Ptolemy.

用牛顿力学处理太阳系，我们需要每颗行星在时间 0 的角动量、半径和旋转相位。运动定律提供其余信息。这意味着每颗行星只需三个数。我忽略太阳质量和引力常数，因为假设它们已知，只考虑每颗行星的额外信息。

When dealing with the solar system via Newtonian mechanics, we need the angular momentum, radius and rotational phase for each planet at time 0. The laws of motion give us all the rest. This amounts to three numbers per planet. I am disregarding knowledge of the mass of the Sun and the gravitational constant, as I am assuming these are given from the outset and we are just considering the extra information needed for each planet.

托勒密模型中，每颗行星绕地球旋转，一个均轮上还有一个本轮旋转。因此至少需要两个半径、两个旋转速率和两个初始相位：每颗行星六个数。所需参数数量是牛顿模型的两倍，形成冗余。因此基于信息经济性，牛顿更可取。他的理论还额外能解释彗星运动，而旧天文学没有相关定律。

If we take the Ptolemaic model, each planet rotates around the Earth with one cycle upon which one epicycle rotates. So for Ptolemy, we needed at least two radii, two rates of rotation and two starting phases: six numbers per planet. That amounts to a twofold redundancy compared to Newton. So on the grounds of the information economy, Newton was to be preferred. His theory had the added bonus that it could also account for the motion of comets – for which no law had existed in the older astronomy.

如果允许本轮叠加本轮，原则上可以把观测数据拟合到任意精度。如图 3.2 所示，托勒密方法相当于把余弦波相加。我们把它简化为向水平轴的投影。对托勒密天文观测，x 轴是天球黄道 360°，他以此为参照绘制行星位置。显然，这与傅里叶分析相同。[12]信号的傅里叶分解为不同频率余弦波，正是 JPEG 和 MPEG 等图像压缩方案使用的方法。观看 YouTube 视频时，你看到的是托勒密方法在运作。计算机把图像分析为不同旋转速度的“本轮”。视频文件包含这些本轮半径，视频编解码器据此重建原始图像的近似。

If we are allowed to add epicycles on epicycles, then it is in principle possible to fit observations, whether of planets or any other data, to any degree of accuracy we want. As Figure 3.2 illustrates, the Ptolemaic approach amounts to adding cosine waves together. We illustrate it as a simple projection down onto the horizontal axis. For Ptolemaic astronomical observations, the x-axis is the 360◦ of the ecliptic in the sky against which he plotted the planetary positions. It is clear that what is going on here is the same as Fourier analysis.[12] Fourier decomposition of a signal into cosine waves of different frequencies is what image compression schemes like JPEG and MPEG use. When you view a YouTube video, you are seeing the Ptolemaic method in action. A computer has analysed the picture into what amounts to epicycles with different rotational velocities. The video file contains what amounts to the radii of these epicycles, from which the video codec can reconstruct a good approximation of the original image.

p. 28–29

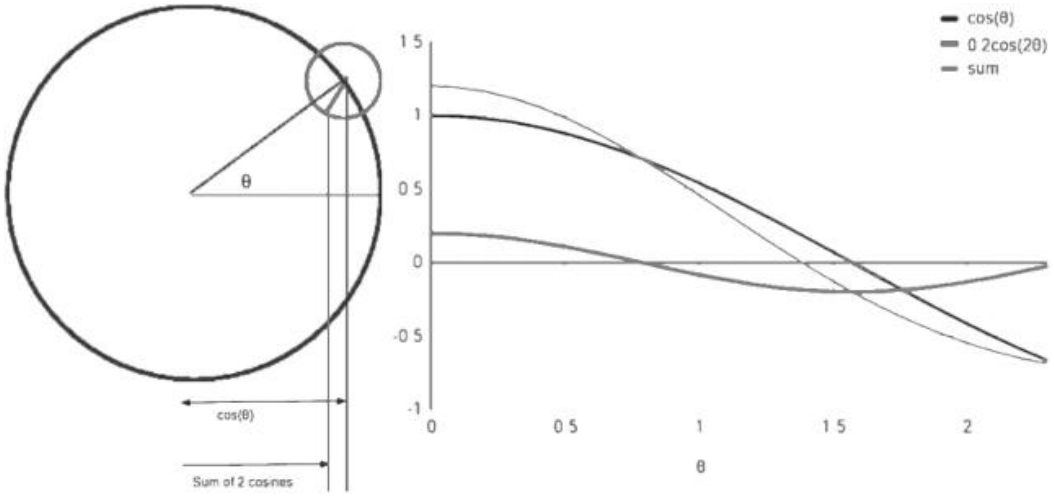


图 3.2 具有本轮运动路径的物体，其轨道在 x 轴上的投影是余弦曲线之和。本轮直径为 0.2，转速是大圆的两倍。右图绘出余弦曲线及其和。作者绘图。

Figure 3.2 The projection of the orbital of a body with an epicyclic path onto the x-axis is a sum of cosine curves. The epicycle has a diameter of 0.2 and rotates at twice the rate of the large circle. The right-hand graph plots the cosines and their sums. Figure by authors.

cos θ：余弦 θ；0.2 cos 2θ：0.2 倍的余弦 2θ；sum：两者之和。坐标数字及 θ 沿用原图。

p. 29

正如莱布尼茨暗示的，我们可以把方法应用于任何数据集。图 3.3 应用于石油销量（百万桶）与每桶价格之间的关系。

We can, as Leibniz implies, apply the method to any data set we want. Figure 3.3 applies it to the relationship between the number of barrels of oil sold (in millions) and the price per barrel at which they were sold.

公式：

The formula

$$\begin{aligned} p = {} & 136\cos(2+\theta) + 27.7\cos(2+2\theta) + 39.5\cos(2+3\theta) \\ & + 19.6\cos(2+4\theta) + 62\cos(2+5\theta) \\ & + 52\cos(2+6\theta) + 50.1\cos(2+7\theta) \\ & + 37\cos(2+8\theta) + 70.5\cos(2+9\theta) + 19\cos(2+10\theta) \end{aligned}$$

p. 29–30

其中 $\theta = \pi(q-80)/10$，p 是油价，q 是销量。这正是莱布尼茨所说的情形：“我认为，完全可能找到一条几何曲线，其运动恒定且均匀，遵循某种规则，使这条曲线按照手写下这些点的顺序依次穿过所有点。”

with $\theta = \pi(q-80)/10$, where p is the price of oil and q the quantity sold, does what Leibniz described when he said: ‘*I maintain that it is possible to find a geometric line whose motion is constant and uniform, following a certain rule, such that this line passes through all the points in the same order in which the hand jotted them down*’.

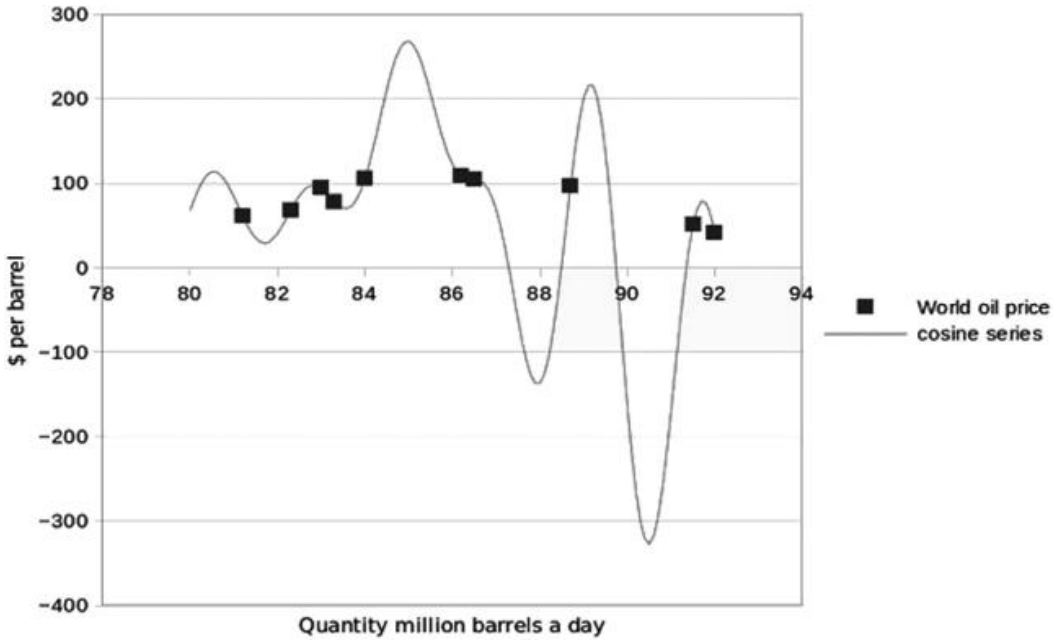


图 3.3 一条连续曲线，穿过代表 2007 年至 2016 年间所采集的世界石油价格 / 出货量数据的各点。此曲线用托勒密天文学所用的本轮方法生成。作者绘图。

Figure 3.3 A continuous curve that goes through the points representing world oil price/ shipment data sampled between 2007 and 2016. The curve was produced by applying the epicycle method used in Ptolemaic astronomy. Figure by authors.
$ per barrel：美元 / 桶；Quantity million barrels a day：每日数量（百万桶）；World oil price：世界石油价格；cosine series：余弦级数。

但显然，把该公式等同于根据供应量调节油价的定律是荒谬的。该规则“极其复杂”，符合它的数据“不规则”。相比之下，牛顿运动定律简单且规律性极强地应用于天文现象。因此它们被视为真正的自然定律。

But it would clearly be absurd to identify this formula with a law regulating the price of oil as a function of quantity supplied. The rule is‘extremely complex’and the data in conformity with it is‘irregular’. In contrast, Newton’s laws of motion are simple and apply with great regularity to astronomical phenomena. As such, they count as real laws of nature.

p. 30

规则 II

RULE II

因此，对相同自然效果，必须尽可能赋予相同原因。

Therefore to the same natural effects we must, as far as possible, assign the same causes.

例如人与动物的呼吸；欧洲与美洲的石头下落；烹饪之火与太阳的光；地球与行星上的光反射。

As to respiration in a man and in a beast; the descent of stones in Europe and in America; the light of our culinary fire and of the Sun; the reflection of light in the Earth and in the planets.

例子虽旧且通俗，但这些例子与这一原则仍被接受。根据光谱估算遥远恒星温度时，我们假设那里与地球适用相同黑体辐射定律。天文学家利用地球矿物样本反照率推导小行星成分时，也应用这一原则。

The examples are a bit old and homely, but both the examples and the principle are still accepted. When temperatures of distant stars are estimated from their spectra, we assume that the same black body radiation laws apply there as on Earth. When astronomers attempt to deduce the composition of asteroids from their albedo, using the albedos of mineral samples here on Earth, they apply this principle.

其背后是刻意排除超自然或宗教原因。牛顿的同时代人会明白，这里指阿那克萨戈拉：他最早提出月球是反射太阳光的岩石，太阳本身是燃烧岩石，因这一不敬神的说法被放逐出雅典。牛顿在规则 II 中隐含站在古代唯物主义者一边。

Underlying it is the deliberate exclusion of supernatural or religious causes. Newton’s contemporaries would have understood that reference was being made to Anaxagoras, who first posited that the Moon was a rock reflecting light of the Sun, itself a burning rock, before

being exiled from Athens for such an impiety. Newton implicitly sides with the ancient materialist in Rule II.

规则 III

RULE III

那些不允许增强或减弱程度、且被发现属于实验范围内所有物体的性质，应被视为所有物体的普遍性质。

The qualities of bodies, which admit neither intension nor remission of degrees, and which are found to belong to all bodies within the reach of our experiments, are to be esteemed the universal qualities of all bodies whatsoever.

科学需要这一规则才能从实验概括。牛顿用摆实验研究动量守恒定律。如果动量守恒在实验中成立，他推断同样适用于月球和行星。阿基米德观察自己入浴时水被排开，推断船只入海也排开一定体积水。

Science needs this rule if it is to be able to generalize from experiment. Newton carried out experiments using pendulums to investigate what he formulated as a law of conservation of momentum. If the conservation of momentum applied in experiments, he inferred that the same applied to the Moon and planets. When Archimedes observed water being displaced as he stepped into a bath, he inferred that a ship likewise displaces a volume of water as it slides into the sea.

规则 IV

RULE IV

p. 30–31 在实验哲学中，我们应把通过一般归纳从现象中收集的命题视为准确或非常接近真实，尽管可能想象出相反假设，直到出现其他现象使它们更精确或表明例外。

In experimental philosophy we are to look upon propositions collected by general induction from phænomena as accurately or very nearly true, notwithstanding any contrary hypotheses that may be imagined, till such time as other phænomena occur, by which they may either be made more accurate, or liable to exceptions.

p. 31 我们必须遵循这一规则，以免归纳论证被假设规避。

This rule we must follow that the argument of induction may not be evaded by hypotheses.

鉴于“假设”一词当前联想，这容易被误解。科学通常被描述为提出假设，然后由观察和实验检验。牛顿为何说“我不做假设”?

This is easily misunderstood given the current associations with the word ‘hypothesis’. A common presentation of science is that it advances hypotheses about reality, which are then put to the test of observation and experiment. Why did Newton famously remark that he made no hypotheses?

在此语境中，他指关于世界必须如何运作的先验假设。例如地球固定在宇宙中心，或力只能通过直接物理接触作用。若坚持其中任何一种，就会规避他在《世界体系》中建立的归纳论证。

In this context, he meant a priori assumptions about how the world must work. Instances of this would have been the hypothesis that the Earth is fixed at the centre of the universe, or the hypothesis that forces could only act by direct physical contact. Either of these would, if insisted upon, have resulted in an evasion of the arguments from induction that he builds up in his *System of the world*.

我们稍后探讨“通过假设规避归纳论证”在历史唯物主义中如何引发争议。

We will later touch on how ‘evading the argument of induction by hypotheses’ has caused controversy in historical materialism.

关于力学的一点说明

A remark on mechanics

我们上文说牛顿力学建立在古典力学基础上；在阿基米德部分，我们叙述几何学本身对古人而言是用

We said above that Newtonian mechanics rested on the foundation of classical mechanics, and in the section on

于建筑的机械艺术。牛顿在《原理》全书中依赖古典几何学。他把方法延伸至微积分，但在研究天体运动时仍以欧几里得为立足点。

Archimedes we recounted how geometry itself was for the ancients a mechanical art to be used in architecture. Newton relies throughout the *Principles* on classical geometry. He extends its methods towards the calculus, but he relies on Euclid as the place to stand while he studies the movement of worlds.

他对机械艺术与几何学关系有有趣观察：

He makes some interesting observations on the relationship between mechanical arts and geometry:

描绘直[13]线和圆是问题，但不是几何问题。这些问题的解决需要力学；几何学展示解决后的应用；几何学的荣耀在于，它仅凭从外部引入的少数原理，便能产生如此多东西。

To describe right[13] lines and circles are problems, but not geometrical problems. The solution of these problems is required from mechanics; and by geometry the use of them, when so solved, is shown; and it is the glory of geometry that from those few principles, brought from without, it is able to produce so many things.

他说，描绘直线和圆——即绘制或标定——不是几何问题，而是制图师或测量员的机械实践问题。公理化演绎几何学利用这些力学简单结果，展示如何组合它们以“产生如此多东西”。

He is saying that describing straight lines and circles, that is to say, describing in the sense of drawing or laying out, is not a problem for geometry. It is a problem for the mechanical practice of the draftsman or the surveyor. Axiomatic and deductive geometry then takes these simple results of mechanics and shows how to compose them to ‘produce so many things’.

因此几何学建立在机械实践上，无非是普遍力学中准确提出并证明测量艺术的部分。但由于手工技艺主要涉及物体运动，几何学通常归于大小，力学归于运动。在这个意义上，理性力学将成为一门科学：准确提出并证明由任何力产生的运动，以及产生任何运动所需的力。

Therefore geometry is founded in mechanical practice, and is nothing but that part of universal mechanics which accurately proposes and demonstrates the art of measuring. But since the manual arts are chiefly conversant in the moving of bodies, it comes to pass that geometry is commonly referred to their magnitudes, and mechanics to their motion. In this sense rational mechanics will be the science of motions resulting from any forces whatsoever, and of the forces required to produce any motions, accurately proposed and demonstrated.

这门科学研究的是物质物体的运动。他将借助几何学，分析运动及引起运动的力。接下来，他定义了自己以几何方式处理物质与运动时将要使用的术语。

The science is to be about the motion of material bodies. It is by geometry that he will analyse the motions and forces which impel motions. He goes on to define the terms he will use in this geometric treatment of matter and motion.

p. 32

定义

Definitions

定义 I

DEFINITION I.

物质的量是同一物质的度量，由密度和体积共同产生。

The quantity of matter is the measure of the same, arising from its density and bulk conjunctly.

因此，密度加倍的空气，在双倍空间中，量是四倍；在三倍空间中，量是六倍。雪、细尘或粉末经压缩或液化而致密，以及所有因任何原因以不同方式致密化的物体，都应如此理解。我在此不考虑任何可能自由渗透物体各部分间隙的介质。今后我无论何处提到物体或质量，都指这个量。它通过每个物体的重量得知；因为重量与它成正比，正如我通过精密摆实验发现的那样，后文将展示。

Thus, air of a double density, in a double space, is quadruple in quantity; in a triple space, sextuple in quantity. The same thing is to be understood of snow, and fine dust or powders, that are condensed by compression or liquefaction; and of all bodies that are by any causes whatever differently condensed. I have no regard in this place to a medium, if any such there is, that freely pervades the interstices between the parts of bodies. It is this quantity that I mean hereafter everywhere under the name of body or mass. And the same is known by the weight of each body; for it is proportional to the weight, as I have found

by experiments on pendulums, very accurately made, which shall be shewn hereafter.

他把重量确立为物质度量，并把重量与体积区分。这在经验上显而易见，但在两个重要意义上具有重要意义：

He is establishing weight as the measure of matter. He is distinguishing the weight of matter from its volume. This is pretty obvious empirically, but it is nevertheless significant in two important senses:

1. 他明确把假设的光以太（luminiferous ether）排除在质量之外，提到“一种介质（如果真有的话），它自由渗透物体各部分之间的间隙”。如果这种以太存在，他说它是无质量或非物质的。2. 他主张惯性质量与引力质量等价。虽然直到后来阐述万有引力定律才明显，但他预先提到用摆实验证明。

1. He explicitly excludes the hypothetical luminiferous ether from mass with his allusion to a ‘medium, if any such there is, that freely pervades the interstices between the parts of bodies’. If this ether exists, he says it is massless or immaterial. 2. He is asserting the equivalence of inertial and gravitational mass. Although this will not become apparent until he later states his law of gravitation, he refers forward to the experiments with pendulums by which he demonstrated this.

定义 II

DEFINITION II.

运动量是同一运动的度量，由速度和物质量共同产生。

The quantity of motion is the measure of the same, arising from the velocity and quantity of matter conjunctly.

整体的运动是各部分运动之和；因此，质量加倍、速度相等时，运动量加倍；在这一质量加倍的物体中，速度也加倍时，运动量为四倍。

The motion of the whole is the sum of the motions of all the parts; and therefore in a body double in quantity, with equal velocity, the motion is double; with twice the velocity, it is quadruple.

牛顿所说的运动即我们现在所说的动量，因此他用文字给出定义方程 $p = mv$ 。《原理》中还没有独立动能概念。我们稍后回到这一点。

What Newton called motion is what we now call momentum, so he is giving in words the defining equation $p = mv$. There is no distinct concept of kinetic energy yet present in the *Principles*. We return to this later.

定义 IV

DEFINITION IV.

p. 33

外力是作用于物体以改变其状态的作用，无论是静止状态还是沿直线匀速运动状态。

An impressed force is an action exerted upon a body, in order to change its state, either of rest, or of moving uniformly forward in a right line.

这是定性定义，由定律 I 和 II 赋予定量意义。它在很大程度上依赖定律获得含义。这是以下内容的引子：

This is a qualitative definition that is given quantitative meaning by his Laws I & II lower down. It does, however, depend in large part on the laws for its meaning. It is a prelude to:

定义 V

DEFINITION V.

向心力是使物体被吸引、推动或以任何方式趋向作为中心的某一点的力。

A centripetal force is that by which bodies are drawn or impelled, or any way tend, towards a point as to a centre.

这里他首次把方向性引入力。有趣的是，在定义 IV 中力还不是向量，没有方向性。引入方向性时，他特指向心吸引力。可能由于未引入参考系，他认为无法定义方向性。一个点允许定义朝向该点的方向性，也允许定义远离该点的力——离心力。全书大部分，牛顿谨慎处理不同力律，考虑与距离成反比、与距离平方成反比、与距离立方成反比等。令

Here he introduces, for the first time, directionality to force. It is interesting that prior to this, in Definition IV, force is not yet a vector, not yet having directionality. When he introduces directionality, it is in the specific sense of a centripetal attractive force. It is possible that absent any introduction of a frame of reference, he felt it impossible to define directionality. One point allows directionality to be defined towards that point, but it also

人惊讶的是，他把定义限制在吸引力，而未定义排斥力。即使当时未研究静电斥力，磁斥力已被研究。

allows the definition of a force away from the point – centrifugal. In most of the book, Newton is careful to deal with different kinds of force laws, considering laws that vary inversely with distance, inversely with square of distance, cube of distance etc. It is surprising that he restricts the definitions to attractive ones rather than defining repulsive ones as well. Even if the repulsive electrostatic force was not studied in his day, repulsive magnetism was.

定义 III

DEFINITION III.

vis insita（内在力），或物质固有力量，是抵抗的力量，每个物体凭此尽力保持当前状态，无论是静止还是沿直线匀速运动。

The vis insita, or innate force of matter, is a power of resisting, by which every body, as much as in it lies, endeavours to persevere in its present state, whether it be of rest, or of moving uniformly forward in a right line.

他定性定义惯性质量。然而在当今公认的牛顿力学表述中，惯性不会被描述为力。

He is here defining in a qualitative way the notion of inertial mass. However, in what is now the accepted presentation of Newtonian mechanics, inertia would not be described as a force.

他还有三个定义（VI–VIII）给出量化向心力的不同方式。第一个定义为“与从中心通过周围空间传播它的原因的效力成正比”，这有些晦涩，因为不清楚如何衡量原因效力及单位。下一个定义把向心力的加速度量定义为与单位时间诱导速度成正比。在 $f = ma$ 关系中，这相当于忽略被加速物体的惯性质量，在处理围绕大质量物体运行的不同小质量时完全合理。此时测量引力的“加速度量”等同于我们现在所说的引力场强度：牛顿每千克。他最后把向心力的动力量定义为与给定时间产生的运动成正比。他用“运动”指动量 mv，因此似乎指以千克米每秒平方为单位的量。

He has three definitions (VI to VIII) giving different ways of quantifying centripetal force. The first defines it as ‘proportional to the efficacy of the cause that propagates it from the centre, through the spaces round about’, which is rather obscure since it is not clear how one is to measure the efficacy of a cause or in what units he intends to measure it. The next defines the accelerative quantity of centripetal forces as being proportional to the velocity they induce per unit time. In the $f = ma$ relation, this amounts to abstracting from the inertial mass of the body being accelerated, something which is perfectly reasonable when dealing with different small masses in orbit around a much larger one. In this case, measuring the ‘accelerative quantity’ of the gravitational force is equivalent to what we now term the intensity of the gravitational field: newtons per kilogram. He finally defines the motive quantity of a centripetal force as proportional to the *motion* it generates in a given time. He uses the term *motion* for what we would now term momentum (mv), so he appears to mean something that in SI units would be measured in kilogram metres per second squared.

p. 33–34

时间与空间

Time and space

“物体与虚空”概念对古典原子论至关重要。对牛顿而言，任何关于物体运动的系统理论都必须先有时间和空间的概念定义。他对时间和空间的定义被广泛引用，因为正是在这一问题上，后爱因斯坦物理学与他分歧。

The conceptual pair (bodies, void) was fundamental to classical atomism. For Newton, a conceptual definition of time and space had to precede any systematic theory of the motion of bodies. His definitions of time and space are widely quoted because it is on this issue that post-Einstein physics departs from him.

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时间

TIME

绝对的、真实的、数学的时间，就其自身和本性而言，均匀流逝，与任何外部事物无关，又名持续时间；相对的、表观的、普通的时间，是借助运动对持续时间作出的一种可感知的外部度量（无论准确还是不均匀），通常用来代替真实时间；例如一小时、一天、一月、一年。

Absolute, true, and mathematical time, of itself, and from its own nature flows equably without regard to anything external, and by another name is called duration: relative, apparent, and common time, is some sensible and external (whether accurate or unequable) measure of duration by the means of motion, which is commonly used instead of true time; such as an hour, a day, a month, a year.

牛顿在此使用隐喻：绝对时间流动。但比喻虽合理，却自相矛盾。溪流流动时在运动，其运动在时间语境中定义。

Newton here applies a metaphor: that absolute time flows. But plausible as the metaphor seems, it is surely contradictory. When a stream flows, it moves. Its motion is defined in the context of time.

我们把树枝扔进桥一侧；几秒后它出现在另一侧，于是得出河水流动。但测量流动需要两个不同时间观察。在 t_0，树枝落入河流；在 $t_0 + 8$ 秒，我们看到树枝从桥另一侧下游 4 米处浮出。于是水流速度为每秒半米。

We throw a twig down one side of a bridge; a few seconds later, it appears on the other side, and we then conclude that the river flows. But to measure the flow, we need two observations at distinct times. At time t_0, the twig lands in the stream, and at time $t_0 + 8$ seconds, we see the twig emerge, 4 metres downstream on the other side of the bridge. So we conclude that the water is flowing at half a metre per second.

这是空间相对于时间的流动。但时间相对于什么流动？

This is a flow in space with respect to time. But what is time supposed to flow with respect to?

谈论“时间流动”在逻辑上与谈论“空间距离流动”一样无意义。

To speak of the flow of time makes no more logical sense than it would to speak of spatial distance flowing.

如果绘制 $y = x^2$，可以说曲线相对于 x 轴向上流动，并求导数 $2x$，它告诉我 y 随 x 变化向上流动多快。但我能这样做，因为涉及两个正交维度。类似地，对下落物体方程 $y = 4.9t^2$ 及导数 $v_y = 9.8t$，可确定物体在时间 t 的空间向下流动。这同样因为方程中时间是正交维度。但时间不能相对于自身流动。

If I plot a graph for $y = x^2$, I can say that the line flows upwards with respect to the x axis, and I can take the derivative of this, $2x$, which tells me how fast y flows up as x changes. But I can only do these things because two orthogonal dimensions are involved. Similarly, given an analogous equation of motion for a falling body, $y = 4.9t^2$ and its derivative $v_y = 9.8t$, I can determine the downward flow through space of the object at time t. Again, this is because time, in the equation, is an orthogonal dimension. But time cannot flow with respect to itself.

但他刚引入流动的绝对时间，就把它搁置，引入实际使用的相对时间。

But he no sooner introduces his flowing absolute time than he puts it to one side to introduce what he actually works with ‑relative time.

p. 34–35 他所说的可感或相对时间由运动衡量，例如年或小时。这些确实由运动衡量：一年是地球绕太阳公转，一天是地球自转。到牛顿时代，人类工艺已制造钟表，其运动模拟地球运动，地球自转一圈时针转两圈。

His sensible or relative time is measured by motions, giving as examples units like the year or the hour. These are indeed measured by motions: the motion of the Earth around the Sun for the year and the spin of the Earth on its axis for the day. By Newton’s day, human arts had constructed clocks whose motions modelled those of the Earth, with two rotations of the hour hand for each rotation of the Earth.

p. 35 据牛顿，运动不能以绝对时间为参照测量，只能相对于其他运动测量。在天文学中，古典天文学家通过测量恒星或参考星相对于天顶的角度，确定月球掩恒星的时刻。

According to Newton, motion cannot be measured in relation to absolute time. Instead, it can only be measured in relation to other motions. In the field of astronomy, a classical astronomer would determine the moment of occultation of a star by the Moon by measuring the angle with respect to the celestial zenith of either the same star or a reference star.

然后转换为恒星时[14]记录。牛顿旨在解释行星运动，在没有独立时间尺度时观察运动与位置相关性。他认为所有运动在概念上相对于时间定义，形成时间维度。地球以绝对时间为基准自转，月球绕地球角运动。牛顿从稳定比例存在推断，这些不同旋转有共同、绝对、客观的时间维度。

The astronomer could then convert this to sidereal time[14] for their records. Newton aimed to explain planetary motion, observing correlations of motions and positions without an independent measure of time. He believed that all motions are conceptually defined with respect to time, creating a time dimension. The Earth rotates with absolute time, while the Moon has an angular rotation around us. Newton inferred from the existence of stable ratios that there is a common absolute objective time dimension for these varied rotations.

空间

SPACE

绝对空间，就其本性而言，与任何外部事物无关，始终相似且不动。相对空间是绝对空间的某种可移动维度或度量；我们的感官通过其相对于物体的位置确定它；通常被当作不动空间；例如地下、空中或天体空间的维度，由相对于地球的位置决定。

Absolute space, in its own nature, without regard to anything external, remains always similar and immovable. Relative space is some movable dimension or measure of the absolute spaces; which our senses determine by its position to bodies; and which is vulgarly taken for immovable space; such is the dimension of a subterraneous, an æreal, or celestial space, determined by its position in respect of the Earth.

牛顿假设的绝对空间在 20 世纪物理学中随相对论被抛弃。但他刚提出绝对空间，就转向相对空间。他知道并接受伽利略相对性：在匀速运动的参考系中，运动定律不变。

Newton’s hypothesized absolute space is something that was dropped by twentieth-century physics with the theory of relativity. But no sooner has Newton raised the idea of absolute space than he immediately moves on to relative space. He was aware of and accepted Galilean relativity, which is to say that the laws of motion remain the same in a constantly moving frame of reference.

他后来用船的例子阐述：

He later elaborates on this relativity, giving the example of a ship:

我们从船实验得到明确证明；无论船静止还是沿直线匀速前进，所有运动以相同方式发生。

A clear proof of which we have from the experiment of a ship; where all motions happen after the same manner, whether the ship is at rest, or is carried uniformly forwards in a right line.

但既然接受伽利略相对性，为何坚持绝对空间？我们认为理由在他对运动的定义最后部分。

But given that he accepts Galilean relativity, why does he persist with the notion of absolute space? The rationale, we think, is provided in the last section of his definition of motion.

运动

MOTION

绝对运动是物体从一个绝对位置向另一个绝对位置的平移；相对运动是从一个相对位置向另一个相对位置的平移。因此在航行船上，物体的相对位置是它占据的船体部分，或它填充的船舱部分，因此随船移动；相对静止是物体持续位于船的同一部分或船舱同一部分。但真实绝对静止是物体持续位于不可移动空间的同一部分，而船本身、船舱及其中一切都在该空间中运动。因此，如果地球确实静止，船上相对静止的物体将以船在地球上的相同速度真实绝对运动。但如果地球也运动，物体真实绝对运动将部分来自地球在不可移动空间中的真实运动，部分来自船在地球上的相对运动；如果物体在船内也相对运动，其真实运动将部分来自地球在不可移动空间中的真实运动，部分来自船在地球上的相对运动。假设地球上船所在部分以 10010 份速度向东运动；船本身在强风满帆下以 10 份速度向西航行；一名水手在船上以所述速度的 1 份向东行

Absolute motion is the translation of a body from one absolute place into another; and relative motion, the translation from one relative place into another. Thus in a ship under sail, the relative place of a body is that part of the ship which the body possesses; or that part of its cavity which the body fills, and which therefore moves together with the ship: and relative rest is the continuance of the body in the same part of the ship, or of its cavity. But real, absolute rest, is the continuance of the body in the same part of that immovable space, in which the ship itself, its cavity and all that it contains, is moved. Wherefore, if the Earth is really at rest, the body, which relatively rests in the ship, will really and absolutely move with the same velocity which the ship has on the Earth. But if the Earth also moves, the

p. 35–36

走；那么水手在不可移动空间中以 10001 份速度真实向东移动，而在地球上以 9 份速度相对向西移动。

〔校注：此段底本在物体也相对于船运动的分句中，仍只列地球与船的运动，表述有所脱落；后面的水手算例明确计入了水手相对于船的速度。译文保留底本而不擅补引文。〕

true and absolute motion of the body will arise, partly from the true motion of the Earth, in immovable space; partly from the relative motion of the ship on the Earth; and if the body moves also relatively in the ship; its true motion will arise, partly from the true motion of the Earth, in immovable space, and partly from the relative motions as well of the ship on the Earth. As if that part of the Earth, where the ship is, was truly moved toward the east, with a velocity of 10010 parts; while the ship itself, with a fresh gale, and full sails, is carried towards the west, with a velocity expressed by 10 of those parts; but a sailor walks in the ship towards the east, with 1 part of the said velocity; then the sailor will be moved truly in immovable space towards the east, with a velocity of 10001 parts, and relatively on the Earth towards the west, with a velocity of 9 of those parts.

p. 36 但他接着解释为何既存在绝对运动又存在相对运动。

But he goes on to explain his belief as to why there is such a thing as absolute motion as well as relative motion.

但我们可以通过性质、原因和效果区分静止与运动、绝对与相对。静止的性质是：真正静止的物体彼此确实静止。因此，尽管在遥远恒星区域或更远处可能存在绝对静止物体；但仅凭我们区域内物体相对位置，无法知道其中任何一个是否与那个遥远物体保持相同位置；因此绝对静止不能由我们区域内物体位置确定。

But we may distinguish rest and motion, absolute and relative, one from the other by their properties, causes and effects. It is a property of rest, that bodies really at rest do rest in respect to one another. And therefore as it is possible, that in the remote regions of the fixed stars, or perhaps far beyond them, there may be some body absolutely at rest; but impossible to know, from the position of bodies to one another in our regions whether any of these do keep the same position to that remote body; it follows that absolute rest cannot be determined from the position of bodies in our regions.

区分绝对与相对运动的效果，是远离圆周运动轴心的离心力。因为纯粹相对圆周运动中没有这种力，但真实绝对圆周运动中有，大小取决于运动量。

The effects which distinguish absolute from relative motion are, the forces of receding from the axis of circular motion. For there are no such forces in a circular motion purely relative, but in a true and absolute circular motion, they are greater or less, according to the quantity of the motion.

最后一点非常重要。

The final point above is very important.

p. 36–37 绝对平移已被迈克尔逊-莫雷实验否定，但牛顿已指出，除旋转外绝对空间不可测量。他通过思想实验证明：用绳子把半桶水悬挂在树上。水最初静止，桶旋转后运动传给水，水开始旋转。水面两侧向上弯曲，表明水最初相对于绝对空间未旋转。桶壁约束产生向心力，使水在边缘堆积。

〔校注：本书英文将实验名拼作 Michaelson-Morely；中文按通常译名“迈克尔逊—莫雷”处理，英文照录。这里关于绝对空间和水桶实验的推论属于原作者论述。〕

Absolute translation is discredited by the Michaelson-Morely experiment, but Newton was already saying that absolute space is immeasurable except for rotation. This is proven through a thought experiment. He demonstrates that absolute space is immeasurable except for rotation by hanging a half-filled bucket with water from a tree with a rope. The water initially remains stationary, but as the bucket spins, its motion is transmitted to the water, which starts spinning. This causes the water’s surface to curve up at the sides, indicating that the water was not initially spinning with respect to absolute space. The constraint of the bucket wall creates a centripetal force, causing the water to pile up at the edges.

p. 37 这个思想实验很有说服力。

This thought experiment is quite convincing.

定律

Laws

他的运动定律为自然法则树立标准。此前已有阿基米德浮体排水定律和伽利略加速度定律[15]，但此前没有哲学家提出完整运动定律。

His laws of motion set the standard for what we understand natural laws to be. There had been previous formulations –Archimedes law of displacement by floating bodies, and Galileo’s law of acceleration[15] – but no

previous philosopher had formulated a complete set of motion laws.

每个物体都保持静止状态或沿直线匀速运动状态，除非施加于它的力迫使它改变这一状态。[16]

Every body perseveres in its state of rest, or of uniform motion in a right line, unless it is compelled to change that state by forces impressed thereon.[16]

运动量的改变始终与施加的驱动力成正比，并沿该力施加的直线方向发生。[17]

The alteration of motion is ever proportional to the motive force impressed and is made in the direction of the right line in which that force is impressed.[17]

对每个作用，总有一个大小相等、方向相反的反作用；或者说，两个物体彼此之间的相互作用总是大小相等、方向相反。

如果你用手指压石头，手指也被石头压。如果马拉绳子系着的石头，马（如果可以这么说）也被同样向石头方向拉回：因为拉伸的绳子努力恢复原状，会把马向石头方向拉，与把石头向马方向拉一样多，并阻碍一方前进，恰如促进另一方前进。[18]

To every action, there is always opposed an equal reaction, or the mutual actions of two bodies upon each other are always equal and directed to contrary parts.

If you press a stone with your finger, the finger is also pressed by the stone. If a horse draws a stone tied to a rope, the horse (if I may so say) will be equally drawn back towards the stone: for the distended rope, by the same endeavour to relax or unbend itself, will draw the horse as much towards the stone as it does the stone towards the horse, and will obstruct the progress of the one as much as it advances that of the other.[18]

定律 III 是明确等价关系，建立在阿基米德力学等价关系上。牛顿用手指压石头例子重申伽利略更有限的观点：

Law III is a clear equivalence relation. As such, it builds on the equivalence relations of Archimidean mechanics. With his example of the pressure of a finger on a stone, Newton is repeating a more limited point made by Galileo

当人手持石头时，他所做的难道不是施加向上推力［virtù impellente］，其大小等于重力向下拉石头的力量［facoltà］吗？只要你握住石头，难道不是持续施加这种力［virtù］吗？

When one holds a stone in his hand does he do anything but give it a force impelling [virtù impellente] it upwards equal to the power [facoltà] of gravity drawing it downwards? And do you not continuously impress this force [virtù] upon the stone as long as you hold it in the hand?

伽利略只在物体与重力平衡的特殊情况给出作用反作用等式。牛顿把它扩展为运动和加速物体的普遍原理，无论力性质如何。

Galileo only gave this equality of action and reaction in the special case of a body balanced against gravity. Newton extended it to a general principle of moving and accelerating bodies, irrespective of the nature of the forces acting on them.

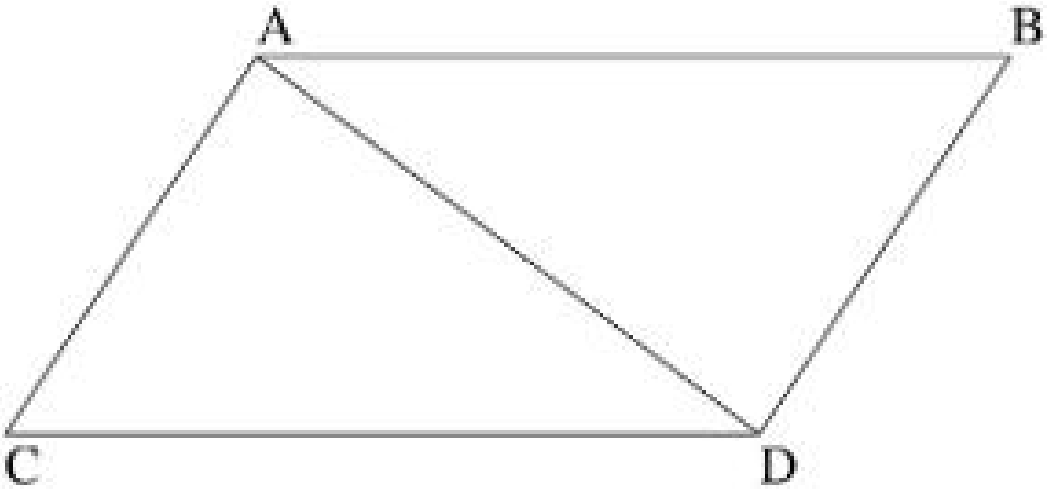
牛顿首次提出变化和运动力学，其中所有变化和运动都受普遍等价关系和约束支配，这些关系可用数学处理。在牛顿那里，与阿基米德和伽利略一样，数学处理主要基于古典几何。他的文本不断穿插几何图示，以说明其论证。论述力合成时，他写道：

Newton is the first to present mechanics of change and motion in which all the changes and motions are subject to certain universal equivalences and constraints, equivalences and constraints which can be treated mathematically. In Newton’s case, as with Archimedes and Galileo, the mathematical treatment is still mainly in terms of classical geometry. His text is constantly interspersed with geometric diagrams to illustrate its arguments. Dealing with the composition of forces, he writes:

物体在两个力共同作用下，会在同一时间内描绘出平行四边形的对角线；这段时间正是两个力分别作用时，物体描绘出相应边所需的时间。

A body by two forces conjoined will describe the diagonal of a parallelogram, in the same time that it would describe the sides, by those forces apart.

p. 38



原书无编号插图：力的平行四边形合成示意。

Unnumbered diagram: parallelogram composition of forces.
A、B、C、D 为原图顶点标记。
校注：此图在原书中没有编号和图注；此处说明为编者补充。

由此解释任意一个直线力 AD 由两个斜向力 AC 和 CD 合成；反之，任意直线力 AD 分解为两个斜向力 AC 和 CD。这种合成与分解在力学中得到充分验证。

And hence is explained the composition of any one direct force AD, out of any two oblique forces AC and CD; and, on the contrary, the resolution of any one direct force AD into two oblique forces AC and CD: which composition and resolution are abundantly confirmed from mechanics.

他提出明确守恒定律。首先是动量守恒：

He formulates a set of explicit conservation laws. First, a conservation of momentum law:

运动量——通过取指向相同方向的运动之和，并取指向相反方向的运动之差而得到——在物体相互作用中不变。[19]

The quantity of motion, which is collected by taking the sum of the motions directed towards the same parts, and the difference of those that are directed to contrary parts, suffers no change from the action of bodies among themselves.[19]

这是守恒定律：物体集合的总动量守恒，由定律推导。回顾牛顿所说的“运动”是 mv，即速度与质量乘积，现在称为动量。他的意思是：对一组物体，若选择任意方向并计算它们在该方向的动量之和，那么物体间任何相互作用，无论相互引力还是碰撞，都不能改变该方向总动量。

The above is a conservation law –conservation of aggregate momentum of a collection of bodies –deduced from the laws. Recall that by‘motion’, Newton means *mv*, the product of velocity and mass in motion, what we would now call momentum. He is saying that given a collection of bodies, if you choose an arbitrary direction and compute the sum of their momenta in that direction, then no interaction of those bodies, whether mutual gravitation or impacts between them, can alter the aggregate momentum of the bodies in the specified direction.

他随后推广到重心。他论证，两个或多个物体的共同重心无论它们如何作用都保持不变，确保所有相互作用物体，排除外部作用和阻碍，要么静止，要么沿直线匀速运动。此外，两个相互作用物体系统中，它们相对于共同重心的相对运动，无论接近还是远离，由于物体之间的相互距离而相等。

〔校注：底本此段将重心“保持不变”与系统匀速运动并列，又称相对运动“相等”，措辞不够严谨。译文照录作者论述；在无外力时应区分质心匀速运动、各物体速度与各自运动量，不能把这些量混为一谈。〕

He next extends this to their centres of gravity. He argues that the common centre of gravity of two or more bodies remains constant regardless of their actions, ensuring that all bodies acting upon each other, excluding outward actions and impediments, are either at rest or moving uniformly in a right line. Moreover in a system of two bodies acting on each other, their relative motions, whether approaching or receding from the common centre of gravity, will be equal due to reciprocal distances between the bodies.

这把阿基米德重心研究扩展到通过相互作用力相互影响的运动物体，后来用于行星轨道问题。例如，这意味着月球绕地球公转时，地月重心不变。他小心限定于物体“相互作用”的效果，因为地球和月球嵌入太阳引力场时，该场影响会改变地月共同重心。

This extends Archimedes’ work on centres of gravity to moving bodies interacting by mutual forces. This will later be used to deal with problems of planetary orbits. It means, for instance, that the centre of gravity of the Earth and the Moon is not altered as the Moon orbits the Earth. He is careful to restrict this to the effect

of the bodies ‘acting upon each other’, since to the extent that the Earth and Moon are embedded in a solar gravitational field, the effect of this will alter the mutual centre of gravity of the Earth-Moon system.

在给定空间内，物体运动彼此相同，无论该空间静止还是沿直线匀速前进且无圆周运动。

The motions of bodies included in a given space are the same among themselves, whether that space is at rest, or moves uniformly forwards in a right line without any circular motion.

p. 39

我们从船实验得到明确证明；无论船静止还是沿直线匀速前进，所有运动以相同方式发生。[20]

A clear proof of which we have from the experiment of a ship; where all motions happen after the same manner, whether the ship is at rest, or is carried uniformly forwards in a right line.[20]

由于地球和月球绕太阳公转时并非“无圆周运动沿直线匀速前进”，他明确把行星-卫星系统排除在规则外。但更广泛地，他主张如今所称的伽利略相对性：惯性参考系中物体相对运动不受参考系匀速运动影响。船的比喻源自伽利略。[21]

Since, in orbiting the Sun, the Earth and Moon do not move‘uniformly forward in a right line without any circular motion’, he is explicitly excluding planet-satellite systems from his rule that the ‘motions of the bodies are the same among themselves’. But more generally, he is asserting what is now known as Gallilean relativity –that is to say that the mutual motions of bodies in an inertial frame of reference are unaffected by the uniform motion of that frame of reference. The analogy with a ship is borrowed from Gallileo.[21]

牛顿明确排除圆周运动，因为在他看来，这种圆周运动通过空间区分相对与绝对运动。他援引伽利略：下落物体经过距离与时间平方成正比：

Newton explicitly excludes circular motion since according to him such circular motion distinguishes relative from absolute motion through space. He explicitly refers to Galilleo for authority on the fact that falling bodies traverse a distance proportional to the square of their time of fall:

根据前两条定律和前两个推论，伽利略发现物体下落遵循时间平方比，抛射体运动呈抛物线；经验与二者相符，除非运动因空气阻力略延迟。当物体下落时，均匀重力在相等时间施加相等力，因此产生相等的速度增量；整个时间施加总力，产生与时间成正比的总速度。成比例时间内描述的距离与速度和时间乘积成正比，即与时间平方成正比。[22]

By the first two Laws and the first two Corollaries, Galileo discovered that the descent of bodies observed the duplicate ratio of the time, and that the motion of projectiles was in the curve of a parabola; experience agreeing with both, unless so far as these motions are a little retarded by the resistance of the air. When a body is falling, the uniform force of its gravity acting equally, impresses, in equal particles of time, equal forces upon that body, and therefore generates equal velocities; and in the whole time impresses a whole force, and generates a whole velocity proportional to the time. And the spaces described in proportional times are as the velocities and the times conjunctly; that is, in a duplicate ratio of the times.[22]

这里的关系是 $v = at$ 和 $d \approx t^2$。随后，他叙述了自己用充作摆锤的钢球相互碰撞所进行的实际实验，以验证这些关系，并仔细计入空气阻力的影响。

〔校注：底本此处印作 d≈t²，但前文文字说距离与时间平方成正比；比例关系通常写为 d∝t²。〕

The relations here are $v = at$ and $d \approx t^2$. He then recounts actual experiments he has undertaken with steel balls as pendulums that collide to verify the relations, with careful allowance being made for the effects of air resistance.

他证明引力的方法

His method of proof of gravity

基于运动基本定律，牛顿在《原理》第一卷考察不同力作用下物体的多种可能轨迹。他不仅研究平方反比力，事实上直到第三卷才引入万有引力定律。他从初始运动定律推导不同力下预期轨道和轨迹，得出概括：

From the basic laws of motion, Newton in book I of the *Principles* examines many possible trajectories of bodies interacting under different types of forces. He does not only look at inverse square law forces like his law of gravitation. Indeed, he does not introduce the law of gravitation until book III. He instead deduces from his initial laws of motion what orbits and trajectories would

be expected under different sorts of forces. He comes up with the following generalizations:

p. 40

如果周期与半径成正比，因此速度相等，向心力与半径成反比；反之亦然。

If the periodic times are as the radii, and therefore the velocities equal, the centripetal forces will be reciprocally as the radii; and the contrary.

如果周期与半径的 3/2 次方成正比[23]，因此速度与半径的平方根成反比[24]，向心力与半径平方成反比；反之亦然。

If the periodic times are in the *sesquiplicate* ratio of the radii,[23] and therefore the velocities reciprocally in the subduplicate ratio of the radii,[24] the centripetal forces will be in the duplicate ratio of the radii inversely, and the contrary.

一般说来，如果周期与半径 R 的任意幂 R^n 成正比，因而速度与半径的幂 R^{n-1} 成反比，那么向心力就与半径的幂 R^{2n-1} 成反比；反之亦然。[25]

And universally, if the periodic time is as any power R^n of the radius R, and therefore the velocity reciprocally as the power R^{n-1} of the radius, the centripetal force will be reciprocally as the power R^{2n-1} of the radius; and the contrary.[25]

他现在有把轨道周期与力性质联系的普遍规则，还能推导开普勒等面积定律作为向心力平方反比时的特例。第三卷引入六个关键事实，即“现象”：

He now has a general rule which relates orbital periods to the nature of forces. He is also able to deduce Kepler’s law that orbital bodies sweep equal areas in equal times as a special case that holds if centripetal forces have an inverse square law. In book III, he introduces six key facts, or, as he puts them, Phænomena:

1. 木星卫星轨道周期与其轨道半径的 3/2 次方成正比。2. 土星卫星轨道周期同样与轨道半径的 3/2 次方成正比。3. 五大行星——水星、金星、火星、木星、土星——绕太阳公转。4. 它们的周期与到太阳平均距离的 3/2 次方成比例。5. 虽然行星以连接太阳与行星的线段为向径扫过面积与时间成正比，但连接行星与地球的半径扫过面积无此关系。6. 月球以连接地球中心与月球的线段为向径扫过面积与时间成正比。

1. That the orbital periods of the moons of Jupiter are in *sesquiduplicate* ratio to their orbital radii. 2. That the orbital periods of the moons of Saturn are similarly in *sesquiduplicate* ratio to their orbital radii. 3. That the five primary planets: Mercury, Venus, Mars, Jupiter and Saturn rotate around the Sun. 4. That they have periodic times in sequiduplicate proportion of their mean distances from the Sun. 5. That while the areas which the planets describe by radii drawn to the Sun are proportional to the times of description, no such relation exists in the areas that are swept by radii connecting the planets to the Earth. 6. That the Moon, by a radius drawn to the Earth’s centre, describes an area proportional to the time of description.

所有情况他都引用当时最佳轨道半径和周期估计。基于这些现象和先前轨道周期结果，他推断木星卫星、主要行星和月球都必须受平方反比力作用而偏离直线运动。他得出结论：

〔校注：底本作 drawn of rectilinear motion，疑漏一个 f；中文依本段轨道论证译为“偏离直线运动”，英文保留底本。〕

In all cases, he cites what were then the best-known estimates of orbital radii and periods to justify his claims. From these phenomena and his previous results about orbital periods, he deduces that the moons of Jupiter, the primary planets and the Moon must all be drawn of rectilinear motion by a force subject to an inverse square law. He concludes

迄今把天体束缚在轨道上的力被称为向心力；但既然已明确它只能是引力，今后称之为重力。

The force which retains the celestial bodies in their orbits has been hitherto called centripetal force; but it being now made plain that it can be no other than a gravitating force, we shall hereafter call it gravity.

p. 40–41

他随后论证引力不仅与距离平方成反比，还与物体质量成正比，无论物体被吸向哪一个天体。他的论证是：若木星及其卫星所受太阳引力与质量不成正比，木星卫星轨道会明显被太阳引力扭曲，但事实并非如此。最终表述为：引力与质量成正比，与距离平方成反比。

He then argues that this gravitational force must not only operate inversely to the square of distances but must also operate in proportion to the masses of the bodies, irrespective of the body towards which the bodies are drawn. His proof is that were it the case that the gravitational attraction towards the Sun experienced by Jupiter and its satellites was not in proportion to their masses, then the orbits of Jupiter’s moons would be very evidently distorted by the Sun’s gravity, which is not the case. His final formulation is a law of attraction that is proportional to the masses and inversely as the square of distances.

p. 41

回顾整个论证结构：

Recapitulating the whole structure of the argument,

• 首先陈述经实验验证的运动定律。

• 通过冗长几何证明，阐明符合这些定律及各种可设想力律的可能轨迹。

• 从当时所有可见卫星和行星观测，证明它们遵循平方反比引力推导的轨道。

• 推断既然所有观测天体遵循该定律，它就是普遍自然定律。

这是循环过程：最初实验提供定律，天文数据帮助推断外部力，进而推导新定律——万有引力。该定律允许演绎重建地球非球形变形和月球轨道不规则等现象，也涉及确定太阳和月亮对海洋的力。

• He initially states laws of motion that have been verified by experiment.

• He then, by lengthy geometrical proofs, demonstrates the possible trajectories consistent with these laws of motion and various conceivable force laws.

• From observation of all the moons and planets then visible, he shows that these follow the trajectories that would follow from inverse square attraction.

• He then infers that since all observed bodies follow this law, it is a universal law of nature.

This is a circular process where initial experiments provide laws, astronomical data helps in infer external forces and a new law, gravity, is inferred. This law allows for deductive reconstruction of phenomena like Earth’s aspherical distortion and Lunar orbit irregularities. The process also involves determining the forces exerted by the Sun and Moon on the sea.

第二卷

Book II

牛顿《原理》中产生持久影响的是包含运动通则和万有引力理论的第一、三卷。第二卷试图推导物体在气体和液体中运动所受阻力定律，后来证明过于简单，无法准确解释流体动力学中的力。

然而，在一点上，他第二卷思想经受时间考验。书中有一节《论通过流体传播的运动》，讨论压缩声波穿过流体，明确采用原子论观点。他把流体视为由原子构成，绘制为球体。他认为脉冲不能直线穿过流体，论据是：若脉冲穿过原子 a、b、c、d、e，只有在原子完美对齐时才能继续。在图中，原子 e 会把脉冲传给 f 和 g，使脉冲分叉为两个分量。由此推断：声波穿过流体时，经过狭窄空腔必然发生衍射。

在这点上他是正确的。

It is books I and III of Newton’s *Principles* –containing the general laws of motion and the theory of gravity – that have had a lasting impact. Book II attempted to deduce the laws of resistance that bodies would experience due to passage through gases and liquids. His treatments have subsequently been proven to be much too simple to account accurately for hydrodynamic forces.

On one point, however, his ideas in book II have stood the test of time. He has a section titled *Of Motion Propagated Through Fluids*, which deals with compressive sound waves passing through a fluid, and his account is explicitly atomistic. He views a fluid as being made up of atoms, which he draws as spheres. He argues that it is impossible for a pulse to pass through a fluid in a straight line, his argument being that if a pulse was being transmitted through atoms *a, b, c, d* and *e*, it would only continue so long as the atoms were perfectly lined up. In his diagram, atom *e* would next pass it on to *f* and *g* causing the pulse to diverge into two components. From this, he was able to deduce that sound waves passing through a fluid must exhibit diffraction when they pass through a narrow cavity.

On this he was right.

黑格尔对牛顿的批判

The Hegelian critique of Newton

德国哲学家格奥尔格 • 威廉 • 弗里德里希 • 黑格尔强烈批判牛顿，把他与“我们伟大的同胞开普勒——这位天赋异禀的天才”进行不利对比。在批判牛顿中，黑格尔声称开普勒本可直接从轨道定律推导万有引力平方反比定律。他写道：

The German philosopher Georg Wilhelm Friedrich Hegel was very critical of Newton, whom he compared unfavourably to ‘our great countryman Kepler, blessed with the gift of genius’. In his critique of Newton, Hegel claims that the inverse square law of gravity could have been deduced by Kepler directly from the latter’s law of orbits. He writes:

p. 42

事实上，他所给出的定律（即作圆周运动的物体的向径所量出的面积与时间成正比），本可以被他转化为物理定律的形式（species）（即引力与相等扇形所属的面积成比例〔in ratione〕）；由于圆 A 和 a 的总面积之比，等于其半径 R 和 r 的平方之比，我们知道 $1/A : 1/a$ 等同于 r^2/R^2。既然 $1/A$ 和 $1/a$ 表示运动量，若愿意也可表示向心力的量，他就可以说，引力或向心力与半径或距离成反比。[26]

Indeed, the law which he gave (ie that the areas measured by the vector radii of the bodies in circular motion are proportional to the times) he would have been able to transmute into the form (species) of *a* physical law (ie that gravity is in proportion (in ratione) to the area belonging to equal sectors); and since the total surfaces of the circles A and a stand in the same ratio as the squares of their radii R and r, we know that $1/A : 1/a$ is equivalent to r^2/R^2. Since $1/A$ and $1/a$ express the quantity of motion, and, if you wish, the quantity of the centripetal force, he could have said that the force of gravitation or centripetal force stands in inverse ratio to the radii, or distances.[26]

这表明他误解牛顿和开普勒。开普勒等面积定律适用于一颗行星椭圆轨道的弧段，意味着角速度在近日点最大。黑格尔错误地把它应用于比较两颗不同行星的两个不同圆形轨道。开普勒定律并未说不同行星在相等时间扫过相等面积。事实上，行星越远，每 24 小时扫过面积越小。观察金星、地球和火星轨道参数可见：

This shows a misunderstanding both of Newton and of Kepler. Kepler’s law of equal areas applies to segments of the elliptical orbit of one planet, meaning that the angular velocity must be greatest at perihelion. Hegel is misapplying it to compare two distinct circular orbits of different planets. Kepler’s law does not state that the planets sweep out equal areas in equal times. In fact, the further out a planet is, the less the area it sweeps out each 24 hours. We can see this by looking at the parameters of the orbits of Venus, Earth and Mars:

行星	半径（天文单位，半长轴）	周期	每日面积（平方天文单位）
金星	0.723	225	0.0202
地球	1	365	0.0172
火星	1.524	687	0.0139

Planet	Radius in AU (semi major axis)	Period	Area per day square AU
Venus	0.723	225	0.0202
Earth	1	365	0.0172
Mars	1.524	687	0.0139

校注：表格及“越远每日面积越小”的叙述均按底本保留。但以表内半径与周期作圆轨道近似，日扫面积 πr²/T 约为 0.00730、0.00861、0.01062 平方天文单位，与原表不符，且随半径增大而增大。原表数值更接近每日路程 2πr/T；这不等同于扫过面积。

牛顿证明，即使引力定律形式不同，仍可得到开普勒定律，但两种情况绝对速度不同。

Newton showed that you can still have Kepler’s law with a gravitational force law that is of different form, but the absolute velocities would be different in the two cases.

牛顿计算了形式为 $f = mMG/r^n$、其中 $n = 1, 2, 3$ 的引力定律下开普勒轨道的性质，发现三种情形均能相容，但所预测的速度不同。

Newton worked out the properties of Keplerian orbits with gravitational laws of the form $f = mMG/r^n$ for $n = 1, 2, 3$ and finds that all three are consistent, but that the predicted velocities differ.

他随后用木星卫星轨道实际观测证明，只有平方反比定律与观测相符。我们可以用简单例子说明开普勒定律与不同引力定律的兼容性。

He then used actual observations of the orbits of the Jovian satellites to show that only the inverse square law was consistent with observation. We can illustrate the compatibility of Kepler’s law with different gravitational laws with a simple example.

考虑卫星绕地球运行，最远点 A（远地点）距离是最近点 P（近地点）两倍。假设开普勒等面积定律成立，若 A 点速度 v，则 P 点速度必须为 $2v$。

Consider a satellite in orbit around the Earth such that the most distant point A (apogee) is twice as far from the Earth as the closest point P (perigee) Assume further that Kepler’s law of equal areas holds, so that if the velocity at point A is *v*, then velocity at point P must be 2v.

设卫星质量为 m，则在 A 点的动能为 $0.5mv^2$，在 P 点为 $2mv^2$。假设牛顿引力成立，地球质量为 M，

Let the satellite have mass m; then kinetic energy at point A is $0.5mv^2$, and at point P, it is $2mv^2$. The dif-

P 与 A 之间的势能差为 $\int_1^2 mMG/r^2\,dr = mMG/2$。换言之，这是引力对径向距离变化的积分；力对距离的积分具有能量的量纲。因此，能量守恒要求 $1.5mv^2 = mMG/2$，于是 $v = \sqrt{MG/3}$。

ference in potential energy between P and A, assuming Newtonian gravity and mass of Earth M, will be $\int_1^2 mMG/r^2\,dr = mMG/2$. In other words this is the integral over the change in radial distance with respect to gravitational force, and an integral of force with distance is of dimension energy. The conservation of energy then requires us to have $1.5mv^2 = mMG/2$, so $v = \sqrt{MG/3}$.

假设引力与距离成反比而非平方反比，但开普勒定律仍成立。

Suppose instead that the gravitational law was such that force was inversely proportional to the distance rather than the square of the distance, but that Kepler's law still holds.

p. 43

在这一情况下，将 A 点的新速度记为 v_1，于是 P 点的速度为 $2v_1$。此时动能之差为 $1.5mv_1^2$，它必须等于另一种力定律下的引力势能差（加速度 $\propto 1/r$），因此 $\int_1^2 mMG/r\,dr = mMG\log(2)$，我们仍可解出速度：$v_1 = \sqrt{MG\log(2)/1.5}$。

Call the new velocity at A in this case v_1, and the velocity at P is now $2v_1$. The kinetic energies now differ by $1.5mv_1^2$, which now has to equal the difference in gravitational potential under a different force law (acceleration $\propto 1/r$) so that $\int_1^2 mMG/r\,dr = mMG\log(2)$ we can again solve for velocity such that $v_1 = \sqrt{MG\log(2)/1.5}$.

由于 $\log(2) \neq \frac{1}{2}$，在新的引力定律下，速度将会不同。

Since $\log(2) \neq \frac{1}{2}$, it follows that velocities will differ with the new gravity law.

速度是模型中的自由变量。只有观测实际速度才能判断哪种引力模型正确。

The velocities are a free variable in the models. Only observation of actual velocities can tell you which model of gravity is correct.

牛顿关于轨道周期的论证

Newton's argument about orbital periods

考虑顺时针圆周运动物体所受向心加速度。

Consider the centripetal acceleration experienced by a body in a clockwise circular orbit.

显然，物体的速度 v 与轨道半径成正比，与轨道周期成反比。如果在 t_0 时刻，物体的速度向量为 $[0, v]$，那么四分之一圈之后，在 t_{90} 时刻，它的速度将为 $[-v, 0]$。它在 X 方向的全部初速度都已被抵消。转动周期越长，这一减速过程越缓慢。因此，向心加速度与半径成正比，与轨道周期的平方成反比。

〔校注：底本所列初速度 [0,v] 的 X 分量本来为零；按通常坐标约定，被抵消的是初始 Y 分量。这里保留底本 X 方向的表述并指出不一致。〕

Clearly, the velocity v of the body is proportional to the radius of the orbit and inversely proportional to period of the orbit. If at time t_0 the body has velocity vector $[0, v]$, then at time t_{90} a quarter rotation later its velocity will be $[-v, 0]$. Its entire initial velocity in the X direction has been cancelled. The longer the period of rotation, the more gradual this deceleration will be. So the centripetal acceleration is directly proportional to radius and inversely proportional to the square of the orbital period.

$$A_c \propto r/p^2$$

该向心加速度必须等于半径 r 处重力加速度。牛顿考虑三种可能重力定律：

This centripetal acceleration has to be equal to the acceleration due to gravity at radius *r*. Newton considers three possible gravity laws:

1. 加速度与距离成反比。鉴于胡克弹性定律也与位移线性相关，这并非不可信：

1. Acceleration is inversely proportional to distance. This would not be implausible given that Hooke's then recently discovered law of elasticity also had a linear dependence on displacement.

$$A_g \propto 1/r$$

2. 加速度与距离平方成反比，如引力在连续球层中扩散。球层表面积随半径平方增加，因此轨道天体截取面积比例呈平方反比：

2. Acceleration is inversely proportional to the square of distance, as seen in gravitational force spreading in successive spheres. The surface of these spheres grows as the square of radius, so the proportion of the area inter-

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$A_g \propto 1/r^2$

3. 加速度与距离立方成反比。若引力场扩展填满更大体积，行星占据半径 r 的太阳引力场体积份额会呈此关系：

$A_g \propto 1/r^3$

如果引力加速度就是向心加速度，则有 $A_c = A_g$。当引力与距离成反比时，就有：

$1/r \propto r/p^2$

比较从金星到土星的行星轨道（图 3.4）与假设引力定律预测，平方反比定律最符合。对最接近圆形的轨道几乎完美。离心率数据见表 3.2。火星是最突出离群点，因为轨道离心率较高。

因此可得 $p^2 \propto r^2$，以及 $p \propto r$。我们可以为三种定律列出类似关系表（表 3.1）。

然后与实际轨道数据比较，尽管每条假设引力定律都有看似合理先验依据，只有平方反比定律与数据吻合良好（表 3.2）。

sected by the orbiting body will have an inverse square relation.

$A_g \propto 1/r^2$

3. Acceleration is inversely proportional to the cube of the distance. This is also a conceivable law if you think of the gravitational field expanding to fill larger volumes; the share of the volume of the Sun’s field at radius *r* occupied by a planet would then have this relation.

$A_g \propto 1/r^3$

If gravitational acceleration is the centripetal acceleration, we have $A_c = A_g$. In the case of gravity being inversely proportional to distance, we then have

$1/r \propto r/p^2$

If you plot of orbits of planets from Venus to Saturn (Figure 3.4) against the predictions of hypothetical gravity laws considered by Newton, it is clear that inverse square laws are the best fit. The fit is almost perfect for the most circular orbits. See the data for eccentricity in Table 3.2. Mars is the most prominent outlier, as it has a relatively more eccentric orbit.

So it follows that $p^2 \propto r^2$ and $p \propto r$. We can construct a table of similar relations for the three laws (Table 3.1).

We can then compare this with actual orbital data for the planets and see that although each of the hypothetical gravity laws had plausible prior justifications, only the inverse square law fits the data well (Table 3.2).

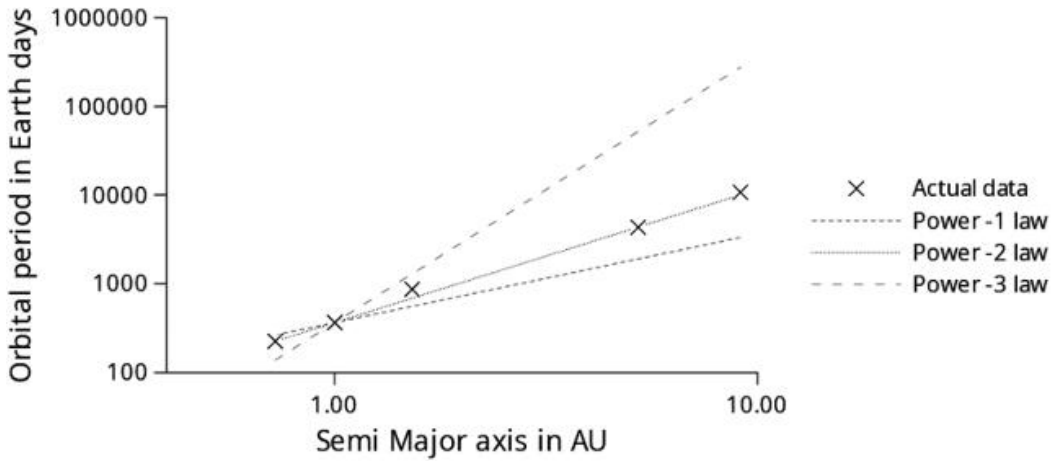


图 3.4 轨道半径与周期的关系。作者绘图。

Figure 3.4 Orbital semi-redius against period. Figure by authors.

纵轴：轨道周期（地球日）；横轴：半长轴（天文单位）；Actual data：实际数据；Power -1/-2/-3 law：-1/-2/-3 次幂定律。

校注：原图注 semi-redius 疑为 semi-radius 误拼；图中横轴明确标为半长轴。英文保留底本。

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表 3.1 牛顿考虑的三种引力定律，以及它们对轨道周期与轨道半径关系的预测。

幂次	向心加速度	引力加速度	周期定律	
-1	$A_c \propto 1/p^2$	$A_g \propto 1/r$	$p^2 \propto r^2$	$p \propto r$
-2	$A_c \propto 1/p^2$	$A_g \propto 1/r^2$	$p^2 \propto r^3$	$p \propto \sqrt{r^3}$
-3	$A_c \propto 1/p^2$	$A_g \propto 1/r^3$	$p^2 \propto r^4$	$p \propto r^2$

Table 3.1 The Three Gravity Laws Considered by Newton and Their Predicted Relations of Orbital Period to Orbital Radius

Power law	Centripental acceleration	Gravity acceleration	Period law	
-1	$A_c \propto 1/p^2$	$A_g \propto 1/r$	$p^2 \propto r^2$	$p \propto r$
-2	$A_c \propto 1/p^2$	$A_g \propto 1/r^2$	$p^2 \propto r^3$	$p \propto \sqrt{r^3}$
-3	$A_c \propto 1/p^2$	$A_g \propto 1/r^3$	$p^2 \propto r^4$	$p \propto r^2$

校注：原表向心加速度栏合并三行，写作 A_c∝1/p²，漏掉前文随半径变化的因子 r。这里保留其公式，并在每行重复展示以便阅读；正文给出的关系为 A_c∝r/p²。

表 3.2 太阳系实际轨道数据与牛顿所考虑的三种可能引力定律的预测比较。

幂律	实际周期	半径	$A_g \propto 1/r$ 预测周期	$A_g \propto 1/r^2$ 预测周期	$A_g \propto 1/r^3$ 预测周期	离心率
金星	225	0.72	264	224	138	0.01
地球	365	1.00	365	365	365	0.02
火星	867	1.52	556	687	1292	0.09
木星	4331	5.22	1906	4355	51960	0.05
土星	10747	9.11	3324	10032	275723	0.05
海王星	59800	30.18	11016	60518	10033936	0.01

Table 3.2 Actual Orbital Data from the Solar System Compared with the Predictions of the Three Possible Gravity Laws Considered by Newton

Power law	Actual period	Radius	Period predicted if $A_g \propto 1/r$	Period predicted if $A_g \propto 1/r^2$	Period predicted if $A_g \propto 1/r^3$	Eccentricity
Venus	225	0.72	264	224	138	0.01
Earth	365	1.00	365	365	365	0.02
Mars	867	1.52	556	687	1292	0.09
Jupiter	4331	5.22	1906	4355	51960	0.05
Saturn	10747	9.11	3324	10032	275723	0.05
Neptune	59800	30.18	11016	60518	10033936	0.01

校注：数据照录底本：火星实际周期在本表为 867，前表为 687；立方反比栏的数值近似按 365r³ 而非表 3.1 给出的 365r² 计算。其他半径及周期数据未在译文中自行替换。本表首列表头原作 Power law，实际列出行星。

相比之下，唯心主义者黑格尔试图从纯粹思想推导运动定律，蔑视实验。他论证牛顿及其追随者通过石头在投石索中旋转的例子确认离心力假说。石头试图远离转动手，拉伸投石索，一旦松手便飞射。但他说：

In contrast, the idealist Hegel wants to deduce the laws of motion from pure thought, disparaging experiment. He argued that Newton and his followers confirm the hypothesis of centrifugal force by giving the example of a stone that is whirled about in a sling. The stone endeavours to recede from the hand that turns it, distending the sling, and as soon as it is let go, it flies away. But he says: p. 44–45

然而，也许哲学先验地推导出实验方法——自称哲学——试图从实验中虚假且不成功地了解的东西；这种方法以盲目热忱和感官为手段，追寻哲学真实概念的幻影。[27]

It may be, however, that philosophy *a priori* deduces what the experimental method, which calls itself philosophy, undertakes to know falsely and with unfelicitous success from experiments, seeking as it does with blind zeal and by means of the senses the simulacrum of the true concepts of philosophy.[27] p. 45

黑格尔批评牛顿为证明惯性质量与引力质量相等而做的摆实验。[28]黑格尔声称牛顿只证明不同物体重量相等。他完全忽略实验要点：我们没有先验理由认为作为重量的引力质量应等于惯性质量，即对加速度的抵抗。二者相同是经验事实，但并非必然。

Hegel criticizes the experiments Newton conducted with pendulums to establish the equality of inertial and gravitational mass.[28] Hegel claims that all that Newton has done is show that the weights of different bodies are equal. Hegel completely misses the point of the experiments, that is, that we have no *a priori* reason to assume that gravitational mass as weight should equal inertial mass: resistance to acceleration. It is an empirical fact that the two are the same, but it does not necessarily have to be the case.

物理学家推测“暗物质”有惯性质量，但仅通过引力与普通物质相互作用。反之，若牛顿错了，有惯性质量的带电粒子可能通过静电力而非引力与普

Physicists speculate that ‘dark matter’ has inertial mass but only interacts with ordinary matter through gravity. Conversely, charged particles with inertial mass p. 45–46

通物质相互作用。但这些粒子存在仍是未解经验问题。牛顿摆实验验证引力质量与惯性质量等价，是历史序列中一环。

could, if Newton was wrong, interact with ordinary matter through electrostatic force, but not via gravity. However, the existence of these particles is still an unanswered empirical question. Newton’s pendulum experiments, which verified the equivalence of gravitational and inertial masses, sit in a historical sequence of such tests.

p. 46 牛顿时代已知除引力外的力，如通过砝码或弹簧秤测量的绳索张力。牛顿用引力做摆实验，但也可用钟表摆轮。摆轮依靠弹簧力与轮质量对抗设定振荡周期。

Forces other than gravity were known in Newton’s time, such as the tension on a cord measured by weight or a spring balance. Newton experimented with pendulums using gravitational force, but clock balance wheels could have been used. They rely on the force of a spring working against the mass of the wheel to set the period of oscillation.

一位在概念分析方面有黑格尔声誉的哲学家，竟完全未理解重量与惯性质量的概念区别，这告诉我们一些东西。

That a philosopher with Hegel’s repute for conceptual analysis should have so completely failed to understand the conceptual distinction between weight and inertial mass tells us something.

他关于行星轨道的论著是胡言乱语和民族偏见的大杂烩，既不懂开普勒、牛顿，也不懂观测事实。例如他声称地球赤道扁平：

His work on planetary orbits is a farrago of nonsense and national prejudice in which he understands neither Kepler nor Newton nor observed facts. He claims, for example, that the Earth is flattened at the equator:

最后，我们可以说，这些现象与地球在赤道以外比赤道更宽的形状惊人吻合，其直径比轴短。[29]

Lastly, let us say that those things agree remarkably well with the figure of an Earth wider elsewhere than at the equator, the diameter of which is shorter than the axis.[29]

除其他怪论外，黑格尔还认为太阳引力不源于巨大质量，而源于它发光：

Among other oddities, Hegel held that it was not the great mass of the Sun that accounted for its gravitational attraction. He instead thought it exerted attraction because it was luminous:

他的论证如下：牛顿认为重心因行星引力轻微偏移，因此不应把重心放在太阳中。但证明曲线运动命题需要轨道中心。牛顿假设吸引与被吸引物体相互吸引，二者仿佛绕共同重心旋转。共同重心是纯数学点。太阳力中心并非必然，而源于巨大的规模，这基于“每种力取决于质量”的假设。物理哲学指出，真正力中心是光源，太阳真正力与力量必须设定于此。[30]

His argument went as follows: Newton believed that the centre of gravity should not be placed in the sun as it shifts slightly due to planetary attractions. However, a centre of orbits is required to prove propositions concerning curvilinear motion. Newton assumed the reciprocal attraction of attracting and attracted bodies, where both revolve as if about a common centre of gravity. The common centre of gravity is a purely mathematical point. The sun’s centre of force is not due to necessity but rather due to its massive size, which is based on the hypothesis that every force depends on mass. Physical philosophy suggests that the true centre of forces is the source of light, where the sun’s true force and power must be posited.[30]

p. 46–47 鉴于这些怪异且违背事实的观点，黑格尔天文学论述很快被遗忘并不奇怪。他在 20 世纪享有声誉，是因为被视为现代共产主义形成中起作用。我们稍后叙述，在俄国革命前后，他被认为为共产主义学说贡献了组成部分。列宁说马克思和恩格斯捍卫哲学唯物主义。他们的观点体现在恩格斯的《路德维希·费尔巴哈》和《反杜林论》中，这些是阶级意识工人的指南。他说马克思扩展 18 世纪唯物主义，用德国古典哲学尤其是导致费尔巴哈唯物主义的黑格尔体系丰富它。列宁说马克思主要成就是辩证法，即发展和人类知识相对性学说。自然科学最新发现，如镭和电子，证实马克思的辩证唯物主义。[31]

Given his odd and counterfactual views, it is no surprise that Hegel’s account of astronomy soon passed into obscurity. Insofar as he had a positive reputation in the twentieth century, it is down to his being seen as having played a role in the formation of modern communism. We will recount later how, in the period around the Russian Revolution, he was credited with having contributed one of the component parts of communist doctrine. Lenin said that Marx and Engels defended philosophical materialism. Their views were exemplified in Engels, *Ludwig Feuerbach* and *Anti-Dühring*, which served as handbooks for class-conscious workers. Marx, he said, expanded on eighteenth-century materialism, enriching it with German classical philosophy, par-

ticularly Hegel’s system, which had led to Feuerbach’s materialism. Lenin said Marx’s main achievement was dialectics, the doctrine of development and the relativity of human knowledge. Recent discoveries in natural science, such as radium and electrons, confirmed Marx’s dialectical materialism.[31]

这种认可促使马克思主义者关注黑格尔，在此过程中，黑格尔对 17、18 世纪唯物主义科学的原本被遗忘的反对意见在 20 世纪部分复兴。我们稍后探讨苏联机械论者与受黑格尔影响的德博林学派争论。在这场争论中，德博林一方呼应黑格尔反对意见，即机械论依赖“外部”力量。在黑格尔那里，反对所谓外部力量的论点具有明确宗教性质，因为上帝行为不是外部、机械、任意或偶然的。根据实验哲学，上帝赋予物质的力量确实存在于物质内部，构成其本质。但力学无法理解，因为它只处理外部原因，不通过理性认识自然。[32]

This stamp of approval encouraged Marxists to pay attention to him, and in the process, Hegel’s otherwise forgotten objections to the materialist science of the seventeenth and eighteenth centuries were partially revived in the twentieth century. We will later look at the debate between the Soviet Mechanists and the Hegel-influenced Deborin school. In this later debate, we see the Deborin side echoing some of Hegel’s objections that mechanics relies on ‘external’ forces. In Hegel’s case, the argument against what he saw as external forces was explicitly religious since God’s actions are not external, mechanical, arbitrary or fortuitous. According to experimental philosophy, the forces that God gave to matter truly dwell within it and constitute its nature. However, mechanics cannot understand this idea since it deals only with external causes and does not conceive nature through reason.[32]

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这种对“外部”力量的反复纠缠在 20 世纪 30 年代共产主义哲学文献中再次出现。例如美国马克思主义者埃默里写道：

This same harping on about ‘external’ forces reappears in the 1930s communist philosophical literature. For example, the American Marxist Emery writes:

原子所经历的正是如此。当发现原子是复杂运动系统时，机械论思维的自然科学家开始寻找系统“最终单位”（电子、亚电子等），以及使系统平衡的力。如果所有物质点只施加一种力，矛盾只能是外部的。因此力学中没有内在矛盾，没有对立统一。所有看似内在的矛盾被一分为二，外化为不同物质粒子体现的力冲突。这是牛顿力学遗产。但作用于某点的力，相对于该点也是外部的。力学承认力是物质点唯一属性，但非必然属性。可以抽去力，从而设想物质本身、完全静止的物质。既然静止被视为物质内在存在方式，运动必须来自外部来源——力。[33]

This was what happened to the atom. When it was discovered that it is a system of complex movements, the mechanically minded natural scientist went on a search for the ‘last unit’ of the system (electrons, subelectrons etc), and the force which holds the system in equilibrium. If all material points exercise only one force, contradictions can only be external. Therefore no inner contradiction, no unity of opposites, is known in mechanics. All contradictions which seem to be immanent are cut in two, externalised into the conflict of forces embodied in different material particles. This is the heritage of Newton’s mechanics. But the force which acts in one point is external in relation to that point, too. Mechanics recognises force as the only attribute of the material point, but not as a necessary attribute. It might be abstracted, and matter as such, matter in complete rest, might be conceived. Since rest is taken as the immanent mode of existence of matter, movement has to come from an outside source –force.[33]

埃默里对牛顿力学的困惑几乎与老黑格尔一样深。他重复黑格尔关于力是外部的旧批评，还错误阐述牛顿对力的定义。[34]牛顿力学不处理“物质点”，而处理物质体。点只是近似轨道天体的数学工具。他所说“物质点”混淆数学处理——用质心点近似物体——与物理学家理解这只是近似。把物体视为点质量只在引力场空间导数的阶数相对较低时成立。

Emery was almost as confused about Newtonian mechanics as old Hegel was. Not only does he echo Hegel’s old criticism about force being ‘external’, he also misstates the Newtonian definition of force.[34] Newtonian mechanics also does not deal with ‘material points’; it deals with material bodies. Points only arise as a mathematical tool for approximating the behaviour of orbiting bodies. His ‘material points’ confuse the mathematical treatment which approximates a body by a point at its centre of mass, with the understanding of physicists that this is just an approximation. The approximation that treats a body as a point mass is only valid up to relatively low spatial derivatives of the gravitational field.

例如卫星在罗希极限内运行，潮汐力会撕裂它——正如形成土星环的固体天体。亚原子物理学也

If a satellite, for example, were to orbit within what is called the Roche limit, then tidal forces would tear it

是半生不熟胡言乱语。过去和现在都没有“亚电子”提案。

apart –as happened to whatever solid bodies gave rise to Saturn’s rings. The subatomic physics was also half-digested nonsense. There was and is no proposal that there are sub-electrons.

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构成世界的物质部分不是过程，而是旧义原子——刚性、孤立点，除拥有的力外无其他属性。每个点只能施加一种力，因此，只有不同点的力才在方向上有所不同。如果假想点施加多种力，机械论者会把它分成多个点。[35]

The material parts of which the world consists are not processes, but atoms in the old sense of the word, rigid, isolated points which have no other attribute besides the forces they possess. Each point can exercise only one force, and therefore, only forces of different points are different in direction. If a hypothetical point would exercise several forces, the mechanist would divide it into several points.[35]

这又是完全胡说。经典力学中质量或电荷能对所有其他质量或电荷施力。处理多体轨道时，每个物体对每个其他物体施力，力可向量相加。例如地球、月球、太阳彼此施力，无需把地球中心分成两个点处理吸引向月球和太阳的力。

This is again complete nonsense. In classical mechanics, masses or charges have the ability to exert forces on all other masses or charges. When dealing with the orbits of multiple bodies, each body exerts a force on each other, and these forces can be combined through vector addition. For instance, the Earth, Moon and Sun each exert forces on one another, and there is no need to divide the centre of the Earth into two points to handle the forces attracting it to the Moon and the Sun.

物理学家发现额外核力解释原子稳定性。他们预测高温氢聚变，实验证实。萨哈罗夫用静电力和强核力计算，而非神秘概念，研制氢弹。忽视核力会阻碍苏联核能进展。

Physicists discovered additional nuclear forces to explain atom stability. They predicted hydrogen fusion at high temperatures, which was confirmed by experiments. Sakharov used calculations of electrostatic and strong nuclear forces, not mystical concepts, to build his H-bomb. Disregard for nuclear forces would have hindered Soviet nuclear energy progress.

我们将在后面章节继续批判黑格尔式马克思解释。

We continue our critique of the Hegelian interpretation of Marx in later chapters.

杜 • 沙特莱

Du Châtelet

杜 • 沙特莱侯爵夫人（1706—1749）完成牛顿著作首部法语译本。她还出版第一本法语物理教科书《物理学基础》（Institutions Physiques）。[36]与伴侣伏尔泰一起，她是法国牛顿体系最重要倡导者之一。当时被视为哲学民族英雄的笛卡尔、牛顿和莱布尼茨之间存在全国性争议。在教科书导言中，她拒绝民族主义：

The Marquise du Châtelet (1706 – 49) produced the first French translation of Newton’s work. She also published the first French physics textbook, *Institutions Physiques.*[36] Along with her companion Voltaire, she was one of the most important advocates of the Newtonian system in France at a time when there existed considerable national controversy between what were seen as national champions in philosophy: Descartes, Newton and Leibniz. In the introduction to her textbook, she rejects this nationalism and writes:

追求真理是唯一不应让爱国之情占上风的事；牛顿和笛卡尔的观点竟成为某种民族事务，这无疑非常不幸。[37]

The search for truth is the only thing in which the love of your country must not prevail, and it is surely very unfortunate that the opinions of Newton and of Descartes have become a sort of national affair.[37]

我们纳入杜 • 沙特莱，是因为她除翻译牛顿外，还阐明当今理解的牛顿物理学。她的教科书仍受莱布尼茨影响，[38]包含许多如今被视为相当形而上学的关于方法论的论点。

We are including du Châtelet in this account because, in addition to translating Newton, she contributed to clarifying Newtonian physics as it is now understood. Her textbook retains considerable influence from Leibniz[38] and includes a number of what would now be seen as rather metaphysical arguments about methodology.

She starts out by defending the use of hypotheses in physics, differentiating herself from Newton in this. She

她开篇为物理学使用假设辩护，在此与牛顿不同。她认为，只有以宗教经院哲学家难以理解且无意义术语形式出现的假设才应摒弃。

例如中世纪经院哲学假设：植物生长因为拥有植物灵魂。她说，该假设给出生长的表面原因，但进一步追问便无意义：因为它不帮助我们理解所寻求的植物生长如何运作。[39]但她认为假设仍不可或缺。针对牛顿学派，她指出天文学进步严重依赖假设。若天文学家等待“真正理论”说明行星运动规律，就不会有进展。相反，天文学家把日心说作为初始工作假设：

〔校注：此处原作者写 heliocentrism（日心说），但紧接着引文说的是被哥白尼放弃的托勒密假说，二者不一致。正文保留原作者用词，不静默改为地心说。〕

因此，他们开始用这一被称为托勒密假设的假说解释和预测现象，直到其推论与观测相比存在无法克服的困难，且无法根据该假说构建与天象相符的星历表，这才促使哥白尼彻底放弃它，检验相反假说；该假说与天象如此吻合，其确凿性目前接近证明；没有天文学家敢再采纳托勒密假说。

因此假设在科学中必须占一席之地，因为它们促进真理发现并提供新视角；因为一旦提出假设，人们常做实验验证它是否成立，若没有假设，这些实验根本不会被想到。如果实验证实它，且它不仅解释原定现象，还使所有推论与观测一致，其概率增长到我们无法拒绝同意，几乎等同于证明。[40]

她为假设实用性制定规则：

1. 不能违背充足理由原则。2. 必须对现有事实有确定知识，以免把假设建立在狭隘基础上；否则假设很快被忽略事实推翻。3. 在获得无可辩驳证据前，不能把假设当作真理本身；必须冷静评估概率，不坚持他人接受。

这些观点今天多数科学家接受。另一方面，她保留莱布尼茨若干本质形而上学思想，认为莱布尼茨提出的连续性原理是物理学基本概念。该原理说，自然不突然变化，任何存在从一种状态到另一种状态必须经过所有中间状态。

像莱布尼茨这样的古典物理学家，把理想曲线的抽象连续性与物质世界属性混淆。“真实”连续体和

argues that it is only hypotheses in the form of the unintelligible and meaningless jargon of the religious schoolmen that must be rejected.

As an example of this, she gives the medieval scholastic hypothesis that plants grow because they have a vegetative soul. This hypothesis, she says, gives an apparent cause for growth, but when one pursues it further, it is meaningless: ‘because it contains nothing that helps us to understand how the vegetation of which I seek the cause operates’.[39] But she held that hypotheses remain essential. To the Newtonians, she objects that advance in astronomy has depended heavily on hypotheses. She says that if astronomers had waited for ‘the true theory’ of what regulated planetary motion, no progress would have been made. Instead, astronomers took heliocentrism as an initial working hypothesis:

p. 48–49

Thus, they began to explain and to predict phenomena by this hypothesis, called Ptolemy’s hypothesis, until the insurmountable difficulties of the consequences that derived from it when compared with observations, and the impossibility of constructing tables according to this hypothesis which were in accord with the phenomena of the sky, brought Copernicus to abandon it entirely and to test the opposite hypothesis, which is so much in agreement with the phenomena, that its certitude is at present not far from demonstration; and that no astronomer dares adopt that of Ptolemy.

p. 49

Hypotheses must then find a place in the sciences, since they promote the discovery of truth and offer new perspectives; for when a hypothesis is once posed, experiments are often done to ascertain if it is a good one, experiments which would never have been thought of without it. If it is found that these experiments confirm it and that it not only explains the phenomenon that one had proposed to explain with it but also that all the consequences drawn from it agree with the observations, its probability grows to such a point that we cannot refuse our assent to it, and that is almost equivalent to a demonstration.[40]

She constructs several rules for hypotheses to be useful.

1. It must not contradict the principle of sufficient reason. 2. You must have certain knowledge of the facts as they stand so as not to base your hypothesis on a narrow foundation. Otherwise, your hypothesis will soon be overthrown by facts that you neglected to pay attention to. 3. A hypothesis must not be passed off as the truth itself before one has irrefutable evidence to support it. You have to be sober in your estimate of the probability of your hypotheses and not insist that others must accept it.

These are views which would be accepted by most scientists today. On the other hand, she retains a number of essentially metaphysical ideas from Leibniz, holding that the principle of continuity, introduced by Leibniz, was a fundamental concept in physics. It stated that nature does not change suddenly, and any being cannot move from one state to another without passing through all the intermediate states between them.

Classical physicists, like Leibniz, confused the abstract continuity of ideal curves with the properties of

连续可微函数是微积分必需的抽象。这些抽象使古典物理学家能准确预测大尺寸物体运动。他们假设物质世界也有理想属性，如数学抽象。

the material world. The‘real’continuum and continuous differential functions are abstractions necessary for differential calculus. These abstractions allowed classical physicists to make accurate predictions about the motion of substantially sized bodies. They assumed that the material world also had ideal properties, like the mathematical abstractions they used.

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20 世纪量子力学表明自然确实发生跳跃，但不能因此指责杜 • 沙特莱缺乏远见。直到 1905 年，大多数科学家都会同意她。

The advent of quantum mechanics in the twentieth century showed that nature does in fact make jumps, but one cannot fairly criticize du Châtelet for her lack of foresight. On this, she was just expressing a view with which most scientists would have agreed until 1905.

杜 • 沙特莱认为，原子既然有限大小，就不可能是物质基本元素，因为基本元素必须是点粒子。

Du Châtelet argues that atoms, insofar as they are of finite size, cannot be the fundamental elements of matter since the fundamental elements would have to be point particles.

现在，由于延展存在物的可分性不矛盾，原子不可分性不能被视为必然；因此必须归结到简单存在物。[41]

Now, as there is no contradiction in the divisibility of extended beings, the indivisibility of atoms cannot be accepted as necessary; thus, one must come to simple beings.[41]

她伟大贡献是提出比牛顿更清晰的能量概念。

Her great contribution was introducing a clearer idea of energy than there is in Newton.

描述摆实验时，牛顿说：因为几何学家熟知，摆体在最低点的速度与它下落所描绘弧的弦成正比。

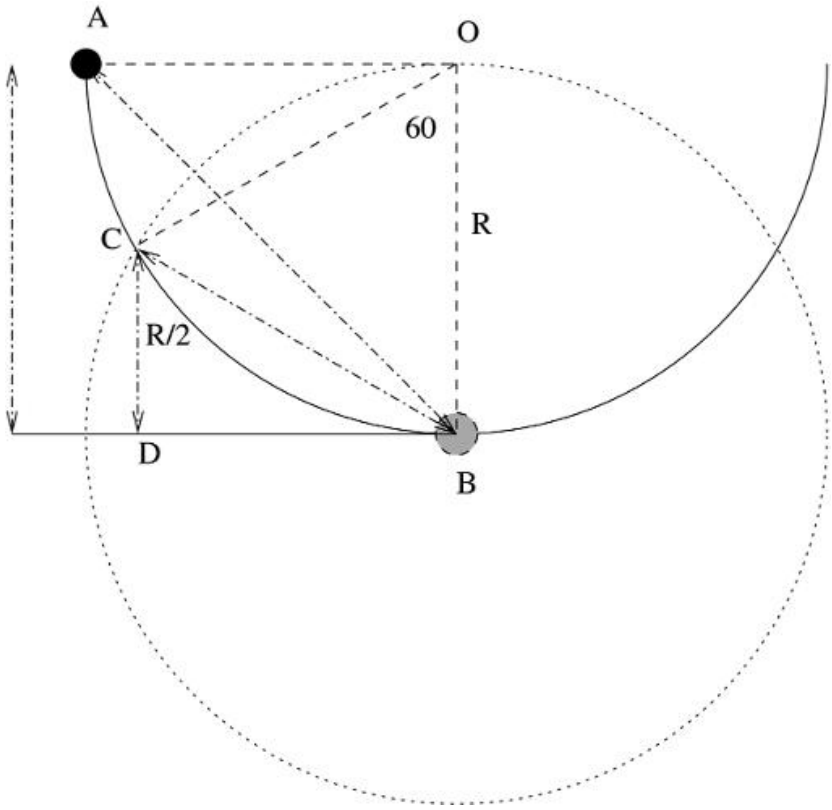
In the course of describing experiments with pendulums, Newton makes the striking remark ‘*For it is a proposition well known to geometers, that the velocity of a pendulous body in the lowest point is as the chord of the arc which it has described in its descent*’.

显然，从弦与速度关系可推导另一等价关系/守恒定律：势能与动能等价。但牛顿未明确表述为守恒定律。利用牛顿在《原理》中的几何论证形式，可轻松证明守恒关系。

It is clear that from this relationship between chords and velocities, one could deduce another equivalence relation/conservation law –that of the equivalence of potential and kinetic energy. But Newton fails to formulate this explicitly as a conservation law. The conservation relation can readily be demonstrated using the form of geometric argument that Newton uses in the *Principles*.

请看下图。[42]

Consider the following diagram.[42]



摆的几何关系（原书无编号插图）。

Pendulum geometry (unnumbered diagram).

A、C：两个起始点；O：摆心；B：最低点；R：摆长；R/2：C 点相对最低点的高度；D：辅助投影点。

校注：原书没有为此图另列图号和图注；上方原注 42 说明由作者绘制。

考虑以 O 为中心的摆。图中两个起始点 A 和 C。

若从 A 开始，角 AOB 为 90 度，则位置 B 速度与弦 AB 成正比。若从 C 开始，角 COB 为 60 度，则速度与弦 CB 成正比。

容易证明，若摆半径 R，则高度 $CD = R/2$ 。

此外，由于 COB 是等边三角形，我们知道 $CB = R$；又由于 AOB 是等腰直角三角形，所以 $AB = \sqrt{2}R$。

因此，两条弦之比为 $AB/CB = \sqrt{2}$，而下落高度之比为 $OB/CD = 2$。

若速度与弦成正比，速度与高度平方根成正比。若假设动能 e（杜 • 沙特莱的“活力”，forces vives），使得 $e \approx v^2$ ，则动能与初始势能成正比。牛顿证明无空气阻力时摆恢复原高度，因此这里有守恒定律。《注释》中关于落体的陈述——“成比例时间内描述的距离与速度和时间乘积成正比，即与时间平方成正比”——也可推导相同守恒关系。

〔校注：底本写 e≈v²，但本段文字谈动能与速度平方成正比。约等关系与比例关系含义不同；这里将底本公式与文字一并保留。〕

Consider the pendulum with centre O. The diagram shows two possible starting points for its swing, A and C. p. 51

If it starts at A with the angle AOB being 90 degrees, it will, according to the above, have a velocity at position B proportional to the chord AB. If it starts at C, with COB being 60 degrees, then it will have a velocity proportional to chord CB.

It is easy to demonstrate that if the radius of the pendulum is R, then the height CD = R/2.

Further, since COB is equilateral, we know $CB = R$, and since AOB is an isosceles right triangle, then $AB = \sqrt{2}R$.

So we have the relation between chords $AB/CB = \sqrt{2}$, but that between heights of fall $OB/CD = 2$.

If the velocity is proportional to the chords, then the velocities are proportional to the square root of the heights. If we postulate kinetic energy *e*, du Châtelet’s *forces vives*, such that $e \approx v^2$ we have the kinetic energy proportional to initial potential energy. Since Newton establishes that pendulums regain their original height in the absence of air resistance, he has a conservation law here. The earlier statement in the Scholium about falling bodies, that ‘the spaces described in proportional times are as the velocities and the times conjunctly; that is, in a duplicate ratio of the times’, could also be used to deduce the same conservation relationship.

牛顿只明确动量守恒，而非能量守恒。他说螺丝施加的力与转动手柄的手施加的力成正比；槌子对楔子施加的力与楔子推进量成正比。这些原理适用于所有机械。更普遍：

Newton only explicitly identifies a conservation of momentum, not a conservation of energy. He said that the force applied by a screw is proportional to the force applied by the hand that turns the handle. The force applied by a mallet to a wedge is proportional to the progress of the wedge. These principles apply to all machines. More generally:

我仅想通过这些例子展示运动第三定律的广泛适用性和确定性。因为如果我们综合作用体的力和速度估算作用，同时综合障碍物各部分速度以及摩擦、内聚力、重量和加速度产生的阻力估算反作用，那么使用各类机械时作用与反作用始终相等。

I was only willing to show by those examples the great extent and certainty of the third Law of motion. For if we estimate the action of the agent from its force and velocity conjunctly, and likewise the reaction of the impediment conjunctly from the velocities of its several parts, and from the forces of resistance arising from the attrition, cohesion, weight, and acceleration of those parts, the action and reaction in the use of all sorts of machines will be found always equal to one another.

这再次接近理想简单机械能量守恒。他说：

This is again close to a conservation of energy principle for ideal simple machines. He is saying

$$f_1 v_1 = f_2 v_2$$

但速度等于距离除以时间，即 d/t，因此，在固定时间间隔内有：

but velocity is distance over time d/t, so over a fixed time interval we have

$$f_1 d_1 = f_2 d_2$$

这与后来功的定义——力乘距离——一致，说明理想简单机械输入功等于输出功（忽略摩擦等）。

which corresponds to the later definition of work as force times distance, saying work in equals work out for an ideal simple machine (disregarding friction etc.).

牛顿有动量守恒和力等价性，但未看到现代功或能量表述。

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What Newton had was conservation of momentum and equivalence of forces, but without seeing through to the modern formulation of work or energy.

杜 • 沙特莱在牛顿基础上进步，倡导所谓“活力”，即我们今天所说的能量。她在此赋予“力”一词多重含义，但正确指出这种“力”必须与速度平方成正比：

Du Châtelet advanced on Newton and was a proponent of what she called *living force*, what we would now call energy. She overloads the term *force* here, but she gets right that it must be proportional to the square of velocity:

活力或许是物理学中唯一尽管有实验证明仍有人异议的论点；因为若问拒绝者，两个质量相等物体撞击两个相同障碍物，速度分别为 4 和 3，会产生什么效果，他们会回答：一个产生 16 的效果，另一个产生 9 的效果。现在容易看出，无论他们随后对这一为真理之力所迫作出的承认加以何种区分和修饰，效果是平方，因此产生效果的力必须是平方的。[43]

Forces vives may be the only point of physics which some still dispute while acknowledging the experiments that prove it; for if you ask those who reject them what would be the effects of two bodies equal in mass on two equal obstacles, but the speeds of which are 4 and 3, they will answer that one will be an effect, as 16 and the other as 9. Now, it is easy to see that, whatever distinction and whatever modification they next bring to this acknowledgment that the force of truth draws from them, it always remains certain that the effect being squared, there must have been a squared force to produce it.[43]

她说牛顿得出结论：运动总体不守恒，相反：

She says that Newton had concluded that there was no overall conservation law regarding motion and that he believed instead that:

宇宙中运动不断减少；最后我们的太阳系终有一天需要造物主重新塑造；这一结论是物质惯性以及牛顿先生所持“力量等于动量”观点的必然结果。[44]

motion is constantly diminishing in the universe; and lastly that our system will some day need to be formed anew by its Author, and this conclusion was a necessary consequence of the inertia of matter, and the opinion held by M. Newton that the quantity of force was equal to the quantity of motion.[44]

她声称牛顿要求持续奇迹维持运动，并反驳：

She claimed that in this Newton was demanding continuous miracles to sustain motion, asserting instead that:

But when the product of the mass by the square of the speed is taken as force, it is easy to prove that the forces vives always

但当把质量与速度平方乘积视为力时，容易证明活力始终不变，尽管宇宙每个瞬间动量可能变化；所有情况下，尤其是我刚引用的牛顿先生论点中，活力保持不变。[45]

remain the same, although the quantity of motion varies perhaps at each instant in the universe, and in all the cases, and especially in that which I have just cited from M. Newton, the forces vives stay invariable.[45]

这是能量守恒原理首次明确表述。卢克莱修“无中不能生有”延伸到运动。杜·沙特莱的继任者拉格朗日（1736—1813）和兰金（1820—1872）扩展该思想，拉格朗日通过动能和势能概念系统化无摩擦系统力学。最终兰金明确把能量守恒表述为自然定律。将其概括为定律的障碍是摩擦损耗。拉格朗日能用势能和动能表述支配行星轨道的定律，但应用于地球系统不明显，因为摩擦使物体减速。认识到热是另一种能量形式，抵消动能损失，才确立能量守恒一般原理。马克思写《资本论》时这些思想已牢固确立，我们稍后展示守恒原理如何支撑他的分析。

This is the first explicit formulation of the principle of energy conservation. Lucretius’ nothing from nothing extended to motion. Du Châtelet’s successors Lagrange (1736–1813) and Rankine (1820–72) extended the idea, Lagrange by systematizing the mechanical dynamics of frictionless systems in terms of kinetic and potential energy. Finally, Rankine explicitly states the conservation of energy as a law of nature. The obstacle to generalizing it as a law had been in dealing with frictional losses. While Lagrange could formulate a law governing the planets’ orbits in terms of potential and kinetic energy, it was far less obvious that it could be applied to systems on the Earth, where friction slowed things down. The recognition that heat was another form of energy which balanced the loss of kinetic energy, allowed a general principle of energy conservation to be established. These ideas had been well established by the time Marx wrote *Capital*, and we will later show how conservation principles underpin his analysis.

机械决定论

Mechanical determinism

亚里士多德把原因分为动力因和目的因。动力因从现在到未来，目的因是未来状态导致现在事件。

Aristotle categorized causes into efficient and final. Efficient causes work from present to future, while final causes are states in the future that cause events in the present.

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牛顿机械唯物主义依赖动力因。所有天体当前位置和动量完全决定未来位置和动量。其他天体存在使计算复杂，但不改变决定论。其他力如静电力和磁力在 19 世纪末发现，但力学不变。

〔校注：原书将静电力与磁力的发现说成 19 世纪末；本章前文已谈古代琥珀和磁石现象，二者存在明显年代不一致，译文照录底本。〕

Newton’s mechanical materialism relies on efficient causes. The current positions and momenta of all celestial bodies fully dictate their future positions and momenta. The existence of other celestial bodies may make the calculations more complex, but they do not alter determinism. Other forces, such as electrostatic and magnetic forces, were discovered in the late nineteenth century, but the mechanics remained the same.

机械决定论悖论：既包含动力因，也包含目的因。地球今天位置和动量决定明天位置，也限定昨天必然位置。t_0 时刻的力决定运动速度。天文学家可向前积分推未来，向后积分推过去。

Mechanical determinism is paradoxical because it implies both efficient and final cause. The position and momentum of the Earth today determine where it will be tomorrow and constrain where it must have been yesterday. Forces t_0 dictate the rate of motion. Astronomers can infer future planetary positions by integrating forward and past positions by integrating backward.

力学定律具有时间对称性。反转时间维度并反转物体运动，时间便倒流。动力因与目的因区别不再重要。太阳系当前状态既决定过去也决定未来。用拉格朗日或哈密顿力学表述——基于完整轨迹——时更清晰。变分法是微分学延伸，用于推导轨迹。

The laws of mechanics possess time symmetry. This means that reversing the time dimension and movements of the bodies cause time to run backward. The distinction between efficient and final cause becomes irrelevant. The present state of the solar system constrains its past and future. This concept becomes clearer when using formulations of mechanics by Lagrange or Hamilton, which reason about complete trajectories. Variational calculus, an extension of differential calculus, is used to reason about these trajectories.

哈密顿和拉格朗日力学借光学类比发展。若光束以一定角度射入水中，会弯曲。反之，把直尺部分浸入水中，直尺看起来弯曲。这是光在空气/水界面折射结果。古典力学家知道，尽管从直尺到眼睛路径弯曲，光“选择”路径使光子穿越距离时间最小。由于水比空气密，光在水中更慢，因此到眼睛最快路径是光子在空气中花费时间比直线传播更多的路径。

The Hamiltonian and Lagrangian mechanics arose by analogy with optics. If a light beam is shone into water at an angle, it bends. Conversely, if you place a straight ruler partially under water, the ruler appears bent. This is a consequence of the bending of light on the air/water boundary. The classical mechanists knew that although light from the ruler to your eye takes a bent path, the path it ‘decides’ to take is such that it minimizes the time a photon will take to traverse the distance. Since water is denser than air, light travels more slowly through it, and in consequence the quickest route to your eye is one in which the photon spends more of its time going through air than it would were it to go on a straight line.

这条光学定律称为费马原理，似乎已动摇朴素时间、因果观念。从直尺顶端出发的光线，怎会知道走哪条路最快到达眼睛？

This law of optics, known as Fermat’s principle, already seemed to undermine naive notions of time, cause and effect. How could the light ray setting out from the tip of the ruler know what path to take to reach your eye fastest?

欧拉变分法可证明[46]光遵循轨迹恰好是可能最快路径。从时间看，光遵循最简单路径。欧拉学生拉格朗日随后意识到，牛顿定律描述的轨迹或许也可同样处理。但此时最简单路径不能只是最快，必须在另一意义上简单。

Euler’s calculus of variation can be used to show[46] that the trajectory the light follows turns out to be the quickest that it could take. From the standpoint of time, light follows the simplest possible path. One of Euler’s students, Lagrange, then had the insight that perhaps the trajectories laid down by Newton’s laws might be treated the same way. But in this case the simplest path could not simply be the quickest. It had to be simple in another sense.

p. 53–54

他得出答案：自然遵循最小作用量原理。作用量是科学隐喻，其精确数学定义至少可以说并非显而易见。[47]拉格朗日力学是牛顿定律数学推论。使用拉格朗日方法，许多问题计算更容易。它显示自然如何对轨迹施加约束，类似光路径。单个粒子轨迹容易想象，但一对或多粒子较难。

The answer he came up with was that nature follows the course of least action. Action here is a scientific metaphor whose precise mathematical definition is non-obvious to say the least.[47] Lagrangian mechanics is a mathematical implication of Newton’s laws. Using Lagrange’s method, computations for many problems are easier. It shows how nature imposes constraints on trajectories, similar to light’s path. While it’s easy to visualize a single particle’s trajectory, it’s harder to understand a pair or a multiplicity of particles.

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学校数学中，我们学习 y 作为 x 函数的轨迹，绘制抛物线，特征方程 $y = ax^2 + b$ 。这简单直观，因为 x、y 代表方格纸水平垂直轴距离。用 x、y 坐标数学描述二维运动也直观。我们直观理解二维、三维。接受生活在三维空间，图画在二维纸张。但六维或十二维空间是什么？

In school maths, one is taught about trajectories in which y is a function of x. We learned to plot parabolas and learned that these have characteristic equations of the form $y = ax^2 + b$. This is all simple and easy to visualize since x, y represent distances along the horizontal and vertical axes on a sheet of graph paper. It is also straightforward to express movement in two dimensions mathematically in terms of x, y coordinates. We have an intuitive notion of what two dimensions or three dimensions means. We accept that we live in three-dimensional space and that we draw our pictures on two-dimensional sheets of paper. But what would a six-dimensional or a twelve-dimensional space be?

十二维轨迹是什么？经典物理学家处理多粒子系统或多个耦合独立运动物体时，借助抽象“位形空间”。

What is a twelve-dimensional trajectory? When the classical physicists dealt with systems of many particles or multiple coupled independently moving bodies, they resorted to an abstract form of space they called *configuration space*.

考虑单个运动粒子。它在三维空间的位置和速度可写成六个数：三个 x、y、z 坐标，另外三个 $\dot{x}, \dot{y}, \dot{z}$

Consider a single moving particle. Its position and velocity in three-dimensional space can be written down as

表示三个基本方向速度分量。数学上是六维系统。两个粒子则十二个数，等等。这组抽象维度称为位形空间。

〔校注：作者这里把位置与速度合在一起称为 configuration space（位形空间）。通常位形空间只包含位置类广义坐标；位置与速度合成的状态空间与之有别。下文沿用作者术语，并保留这一辨析。〕

six numbers, three to give its x, y, z coordinates and another three, $\dot{x}, \dot{y}, \dot{z}$, which mean its velocity components in the three cardinal dimensions. Mathematically, this is a six-dimensional system. If you have two particles, you have twelve numbers etc. This set of abstract dimensions is called configuration space.

力学抽象空间与欧几里得真实空间有何共同点？

What does the abstract space of mechanics have in common with the real space that we learn about from Euclid?

当我们实际只有纸上符号 $x_1, y_1, z_1, \dot{x}_1, \dot{y}_1, \ldots \dot{z}_n$ 时，从何意义谈维度和轨迹？这些符号与真实空间如何关联？回到二维例子，可见方格纸位置与数对等价性有操作基础。可数二维网格线，把纸上点转换为 x、y 坐标；反之，根据两个数字在纸上标点。建筑和测量实践在古典几何与物理世界间建立联系。建筑图纸用于建造与图纸保持欧几里得相似关系的更大尺度建筑。对应关系通过实践确立。

In what sense can we speak of dimensions and trajectories when all we actually have are signs on paper, $x_1, y_1, z_1, \dot{x}_1, \dot{y}_1, \ldots \dot{z}_n$? How do the signs and squiggles relate to real space? If we return to the two-dimensional example, we can see that there is an operational basis for the equivalence of positions on graph paper and pairs of numbers. You can count grid lines in the two dimensions to convert a dot on the paper to x, y coordinates or, conversely, you learn to plot a point on the paper given two written numbers. Architectural and surveying practice created a link between classical geometry and the physical world. Architectural diagrams were used to build larger buildings sharing Euclidean similarity. This correspondence was established through practice.

《原理》中牛顿仍用古典欧几里得几何。到 18 世纪中叶，用笛卡尔坐标推理物理成为标准。这涉及建模物质实践转变：不再用直尺圆规，而用代数。欧几里得方法可以得到模拟式结果。按特定半径几何作图，沿新画线测量得结果。若想比直尺更精确，必须设计基于直角三角形的毕达哥拉斯过程，再明确计算平方和平方根。

In the *Principia*, Newton still used classical Euclidean geometry. By the mid-eighteenth century it becomes standard to reason about physics in terms of Cartesian coordinates. This involves a shift in the material practice of modelling. Instead of using the ruler and compass, one uses algebra. With Euclidean techniques one can get analogue results. You follow the geometric construction using specific radii and then obtain a result measuring along new lines that you have drawn. If you want the result to be more accurate than you can manage with a ruler, you have to devise some ultimately Pythagorean process of construction in terms of right triangles, after which you explicitly compute squares and square roots.

物理学家用笛卡尔方法推导行星轨道方程。天文学家使用拉格朗日方法参数化实际轨道。方程与轨道关系不立即清楚，但可学会比旧几何更精确高效的方法。两种情况下，我们都有在地球上制作的、关于月球运动的实体纸面模型。

The physicist used the Cartesian method to derive an equation for a planetary trajectory. The astronomer used Lagrangian methods to parameterize the actual orbit. While the relationship between equation and orbit isn’t immediately clear, we can learn to use methods that are more accurate and efficient than old geometric approaches. In both cases, we have a physical paper model of the Moon’s motion on Earth.

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回到维度含义。人类实践语境中，维度与身体运动相关。看 18 世纪爱丁堡新城地图。[48]这需要启蒙运动所培育的感受方式按笛卡尔几何公设构建城市环境，但同样方法更大规模用于绘制整个国家地图，把每个道路交叉口、山丘或桥梁标在笛卡尔网格上。

Let us return to what a dimension means. In the context of human practice, dimensions are related to bodily movement. Look at a map of the eighteenth-century New Town of Edinburgh.[48] It took an Enlightenment sensibility to construct the urban environment according to the geometric postulates of Descartes, but the same procedure, writ large, was to be employed to map whole nations, plotting every road junction, hill or bridge on Cartesian grids.

沿乔治街向东走，获得一个维度。左转进入北城堡街，获得第二维度——南北。使维度区别的是独立性：向北走不向东移动。若目的地圣安德鲁广场，

Walking east along George Street gives me one dimension. I can then turn left into North Castle Street. This gives me a second dimension, the north-south one.

仍需沿皇后街三个街区，然后左转北圣大卫街。向北步行未减少向东还需街区数。

What makes the dimensions distinct is that walking north does not move me in any way east. If my destination was St Andrew Square, I would still have to go three blocks along Queen Street and then turn left into North St David Street. My walk to the north did not reduce the number of blocks I still had to go east.

南北维度与东西维度之所以不同，在于二者的独立性。沿一个轴运动，不会使我沿另一个轴向前。两个维度就是我沿街漫步时的自由度。拉格朗日方程中的变量 $x_1, y_1, z_1, \dot{x}_1, \dot{y}_1, \ldots x_2, y_2, \ldots \dot{z}_n$，也是多粒子系统的独立自由度。其中 x_1 表示第一个粒子在 x 方向的位置，y_2 表示第二个粒子在 y 方向的位置，而 $\dot{z}_n$ 表示第 n 个粒子在 z 方向的运动。所有这些量都可以独立变化，因此，在一种抽象的比喻意义上，它们就是维度。所以，位形空间中的一个点，就是所有粒子的位置和速度的一种特定组合；位形空间中的一条轨迹，必定是这些点的序列。

What makes the north-south and east-west dimensions distinct is their independence. Movement on one axis gets me no further along towards the other. The two dimensions are my degrees of freedom in a street stroll. The variables in the equations of Lagrange, $x_1, y_1, z_1, \dot{x}_1, \dot{y}_1, \ldots x_2, y_2, \ldots \dot{z}_n$, are also independent degrees of freedom for the multi-particle system. The x_1 denotes position of the first particle in the *x* direction, y_2 the position of the second particle in the *y* direction, and $\dot{z}_n$ the motion of the nth particle in the *z* direction. All of these can vary independently, and so in an abstract metaphorical sense they are dimensions. A point in configuration space is therefore a particular combination of positions and velocities for all particles. A trajectory in configuration space must then be a sequence of these points.

关键点：拉格朗日方程规定整个多粒子系统在多维位形空间中唯一时间轨迹。拉格朗日决定论对全部运动的限定，与对单个物体运动的限定同样严格。整个宇宙有位形空间——所有原子位置和速度。穿越位形空间的宇宙轨迹是实空间中所有个体轨迹的乘积。

Now here is the crucial point: *the equations of Lagrange specify a unique time trajectory of the entire multi-particle system through multi-dimensional configuration space.* Lagrangian determinism specifies the totality of all movement just as strongly as it specifies the movement of one body. The entire universe has a configuration space – the positions and velocities of all the atoms. The universal trajectory through configuration space is the product of all the individual trajectories in real space.

若刚接触力学，需要时间理解。

If you are new to mechanics and the mechanical approach, this takes some time to sink in.

由此，宇宙在位形空间中也有唯一轨迹，即贯穿时间的最小作用量路径。这种决定论足以令加尔文或诺克斯汗颜。

It follows that the universe too has a unique trajectory through configuration space, a path of least action throughout time. This is a determinism to shame Calvin or Knox.

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2013 年 2 月 15 日，俄罗斯中部工业城市车里雅宾斯克上空出现炽热火球。比太阳亮，户外市民感到皮肤灼热。幸运的是，当量为 500 千吨的爆炸发生在约 30,000 米高空。尽管如此，许多人受伤住院。报告闪光盲、辐射烧伤、皮肤剥落、爆炸创伤和飞溅碎片割伤。冲击波破坏数百平方公里。数千栋公寓楼、学校、医院和工厂需修复。

On 15 February 2013, over the major industrial city of Chelyabinsk in central Russia, an intense ball of fire appeared in the sky. Brighter than the Sun, citizens out in the open could feel its intense heat on their skin. Luckily for the residents of Chelyabinsk, the 500 kiloton detonation took place at an altitude of about 30,000 metres. Nonetheless, many were injured and hospitalized. Flash blindness, radiation burns, peeling the skin off, blast injuries and cuts from flying debris were reported. Shockwave damage spread over hundreds of square kilometres. Thousands of blocks of flats, schools, hospitals and factories had to be repaired.

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原因？不是自 1960 年代以来瞄准车里雅宾斯克的美国民兵导弹，而是来自小行星带的 20 米岩石。它沿最小作用量路径在太阳系运行无数时代，最终在该市上空终结。目击者见它从西方低空、朝阳方向出现。受开普勒定律约束，它在返回小行星带途中离开近日点，速度极快。以 60,000 km/h 沿双曲线路径，若未擦过大气，会飞回外层空间。但动能瞬间

The cause? Not one of the American Minuteman missiles pointed at Chelyabinsk since the 1960s, but a 20-metre lump of rock from the asteroid belt. Its path of least action, looping through the solar system for countless ages, terminated over the city. Witnesses saw it emerge in the low Western sky, from the direction of the rising sun. Fleeing its closest solar approach on its way

转化为热、辐射和冲击波。若入射角更陡、爆炸更低，车里雅宾斯克会被夷平。

〔校注：底本将“西方低空”与“日出方向”并列，方向表述存在矛盾；译文保留两者，不擅自替作者改写。〕

back to the asteroid belt, bound by Kepler’s law, it had to be travelling fast. It raced along at 60,000 km/h on a hyperbolic path that, had it not grazed the atmosphere, would have led it back into outer space. Instead, kinetic energy flashed into heat, radiation and blast. Had the angle of approach been steeper and the explosion lower, Chelyabinsk would have been flattened.

但轨道动力学中没有“本可能”。它的路径不可能不同。用古典语言，它注定与城市擦肩而过。尽管车里雅宾斯克幸免，仍是警示。宙斯或许投掷雷霆，法埃顿也可能无预警坠落。[49]

But in orbital dynamics, there are no might-have-beens. There was no way that its path could have been any different. To revert to classical language, it was *fated* to narrowly spare the city. Though Chelyabinsk lived, it was still a salutary reminder. Zeus may hurl his bolts, or the Phaethon may fall without warning.[49]

NASA 试图预警。“哨兵地球”项目搜寻危险小行星。但车里雅宾斯克超火流星未被 NASA 发现。

NASA tries to grant warnings. Their ‘Sentry Earth’ project searches out dangerous asteroids. Yet Chelyabinsk’s superbolide had not been spotted by NASA.

搜索受望远镜限制。即使已探测天体，也无法精确预测，只能给概率。可访问网站查看风险。例如小行星 2020 VV 估计 2044 至 2111 年间撞击概率 0.23%。幸运的是仅 36 千吨。但既然轨道确定性，为何只给概率？2021 年 NASA 用拉格朗日力学送凤凰号着陆火星，准时着陆，距目标几米。既然 NASA 能做到，为何小行星撞击像掷骰子？

〔校注：本段风险概率、时间范围、当量和任务名称 / 年份均为原书陈述，按底本保留，未作为当前 NASA 风险评估或任务记录重新核验。〕

Searches are bounded by the available telescopes. Even for the ones they have spotted, no exact predictions are made. They instead give probabilities. You can go to their website and view the risks they have spotted. For example, an asteroid called 2020 VV is estimated to have a 0.23 per cent chance of striking us sometime between 2044 and 2111. Luckily, this would only be a 36 kiloton hit. But why, given the determinism of orbits, is the estimate only in terms of probabilities? In 2021, NASA sent Phoenix Lander to Mars using Lagrangian mechanics. It landed on time, within a few metres of its intended destination. If NASA can do that, why are asteroid collisions a game of dice?

2020 年 10 月至 12 月，NASA 对 12 米天体 2020 VV 观测 79 次，用数百万公里外望远镜。NASA 必须根据观测估计位置和速度，但屏幕确切位置不确定。初始信息模糊，预测也模糊。

NASA has made seventy-nine observations of 2020 VV, an object 12 metres across, between October and December 2020. These observations were made using telescopes from millions of kilometres away. NASA must estimate the object’s positions and velocity based on these observations, but the exact position on the screen is uncertain. As a result, the initial information is fuzzy, and the predictions made based on this information will also be fuzzy.

据我们理解，宏观运动定律确定性。但鉴于无知——未观测现象和观测不精确——似乎存在偶然。未观测流星可能任何瞬间撞击。已观测流星最多知道某天经过的窗口。

To the best of our understanding, the macroscopic laws of motion are deterministic. But given our ignorance, both in terms of things we have not yet observed and in terms of the imprecision of our observations, there appears to be an element of chance. An unobserved meteor may hit at any instant. For the observed ones, we know at most a window through which they will pass on a given day.

拉格朗日唯物主义绝对决定论，表面随机性是无知产物。现代唯物主义是否必然蕴含决定论，以及决定论意味着什么，我们稍后回到。现在考察经典力学构建的认知结构。

Lagrangian materialism is absolutely deterministic, and apparent randomness is the product of our ignorance. Whether modern materialism necessarily implies determinism, and what such determinism means, is something we will return to. For now, let us examine the structure of knowing created by classical mechanics.

我们面对多层次算法建模过程。微分学和变分法是转换公式的程序。借助它们，牛顿初始定律可转化为包含势能和动能的欧拉-拉格朗日运动方程。现代术语中，这些方程是一类软件。

What we have is a multilayered process of algorithmic modelling. The differential calculus and the calculus of variations are procedures by which formulae can be transformed. Using them, Newton’s initial laws could

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be transformed into Euler-Lagrange equations of motion, including terms for potential and kinetic energy. In modern language, these equations are a class of software.

它们相对高级，因为涉及泛函——转换其他函数的函数。规则作用：把广义坐标表示动能和势能的函数转换为新运动函数。

〔校注：原书把 functionals 解释为“变换其他函数的函数”，这里按原文翻译。严格数学用法中，泛函通常是把函数映射到标量的映射；把函数映射到函数通常称算子。〕

They are relatively high-level software since they involve functionals, or functions which transform other functions. They are rules for taking functions which give kinetic and potential energies in terms of generalized coordinates and transforming these into new functions of motion.

新运动函数本身需初始边界条件参数化。完成参数化后，推导仅含一个自由变量——时间的函数。

The new functions for motion themselves need to be parameterized by initial boundary conditions. Once the full parameterization has occurred, you then derive functions with only one free variable –time.

此时，可以把算法运算从欧拉的代数式重写，切换到我们在学校学到的、作为笔算乘法和加法的原初算法（algorithismes）。[50]运用这些熟悉的方法，人们以十进制算术进行详细计算。

At this point, you can switch your algorithmic operations from the algebraic term rewriting of Euler to the original *algorithismes* that we are taught in school as long multiplication and addition.[50] With these familiar techniques, detailed calculations were done using decimal arithmetic.

如今，参数化运动函数用 Fortran 或类似语言编码，NASA 计算机用代码计算轨道。

Nowadays, the parameterized functions of motion would be encoded in Fortran or some similar language, and NASA computers would use the code to compute trajectories.

结合物质计算过程——人类数学家纸笔劳动或电子计算机循环——参数化函数构成所研究系统模型，如木星卫星、近地小行星。

When combined with a process of material computation –either the paper and pencil labour process of a human mathematician or the cycling of an electronic computer –the parameterized functions constitute a model for the system under study, be it Jovian moons, near-Earth asteroids etc.

建模关系存在于卫星或小行星实时运动与计算机器或过去从事计算工作的人模拟计算时间之间。

The modelling relationship that exists is between the real-time motion of the moons or asteroids and the simulated computational time of the computing machine or old-style human computer.

两种情况下都有物质系统：一侧卫星和行星，另一侧计算系统。都随时间演化。计算系统时间演化通常比现实快。模型/系统等价性本身在观测劳动过程精度约束下确立。人们用工具——望远镜、笔记本、键盘、处理器——确定初始位置并核对最终位置。

In both cases, we have material systems, with moons and planets on one side and computational systems on the other. They both evolve through time. The time evolution of the computational system will typically be faster than that of the real world. The model/system equivalence is itself established within certain accuracy constraints by observational labour processes. People use tools –telescopes, notepads, keyboards, processors etc. –to determine initial positions and check final positions.

科学家通过实践验证知识。火箭科学中，实践验证了被模拟系统在本体论上的实在性。火星曾是红点，如今我们看岩石沙丘视频，听火星微风。

Scientists use practice to verify knowledge. In rocket science, practice has verified the ontological reality of simulated systems. Mars was once just a red dot in the sky, but now we watch videos of his rocks and dunes up close and hear Martian breezes.

对现实建模，构建拉格朗日量或哈密顿量的程序是模型生产手段，不是模型本身。实际模型通过集体劳动过程和观测构建。它们不是单纯“思想对象”，而是消耗能量、在计算进行时随时间演化的、发生于具体地点的物理过程。

For modelling the real world, procedures for constructing Lagrangians or Hamiltonians are means of production for models, not models themselves. Actual models are built through collective labour processes and observation. They are not mere ‘thought objects’, but localized physical processes that consume energy and evolve through time as computations are performed.

第 3 章原注 / Chapter 3 Notes

↑1. 阿基米德：《阿基米德著作集：以现代符号编订并附导论各章》（The Works of Archimedes: Edited in Modern Notation with Introductory Chapters），Thomas L. Heath 编，Cambridge Library Collection – Mathematics（剑桥大学出版社，2009），https://doi.org/10.1017/CBO9780511695124。

1 Archimedes, *The Works of Archimedes: Edited in Modern Notation with Introductory Chapters*, ed. Thomas L. Heath, Cambridge Library Collection – Mathematics (Cambridge University Press, 2009), https://doi.org/10.1017/CBO9780511695124. p. 237

↑2. Lucio Russo：《被遗忘的革命：科学如何诞生于公元前 300 年，以及为何必须重新诞生》（The Forgotten Revolution: How Science Was Born in 300 BC and Why it Had to be Reborn）（Springer Science & Business Media，2013）。

2 Lucio Russo, *The Forgotten Revolution: How Science Was Born in 300 BC and Why it Had to be Reborn* (Springer Science & Business Media, 2013). p. 237

↑3. Rocco Leonardis：“古希腊建筑师对几何学的运用”（The Use of Geometry by Ancient Greek Architects），A Companion to Greek Architecture（2016），第 92–104 页，https://doi.org/10.1002/9781118327586。

3 Rocco Leonardis, ‘The Use of Geometry by Ancient Greek Architects’, *A Companion to Greek Architecture* (2016): 92–104, https://doi.org/10.1002/9781118327586. p. 237

↑4. Galen、Arthur John Brock:《盖伦论自然机能，附英译》（Galen On the Natural Faculties with an English Translation）（W. Heinemann，1916），第 1 卷第 13 章。

4 Galen and Arthur John Brock, *Galen On the Natural Faculties with an English Translation* (W. Heinemann, 1916), Book I, Chapter XIII. p. 237

↑5. Galen、Brock：《盖伦论自然机能，附英译》（Galen On the Natural Faculties with an English Translation），第 1 卷第 14 章。

5 Galen and Brock, *Galen On the Natural Faculties with an English Translation*, Book I, Chapter XIV. p. 237

↑6. pneuma 一词在希腊语中意为“空气”。

6 The term pneuma was Greek for air. p. 237

↑7. C. E. Quin：“盖伦之后关于神经系统功能的灵魂与气”（The Soul and the Pneuma in the Function of the Nervous System after Galen），Journal of the Royal Society of Medicine，第 87 卷第 7 期（1994），第 393–395 页，ISSN：0141-0768，https://europepmc.org/articles/PMC1294649。

7 C. E. Quin, ‘The Soul and the Pneuma in the Function of the Nervous System after Galen’, *Journal of the Royal Society of Medicine* 87, no. 7 (1994): 393–5, issn: 0141-0768, https://europepmc.org/articles/PMC1294649. p. 237 p. 237

↑8. 这与 19 世纪弗兰肯斯坦博士利用新发现的电能使尸体获得生命的观念，有相似之处。

8 There is a similarity here with the nineteenth-century notion of Dr Frankenstein animating a corpse with the newly discovered electric energy. p. 237

↑9. 鲁索：《被遗忘的革命：科学如何诞生于公元前 300 年，又为何必须重新诞生》。

9 Russo, *The Forgotten Revolution: How Science Was Born in 300 BC and Why It Had to Be Reborn.* p. 237

↑10. 艾萨克 • 牛顿：《自然哲学的数学原理……安德鲁 • 莫特译……附牛顿〈世界体系〉；W. 埃默森对〈原理〉的简短评注与辩护；以及约翰 • 梅钦依据引力所论月球运动定律……（先生的序言）》（1848）。

〔校注：底本书目题名末尾“(The Preface of Mr)”自身不完整；按原样保留，不擅自补人名。〕

10 Isaac Newton, *The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which Are Added Newton’s System of the World; a Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . . (The Preface of Mr)* (1848). p. 237–238

↑11. G. W. 莱布尼茨：《形而上学论》，1686 年，http://www.earlymoderntexts.com/assets/pdfs/leibniz1686d.pdf。

11 G. W. Leibniz, ‘Discourse on Metaphysics’, 1686, http://www.earlymoderntexts.com/assets/pdfs/leibniz1686d.pdf. p. 238

↑12. 鲁索在《被遗忘的革命：科学如何诞生于公元前 300 年，又为何必须重新诞生》中，解释了托勒密天文学与傅里叶分解之间的相似性。

12 Russo, *The Forgotten Revolution: How Science Was Born in 300 BC and Why It Had to Be Reborn,* explains the similarity between Ptolemaic astronomy and Fourier decomposition. p. 238

↑13. 我们现在会用 straight（直的）这个词。

13 We would say straight. p. 238

↑14. 恒星日以恒星相对于地球的视旋转来衡量，比太阳日短约 4 分钟。 p. 238

14 A sidereal day is measured in terms of the apparent rotation of the stars with respect to the Earth and is about 4 minutes shorter than a solar day.

p. 238 ↑15. “物体从静止开始作匀加速下落时，所经过的距离之比，等于通过这些距离所用时间间隔的平方之比。”定理 II，命题 2。

15‘The spaces described by a body falling from rest with a uniformly accelerated motion are to each other as the squares of the time-intervals employed in traversing these distances’, Theorem II, Proposition 2.

p. 238 ↑16. 牛顿：《自然哲学的数学原理……安德鲁 • 莫特译……附牛顿〈世界体系〉；W. 埃默森对〈原理〉的简短评注与辩护；以及约翰 • 梅钦依据引力所论月球运动定律……（先生的序言）》，定律 I。

〔校注：底本书目题名末尾“(The Preface of Mr)”自身不完整，按原样保留。〕

16 Newton, *The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which are Added Newton’s System of the World; a Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . . (The Preface of Mr)*, Law I.

p. 238 ↑17. 牛顿：《自然哲学的数学原理……安德鲁 • 莫特译……附牛顿〈世界体系〉；W. 埃默森对〈原理〉的简短评注与辩护；以及约翰 • 梅钦依据引力所论月球运动定律……（先生的序言）》，定律 II。

〔校注：底本书目题名末尾“(The Preface of Mr)”自身不完整，按原样保留。〕

17 Newton, *The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which are Added Newton’s System of the World; a Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . . (The Preface of Mr)*, Law II.

p. 238 ↑18. 牛顿：《自然哲学的数学原理……安德鲁 • 莫特译……附牛顿〈世界体系〉；W. 埃默森对〈原理〉的简短评注与辩护；以及约翰 • 梅钦依据引力所论月球运动定律……（先生的序言）》，定律 III。

〔校注：底本书目题名末尾“(The Preface of Mr)”自身不完整，按原样保留。〕

18 Newton, *The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which are Added Newton’s System of the World; a Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . . (The Preface of Mr)*, Law III.

p. 238 ↑19. 牛顿：《自然哲学的数学原理……安德鲁 • 莫特译……附牛顿〈世界体系〉；W. 埃默森对〈原理〉的简短评注与辩护；以及约翰 • 梅钦依据引力所论月球运动定律……（先生的序言）》，推论 III。

〔校注：底本书目题名末尾“(The Preface of Mr)”自身不完整，按原样保留。〕

19 Newton, *The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which are Added Newton’s System of the World; a Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . . (The Preface of Mr)*, Corrollary III.

p. 238 ↑20. 牛顿：《自然哲学的数学原理……安德鲁 • 莫特译……附牛顿〈世界体系〉；W. 埃默森对〈原理〉的简短评注与辩护；以及约翰 • 梅钦依据引力所论月球运动定律……（先生的序言）》，推论 V。

〔校注：底本书目题名末尾“(The Preface of Mr)”自身不完整，按原样保留。〕

20 Newton, *The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which are Added Newton’s System of the World; a Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . . (The Preface of Mr)*, Corrollary V.

p. 238 ↑21. 相对于不具有某种运动的事物而言，以及在一切同等分享该运动的事物之间，该运动不起作用，就仿佛不存在一样。因此，一艘满载货物的船驶离威尼斯，途经科孚、克里特、塞浦路斯，前往阿勒颇。威尼斯、科孚、克里特等地静止不动，不随船运动；但对于船上装载的袋子、箱子和包裹，以及船本身而言，从威尼斯到叙利亚的运动等于没有运动，丝毫不会改变它们彼此之间的关系。原因就在于，这一运动为它们所共有，它们全都同等地分享这一运动。伽利略 • 伽利莱：《关于两大世界体系的对话》（1632）（Robert C. Kuhmann，2006），第 41 页，`http://www.kuhmann.com/starstuff/DialogueConcerningtheTwoChiefWorldSystems.pdf`。对于一个简单的可动物体，只能有一种、而不能有更多的运动天然适合它；其他运动，它只能偶然地、

21 Relatively to things that lack it, and among things which all share equally in any motion, it does not act and is as if it did not exist. Thus the goods with which a ship is laden leaving Venice, pass by Corfu, by Crete, by Cyprus and go to Aleppo. Venice, Corfu, Crete etc., stand still and do not move with the ship; but as to the sacks, boxes and bundles with which the boat is laden and with respect to the ship itself, the motion from Venice to Syria is as nothing, and in no way alters their relations among themselves. This is so because it is common to all of them and all share equally in it. Galileo Galilei, *Dialogue Concerning the Two Chief World Systems*, (1632) (Robert C. Kuhmann, 2006), 41, `http://www.kuhmann.com/starstuff/DialogueConcerningtheTwoChiefWorldSystems.pdf`. For

通过参与而具有。因此，当一个人沿船甲板行走时，他自身的运动是步行，而把他带到港口的运动则是他所参与的运动；因为若不是船通过自身运动将他带到那里，他单靠步行永远无法抵达。伽利莱：《关于两大世界体系的对话》，第 45 页。

a simple movable body, there can be only a single motion, and no more, which suits it naturally; any others it can possess only incidentally and by participation. Thus, when a man walks along the deck of a ship, his own motion is that of walking, while the motion which takes him to port is his participation; for he could never arrive there by walking if the ship did not take him there by means of its motion. Galilei, *Dialogue Concerning the Two Chief World Systems*, 45.

†22. 牛顿：《自然哲学的数学原理……安德鲁 • 莫特译……附牛顿〈世界体系〉；W. 埃默森对〈原理〉的简短评注与辩护；以及约翰 • 梅钦依据引力所论月球运动定律……（先生的序言）》，第 183 页。

〔校注：底本书目题名末尾“(The Preface of Mr)”自身不完整，按原样保留。〕

22 Newton, *The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which are Added Newton’s System of the World; a Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . . (The Preface of Mr)*, 183. p. 239

†23. sesquiduplicate ratio 意为 $r^{2.5}$。

〔校注：底本确实印为 r 的 2.5 次方；但正文所述周期—半径关系对应 3/2 次方，即 1.5 次方。此处保留原注并指出不一致。〕

23 Sesquiduplicate ratio means $r^{2.5}$. p. 239

†24. 这意味着平方根之比：$r_1 : r_2$ 的 subduplicate ratio 为 $\sqrt{r_1} : \sqrt{r_2}$；所以，若原比为 18:16，subduplicate ratio 就是 9:4。

〔校注：底本 18:16 开平方后的比写成 9:4，计算有误；正确为 √18:√16，即 3√2:4。公式中的等号和正文两处异常句点均为底本所有。〕

24 Th. is means in the ratio of the square roots, the subduplicate ratio of $r_1 : r_2 = \sqrt{r_1} : \sqrt{r_2}$ so if the ratio were 18:16, the. subduplicate ratio would be 9:4. p. 239

†25. 牛顿：《自然哲学的数学原理……安德鲁 • 莫特译……附牛顿〈世界体系〉；W. 埃默森对〈原理〉的简短评注与辩护；以及约翰 • 梅钦依据引力所论月球运动定律……（先生的序言）》，命题 IV，定理 IV。

〔校注：底本书目题名末尾“(The Preface of Mr)”自身不完整，按原样保留。〕

25 Newton, *The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which are Added Newton’s System of the World; a Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . . (The Preface of Mr)*, PROPOSITION IV. THEOREM IV.. p. 239

†26. 格奥尔格 • 威廉 • 弗里德里希 • 黑格尔：“关于行星轨道的哲学论文（1801），附 1801 年 8 月 27 日答辩的十二条论题”（Philosophical Dissertation on the Orbits of the Planets (1801), Preceded by the 12 Theses Defended on August 27, 1801），Graduate Faculty Philosophy Journal，第 12 卷第 1/2 期（1987），第 282—283 页。

26 Georg Wilhelm Friedrich Hegel, ‘Philosophical Dissertation on the Orbits of the Planets (1801), Preceded by the 12 Theses Defended on August 27, 1801’, *Graduate Faculty Philosophy Journal* 12, no. 1/2 (1987): 282–3. p. 239

†27. 黑格尔：“关于行星轨道的哲学论文（1801），附 1801 年 8 月 27 日答辩的十二条论题”，第 285 页。

27 Hegel, ‘Philosophical Dissertation on the Orbits of the Planets (1801), Preceded by the 12 Theses Defended on August 27, 1801’, 285. p. 239

†28. 黑格尔：“关于行星轨道的哲学论文（1801），附 1801 年 8 月 27 日答辩的十二条论题”，第 285 页。

28 Hegel, ‘Philosophical Dissertation on the Orbits of the Planets (1801), Preceded by the 12 Theses Defended on August 27, 1801’, 285. p. 239

†29. 黑格尔：“关于行星轨道的哲学论文（1801），附 1801 年 8 月 27 日答辩的十二条论题”，第 290 页。

29 Hegel, ‘Philosophical Dissertation on the Orbits of the Planets (1801), Preceded by the 12 Theses Defended on August 27, 1801’, 290. p. 239

†30. 黑格尔：“关于行星轨道的哲学论文（1801），附 1801 年 8 月 27 日答辩的十二条论题”，第 296 页。

30 Hegel, ‘Philosophical Dissertation on the Orbits of the Planets (1801), Preceded by the 12 Theses Defended on August 27, 1801’, 296. p. 239

†31. 弗拉基米尔 • 伊里奇 • 列宁：《马克思主义的三个来源和三个组成部分》，第 1 卷（外文出版社，1977）。

31 Vladimir Ilich Lenin, *The Three Sources and Three Component Parts of Marxism*, vol. 1 (Foreign Languages Press, 1977). p. 239

†32. 黑格尔：“关于行星轨道的哲学论文（1801），附 1801 年 8 月 27 日答辩的十二条论题”，第 294 页。

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32 Hegel,‘Philosophical Dissertation on the Orbits of the Planets (1801), Preceded by the 12 Theses Defended on August 27, 1801’, 294.

p. 239 ↑33. A. 埃默里：“辩证法与机械论：一场关于科学方法的共产主义论争”（Dialectics versus Mechanics. A Communist Debate on Scientific Method），Philosophy of Science，第 2 卷第 1 期（1935），第 9—38 页。

33 A. Emery,‘Dialectics versus Mechanics. A Communist Debate on Scientific Method’, *Philosophy of Science* 2, no. 1 (1935): 9–38.

p. 239 ↑34. “外力是施加于物体以改变其状态的一种作用，无论该状态是静止，还是沿直线匀速向前运动。”

34 ‘An impressed force is an action exerted upon a body, in order to change its state, either of rest, or of moving uniformly forward in a right line.’

p. 239 ↑35. 埃默里：“辩证法与机械论：一场关于科学方法的共产主义论争”。

35 Emery,‘Dialectics versus Mechanics. A Communist Debate on Scientific Method’.

p. 239 ↑36. 英译见埃米莉•杜•沙特莱：《哲学与科学著作选》（Selected Philosophical and Scientific Writings）（芝加哥大学出版社，2009）。

36 Available in English in Emilie Du Châtelet, *Selected Philosophical and Scientific Writings* (University of Chicago Press, 2009).

p. 239 ↑37. 杜 • 沙特莱：《哲学与科学著作选》（Selected Philosophical and Scientific Writings），《物理学基础》，序言 IV。

37 Du Châtelet, *Selected Philosophical and Scientific Writings*, Foundations of Physics, Preface IV.

p. 239 ↑38. Ruth Hagengruber：《莱布尼茨与牛顿之间的埃米莉 • 杜 • 沙特莱》（Emilie du Châtelet Between Leibniz and Newton）（Springer，2012）。

38 Ruth Hagengruber, *Emilie du Châtelet Between Leibniz and Newton* (Springer, 2012).

p. 239 ↑39. 杜 • 沙特莱：《哲学与科学著作选》（Selected Philosophical and Scientific Writings），第 132 页。她反对植物灵魂的论证，显然也适用于其他灵魂变体和类似宗教解释：精神、自由意志、主体、性别认同等等。

39 Du Châtelet, *Selected Philosophical and Scientific Writings*, 132, her argument against vegetative souls can clearly be applied to other variants of souls and similar religious explanations: the spirit, free will, the subject, gender identities etc.

p. 239 ↑40. 杜 • 沙特莱：《哲学与科学著作选》（Selected Philosophical and Scientific Writings），第 149 页。

40 Du Châtelet, *Selected Philosophical and Scientific Writings*, 149.

p. 239 ↑41. 杜 • 沙特莱：《哲学与科学著作选》（Selected Philosophical and Scientific Writings），第 166 页。

41 Du Châtelet, *Selected Philosophical and Scientific Writings*, 166.

p. 239 ↑42. 作者绘图。

42 Figure by authors.

p. 239 ↑43. 杜 • 沙特莱：《哲学与科学著作选》（Selected Philosophical and Scientific Writings），第 188 页。

43 Du Châtelet, *Selected Philosophical and Scientific Writings*, 188.

p. 239 ↑44. 杜 • 沙特莱：《哲学与科学著作选》（Selected Philosophical and Scientific Writings），第 199 页。

44 Du Châtelet, *Selected Philosophical and Scientific Writings*, 199.

p. 239 ↑45. 杜 • 沙特莱：《哲学与科学著作选》（Selected Philosophical and Scientific Writings），第 199 页。

45 Du Châtelet, *Selected Philosophical and Scientific Writings*, 199.

p. 239 ↑46. 证明参见 John R. Taylor 等：《经典力学》（Classical Mechanics）（University Science Books，2005），第 217 页。

46 See the proof in John R. Taylor, et al., *Classical Mechanics* (University Science Books, 2005), .217.

p. 240 ↑47. 作用量定义为 $S = \int \mathcal{L}\, dt$，其中 $\mathcal{L}$ 为拉格朗日量，表示动能与势能之差。详细论述参见 Taylor 等：《经典力学》，第 6、7、13 章。

47 Action is defined as $S = \int \mathcal{L}\, dt$ where $\mathcal{L}$. the Lagrangian, stands for the difference between kinetic and potential energy. For a detailed treatment, see see Taylor et al., *Classical Mechanics*, chap. 6, 7, 13.

p. 240 ↑48. 例如参见 2015 年 6 月 6 日“La ciudad como reivindicación identitaria: el caso de la New Town de Edimburgo”（作为身份认同诉求的城市：爱丁堡新城案例），`https://urban-networks.blogspot.com/2015/06/la-ciudad-como-reivindicacion.html`。

48 See for instance *6 jun 2015 La ciudad como reivindicación identitaria: el caso de la New Town de Edimburgo.* at `https://urban-networks.blogspot.com/2015/06/la-ciudad-como-reivindicacion.html`.

p. 240 ↑49. Kord Ernstson 等：“基姆高陨石坑散布区：德国巴伐利亚东南部全新世大型撞击事件的证据”（The Chiemgau Crater Strewn Field: Evidence of a Holocene

49 Kord Ernstson et al.,‘The Chiemgau Crater Strewn Field: Evidence of a Holocene Large Impact Event in Southeast Bavaria, Germany’, *Journal of Siberian Fed-*

Large Impact Event in Southeast Bavaria, Germany），Journal of Siberian Federal University. Engineering & Technologies，第 1 卷第 3 期（2010），第 72—103 页；Barbara Rappenglück 等：“法埃顿之坠落：一个希腊—罗马地质神话保存了巴伐利亚（德国东南部）陨石撞击的记忆”（The Fall of Phaethon: A Greco-Roman Geomyth Preserves the Memory of a Meteorite Impact in Bavaria (South-East Germany)），Antiquity，第 84 卷第 324 期（2010），第 428—439 页。

eral University. *Engineering & Technologies* 1, no. 3 (2010): 72–103; Barbara Rappenglück et al.,‘The Fall of Phaethon: A Greco-Roman Geomyth Preserves the Memory of a Meteorite Impact in Bavaria (South-East Germany)’, *Antiquity* 84, no. 324 (2010): 428–39.

↑50. 这些算法以花剌子密（al-Khwārizmī）命名，他的算术方法通过斐波那契的《算盘书》（Liber Abaci）传入欧洲。

50 These were named after al-Khwārizmī, whose method of arithmetic was introduced to Europe in Fibonnaci’s *Liber Abaci*.

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第 4 章辩证法、唯物主义与变化：从伊壁鸠鲁经亚里士多德到马克思

Chapter 4. DIALECTICS, MATERIALISM, CHANGE FROM EPICURUS TO MARX VIA ARISTOTLE

黑格尔重新构造之前的辩证法

Dialectics before its Hegelian reinvention

p. 59

想必这是一个广为人知的事实：在古希腊，术语“辩证法”（dialektikē）指的是亚里士多德在《修辞学》[1]中所说的正确思维的“艺术”，它应当是科学或哲学探究的先决条件和必然特征。[2]任何可能导致真实或准确结论的考察——用亚里士多德的话说，即“逻辑的”或“三段论的”结论——[3]任何能够自称哲学的真理追求，都基于我们今天所称的逻辑前提。亚里士多德会称之为辩证法——“logikē”一词直到古代晚期才出现，而且即便出现，其含义也仍不同于现代用法。在《修辞学》中，亚里士多德特意把诡辩意义上的良好论证——因此未必由“对智慧的热爱”驱动——与真正的辩证论证区分开。后者关乎价值，更确切地说，关乎伦理学；前者则关乎准确论证本身作为一种技艺或艺术——亚里士多德会称之为 technē——[4]这种能力任何个体都能掌握，而不必是职业哲学家（或古希腊的等同者）。

It is presumably a widely known fact that, in Greek antiquity, the term *dialectics* (*dialektikê*) referred to the ‘art’, as Aristotle calls it in *Rhetoric*,[1] of correct thinking, one that ought to be the prerequisite and the inevitable characteristic of scientific or philosophical inquiry.[2] Any examination of a matter that could lead to a truthful or accurate conclusion, a ‘logical one’ (or ‘syllogical’) as Aristotle would put it,[3] any pursuit of truth that could call itself philosophical, operates on the premises of what we would nowadays call logic. Aristotle would have called it *dialectics* – *logikê* does not appear until late antiquity, and when it does, it is still not in the sense we use the word in our era of modernity. In *Rhetoric*, Aristotle makes sure to distinguish a good argument in the sophist sense – thus not necessarily driven by ‘the love of wisdom’ – from a properly dialectical argument. The latter is concerned with values, or ethics, to be more precise, the former with the skill or art – the *technē*, Aristotle would say[4] – of accurate argumentation itself, which can be entertained by any individual and not necessarily a professional philosopher (or its equivalent in Greek antiquity).

在本章中，我们将尝试重建辩证法作为本体论范畴在欧洲哲学舞台上的出现。似乎我们在共产主义学说“diamat”中所见的唯物辩证法，其起源可以追溯到黑格尔，但并不完全源于马克思本人。这似乎有悖直觉，或与我们所受的教育以及在马克思主义、马克思本人著作及其思想遗产方面习惯性的思维方式背道而驰。然而，若仔细审视马克思关于唯物主义与辩证法这一双重主题的著作——既把它视为统一体，也把它视为两个独立范畴——就会发现，他对辩证法的处理始终更接近古代人，而非黑格尔。

In this chapter, we will attempt to reconstruct the appearance of dialectics as an ontological category on the scene of European philosophy. It seems that the origins of materialist dialectics as we find it in the communist doctrine of Diamat can be traced back to Hegel but not quite to Marx himself. This may seem counterintuitive, or as something that goes against everything we have been taught and have gotten used to thinking on the matter regarding Marxism and Marx’s own body of work and legacy. Nonetheless, a closer look at Marx’s writings on the dual topic of materialism and dialectics, seen as a unity as well as two separate categories, will show us that his treatment of dialectics has always been closer to that of the Ancients than to that of Hegel.

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我们必须注意他的博士论文中的一个微小例外：其中我们可以发现一种推理方式，在某种（且值得商榷的）程度上与黑格尔式辩证法模式相似；而他基于原子论的唯物主义，则与受黑格尔启发的“diamat”逻辑相矛盾。马克思对古希腊原子论的解读，塑造了他毫不妥协的唯物主义立场，我们将在本章后文更深入探讨。在此只需指出，根据青年马克思，支撑伊壁鸠鲁原子论的矛盾，是定义原子物质属性的特性与作为原子概念规定的形式之间的产物。然而，后者并不构成更高的真理形式，也不是存在的目的（causa finalis）。它无非是物质现象的概念——原子，始终已经具象化，从未被赋予理想性，也从未参与任何更宏大且包罗万象的观念。[5]因此，理想性并非观念，而是物质的完美形式；若以属性来表述这种形式，必然偏离理想形式——或者简而言

We must note a slight exception to this rule in his doctoral thesis, in which we can detect a form of reasoning that operates to a certain (and arguable) degree similarly to the Hegelian model of dialectics, whereas his materialism grounded in the atomist theory contradicts the logic of the Hegel-inspired Diamat. Marx’s reading of Greek atomism informs his uncompromisingly materialist stance, something we will inspect more closely further on in the chapter. At this point, suffice it to say that the contradiction that underpins Epicurean atomism, according to young Marx, is the product of the properties that define an atom materially and the form that is the atom’s conceptual determination. The latter, however, does not constitute a higher form of truth or purpose of existence, *causa finalis*. It is nothing that would go be-

之，正如马克思所言，偏离概念本身。概念虽从具体中抽象出来，却并不存在于“理想性”领域，也无法栖居于物质性之外的任何其他宇宙——因为根本不存在这样的世界。

yond quite simply being the concept of a material phenomenon –the atom, always already embodied and never endowed with an ideality or participation in a greater and all-encompassing idea.[5] Ideality is thus not an idea but a perfect form of the material whose predication in terms of property is necessarily an aberration from the ideal form –or simply the concept, as Marx puts it. The concept is abstracted from the concrete but does not reside in a realm of ideality, and it cannot inhabit any other universe but that of materiality –because there is no such world.

因此，我们将把唯物主义和辩证法这两个范畴视为彼此区分、相互处于单边关系之中[6]，摒弃马克思主义思想传统中——无论是“科学”的（马克思主义）还是哲学层面的——对二者统一性的教条式预设。马克思将引导我们回溯到古希腊原子论——更确切地说，是伊壁鸠鲁及其著名的伊壁鸠鲁学派代表卢克莱修·卡鲁斯——去探寻唯物主义的科学起源，或者说，探寻那种更接近科学探究方式而非哲学—神学探究方式的唯物主义的起源。在此之前，让我们重新审视古希腊时期的辩证法问题，以及它演变为黑格尔对该术语理解的可能路径。

We will, therefore, look at the two categories, materialism and dialectics, as distinct and unilateral to one another,[6] without the canonically presumed unity of the two when it comes to the Marxist tradition of thought, be it ‘scientific’ (Marxism) or philosophy. Marx will lead us retrospectively to look for the scientific origins of materialism –or for the origins of the materialism that is closer to a scientific mode of inquiry than to a philosophical-theological one –in Greek atomism, more specifically that of Epicurus and the famous Epicurean Lucretius Carus. Before we do so, let us revisit the question of dialectics in Greek antiquity and the possible routes of its evolution to the Hegelian understanding of the term.

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Dialektikē、认知的机械性与物质性

Dialektikê, mechanicity and materiality of cognition

在接触关于希腊“辩证法”概念的文献学与哲学讨论时——尤其是亚里士多德著作中所使用的这一概念——人们往往会感到困惑。希腊城邦中典型的“辩论”实践，或者更确切地说，那种公开论战的文化，怎么会与“逻辑”（即配得上“哲学”这一属性、或当今所称“科学”的思维艺术）混为一谈呢？然而，我们认为这并非混淆，而是通过对辩证交流实践规则的抽象——或者说，通过对准确思考或符合对真理一贯追求的思维规律的抽象——对这种被观察到的实践进行形式化。因此，类似于马克思对“原子概念”——或其形式——的处理，亚里士多德抽象出了我们今天所称的逻辑思维的理想形式。这仿佛是把观察到的物质现实的“模具”从其基础中抽离出来，并通过运用辩证思维的“艺术”，将其形象（或形式）进一步形式化为一个“概念”（如同马克思的原子概念），这与维特根斯坦在《逻辑哲学论》中提出的 Maßstab（尺度）极为相似。[7]正如亚里士多德在《物理学》（214a12–214a16）中所论述的，概念是“正确思维方式”的产物，亦即下文将通过对亚里士多德《修辞学》相关段落的细读所探讨的辩证或三段论式的思维方式。这门艺术的精髓在于能够辨别“真实的三段论”——即属于辩证法（dialektikē）的三段论——与属于修辞学的虚幻三段论：ὥσπερ καὶ ἐπὶ τῆς διαλεκτικῆς συλλογισμόν τε καὶ φαινόμενον συλλογισμόν（引自亚里士多德《修辞学》1:1）[8]，或译为英文（W. Rhys Roberts 译本）：“辨别真实与表象的劝说手段的艺术，正如辨别真实与表象的三段论是辩证法的职能一样。”[9]

In encountering philological-philosophical discussions of the Greek notion of dialectics, in particular the one in use in Aristotle’s works, one often becomes bewildered. How could the practice of ‘debating’, or rather the culture of public polemical exchanges typical of the Greek *polis*, be confused with ‘logic’ (the art of thinking worthy of the attribute philosophical, or nowadays scientific)? Yet we do not believe it is a matter of confusion but a formalization of the observed practice of dialectical exchanges through the abstraction of its rules or simply the laws of thinking accurately or in line with a consistent pursuit of truth. Thus, similarly to Marx’s treatment of the ‘concept of the atom’ –or its form –Aristotle abstracts the ideal form of what we would nowadays call logical thinking. It is as if a mould of the observed material reality is detached from its foundation and, by way of applying ‘the art’ of dialectical thinking, its image (or form) further formalized as a ‘concept’ (like Marx’s concept of the atom), very much like Wittgenstein’s Maßstab in the *Tractatus*.[7] The concept is the product of a ‘correct way of thinking’, as Aristotle argued in *Physics* (214a12–214a16), or the dialectical or syllogistic way of thinking discussed below through a close reading of passages from Aristotle’s *Rhetoric*. The art consists in being capable of discerning the ‘real syllogisms’, which are those of dialektikê, from the illusionary syllogism, which belongs to rhetoric: ὥσπερ καὶ ἐπὶ τῆς διαλεκτικῆς συλλογισμόν τε καὶ φαινόμενον συλλογισμόν (from Aristotle’s *Rhetoric* 1: 1),[8] or in En-

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glish (W. Rhys Roberts’translation): ‘art to discern the real and the apparent means of persuasion, just as it is the function of dialectic to discern the real and the apparent syllogism’.[9]

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亚里士多德所指的辩证法，涉及“三段论”思维的纯粹形式，可普遍适用于任何主题。如前所述，这是一种元科学，他实际上将其称为一种能够引导人得出真实结论的思维“艺术”（技艺或技能——technē）。若剥离对真理的追求，它仅仅是修辞学；而使其具有辩证性并适于服务于正当科学探索的，正是其伦理关切——即对真实阐释周围现实事物的兴趣。

Dialectics, as used in Aristotle, concerns the pure form of ‘syllogical’ thinking, universally applicable to any subject matter. As already said, it is a meta-science that he in fact calls ‘art’ (craft or skill –*technē*) of thinking that can lead one to truthful conclusions. Emptied of the pursuit of truth, it will remain rhetoric, whereas what makes it dialectical and apt to serve proper scientific exploration would be its ethical concern – the interest in truthfully explaining matters of surrounding reality.

因此，存在这三种实现说服的手段。显然，要掌握这些手段的人必须能够（1）进行逻辑推理［原文：λαβεῖν τοῦ συλλογίσασθαι δυναμένου；换言之——通过三段论的方式］，（2）理解人类品性与善的各种形式，以及（3）理解情感——即能够命名和描述它们，了解其成因以及被激发的方式。由此可见，修辞学既是辩证法的分支，也是伦理学研究的延伸。伦理学研究完全可以被称为政治学；正因如此，修辞学常伪装成政治学，其教授者也常自诩为政治专家——有时是因为缺乏教育，有时是为了炫耀，有时则源于其他人性弱点。事实上，正如我们开篇所言，它属于辩证法的一个分支，并与之相似。无论是修辞学还是辩证法，都不是对任何单一独立主题的科学研究：二者都是提供论据的能力。（亚里士多德，《修辞学》1:2）[10]

There are, then, these three means of effecting persuasion. The man who is to be in command of them must, it is clear, be able (1) to reason logically [in the original: λαβεῖν τοῦ συλλογίσασθαι δυναμένου; or, in other words –by way of syllogism], (2) to understand human character and goodness in their various forms, and (3) to understand the emotions –that is, to name them and describe them, to know their causes and the way in which they are excited. It thus appears that rhetoric is an offshoot of dialectic and also of ethical studies. Ethical studies may fairly be called political; and for this reason rhetoric masquerades as political science, and the professors of it as political experts –sometimes from want of education, sometimes from ostentation, sometimes owing to other human failings. As a matter of fact, it is a branch of dialectic and similar to it, as we said at the outset. Neither rhetoric nor dialectic is the scientific study of any one separate subject: both are faculties for providing arguments. (Aristotle, *Rhetoric* 1: 2)[10]

伦理投入的存在——或其缺失——不仅影响三段论的内容和论证艺术本身，也影响其形式，这使得修辞学成为一种逊于辩证法的艺术，孕育出诡辩家而非哲学家：

The presence of ethical investment – or the lack of it – affects not only the content of a syllogism and of the art of argumentation itself but also its form, rendering rhetoric an art inferior to that of dialectics, producing sophists instead of philosophers:

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因此，很明显，修辞学并不局限于某一特定类别的主题，而是像辩证法一样具有普遍性；同样明显的是，它具有实用价值。此外，很明显，它的功能不仅仅在于成功地说服他人，而在于探索在每个具体案例的条件下，尽可能接近这种成功的方法。在这方面，它与所有其他艺术相似。例如，医学的功能并非仅仅是使人完全健康，而是尽可能地将人引向健康的道路；即便是那些永远无法享有强健体魄的人，也完全可能获得极佳的治疗。此外，很明显，辨别真实的和表象的说服手段，正是同一门艺术的功能，正如辨别真实的三段论和表象的三段论，正是辩证法的功能。使一个人成为“诡辩家”的不是他的能力，而是他的道德目的。然而，在修辞学中，“修辞家”一词既可以指演讲者对这门艺术的掌握，也可以指他的道德目的。在辩证法中则不同：一个人之所以是“诡辩家”，是因为他怀有某种道德目的，而称他为“辩证家”，依据的不是他的道德目的，而是他的能力。（亚里士多德，《修辞学》1:1）[11]

It is clear, then, that rhetoric is not bound up with a single definite class of subjects, but is as universal as dialectic; it is clear, also, that it is useful. It is clear, further, that its function is not simply to succeed in persuading, but rather to discover the means of coming as near such success as the circumstances of each particular case allow. In this it resembles all other arts. For example, it is not the function of medicine simply to make a man quite healthy, but to put him as far as may be on the road to health; it is possible to give excellent treatment even to those who can never enjoy sound health. Furthermore, it is plain that it is the function of one and the same art to discern the real and the apparent means of persuasion, just as it is the function of dialectic to discern the real and the apparent syllogism. What makes a man a ‘sophist’ is not his faculty, but his moral purpose. In rhetoric, however, the term *rhetorician* may describe either the speaker’s knowledge of the art, or his moral purpose. In dialectic it is different: a man is a ‘sophist’ because he has a certain kind of moral purpose, a ‘dialectician’ in respect, not of his moral purpose, but of his faculty. (Aristotle, *Rhetoric*: 1: 1)[11]

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亚里士多德开创了抽象的技艺及其过程的形式化——例如三段论，后来演变为算法——将其作为一种纯粹且在语义上空洞的思维形式的实践。他开创

Aristotle invents the craft of abstraction and the formalization of its processes –such as syllogisms, later to become algorithms –as the practice of pure and semanti-

了算法思维的技艺。无论是他对抽象（或形式）的运用，还是他的辩证法，都属于技艺（technē）的范畴，而非本体论——外部现实无需具有辩证性或“三段论”性质，但我们对其的推理必须如此，正如算法无需反映外部现实的任何本质，却依然能够传达并模仿它。形式思维的发明，以及逻辑和对思维的算法规训，都是从物质和具体中提炼出来的实践产物。这一认识历经数千年才被重新发现并阐明，首先是由马克思（他在《经济学手稿》中捍卫了“具体”的思维，下文将讨论），随后是索绪尔，他在《普通语言学教程》中感叹道：

cally empty form of thought. He invents the *technē* of algorithmic thinking. Both his use of abstraction (or form) and his dialectics are a matter of *technē*, not of ontology – the outside reality does not need to be dialectical or ‘syllogical’, but our reasoning of it must be, just as an algorithm does not need to reflect any nature of outside reality but can nonetheless convey it and mime it too. The invention of formal thinking and of logic and algorithmic disciplining of thought is the product of a practice extracted from the material and the concrete. It took thousands of years for this realization to be rediscovered and elucidated, first by Marx (defending the thought of the concrete in *Grundrisse*, discussed below) and then by Saussure, who exclaims in his *Course in General Linguistics*:

我们第一次摆脱了抽象。现在，我们第一次找到了占据特定位置、并与口语链中某个转折点相对应的具体、不可还原的单位：p 只不过是一个抽象单位，它连接了 p> 和 p< 的共同特征，而 p> 和 p< 才是唯一实际存在的单位。同样，“唇音性”这一更高级的抽象概念将 P、M 和 B 联系起来。我们可以将 P 视为一种动物学物种；该物种有雄性和雌性代表，但并不存在理想标本。此前，我们一直在筛选和分类这些抽象概念；但我们必须超越抽象，才能抵达具体。音系学犯了一个重大错误，即在未更仔细考察“单位”定义的情况下，便将抽象概念视为真实的单位。（1959, 53）[12]

For the first time we have broken away from abstraction. Now for the first time we have found the concrete, irreducible units that occupy a place and correspond to a bent in the spoken chain: p was nothing except an abstract unit linking the common characters of p> and p<, the only units that actually exist. In the same way, the still higher abstraction of ‘labiality’ links together P, M and B. We may speak of P as if it were a zoological species; there are male and female representatives of the species, but there is no ideal specimen. Before, we had been singling out and classifying the abstractions; but we had to go beyond the abstract to reach the concrete. Phonology made a great mistake in considering abstractions real units without examining more carefully the definition of the unit. (1959, 53)[12]

索绪尔不仅证明了任何形式化的分析程序——即通向科学真理的道路——都必须从物质出发，还证明了物质是具体的，且不应将抽象概念误认为独立存在的现实。这既是对物质性的辩护，也是对作为科学考察的（物质）实践的抽象的辩护，最终更是对科学思维中必然存在的机械论程序主义的辩护。然而，这也揭示了所有有机性——在此指语言——的机械论方面。马克思对真正唯物主义探究的本质及其政治经济学方法的阐释与此非常相似：

Saussure demonstrates not only that any formal procedure of analysis, which is the path to scientific truth, must depart from the material but also that the material is concrete and that the abstracted concepts ought not to be mistaken for self-standing realities. This is a vindication of materiality, of abstraction as the (material) practice of scientific examination, and, finally, of the necessarily mechanistic proceduralism of scientific thought. However, it also discloses the mechanistic aspect of all organicity –in this case, language. Marx’s explanation of the nature of true materialist inquiry and the method of his political economy is very similar:

如果我们从人口出发，这将是对整体的一种混沌表象［Vorstellung］，然后，我们将通过进一步的规定，以分析的方式向越来越简单的概念［Begriff］推进，从“想象中的具体”走向越来越细微的抽象，直到抵达最简单的规定。从那里开始，我们必须回溯这一过程，直到最终再次抵达“人口”这一概念，但这次它不再是关于整体的混乱观念，而是一个由众多规定性和关系构成的丰富总体。［……］具体之所以是具体的，是因为它是多种规定的集中，因而也是多样性的统一。（《经济学手稿》，41）[13]

if we were to begin with the population, this would be a chaotic conception [Vorstellung] of the whole, and we would then, by means of further determination, move analytically towards ever more simple concepts [Begriff], from the Imagined concrete towards ever thinner abstractions until we had arrived at the simplest determinations. From there the journey would have to be retraced until we had finally arrived at the population again, but this time not as the chaotic conception of a whole, but as a rich totality of many determinations and relations. [. . .] The concrete is concrete because it is the concentration of many determinations, hence unity of the diverse. (*Grundrisse*, 41)[13]

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从手头的物质研究对象中推导出形式的机械性行为（马克思，部分源自亚里士多德），或是“根本二元组”这一最基本层面的意指行为——此处我们指的是两个能指之间的关系，这种关系并不一定需要被视为一种统一体、一对或二元对立，正如传统结构主义语言学所处理的那样——归根结底不过是单

Mechanical acts of extrapolating form from a material subject of study at hand (Marx, partly Aristotle) or the act of signification in its most basic instance of ‘a radical dyad’ –here we are referring to the relation between two signifiers, which does not necessarily need to be considered a unity, a couple or a binary, as conven-

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纯的物质差异，始终已经借助两个能指单位之间的物质标记（音素、书写痕迹或其他）来实现。我们在其他地方曾论述过，在“根本二元组”（这一术语借自弗朗索瓦·拉鲁埃尔[14] ）中，两个要素之间是单向的关联，其中一方对另一方而言是“实在”与“外部性”。所谓“二者的统一”，即二元对立作为意义生成首要要素的观念，用拉鲁埃尔的话来说，不过是哲学化或哲学幻觉罢了；因为链条内部的关系实为一个单位与另一个单位之间、一与零之间的差异，而在意义实践的物质性中，它们并不以二元对立的形式存在。引用保罗·科克肖特在 2020 年 8 月我们私人通信中的话：“简单的二元对立甚至算不上一种关系，因为它没有顺序。它就是 {＋, －} ＝ {－, ＋}，仅仅是一个包含两个可区分符号的集合。这与 {α,ω} 或 {■,▫} 等并无二致。若无其他语义支撑，它们仅仅是两个可区分的标记。”尽管如此，这些机械性行为却构成了多种形式的有机性：自然语言与人工语言中的能指自动机，或者正如许煜（Yuk Hui）最近所论证的，甚至存在于自然本身之中。

tional structuralist linguistics would treat it – come down to mere material differentiation, always already using a material mark (phonemes, writing traces or otherwise) between two signifying units. We have argued elsewhere that within the ‘radical dyad’, a term borrowed from François Laruelle,[14] the two elements are unilaterally related to one another, the one being the real and outsideness to the other. The idea of a unity of two, of a binary as the primary element of signification, is a matter of philosophization or philosophical hallucination, to paraphrase Laruelle, as the relation within a chain is differentiation between one and another unit, between one and zero, and they do not exist as a binary in the materiality of signifying practice. To quote Paul Cockshott from our private correspondence in August 2020: ‘A simple binary is not even a relation since it is unordered. It is { +,-} == {-,+}, simply a set with two distinguishable signs. No different from {α,ω} or {■,▫} etc. Without some other semantics, there are just two distinguished marks.’ Nonetheless, these acts of mechanicity amount to multiple forms of organicity: the signifying automata of natural and artificial languages, or, as Yuk Hui has argued recently, even in nature itself.

在许煜的《偶然性与递归性》一书中，后者也被视为一种指涉过程；正如许煜所论证的，正是有机生命中同样存在的递归程序，使得机械性总是在总是已经转化为有机性——二者不过是莫比乌斯带的两面，而非相互排斥的范畴。偶然性缺乏目的（telos）；它几乎毫无意义，而机械性作为机械性本身也是无意义的（它就是“管用就行”，这在本性上是偶然的）。自然借助递归来包容偶然性，正如人工智能所做的那样。据许煜所言，只有一种非目的论的目的，即生命自我维持的循环性：

In Hui’s *Contingency and Recursivity*, the latter is also treated as a process of signification, and, as Hui argues, it is the procedure of recursion one also finds in organic life that enables mechanicity to always already turn into organicity, the two being mere sides of a Moebius stripe rather than mutually exclusive categories. Contingency is bereft of telos; it is almost nonsensical, and mechanicity as mechanicity is senseless too (it is ‘whatever works’, which is in its nature contingent). Nature resorts to recursion in order to include the contingent, just as AI does. According to Yuk Hui, there is only a non-teleological telos, circularity of life sustaining itself:

> 自然的终极目标是无法被客观观察到的。我们可以看到某棵树或某种动物，但无法通过机械规则把握自然整体。理性只能通过反思性判断来理解自然的终极目标，这意味着它会递归地抵达一个自我组织的实体。从这个意义上说，目的论思维是循环的：A→B→C→A。（Hui, 41）[15]

> The natural end is something that cannot be observed objectively. We can see such and such a tree or such and such an animal, but we cannot grasp nature as a whole through mechanical rules. Reason can only understand the natural end through reflective judgment, meaning that it recursively arrives at a self-organising being. Teleological thinking is in this sense circular: A→B→C→A. (Hui, 41)[15]

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机械性与唯物主义互为条件，而我们在此使用该术语的方式，与笛卡尔将身体还原为无生命的机器时所采用的用法截然不同，甚至可说是截然相反——在笛卡尔的观点中，心灵（res cogitans）是至高无上的，且在任何意义上都不属于物质领域。因此，一切看似充满生命气息的事物，皆由心灵所驱动；所有内在的、看似自发运动的物质，是由意识所驱动的；而有机体本身就是精神的。那种源自原子论和马克思、源自索绪尔、图灵和玻尔兹曼的唯物主义，本质上是一种机械论（Cockshott Turing 在此指向他关于“机械性”的章节，以及关于图灵的论文）。[16]然而，正是物质中不可分割的最小粒子在持续的机械性碰撞中产生的运动——这与伊壁鸠鲁所说的“偏转”所引发的持续碰撞相似——能够转化为波、人类与神明所使用的语言、社会关系以及人类的生产力。一切有机的、自我调节的事物，其起源

Mechanicity and materialism are conditioned upon one another, and we are using the term here quite differently from, if not in the opposite sense to, that deployed by Descartes in reducing the body to a lifeless machine, while mind – *res cogitans* – is superior and not of the realm of materiality in any way whatsoever. Therefore, everything that seems to breathe life is moved by Mind; all immanent, seemingly moved by and of itself, is moved by mind; and the organic is spiritual. Materialism that draws from atomism and Marx, from Saussure, Turing and Boltzmann, is mechanistic (Cockshott Turing refers to his chapter here on Mechanicity plus the paper/s on Turing).[16] Yet it is the movement of the uncuttable smallest particles of matter enabled by their constant mechanic encounters – similarly to the constant collisions caused by Epicurus’ swerve – that can transform itself

或前提皆在于粗糙物质性所具有的原始机械性。我们在本书的其他部分论证道，那些我们认为未受唯心主义哲学化影响的唯物主义形式，要么是机械论的，要么是原子论的，或者两者兼而有之：牛顿和拉格朗日，以及玻尔兹曼和马尔可夫，仅举几例。如果将机械性简化为一种低级现实——即物质性，从而容许存在一个既无法通过机械方式也无法通过有机方式（即通过“递归”整合机械默认状态下“无意义的事件”）加以解释的完整存在领域，那么我们面对的不仅是唯心主义，还是一种无法作为唯物主义主要模态发挥作用的机械性概念。很显然，我们所探讨的唯物主义并非作为形而上学基础，用于追寻万物之“始源”（有时为复数形式）。在此，某位作者在“形而上学”一词的用法上可能与本书的其他合著者有所不同。作为推动思想的前科学动力，乃至整个人类生产力的推动者——其核心在于诸如“我们为何存在于这个星球上？”“什么是善？”“为何存在‘外部世界’，而我又与之分离？”“宇宙是否有开端？”——任何形而上学问题都有相关性，且可以通过一种绕开哲学及“充分哲学原则”影响的方式加以探讨。[17]机械性——通常与原子论相结合，并将运动与变化提升为其第一原理——正是定义真正唯物主义的核心。正如马克思在《德意志意识形态》中论及费尔巴哈的段落及其他著作中所论述的，理性主义和哲学的埃利亚学派渊源所留下的残余——即在不变且不朽的事物中寻求原理或始原（archai）——在所有形式的唯心主义中都是不可避免的，包括受哲学影响的唯物主义。请考虑以下段落：

〔校注：英文括号“Cockshott Turing refers to his chapter here on Mechanicity plus the paper/s on Turing”形似未清理的编写提示，按底本保留；原注 16 实际列科克肖特讨论图灵的文章。本文其他作者可能对“形而上学”采用不同定义，这是本段原作者明确指出的差异。〕

into waves, languages spoken by humans and gods, social relations and human productivity. All that is organic and self-regulated is originated or premised on the crude mechanicity of coarse materiality. We argue elsewhere in the book that the forms of materialism we consider uncompromised by idealist philosophization are either mechanicist or they embrace atomism, or both: Newton and Lagrange, and Boltzmann and Markov, to name a few. If mechanicity is reduced to materiality as an inferior reality allowing for an entire realm of existence that is explained neither mechanically nor organically (by way of ‘recursive’ integration of ‘nonsensical occurrence’ of mechanical default), we are dealing not only with idealism but also with a notion of mechanicity that does not play the role of the primary modality of materialism. We are quite evidently not dealing with materialism as the *metaphysical* foundation in the search of the *archē* (sometimes in plural) of the all-existing. Here, one author might differ from the co-authors of this book in the use of the term *metaphysical*. As a proto-scientific mover of thought, and of all human productivity for that matter –consisting in questions such as *Why do we exist on this planet? What is good? Why is there an outsideness and am I separated from it?* and *Does the universe have a beginning?* –any *metaphysical* question is relevant and can be tackled in a way that circumvents philosophy and the effects of the ‘principle of sufficient philosophy’.[17] Mechanicity –very often combined with atomism, elevating movement and change among its first principles – is what defines materialism proper. The residues of the Eleatic origin of rationalism and philosophy, which consist in seeking the principles or the *archai* in what is unchanging and undying, are unavoidable in all forms of idealism, including materialism indebted to philosophy, as argued by Marx in his passages on Feuerbach in *German Ideology* and elsewhere. Consider the following passage:

迄今为止一切唯物主义——包括费尔巴哈的唯物主义——的主要缺陷在于：事物、现实、感性，仅被理解为对象或观照的形式，而未被理解为一种感性的人类活动、实践，亦未从主体方面加以理解。因此，与唯物主义相反，唯心主义抽象地发展了能动的一面——当然，唯心主义本身并不认识作为这种活动的真实、感性的活动。（马克思，《关于费尔巴哈的提纲》）[18]

The chief defect of all hitherto existing materialism –that of Feuerbach included –is that the thing, reality, sensuousness, is conceived only in the form of the object or of contemplation, but not as a sensuous human activity, practice, not subjectively. Hence, in contradistinction to materialism, the active side was developed abstractly by idealism –which, of course, does not know real, sensuous activity as such. (Marx, *Theses on Feuerbach*)[18]

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上述所有唯物主义者有一个共同点，即一个跨越数个世纪的共同特征：与他们所处时代绝大多数权威哲学不同，他们主张，运动并非一种“幻象”，其背后隐藏着静止的实体——即“存在”的真理，而“存在”本身是静止不变且永恒的——而是存在既显而易见又隐而不显的真理：即不断的运动与变化。

What all of the materialists mentioned have in common, a shared constant across centuries, is that unlike the greatest part of the authoritative philosophy of their era, they argued that movement is not an ‘apparition’ behind which a static substance resides –the truth of Being, with the Being itself unmovable and eternal –but rather that which is both the apparent and the hidden truth of existence: constant movement and change.

激进运动与历史唯物主义，而非辩证唯物主义

Radical movement and historical materialism rather than dialectical materialism

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历史唯物主义应被视为对运动与变化的激进唯物主义理解，其思想渊源更多地源自早期原子论者，而非黑格尔。在对伊壁鸠鲁原子论的阐释中，马克思展现出黑格尔主义的影响，因为他试图通过辩证法来解决原子概念与其作为物质的具体属性之间的矛盾。我们在开篇讨论过马克思解决这一矛盾的方式，这构成了他基于伊壁鸠鲁思想遗产的唯物主义哲学的基础。据托马斯·奈尔（Thomas Nail）所言，马克思在接触黑格尔之前就开始发展他的唯物主义哲学和科学世界观，这使他为奈尔所称的“运动唯物主义（kinetic materialism）”（而非“辩证唯物主义”）奠定了基础：

Historical materialism should be seen as a radically materialist understanding of movement and change, more indebted to the early atomists than to Hegel. In his interpretation of Epicurean atomism, Marx displays a Hegelian influence insofar as he seeks a dialectical solution to the contradiction between the concept of the atom and the concrete properties that make it material. We discussed at the beginning the way Marx solves this contradiction, which amounts to the foundation for his materialist philosophy, building on the legacy of Epicurus. According to Thomas Nail, Marx began to develop his materialist philosophy and scientific world view prior to his exposure to Hegel, leading him to lay the foundations for what Nail calls kinetic materialism (rather than dialectical materialism):

> 马克思的运动理论，在其博士论文和《伊壁鸠鲁笔记》这些最早的著作中表现得最为鲜明和集中。早在阅读黑格尔之前，马克思就已经通过古代原子论展开对宗教的批判，并形成一种新的唯物主义理论。接触黑格尔之后，马克思在博士论文中同时批判了对古代原子论的反动现代主义解释和启蒙主义解释、唯物主义、唯心主义的自由理论、宗教，以及黑格尔的自然哲学。因此，马克思在博士论文中提出的运动理论和唯物主义，不仅仅是黑格尔思想的残余；这是他第一次尝试发展一种唯物主义的、以运动为基础的辩证法理论。（Nail，第 20—21 页）[19]

> Marx’s theory of motion is nowhere more evident and focused than in the earliest writings of his doctoral dissertation and Epicurean notebooks. Long before Marx had read Hegel, he had already been working out a critique of religion and a new theory of materialism through ancient atomism. After his exposure to Hegel, Marx used his dissertation to simultaneously critique the reactionary modernist and enlightenment interpretations of ancient atomism, materialism, idealist theories of freedom, religion, and the Hegelian philosophy of nature. Thus, the theory of motion and materialism put forward in Marx’s dissertation is not just a Hegelian residue; it is his first effort to develop a materialist and kinetic theory of dialectics. (Nail, 20–1)[19]

我们赞同奈尔的观点，即马克思思想中确实存在“运动唯物主义”，且其源于伊壁鸠鲁原子论；因为正是这种原子论的运动理论——以及“偏离概念”（即“偏转”，意指运动的不完美）这一概念的提出——才为其哲学以及后来发展为科学的理论奠定了纯粹的唯物主义基础。然而，我们认为马克思采用唯物主义并非为了提供一种辩证法理论，而是恰恰相反：他运用辩证法方法——主要是由古希腊人，特别是亚里士多德所发展的方法——来构建一种以“永不停息的运动”为前提的唯物主义，或者如奈尔所称的“运动唯物主义”。如前所述，当辩证法不再仅仅是一种方法，而是同时构建一种本体论论证时，便能察觉到黑格尔的影响。例如，概念与原子的物质决定因素或属性之间的张力，被呈现为物质的，或者至少是真实的——这符合马克思主义认识论家阿尔弗雷德·索恩-雷特尔所提出的“现实抽象”概念。[20]“概念”确实总是已经物质化了，这是不可避免的，因为正如马克思所解释的（如上所述），如果没有物质属性，就不会有概念，然而它似乎仍被视为一种能动主体——本体论实体：

We concur with Nail that we can speak of kinetic materialism in Marx and that it is derived from Epicurean atomism, as it is precisely the atomist theory of movement –and the invention of the ‘declination from the concept’, or the swerve, meaning the imperfection of movement –that enables a purely materialist grounding for his philosophy and what is to become science. However, we argue that Marx did not arrive at materialism to provide a theory of dialectics but instead quite the opposite: he used the dialectical method, mainly as developed by the Greeks and Aristotle in particular, to create a materialism premised on the idea of incessant movement, or as Nail has termed it, kinetic materialism. As already noted, one can detect Hegel’s influence when dialectics is no longer only a method but also builds an ontological argument. For example, the tension between the concept and the material determinants or properties of the atom is presented as material, or at least real, in the sense of the notion of ‘real abstraction’ developed by the Marxist epistemologist Alfred Sohn-Rethel.[20] The ‘concept’ is indeed always already materialized, inescapably so, since, as Marx explains (as discussed above), there would be no concept without the material properties, and it nonetheless seems to be treated as an active agency –ontological entity:

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> 存在与本质之间、物质与形式之间的矛盾——这种矛盾内在于原子概念之中——一旦原子被赋予了属性，便在单个原子本身中显现出来。通过这种属性，原子与其概念相异化，但同时在其构造上得以完善。正是源于排斥以及随

> The contradiction between existence and essence, between matter and form, which is inherent in the concept of the atom, emerges in the individual atom itself once it is endowed with qualities. Through the quality the atom is alienated from its concept, but at the same time is perfected in its construction.

之而来的具有属性的原子的聚合，现象世界才得以产生。[21]

It is from repulsion and the ensuing conglomerations of the qualified atoms that the world of appearance now emerges.[21]

马克思和伊壁鸠鲁并不设想彼此独立的、相互分离的领域。这种矛盾是物质性的，其辩证解决并非消除矛盾；矛盾是永恒的，正是它使物质现实处于不断的运动之中。原子是“不可见的”，但它的运动及其它属性——例如重力——却造就了可见世界（或呈现在我们眼前的世界），即一种无时无刻不在运动、无时无刻不在变化的存在现象。在马克思的诠释中，他与欧洲哲学的主流及其固有的唯心主义分道扬镳，因为他并未将可见世界贬低为虚假、幻象和错觉的领域——它与不可见世界一样“真实”。恰恰相反，“不可见的原子”只有通过其现象形式（包括运动在内的各种属性）才能得以实现。真正的唯物主义者既肯定现象世界的现实性，又将其视为通向不可见世界的出发点——两者在现实性和真实性上并无高低之分：

Marx and Epicurus do not postulate separate realms that are independent from one another. The contradiction is material, and its dialectical solution is not a resolution; the contradiction is perpetual, and it is what maintains material reality in constant movement. The atom is ‘invisible’, yet its movement and other properties, such as *gravity*, enable the visible world (or the one that appears to us), the phenomenon of an all-moving and all-changing existence. In his interpretation, Marx parts ways with the greater share of European philosophy and its inherent idealism, as he does not relegate the visible world to a realm of falsities, apparitions and hallucinations –it is ‘no less true’ than the invisible one. Quite to the contrary, ‘the invisible atoms are realised’ only through their phenomenal form (the properties that include movement). A true materialist affirms the reality of the phenomenal world and also treats it as the departure point for accessing the invisible one –the one is no less real or true than the other:

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其中一位［德谟克利特］是怀疑论者，另一位［伊壁鸠鲁］是教条主义者；前者将感性世界视为主观幻象，后者则视为客观表象。将感性世界视为主观幻象的人，致力于经验自然科学和实证知识，体现出无处不在的观察、实验和学习的热忱，足迹遍布广阔无垠的世界。而那位将现象世界视为真实的哲学家，则鄙视经验主义；他身上体现了自足而安宁的思想，以及那种从内在原则（ex principio interno）汲取知识的自给自足。但这种矛盾还远不止于此。那位将感性自然视为主观幻象的怀疑论者和经验论者，是从必然性的角度来审视它，并致力于解释和理解事物的真实存在。另一方面，那位将现象视为真实的哲学家和教条主义者，则处处只看到偶然性，而他的解释方法，反而倾向于否定自然界的一切客观现实。这些矛盾似乎蕴含着某种荒谬性。[22]

The one [Democritus] is a sceptic, the other [Epicurus] a dogmatist; the one considers the sensuous world as subjective semblance, the other as objective appearance. He who considers the sensuous world as subjective semblance applies himself to empirical natural science and to positive knowledge, and represents the unrest of observation, experimenting, learning everywhere, ranging over the wide, wide world. The other, who considers the phenomenal world to be real, scorns empiricism; embodied in him are the serenity of thought satisfied in itself, the self-sufficiency that draws its knowledge *ex principio interno*. But the contradiction goes still farther. The sceptic and empiricist, who holds sensuous nature to be subjective semblance, considers it from the point of view of necessity and endeavours to explain and to understand the real existence of things. The philosopher and dogmatist, on the other hand, who considers appearance to be real, sees everywhere only chance, and his method of explanation tends rather to negate all objective reality of nature. There seems to be a certain absurdity in these contradictions.[22]

怀疑论以及任何形式的主观唯心主义——我们在此将后结构主义对“实在的彻底封闭”的理解也包括在内——都建立在这样一种预期之上：即人类思想能够完全接触到实在，甚至与之融合，这相当于将真理等同于实在（反之亦然）。拉鲁埃尔认为，这正是构成所有哲学的根本谬误之处。[23]在《哲学与非哲学》（[1989]；2013）的开篇几页中，拉鲁埃尔将上述谬误称为“实在与思想的两义混淆”[24]，其最典型的原则正是所有欧洲哲学的原则——即“存在”（to on，τὸὄν）作为真理与实在的统一体这一观念——也被称为“充分哲学原则”（PSP）。[25]马克思推崇伊壁鸠鲁，因为他没有落入拉鲁埃尔所指认的、具有自我挫败性、自相吞噬的哲学基础——即不将“表象”世界视为欺骗性的现象，而是肯定其物质性正是（原子）不可见基础的唯一蓝图，或许也是其实现的必然性。可见世界的“幻象”或“欺骗性现象”（正如巴门尼德、柏拉图、贝克莱、黑格尔、朱迪斯·巴特勒以及所有哲学家——至少根据我们在

Scepticism and any variant of subjective idealism –we would include here the poststructuralist understanding of the ‘Real’s radical foreclosure’–are predicated on the expectation that human thought has full access to, if not fusion with, the real, amounting to the equation truth = real (and vice versa). This, according to Laruelle, is what constitutes the grounding fallacy of all philosophy.[23] In the opening pages of *Philosophy and Non-Philosophy* ([1989]; 2013), Laruelle terms said fallacy the amphibology of real and thought[24] whose principle *par excellence* is the principle of all European philosophy –the idea of *being* or tò ón (τὸ ὄν) as the unity of truth and reality –also termed as the principle of sufficient philosophy (PSP).[25] Marx admires Epicurus for not falling into the trap of the same principle that Laruelle identifies as the self-defeating, autophagist foundation of philosophy –dismissing the world of ‘appearances’ as phenomena deceiving instead of affirming that its materiality is the

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此认同的拉鲁埃尔和马克思的分析——所论证的那样）的“显现”或“欺骗性现象”，正是马克思所写的“感性世界”，也是真正的唯物主义科学——或者我们补充说，是摆脱了“PSP”的哲学——应当立足的唯一基础。“客观表象”是“自身之为自身”的实在的一个实例；它与社会关系或价值观一样客观，正如阿尔弗雷德·索恩-雷特尔在唯物主义认识论细节中阐述的那样，并以“现实抽象”的概念为例。[26]在该段结尾，马克思得出结论：支撑伊壁鸠鲁哲学的矛盾中存在着一种荒谬性。这些矛盾是可以解释的，但正如马克思在其关于费尔巴哈、德国意识形态及其预期的著作中所阐明的，其根源在于所有哲学中潜藏的狡猾的唯心主义。如前所述，形式或概念与原子的属性——或者简而言之，一切事物的无形物质基础和“始原”（ἀρχή）的属性——构成了辩证的连续性，这种辩证法表现为偶然者（底本作continent）或“偶然”的永恒递归，而非等同于超越和黑格尔式辩证法理解的扬弃（das Aufgehobene）。

〔校注：英文底本末句作 continent，按语境疑为 contingent（偶然者），中文加括注注明底本用词。底本以 sublimation 对应 das Aufgehobene，中文按其黑格尔语境译“扬弃”；此处不是地理上的大陆或精神分析的升华。〕

only blueprint of the invisible foundation (of the atoms) and perhaps also the necessity of their realization. The ‘apparition’ or the ‘deceiving phenomena’ of the visible world (as Parmenides, Plato, Berkeley, Hegel, Judith Butler and, in one way or another, all philosophy has argued, at least as per the analysis of Laruelle and Marx we subscribe to here) are the ‘sensuous world’ Marx writes about as the only foundation a true materialist science – or a philosophy rid of the PSP, we would add – should ground itself in. ‘Objective appearance’ is an instance of the real in and of itself; it is as objective as social relations or values, as elaborated in materialist epistemological detail by Alfred Sohn-Rethel, exemplified in the notion of ‘real abstraction’.[26] At the end of the paragraph, Marx concludes that there is an absurdity in the contradictions underpinning Epicurus’ philosophy. These contradictions can be explained, but through the insidious idealism in all philosophy, as Marx explains in his writings on Feuerbach and the German ideology and its expectations. As already noted, form or concept and the atom’s properties – or simply the properties of the invisible material foundation and *archē* (ἀρχή) of everything – constitute a dialectical continuity, a dialectics that works as a perpetual recursion of the continent or ‘chance’ rather than a sublimation (das Aufgehobene) amounting to transcendence and the Hegelian understanding of dialectics.

辩证唯物主义（diamat）的基础及其矛盾

The foundations of Diamat and their contradictions

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在马克思主义发展一个多世纪、且辩证唯物主义（或称“diamat”教条）始终占据核心地位之后[27]，要证明“辩证法与唯物主义”这一复合概念并非马克思主义在其原始形态——即马克思的形态——中的基石，并且马克思将辩证法与唯物主义视为两个独立范畴，二者并不必然且不可避免地构成统一体，这无疑是一项巨大挑战。他那种从抽象（实际上是一种模糊的哲学概括）出发，抵达具体，进而推导出另一种（由具体所决定的）新抽象形式的方法，既不允许方法上的混淆与模棱两可，也不允许对现实的本体论基础——即对辩证法和方法各自的本体论基础——作出主张。我们认为，这种偏离马克思初衷的奇怪迂回，其后续发展很大程度上可能源于“辩证唯物主义”（diamat）的遗产——无论是共产国际所确立的，还是列宁去世后作为苏联官方党的学说所确立的。

After more than a century of Marxism and the centrality of the notion of dialectical materialism (or the dogma of Diamat),[27] it is quite a challenge to demonstrate that the compound notion of dialectics and materialism was not the cornerstone of Marxism in its original form, that is, Marx’s form, and that he viewed dialectics and materialism as distinct categories not necessarily and unavoidably constituting a unity. His method of moving away from an abstraction, which is in fact a vague philosophical generalization, to arrive at the concrete, only to extrapolate another new form of abstraction (determined by the concrete), would not allow for a mixture and an amphibology of method, nor a claim about an ontological foundation of reality, that is, of dialectics and method, respectively. Much of what followed in this strange detour from what we argue was Marx’s original intention is probably indebted to the legacy of Diamat (dialectical materialism), both as consolidated by the Comintern and as the official party doctrine of the USSR consolidated after Lenin’s death.

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在《唯物主义和经验批判主义》[28]中，列宁赞扬约瑟夫·狄慈根创造了“辩证唯物主义”这一概念，从而产生了一种真正的唯物主义——一种通过辩证法方法实践的，或称为唯物主义辩证法——这被视为对马克思和恩格斯原始学说的重要补充。[29]在列宁撰写的这部论著中，有一点非常明确——马克思并非被呈现为发展了辩证唯物主义的思想家，而是

In *Materialism and Empirio-criticism*,[28] Lenin praises Joseph Dietzgen for coining the notion of ‘dialectical materialism’, thus having produced a proper materialism, one practiced by way of the method of dialectics, or as materialist dialectics, to be considered as an important addition to Marx’s and Engel’s original doctrine.[29] In this treatise penned by Lenin, we see one thing

被描绘为为狄慈根发展该理论奠定基础的人。然而，列宁对“辩证唯物主义”方法的运用，并非以构建一种古典的、斯大林主义式的“辩证唯物主义”本体论为中心，而是旨在驳斥主观主义（因其本质上属于唯心主义），并捍卫马克思主义作为毫不妥协的唯物主义。“正题、反题、合题”这一三元结构，以及由此实现的唯物主义与唯心主义的调和，既非列宁关注的焦点，也非他在这部著作中所追求的目标：

very clearly –Marx is not presented as the thinker who developed dialectical materialism but rather as the one who offered grounds for it to be developed by Dietzgen. However, Lenin’s use of the method of ‘dialectical materialism’ is not centred on constituting an ontology of the classical, Stalinist Diamat type but rather serves to refute subjectivism as inherently idealist and proffer a defense of Marxism as uncompromising materialism. The triad of thesis, antithesis and synthesis, and thus the reconciliation of materialism and idealism, is neither Lenin’s object of interest nor his objective in the title at issue:

贯穿所有马赫主义者所有著作的一条红线，就是他们那种愚蠢的论调，声称自己已经“超越”了唯物主义和唯心主义，已经超越了这种“过时”的对立；但事实上，这整个团体正在不断滑向唯心主义，并且对唯物主义进行着持续不断的斗争。[30]

A red thread that runs through all the writings of all the Machists is the stupid claim to have ‘risen above’ materialism and idealism, to have transcended this ‘obsolete’ antithesis; but in fact this whole fraternity is continually sliding into idealism and it conducts a steady and incessant struggle against materialism.[30]

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列宁的《唯物主义和经验批判主义》首先是一部对唯物主义的坚定捍卫之作，而很难说——即使勉强可以这样说——是对当时仍属新概念的“辩证唯物主义”的捍卫，因为马克思及其亲密圈子从未真正接受过这一概念。

Lenin’s *Materialism and Empirio-criticism* is first and foremost a stark defense of materialism, and it is hardly –if at all –a defense of the notion, then still novel, called ‘dialectical materialism’, which was never really accepted by Marx and his immediate circle.

埃瓦尔德·伊连科夫对马克思和列宁著作的解读证明，马克思——以及列宁——唯物主义的核心在于构建一种知识型（episteme），它既允许以具体为出发点进行思考，又构成了客观性思想的基础。

Evald Ilyenkov’s reading of both Marx and Lenin proves that the kernel of Marx’s –and, for that matter, Lenin’s –materialism is the creation of an episteme allowing for thinking in terms of the concrete and constituting thought of objectivity.

马克思对“具体”的定义中最重要的一点在于：具体首先被视为事物的客观特征，这一特征完全独立于认知主体可能发生的任何演变。[……] 具体性并非在主体反映对象的过程中产生的，无论是在感性阶段还是在理性—逻辑阶段。换言之，“具体”首先与唯物辩证法的任何其他范畴——如“必然”与“偶然”、“本质”与“现象”——同属一类客观范畴。它表达了自然、社会和思维的一种普遍发展形式。在马克思的理论体系中，“具体”绝非感性给定、可直接观照之物的同义词。[31]

The most important aspect of Marx’s definition of the concrete is that the concrete is treated first of all as an objective characteristic of a thing considered quite independently from any evolutions that may take place in the cognising subject. [. . .] Concreteness is not created in the process of reflection of the object by the subject either at the sensual stage of reflection or at the rational-logical one. In other words, ‘the concrete’ is first of all the same kind of objective category as any other category of materialist dialectics, as ‘the necessary’ and ‘the accidental’, ‘essence, and ‘appearance’. It expresses a universal form of development of nature, society, and thinking. In the system of Marx’s views, ‘the concrete’ is by no means a synonym for the sensually given, immediately contemplated.[31]

因此，在伊连科夫那里，“具体”是一种概念化形式，它由物质所决定并客观地塑造，是通过按照马克思的方法采取第三方的视角来实现的，而其“归根到底的同一性”[32] 可以是一种“抽象”，类似于索恩-雷特尔的“现实抽象”概念（如上所述）。换言之，用拉鲁埃尔的术语来说，其物质可以具有先验的性质，类似于拉鲁埃尔所说的先验或哲学上的“物质”[33]，而其效应则是真实的。这些效应不仅是真实的，而且还由一组物质决定因素所支撑，这些因素奠定了意义结构的基础，这类似于维特根斯坦所说的 Maßstab，即英语中的尺度（“实在的尺度”）[34]，或者类似于拉鲁埃尔在《非马克思主义导论》中提出的“实在的句法”概念。[35]这就是伊连科夫在上述引文中提到的那种辩证法，其中认知与被认知的具体对象之间的鸿沟保持着生产性矛盾的地位，并蕴含着一种扬弃的运动——这种运动应按照马克思博士论文中的理解来理解，而不是将矛盾实例（通过统一）进行救赎式的调和。换言之，具体事物应当

In Ilyenkov, concrete is, therefore, a form of conceptualization, one determined by the material and shaped objectively, which is achieved through assuming a third party’s perspective as per Marx’s method, while its ‘identity of the last instance’ [32] can be an ‘abstraction’, akin to Sohn-Rethel’s notion of real abstraction (discussed above). Put differently, in Laruellian terms, its material can be of transcendental nature, similar to the notion of Laruelle’s transcendental or philosophical ‘material’,[33] whereas its effects are real. Not only are they real, they are also underpinned by a compound of material determinants grounding a structure of meaning similar to what Wittgenstein would call Maßstab, or, in English, scale (‘of the real’),[34] or to the notion of ‘syntax of the real’ as developed by Laruelle in *Introduction to Non-Marxism*.[35] This is the type of dialectics Ilyenkov refers to in the passage cited above, whereby the gap between cognition and the concrete object of cognizing re-

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保持其具体性，而非经历自我超越，最终升华为某种形式的理想性——即使这种理想性被包装成唯物主义，例如狄慈根的综合理论中所呈现的那样。[36]伊连科夫的以下段落进一步印证了上述比较性解读：

tains its status of a productive contradiction and implies a movement of sublation understood as it is in Marx’s doctoral dissertation rather than as a redeeming reconciliation (through unity) of contradicting instances. In other words, the concrete ought to remain concrete instead of undergoing self-transcendence culminating in some form of ideality, even if presented as materialist, such as in Dietzgen’s synthesis.[36] The following passage from Ilyenkov offers further corroboration of the comparative interpretation just presented:

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这种对“抽象”一词的用法绝非马克思的术语任性：它与马克思逻辑观点的本质密切相关，与对思维形式与客观现实形式之间关系的辩证诠释相关，也与将实践（涉及客体的感性活动）视为思维抽象真理标准的观点相关。更不能将这种用法解释为“倒退回黑格尔主义”：马克思的论断正是针对黑格尔提出的，即在“最简单的经济范畴，例如交换价值……只能作为已经存在的具体有机整体的一种抽象的、单方面的关系而存在”。[37]

This use of the term‘abstract’is not a terminological whim of Marx’s at all: it is linked with the very essence of his logical views, with the dialectical interpretation of the relation of forms of thinking and those of objective reality, with the view of practice (sensual activity involving objects) as a criterion of the truth of the abstractions of thought. Still less can this usage be explained as ‘a throwback to Hegelianism’: it is against Hegel that Marx’s proposition is directed to the effect that ‘the simplest economic category, eg exchange value . . . cannot exist except as an abstract, unilateral relation of an already existing concrete organic whole’.[37]

唯物主义者所思考的范畴，根据定义且不可避免地是“抽象”的，但这种抽象既不能取代现实，现实与作为抽象（根据定义）的思想，也不构成现实与真理的统一体——即对“存在”（无论是物质的还是理想的）的哲学实现或实际化。正如伊连科夫和列宁[38]所阐明的，具体性并非感性的，而是概念性的。另一方面，抽象性并非理想的，而是人类认知的产物，一种先验层面的实体，它能够相对于具体性——以及通过具体性相对于现实——被单方面定位，因为具体性最接近上文提到的“现实尺度”。在伊连科夫的分析中，我们可以看到这种单边二元性在起作用，这与我们在马克思本人的著作中——更具体地说，是在他的博士论文中——所指出的，以及我们在上文讨论的拉鲁埃尔的方法中观察到的，是同一种二元性。

The categories in which materialists think are, by definition and unavoidably,‘abstract’, but that abstraction is neither substituted for the real nor do they, the real and thought insofar as abstraction (by definition), constitute a unity of reality and truth –a philosophical realization or actualization of the Being, be it material or ideal. The concrete is not sensual, as both Ilyenkov and Lenin[38] explain, but conceptual. On the other hand, the abstract is not ideal but rather a human product of cognition, an entity of the transcendental order which can be unilaterally positioned vis-à-vis the concrete, and through it the real as the concrete is the closest to the ‘scale of reality’ mentioned above. In Ilyenkov’s analysis, we can see the unilateral duality at work, which is the same one we already noted in Marx’s own writings, his dissertation more specifically, as we did in Laruelle’s method too, discussed above.

总而言之，我们想指出这样一个事实：在后结构主义知识范式崛起的时代，我们目睹了正是索恩-雷特尔（伊连科夫亦然）曾试图推翻的逻辑仍在延续：无需讨论具体作者，我们完全可以断言，体力劳动（物质的）与脑力劳动（非物质的）之间的旧有划分不仅得以强化，而且后者在解放潜力方面的优越性这一隐含观点也随之强化。我们指的是阿尔贝托·托斯卡诺（Alberto Toscano）在论文《现实抽象的公开秘密》中讨论的索恩-雷特尔的接受史，该文主要聚焦于保罗·维尔诺（Paolo Virno）和洛伦佐·西利亚里奥（Lorenzo Cillario）。[39]“一般智力”似乎正在超越资本主义的价值形式，而实践则因将具体性还原为感性、物理和感官层面而被摒弃，这犯了伊连科夫所警告的谬误，如上所述。索恩-雷特尔的研究项目与伊连科夫的类似，其目标是准确解读马克思理论体系的认识论基础，并在此基础上进行富有成效的拓展。在托斯卡诺引用的致阿多诺的一封信中，索恩-雷特尔写道：

In conclusion, we would like to point out to the fact that, in the era of the rise of the poststructuralist episteme, we witness a perpetuation of the very logic that Sohn-Rethel (and similarly, Ilyenkov too) sought to defeat: without discussing individual authors, we can safely argue that the old division between manual (the physical) and intellectual (immaterial) labour was reinforced, as well as the implication of the superiority of the latter in terms of its emancipatory potential. We are referring to the reception of Sohn-Rethel discussed in Alberto Toscano’s paper, *The Open Secret of the Real Abstraction*, focusing mainly on Paolo Virno and Lorenzo Cillario.[39]‘General intellect’seems to be transcending the value form of capitalism, whereas praxis is dismissed by reducing the concrete to the sensuous, physical and sensory, committing the fallacy Ilyenkov warns against, as presented above. The objective of Sohn-Rethel’s project, similarly to Ilyenkov’s, was to convey an accurate reading, and, based on that, a productive expansion,

逻辑的拜物教概念与价值的拜物教概念在社会指涉上有所不同。后者指向资本与劳动之间的对立，前者则指向脑力劳动与体力劳动之间的对立。[40]

正如索恩-雷特尔所言，二者是在发生学意义上相互关联的[41]，而在如此论证时，他始终保持着对马克思的正统忠诚：交换价值形式被确认为一种抽象，尽管将其视为一种决定性的现实（例如，类似于“社会关系”的概念）。所说的这种抽象并非通过任何形式的（一般）“智力”被客体化或实现而成为现实。这种推理将暗示物化谬误。使用价值与剩余价值之间的辩证关系是资本主义的核心；它决定了雇佣劳动作为商品化的存在。事实上，它决定了每一件商品，而其自我加速的逻辑（M–M′ 取代了 M–C–M）表明，它最终将演变为超越商品或任何物质形式的纯粹价值交换（想想投机经济）。[42]这便是资本主义宇宙最终层面的决定性因素。经济生产正是以此方式决定着社会中的万事万物——它是社会关系宇宙中跳动的心脏，在从艺术、科学到亲属关系与亲密关系的组织等所有形式中，不断再现着同一套主体化模式。

〔校注：本段底本确实写“使用价值与剩余价值”，并写 M-M′ 取代 M-C-M；这里按底本保留，不以通常的资本循环公式替换作者表述。〕

of the epistemological foundation of Marx’s project. In a letter to Adorno, cited by Toscano, Sohn-Rethel writes:

‘fetish-concept of logic has a different social referent with regard to the fetish-concept of value. The latter refers to the antagonism between capital and labor, the former to the antithesis between intellectual and manual labor.[40]

p. 70

The two are connected in a ‘genetic’ sense, as Sohn-Rethel puts it,[41] and in arguing so, he retains orthodox fidelity to Marx: the exchange value form is affirmed as abstraction, albeit treating it as a determining reality (akin to the notion of ‘social relations’, for example). The abstraction in question does not become real through any kind of (general) ‘Intellect’ being objectified or realized. Such reasoning would imply the fallacy of reification. The dialectics between use and surplus value is at the core of capitalism; it determines wage labour as commodified. It determines every commodity for that matter, and its very logic of auto-acceleration (M-M’ superceding M-C-M) demonstrates that it culminates in the exchange of pure value that transcends commodity or anything material (think of speculative economy).[42] This would be the determination of the last instance of capitalist universe. It is in that way indeed that economic production determines everything in society – it is the beating heart of a universe of social relations that recreates the same model of subjectivation in all forms, ranging from art and science to the organization of kinship and intimacy.

第 4 章原注 / Chapter 4 Notes

p. 240 ↑1. 亚里士多德：《修辞学》（Ars Rhetorica），W. D. Ross 编（Clarendon Press，1959），1:1。

1 Aristotle, *Ars Rhetorica,* ed. W. D. Ross (Clarendon Press, 1959), 1: 1.

p. 240 ↑2. Matthewe Duncombe、Catarina Dutilh Novaes：“亚里士多德及其传统中的辩证法与逻辑”（Dialectic and Logic in Aristotle and his Tradition），History and Philosophy of Logic，第 37 卷第 1 期（2015），第 1—8 页。

2 Matthewe Duncombe and Catarina Dutilh Novaes, ‘Dialectic and Logic in Aristotle and his Tradition’, *History and Philosophy of Logic* 37, no. 1 (2015): 1–8.

p. 240 ↑3. 亚里士多德：《修辞学》，W. Rhys Roberts 译（Bexar County, TX：Bibliotech Press，2012），引自 Internet Classics Archive，`http://classics.mit.edu/Aristotle/rhetoric.1.i.html`（访问日期：2021 年 11 月 27 日），第 1 卷第 1 部分。

3 Aristotle, *Rhetoric*, trans. W. Rhys Roberts (Bexar County, TX: Bibliotech Press, 2012), retrieved from *The Internet Classics Archive*, `http://classics.mit.edu/Aristotle/rhetoric.1.i.html` (accessed on 27 November 2021), Book 1: Part 1.

p. 240 ↑4. 下文将更详细地讨论这里的区分，并提供英文译文及原文引证。

4 The distinction at issue is discussed in more detail below, where citations are provided, both in English translation and the original.

p. 240 ↑5. 出自马克思博士论文关于矛盾、形式的对象化和关系的对象化的论述：“原子通过性质获得了一种与其概念相矛盾的存在；它被设定为不同于自身本质的、外化了的存在。伊壁鸠鲁主要关心的正是这一矛盾。因此，他一旦设定一种属性，从而得出原子的物质本性的后果，就同时与之相对地设定一些规定，这些规定又在该属性自身范围内破坏它，转而使原子的概念得到确认。因此，他以使各种属性自相矛盾的方式规定它们。相反，德谟克利特从不联系原子本身来考察属性，也不把其中固有的概念与存在之间的矛盾对象化。”引自卡尔·马克思：《德谟克利特的自然哲学和伊壁鸠鲁的自然哲学的差别》，《马克思恩格斯全集》英文版第 1 卷（Progress Publishers，1975），无页码，引自 Karl Marx Internet Archive，`https://www.marxists.org/archive/marx/works/1841/dr-theses/index.htm`（访问日期：2021 年 11 月 27 日）。

5 From Marx’s dissertation on contradiction, objectification of form and objectification of relation: ‘Through the qualities the atom acquires an existence which contradicts its concept; it is assumed as an externalised being different from its essence. It is this contradiction which mainly interests Epicurus. Hence, as soon as he posits a property and thus draws the consequence of the material nature of the atom, he counterposits at the same time determinations which again destroy this property in its own sphere and validate instead the concept of the atom. He therefore determines all properties in such a way that they contradict themselves. Democritus, on the other hand, nowhere considers the properties in relation to the atom itself, nor does he objectify the contradiction between concept and existence which is inherent in them’. Citation from Karl Marx, *The Difference Between the Democritean and Epicurean Philosophy of Nature*, Marx-Engels Collected Works Volume 1 (Progress Publishers, 1975), s.p., retrieved from the Karl Marx Internet Archive, `https://www.marxists.org/archive/marx/works/1841/dr-theses/index.htm` (accessed on 27 November 2021).

p. 240 ↑6. 这里我们采用弗朗索瓦·拉鲁埃尔阐发的单边化（unilateralization）或双重分析（dualysis）方法。它以拆解哲学“世界”为前提：哲学按定义设定实在，并把这一设定当作实在本身。这涉及歧义（amphibology）与“决断主义”的问题。拉鲁埃尔在许多著作中将其问题化，尤其在《哲学与非哲学》（Philosophy and Non-philosophy），Taylor Adkins 译（明尼苏达大学出版社，2013）的开篇数章中作了更细致的讨论。

6 We will resort here to the method of unilateralization or dualysis elaborated by François Laruelle which is premised on dissembling philosophical ‘worlds’–per definition postulating the real and treating that postulate as the real itself, the problem of amphibology and ‘decisionism’, Laruelle problematizes in many of his works, in particular more closely in the opening chapters of his Philosophy and Non-philosophy, trans. Taylor Adkins (University of Minnesota Press, 2013).

p. 240 ↑7. 路德维希·维特根斯坦：《逻辑哲学论》（Tractatus Logico-Philosophicus）（Kegan Paul, Trench, Trubner & Co., Ltd.，1922），Gutenberg Project（访问日期：2021 年 11 月 27 日），2.150、2.511、2.1512 及其他多处。

7 Ludwig Wittgenstein, *Tractatus Logico-Philosophicus* (Kegan Paul, Trench, Trubner & Co., Ltd., 1922), Gutenberg project (accessed on 27 November 2021): 2.150, 2.511, 2.1512, et passim.

p. 241 ↑8. 亚里士多德：《修辞学》（Ars Rhetorica），1:1，第 14 节，`http://www.perseus.tufts.edu/hopp`

8 Aristotle, *Ars Rhetorica*, 1:1, section 14, `http://www.perseus.tufts.edu/hopper/text?doc=Pe`

er/text?doc=Perseus%3Atext%3A1999.01.0059%3Abook%3D1%3Achapter%3D1%3Asection%3D14（访问日期：2021 年 11 月 27 日）。

[↑]9. W. Rhys Roberts 的译文引自 MIT 的 Internet Classics Archive，http://classics.mit.edu/Aristotle/rhetoric.1.i.html（访问日期：2021 年 11 月 27 日）。

[↑]10. 引文出自 W. Rhys Roberts 的译本；其希腊原文为：ἐπεὶ δ᾽ αἱ πίστεις διὰ τούτων εἰσί, φανερὸν ὅτι ταύτας ἐστὶ λαβεῖν τοῦ συλλογίσασθαι δυναμένου καὶ τοῦ θεωρῆσαι περὶ τὰ ἤθη καὶ περὶ τὰς ἀρετὰς καὶ τρίτον τοῦ περὶ τὰ πάθη, τί τε ἕκαστόν ἐστιν τῶν παθῶν καὶ ποῖόν τι, καὶ ἐκ τίνων ἐγγίνεται καὶ πῶς, ὥστε συμβαίνει τὴν ῥητορικὴν οἷον παραφυές τι τῆς διαλεκτικῆς εἶναι καὶ τῆς περὶ τὰ ἤθη πραγματείας, ἣν δίκαιόν ἐστι προσαγορεύειν πολιτικήν. διὸ καὶ ὑποδύεται ὑπὸ τὸ σχῆμα τὸ τῆς πολιτικῆς ἡ ῥητορικὴ καὶ οἱ ἀντιποιούμενοι ταύτης τὰ μὲν δι᾽ ἀπαιδευσίαν, τὰ δὲ δι᾽ ἀλαζονείαν, τὰ δὲ καὶ δι᾽ ἄλλας αἰτίας ἀνθρωπικάς: ἔστι γὰρ μόριόν τι τῆς διαλεκτικῆς καὶ ὁμοίωμα, καθάπερ καὶ ἀρχόμενοι εἴπομεν: περὶ οὐδενὸς γὰρ ὡρισμένου οὐδετέρα αὐτῶν ἐστιν ἐπιστήμη πῶς ἔχει, ἀλλὰ δυνάμεις τινὲς τοῦ πορίσαι λόγους. 希腊文出处：亚里士多德《修辞学》（Ars Rhetorica），引自 Perseus Tufts 网络文献库，http://www.perseus.tufts.edu/hopper/text?doc=Perseus%3Atext%3A1999.01.0059（访问日期：2021 年 11 月 21 日），1:2，第 7 节。

[↑]11. 引文出自 W. Rhys Roberts 的译本，引自 MIT 的 Internet Classics Archive，http://classics.mit.edu/Aristotle/rhetoric.1.i.html（访问日期：2021 年 11 月 21 日）；其希腊原文为：ὅτι μὲν οὖν οὐκ ἔστιν οὐθενός τινος γένους ἀφωρισμένου ἡ ῥητορική, ἀλλὰ καθάπερ ἡ διαλεκτική, καὶ ὅτι χρήσιμος, φανερόν, καὶ ὅτι οὐ τὸ πεῖσαι ἔργον αὐτῆς, ἀλλὰ τὸ ἰδεῖν τὰ ὑπάρχοντα πιθανὰ περὶ ἕκαστον, καθάπερ καὶ ἐν ταῖς ἄλλαις τέχναις πάσαις （οὐδὲ γὰρ ἰατρικῆς τὸ ὑγιᾶ ποιῆσαι, ἀλλὰ μέχρι οὗ ἐνδέχεται, μέχρι τούτου προαγαγεῖν: ἔστιν γὰρ καὶ τοὺς ἀδυνάτους μεταλαβεῖν ὑγιείας ὅμως θεραπεῦσαι καλῶς): πρὸς δὲ τούτοις ὅτι τῆς αὐτῆς τό τε πιθανὸν καὶ τὸ φαινόμενον ἰδεῖν πιθανόν, ὥσπερ καὶ ἐπὶ τῆς διαλεκτικῆς συλλογισμόν τε καὶ φαινόμενον συλλογισμόν: ἡ γὰρ σοφιστικὴ οὐκ ἐν τῇ δυνάμει ἀλλ᾽ ἐν τῇ προαιρέσει: πλὴν ἐνταῦθα μὲν ἔσται ὁ μὲν κατὰ τὴν ἐπιστήμην ὁ δὲ κατὰ τὴν προαίρεσιν ῥήτωρ, ἐκεῖ δὲ σοφιστὴς μὲν κατὰ τὴν προαίρεσιν, διαλεκτικὸς δὲ οὐ κατὰ τὴν προαίρεσιν ἀλλὰ κατὰ τὴν δύναμιν. 希腊文出处：亚里士多德《修辞学》（Ars Rhetorica），引自 Perseus Tufts 网络文献库，http://www.perseus.tufts.edu/hopper/text?doc=Perseus%3Atext%3A1999.01.0059（访问日期：2021 年 11 月 21 日），1:1，第 14 节。

[↑]12. 费迪南·德·索绪尔：《普通语言学教程》（Course in General Linguistics），Wade Baskin 译，Charles Bally 与 Albert Reidlinger 编（Philosophical Library，1959），第 53 页。

rseus%3Atext%3A1999.01.0059%3Abook%3D1%3Achapter%3D1%3Asection%3D14 (accessed on 27 November, 2021).

9 W. Rhys Roberts’ translation is retrieved from the Internet Classics Archive of MIT at http://classics.mit.edu/Aristotle/rhetoric.1.i.html (accessed on 27 November 2021). p. 241

10 The quote is from W. Rhys Roberts’ translation; in its Greek original, it reads:ἐπεὶ δ᾽ αἱ πίστεις διὰ τούτων εἰσί, φανερὸν ὅτι ταύτας ἐστὶ λαβεῖν τοῦ συλλογίσασθαι δυναμένου καὶ τοῦ θεωρῆσαι περὶ τὰ ἤθη καὶ περὶ τὰς ἀρετὰς καὶ τρίτον τοῦ περὶ τὰ πάθη, τί τε ἕκαστόν ἐστιν τῶν παθῶν καὶ ποῖόν τι, καὶ ἐκ τίνων ἐγγίνεται καὶ πῶς, ὥστε συμβαίνει τὴν ῥητορικὴν οἷον παραφυές τι τῆς διαλεκτικῆς εἶναι καὶ τῆς περὶ τὰ ἤθη πραγματείας, ἣν δίκαιόν ἐστι προσαγορεύειν πολιτικήν. διὸ καὶ ὑποδύεται ὑπὸ τὸ σχῆμα τὸ τῆς πολιτικῆς ἡ ῥητορικὴ καὶ οἱ ἀντιποιούμενοι ταύτης τὰ μὲν δι᾽ ἀπαιδευσίαν, τὰ δὲ δι᾽ ἀλαζονείαν, τὰ δὲ καὶ δι᾽ ἄλλας αἰτίας ἀνθρωπικάς: ἔστι γὰρ μόριόν τι τῆς διαλεκτικῆς καὶ ὁμοίωμα, καθάπερ καὶ ἀρχόμενοι εἴπομεν: περὶ οὐδενὸς γὰρ ὡρισμένου οὐδετέρα αὐτῶν ἐστιν ἐπιστήμη πῶς ἔχει, ἀλλὰ δυνάμεις τινὲς τοῦ πορίσαι λόγους.The source in Greek is: Aristotle, *Ars Rhetorica,* and it is retrieved from the Perseus Tufts internet archive available at http://www.perseus.tufts.edu/hopper/text?doc=Perseus%3Atext%3A1999.01.0059 (accessed on 21 November 2021), 1: 2, section 7. p. 241

11 The quote is from W. Rhys Roberts’ translation, retrieved from the Internet Classics Archive of MIT, available at http://classics.mit.edu/Aristotle/rhetoric.1.i.html (accessed on 21 November 2021); in its Greek original, it reads:ὅτι μὲν οὖν οὐκ ἔστιν οὐθενός τινος γένους ἀφωρισμένου ἡ ῥητορική, ἀλλὰ καθάπερ ἡ διαλεκτική, καὶ ὅτι χρήσιμος, φανερόν, καὶ ὅτι οὐ τὸ πεῖσαι ἔργον αὐτῆς, ἀλλὰ τὸ ἰδεῖν τὰ ὑπάρχοντα πιθανὰ περὶ ἕκαστον, καθάπερ καὶ ἐν ταῖς ἄλλαις τέχναις πάσαις （οὐδὲ γὰρ ἰατρικῆς τὸ ὑγιᾶ ποιῆσαι, ἀλλὰ μέχρι οὗ ἐνδέχεται, μέχρι τούτου προαγαγεῖν: ἔστιν γὰρ καὶ τοὺς ἀδυνάτους μεταλαβεῖν ὑγιείας ὅμως θεραπεῦσαι καλῶς): πρὸς δὲ τούτοις ὅτι τῆς αὐτῆς τό τε πιθανὸν καὶ τὸ φαινόμενον ἰδεῖν πιθανόν, ὥσπερ καὶ ἐπὶ τῆς διαλεκτικῆς συλλογισμόν τε καὶ φαινόμενον συλλογισμόν: ἡ γὰρ σοφιστικὴ οὐκ ἐν τῇ δυνάμει ἀλλ᾽ ἐν τῇ προαιρέσει: πλὴν ἐνταῦθα μὲν ἔσται ὁ μὲν κατὰ τὴν ἐπιστήμην ὁ δὲ κατὰ τὴν προαίρεσιν ῥήτωρ, ἐκεῖ δὲ σοφιστὴς μὲν κατὰ τὴν προαίρεσιν, διαλεκτικὸς δὲ οὐ κατὰ τὴν προαίρεσιν ἀλλὰ κατὰ τὴν δύναμιν. The source in Greek is: Aristotle, *Ars Rhetorica*, and it is retrieved from the Perseus Tufts internet archive available at http://www.perseus.tufts.edu/hopper/text?doc=Perseus%3Atext%3A1999.01.0059 (accessed on 21 November 2021), 1: 1, section 14. p. 241

12 Ferdinand de Saussure, *Course in General Linguistics*, trans. Wade Baskin, ed. Charles Bally and Albert Reidlinger (Philosophical Library, 1959), 53. p. 241

p. 241 ↑13. 卡尔•马克思:《政治经济学批判大纲》(Grundrisse: Outlines of the Critique of Political Economy),Martin Nicolaus 译(Penguin Books,1973),第 41 页。

13 Karl Marx, *Grundrisse: Outlines of the Critique of Political Economy*, trans. Martin Nicolaus (Penguin Books, 1973), 41.

p. 241 ↑14. 拉鲁埃尔:《哲学与非哲学》(Philosophy and Non-Philosophy),第 91—126 页。

14 Laruelle, *Philosophy and Non-Philosophy*, 91–126.

p. 241 ↑15. 许煜(Yuk Hui):《递归性与偶然性》(Recursivity and Contingency)(Rowman & Littlefield International,2019),第 41 页。

15 Yuk Hui, *Recursivity and Contingency* (Rowman & Littlefield International, 2019), 41.

〔校注:正文将题名写作 Contingency and Recursivity,原注作 Recursivity and Contingency,正文与原注各按底本保留。〕

p. 241 ↑16. Paul Cockshott:“图灵:唯物主义闯入思想”(Turing: The Irruption of Materialism into Thought),Oxford University Press's Blog: Academic Insights for the Thinking World(2012),https://blog.oup.com/2012/06/turing-the-irruption-of-materialism-into-thought/(访问日期:2021 年 11 月 27 日)。

16 Paul Cockshott, 'Turing: The Irruption of Materialism into Thought', Oxford University Press's Blog: Academic Insights for the Thinking World (2012), https://blog.oup.com/2012/06/turing-the-irruption-of-materialism-into-thought/ (accessed on 27 November 2021).

p. 241 ↑17. Katerina Kolozova:《走向社会主义的激进形而上学:马克思与拉鲁埃尔》(Toward a Radical Metaphysics of Socialism: Marx and Laruelle)(Punctum Books,2015),第 8—10 页。

17 Katerina Kolozova, *Toward a Radical Metaphysics of Socialism: Marx and Laruelle* (Punctum Books, 2015), 8–10.

p. 241–242 ↑18. 卡尔 • 马克思:《关于费尔巴哈的提纲》,W. Lough 自德文译出(Progress Publishers,1969),无页码,https://www.marxists.org/archive/marx/works/1845/theses/theses.htm(访问日期:2019 年 3 月 22 日)。本引文取自无页码的在线来源。

18 Karl Marx, *Theses on Feuerbach*, translated from the German by W. Lough (Progress Publishers, 1969), s.p., https://www.marxists.org/archive/marx/works/1845/theses/theses.htm (accessed on 22 March 2019). This citation is retrieved from an online source without pagination.

p. 242 ↑19. Thomas Nail:《运动中的马克思》(Marx in Motion)(Oxford University Press,Kindle 版,2020),第 20—21 页。

19 Thomas Nail, *Marx in Motion* (Oxford University Press. Kindle Edition, 2020), 20–1.

p. 242 ↑20. Alfred Sohn-Rethel:《脑力劳动与体力劳动:认识论批判》(Intellectual and Manual Labor: Critique of Epistemology)(伦敦:Macmillan,1978)。

20 Alfred Sohn-Rethel, *Intellectual and Manual Labor: Critique of Epistemology* (London: Macmillan: 1978).

p. 242 ↑21. 卡尔 • 马克思:“德谟克利特的自然哲学和伊壁鸠鲁的自然哲学的差别”,《马克思恩格斯全集》英文版第 1 卷(莫斯科:Progress Publishers,1975),无页码,引自 Karl Marx Internet Archive,https://www.marxists.org/archive/marx/works/1841/dr-theses/index.htm(访问日期:2021 年 11 月 21 日)。在线版本无页码(s.p.);引文出自博士论文第二部分第三章。

21 Karl Marx, 'The Difference Between the Democritean and Epicurean Philosophy of Nature', *Marx-Engels Collected Works* Vol. 1 (Moscow: Progress Publishers, 1975), s.p., retrieved from the Karl Marx Internet Archive, https://www.marxists.org/archive/marx/works/1841/dr-theses/index.htm, accessed on 21 November 2021. The online version is without pagination (s.p.); the quote is from Part 2 the Chapter 3 of the Dissertation.

p. 242 ↑22. 马克思:博士论文,无页码:“德谟克利特的自然哲学和伊壁鸠鲁的自然哲学的同一性所引起的困难”。

22 Marx, *Dissertation*, s.p.: 'Difficulties Concerning the Identity of the Democritean and Epicurean Philosophy of Nature'.

p. 242 ↑23. 拉鲁埃尔:《哲学与非哲学》,第 51—53、227、228 页。

23 Laruelle, *Philosophy and Non-Philosophy*, 51–3, 227, 228.

p. 242 ↑24. 拉鲁埃尔:《哲学与非哲学》,第 10 页。

24 Laruelle, *Philosophy and Non-Philosophy*, 10.

p. 242 ↑25. 拉鲁埃尔:《哲学与非哲学》,第 12—13、20、24—29 页及其他多处。

25 Laruelle, *Philosophy and Non-Philosophy*, 12–13, 20, 24–29 et passim.

p. 242 ↑26. Sohn-Rethel:《脑力劳动与体力劳动》,第 13—35 页。

26 Sohn-Rethel, *Intellectual and Manual Labor*, 13–35.

p. 242 ↑27. 鉴于苏联国家及其机构和执政党,以及通过它们发挥作用的共产国际,曾谨慎筛选并维护对马

27 We dare call it a dogma considering the fact that the USSR as a state, its institutions and the ruling party, and

克思、恩格斯原著的一种特定解读，我们才敢称之为教条。这种解读主要依据列宁《唯物主义和经验批判主义》（初版于 1909 年）。

through that the Comintern, carefully curated a particular reading of Marx’s and Engel’s original writing, mainly based on Lenin’s *Materialism and Empirio-criticism* (originally published in 1909).

↑28. 弗拉基米尔 • 伊里奇 • 列宁：《全集》第 14 卷：《唯物主义和经验批判主义》（Progress Publishers，1977）。

28 Vladimir Ilych Lenin, *Collected Works Vol. 14*: *Materialism and Empirio-criticism* (Progress Publishers, 1977). p. 242

↑29. 列宁：《唯物主义和经验批判主义》，第 117—126 页。

29 Lenin, *Materialism and Empirio-criticism*, 117–26. p. 242

↑30. 列宁：《唯物主义和经验批判主义》，第 341 页。

30 Lenin, *Materialism and Empirio-criticism*, 341. p. 242

↑31. Evald Ilyenkov：《马克思〈资本论〉中抽象与具体的辩证法》（The Dialectics of the Abstract & the Concrete in Marx’s Capital），Sergei Kuzyakov 译（Progress Publishers，1982），`https://www.marxists.org/archive/ilyenkov/works/abstract/index.htm`（访问日期：2022 年 2 月 27 日），第 1 章，无页码。

31 Evald Ilyenkov, *The Dialectics of the Abstract & the Concrete in Marx’s Capital*, trans. Sergei Kuzyakov (Progress Publishers, 1982), `https://www.marxists.org/archive/ilyenkov/works/abstract/index.htm` (accessed on 27 February 2022), Ch. 1, s.p. p. 242

↑32. 弗朗索瓦 • 拉鲁埃尔：《同一性理论》（Theory of Identities），Alyosha Edlebi 译（哥伦比亚大学出版社，2016），第 59、120、135—140 页。

32 François Laruelle, *Theory of Identities*, trans. Alyosha Edlebi (Columbia University Press, 2016), 59, 120, 135–40. p. 242

↑33. 拉鲁埃尔：《哲学与非哲学》，第 22 页及其他多处。

33 Laruelle, *Philosophy and Non-Philosophy*, 22 et passim. p. 242

↑34. 维特根斯坦：《逻辑哲学论》，Frank P. Ramsey 和 Charles Kay Ogden 译，2.1512；或：“施用于实在的尺度”，德文为 Es ist wie ein Maßstab an die Wirklichkeit angelegt（2.1512）。

34 Wittgenstein, *Tractatus Logico-Philosophicus*, trans. Frank P. Ramsey and Charles Kay Ogden, 2.1512; or: ‘scale applied to reality’, in German: Es ist wie ein Maßstab an die Wirklichkeit angelegt (2.1512). p. 242

↑35. 弗朗索瓦 • 拉鲁埃尔：《非马克思主义导论》（Introduction to Non-Marxism），Anthony Paul Smith 译（University of Minnesota Press-Univocal Publishing，2014）。

35 François Laruelle, *Introduction to Non-Marxism*, trans. Anthony Paul Smith (University of Minnesota Press-Univocal Publishing, 2014). p. 242

↑36. Tony Burns：“约瑟夫 • 狄慈根与马克思主义史”（Joseph Dietzgen and the History of Marxism），Science & Society，第 66 卷第 2 期（2002），第 202—227 页。

36 Tony Burns, ‘Joseph Dietzgen and the History of Marxism’, *Science & Society* 66, no. 2 (2002): 202–27. p. 242

↑37. Ilyenkov：《马克思〈资本论〉中抽象与具体的辩证法》，第 1 章，无页码。

37 Ilyenkov, *The Dialectics of the Abstract & the Concrete in Marx’s Capital*, Ch. 1, s.p. p. 242

↑38. 列宁：《唯物主义和经验批判主义》，第 45—46 页；参见：“在这个特定问题上，唯物主义与‘马赫主义’的区别在于：唯物主义完全符合自然科学，把物质看作第一性的，把意识、思想、感觉看作第二性的，因为在明确的形式下，感觉只同物质的较高形式（有机物质）相联系，而‘在物质结构的基础中’，只能推测存在着一种类似于感觉的能力。”（第 46 页）

38 Lenin, *Materialism and Empirio-Criticism*, 45–6; cf.: ‘The difference between materialism and “Machism” in this particular question thus consists in the following. Materialism, in full agreement with natural science, takes matter as primary and regards consciousness, thought, sensation as secondary, because in its well-defined form sensation is associated only with the higher forms of matter (organic matter), while “in the foundation of the structure of matter” one can only surmise the existence of a faculty akin to sensation’ (46). p. 242

↑39. Alberto Toscano：“现实抽象的公开秘密”（The Open Secret of Real Abstraction），Rethinking Marxism，第 20 卷第 2 期（2008），第 282—287 页。

39 Alberto Toscano, ‘The Open Secret of Real Abstraction’, *Rethinking Marxism* 20, no. 2 (2008): 282–7. p. 242

↑40. 索恩-雷特尔致阿多诺的信，转引自 Toscano：“现实抽象的公开秘密”，第 281 页。

40 Sohn-Rethel to Adorno cited in Toscano’s ‘The Open Secret of the Real Abstraction’, 281. p. 243

↑41. Toscano：“现实抽象的公开秘密”，引索恩- p. 243

雷特尔致阿多诺的信：Alfred Sohn-Rethel、Theodor Adorno，Carteggio 1936/1969（manifestolibri，2000），第 112 页。

p. 243

↑42. 卡尔 • 马克思:《资本论》第 1 卷第 2 篇第 4 章，I. Lasker 译（Progress Publishers，1956），https://www.marxists.org/archive/marx/works/1885-c2/（访问日期：2022 年 5 月 12 日）。

〔校注：底本文献写第 1 卷，但所列网址路径含 1885-c2；这里保留书目信息及链接，不擅自更换。〕

41 Toscano, ‘The Open Secret of Real Abstraction’, quoting Sohn-Rethel to Adorno: Alfred Sohn-Rethel and Theodor Adorno, *Carteggio 1936/1969* (manifestolibri, 2000), 112.

42 Karl Marx, *Capital*, vol. I, Part 2: Chapter 4, trans. I. Lasker (Progress Publishers, 1956), https://www.marxists.org/archive/marx/works/1885-c2/ (accessed on 12 May 2022).

第 5 章历史唯物主义者与机械唯物主义者

Chapter 5. HISTORICAL AND MECHANICAL MATERIALISTS

斯密与恩格斯

Smith and Engels

迄今为止我们讨论的唯物主义者关注自然世界。他们描述了一个由原子构成的世界，原子运动受自然规律支配。我们今天称这些规律为经典力学，因此可以把这些思想家视为机械唯物主义者。在苏联文献中，“机械唯物主义”一词往往带有相当负面的含义，被视为一种原始理论，后来被辩证唯物主义和历史唯物主义取代。随着本书展开，我们将论证这些负面含义没有根据。暂且不论这一点，本节关注两位最有影响的历史唯物主义者：亚当·斯密和弗里德里希·恩格斯。

The materialists we have dealt with up to now concerned themselves with the natural world. They described a world made up of atoms whose movements were governed by natural laws. We now call these laws classical mechanics, and we may thus consider these thinkers to have been mechanical materialists. In Soviet literature, the term ‘mechanical materialism’ tended to have a rather negative connotation, as something primitive that was later to be replaced by dialectical and historical materialism. As this book goes on, we will argue that these negative connotations are unjustified. Leaving that aside, this section will concern itself with the ideas of two of the most influential Historical Materialists: Adam Smith and Frederick Engels. p. 71

为什么我们把分别被视为资本主义和共产主义首席倡导者的两位作者归为一类？

Why do we group together two authors who are seen respectively as the arch advocates of capitalism and communism?

因为他们对历史的基本唯物主义方法几乎相同。恩格斯最有影响的著作是《家庭、私有制和国家的起源》[1]，1884 年出版。斯密关于这一主题的主要著作是《法理学讲义》，可追溯到 1760 年代，但直到 1978 年才出版。[2]因此，两位思想家推理中的相似之处是独立发现的结果，尽管恩格斯无疑对斯密的《国富论》有所了解。

It is because their basic materialist approach to history was almost the same. Engel’s most influential book was *The Origin of the Family Private Property and the State*,[1] which was published in 1884. Smith’s main work on the subject is his *Lectures on Jurisprudence*, which date from the 1760s but were not published until 1978.[2] The similarities in the reasoning of the two thinkers were thus the result of independent discovery, though Engels had undoubtedly some familiarity with Smith’s *Wealth of Nations*.

国家源于私有财产

State derives from private property

斯密说，政府因阶级不平等而产生。

Government, Smith said, arises due to class inequality.

斯密把社会划分为四个阶段：狩猎、畜牧、农耕和商业社会。前三个阶段与恩格斯所说的“蒙昧”“野蛮”和“文明”阶段相当吻合。到 1880 年代，恩格斯已经接触到达尔文著作以及至少部分考古学，因此把狩猎社会划分为几个子阶段。第一子阶段，人类仍生活在热带森林中，至少主要还是树栖。第二阶段，石器和火被使用。狩猎采集社会最后阶段，人们已掌握手工织布和弓箭。

Smith conceives of four stages of society: hunting, herding, farming and commercial society. The first three correspond fairly closely to the stages that Engels referred to as savagery, barbarism and civilization. Engels, having access to Darwin and at least some archaeology by the 1880s, divides the hunting stage of society into sub-phases. In the first, man still lived in tropical forests and was primarily at least a tree dweller. In the second phase, stone tools and fire are put to use. In the final phase of hunter-gatherer society, hand weaving and bows and arrows are known.

斯密把从狩猎向畜牧的转变视为第一次重要过渡。恩格斯在旧大陆方面同意，但指出美洲缺乏驯化绵羊和牛，导致东西半球走上不同发展道路。

Smith saw the shift from hunting to herding as the first important transition. Engels agrees with this with respect to the Old World but says that the lack of domesticated sheep and cattle in the Americas led to distinct development paths in the Eastern and Western Hemispheres. p. 72

两人都认为，在生计方式尚无畜牧或农耕的人群中，国家不可能产生。

Both agreed that the state cannot arise among a population whose mode of subsistence lacks herding or farming.

人口极其稀少；仅在部落定居点处较密集，其周围呈大范围圆圈分布着狩猎场，外围则是将该部落与其他部落隔开的中立森林保护带。劳动分工纯粹原始，仅限于两性之间。男子在战争中作战，狩猎捕鱼，获取食物原料及所需工具。妇女负责家务、准备食物和缝制衣物，做饭、织布、缝纫。[3]

The population is extremely sparse; it is dense only at the tribe’s place of settlement, around which lie in a wide circle first the hunting grounds and then the protective belt of neutral forest, which separates the tribe from others. The division of labor is purely primitive, between the sexes only. The man fights in the wars, goes hunting and fishing, procures the raw materials of food and the tools necessary for doing so. The woman looks after the house and the preparation of food and clothing, cooks, weaves, sews.[3]

斯密同样强调人口密度有限的影响。

Smith similarly emphasizes the effect of a limited population density.

在一个以狩猎和捕鱼为生的民族中，很少有人能聚居，因为短时间内任何数量稍多的人群都会耗尽该地区全部猎物，从而失去生存手段。二十到三十个家庭是能聚居的上限，这些家庭共同构成一个村落。但由于他们聚居是为了相互防御和互助，各村落之间距离并不遥远。当不同村庄的人发生争端时，由两个村庄全体大会裁决。正如每个村庄都有自己的首领，整个民族共同体也有一位首领。这一共同体是由不同村庄组成的联盟，各部落首领对决议有重大影响。[4]

In a nation of hunters and fishers few people can live together, for in a short time any considerable number would destroy all the game in the country, and consequently would want a means of subsistence. Twenty or thirty families are the most that can live together, and these make up a village. But as they live together for their mutual defence, and to assist one another, their villages are not far distant from each other. When any controversy happens between persons of different villages, it is decided by a general assembly of both villages. As each particular village has its own leader, so there is one who is the leader of the whole nation. The nation consists of an alliance of the different villages, and the chieftains have great influence on their resolutions[4]

双方都认为，正是向畜牧的转变导致不平等。斯密描述如下：

Both agree that it is the transition to herding that gives rise to inequality. Smith described this process as follows:

在一个狩猎民族中，严格来说根本不存在政府。这个社会由少数独立家庭组成，他们居住在同一个村庄，说着同一种语言，并为了共同安全而商定团结在一起，但彼此之间并无权威。整个社会都会关注任何犯罪行为；如果可能，他们会让当事人和解；如果无法和解，就会将犯罪者驱逐出社会，处死，或交由受害者发泄怨恨。但这并非正规政府，因为尽管其中可能有人备受尊敬并在决策中有很大影响，但未经全体同意，他绝不能做任何事。

In a nation of hunters there is properly no government at all. The society consists of a few independent families who live in the same village and speak the same language, and have agreed among themselves to keep together for their mutual safety, but they have no authority one over another. The whole society interests itself in any offence; if possible they make it up between the parties, if not they banish from their society, kill or deliver up to the resentment of the injured him who has committed the crime. But this is no regular government, for though there may be some among them who are much respected, and have great influence in their determinations, yet he never can do anything without the consent of the whole.

p. 72–73

正是牲畜的私有化导致财富不平等，这也正是正规政府产生的最初原因。在出现私有财产之前，不可能有政府，而政府的最终目的正是保障财富，并保护富人免受穷人侵害。[5]

The appropriation of herds and flocks which introduced an inequality of fortune, was that which first gave rise to regular government . Till there be property there can be no government, the very end of which is to secure wealth, and to defend the rich from the poor.[5]

p. 73

最后一句——政府职能是保障财富并保护富人免受穷人侵害——对历史唯物主义的阐述之清晰，堪比列宁。[6]

That last phrase that the function of government is to secure wealth and defend the rich from the poor is as clear an expression of historical materialism as you would get from Lenin.[6]

斯密把动物驯化与私有财产直接联系起来，他说：“富人已把这些如今已驯服的猎物据为己有。”正是使原本野生的牛群被驯化的过程，消除了它们作为猎人潜在猎物的可能性。斯密在此运用了“原始积累”概念——一个阶级掠夺财富，另一个阶级

Smith directly links the domestication of animals to private property saying: ‘the rich had made the game, now become tame, their own property’. The very process that makes previously wild cattle domesticated eliminates them as potential prey animals for hunters.

因此陷入贫困——这一概念将在马克思对资本主义的分析中再次出现。

Smith is deploying a concept of ‘primitive accumulation’ as simultaneously appropriation by one class and pauperizing another that will recur in Marx’s analysis of capitalism.

根据《讲义》，在畜牧社会中，财富皆可携带，并以牲畜形式积累。与后来能把财富消耗为奢侈品的统治者不同，一个没有定居制造业的社会几乎无法提供奢侈品。因此，酋长的财富表现为对那些自身缺乏足够牲畜的依附者的影响力。穷人完全依赖酋长恩惠。游牧生活与酋长权威结合，导致无休止战争。

According to the *Lectures*, in a herding society, wealth is all portable and accumulates in the form of animals. Unlike later rulers who can consume wealth as luxuries, a society without settled manufactures provides few luxury goods. The wealth of the chiefs thus takes the form of influence over dependents who lack sufficient animals of their own. The poor are fully dependent on the benevolence of the chiefs. The combination of nomadism and chiefly authority generates incessant warfare.

猎人的壮举，尽管勇敢英武，却从未十分惊人。由于他们很少能集体行军，人数极少超过 200 人，即便这 200 人也难以维持超过十四天。因此，一个由猎人组成的民族几乎不会构成威胁。我们的殖民地对他们极为恐惧，却毫无正当理由。他们确实可能通过突袭和远征给殖民地带来一些麻烦，但绝不会构成真正威胁。另一方面，数量庞大的牧民却能聚居生活。同一个村庄里可能有上千户人家。阿拉伯人和鞑靼人——他们历来以牧业为生——曾多次造成极其惨烈的破坏。鞑靼首领极其可怕，当其中一人战胜另一人时，总会引发最可怕、最激烈的动乱。他们会带着全部牛羊群随行出征，凡战败者，既失去部族，也失去财富。获胜一方则追随其畜群，继续征服；如果他们带着如此庞大的人数进入耕作区，那简直无人能挡。穆罕默德正是以这种方式蹂躏了整个亚洲。[7]

The exploits of hunters, though brave and gallant, are never very considerable. As few of them can march together, so their number seldom exceeds 200 men, and even these cannot be supported above fourteen days. There is therefore very little danger from a nation of hunters. Our colonies are much afraid of them without any just grounds. They may indeed give them some trouble by their inroads and excursions, but can never be very formidable. On the other hand a much greater number of shepherds can live together. There may be a thousand families in the same village. The Arabs and Tartars, who have always been shepherds, have on many occasions made the most dreadful havoc. A Tartar chief is extremely formidable, and when one of them gets the better of another, there always happens the most dreadful and violent revolutions. They take their whole flocks and herds into the field along with them, and whoever is overcome loses both his people and wealth. The victorious nation follows its flocks, and pursues its conquest, and if it comes into a cultivated country with such numbers of men, it is quite irresistible. It was in this manner that Mahomet ravaged all Asia.[7]

因此，在畜牧业作为主要财富形式的时代，各酋长国往往倾向于掠夺和无休止小规模战争。在他看来，这不仅符合鞑靼人（蒙古人）和阿拉伯人的历史社会，也符合荷马所描述的希腊英雄社会。他指出，荷马所描述的所有财产纠纷都与牲畜有关。

So during the ages when cattle herding was the predominant form of wealth, you had chiefdoms that were prone to raiding and incessant minor warfare. In his opinion, this corresponded not only to the historic societies of the Tartars (Mongols) and the Arabs but also to the heroic Greek society described by Homer. He notes that all property disputes described by Homer are over cattle.

随着农业地位日益重要，游牧生活变得难以维系，但牛群仍是最重要财富储备。社会现在能生产剩余产品，理论上可与邻近部落交易，但牛群始终面临被邻近部落盗窃的风险。这迫使人们建造设防聚落，以便冲突时期将人和牲畜迁入其中。设防城镇的人口比当地酋长所统治的聚落更多，通常由几个小部落组成。虽然该城镇最初政体由一位酋长居于至高地位，但很快被共和制取代。

As agriculture becomes more significant, nomadism becomes impossible, but the most significant store of wealth is still cattle. Society is now able to produce a surplus, which can potentially be traded with neighbouring tribes, but cattle are constantly subject to the risk of being stolen by neighbouring tribes. This forces the construction of fortified settlements into which the people and beasts can move in times of conflict. A fortified town has a bigger population than those that local chiefs ruled over, typically comprising several small tribes. While the first form of government of the town places one chief in a supreme position it is soon replaced by a republican form of state.

p. 73–74

居住在这样国家的人民，当土地划分完成且耕作普遍开展时，自然会把产品剩余部分分销给邻近部落，这将刺激他们的生产热情。但与此同时，这也可能诱使邻近部落对他们发动侵袭。因此，他们必须想出某种方法来保障自

A people inhabiting such a country, when the division of land came to take place and the cultivation of it to be generally practised, would naturally dispose of the surplus of their product among their neighbours, and this would be a spur to

p. 74

身安全，并守护那些曾耗费巨大心力才获得的成果。在交通便利之处修筑一座城池，比加固整个国家边境要容易得多，因此他们便采用这种方法。他们在最便利的地方修建设防城池，每当遭到入侵时，便带着牲畜和动产躲进城中，并在那里发展工艺和科学。与此相符的是，我们发现忒修斯加固了雅典防御，并命令阿提卡人将所有财物运入城内，这不仅增强了他对民众的控制力，也提升了该城邦相对于其他城邦的权威。当人们以这种方式同意定居城镇后，各氏族酋长很快就会失去权威，政体也会转向共和制，因为他们的收入微薄，无法使自己显得如此显赫、与众不同，以致于能让民众继续依赖他们。公民财富逐渐增加，随着其地位日益接近酋长，便开始嫉妒酋长的权威。因此，我们可以看到，忒修斯本人最终也被赶下了台。

their industry. But at the same time it would be a temptation to their neighbours to make inroads upon them. They must therefore fall upon some method to secure themselves from danger, and to preserve what it formerly cost them so much trouble to procure. It would be more easy to fortify a town in a convenient place than to fortify the frontiers of the whole country, and accordingly this was the method they fell upon. They built fortified towns in the most convenient places, and whenever they were invaded took shelter in them with their flocks and moveable goods, and here they cultivated the arts and sciences. Agreeable to this, we find that Theseus fortified Athens and made the people of Attica carry into it all their goods, which not only increased his power over them, but also the authority of that state above others. When people agreed in this manner to live in towns, the chieftains of the several clans would soon lose their authority, and the government would turn republican, because their revenue was small, and could not make them so conspicuous and distinguished above others as to retain them in dependence. The citizens gradually increase in riches, and coming nearer the level of the chieftain, become jealous of his authority. Accordingly we find that Theseus himself was turned out.

此后，设立了九位摄政官，起初他们拥有终身权力，但后来任期被缩短为十年。就这样，雅典以及所有希腊城邦，都从酋长制演变为类似君主制的政体，进而发展为贵族制。[8]

After this nine regents were set up who were at first to have authority for life, but were afterwards continued only for ten years. Thus Athens, and in like manner all the Greek states, came from a chieftainship to something like monarchy, and from thence to aristocracy.[8]

恩格斯描述了同样过程，尽管他遗漏了斯密指出的某些经济和军事演变。须知，尽管斯密著作问世较早，但恩格斯当时无法接触到这些著作，而且恩格斯并非像马克思或斯密那样具有重要地位的经济学家。

Engels describes the same process, though he misses out on some of the economic and military processes that Smith identified. Remember that although Smith's work was earlier, Engels did not have access to it, and Engels was not an economist of the stature of Marx or Smith.

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据称由忒修斯制定的宪法由此问世。该宪法带来的主要变革是在雅典建立中央权威——即，此前由各部落独立管理的部分事务被宣布为公共事务，并委托给设在雅典的共同议会处理。通过采取这一步骤，雅典人走得比美洲任何土著民族都更远：他们没有像邻近部落那样仅仅组成一个简单邦联，而是融合成一个单一民族。由此诞生了共同的雅典民法，其地位凌驾于各部落和氏族的法律习俗之上。

The constitution ascribed to Theseus was introduced. The principal change which it made was to set up a central authority in Athens –that is, part of the affairs hitherto administered by the tribes independently were declared common affairs and entrusted to the common council sitting in Athens. In taking this step, the Athenians went further than any native people of America had ever done: instead of neighboring tribes forming a simple confederacy, they fused together into one single nation. Hence arose a common Athenian civil law, which stood above the legal customs of the tribes and gentes.

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据称，忒修斯采取的第二项措施是把全体人民——无论所属氏族、胞族或部落——划分为三个阶层：贵族（eupatridai）、农民（geomoroi）和手工业者（demiourgoi），而担任公职的权利则完全归属于贵族阶层。除了贵族担任公职这一规定外，这种划分并未产生实际作用，因为它并未在阶级之间建立任何其他法律上的区别。然而，这一划分之所以重要，是因为它揭示了那些此前未被察觉的新兴社会要素。这表明，某些家族成员惯常被任命担任氏族公职的做法，已经演变为这些家族几乎无可争议的任职权利；这表明，这些家族凭借财富已然强大，开始在氏族之外形成群体，成为一个独立的特权阶级，而正在成形的国家则认可了这种僭越。[9]

By a second measure ascribed to Theseus, the entire people, regardless of gens, phratry or tribe, was divided into three classes: eupatridai, or nobles, geomoroi, or farmers, and demiourgoi, or artisans, and the right to hold office was vested exclusively in the nobility. Apart from the tenure of offices by the nobility, this division remained inoperative, as it did not create any other legal distinctions between the classes. It is, however, important because it reveals the new social elements which had been developing unobserved. It shows that the customary appointment of members of certain families to the offices of the gens had already grown into an almost uncontested right of these families to office; it shows that these families, already powerful through their wealth, were beginning to form groupings outside their gentes as a separate, privileged class, and that the state now taking form sanctioned this presumption.[9]

据斯密，尽管希腊人和罗马人最初实行贵族制政府——其贵族家族可追溯至城市化之前的酋长阶层——但有两个经济进程促使国家逐渐向更共和的形式转变。

According to Smith, while the Greeks and the Romans initially had aristocratic forms of government, with the aristocratic families inherited from the pre-urban class of chiefs, two economic processes tended to shift the state towards a more republican form.

其一是奴隶制存在。这使自由但非贵族的公民能出席公民大会，而他们的奴隶则负责照料农场或作坊。他把此与更现代的意大利和荷兰共和国对比，指出后者完全由贵族阶层统治。

The first was the existence of slavery. This allowed free, but non-aristocratic citizens to attend the assembly while their slaves minded the farm or workshop. He contrasts this with the more modern Italian and Dutch republics which, he says, were exclusively run by the aristocracy.

其次，劳动分工的发展和奢侈品生产意味着贵族养活的随从减少，转而把财富用于奢侈消费。由于减少了对依附者的开支，他们在公民大会上左右投票结果的能力也随之减弱。民粹主义者推动允许平民当选公职。由于这两大经济进程，在罗马和希腊，民选公职都向非贵族公民开放了。

Secondly, the development of the division of labour and the production of luxury goods meant that the aristocrats supported fewer retainers, choosing instead to spend their wealth on luxuries. Since they spent less on their clients, their ability to swing votes in the assembly diminished. Populists pushed to allow plebs to be elected to office. As a result of these two economic processes, in both Rome and Greece, elected offices became open to non-noble citizens.

我们在此看到一种社会模式：政治形式因物质生活方式的变迁而产生并演变。不仅如此，用恩格斯和马克思的话来说，这些城市文明社会的历史，就是一部阶级斗争史。

What we have here is a model of society in which political forms arise and change as a result of the mode of material life. Not only that, the history of these urban, civilized societies is, to paraphrase Engels and Marx, a history of class struggles.

斯密还意识到，任何形式的宪法或政府在历史上都是暂时的。每一种政治和经济形式都会在其内部孕育出最终导致其覆灭的进程。谈到罗马帝国覆灭时，他写道：

Smith was also aware that any form of constitution or government was historically transitory. Each political and economic form generated within itself the processes which would ultimately lead to its downfall. Speaking of the fall of the Roman Empire, he wrote:

> 我们现在要说明，这种军事君主制如何也遭遇了等待每一个国家和宪法的命中注定的解体。[10]

> We come now to show how this military monarchy came to share that fated dissolution that awaits every state and constitution whatever.[10]

每个时期导致解体的具体冲突各不相同，但某些共同主题反复出现。劳动分工的发展使上层阶级能把更多资金用于奢侈品。在封建制度下，这种奢侈消费增长意味着贵族用于维持随从的开支减少。随着随从队伍缩减，他们的军事影响力也减弱，这为近代早期专制君主制兴起创造了条件。他的论点是，王室收入如此庞大，以至于国王很难把所有收入都花在奢侈品上。总会有部分资金留作军费开支，而随着男爵们转而建造豪华宅邸，而非在简朴大厅里养活大批随从，王室军事力量便取得胜利。

For each period, the specific conflicts that led to the dissolution were different, but certain common themes recur. The development of the division of labour allows the upper classes to spend more on luxuries. Under feudalism, this growth in luxury consumption meant that the aristocrats had less to spend on retainers. As their retinues declined, so too did their military influence, allowing the growth of the absolute monarchies of the early modern period. His argument being that the revenues of the Crown were so large that it was hard for a king to spend it all on luxuries. Some was always left over for the army, and as the barons built luxury homes rather than maintaining hordes of retainers in simple halls, the military Power Royal was triumphant.

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在论述罗马衰落时，他也运用了这一“经济发展导致军事变革”的主题。随着社会日益“文明”，人们逐渐习惯家庭生活中的奢华，“他们便不再热衷于打仗”。一个经济发达的国家会发现，若把从事制造业的人口动员起来参战，其税收收入将受到影响；而一个蛮族社会或纯粹农业社会，则可以在播种与收获之间的夏季空档期把男子送上战场。游牧牧民最能动员全体民众进行征服，因为他们没有不动产，整个民族都可以出征征服。正因如此，从邻近蛮族国家征召士兵比从罗马公民中征召更廉价，而且这

This same theme of economic development leading to military change was deployed in his account of the fall of Rome. As society became more‘polite’and people became acquainted with domestic luxury,‘they become less fond of going to war’. An economically developed state finds that its tax revenues are hurt if it mobilizes its manufacturing population for war, while a barbarian society or a purely agricultural one can send its men off to war during the summer gap between sowing and harvest. Nomadic herdsmen were the most able to mobilize

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样做不会损害罗马工业。蛮族军队首领们最终发现，他们可以把武器对准罗马政府，并成为这个国家的统治者。[11] 他声称，这一过程反复发生：

their entire populations for conquest, since they had no immovable possessions and the entire nation could venture forth to conquer. So it was that soldiers could be recruited more cheaply from among the barbarian nations nearby than from among the Roman citizens, and this could be done without harm to the industries of the Romans. The leaders of the barbarian armies, in due course, found that they could turn their arms against the Roman government and make themselves masters of the country.[11] This process, he claimed, happened repeatedly:

所有亚洲政权都是以同样方式瓦解的。他们的士兵是从鞑靼地区雇来的，手工业和制造业得以持续，人民靠经商获得的收益比参战更多。

In the same manner all the Asiatic governments were dissolved. Their soldiers were hired from Tartary, arts and manufactures were carried on, the people made more by their trades than by going to war.

关于“亚洲政权”覆灭的叙述中，可见哈勒敦[12] 的回响；然而，斯密的分析植根于更广阔的历史唯物主义理论中，其视野从狩猎社会延伸至资本主义，这一宏大视角在恩格斯和马克思著作中再次显现。当一位唯物主义者似乎在延续前人已开始探索的思想时——无论是阿基米德和牛顿对无穷小的运用，哈勒敦和斯密对劳动价值的探讨，还是斯密和恩格斯关于生产方式演进的论述——人们往往会倾向于认为这是一种思想传承，而非重新发明。[13] 鉴于《导论》在 18 世纪仅有阿拉伯语版本，而《法理学讲义》直到恩格斯去世很久之后才面世，这一推测似乎缺乏依据。

There are echoes of Khaldun[12] in the account of the fall of ‘Asiatic governments’, Smith’s analysis, however, is set within a much broader historical materialist theory, going from hunting society to capitalism, a vast perspective that reappears in the work of Engels and Marx. When one materialist appears to extend ideas that an earlier one started to work on, whether it be the use of infinitesimals by both Archimedes and Newton, labour value in Khaldun and Smith or the progression of modes of production in Smith and Engels, there is a temptation to suppose that transmission not reinvention was involved.[13] Since the Al-Muqaddimah was only available in Arabic in the eighteenth century, and the Lectures on Jurisprudence only became available long after Engels was dead, this seems unsupported.

狄德罗的非人类中心主义唯物主义世界观

Diderot’s non-anthropocentric materialist world view

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狄德罗的唯物主义可以被称为——沿用托马斯·奈尔的术语——“运动唯物主义”：运动是有机物质产生的驱动力，也是惰性物质向有机（生命）物质转变的动力。与包括亚里士多德在内的传统哲学观点不同，狄德罗并未把运动和变化定义为“空间位移”——这种定义以把空虚空间等同于本体论虚无为前提，从而排除了（或如亚里士多德那样，瘫痪了）任何关于变化与运动的思考尝试（本书第二章对此有论述）。与热相关的物理过程——这一特定“热力学”论点[14]，正是狄德罗与达朗贝尔对话的核心——催生了日益复杂的有机生命，而人类与动物之间的差异仅被视为程度问题。这一切的基础是物质，而第一且不可分割的构成单元是原子。[15]

Diderot’s materialism is what one might call, following Thomas Nail’s use of the term, kinetic materialism: *motion* is behind the creation of organic matter, or the transition from inert to organic (living) matter. Differently from the traditional philosophical views, including Aristotle’s, Diderot does not define *motion* and change as ‘displacement in place’ premised on the equation of empty space and ontological nothingness which precluded, or paralysed as in the case of Aristotle, any attempt to think about change and movement (discussed in Chapter 2 of this book). Physical processes linked to heat –a particular ‘thermodynamic’ thesis,[14] at the centre of Diderot’s dialogue with D’Alembert –lead to the creation of organic life in its ever-increasing complexity, while the difference between human and animal is seen only as a matter of degree. The foundation of it all is material, and the first and indivisible building bloc is the atom.[15]

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但且看一枚蛋。这正是对世间所有神学派别和所有神庙的驳斥。这枚蛋是什么？在胚胎被植入之前，它只是一团无感的物质；而当胚胎被植入后，它又变成了什么？依然是一团无感的物质，因为胚胎本身也仅仅是一种惰性的、原始的液体。这团物质是如何形成不同的组织结构，

But take an egg. This is what refutes all the schools of theology and all the temples on earth. What is this egg? A mass that is insensible until the embryo is introduced into it, and when this embryo is introduced, what is it then? An insensible mass, for in its turn, this embryo is only an inert and crude liquid. How does this mass arrive at a different organisation, arrive

从而获得感知和生命的？靠的是热量。而热量又由什么产生？运动。[16]

at sensibility and life? By means of heat. And what produces heat? Motion.[16]

此外，狄德罗在与达朗贝尔的上述对话中指出，感觉内在于物质本身。简而言之，狄德罗解释道，一切认知都取决于物质性生命成分的组织结构——这种组织结构源于感觉，进而形成具有不同认知能力的各类有机体（例如动物与人类）。引用狄德罗本人的话（也是作为引述对话中的人物，对达朗贝尔的回应）：“一个能解释一切的简单假设，即感觉能力是物质的普遍属性，或是其组织结构的产物。”[17]至于人类与动物的区别，就认知而言，狄德罗的表述十分明确：

Furthermore, sensation, argues Diderot in the already cited dialogue with D’Alembert, is built into matter itself. In short, Diderot explains that all cognition depends on the organization of the material living components as they ensue from sensation, which amounts to different forms of organisms (animal, for example, vversus humans) with different cognitive faculties. To quote Diderot himself (also as the character appearing in the cited dialogue, responding to D’Alembert): ‘a simple supposition which explains everything, namely, that the faculty of sensation is a general property of matter, or a product of its organisation’.[17] As far as the difference between human and animal is concerned, as far as cognition is concerned, Diderot is explicit:

你是否会像笛卡尔那样坚持认为，这不过是一台简单的模仿机器？小孩子都会嘲笑你，而哲学家们则会反驳说：如果这是台机器，那么你自己也是一台机器。如果你承认这些动物与你的区别仅仅在于组织形式，你就证明了自己的常识和睿智，你就是对的。但这将引出一个对你不利的结论，即：由以某种方式组织起来的无生命物质，在热和运动的作用下，被另一部分无生命物质渗透，从而产生了感觉、生命、记忆、意识、情感和思想。狄德罗接着说，二者必居其一：要么承认蛋中存在某种“隐秘元素”，它在发育的某个阶段以未知方式渗入其中——这种元素究竟是否占据空间、是否具有物质性，抑或是否专为此目的而创造，目前尚无定论——这与常识相悖，并会导致自相矛盾和荒谬；要么我们必须提出一个简单的假设，这个假设能解释一切，即感觉能力是物质的普遍属性，或是其组织结构的产物。（狄德罗致达朗贝尔）[18]小孩子会嘲笑你，而哲学家们会反驳说：如果这是台机器，那么你也是台机器。如果你承认这些动物与你的区别仅在于组织结构，你将证明你的常识和机智，你也是对的。但由此将得出一个对你不利的结论：即，由以某种方式组织起来的惰性物质，通过热和运动与另一部分惰性物质相互作用，从而产生了感觉、生命、记忆、意识、情感和思想。

〔校注：原注 18 之后重复了前面关于机器、组织及感觉产生的一段引文。已核对源 PDF，确为底本重复，正文保留，不将其当作译稿误贴而删去。〕

Would you maintain with Descartes that this is a simple imitating machine? Little children will laugh at you, and the philosophers will reply that if this be a machine then you too are a machine. If you admit that the difference between these animals and you is only one of organisation, you will prove your common sense and sagacity, you will be right. But from this will follow a conclusion against you; namely, that from inert matter organised in a certain way, impregnated with another bit of inert matter, by heat and motion—sensibility, life, memory, consciousness, emotion, and thought are generated. One of the two, continues Diderot, either admit some ‘hidden element’ in the egg, that penetrates to it in an unknown way at a certain stage of development, an element about which it is unknown whether it occupies space, whether it is material or whether it is created for the purpose –which is contradictory to common sense, and leads to inconsistencies and absurdities; or we must make a simple supposition which explains everything, namely, that the faculty of sensation is a general property of matter, or a product of its organisation. (Diderot to D’Alembert)[18]Little children will laugh at you, and the philosophers will reply that if this be a machine then you too are a machine. If you admit that the difference between these animals and you is only one of organisation, you will prove your common sense and sagacity, you will be right. But from this will follow a conclusion against you; namely, that from inert matter organised in a certain way, impregnated with another bit of inert matter, by heat and motion—sensibility, life, memory, consciousness, emotion, and thought are generated.

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当代实验哲学借鉴认知科学（包括进化生物学）的成果，证明“抽象”或模式识别不仅发生在蛋白质物质和细胞组织层面，[19]也发生在其他层面，例如蜜蜂的计数能力。[20]借用狄德罗的话来说，高等“物理组织”形式——即人类以外的动物——具备“抽象”或识别数学模式甚至计数的能力，这在 21 世纪绝非一个有争议的观点。[21] 除了所谓新理性主义者对“理性”的形而上学遐想之外，[22] 几乎没有任何科学哲学或与认知科学这一跨学科、多学科领域相关的科学本身，会把“理性”视为人类独有能力。正如“自然”或“情感”等与其相对的概念一样，“理性”这一概念已被当代科学视为过时概念——它们都是同一信息生态系统连续体的一部分。

Contemporary experimental philosophy, operating with knowledge from the cognitive sciences including evolutionary biology, demonstrates that ‘abstracting’ or pattern recognition happens on the level of protein mass and cell organization,[19] as it does on the level of, for example, bee’s ability of counting.[20] The ability of the higher forms of ‘physical organization’, to paraphrase Diderot, that is, of the animals other than humans, to ‘abstract’ or to recognize mathematical patterns and even count, in the twenty-first century, is by no means a controversial idea.[21] Apart from the metaphysical reveries of Reason among the so-called neo-rationalists,[22] hardly any philosophy of science or the sciences themselves linked with the interdisciplinary

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and multidisciplinary area of cognitive sciences believes in Reason as the exclusive faculty of humanity. Reason has been rendered obsolete as a notion by contemporary science just as its opposites, such as nature or emotion, have been –they are all part of the same continuum of informational ecosystems.

感觉、认知和心理设计源于物质基础，因此，借用马克思的术语来说[23]，“人类作为类存在”的社会关系的“目的”，正是物质必然性的产物，旨在维持和扩展生命，或者说其“形式”——即亚里士多德所定义的特定“自动机”，一种自运动、自维持的系统（这是我对马克思定义的转述，或者更确切地说，是我的诠释）。用马克思自己的话来说：

Sense, cognition and mental design are derived from the material foundation, and thus the ‘purpose’ of social relations of the ‘species being of humanity’, to borrow Marx’s term,[23] is the product of material necessity to sustain and expand life or its ‘form’ as design of particular ‘automaton’, self-moved and self-sustained system, as Aristotle would define it (my paraphrase or rather interpretation of Marx’s definition). In Marx’s own words:

p. 78–79

人之所以是“类存在”（Gattungswesen），不仅因为他在实践和理论中把物种（包括他自身以及其他事物的物种）作为其对象，而且——这只是另一种表述方式——还因为他把自己视为实际的、活生生的类；因为他把自己视为一种普遍的、因而也是自由的存在。无论在人类还是动物身上，物种的生命在生理上都体现在这样一个事实：人（如同动物一样）依赖有机自然而生存；而人（或动物）越是普遍，他赖以生存的无机自然领域就越普遍。正如植物、动物、石头、空气、光等在理论上构成人类意识的一部分——部分作为自然科学对象，部分作为艺术对象——即他必须先加以加工才能使其可口且易于消化的精神上的无机自然、精神上的养分——同样，在实践领域中，它们也构成人类生活和人类活动的一部分。从生理上讲，人仅靠这些自然产物维生，无论它们以食物、取暖、衣物、住所等形式出现。人的普遍性在实践中恰恰体现在这种普遍性上，即使整个自然成为他的无机躯体——既因为自然是（1）他直接的生活手段，也是（2）他生活活动的物质、对象和工具。自然是人的无机躯体——即自然本身并非人体的那一部分。人依靠自然生存——这意味着自然就是他的身体，若不想死亡，他就必须与之保持持续交流。人的肉体和精神生活与自然相联，这仅仅意味着自然与自身相联，因为人是自然的一部分。[24]

Man is a species-being [Gattungswesen], not only because in practice and in theory he adopts the species (his own as well as those of other things) as his object, but –and this is only another way of expressing it –also because he treats himself as the actual, living species; because he treats himself as a universal and therefore a free being. The life of the species, both in man and in animals, consists physically in the fact that man (like the animal) lives on organic nature; and the more universal man (or the animal) is, the more universal is the sphere of inorganic nature on which he lives. Just as plants, animals, stones, air, light etc constitute theoretically a part of human consciousness, partly as objects of natural science, partly as objects of art –his spiritual inorganic nature, spiritual nourishment which he must first prepare to make palatable and digestible –so also in the realm of practice they constitute a part of human life and human activity. Physically man lives only on these products of nature, whether they appear in the form of food, heating, clothes, a dwelling etc. The universality of man appears in practice precisely in the universality which makes all nature his inorganic body –both inasmuch as nature is (1) his direct means of life, and (2) the material, the object, and the instrument of his life activity. Nature is man’s inorganic body –nature, that is, insofar as it is not itself human body. Man lives on nature –means that nature is his body, with which he must remain in continuous interchange if he is not to die. That man’s physical and spiritual life is linked to nature means simply that nature is linked to itself, for man is a part of nature.[24]

p. 79

作为生命的物质活动，把有机和无机物质（自然或“自然”——这一自然概念既远离德国唯心主义的浪漫理解，也不同于费尔巴哈的唯物主义）视为人类本质的延伸或定义，或者说与之构成一种生态统一体。因此，本质与目的——即物种的存在——在于维持和扩展生命；而生命既包含有机部分也包含无机部分，它既由人类活动决定，也由其非人类的延伸或扩展——即向自然与技术共同决定的生态系统中的延伸或扩展——所决定。生命是机械的，因此从这个意义上说它是无机的，而非笛卡尔那种把动物等同于可鄙钟表式机器的还原论等式所指的无机（与有机、自然的——仿佛被一种崇高或神性实体驱动——相对）。它是“mekhane”或“techne”的产物，即拼接任何形式物质所固有的并持续执行的信息片段——从蛋白质折叠到所谓自然语言的语音学——因此，“有机”仅仅是程度问题，而非与“机

The material activity which is life treats organic and inorganic matter (nature or ‘nature’, as this notion of nature is far from the romantic understanding of German idealism, as well as from Feuerbach’s materialism) as an extension of the substance or definition of humanity or rather constitutes an ecological unity with it. Thus, the essence and the purpose –the being of the species –consist in sustaining and expanding life, whereas life is both organic and inorganic and it is determined by activity both human and its inhuman extension or expansion into the co-determining ecosystem of nature and technology. Life is mechanic, and it is thus inorganic in that sense and not in the sense of the Cartesian reductionist equation of the animal and the contempt-worthy clock-like machine (as opposed to the organic, natural as if moved by a sublime or godlike substance). It is the prod-

械”在实质上的差异。尽管如此，在我们凭常识以及某些更古老科学语言所指代的“生命”之间，既存在连续性，也存在仅是程度上的差异而非本质上的差异。这正是当代科学所教导我们的，正如原子论者——尤其是伊壁鸠鲁学派的卢克莱修——以及青年马克思和列宁所揭示的那样。[25]

〔校注：底本中 rganic 漏首字母 o，中文按 organic 译有机，英文保留原样。〕

uct of the *mekhane* or *techne* of piecing bits of information any form of matter is endowed with and perpetually executes –ranging from protein folding to phonetics of the so-called natural languages –and thus the ‘rganic’ is merely a matter of degree, not of substantive difference from the ‘mechanic’. Still, there is a continuity and also difference of mere degree and not of substance between what we commonsensically and in some more archaic scientific language mean by‘life’. This is what the contemporary science teaches us as did the atomists, in particular the Epicurean Lucretius, as well as young Marx and Lenin.[25]

在此意义上所说的“目的”或“telos”，显然并非线性的，它并不作为“目的因”（causa finalis）发挥作用，因此我们无法确定能否将其称为“telos”。鉴于目前尚无更恰当术语，我们将暂且使用这一术语。在《偶然性与递归性》中，许煜解释说，按照德国唯心主义（特别是谢林）的理解，有机体或自然的“telos”归根结底是一种永恒的偶然性与递归性，这与我们视为计算自动机奠基原则的那些特性如出一辙。尽管许煜秉持谢林和唯心主义立场，但其分析所揭示的运作机制表明，“理想”与自然密不可分。此外，人们可以把他的分析解读为揭示了机械性与有机性之间某种连续性，这种二分法是多余的，在分析中毫无意义。[26]或者用许煜自己的话来说：

The ‘purpose’or the telos one might speak of in this sense is apparently not linear, it does not operate as *causa finalis*, and we are, therefore, not certain that one can speak of ‘telos’. We will use the term on a provisory basis, in the absence of a more fitting one. In *Contingency and Recursivity*, Yuk Hui explains that the‘telos’ of the organic or nature, understood in line with German idealism and Schelling in particular, comes down to a perpetual contingency and recursion akin to the ones we find as founding principles of computing automata. Despite Hui’s allegiance with Schelling and idealism, the operations his analysis reveals demonstrate that the ‘ideal’is inextricable from nature. Also, one can read his analysis as demonstrating a certain continuity between mechanicity and organicity, the dichotomy is redundant and serves no purpose in the analysis.[26] Or in Hui’s own words:

自然目的是一种无法客观观察的事物。我们可以看到某棵树或某种动物，但无法通过机械规则把握自然整体。理性只能通过反思性判断来理解自然目的，这意味着它会递归地抵达一个自我组织的存在。从这个意义上说，目的论思维是循环的：*A→B→C→A。[27]

The natural end is something that cannot be observed objectively. We can see such and such a tree or such and such an animal, but we cannot grasp nature as a whole through mechanical rules. Reason can only understand the natural end through reflective judgment, meaning that it recursively arrives at a self-organising being. Teleological thinking is in this sense circular: *A→B→C→A[27]

那套旨在维持人类“类存在”（形式与功能）的设计——该设计由社会关系构成，旨在维持其存在并提升其效能——所包含的一系列功能，本身具有循环性。

The set of functions that amount to maintaining the design (form and function) of the ‘species being’of humanity, which is built of social relations and with the goal of sustaining its existence and enhancing its effectiveness, is circular.

p. 80

换言之，该“存在”在生产与再生产过程中持续进行自我组织，同时改善着技术与自然之间的生态系统（即非人工建造的、非人类“技艺”（technē）或巧思所创造的——此处“技艺”（technē）取其字面含义）。这里似乎是在谈论一种自动机，即亚里士多德所指的那种——一种不受外部原因或推动者驱动的存在，但同时也缺乏“目的因”（causa finalis）；或者，正如亚里士多德所解释的，它运作起来仿佛拥有一个“目的”（telos），尽管它实际上并没有（因为它缺乏理性）。[28]唯有“不动的推动者”是自运动的，它是一种理念，[29]是一切事物的“为什么”——这就是为什么自动机——这种自运动的物质现实——在亚里士多德宇宙中，看似具有目的性和意义，却并非如此。

In other words, the ‘being’is perpetually self-organizing in production and reproduction while improving the ecosystem between technology and nature (that which is not artificially built, which is not the product of the human *technē* or artfulness, the literal meaning of *technē*). One seems to be talking of an automaton here, in Aristotle’s sense –a being unmoved by an external cause, or by a mover, but also bereft of *causa finalis*, or, as Aristotle explains, it operates as if it had a telos nonetheless it doesn’t have one (as it is void of Reason).[28] Only the ‘unmoved mover’is self-moved, and it is an Idea,[29] the ‘why’of it all –that is why the automaton, the self-moved material reality resembles a purposeful and meaningful reality, yet it is not –in Aristotle’s universe.

人们常说亚里士多德对马克思影响相当大，如果抽去“自我观照的完美理念”或“不动的推动者”这

It is often said that Aristotle’s influence on Marx has been considerable, and if one abstracts the notion of

一概念，亚里士多德所发展的形式与物质之间的辩证关系，很大程度上可以在马克思思想中看到反映，包括他的博士论文。在该论文中，马克思提出以下论点：

原子通过其属性获得了一种与其概念相矛盾的存在；它被假定为一种与本质相异的、外化的存在。正是这种矛盾最令伊壁鸠鲁感兴趣。因此，一旦他设定某种属性并由此推导出原子的物质性质，他便同时反设一些规定，这些规定又在其自身范畴内摧毁了该属性，转而验证了原子的概念。因此，他以一种使所有属性自相矛盾的方式来界定它们。另一方面，德谟克利特在任何地方都没有把属性与原子本身联系起来考虑，也没有把属性中固有的概念与存在之间的矛盾对象化。[30]

这里并不存在某种外部理念值得向往或寻求实现——如同黑格尔设想那样，通过物质的自我消亡回归到该理念——而是原子的“概念”不可避免地通过其物质属性的对象化而得以物质化并作为物质存在而得到验证。这种“形式”（概念）与物质的辩证关系，与其说是黑格尔或其他唯心主义者的，不如说更接近亚里士多德形而上学（在其原始的亚里士多德意义上）。同样，社会（包括技术）设计是无产阶级直接的、物质的需求和阶级利益的产物——它在这个词的任何意义上都不是观念性的（或唯心主义的），包括“伦理的”或道德的（或道德主义的）意义。[31]

唯心主义是以主体性为中心的，是人类中心主义的，这不仅符合马克思观点，也符合狄德罗观点：

p. 80–81 那些哲学家被称为唯心主义者，因为他们只意识到自己的存在以及自身内部接连不断的感觉，而不承认其他任何东西。这是一个荒诞的体系，依我之见，唯有盲人才可能提出；这一体系虽是所有体系中最荒谬的，却也是最难反驳的，这令人类智慧和哲学蒙羞。[32]

p. 81 在《对黑格尔哲学的总体批判》（《1844 年经济哲学手稿》的一部分）[33] 中，马克思讨论了“主体性”概念，认为正是哲学的定义所在，而诉诸主体性则是破坏任何“马克思主义”思想试图确立其科学性的关键因素。[34]马克思所倡导的客观思维方式与奥古斯特·孔德的实证主义截然不同。恰恰相反，它以把思维从（个体）人类主体中“去中心化”为前提。引用马克思的话：

作为对象性的、自然的、感性的存在，同时在自身之外拥有对象、自然和感性，或者自身对于第三者而言就是对象、自然和感性，乃是一回事。[35]

“感性”和“自然”这两个术语的使用应与狄德罗的学术传统联系起来，而非德国活力论者，因为这种用法是反人类中心的——即使在激进人文主义语境下也是如此——而且当马克思进行唯物主义论证

the perfect Idea contemplating itself or the ‘unmoved mover’, much of the dialectics between form and matter developed by Aristotle can be seen as reflected in Marx’s thought, including his doctoral dissertation. In said thesis, Marx puts forward the following argument:

Through the qualities the atom acquires an existence which contradicts its concept; it is assumed as an externalised being different from its essence. It is this contradiction which mainly interests Epicurus. Hence, as soon as he posits a property and thus draws the consequence of the material nature of the atom, he counterposits at the same time determinations which again destroy this property in its own sphere and validate instead the concept of the atom. He therefore determines all properties in such a way that they contradict themselves. Democritus, on the other hand, nowhere considers the properties in relation to the atom itself, nor does he objectify the contradiction between concept and existence which is inherent in them.[30]

There is no external idea to aspire to or seek to fulfil –by returning to it through self-annihilation of the material as in Hegel’s vision –but rather ‘the concept’ of the atom is unavoidably materialized and validated as such through its objectivization via material properties. This dialectic of ‘form’ (concept) and matter resembles Aristotle’s metaphysics (in its original, Aristotelian sense) far more than that of Hegel or the other idealists. Similarly, social (including technological) design is the product of the immediate and material needs and class *interests* of the proletariat –it is not ideal(ist) in any sense of the word, including ‘ethical’ or moral(ist).[31]

Idealism is subjectivity-centred, it is anthropocentric, not only according to Marx but also according to Diderot:

Those philosophers are called idealists who, being conscious only of their existence and of the sensations which succeed each other within themselves, do not admit anything else. An extravagant system which, to my thinking, only the blind could have originated; a system which, to the shame of human intelligence and philosophy, is the most difficult to combat, although the most absurd of all.[32]

In ‘Critique of Hegel’s Philosophy in General’ (part of *The Economic and Philosophic Manuscripts of 1844*)[33] Marx discusses the concept of subjectivity as being what defines philosophy, and a resort to subjectivity as the key factor of undermining any attempt of ‘Marxist’ thought to establish itself as scientific.[34] The type of objective thought Marx advocates is in no way similar to August Comte’s positivism. Quite the opposite, it is premised on decentring thinking from the (individual) human subject. To quote Marx:

To be objective, natural and sensuous, and at the same time to have object, nature and sense outside oneself, or oneself to be object, nature and sense for a third party, is one and the same thing.[35]

The use of ‘sensuous’ and ‘natural’ should be linked to Diderot’s heritage rather than the German vitalists as it is anti-anthropocentric –even when radically humanist –and the use of both terms, including that of ‘the physical

时，这两个术语（包括“物理的”和“真实的”）的使用远比“物质”或“材料”更为常见。我认为这种术语选择的原因在于他避免被等同于另一种青年黑格尔派唯物主义——即费尔巴哈的唯物主义，这个问题我在我的著作《迈向社会主义的激进形而上学》（卡塔琳娜·科洛佐娃，Punctum Books，2015）中进行了详细讨论。[36]让我们注意狄德罗这段话结语（即前文引用的与达朗贝尔的对话）中的相似之处：

and the real’is far more common in Marx when he makes a materialist argument (rather than ‘matter’or ‘material’). I find the reasons for this terminological choice in his avoidance to be equated with the other young-Hegelian materialism, that of Feuerbach, a question I discuss in detail in my bookKaterina Kolozova, (Punctum Books, 2015).[36] *Toward Radical Metaphysics of Socialism*. Let us note the similarity in this closing note from Diderot (the already cited dialogue with d’ Alembert):

〔校注：底本将 bookKaterina Kolozova、出版社及书名错排在正文中；中文据本段及对应原注整理，未把作者名当书名。〕

达朗贝尔：例如，你的体系并未阐明我们如何形成三段论或作出推论。狄德罗：我们并不作出这些推论；它们全是由自然推导出来的。我们只是陈述相互关联的现象的存在，这些现象是我们通过实践所知的，这些现象的存在可能是必然的，也可能是偶然的；在数学、物理学及其他精确科学中是必然的；在伦理学、政治学及其他推测性科学中则是偶然的。[37]

d’Alembert: For instance, your system doesn’t make it clear how we form syllogisms or draw inferences. Diderot: We don’t draw them; they are all drawn by nature. We only state the existence of connected phenomena, which are known to us practically, by experience, whose existence may be either necessary or contingent; necessary in the case of mathematics, physics, and other exact sciences; contingent in ethics, politics and other conjectural sciences.[37]

我们认为，在这些专门论述狄德罗唯物主义及其与马克思联系的段落中，我们已经证明：二者既不是哲学意义上的自然主义，也不是某种泛灵论式的活力论，而是非人类中心的唯物主义世界观。与此同时，他们确实提出了“人道主义”理想，以及关于一个更公正的（人类）社会关系宇宙的愿景，但两位作者都极其明确地强调，这些理想和愿景是超越资本主义对“自然资源”剥削态度的生态系统的产物。

We believe we have demonstrated in these passages dedicated to Diderot’s materialism and its links with Marx that both are neither naturalism in the philosophical sense nor some sort of animistic vitalism, but rather non-anthropocentric materialist world views. At the same time, they do propose ‘humanist’ideals and visions of a more just universe of (human) social relations, but the emphasis on them being the result of ecosystems that transcend the capitalist exploitative attitude towards ‘natural resources’is more than clear in both authors.

牛顿式的马克思

The Newtonian Marx

一个国家可以也应该向其他国家学习。甚至当一个社会已经走上发现其运动自然规律的正确轨道时——而本著作的最终目标，正是揭示现代社会的经济运动规律——它既无法通过大胆飞跃来跨越，也无法通过立法来消除其正常发展各阶段所面临的障碍。但它可以缩短并减轻这些分娩的阵痛。[38]

One nation can and should learn from others. And even when a society has got upon the right track for the discovery of the natural laws of its movement –and it is the ultimate aim of this work, to lay bare the *economic law of motion* of modern society –it can neither clear by bold leaps, nor remove by legal enactments, the obstacles offered by the successive phases of its normal development. But it can shorten and lessen the birth-pangs.[38]

p. 81–82

上述引自《资本论》序言的目标，是确立现代社会的运动规律。换言之，就是对现代社会做牛顿对自然界所做的工作。马克思在《资本论》中的方法如何契合牛顿式科学实践？

The aim, stated above from the preface to Capital, was to establish a law of motion for modern society. In other words, to do for modern society what Newton had done for the natural world. How does the method of Marx in Capital fit into the Newtonian scientific practice?

p. 82

因为哲学的一切困难似乎都在于这一点——从运动的现象出发去探究自然力，然后又从这些自然力出发去推导出其他现象。（牛顿，《自然哲学的数学原理》序言）

for all the difficulty of philosophy seems to consist in this –from the phænomena of motions to investigate the forces of nature, and then from these forces to demonstrate the other phænomena. (Newton, introduction to the *Principia*)

牛顿的方法是：

1. 从实验和观察出发，表述运动定律。

2. 借助几何学进行演绎，进而得出向心力的普遍性质。

The method of Newton was,

1. From experiment and observation to formulate laws of motion.

2. By deduction, via geometry to then derive general properties of centripetal forces.

3. 根据木星卫星的周期和轨道半径，推导出引力定律必须遵循平方反比关系。

4. 从万有引力平方反比定律推导出关于行星轨道的更精确预测。

当我们审视马克思从抽象到具体的论证过程时，会发现其中存在非常相似的逻辑路径。

这两种情况下都提出了守恒定律。

牛顿：作用与反作用相等且方向相反，动量守恒；

马克思：交换中价值相等，贸易中价值总量守恒；

这些初始公理均基于观测。

两者都从抽象定律推导出关于运动的具体结论。

牛顿：向心力在等时间内扫过等面积，引力遵循平方反比定律；• 马克思：随着资本积累加深，利润率下降。

〔校注：底本写作“向心力在相等时间内扫过相等面积”，此处按原文保留其表述；通常扫过面积的是运动物体与力心的连线。〕

从《资本论》一开始，马克思就致力于阐明商品交换涉及一种守恒关系：

p. 82–83 让我们以两种商品为例，例如谷物和铁。无论它们的交换比例如何，总可以用一个等式来表示这个比例，使一定量的谷物与一定量的铁相等：例如，1 夸特谷物＝ x 英担铁。这个等式告诉我们什么？

p. 83 它告诉我们，在两种不同的东西中——在 1 夸特谷物和 x 英担[39]铁中——存在着某种两者共有的、数量相等的东西。因此，这两种东西必定等于第三种东西，而这第三种东西本身既不是前者，也不是后者。因此，两者中的每一种，就其作为交换价值而言，都必定能够还原为这第三种东西。

一个简单的几何图解可以说明这一点。为了计算和比较直线图形的面积，我们将它们分解为三角形。但三角形本身的面积，是用一种与其可见形状完全不同的东西来表示的，即底边与高之积的一半。同样地，商品的交换价值必须能够用它们所共有的某种东西来表示，而它们所代表的正是该东西的多少。[40]

因此，如果发生 $\frac{1}{4}$ 英担谷物＝ x 英担铁这样的交换，双方就包含了同样数量的某种东西：这就是牛顿的原理，即作用会产生大小相等、方向相反的反作用。

这种等价性甚至可以用阿基米德所采用的希腊几何方法来解释——即通过把面积分解为多个三角形来证明其等价性。阿基米德利用这一方法推导出圆的面积为 πr^2，尽管从表面上看，圆和三角形截然不同。马克思的论点直接承袭了阿基米德/牛顿传统。

3. From the periods and orbital radii of the satellites of Jupiter to deduce that the gravitational force must be inverse square.

4. From an inverse square law of gravitation to deduce more accurate predictions about the planetary orbits.

When we look at the process of argument from the abstract to the concrete in Marx, we see a very similar process.

Formulation of conservation laws in both cases.

−Newton: equal and opposite action, conservation of momentum,

−Marx: equal values in exchange, sum of values conserved in trade

−These initial axioms are observationally based.

Movement from abstract laws to concrete conclusions about motion in both cases.

−Newton: centripetal forces sweep equal area in equal time, gravitational force inverse square −Marx: rate of profit falls as capital accumulation intensifies

Right from the start of Capital, Marx is concerned to establish that commodity exchange involves a conservation relation:

Let us take two commodities, eg corn and iron. The proportions in which they are exchangeable, whatever those proportions may be, can always be represented by an equation in which a given quantity of corn is equated to some quantity of iron: eg 1 quarter corn = x cwt. iron. What does this equation tell us?

It tells us that in two different things – in 1 quarter of corn and x cwt.[39] of iron, there exists in equal quantities something common to both. The two things must therefore be equal to a third, which in itself is neither the one nor the other. Each of them, so far as it is exchange value, must therefore be reducible to this third.

A simple geometrical illustration will make this clear. In order to calculate and compare the areas of rectilinear figures, we decompose them into triangles. But the area of the triangle itself is expressed by something totally different from its visible figure, namely, by half the product of the base multiplied by the altitude. In the same way the exchange values of commodities must be capable of being expressed in terms of something common to them all, of which thing they represent a greater or less quantity.[40]

So, if an exchange takes place between $\frac{1}{4}$ cwt of corn = x cwt Iron both sides contain the same amount of something: Newton’s principle of action producing an equal and opposite reaction.

The equivalence is even explained in terms of Greek geometric approach, used by Archimedes, of proving the equivalence of areas by breaking them down into triangles. Archimedes used this to derive the area of a circle as being πr^2 even though, at a surface level, a circle and

交换价值的形式是经验观察所得的。马克思从这一观察的性质推导出，这里存在一个守恒过程。在等价物的交换中，有某种实体——价值——保持守恒。这与牛顿从相互碰撞的摆的运动形式推导出运动守恒（用我们今天的话说，就是动量守恒）相对应。

他指出，交换是一种等价关系。从形式上讲，我们如今会说：等价关系是一种二元关系，其性质为自反性（$a \equiv a$）、对称性（$a \equiv b \Rightarrow b \equiv a$）和传递性（$a \equiv b, b \equiv c \Rightarrow a \equiv c$）。由于马克思著作成于现代符号逻辑记号出现之前，他并未把“交换是等价关系”这一论点精确表述为上述形式，但他通过自己创造的记号，表达了相同意思。他用自己所称的“价值形式 B”来阐明传递性，其符号表示如下：

B. 总和的或扩大的价值形式 z 商品 A ＝ u 商品 B 或＝ v 商品 C 或＝ w 商品 D 或＝ x 商品 E 或＝等等。[41]

他在随后段落中明确论证了自反性，该段落直接用文字表述了蕴含关系 $a \equiv b \Rightarrow b \equiv a$。

〔校注：底本此处称 reflexivity（自反性），但所列蕴含式表达的是对称性；译文保留底本用词，不静默更改。〕

然而，这种扩大的相对价值形式无非是第一类简单的相对价值表达式或等式的总和，例如：

20 码亚麻布＝ 1 件外套

20 码亚麻布＝ 10 磅茶叶，等等。

上述每条等式都隐含相应的逆等式：

1 件外套＝ 20 码亚麻布

10 磅茶叶＝ 20 码亚麻布，等等。[42]

对称关系（x 商品 A ＝ x 商品 A）过于显而易见，无需赘述。

〔校注：底本此处称 symmetry（对称性），但所列同一商品等于自身的式子表达的是自反性；这与上一段的术语倒置均按底本保留并说明。〕

他从这些观察到的性质中推导出：既然存在等价关系，那么在等价关系背后必然存在某种抽象的第三种东西。等价关系是守恒的标志。关于这种共同实体存在的推导，在结构上等同于以下推导：在行星绕太阳公转过程中，一种共同实体——能量——会在势能和动能形式之间交换。我们仅通过能量强加于运动上的等价关系，才知晓能量存在。

在马克思撰写《资本论》之时，关于能量守恒的研究早已远远超出动能和势能范畴。当时他能接触到的科学文献，正在探讨电能、化学能、热能和光能中的能量守恒问题。关于“共同实体”——即以多种形式存在的能量——这一概念，在撰写《资本论》之前，马克思与恩格斯之间就已展开热烈讨论。

a triangle are very different. Marx’s argument is directly in the Archimedes/Newton tradition.

The form of exchange *value* is something that is empirically observed. From the properties of this observation, Marx deduces that there is a conservation process involved. There is some substance, *value*, that is conserved in an exchange of equivalents. This maps onto Newton’s deduction of the conservation of *motion*, or as we now say momentum, from the form of *motion* of colliding pendulums.

He says exchange is an equivalence relation. Formally, we would now say that an equivalence relation is a binary relation that is reflexive $a \equiv a$, symmetric $a \equiv b \Rightarrow b \equiv a$ and transitive $a \equiv b, b \equiv c \Rightarrow a \equiv c$. Writing before modern symbolic logic notation, Marx does not set his argument for exchange being an equivalence relation in exactly this form, but using his own invented notation, he says the same things. He states transitivity with what he terms his Form B of value, which he writes down symbolically as:

B. Total or Expanded Form of value *z* Com. A = *u* Com. B or = *v* Com. C or = *w* Com. D or = *x* Com. E or = etc.[41]

He explicitly establishes reflexivity in a subsequent passage that directly gives in words the implication $a \equiv b \Rightarrow b \equiv a$.

The expanded relative value form is, however, nothing but the sum of the elementary relative expressions or equations of the first kind, such as: p. 83–84

20 yards of linen = 1 coat

20 yards of linen = 10 lbs of tea, etc.

Each of these implies the corresponding inverted equation,

1 coat = 20 yards of linen

10 lbs of tea = 20 yards of linen, etc.[42]

The symmetry relation (*x* commodity *A* = *x* commodity *A*) was too obvious to need stating. p. 84

His deduction from these observed properties is that because there is an equivalence relation, there is some abstract third thing behind the equivalence. An equivalence relation is an indicator of conservation. The deduction that this common substance exists is structurally equivalent to the deduction that a common substance *energy* is exchanged between potential and kinetic forms during a planet’s orbit around the Sun. We only know of the existence of *energy* via the equivalence relationships that it imposes on motions.

By the time Marx was writing Capital, work on the conservation of energy had gone much further than just kinetic and potential energy. The science available to him was discussing energy conservation in electricity, chemical energy, heat and light. The idea of a common substance, energy present in multiple forms, was being

actively discussed between him and Engels prior to writing Capital.

另一个会让老黑格尔欣喜若狂的发现，是物理学中的力之关联，即在特定条件下，机械运动（即机械力，例如通过摩擦产生的力）转化为热能，热能转化为光能，光能转化为化学亲和力，化学亲和力（例如在伏打堆中）转化为电能，而电能又转化为磁力。这些转化过程也可能以不同方式发生，既可以是正向的，也可以是逆向的。一位我记不清名字的英国人［焦耳］现已证明，这些力以非常具体的定量比例相互转化，因此，例如，某一种力（如电）的特定量，对应于其他每种力量（如磁、光、热、化学亲和力（正或负——结合或分离）以及运动）的特定量。[43]

Another result that would have delighted old Hegel is the correlation of forces in physics, or the law whereby mechanical motion, ie mechanical force (eg through friction), is, in given conditions, converted into heat, heat into light, light into chemical affinity, chemical affinity (eg in the voltaic pile) into electricity, the latter into magnetism. These transitions may also take place differently, backwards or forwards. An Englishman [Joule] whose name I can't recall has now shown that these forces pass from one to the other in quite specific quantitative proportions so that eg a certain quantity of one, eg electricity, corresponds to a certain quantity of each of the others, eg magnetism, light, heat, chemical affinity (positive or negative — combining or separating) and motion.[43]

从交换中必定有一种共同实体保持守恒这一推论出发，马克思立即把这一实体认定为劳动。

From the deduction that there must be a common substance conserved in exchange, Marx immediately identifies this with labour.

这种共同的“某种东西”既不能是商品的几何属性，也不能是化学属性，更不能是任何其他自然属性。这些属性只有在影响商品效用、使商品成为使用价值时，才值得我们关注。但商品交换显然是一种完全抽去使用价值的行为……

This common 'something' cannot be either a geometrical, a chemical, or any other natural property of commodities. Such properties claim our attention only in so far as they affect the utility of those commodities, make them use values. But the exchange of commodities is evidently an act characterised by a total abstraction from use value. . . .

因此，如果我们不考虑商品的使用价值，它们就只剩下一项共同属性，即作为劳动产品的属性。……

If then we leave out of consideration the use value of commodities, they have only one common property left, that of being products of labour . . .

p. 85

随着产品本身有用性质的消失，体现在产品中的各种劳动的有用性质及其具体形式也被撇开了；剩下的只有它们所共有的东西；它们全都还原为同一种劳动，即抽象的人类劳动。现在让我们考察这些产品各自的残余：每一种产品中都留下同样的无形的实在，不过是同质人类劳动的凝结，是不管以何种方式耗费的人类劳动力的凝结。这些东西现在仅仅表明，在它们的生产中耗费了人类劳动力，其中凝结了人类劳动。作为这种共同社会实体的结晶来考察，它们就是——价值。

Along with the useful qualities of the products themselves, we put out of sight both the useful character of the various kinds of labour embodied in them, and the concrete forms of that labour; there is nothing left but what is common to them all; all are reduced to one and the same sort of labour, human labour in the abstract. Let us now consider the residue of each of these products; it consists of the same unsubstantial reality in each, a mere congelation of homogeneous human labour, of labour power expended without regard to the mode of its expenditure. All that these things now tell us is, that human labour power has been expended in their production, that human labour is embodied in them. When looked at as crystals of this social substance, common to them all, they are –Values.

受边际经济学熏陶的现代评论家们认为，这种推论过于牵强。尽管他们或许愿意接受马克思关于“交换是一种等价关系”的结论，但他们认为马克思关于“劳动是守恒量”的结论站不住脚。如今，我们已拥有大量实证证据表明马克思是正确的。[44]这些新证据依赖于投入产出表等现代经济统计数据。马克思当时虽未掌握这些数据，但在他著述之际，劳动决定价值已被视为政治经济学中已经确立的真理。这种劳动决定价值理论，早在斯密《国富论》中便已确立。[45]正如牛顿把伽利略的发现视为既定事实一样，马克思也可以假定劳动含量决定价格这一原理已被广泛理解。直到马克思分析所蕴含的颠覆性影响被广泛认可之后，经济学界的共识才逐渐偏离斯密的结论。对当代读者而言，把劳动视为共同要素的论述似乎有些武断。然而，马克思最初证明交换是一种具有守恒实体的等价关系的步骤，对于他后续论证阶段至关重要。

Modern commentators, raised on marginal economics, see this as a leap too far. While they may be willing to accept his conclusion that exchange is an equivalence relation, they believe Marx's conclusion that labour is the conserved quantity to be unsound. We now have considerable empirical evidence that Marx was correct.[44] This new evidence relies on modern economic statistics like I/O tables. Marx did not have this data, but at the time he was writing, the determination of value by labour was already taken as an established truth of political economy. The determination of value by labour had been established by Smith in the Wealth of Nations.[45] Just as Newton accepted Galileo's findings as a given, Marx could assume the determination of prices by labour content was well understood. It was only after the disruptive implications of Marx's analysis became widely recognized that

他用以下公式对商品流通进行了符号分析：

$C_1 \to M \to C_2$

卖方起初持有商品 1（C_1），将其换成等值的货币 M，再用这笔货币购买一定量的商品 2（C_2）。由于交换是一种等价关系，因而具有传递性，起点和终点的价值就相同，即 $C_1 \equiv C_2$。一切看起来都公平而妥当，直到马克思指出：资本家所做的并不是这样。他们的活动采取如下形式：

$M \to C \to M'$

其中，最终的货币量大于起初的货币量：$M' > M$。差额 $S = M' - M$ 就是马克思所说的资本家的剩余价值。显然，$M \to C \to M'$ 违反了商品交换的等价性质。

先前的经济学家并未把商品交换视为一个受守恒定律约束的形式系统。通过形式化分析，马克思得以发现利润存在所固有的矛盾或对称性破缺。[46]这怎么可能呢？

如果价值在交换中得以保存，那么利润或剩余价值就无法在商品交换层面得到解释。它必须由交换之外的因素来解释，具体而言，就是商品生产的实际过程。资本家把初始货币 M 用于两部分，马克思把它们标记为不变资本 c 和可变资本 v。因此 $M = c + v$。不变资本由资本家购入的典型商品组成，例如棉纱和蒸汽机所需煤炭。可变资本这一术语由马克思首创，指资本家购买的劳动力。之所以被称为“可变”，是因为它是资本中价值会增加的组成部分；与不变资本不同，其价值并非固定不变，而是会增长。不变资本的价值得以保留，只是把它现有价值原封不动地转移到最终产品上。

关于不变资本，马克思指出：

这不会对价值创造或价值量变化产生任何影响。卢克莱修所言不言自明：“nilposse creari de nihilo”，即“无中不能生有”。价值的创造就是把劳动力转化为劳动。劳动力本身是通过营养物质转移到人体内的能量。[47]

多么精彩的段落！它把马克思博士论文的研究主题——原子论，与“功率”“功”和“能量”等概念融为一体。

the economic consensus moved away from Smith’s conclusions. To contemporary readers, the reference to labour as the common element may appear arbitrary. However, Marx’s initial step of proving that exchange is an equivalence relation with a conserved substance is crucial for the subsequent phase of his argument.

He symbolically analyses the circulation of commodities with the formula:

$C_1 \to M \to C_2$

A seller starts out with commodity 1 (C_1) exchanges it for money *M* of equal value, and then uses the money to purchase an amount of commodity 2 (C_2). Since exchange is an equivalence relation and as such is transitive, the value of the start and finish are the same $C_1 \equiv C_2$. All is fair and good until he points out that this is not what capitalists do. Their activity takes the form

$M \to C \to M'$

Where the final quantity of money is greater than what he starts out with: $M' > M$. The difference $S = M' - M$ is what Marx calls the capitalist’s *surplus value*. It is evident that $M \to C \to M'$ violates the equivalence property of commodity exchange.

Previous economic writers had not analysed commodity exchange as a formal system subject to conservation laws. By using a formal analysis, Marx was able to spot a contradiction or symmetry breaking inherent in the existence of profit.[46] How is this possible? p. 85–86

If value is preserved in exchanges, then profit or surplus value cannot be accounted for at the level of commodity exchange. It must be explained by factors external to the exchange, specifically, the actual process of commodity production. The capitalist spends their initial *M* on two components that Marx labels constant capital *c* and variable capital *v*. So $M = c + v$. Constant capital consists of the typical commodities that a capitalist acquires, such as cotton yarn and coal for the steam engine. Variable capital, a term coined by Marx, refers to the labour power purchased by the capitalist. It is deemed ‘variable’ because it is the component of the capital that increases in value; unlike constant capital, its value is not fixed but grows. The value of constant capital remains preserved, simply transferring its existing value to the final product without alteration. p. 86

Of constant capital, he says:

this will have no influence on the creation of value or on the variation in the quantity of value. What Lucretius says is self-evident; ‘nil posse creari de nihilo,’ out of nothing, nothing can be created. Creation of value is transformation of labour-power into labour. Labour-power itself is energy transferred to a human organism by means of nourishing matter.[47]

What a great passage! It brings together Atomism that was the subject of Dr Marx’s PhD with concepts of power, work and energy.

像斯密这样的 18 世纪历史唯物主义者仅把工资视为劳动报酬，而马克思能区分“功率”与“功”。如果劳动是一种商品，且商品按其价值——即其劳动含量——进行交换，那么我们就面临一个矛盾。

Where a historical materialist of the eighteenth century like Smith just saw in wages a payment for labour, Marx could make the distinction between power and work. If labour were a commodity and if commodities exchanged in proportion to their value, their labour content, then we have a contradiction.

劳动本身的价值究竟是什么？

What is the value of labour itself?

功与功率

Work and power

乍看之下，人们可能会说一小时劳动的价值仅仅是一小时。然而，政治经济学家们认为，其价值实际上由生产工人该小时所需生活资料——食物和衣物——所需要的劳动量决定。这种观点在未承认其内在矛盾的情况下一直被沿用。

At first glance, one might say the value of an hour’s labour is simply one hour. However, political economists argue that its value is actually determined by the amount of labour needed to produce the sustenance –food and clothing –that a worker requires for that hour. This perspective was maintained without acknowledging the inherent contradiction.

亚当·斯密及其继承者李嘉图提出理论时，劳动主要依靠人力。马克思后来提出的把“劳动”与“劳动力”区分开来的概念，在当时难以想象。当马克思著书立说之时，经过一个多世纪机械技术进步，蒸汽机已成为英国工业主要能源。为了区分产品，蒸汽机制造商以马力为单位对机器进行评级，从而提供一种与竞争对手发动机进行比较的标准衡量方式。

Adam Smith and his successor, Ricardo, developed their theories during a time when labour was predominantly human-powered. The concept that Marx later introduced, distinguishing labour from labour power, was inconceivable then. By the time Marx was writing, after over a century of mechanical progress, steam engines had become the primary source of energy for British industry. To differentiate their products, steam engine manufacturers rated their machines in terms of horsepower, providing a standard measure to compare against competitors’ engines.

p. 87

蒸汽机最初任务是从矿井中运出煤炭，或把矿井中水抽出来。据估算，一匹马在绞盘上每秒可把 550 磅重量提升 1 英尺。如果一台蒸汽机能在 1 秒内把同等重量煤炭提升 10 英尺，那么它就是一台 10 马力蒸汽机，以此类推。

The initial task of steam engines was hauling coal from mines, or pumping water from mines. It was reckoned that a horse on a windlass could raise a 550 lb weight at 1 foot a second. If a steam engine could raise the same weight of coal 10 feet in a second then it was a 10 horsepower engine, etc.

随着机械工程实践发展，所做功与作为做功能力的功率之间的区别变得显而易见。通过简单量纲分析可发现，所做功等同于能量。[48]

As a result of practical mechanical engineering, the distinction between work done and power as the ability to do work became evident. A little dimensional analysis revealed that work done was the same as energy.[48]

功被等同于能量，而功率被定义为在特定时间单位内所做的功。在维多利亚时期英国，能量或功以英尺-磅为单位量化，而功率以马力为单位衡量。一个专门工程领域应运而生，旨在通过示功图准确测量蒸汽机功率输出。这些图是通过把发动机活塞连接到纸张 x 轴上自动绘制的曲线图，同时与压力表相连的铅笔垂直移动以记录压力变化。这些图被用来向持怀疑态度的买家证明发动机功率。

Work was equated with energy, and power was defined as the amount of work done in a specific time unit. In Victorian England, energy or work was quantified in foot-pounds, while power was measured in horsepower. A specialized field of engineering emerged to accurately measure the power output of steam engines through indicator diagrams. These diagrams were graphs automatically drawn by connecting the engine’s piston to the x-axis of a paper, while a pencil, linked to the pressure gauge, moved vertically to record the variations. These diagrams were used to prove, to dubious buyers, the power of the engine.

在 21 世纪，解读示功图或衡量蒸汽机功率似乎是冷门话题。然而在 19 世纪，蒸汽技术是当时最前沿技术。对蒸汽技术的了解，就如同我们今天对计算机的理解一样，蒸汽时代的概念和计量单位对维多

In the twenty-first century, interpreting indicator diagrams or gauging the power of steam engines may seem like obscure subjects. However, in the nineteenth century, steam was the cutting-edge technology of the era. Knowledge of steam was akin to our understanding of

利亚时代的人来说，就像今天的千兆字节和像素一样熟悉。

马克思对“劳动力”（即劳动能力）与“劳动本身”（即所完成的劳动）的区分，对于 1860 年代读者来说立即可理解。在机械工程语境中，这些区分司空见惯。恩格斯拥有一家蒸汽动力工厂，想必对这些概念了如指掌。马克思和恩格斯，本质上就是蒸汽朋克式共产主义者，他们用能量、力与功率的术语来阐述思想。

以一位租用 10 马力蒸汽机的矿主为例。他每小时能否实际完成 $10 \times 60 \times 60 \times 550 = 19{,}800{,}000$ 英尺‑磅的煤炭提升功，取决于他让蒸汽机运转的强度。如果运到竖井底部的煤车数量不足，就会出现蒸汽机闲置时段。因此，蒸汽机功率与实际所做的功被理解为两件不同事情。

从马克思把劳动，即所做的功，引入为价值实体的那一刻起，他就把它等同于劳动力——劳动能力——的耗费。由此，他得出对 $M \rightarrow C \rightarrow M'$ 过程所隐含矛盾的解决之道。马克思认为，资本家雇佣一名工人工作一天，劳动力价值由维持该工人在此期间生存所需生活资料决定。然而，实际工作时长可能不同。正如矿主会尽可能长时间利用蒸汽机一样，资本家对工人也是如此。他们支付报酬仅够工人维持每日生存。马克思指出，如果六小时劳动就足以生产工人生存所需必需品，那么资本家就不受任何必须在六小时后让工人下班的强制。相反，资本家会设法把工作日延长到最大程度。

因此，工作日并非一个常量，而是一个变量。当然，其中一部分由劳动者自身劳动力再生产所需工作时间决定。但其总时长则随剩余劳动的持续时间而变化。因此，工作日虽然可以确定，但就其本身而言却是未定的。

这一上限由两方面条件决定。首先，是劳动力的生理极限。在自然日 24 小时内，一个人只能消耗一定量生命力。同样，一匹马每天也只能工作 8 小时。在一天中的部分时间里，这种生命力必须休息、睡眠；在另一部分时间里，人必须满足其他生理需求，如进食、洗漱和穿衣。除了这些纯粹生理限制外，工作日长度延长还面临道德限制。劳动者需要时间满足其智力和社会需求，而这些需求的范围和数量取决于社会进步总体状况。因此，工作日长度变化在生理和社会界限内波动。但这两种限制性条件都具有极大弹性，允许最大自由度。于是，我们发现工作日有 8、10、12、14、16、18 小时之长，即长度千差万别。[49]

computers, with steam-era concepts and metrics as familiar to Victorians as gigabytes and pixels are to us today.

Marx’s differentiation between labour power ‑the capacity to work ‑and labour itself ‑the work performed ‑would have been instantly comprehensible to a readership in the 1860s. These were commonplace distinctions within the context of mechanical engineering. Engels, owning a steam-powered mill, would have been well-versed in these concepts. Marx and Engels, in essence, were the steampunk communists, articulating in a vernacular of energy, force and power.

Take the example of a mine owner who hired a 10-horsepower steam engine. Whether he actually got $10 \times 60 \times 60 \times 550 = 19{,}800{,}000$ foot-pounds of coal lifting done each hour would depend on how intensively he worked the engine. If there were not enough truckloads of coal brought to the foot of the shaft, there would be periods when the engine stood idle. So the power of the engine and the work actually done were understood to be two different things.

From the very moment that Marx introduces labour, work done, as the substance of value he equates this with the expenditure of labour power ‑the ability to work. From this, he derives his resolution to the contradiction implicit in $M \rightarrow C \rightarrow M'$. According to Marx, a capitalist employs a labourer for a day, with the labour power’s value determined by the sustenance needed to keep the worker alive for that period. However, the actual hours worked can vary. Similar to a mine owner utilizing a steam engine for the longest possible duration, the capitalist does the same with his workers. They are paid just enough for daily survival. Marx suggests that if six hours of labour suffice to produce the necessities for a worker’s survival, there is no compulsion for the capitalist to release the workers after six hours. Instead, the capitalist aims to extend the working day to its fullest extent.

The working day is thus not a constant, but a variable quantity. One of its parts, certainly, is determined by the working-time required for the reproduction of the labour-power of the labourer himself. But its total amount varies with the duration of the surplus labour. The working day is, therefore, determinable, but is, per se, indeterminate. p. 88

This maximum limit is conditioned by two things. First, by the physical bounds of labour-power. Within the 24 hours of the natural day a man can expend only a definite quantity of his vital force. A horse, in like manner, can only work from day to day, 8 hours. During part of the day this force must rest, sleep; during another part the man has to satisfy other physical needs, to feed, wash, and clothe himself. Besides these purely physical limitations, the extension of the working day encounters moral ones. The labourer needs time for satisfying his intellectual and social wants, the extent and number of which are conditioned by the general state of social advancement. The variation of the working day fluctuates, therefore, within physical and social bounds. But both these limiting conditions are of a very elastic nature, and allow the greatest latitude. So we

find working days of 8, 10, 12, 14, 16, 18 hours, ie of the most different lengths.[49]

马克思把工人除赚取工资所需劳动时间之外必须额外劳动的时间定义为“剩余劳动”。他断言，所做剩余劳动与资本家攫取的剩余价值之间存在直接关联。基于这一基本原理，马克思推导出他所观察到的资本主义社会的一系列经验特征。

Marx defined the additional time a worker must labour beyond what is needed to earn their wages as surplus labour. He asserted that there is a direct correlation between the surplus labour done and the surplus value seized by the capitalist. From this fundamental principle, Marx inferred a range of empirical characteristics of the capitalist society he observed.

• 由于资本家试图把利润最大化，而利润又取决于剩余劳动，因此他们不断推动把工作日延长至人类耐力极限。马克思把此称为绝对剩余价值。

• Since the capitalists try to maximize profit, and since this depends on surplus labour, they constantly push to extend the working day to the limits of human endurance. This he termed absolute surplus value.

• 资本家实现利润最大化的另一种途径是减少生产工资购买的生活资料所需时间。如果通过织布机械化，为工人提供服装所需工作可以在更短时间内完成，那么在任何给定工作日中，归属于资本的部分就会增加。由此产生不断推动机械化的动力。马克思把此称为相对剩余价值。

• The other way they can maximize profit is by reducing the hours required to produce wage goods. If, through mechanization of weaving, the work required to clothe the workers can be performed in fewer hours, then the share of any given working day falling to capital will increase. From this stemmed the constant drive towards mechanization. This he called relative surplus value.

• 类似于牛顿的轨道力学，马克思表述了一个维持资本对劳动支配的循环过程（见图 5.1，该图根据 Cockshott、Cottrell 与 Michaelson 计算的数据绘制[50]）。他把这称为资本积累的一般规律。这是支配资本主义社会的运动规律。

• Analogous to Newton’s orbital mechanics, he formulates (see Figure 5.1 drawn from data computed in Cockshott, Cottrell and Michaelson[50]) a rotating process by which the dominance of capital over labour is maintained. This he called the General Law of Capital Accumulation. It was the governing law of motion of capitalist society.

牛顿力学所研究的行星系统演变在很大程度上是周期性的。各种周期相互交织：地球自转周期影响每日潮汐，而月球轨道改变高潮发生时间。年度周期决定四季更替，更长周期控制地球自转轴及其轨道长轴岁差。如今人们认识到，正是这些漫长米兰科维奇周期引发冰河时代。

The progression of planetary systems, as studied by Newtonian mechanics, was largely cyclical. Various cycles overlapped: the Earth’s rotation cycle influenced daily tides, while the lunar orbit altered the timing of high tides. The annual cycle dictated the seasons, and even longer cycles controlled the precession of Earth’s axis and its orbital major axis. It is now understood that these extended Milankovitch cycles are responsible for initiating Ice Ages.

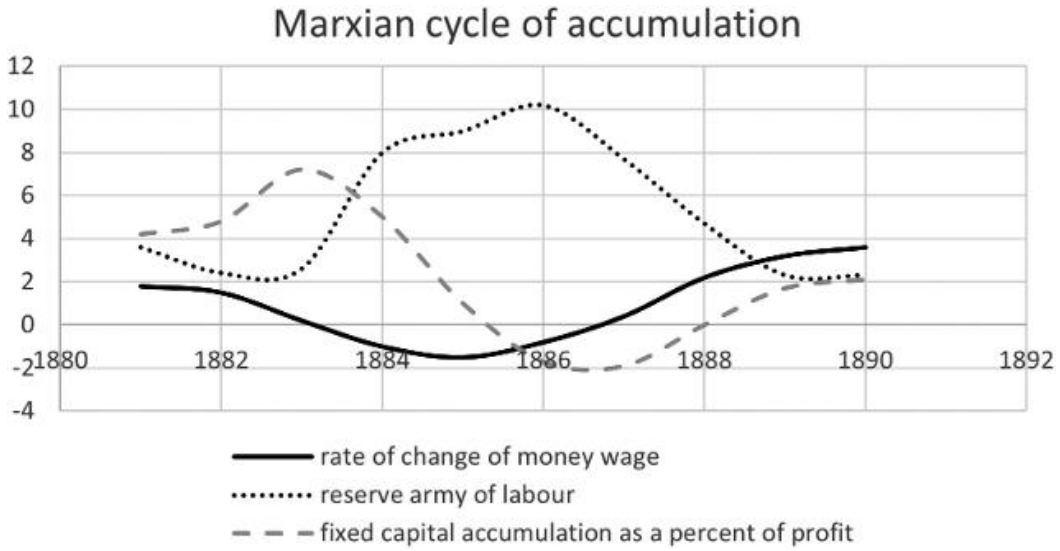


图 5.1 以维多利亚时期的一个经济周期说明马克思的资本积累一般规律。该规律描述了一个循环过程：相对充分就业的时期（1881—1883 年）使工资得以上涨。资本家的应对是投资于节省劳动的机器（1882—1885 年）。这使工人加入劳动后备军，并导致工资下降（1883—1887 年）。在这一较低工资水平上，资本家再次雇佣更多工人，工资开始上涨，循环重新开始。这样的周期通常略短于十年。作者绘图。

Figure 5.1 Illustration of Marx’s General Law of Capital Accumulation using a Victorian economic cycle. The law describes a cyclical process whereby a period of relatively full employment (1881–3) allows wages to rise. The capitalists respond by investing in labour-saving machinery (1882–5). This throws workers onto the reserve army of labour and causes wages to fall (1883–7). At this lower wage, the capitalists again take on more workers, wages start to rise and the cycle repeats. The cycle typically took just under a decade. Figure by authors.

Marxian cycle of accumulation：马克思式积累周期；实线 rate of change of money wage：货币工资变化率；点线 reserve army of labour：劳动后备军；虚线 fixed capital accumulation as a percent of profit：固定资本积累占利润的百分比；横轴数字为年份。

对马克思而言，经济世界也是类似周期叠加：商品周期 $C \to M \to C$ 以及基本资本周期 $M \to C \to M'$ 相互交织，且速度极快，通常以天或周为单位。在此之上，还有由他的“一般规律”驱动的周期，持续时间约十年。除此之外，还存在一个漫长过程，其间其他趋势逐渐显现：利润率下降以及工业无产阶级社会和政治影响力稳步增强。资本主义制度兴衰时间线——从 16 世纪所谓“原始积累”到最终革命危机——跨越数百年。各种生产方式都融入一个“大周期”，从原始共产主义演进到高级共产主义。对马克思而言，这些周期是向上螺旋的，未来共产主义将超越其前身，而推动这一进程的正是他用另一个机械隐喻所称的“生产力”。

For Marx, the economic world was a similar superposition of cycles: the commodity cycle $C \to M \to C$ and the basic capital cycle $M \to C \to M'$ interleaved and were fast, of the order of days or weeks. On top of this came the cycle driven by his General Law, which lasts a decade or so. Beyond that, there was a protracted process through which other trends emerged: a declining rate of profit and the steadily increasing social and political influence of the industrial working class. The timeline for the rise and fall of the capitalist system, from the so-called primitive accumulation in the sixteenth century to a final revolutionary crisis, spanned hundreds of years. The modes of production fit into a Grand Cycle, evolving from primitive to advanced communism. For Marx, these cycles were upward spirals, with the future communism surpassing its predecessor, propelled by what he termed, in another mechanical metaphor, the productive forces.

任何社会制度，在其中所能容纳的所有生产力都得到发展之前，绝不会消失；而新的、更高的生产关系，在支撑其存在的物质条件于旧社会子宫中成熟之前，也绝不会出现。因此，人类总是只去解决那些它能够解决的问题，因为如果我们更仔细地观察，就会发现：一个问题本身只有在解决该问题所必需的物质条件已经存在，或者至少正在形成过程中，才会出现。从大体上讲，我们可以把亚细亚的、古代的、封建的和现代资产阶级的生产方式，视为社会经济形态演进过程中的若干阶段。资产阶级生产关系是社会生产过程中最后一种对立形式——这种对立并非指个体间对立，而是源于个体在社会中生活所处环境条件；与此同时，在资产阶级社会内部发展起来的生产力，

No social order ever dissappears before all the productive forces, for which there is room in it, have been developed; and new, higher relations of production never appear before the material conditions of their existence have matured in the womb of the old society. Therefore, humanity always takes up only such problems as it can solve, since, looking at the matter more closely we will always find that the problem itself arises only when the material conditions necessary for its solution already exist or are at least in the process of formation. In broad outline we can designate the Asiatic, the ancient, the feudal, and the modern bourgeois methods of production as so many epochs in the process of the economic formation of society. The bourgeois relations of production are the last antagonistic

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为解决这种对立创造了物质条件。因此，这种社会形态构成了人类社会史前阶段的最后一章。[51]

form of the social process of production –antagonistic not in the sense of individual antagonism, but of one arising from conditions surrounding the life of individuals in society; at the same time the productive forces developing in the womb of bourgeois society create the material conditions for the solution of that antagonism. This social formation constitutes, therefore, the closing chapter of the prehistoric stage of human society.[51]

修辞与发现的区别

Rhetoric versus discovery

p. 90

熟悉当前马克思主义哲学辩论的读者会发现，我们所阐述的牛顿式和机械论式马克思，与近几十年来逐渐占据上风的某些新黑格尔主义对马克思的诠释截然不同。这类诠释多种多样，但其中一个特别清晰的例子见于……[52]

〔校注：底本句末 given by 后未写作者姓名而直接接原注 52；译文保留省略形式，由原注标明所引文献。〕

Those of our readers who are familiar with current debates in Marxian philosophy will see that the account we have given of a Newtonian and mechanistic Marx is radically different from some neo-Hegelian presentations of Marx that have gained ground in recent decades. These accounts are multiple, but a particularly clear example is given by.[52]

斯塔罗斯塔（Starosta）很坦率。他仅声称自己在探讨马克思在《资本论》中的论述，而非其思想来源。这对于该书某一章的表述形式而言，是一份相当称职的阐释。与非黑格尔式马克思论述联系最紧密的马克思主义哲学家是阿尔都塞，他主张马克思在 1840 年代与黑格尔主义发生根本性概念断裂。作为反对阿尔都塞“在《德意志意识形态》时期就已与黑格尔发生认识论断裂”这一论点的论者，斯塔罗斯塔毫无用处。他只是轻率地假定了一种连续性，例如：

Starosta is honest. He only claims to be dealing with Marx’s exposition in Capital, not the sources of his ideas. It is a reasonably competent account of the form of presentation in one chapter of that book. The Marxist philosopher most strongly associated with non-Hegelian accounts of Marx was Althusser with his claim that there was a fundamental conceptual break from Hegelianism in the 1840s. As an arguer against the Althusserian thesis that there was an epistemological break from Hegel at the time of the German Ideology, Starosta is useless. He just blithely assumes a continuity, for example:

> 正如他在《阿道夫 • 瓦格纳边注》中明确指出，马克思既没有把政治经济学概念，也没有把任何其他概念作为出发点（马克思，1975，198），以此来发现异化劳动。[53]

> and as he clearly states in the Marginal Notes on Adolf Wagner, Marx takes as a point of departure neither the concepts of political economy nor any concept whatsoever (Marx, 1975, 198), in order thereby to discover alienated labor[53]

然而，在关于瓦格纳的笔记中，根本没有提到异化劳动。[54]这篇笔记最初于 1972 年由一份阿尔都塞主义期刊（《理论与实践》）翻译成英文，目的正是为了证明所代表的认识论断裂是彻底的。作者若能阅读阿萨尔 • 侯赛因（Athar Hussain）为该英文译本所作原始导言，本会明智得多。[55]

Well, there is no mention whatsoever of alienated labour in notes on Wagner.[54] This was originally translated into English in 1972 by an Althusserian journal (Theoretical Practice) precisely in order to demonstrate the completeness of the epistemological break. The author would have been wise to read the original introduction to the English translation by Athar Hussain.[55]

随后，他便简单假设“异化劳动”这一概念在《资本论》中被使用：

He then goes on to simply assume that the notion of alienated labour is used in Capital:

> 从这个意义上说，马克思在《资本论》中的论述并非朝着发现异化方向推进，而是从辩证探究的分析阶段所揭示的、作为其最抽象和最一般形式的异化出发。

> In this sense, Marx’s exposition in Capital does not advance towards the discovery of alienation but starts from what the analytic stage of the dialectical inquiry revealed as its most abstract and general form

在《资本论》第一卷的英译本中，alienation 一词只以亚当 • 斯密所用的旧苏格兰法律含义出现，即财产的出售或赠与。卖掉房屋，就是将它让渡出去（alienate）。这就是该词在《资本论》中的含义。例如，以下几段引自该书：[56]

In the English translation of Capital I the word ‘alienation’ only occurs in the old Scottish legal sense that Adam Smith uses –meaning sale or gift of property. You alienate your house if you sell it. This is the sense in Capital. For example, these passages from[56]

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> 它通过商品的普遍让渡，通过实际上与其作为有用物的自然形式互换位置，从而在现实中成为其价值的化身，才变成了真正的货币。

> It became real money, by the general alienation of commodities, by actually changing places with their natural forms as useful objects, and thus becoming in reality the embodiment of their values.

但随着流通发展，会出现这样一种情况：商品的出让与价格的实现之间，因时间间隔而分离。

《资本论》第三章德语原文使用的是“Veräußerung”（出售），而非“Entfremdung”——后者在马克思早期著作英译本中被译为“alienation”（异化）或“estrangement”（疏离）。这绝非《1844 年经济学哲学手稿》（EPM）中“异化劳动”（entfremdete Arbeit）所涉及的问题，即使它们在英文中可能被译为相同词。

斯塔罗斯塔还宣扬了一种目前虽流行但颇为奇怪的解释，即认为马克思在讨论商品时，仅指资本主义社会：

正如如今已得到广泛承认的那样，这个出发点并不是源自恩格斯、又经斯威齐和米克等作者普及的正统解释所认为的那种简单商品生产社会——无论它是理想类型，还是更糟，是历史上实际存在过的社会。

这一点极具争议。正如米克尔（Meikle）所指出的，马克思提出的使用价值与交换价值之间的基本区分源自亚里士多德。[57]马克思在脚注和正文中明确把许多思想归功于亚里士多德。特别值得一提的是，由于这直接关系到价值形式，马克思非常明确指出，他在《资本论》中提出的等价形式源自亚里士多德：

如果我们回溯到那位最先分析了如此多种形式——无论是思想、社会还是自然的形式，其中也包括价值形式——的伟大思想家，那么等价形式的后两项特殊性就会变得更易理解。我指的是亚里士多德。

首先，他明确指出，商品的货币形式仅仅是简单价值形式的进一步发展——即用任意另一种商品来表示一种商品的价值；因为他说：

5 张床＝ 1 栋房子与“5 张床＝若干货币”并无区别。

他进一步指出，产生这种表达方式的价值关系，要求房屋在质上必须与床等同；若无这种等同，这两种显然不同的事物便无法作为可通约的量相互比较。他说：“没有相等就不能交换，没有可通约性就不能相等。”

……然而，他到此便停下来，放弃了对价值形式的进一步分析。“然而，实际上，如此不同的事物不可能具有可通约性”——即在质上相等。这种等同只能是与其真实本质相悖的，因此仅是“一种为实际目的而采取的权宜之计”。

因此，亚里士多德本人告诉我们，是什么阻碍了他进一步分析；那就是缺乏任何价值概念。究竟是什么等价物、什么共同实体，使得床的价值能够用一所房子来表达？亚里士多德说，事实上，这样的东西不可能存在。为什么呢？

But with the development of circulation, conditions arise under which the alienation of commodities becomes separated, by an interval of time, from the realisation of their prices.

The German original of Capital Chapter 3 uses veräußerung or sale, not entfremdung, the word that is translated as alienation or estrangement in English translations of the early writings. This is by no means the problematic of ‘alienated labour’ ‘entfremdete Arbeit’ in the EPM, even if they may be given the same English translation.

Starosta also propagates the currently fashionable, but rather odd, interpretation that when discussing commodities, Marx was only talking about capitalist society:

As has now been widely acknowledged, this starting point is not an ideal-typical -or worse, historically existent -simple commodity - producing society, as in the orthodoxy derived from Engels and popularised by authors such as Sweezy and Meek.

That is highly disputable. The basic distinction between use value and exchange value that Marx presents is drawn from Aristotle as was shown by Meikle.[57] Marx explicitly credits many of his ideas to Aristotle in footnotes and body text. Particularly relevant, because it relates directly to the value form, Marx was quite specific that the equivalent form of value that he presents in Capital was derived from Aristotle:

The two latter peculiarities of the equivalent form will become more intelligible if we go back to the great thinker who was the first to analyse so many forms, whether of thought, society, or Nature, and amongst them also the form of value. I mean Aristotle.

In the first place, he clearly enunciates that the money form of commodities is only the further development of the simple form of value -ie of the expression of the value of one commodity in some other commodity taken at random; for he says:

5 beds = 1 house is not to be distinguished from 5 beds = so much money.

He further sees that the value relation which gives rise to this expression makes it necessary that the house should qualitatively be made the equal of the bed, and that, without such an equalisation, these two clearly different things could not be compared with each other as commensurable quantities. ‘Exchange,’ he says, ‘cannot take place without equality, and equality not without commensurability’.

. . . Here, however, he comes to a stop, and gives up the further analysis of the form of value. ‘It is, however, in reality, impossible , that such unlike things can be commensurable’ -ie qualitatively equal. Such an equalisation can only be something foreign to their real nature, consequently only ‘a makeshift for practical purposes.’

Aristotle therefore, himself, tells us what barred the way to his further analysis; it was the absence of any concept of value. What is that equal something, that common substance, which admits of the value of the beds being expressed by a house? Such a thing, in truth, cannot exist, says Aristotle. And why

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与床相比，房子确实代表了与床等价的东西，因为它代表了床和房子中真正等价的东西。那就是——人的劳动。

然而，有一个重要事实阻碍了亚里士多德认识到：赋予商品以价值，仅仅是把一切劳动表述为等值的人类劳动，进而表述为质上相等的劳动的一种方式。希腊社会建立在奴隶制基础上，因此其自然基础就是人与人之间及其劳动能力的不平等。价值的表达之谜——即各种劳动之所以平等且等价，是因为且仅在它们作为一般人类劳动范围内——只有当人类平等观念已经取得一种普遍成见的牢固性之后，才能被破译。

因此，马克思指出：

• 他对价值形式矛盾的分析源自亚里士多德。

• 商品生产和商品价值早在资本主义出现之前就已存在于古代雅典。

• 价值的奥秘——即价值本质上就是人类劳动——对于当时奴隶主阶级成员来说不可想象——尽管即使在当时，价值来源也是人类劳动。正是作为奴隶制拥护者的阶级盲目性，阻碍了他认识到这一点。

马克思和亚里士多德把交换视为等价关系是合理的，因为他们认识到在不同商品交换中必须存在一个共同要素。针对马克思未能把劳动确立为这种共同实体的批评，斯塔罗斯塔提出异议，他认为马克思确实指出了共同实体，并从辩证角度引入了劳动这一概念，其依据正是为社会需求而进行的私人劳动的特性。然而，马克思这种做法可被视为一种辩证策略，虽然看似合理，但未必在科学上严谨。历史实例，如解释燃烧的“燃素”[58]理论，表明看似合理且符合唯物主义的解释最终仍可能被证明是错误的：

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易燃矿物[59]富含燃素，不与水结合，但纯净时可溶于油脂；遇火时，它们会冒烟，通常会燃烧，大部分会被烧尽，有时甚至完全消失。

〔校注：本段及以下历史化学引文含 phiogifton、confumed、diffolve、fix、finell 等底本拼写，英文按原书照录；中文结合上下文及原注翻译，不将其当作现代术语。〕

金属在纯净状态下完全不溶于水；只有少数几种能溶于油，且仅在部分失去其燃素时才可溶解。它们是已知物质中最重的，其中最轻的重量也超过同等体积水的六倍。

燃素化的矾酸（挥发性矾酸[60]）经常从火山口喷出；其气味令人窒息而刺鼻。与燃素及热质的结合使它成为气态，但并不妨碍它与水结合。

有些人把硝酸排除在矿物界之外，因为他们认为它是有机体腐烂产生的。但这些有机体一旦失去生命，又会被

not? Compared with the beds, the house does represent something equal to them, in so far as it represents what is really equal, both in the beds and the house. And that is ‑human labour.

There was, however, an important fact which prevented Aristotle from seeing that, to attribute value to commodities, is merely a mode of expressing all labour as equal human labour, and consequently as labour of equal quality. Greek society was founded upon slavery, and had, therefore, for its natural basis, the inequality of men and of their labour powers. The secret of the expression of value, namely, that all kinds of labour are equal and equivalent, because, and so far as they are human labour in general, cannot be deciphered, until the notion of human equality has already acquired the fixity of a popular prejudice.

So Marx is saying:

• That his analysis of the contradiction of the value form is derived from Aristotle.

• That commodity production and commodity value existed in ancient Athens ‑long before capitalism.

• That the secret of value, that it is human labour, was inconceivable to a member of the slaveholding class of that time ‑even though the source of value even then was human labour. It was the class blindness of an advocate of slavery that blocked him from realizing this.

Marx and Aristotle were justified in treating exchange as an equivalence relation because they recognized the necessity of a common element in the exchange of dissimilar commodities. The criticism that Marx fails to establish labour as this common substance is contested by Starosta, who argues that Marx does indeed identify a common substance and dialectically introduces labour as such, justified by the characteristics of labour conducted privately for societal needs. However, this approach by Marx can be seen as a dialectical manoeuvre that, while plausible, is not necessarily scientifically rigorous. Historical examples, such as the phlogiston[58] theory of combustion, demonstrate that plausible and materialist explanations can ultimately be proven incorrect:

INFLAMMABLE fossils[59] abound with phiogifton, do not unite with water, but when pure dissolve in oils; exposed to the fire, they smoke, generally inflame, are for the most part confumed, and sometimes, totally vanish.

METALS when perfect do not diffolve at all in water ; only a few of them in oils, and then only when in part deprived of their phlogiston. They are the heaviest of all known substances, the lightest of them weighing more than fix times its bulk of water.

Phlogisticated vitriotic ACID (volatile vitriolic àcid[60]) is frequently thrown out by the craters of volcanoes ; its finell suffocating and penetrating, The union to phlogiston and the matter of heat gives it an aerial form, but does not prevent its union with water.

Nitrous ACID is by some excluded from the foffil kingdom, because they suppose it to be produced from the putrefaction

重新纳入矿物界，而它们原本较不易挥发的部分正是源自那里。

〔校注：此处及下文 nitrous acid 是引文所用旧称；原注 61 明确以现代发烟硝酸解释。底本 folds、miore 等拼写原样保留于英文。〕

在人工能制备出的最高浓度状态下，其比重为 1.580。纯净时无色；但因其对燃素的强烈吸引力，必须采取特殊处理手段才能使其呈现这种无色状态。与不同比例燃素结合时，它会形成燃素酸和亚硝气。

〔校注：底本在 attraction 和 so 后留有星号，但本书此处没有相应星号注释；英文保留星号。〕

颜色最深且冒烟的硝酸[61]若在敞口容器中快速煮沸，便能轻易使其无色。部分酸会挥发，同时把过量燃素以亚硝气形式带走。[62]

18 世纪化学家给出的论证在结构上与马克思辩证论述极为相似。共同性质（可燃性、可交换性）和表现形式（火焰、以黄金表示的价格）——由此辩证推导出必然存在一种共同实体（燃素、价值）。但我们知道，燃素说是不正确的，可燃物之间并不存在共同实体；相反，火是由燃料氧化产生的。普里斯特利的“纯燃素”实际上是氢。金属与其“氧化物”之间的差异源于氧化，而非燃素损失。最终发现的共同特征是：可燃物质可以通过氧化转变为较低能量状态。

虽然马克思可能把价值视为可交换商品中的共同实体，并论证劳动在抽象层面上可量化且是商品存在必要条件，但仅凭这一点并不能证实劳动就是那种共同实体。

这并非要质疑马克思正确性；而是可以设想，可能存在另一种被他忽略的共同实体。能量是一个首要候选者，它既具备抽象可量化特性，又能以热能、电能、动能和化学能等多种具体形式表现出来。

马克思推导的正确性并不取决于其理论上的合理性，而在于商品价值是否确实与劳动含量成正比。如果经实测发现，价格与生产商品所需瓦特秒（即焦耳）数量的相关性，远高于与人类劳动人秒数量的相关性，那么马克思观点就是错误的。如今我们已知，从经验上看马克思是正确的。商品价格与劳动人秒相关性远高于与所耗能量焦耳相关性。若是那种情况，马克思关于资本主义运动的定律就只会是能量守恒定律的一个特例，而非更高阶统计力学定律。[63]

〔校注：底本在陈述劳动相关性更强之后，紧接 In that case ... would ...，其指代与段内次序存在歧义。中文保留假设语气；未调换原句顺序或删去该句。〕

当然，马克思并非随意选择劳动作为价值实体。他遵循了从哈勒敦[64]到李嘉图[65]的学术传统——他

of organic bodies. But these bodies when deprived of life are again received amongst the folds, from whence their miore fixed parts were originally derived.

In the most concentrated state that art can procure it, its specific gravity is 1, 580. Colourless when pure; but its strong attraction*to phlogiston renders particular management necessary to procure it so *. With different proportions of phlogiston it forms phlogisticated acid and nitrous air.

The most highly coloured and fuming nitrous acid[61] may readily be rendered colourlets by boiling it hastily in an open veffel. Part of the acid fies off, carrying the fuperabundant phlogiston along with it, in the form of nitrous air[62]

The justification given by eighteenth-century chemists is structurally very similar to Marx’s dialectical presentation. Common properties (combustibility, exchangeability) and forms of expression (flame, price in gold) – dialectical inference there must be a common substance (phlogiston, value). But we know that the phlogiston theory was wrong, that there is no common substance to things that combust; instead, fire is produced by the oxidation of the fuel. Priestly’s pure phlogiston was actually hydrogen. The difference between a metal and its ‘calc’ is due to oxidation, not the loss of phlogiston. The shared characteristic eventually discovered was that combustible substances could transition to a lower energy state through oxidation.

While Marx may have identified value as a common substance in exchangeable commodities and argued that labour is abstractly quantifiable and essential for a commodity’s existence, this alone does not confirm labour as that substance.

It is not a matter of contesting Marx’s correctness; rather, it is conceivable that there exists another common substance he overlooked. Energy is a prime candidate, possessing the qualities of abstract quantifiability and the ability to manifest in various concrete forms, such as heat, electricity, kinetic energy and chemical energy.

The correctness of Marx’s derivation rests not on its intellectual plausibility, but on whether in fact commodity values are proportional to labour content. If it turns out when we measure things that prices correlate much more closely with the number of watt-seconds = Joules needed to make a commodity rather than the number of person-seconds of human effort, then Marx would be wrong. We now know that empirically Marx was right. Commodity prices do correlate much more closely with person-seconds of labour than with Joules of energy used. In that case, Marx’s laws of capitalist motion would just be a special case of the conservation of energy rather than higher order statistical mechanics laws.[63]

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Of course, Marx was not arbitrarily selecting labour as the value substance. He was following a tradition from Khaldūn[64] to Ricardo[65] who had observed, by rough and

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们曾通过粗略而简便的方法观察到，这种必要经验关系确实存在。

作为一种科学方法，马克思做法确实是一条捷径。它未达到现代科学的严谨性标准，依赖辩证法“障眼法”。理想情况下，他本应提供自己的实证数据来证实“劳动是价值本质”这一主张，或者至少援引其前辈的观察论据，以支持“劳动含量决定价格”这一观点。

这些来源——即他的前辈们——并未被隐瞒。《资本论》第四卷题为《剩余价值理论》，整卷都献给了他的前辈及其思想。从中，我们可以辨别出他认为哪些观点正确，他们之间存在哪些概念争论，以及他认为他们的推理在哪些地方存在缺陷。

然而，对于像《资本论》这样兼具科学性和论战性的著作，辩证法或许大有裨益。当目标是说服读者时，辩证修辞可被视为一种有效策略。问题在于，后世把这些方法视为值得效仿的研究标准时，问题便出现了。

马克思的论点之所以成立，是因为他站在了前辈们“巨人的肩膀”上。[66]他得以把斯密和李嘉图此前已奠定的理论基础融入自己的辩证论述之中。

这也是为何此后进一步尝试应用辩证法都是徒劳的。马克思主义经济学中的黑格尔学派未能通过这种方法进行创新或产生新知识，因为该方法本身并非为此目的而设计。它仅仅是把现有知识重新表述为修辞结构罢了。

这一点在关于自然辩证法的那些荒谬论述中尤为明显。这些理论的支持者并未发现任何新东西。相反，他们只是把通过常规研究方法获得的科学知识，用“对立统一”和“量变到质变”等辩证概念术语重新包装了一下。

恩格斯[67]之所以能把水转化为蒸汽作为“量变到质变”以及“辩证飞跃”的例证，完全是因为此前热力学领域已有相关研究。布莱克及其助手瓦特的研究所揭示的，正是热本身可以量化[68]，并且汽化潜热这一现象确实存在。辩证法并非他们这些发现的必要条件。事实上，布莱克教授早在 1761 年就发现了潜热，这比黑格尔时代要早得多。我们提及布莱克有两个原因：其一，他的思想后来被恩格斯作为自然辩证法范例加以阐述；其二，这些思想与马克思在《资本论》中后续论述的一个阶段——剩余价值奥秘——密切相关。

p. 95 马克思通过把劳动视为价值的共同本质，来回应亚里士多德关于使用价值与交换价值的二分法——这一见解源自他从早期政治经济学家那里汲取的智

ready means, that the requisite empirical relation actually existed.

Marx’s approach, as a scientific procedure, was indeed a shortcut. It lacked the modern standards of scientific rigour, relying on dialectical sleight of hand. Ideally, he should have provided his own empirical data to substantiate the assertion that labour is the essence or at least referenced the observational arguments of his predecessors that supported the notion that labour content dictates price.

These sources, his predecessors, are not concealed. The entire fourth volume of Capital, titled Theories of Surplus Value, is dedicated to his predecessors and their thoughts. Within, we can discern which ideas he deemed correct, the conceptual disputes among them, and where he found their reasoning flawed.

However, dialectical methods might offer benefits in a work that is both scientific and polemical, such as Capital. When the goal is to persuade readers, dialectical rhetoric can be judged as an effective strategy. The issue arises when later generations regard these methods as a standard for investigation worthy of emulation.

Marx’s argument only works because he stood on the giant shoulders of his predecessors.[66] He could introduce into his dialectical argument something which had been prepared earlier by Smith and Ricardo.

This is also why further attempts to apply the dialectical method are futile. The Hegelian school of Marxist economists has failed to innovate or generate new knowledge through this method, as it is not designed for such purposes. It merely reformulates pre-existing knowledge into rhetorical constructs.

This is particularly evident in the non-sensical writings on natural dialectics. Proponents of these theories have not uncovered anything novel. Instead, they have repackaged scientific knowledge, obtained through conventional research methods, in the terminology of dialectical concepts like the unity of opposites and the transition from quantity to quality.

Engels[67] is only able to use the transition of water into steam as an example of a change of quantity into quality and as an example of dialectical leap because of prior work in thermodynamics. The investigations of Black and his assistant Watt had to have shown that heat itself was quantifiable[68] and that there was such a thing as a latent heat of steam vapourization. The dialectical method was not essential for their discoveries. Indeed, Professor Black discovered latent heat in 1761, well before Hegel’s time. We reference Black for two reasons: his ideas were later showcased by Engels as examples of natural dialectics, and they pertain to a subsequent stage in Marx’s discourse in ‘Capital’—the mystery of surplus value.

Marx addresses Aristotle’s dichotomy between use value and exchange value by positing labour as the shared essence of value, an insight he gained from ear-

慧。随后，他解决了另一个重大辩证矛盾：剩余价值谜题。如你所知，他首先提出一个悖论：资本的 $m-c-m'$ 循环过程确实存在，但这似乎与商品等价交换的一般原则相冲突。他问道：如果沿途所有商品交换都是等值交换，那么 $m < m'$ 的情况如何可能发生？

lier political economists. He then resolves another significant dialectical contradiction: the enigma of surplus value. If you recall, he first presents the paradox that the circuit of capital m-c-m’ exists, which appears to conflict with the general principle of equivalent exchange of commodities. How, he asks, can you have m<m’ if all commodity exchanges along the way are exchanges of equal value?

正如我们在第 5.3.1 节中所解释的，马克思通过在“劳动”与“劳动力”之间作出概念区分来解决这一问题。他把劳动力定义为劳动能力。马克思认为，资本家用工资购买的是劳动力，而不是劳动本身。剩余价值奥秘在于：劳动力价值由其再生产劳动成本决定——这与所有商品一样——但维持工人一天生活所需劳动量却少于一整天劳动。这部分多余时间，即工作日长度与工资购买的生活资料的劳动价值之间的差额，便成为剩余价值来源。

As we explain in Section 5.3.1, Marx addressed the issue with a conceptual distinction between labour and labour power. He defined labour power as the capacity to work. According to Marx, it is labour power, not labour itself, that the capitalist purchases with wages. The mystery of surplus value is unveiled by the fact that the value of labour power is set by its labour cost of reproduction, similar to all commodities, yet it requires less than a full day’s labour to sustain the worker for a day. The excess time, which is the difference between the working day’s length and the labour value of the wage goods, becomes the source of surplus value.

这种把劳动与劳动力区分开来的观点，是政治经济学领域一项创新。斯密和李嘉图等前辈曾把工资等同于劳动本身价值，其中李嘉图认为，工资与维持工人生活所需劳动量相等。此外，李嘉图还指出工资与构成资本家利润的价值份额之间的反向关系。

This differentiation between labour and labour power was an innovation in political economy. Predecessors like Smith and Ricardo had equated the wage with the value of labour itself, with Ricardo determining that the wage matched the labour needed to sustain the worker. Furthermore, Ricardo identified the inverse relationship between wages and the proportion of value that forms the capitalist’s profit.

马克思在此处的创新有两方面：

1. 从利润推广到剩余价值——其中还包括地租和利息。

2. 把“劳动”与“劳动力”区分开来。

Marx’s innovations here were twofold:

1. Generalization from profit to surplus value –which included also rent and interest.

2. The distinction between labour and labour power.

可以认为，《资本论》中对这一问题的阐述采用了黑格尔式修辞技巧：先从“$c-m-c$”与“$m-c-m'$”之间推导出矛盾，然后通过“劳动力”这一新概念来化解矛盾。但这并不能解释，这种辩证“魔术”究竟是从何处衍生出“劳动力”这一概念的。

It can be argued that the mode of presentation of the issue in Capital uses Hegelian rhetorical tricks: deriving a contradiction between c-m-c and m-c-m’ and then resolving the contradiction in a new concept, labour power. But that does not explain where the dialectical conjuration produces the idea of labour power from.

这一概念显然并非源自黑格尔，在斯密或李嘉图著作中也找不到。相反，如前所述，它源于瓦特（Watt）的概念。瓦特研究为马克思理论奠定了双重基础。瓦特与布莱克（Black）共同发现蒸汽具有潜热后，意识到通过添加独立冷凝器可以显著提高蒸汽机效率。这一洞见不仅确保了他作为工程师的成功，也推动了资本主义工业化进程。马克思指出，手摇磨坊催生了封建领主，而蒸汽磨坊则催生了工业资本家。实际上，瓦特发明对产生马克思所称的“相对剩余价值”起到了关键作用。它促成了蒸汽动力机械的广泛应用，取代了人力劳动，并为马克思提供了揭开剩余价值之谜所需的理论工具。马克思指出，亚里士多德由于受奴隶社会意识形态束缚，未能认识到劳动是价值的共同实体。只有随着经济发展，斯密才看清了这一点。

This idea certainly did not originate from Hegel, nor can it be found in Smith or Ricardo. Rather, as previously mentioned, it stemmed from Watt’s concepts. Watt’s work laid a dual foundation for Marx’s. Watt, having discovered with Black that steam had latent heat, realized he could significantly enhance steam engine efficiency by adding separate condensers. This insight not only secured his success as an engineer but also propelled capitalist industrialization. Marx noted that the hand mill yielded the feudal lord, while the steam mill brought forth the industrial capitalist. Practically, Watt’s invention was pivotal in generating what Marx termed relative surplus value. It enabled the widespread adoption of steam-powered machinery, supplanting human labour and providing Marx with the theoretical tools needed to unravel the mystery of surplus value. Marx observed that Aristotle failed to recognize labour as the common substance of value due to the ideological con-

straints of slave society. It was only through economic advancement that this became apparent to Smith.

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因此，到马克思著书立说之时，工程实践已把“功”与“功率”之间、以及“功”与“做功能力”之间的概念区分，纳入了工业化英国的普遍基础知识范畴。同样，资本主义商业实践中蒸汽机租赁行为，也确立了按月费出租这些机器做功能力的原则。

So by the time Marx is writing, engineering practice has made the conceptual distinction between work and power, between work and the ability to perform work, part of the common basic knowledge of industrial England. Similarly, capitalist commercial practice in the renting of steam engines had established the principle of hiring out the ability of these engines to perform work for a monthly rate.

当斯密著书时，他的熟人瓦特仍在捣鼓原型机，[69]甚至在李嘉图时代，与水磨相比，蒸汽机也相对罕见。但到马克思著书之时，热力学和资本主义文明实践已经为他提供了必要的概念抽象，使他能够区分劳动与劳动力，从而解决了剩余价值问题。

When Smith was writing, his acquaintance Watt was still tinkering with his prototypes,[69] and even in Ricardo’s time steam engines were relatively rare compared to water mills. But by the time Marx was writing, thermodynamics and the practice of capitalist civilization had provided the conceptual abstractions necessary for him to make the distinction between labour and labour power and thus solve the problem of surplus value.

此外，还有一个平行概念——剩余价值率对应于蒸汽机的煤炭回报率：即消耗煤炭与运出煤炭之比。这被称为能源投资回报率，代表了工业资本主义社会在热力学或能源方面的本质限制。正的剩余对于正的相对剩余价值率至关重要。随着我们日益接近化石燃料资本主义危机，这些比率之间关联值得政治经济学家进一步研究。

〔校注：底本将此比率写为“消耗煤炭与运出煤炭之比”，随后称 energy return on investment；本译保留其比率方向，不擅自倒置。〕

Additionally, there is a parallel concept –the rate of surplus value corresponds to the steam engine’s coal return rate: the proportion of coal consumed versus coal hauled. Known as the energy return on investment, this represents the essential thermodynamic or energy limitation of industrial capitalist society. A positive surplus is crucial for a positive rate of relative surplus value. The correlation between these rates warrants further investigation by political economists as we near the fossil fuel capitalism crisis.

此外，我们不应把辩证法修辞策略与实际历史或观念结构混为一谈。辩证法家的成功取决于能否利用现有概念。其方法有效性在于按适当顺序揭示这些概念，并假定听众对所呈现思想有一定了解。在 19 世纪中叶，“劳动作为价值来源”以及“功”与“功率”的区分等概念，对《资本论》普通读者而言，大概已经足够熟悉，足以理解马克思论点。

Furthermore, we should not confuse the dialectical method’s rhetorical manoeuvres with actual historical or ideational structures. A dialectician’s success hinges on the availability of pre-existing concepts to utilize. The effectiveness of their method depends on unveiling these concepts in an appropriate sequence and assumes the audience has some understanding of the presented ideas. In the mid-nineteenth century, concepts such as labour as a value source and the distinction between work and power were presumably familiar enough for the typical reader of ‘Capital’ to grasp Marx’s points.

马克思把某些概念融入政治经济学的独创性并不受质疑。然而，这些基础思想并非源自黑格尔，而是源于更早期的社会研究和工程研究。当今要推进马克思主义科学，就必须研究经济和社会数据中的具体模式，并广泛借鉴各门科学和工程学中的分析工具。辩证法研究，充其量只能提供一种围绕研究发现构建有说服力叙事的方式。然而，这种方法风险重重，因为要取得成功，需要具备与马克思相媲美的非凡修辞才能。若由那些缺乏马克思博士写作造诣和广博知识的人来操作，结果很可能平淡无奇、索然无味。

Marx’s originality in adapting certain concepts to political economy is not under scrutiny. However, the foundational ideas originated not from Hegel but from earlier social and engineering studies. Advancing Marxist science today necessitates studying tangible patterns in economic and social data and drawing extensively from the sciences and engineering for analytical tools. Dialectics study, at best, offers a way to craft compelling narratives around your findings. However, this approach is fraught with risk, as it requires an extraordinary rhetorical talent akin to that of Marx to succeed. In the hands of individuals lacking Dr Marx’s writing prowess and extensive knowledge, the outcome is likely to be dull and uninspiring.

达尔文与目的论的终结

Darwin and the end of teleology

顺便提一句，我正在阅读的达尔文著作绝对精彩绝伦。目的论曾有一个方面尚未被推翻，而如今这一工作已然完成。此前从未有人以如此宏大尝试来论证自然界中的历史演化，更遑论取得如此卓著效果了。[70]

Darwin, by the way, whom I'm reading just now, is absolutely splendid. There was one aspect of teleology that had yet to be demolished, and that has now been done. Never before has so grandiose an attempt been made to demonstrate historical evolution in Nature, and certainly never to such good effect.[70]

达尔文著作至关重要，且恰好符合我的目的，因为它为历史性阶级斗争提供了自然科学基础。……尽管存在种种不足，正是在此处，自然科学中的“目的论”不仅首次遭受致命打击，其理性含义也得到了经验性阐释。[71]

Darwin's work is most important and suits my purpose in that it provides a basis in natural science for the historical class struggle. . . . Despite all shortcomings, it is here that, for the first time, 'teleology' in natural science is not only dealt a mortal blow but its rational meaning is empirically explained.[71]

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达尔文著作中有两个重要主题特别值得关注。第一个是通过自然选择理论对目的论的解构，这一点曾被恩格斯和马克思强调。其次是挑战人类例外论，把人类单纯视为哺乳动物中的又一物种。达尔文基本方法是通过动物行为视角来解读人类行为。他并不回避把类似人类的特质归于动物，而且提出：要理解人类特征，就必须承认我们共同的哺乳动物血统。

Two significant themes in Darwin's work are particularly noteworthy. The first is the dismantling of teleology through his theory of natural selection, a point that Engels and Marx emphasized. The second is the challenge to human exceptionalism, viewing humans simply as another species of mammal. Darwin's fundamental approach was to interpret human behaviour through the lens of animal behaviour. Rather than shying away from attributing human-like qualities to animals, he posited that understanding human traits requires acknowledging our shared mammalian heritage.

对这一观点的抵制很可能源于与早期经济体系相关的意识形态。例如，亚里士多德关于某些个体天生就是奴隶的论断，反映了当时社会状况。同样，依赖马、牛等动物劳动的经济体系，需要明确划分人类与动物界限。人类，如同奴隶主一样，被视为理性生物，而动物或奴隶据称则缺乏这种特质。

The resistance to this perspective likely stems from the ideologies tied to earlier economic systems. For instance, Aristotle's assertion that some individuals were naturally slaves mirrored the societal conditions of his time. Similarly, an economy reliant on animal labour, such as that of horses and oxen, necessitated a clear demarcation between humans and animals. Humans, akin to slave masters, were deemed rational beings, a quality supposedly absent in animals or slaves.

达尔文出身于当时正在崛起的贵格会工业阶级家庭，是奴隶制的坚定反对者，[72]他在撰写相关著作时，蒸汽动力正逐渐取代马力，成为社会驱动力。他既致力于主张白人与黑人的共同人性，也致力于阐明人类与动物作为哺乳动物的共同特征。

Darwin, whose family background was in the rising Quaker industrial class, was a passionate opponent of slavery,[72] and he wrote when steam was displacing horsespower as the motive force of society. He could work both to assert the common humanity of white and black and the common mammalian character of human and animal.

终结目的论

Ending teleology

19 世纪中叶，如何解释漫长地质时间以及地球物种曾多次经历彻底变迁这一事实，成为一项重大挑战。这一过程显然具有方向性，而非循环往复。于是问题随之而来：这该如何解释？同样，从资本主义工业文明角度来看，历史似乎也具有方向性。究竟哪种因果模型能够解释这一现象？

In the mid-nineteenth century, there was a significant challenge in explaining the vast expanse of time and the realization that Earth's species had undergone complete transformations repeatedly. This process was clearly directional, not cyclical. The question arose: how could this be explained? Similarly, from the perspective of capitalist industrial civilization, history appeared to have a direction. What causal model could explain this phenomenon?

目的论是其中一种解决方案。马克思和恩格斯把达尔文对目的论的否定视为一个关键性进展。要理

Teleological theories were one solution. Marx and Engels saw Darwin's rejection of teleology as a pivotal development. To comprehend this, it's essential to under-

解这一点，必须了解达尔文提出其思想时的思想背景。他否定了何种目的论概念？

stand the intellectual milieu in which Darwin formulated his ideas. What teleological concept did he dismiss?

关于物种历史演化的目的论理论中最成熟版本，是由钱伯斯在其 1844 年首次出版的《创世自然史遗迹》中提出的。《创世自然史遗迹》一书极具争议性，因为它普及了赫顿、米勒和莱尔的地质深时观，并将其与基督教视角相调和，从而彻底打破了厄舍主教关于“公元前 4004 年创世”的说法。钱伯斯很难为该书争取到书评，但该书为《物种起源》问世铺平了道路。

The most developed version of a teleological theory of the historical evolution of species had been put forward by Chambers in his *Vestiges of the Natural History of Creation*, first published in 1844. Vestiges was extremely controversial as it popularized Hutton/Miller/Lyell’s geological deep time reconciled with a Christian perspective, breaking decisively with Bishop Usher’s creation in 4004 BC. Chambers found it hard to get it reviewed, but it paved the way for Origin of Species.

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这本书有趣之处在于，当时人们对天文学和地质学了解程度之深。钱伯斯不仅对太阳系规模，而且对半人马座 α 星距离都有相当准确估算，并对我们银河系形状也有相当清晰认识。他知道还有其他许多星系。他对太阳系形成过程描述，与如今学校天文学教科书中内容相差无几。

What is interesting about the book is just how much was known about astronomy and geology at this point. Chambers had a pretty good estimate of the size not only of the solar system but also of the distance to Alpha Centauri, a fair idea of the shape of our galaxy. He was aware that there were multiple other galaxies. His account of how the solar system was formed was not so far off from what you would now get in a school astronomy book.

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书中概述了地质年代大致脉络。令人着迷的是，坎布里亚纪（Cumbrian，并非寒武纪 Cambrian）、泥盆纪、志留纪、石炭纪和白垩纪这些地质年代，并未像如今那样被呈现为带有起止日期的抽象时期，而是直接与英国特定地区岩层相关联——例如泥盆纪位于德文郡，坎布里亚纪位于威尔士，白垩纪位于苏塞克斯郡。

〔校注：底本明确写 Cumbrian (not Cambrian)，并在后文将其对应威尔士；中文保留底本这一区分与地理表述。〕

The broad outlines of geological time are recounted. It is fascinating how the ages: Cumbrian (not Cambrian), Devonian, Silurian, Carboniferous and Cretaceous are not presented as abstract epochs with start and end dates as they are now, but rather are directly related to the rocks in specific parts of Britain, such as Devonian in Devon, Cumbrian in Wales and Cretaceous in Sussex.

他对主要动植物类群出现时间顺序有大致了解，但在地质成因方面，他持有一些当时看似合理、如今却显得荒诞不经的观点。例如，他认为沉积“古红砂岩”的海域深度达数百英里，且直到石炭纪之前都不存在陆地。他提出一个引人入胜假说：在石炭纪之前，二氧化碳 CO_2 浓度过高，陆地动物无法生存。根据他的理论，碳的封存使得陆地动物得以在他所称的“新红砂岩”时期出现，而我们现在把这一时期认定为二叠纪至侏罗纪。

He had a rough understanding of the chronological appearance of major animal and plant classes, but he held what might have seemed plausible, yet now appears to be wild ideas about geological causes. For example, he believed that the seas where the Old Red Sandstone was deposited were hundreds of miles deep and that dry land did not exist until the Carboniferous period. He proposed the intriguing hypothesis that before the Carboniferous, CO_2 levels were too high for land animals to exist. According to his theory, the sequestration of carbon allowed land animals to emerge during what he referred to as ‘the new red sandstone’, which we now identify as the Permian to Jurassic periods.

他提出观点中明显体现了目的论：形成煤层的森林既是为了净化空气以维持动物生命，也是为了提供我们文明所需燃料。

The teleology is evident in his suggestion that the *function* of the forests giving rise to the coal beds was both to purify the air for animal life and to provide the fuel needed for our civilization.

这与他关于动物形态如何产生的理论密切相关。他强调，用他的话说，生命会抓住任何机会。在现代人听来，这似乎是生命为适应生态位而快速进化的观念，但在他看来，这被视为形态规律的发展。他是一位鲜明的自然主义者。生命遵循自然法则而产生，生命形态同样受这些法则驱动。早在 19 世纪初，他就确信电是生命源泉。这与玛丽·雪莱的《弗兰肯斯坦》所探讨主题遥相呼应。

This was tied to his theory of how animal forms arise. He emphasized that life as he put *it presses in* on any opportunities. To a modern ear, this sounds like the idea of life rapidly evolving to fit ecological niches, but in his case, it was seen as the development of the laws of form. He was a pronounced naturalist. Life arises by natural laws and the forms of life are driven by them as well. He was convinced by the early nineteenth century that electricity was the source of life. This echoed the same themes as Mary Shelly’s Frankenstein.

例如，他声称电放电中出现的树枝状或分支图案与树木形态相同，因为在带负电地球与带正电大气之间，起作用的是相同电学定律。这种电势推动了植物生长，并造就了树木、棕榈等植物特征形态。他甚至引用了克罗斯先生进行的一项类似《弗兰肯斯坦》的实验——该先生声称，通过向“硅酸钾”通入强电流，观察到了昆虫生命自发生成。据称，威克斯先生使用“亚铁氰化钾”也成功复现了这一实验。

As an example, he claimed that the dendritic or branching pattern that you get in electric discharges was the same as you get in trees because the same electric laws operate between the negatively charged Earth and the positively charged atmosphere. This electric potential drove plant growth and gave rise to the characteristic shapes of trees, palms, etc. He even cited a Frankenstein-like experiment by a Mr Crosse, who claimed to have observed the spontaneous generation of insect life by passing strong electric currents through ‘silicate of potash’. This was apparently replicated by Mr Weekes using ‘ferro-cyanet of potash’.

这些昆虫据说非常微小，“是一种微小的、半透明的、长有长刚毛的 ascuraus”。

The insects were apparently rather minute, ‘a species of ascuraus minute and semi transparent and furnished with long bristles’.

这一系列实验的引用并不十分令人信服：“参见威克斯先生于 1842 年发行的一份小册子”。

The citation for this series of experiments is not very convincing: ‘see a pamphlet circulated by Mr. Weekes in 1842’.

钱伯斯把自然法则与“摩西经文”相调和的基本方式是：上帝意欲让生命存在，而非亲手创造了每一类生物。相反，宇宙的大建筑师设计了自然法则，以确保行星首先形成，随后生命会通过电作用从无机化合物中自发产生。然后，一旦出现新的物理条件，相应生命形式就会自发发展起来以填补这一空间。

The basic way Chambers reconciles natural law with ‘the Mosaic texts’, is to say that God willed that there be life, not that he created each line of organisms. Instead, natural laws were so designed by The Great Architect of the Universe as to ensure that first planets would come into existence, and then that life would spontaneously be generated by electricity from inorganic compounds. Then, as soon as new physical conditions arose, the appropriate form of life would spontaneously develop to fill the space.

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对现代读者而言，他反复使用“物种作为过渡形式”这一概念令人困惑。如今，我们从达尔文主义角度理解“过渡形式”——始祖鸟作为爬行动物与鸟类之间过渡形式，是指它源自爬行动物，而后来的鸟类则源自它（或其近亲之一）。

To the modern reader, it is disconcerting that he repeatedly uses the idea of species being links. We now think of links in Darwinian terms – *Archaeopteryx* being the link between reptiles and birds in the sense that it was descended from reptiles and that subsequent birds descend from it (or one of its close relatives).

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然而，他所指的“过渡形式”如今已被认为可信度极低。他认为，具有骨质外层的泥盆纪盾皮鱼是甲壳类与鱼类之间过渡形式。他提出，鸵鸟——其膀胱和膈肌与哺乳动物相似，羽毛则类似于毛发——是哺乳动物与鸟类之间过渡形式。他还视鸭嘴兽为鸟类与哺乳动物之间过渡形式。他提出的演化序列是：鱼类、爬行动物（包括两栖动物）、鸟类，然后是哺乳动物。因此，鸵鸟正在演化为哺乳动物的观点与这一序列相吻合。他经常使用时间顺序术语，把鱼类置于首位，而把爬行动物置于更高位置或随后位置，这反映了一种目的论序列，与物种出现的地质记录相吻合。他作品中核心的目的论主题是：存在一种与生俱来的“发展法则”，它引导着更高等生命形式的产生。钱伯斯以蜜蜂为例——蜜蜂幼虫喂养方式决定了它们长大后是成为蜂王、工蜂还是雄蜂——据此推测，胚胎环境条件改变将导致后代发育得更为完善。他声称，一旦摆脱野蛮状态艰辛，生活在欧洲环境中的非洲人不仅会同化于欧洲人的生活方式，而且外貌也会趋同于欧洲人：

However, the links he refers to are now considered to have little credibility. He believes that Devonian placoderms with bone coverings are a transitional form between crustaceans and fish. He suggests that ostriches, with their urinary bladder and diaphragm similar to mammals and feathers resembling hair, are a link between mammals and birds. He also views the platypus as an intermediary between birds and mammals. His proposed sequence is Fish, Reptiles (including amphibians), Birds and then Mammals. Thus, the notion that ostriches are evolving into mammals aligns with this sequence. He frequently employs temporal terms, placing fish first and reptiles either higher or subsequent, reflecting a teleological sequence that coincides with the geological record of species emergence. The central teleological theme of his work was the notion that there was an innate ‘law of development’ that led to the generation of higher forms of life. Using the analogy with honey bees, the feeding of whose larvae determines whether they emerge as queens, workers or drones, Chambers hypothesized that changed conditions in embryo will lead to more developed offspring. He claimed that once removed from the hardships of barbarism, Africans who lived in European

黑人种族的粗犷五官及其他结构上的特殊性，仅在这些人生活在通常与野蛮状态相关联的环境中时才会持续存在。在气候更为温和、社会状态更高级的环境中，面部和体态会变得极为精致。那些拥有一定文明的少数非洲民族，其外貌也呈现出接近欧洲人的特征。[73]

conditions would assimilate not only to the mode of life of Europeans but also to their appearance:

The coarse features, and other structural peculiarities of the negro race only continue while these people live amidst the circumstances usually associated with barbarism. In a more temperate clime, and higher social state, the face and figure become greatly refined. The few African nations which possess any civilisation also exhibit forms approaching the European.[73]

他是一位极端环境决定论者。他认为，居住在欧洲的非洲人第三代后裔之所以看起来更“精致”，并非归因于通婚，而是归因于生活条件。钱伯斯毫不犹豫地断定，他在本国同胞身上最熟悉的那些特征才是最先进的。在他看来，整个自然发展法则似乎以孕育出一位营养充足的英国上层阶级成员为最终目标。

He was an extreme environmentalist. That third-generation descendants of Africans resident in Europe might look more ‘refined’ was not put down to intermarriage, but to living conditions. Chambers unhesitatingly assumed that those features he was most familiar with among his own countrymen were the most advanced. The whole law of development of nature aimed, it seemed to him, to produce as its goal a well-fed member of the British upper classes.

他利用哺乳动物、鱼类、爬行动物等之间胚胎学相似性，论证哺乳动物胚胎分化是内在发育规律的结果，只要条件允许，鱼类胚胎也具备这种能力。

Using the embryological similarity between mammals, fish, reptiles, etc., he argues that the differentiation of a mammalian embryo is the result of an inbuilt developmental law, which, given the appropriate opportunities, a fish embryo is also capable of.

p. 99–100

人类胎儿的躯体中，往往有一处最重要部位发育不完全：例如，心脏仅停留在三腔结构阶段，因此它实际上是爬行动物的心脏。甚至还有这样的例子：该器官仍保留着双腔或鱼类形态。此类缺陷纯粹是由于母体机体发育能力不足所致，而这种不足又源于体质虚弱或生活困苦。在这里，我们显然看到了一种与那些使物种向另一物种转变的条件相反的情况——至少就这一器官而言是如此。既然在这个特定点上出现了完全的物种退化，那么不难想象，一旦有利条件充分增加，便足以逆转这一现象，使鱼类母亲发育出爬行动物心脏，或使爬行动物母亲发育出哺乳动物心脏。推测如下并非过于大胆：若这种“不足”程度被“过度补偿”（前者与后者似乎同样是自然现象），就足以使一只鹅的后代拥有老鼠身体，从而产生鸭嘴兽；或者，使鸭嘴兽后代拥有真正啮齿动物嘴巴和脚，从而在两个阶段内完成从鸟类到哺乳类转变。[74]

A human foetus is often left with one of the most important parts of its frame imperfectly developed: the heart, for instance, goes no farther than the three-chambered form, so that it is the heart of a reptile. There are even instances of this organ being left in the two-chambered or fish form. Such defects are the result of nothing more than a failure of the power of development in the system of the mother, occasioned by weak health or misery. Here we have apparently a realisation of the converse of those conditions which carry on species to species, so far, at least, as one organ is concerned. Seeing a complete specific retrogression in this one point, how easy it is to imagine an access of favourable conditions sufficient to reverse the phenomenon, and make a fish mother develop a reptile heart, or a reptile mother develop a mammal one. It is no great boldness to surmise that a super-adequacy in the measure of this under-adequacy (and the one thing seems as natural an occurrence as the other) would suffice in a goose to give its progeny the body of a rat, and produce the ornithorynchus, or might give the progeny of an ornithorynchus the mouth and feet of a true rodent, and thus complete at two stages the passage from the aves to the mammalia.[74]

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他深信，生命不仅存在于火星和木星上，而且由于这一发育定律，这些生命与地球上生命极为相似；在其他恒星周围无数行星上，也存在着类似生物。这种自负观点，在当今时代，也被《星球大战》制片人所共享。

He was convinced that life existed not only on Mars and Jupiter but that, thanks to this developmental law, it looked very much like Earth life, and that similar organisms exist on countless planets around other stars. A conceit which is shared, in our day, by the producers of Star Wars.

钱伯斯著作描绘了一幅生物在内在发展规律引导下，不断、渐进地向上发展图景——这一概念无疑具有目的论色彩，尤其是考虑到钱伯斯把它解释为上帝设计体现。然而，这种进步观念与某些对马克思主义的常见诠释颇为相似：其中，从鱼类到爬行动物、鸟类，最终到哺乳动物的必然演进，恰与不同生产方式演进相呼应。例如，塞耶斯[75]指出：

Chambers’ book presents a vision of a relentless, progressive ascent of organisms, guided by an inherent law of development –a concept that is undeniably teleological, especially considering Chambers’ interpretation of it as the manifestation of God’s design. However, this notion of progress bears a resemblance to certain common portrayals of Marxism, where the inevitable advancement from fish to reptiles, birds and ultimately mammals

马克思主义把历史发展的进步模式理解为阶级斗争以及生产力与生产关系之间冲突的结果。马克思——就像达尔文一样——最恰当的理解并非在于他否定了目的论观念，而在于他采用了一种与现代科学相一致的自然主义版本的目的论。

这种观点的问题在于，钱伯斯（Chambers）那种颇具影响力的达尔文之前理论，同样可以被描述为提出了一种符合 19 世纪中叶现代科学的自然主义目的论。马克思赞扬达尔文打破了这一范式，这表明他认为自己和达尔文研究都与目的论有着根本性区别。

查尔斯 • 达尔文提出的因果机制概要如下：

• 所有动植物物种中都存在变异。

• 可变性状可以遗传。

• 对可变性状的选择会增加其出现频率，这一点从农民和园艺师对动植物选育中即可得到印证。

• 生存竞争会筛选出最适应环境、从而能繁育后代的个体。

• 这可能导致足够大变化，从而产生新物种。

他在《物种起源》第三章中用一段话很好总结了自己的理论：

由于这种生存斗争，任何变异——无论多么微小，无论源于何种原因——只要在某一物种的个体与其他生物及外部自然环境的无限复杂关系中，能给该个体带来任何程度的益处，就会有助于该个体保存，并且通常会遗传给其后代。后代也将因此获得更大生存机会，因为在任何物种周期性诞生的众多个体中，只有少数能够存活下来。我把这一原则——即任何微小变异，只要是有益的，就会被保留下来——称为“自然选择”，以此标明它与人类选择能力之间关系。我们已经看到，人类通过人工选择确实能够取得巨大成果，并能利用大自然赋予的微小但有益的变异，通过其积累，使生物适合自身需求。但正如我们今后将看到的，自然选择是一种随时准备发挥作用的力量，它比人类微弱努力要高明无数倍，正如自然之作远胜于人工之作。[76]

达尔文理论常与拉马克进化论相互对照，人们认为两者关键区别在于后代能否继承获得性状——例如海豹增强的游泳能力。然而，把这种遗传现象称为“拉马克主义”，而把其否定视为“达尔文主义”，实属时代错置。在遗传学被理解之前，后天习得性状与遗传性状是否截然不同并不明显，而且达尔文从未排除获得性状遗传可能性。在某些地方，[77]他提到后代会继承后天习得的情感表达习惯。

parallels the progression through different modes of production. Sayers[75] for example, claims that

Marxism comprehends the progressive patterns of historical development as the outcome of class struggle and the conflict between the forces and relations of production. Marx –like Darwin –is best understood, not as repudiating teleological notions, but rather as using a naturalistic version of them that is consistent with modern science.

The problem with this is that the influential pre-Darwinian account of Chambers could also be described as presenting a naturalistic version of teleology consistent with modern science as it stood in the mid-nineteenth century. That Marx applauded Darwin for breaking with this paradigm indicates that he saw both his own work and Darwin’s as radically distinct from teleology.

An outline of the causal mechanism proposed by Charles Darwin is:

• Variation exists in all animal and plant species.

• Variable traits can be inherited.

• Selection for variable traits increases their frequency, as evidenced by the selective breeding of animals and plants by farmers and gardeners.

• The struggle for survival selects those best adapted to produce offspring.

• This can lead to sufficient change that new species arise

He gives a good one-paragraph summary of his theory in the third chapter of *Origin of the Species*:

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Owing to this struggle for life, any variation, however slight and from whatever cause proceeding, if it be in any degree profitable to an individual of any species, in its infinitely complex relations to other organic beings and to external nature, will tend to the preservation of that individual, and will generally be inherited by its offspring. The offspring, also, will thus have a better chance of surviving, for, of the many individuals of any species which are periodically born, but a small number can survive. I have called this principle, by which each slight variation, if useful, is preserved, by the term of Natural Selection, in order to mark its relation to man’s power of selection. We have seen that man by selection can certainly produce great results, and can adapt organic beings to his own uses, through the accumulation of slight but useful variations, given to him by the hand of Nature. But Natural Selection, as we shall hereafter see, is a power incessantly ready for action, and is as immeasurably superior to man’s feeble efforts, as the works of Nature are to those of Art.[76]

Darwin’s theory is frequently contrasted with Lamarckian evolution, where the key distinction is thought to be the inheritance of acquired traits, such as a seal’s improved swimming ability, by offspring. Labelling this inheritance as Lamarckian and its rejection as Darwinian is, however, anachronistic. Before genetics was understood, it wasn’t obvious that learnt traits were separate from inherited ones, and Darwin never ruled out the in-

heritance of acquired characteristics. In places, in[77] he speaks of learnt habits of emotional expression being inherited by offspring.

在[78]中，他强调自己所要求仅仅是变异存在，无论导致或引发变异具体情况如何。变异起源：

In[78] he emphasizes that all he requires is that variation exists, irrespective of the circumstances that cause it or give rise to it. Origins of variation:

关于变异原因——无论其具体为何——通常在生命周期哪个阶段起作用，一直存在争议：是在胚胎发育早期还是晚期，抑或是在受精瞬间。杰弗鲁瓦·圣伊莱尔实验表明，对胚胎进行非自然处理会导致畸形；而畸形与单纯变异之间无法用任何明确界限加以区分。但我强烈倾向于认为，变异最常见原因或许可归因于雄性和雌性生殖细胞在受精行为发生之前已受到影响。……事实上，“变异体”支持我的观点，即变异性很大程度上可归因于胚珠或花粉，或二者，在受精行为发生前受到亲本所受环境影响。[79]

It has been disputed at what period of life the causes of variability, whatever they may be, generally act; whether during the early or late period of development of the embryo, or at the instant of conception. Geoffroy St. Hilaire’s experiments show that unnatural treatment of the embryo causes monstrosities; and monstrosities cannot be separated by any clear line of distinction from mere variations. But I am strongly inclined to suspect that the most frequent cause of variability may be attributed to the male and female reproductive elements having been affected prior to the act of conception. . . . in fact, ‘sports’ support my view, that variability may be largely attributed to the ovules or pollen, or to both, having been affected by the treatment of the parent prior to the act of conception.[79]

“变异体”是植物品种中我们现在所称“突变”的旧称。尽管当时尚未有“基因突变”这一概念，但这一论述在功能上具有相似性。他的意思是，影响物种性器官的环境压力可能会诱发变异。如果我们将暴露于背景辐射等因素视为此类压力，那么这一观点仍然成立。

Sports is the old term for what we now call mutations in plant varieties. Although the concept of genetic mutation was not yet available, this account is functionally analogous. He is saying that environmental stress affecting the sex organs of a species may induce variation. If we rate exposure to background radiation, etc., as such stress, then this would still be acceptable.

p. 102

如果后天习得习惯也能诱导出额外变异，这对他的选择理论影响甚微。无论变异来源是什么，关键在于选择过程。

If additional variation could be induced by learnt habits, that makes little difference to his theory of selection. Whatever the source of the variation, the important thing is the selection process.

他在驯化动植物背景下引入了变异与选择概念。其前几章列举大量实例，展示驯化物种中存在着多么广泛变异。他强调，例如家犬变异程度远比野犬更为广泛。他甚至加入了鸽友协会和鸽育种协会，以深入了解这些鸟类变异范围及其育种者经验之谈。每当他试图阐明一个理论观点时，都会援引大量支持性观察结果。其中一些观察结果对于 19 世纪著作而言显得格外现代。例如，他早已意识到家养动物身上存在某些特征，而这些特征直到 20 世纪末才得到证实。

He introduces variation and selection in the context of domesticated animals and plants. His first chapters contain a wealth of examples showing what vast range of variation there is in our domestic species. He emphasizes that domestic dogs, for example, vary far more widely than wild ones. He went so far as to join pigeon fancier and pigeon breeding societies to educate himself on the range of variation of these birds and the lore of their breeders. Whenever he attempts to establish a theoretical point, he brings in a wealth of supporting observations. Some of his observations seem strikingly modern for nineteenth-century books. For instance, he was aware of traits associated with domesticated animals that were confirmed in the late twentieth century.

举不出任何一种家养动物，它在所有国家都不出现垂耳；一些作者提出，垂耳是因为这些动物不常受到危险惊吓，导致耳部肌肉不用，这一看法似乎是可信的。[80]

Not a single domestic animal can be named which has not in some country drooping ears; and the view suggested by some authors, that the drooping is due to the disuse of the muscles of the ear, from the animals not being much alarmed by danger, seems probable.[80]

这一经验观察结果得到了现代研究的证实，[81]但达尔文关于家养性状成因理论并未得到支持。垂耳、短吻、幼态持续行为和斑驳毛色等性状，如今被统称为“驯化综合征”（DS）。20 世纪 50 年代末在苏联启动的长期实验——该实验通过选择圈养狐狸进行驯化——为这一过程提供了新见解。经过五十年的选择性育种，由柳德米拉·特鲁特（Lyudmila Trut）领导的研究项目表明，即使在没有驯化历史的物种

The empirical observation is borne out by modern research,[81] but Darwin’s theory of the causes of domestic traits is not supported. Floppy ears, shorter muzzles, neotonous behaviour and patchy coat are all traits that in combination are now called the domestication syndrome (DS). Long-term experiments begun in the USSR in the late 1950s, which involved selecting captive foxes for domestication, have provided insight into this pro-

中，仅通过选择温顺性，也能迅速出现一整套 DS 性状。当同一研究团队对大鼠和貂施加相同选择压力时，也观察到了类似结果。[82]

cess. Over fifty years of selective breeding, the research programme led by Lyudmila Trut has shown that a full suite of DS traits can quickly emerge through selecting solely for tameness in a species without prior domestication history. Comparable results have been seen in rats and mink when subjected to the same selection pressures by the same research team.[82]

通过反复选择所需性状，育种者成功培育出了变异性极强的家养种群。这种变异程度之大，以至于如果把不同犬种置于野外环境中观察，它们看起来会像是不同物种。如果人为选择能制造出物种形成表象，那么何种形式选择才会导致真正物种形成？

By repeatedly selecting for desired traits, breeders have been able to create enormously variable domestic stock. The variation is so great that were the different breeds of dogs to be observed in the wild, they would appear to be different species. If deliberate selection can create the appearance of speciation, what form of selection produces actual speciation?

随后，他引入了自然界中的生存斗争。其根本在于任何生物都具有指数级繁殖潜力。

He then introduces the struggle for existence in nature. At its base is the exponential reproductive potential of any living organism.

“每个有机体自然增长的速度都高得惊人，若不被消灭，地球很快就会被一对生物的后代所覆盖”——这一规律没有例外。[83]

There is no exception to the rule that every organic being naturally increases at so high a rate, that if not destroyed, the Earth would soon be covered by the progeny of a single pair.[83]

如果指数增长持续下去，生物数量将很快超出其栖息地承载极限。即便最肥沃环境也是有限。对于每一个生物来说，都存在潜在竞争者。竞争者可能试图捕食你，也可能与你争夺食物或空间。正如他所说，大自然面容常常洋溢着喜悦，而且食物似乎俯拾皆是；然而我们却忽略了一个事实：周围那无忧无虑鸟鸣，源自那些以昆虫或种子为生、不断终结其他生命的生物。我们也忽视了这些歌者、它们的卵或幼崽在多大程度上沦为其他掠食者猎物。此外，我们未能意识到，这种食物丰裕并非在一年四季中始终如一。[84]

If exponential growth continues, an organism will quickly outgrow the limits of its habitat. Even the most fertile environments are finite. For every organism, there is a potential competitor. The competitor may be trying to eat you, or may compete for food or space. As he said, the face of nature often beams with joy, and food seems plentiful; yet we overlook the fact that the carefree birdsong around us comes from creatures that survive on insects or seeds, perpetually ending lives. We also disregard the extent to which these singers, their eggs or their young fall prey to other predators. Moreover, we fail to remember that this abundance of food is not consistent throughout the year’s cycles.[84]

p. 102–103

生命是一场永无止境斗争。动物与环境抗争。它们面临周期性干旱或霜冻威胁。它们与竞争物种争夺生态位。同一物种个体则在该生态位内相互竞争，以期让后代繁衍其中。

Life is a ceaseless struggle. Animals struggle against the environment. They are threatened by periodic drought or frost. They compete against rival species for their ecological niche. Individuals of a species compete within that niche to populate it with their offspring.

p. 103

只要稍微减轻任何制约，哪怕只是微乎其微地缓解破坏，该物种数量几乎会瞬间激增到任意规模。大自然面貌可以比作一块柔韧表面，上面密密麻麻堆满了一万个锋利楔子，它们被无休止打击向内驱入，有时击中一个楔子，接着又以更大力气击中另一个。[85]

Lighten any check, mitigate the destruction ever so little, and the number of the species will almost instantaneously increase to any amount. The face of Nature may be compared to a yielding surface, with ten thousand sharp wedges packed close together and driven inwards by incessant blows, sometimes one wedge being struck, and then another with greater force.[85]

达尔文虽然没有使用“生态位”“栖息地”或“食物链”等现代生态学术语，但他的论点仍然基于类似概念。他描述了熊蜂为红三叶草授粉过程；熊蜂数量丰富，三叶草便生长茂盛。然而，田鼠以熊蜂地下蜂巢为食，随着田鼠数量增加，三叶草数量便随之减少。在村庄里，家猫捕食田鼠，从而使村庄花园里熊蜂数量丰富，三叶草生长茂盛。因此，在每一个生态系统中，错综复杂相互作用施加着持续压力，只有少数后代能够存活。在每个物种中，适应性最强后代得以存活并繁衍后代。

Darwin, while not using modern ecological terms like niche, habitat or food chain, still bases his arguments on similar concepts. He describes the pollination of red clover by bumblebees; an abundance of these bees leads to thriving clover. However, field mice prey on the bees’ underground hives, reducing the clover as their numbers increase. In villages, domestic cats hunt the mice, resulting in bountiful bumblebees and flourishing clover in village gardens. Thus, in every ecosystem, intricate interactions exert constant pressure, allowing only a few offspring to survive. In each species, the best adapted offspring survive to reproduce themselves.

既然对人类有益变异无疑已经发生，那么在漫长而复杂的生存斗争中，对每个生物都以某种方式有益的其他变异，在数千代演变过程中偶尔出现，难道就难以想象吗？如果确实发生了这样变异，我们岂能怀疑（须知出生个体远多于能够存活数量），那些比其他个体具有任何优势——无论多么微小——的个体，拥有最大生存机会并繁衍后代？[86]

Can it, then, be thought improbable, seeing that variations useful to man have undoubtedly occurred, that other variations useful in some way to each being in the great and complex battle of life, should sometimes occur in the course of thousands of generations? If such do occur, can we doubt (remembering that many more individuals are born than can possibly survive) that individuals having any advantage, however slight, over others, would have the best chance of surviving and of procreating their kind?[86]

自然选择不断作用于每个物种，以优化其适应性。随着时间推移，适应性不断提升，物种随之演变。旧物种逐渐灭绝，新的、适应性更强物种取而代之。

Natural select acts constantly to hone the fitness of every species. Over time, fitness improves and species change. Old species die out as new better-adapted ones replace them.

p. 103–104

可以说，自然选择每天、每时每刻都在全世界范围内审视着每一个变异，哪怕是最微小变异；淘汰不良，保留并积累所有有益；只要机会出现，无论何时何地，它都在默默无闻、悄无声息地运作，致力于使每个有机体在其有机和无机生活条件下适应性得到提升。在这些缓慢变化进行过程中，我们一无所见，直到时间之手标记了漫长岁月流逝，而到那时，我们对久远地质时代的洞察仍如此不完善，以至于只能看到，如今生命形态与过去已不相同。[87]

It may be said that natural selection is daily and hourly scrutinising, throughout the world, every variation, even the slightest; rejecting that which is bad, preserving and adding up all that is good; silently and insensibly working, whenever and wherever opportunity offers, at the improvement of each organic being in relation to its organic and inorganic conditions of life. We see nothing of these slow changes in progress, until the hand of time has marked the long lapse of ages, and then so imperfect is our view into long past geological ages, that we only see that the forms of life are now different from what they formerly were.[87]

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但我们究竟能从何种意义上谈论“方向”呢？过去那种把动物划分为“高等”和“低等”的说法，难道不是一种以人类为中心投射吗？

我们是否真的能在某种意义上说，今天周围野生动物比五千万年前那些更“先进”或更好？

也许进化中看似存在方向，不过是我们基于对祖先谱系了解而产生错觉。马真的比始马“更好”吗？

难道始马（Eohippus）不正是完美适应了它所栖息的始新世森林世界吗？

如果我们采纳这种观点，自然选择就不会赋予进化以长期方向。并不存在整体上进步，只有对气候变化或快或慢适应。

达尔文坚信“改良”是真实存在。他认为在任何特定环境中，都存在逐步适应空间。他援引入侵物种如何压倒本土物种例子，作为存在改良空间证据。

But in what sense can we speak of direction? Is the old language of higher and lower orders of animals not just an anthropocentric projection?

Is there any real sense in which we can say that wild animals around today are more ‘advanced’ or better than those from fifty million years ago?

Perhaps the apparent direction of evolution is just an illusion prompted by our knowledge of ancestral lines. Is the horse really ‘better’ than *Eohippus*?

Might it not be that Eohippus was perfect for the kind of forested Eocene world that it inhabited?

If we take that view, natural selection gives no long-term direction. There is no overall improvement, just more or less rapid adaptation to climate change.

Darwin was convinced that improvement is real. That in any given environment there is room for progressive adaptation. He cites the way that invasive species outcompete natives as proof of the room for improvement.

没有任何一个国家能被列举出来，其所有本土物种如今都已彼此之间以及与所处自然环境如此完美地适应，以至于其中任何一种都无法再得到任何改善；因为在所有国家，本土物种都已被归化物种所征服，以至于它们允许外来物种牢牢占据了这片土地。既然外来物种在各地都已击败了部分本土物种，我们便可确信地推断，本土物种本可通过改良获得优势，从而更好地抵御此类入侵者。[88]

No country can be named in which all the native inhabitants are now so perfectly adapted to each other and to the physical conditions under which they live, that none of them could anyhow be improved; for in all countries, the natives have been so far conquered by naturalised productions, that they have allowed foreigners to take firm possession of the land. And as foreigners have thus everywhere beaten some of the natives, we may safely conclude that the natives might have been modified with advantage, so as to have better resisted such intruders.[88]

古尔德[89]认为这一证据无可辩驳。他以南美洲有袋类物种为例，指出自然选择促进了马类生物和类似剑齿虎食肉动物趋同进化。然而，随着巴拿马地峡形成，这些物种被更具优势北美动物群所取代并

Gould[89] argues that the evidence is incontestable. He refers to the marsupial species of South America, where natural selection has fostered the convergent evolution of horse-like creatures and carnivores akin to sabre-toothed

灭绝。真正猫科动物确实优于其有袋类对应物。引入澳大利亚欧洲兔数量激增，已泛滥成灾。相反，被放养到欧洲的小袋鼠仅能在孤立小岛上生存。

cats. However, following the creation of the Panama isthmus, these species were outcompeted and eradicated by the more dominant North American fauna. The true felines were indeed superior to their marsupial counterparts. European rabbits introduced in Australia proliferated to pestilential levels. Conversely, wallabies released in Europe only survive on isolated, small islands.

因此，我们观察到改进、提升以及设计般表象，却没有目的论或设计者。达尔文生物学界定了几个层次：谱系、营养级和适应性。从纯粹达尔文主义角度来看，尽管存在改进，但不能把动物分为低等和高等。人们之所以不愿使用“高等”和“低等”等术语，是因为这些词可能暗示“高等”物种比“低等”物种更擅长生存。通常，人们认为海鸥优于海胆。尽管海鸥可能在海胆上方翱翔并以海胆为食，但它们并不争夺相同生态位。生物学家现在更倾向于把棘皮动物描述为比鸟类更“基底”的物种。“基底”指的是在祖先树上位置。总体而言，海胆和棘皮动物比鸟类更接近棘皮动物与脊椎动物共同祖先。海鸥在食物链中地位更高，在演化谱系上也更接近顶端，但它们对自身独特生活方式适应程度并不比海胆更好。

Thus, we observe improvement, enhancement and the semblance of design, yet without teleology or a designer. Darwinian biology delineates several hierarchies: lineage, trophic levels and adaptability. From a pure Darwinian standpoint, despite the presence of improvement, one cannot categorize animals into lower and higher orders. The hesitation to use terms like ‘higher’ and ‘lower’ stems from the potential insinuation that ‘higher’ species are more adept at survival than ‘lower’ ones. Commonly, seagulls are perceived as superior to sea urchins. Although gulls may soar over, feed on, sea urchins, they do not compete for the same ecological niche. Biologists now prefer to describe echinoderms as more basal than birds. Basal refers to a position on an ancestral tree. Sea urchins and echinoderms, in general, are closer to the common ancestor of echinoderms and vertebrates than birds are. Gulls are higher in the food chain and less basal in terms of descent, but they are no better adapted to their own peculiar mode of life than sea urchins.

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针对达尔文的“目的论”指控在“有袋类障碍”上碰壁了。那么，斯密或恩格斯提出的“生产方式的等级体系”又该如何解释呢？

The accusation of teleology against Darwin falls on the marsupial hurdle. What about the hierarchy of modes of production advanced by Smith or Engels?

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回顾一下，斯密的序列是从狩猎社会到牧业社会，再到农业文明。恩格斯的序列是：蒙昧、野蛮、奴隶社会、封建主义、资本主义和共产主义。他对前两个术语的使用与摩根相同。[90]蒙昧大致对应于狩猎社会。野蛮则对应于从事园圃种植或畜牧、但尚未实行犁耕农业的经济形态。

Recall that Smith’s sequence went from hunting society to pastoral society to agricultural civilization. Engels had the sequence: savagery, barbarism, slave society, feudalism, capitalism and communism. The first two terms he used in the same sense as Morgan.[90] Savagery maps roughly onto hunting society. Barbarism maps onto those economies that practice horticulture or herding but not plough agriculture.

恩格斯的框架把社会发展同时看作谱系上的承续和进步。资本主义被视为比封建主义更先进，野蛮状态则比蒙昧状态更先进。现代人的敏感意识往往排斥“野蛮人”或“蒙昧人”等词，因为其中带有欧洲中心主义或中国中心主义的偏见，甚至可能接近种族主义。然而，从马克思主义视角来看，从事园圃种植和畜牧的野蛮社会，相较于仅依赖狩猎的社会，其进步性不可否认。同样，资本主义也被视为比封建主义更先进。这基于古尔德指出的同一标准：竞争能力。历史上，处于野蛮阶段的民族与处于蒙昧阶段的民族接触时，常常取代后者，占据其原有领土的大片区域。来自西非和安纳托利亚、拥有家畜和农业的人群迫使更早的狩猎采集社会退却，就体现了这一点。

Engels’s framework views societal development as both ancestral and progressive. Capitalism is seen as more advanced than feudalism, and barbarism as more advanced than savagery. Modern sensitivity often rejects terms like ‘barbarian’ or ‘savage’ due to their Euro-centric or Sino-centric biases, which can border on racism. However, from a Marxist perspective, the advancement of a barbarian society engaged in horticulture and animal husbandry over a society solely reliant on hunting is undeniable. Similarly, capitalism is deemed more advanced than feudalism. This is based on the same criterion Gould identifies: the ability to compete. Historically, when barbarian nations came into contact with savage ones, they often displaced them, taking over large areas of their original territories. This was evident as populations from West Africa and Anatolia, equipped with domesticated animals and agriculture, pushed back the earlier hunter-gatherer societies.

工业资本主义同样席卷全球，征服了处于野蛮阶段的民族，并迫使处于封建阶段社会实现现代化。

Industrial capitalism likewise spread out across the world, subduing barbarian nations and forcing those at

更先进社会秩序取代了较落后社会秩序，正如胎盘类猫科动物在南美洲取代了有袋类博尔海尼科动物一样。

让我们从目的论问题转向“何为人类”这一议题。达尔文主义通过确立人类与猿类共同祖先，动摇了唯心主义或宗教世界观。取代了“按上帝形象创造的理性人类”与“无理性的动物”之间那种绝对区分的，是“裸猿”这一人类概念。那些看似为人类独有的特征，往往只是早期动物特征演化结果。

在考察任何人类特征时，达尔文出发点是把它置于我们哺乳动物祖先语境中审视。无论是讲述黑猩猩使用简单石器，还是埃塞俄比亚狒狒使用投掷武器，[91]他都不断强调，那些被认为是人类独有的特征，不过是共同遗产一部分。

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人们常说，没有动物会使用任何工具；但在自然状态下，黑猩猩会用石头敲开一种类似核桃的当地水果。（37. Savage 和 Wyman 在《波士顿自然史杂志》第 IV 卷，1843—44 年，第 383 页。）伦格（Rengger）（38.《巴拉圭哺乳动物》，1830 年，第 51—56 页。）轻松教会了一只美洲猴子用这种方法敲开坚硬棕榈果；此后，这只猴子便主动用石头打开其他种类坚果以及盒子。它还去除了味道难闻水果软皮。另一只猴子被教导用棍子撬开一个大箱子盖子，之后它便把棍子当作杠杆来移动重物；而我本人也曾亲眼目睹一只幼年猩猩把一根棍子插入缝隙中，把手伸到另一端，并恰当地把它作为杠杆使用。……布雷姆（40.《动物生活》，第一卷，第 79、82 页）援引著名旅行家辛珀记载，在阿比西尼亚，当一种狒狒（C. gelada）成群结队从山上下来掠夺农田时，有时会遇到另一种狒狒（C. hamadryas）群体，随后便会爆发战斗。格拉达狒狒把巨石滚下山坡，哈马德里亚斯狒狒则试图躲避，随后两个物种在喧嚣声中狂暴地向对方冲去。布雷姆曾陪同科堡—哥达公爵，在阿比西尼亚门萨山口协助用火器袭击了一群狒狒。作为反击，狒狒们把大量巨石——其中有些甚至大如人的头颅——滚下山去，迫使攻击者仓皇撤退；该山口甚至一度对商队关闭。值得注意的是，这些狒狒当时表现出了高度协同行动。

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在探讨动物心理学时，他同样把人类情感全部谱系——羞耻、愤怒、勇气和亲情——归因于动物。

我认为，毫无疑问，狗会感到羞耻——这与恐惧不同——而且当它们过于频繁乞食时，还会表现出一种非常类似于羞怯情绪。一只大狗会轻蔑地忽视小狗咆哮，这可以称为胸襟宽广。

针对这属于拟人化的偏见，他提出相反的原则：只要我们拥有情感生活，我们之所以能够感受这些情绪，就只能是因为我们共有的哺乳动物遗产。情绪

a feudal level to modernize. The more advanced social order displaced the less advanced just as the placental felids replaced the marsupial borhyaenids in South America.

Let’s go from the question of teleology to what it means to be human. Darwinism undermined the idealist or religious world view by establishing the common descent of humans and apes. In the place of a categorical distinction between rational humans made in the image of God and brute animals, there came the conception of humans as naked apes. What seems specially human is often just a development of earlier animal traits.

When looking at any human trait, Darwin’s starting point is to treat it in the context of our mammalian ancestors. Whether he is recounting the use of simple stone tools by chimps or projectile weapons by baboons in Ethiopia,[91] he constantly emphasizes that supposedly unique human traits are just part of a common heritage.

It has often been said that no animal uses any tool; but the chimpanzee in a state of nature cracks a native fruit, somewhat like a walnut, with a stone. (37. Savage and Wyman in ‘Boston Journal of Natural History’, vol. iv. 1843-44, p. 383.) Rengger (38. ‘Saugethiere von Paraguay’, 1830, s. 51-56.) easily taught an American monkey thus to break open hard palm-nuts; and afterwards of its own accord, it used stones to open other kinds of nuts, as well as boxes. It thus also removed the soft rind of fruit that had a disagreeable flavour. Another monkey was taught to open the lid of a large box with a stick, and afterwards it used the stick as a lever to move heavy bodies; and I have myself seen a young orang put a stick into a crevice, slip his hand to the other end, and use it in the proper manner as a lever. . . . Brehm (40. ‘Thierleben’, B. i. s. 79, 82.) states, on the authority of the well-known traveller Schimper, that in Abyssinia when the baboons belonging to one species (C. gelada) descend in troops from the mountains to plunder the fields, they sometimes encounter troops of another species (C. hamadryas), and then a fight ensues. The Geladas roll down great stones, which the Hamadryas try to avoid, and then both species, making a great uproar, rush furiously against each other. Brehm, when accompanying the Duke of Coburg-Gotha, aided in an attack with fire-arms on a troop of baboons in the pass of Mensa in Abyssinia. The baboons in return rolled so many stones down the mountain, some as large as a man’s head, that the attackers had to beat a hasty retreat; and the pass was actually closed for a time against the caravan. It deserves notice that these baboons thus acted in concert.

When discussing the psychology of animals, he likewise attributes to them the full gamut of human emotions:, shame, anger, courage and affection.

There can, I think, be no doubt that a dog feels shame, as distinct from fear, and something very like modesty when begging too often for food. A great dog scorns the snarling of a little dog, and this may be called magnanimity.

Against the prejudice that this is anthropomorphism, he counterposes the opposite principle. Insofar as we have an emotional life, we can only feel these emotions because of our common mammalian heritage. Emotions

本身被视为身体的状态和姿态。他认为，情绪表达受对立原则支配：对立的情绪调动相反的肌肉群。

themselves are treated as states and postures of the body. Their expression, he claims, is govererned by the principle of opposition. Opposing emotions bring into play opposing sets of muscles.

他以猫或狗行为为例加以说明。当猫感到攻击性时，它会采取蹲伏姿势，背部放平，耳朵向内收拢，尾巴猛烈摆动。这些姿势具有生存优势：低矮姿态略微改变了扑击前姿势，既不引人注意，又让后腿随时准备跃起；贴平耳朵能防止被咬，而甩动尾巴和嘶嘶声则起到威慑作用。相反，友善猫会挺直身子，四肢伸展，背部拱起，尾巴竖起。情绪与神经肌肉活动有着内在联系；如果没有相应身体表现，情绪就无法显现。达尔文先是利用动物行为来阐释这一概念，随后证明其对人类情绪和表情适用性，从而提出了一种根本上的唯物主义观点。

He illustrates this with the behaviour of cats or dogs. When a cat feels aggressive, it assumes a crouching position, flattens its back, tucks its ears and lashes its tail. These postures offer a survival benefit. The low stance slightly alters the pre-pounce posture, staying inconspicuous with hind legs ready to spring. Flattened ears prevent bites, while a thrashing tail and hissing serve as deterrents. Conversely, a friendly cat stands tall, with extended legs, an arched back and a raised tail. Emotions are intrinsically linked to neuromuscular actions; without corresponding physical expressions, emotions do not manifest. Darwin used animal behaviours to illustrate this concept before demonstrating its applicability to human emotions and expressions, presenting a fundamentally materialistic viewpoint.

这一观点得到了介于盖伦与赫布[92]之间的一种神经系统观念支持。达尔文写道，神经力量沿着神经并穿过神经传递。这与盖伦的“精神之气”（psychic pneuma）仅有一步之遥。

It is backed up by a conception of the nervous system that sits halfway between Galen and Hebb.[92] Darwin writes of *nerve-force* being transmitted along and through the nerves. This is only one step removed from Galen’s *psychic pneuma*.

但这一观点结合了对反射认识，以及“习惯性动作会强化神经系统各部分之间相互连接”这一概念。

But it is combined with an awareness of reflexes and the concept that habitual action reinforces interconnection between parts of the nervous system.

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经验表明，每当脑脊髓系统受到刺激时，神经力就会产生并释放出来。这种神经力所遵循方向必然由神经细胞之间以及神经细胞与身体各部位之间连接线路所决定。但该方向同样在很大程度上受到习惯影响；因为神经力会顺着习惯通道轻松地传递。[93]

Experience shows that nerve-force is generated and set free whenever the cerebro-spinal system is excited. The direction which this nerve-force follows is necessarily determined by the lines of connection between the nerve-cells, with each other and with various parts of the body. But the direction is likewise much influenced by habit; inasmuch as nerve-force passes readily along accustomed channels.[93]

达尔文对遗传和神经活动有着正确直觉，尽管他并不了解其背后实际机制：DNA 和突触。达尔文使用“力”这一非常物理化术语来描述神经系统。但在盖伦所处的时代，他的“气”（pneuma）本身就是一种关于神经系统的物理理论。随着时间推移，盖伦这一唯物主义理论逐渐演变为与其截然相反观点——当它被翻译成拉丁语 spiritus 并由神职人员宣扬时，便成为了一种神秘主义概念。在达尔文与赫布之间时期，我们也能看到同样潜在歧义。卡哈尔和巴甫洛夫探索了机械唯物主义道路，阐明了基本机制，从而使赫布能够断言：

Darwin had the right basic intuitions on heredity and nervous action, though he did not know the actual mechanisms by which it came about: DNA and the synapse. Darwin uses a language of ‘force’, which is very physical, to describe the nervous system. But in its own day, Galen’s *pneuma* was a physical theory of the nervous system. In time, this materialist theory of Galen became its opposite, so that translated into the Latin *spiritus* and intoned by priests, it became something mystical. We can see the same potential ambiguity in the period between Darwin and Hebb. Cajal and Pavlov explored the mechanical materialist path, elucidating basic mechanisms leading to the point where Hebb could declare:

现代心理学完全理所当然地认为，行为与神经功能是完美相关的，一方完全由另一方所决定。并不存在独立灵魂或生命力，时不时地向大脑“插一脚”，迫使神经细胞做它们原本不会做事情。实际上，当然，这仅在行为中存在未解之谜时才是一种工作假设。……我们迄今未能解决一个问题，并不意味着它无法解决。从逻辑上讲，一个人不可能在物理、化学和生物学领域是决定论者，而在心理学领域却成为神秘主义者。[94]

Modern psychology takes completely for granted that behaviour and neural function are perfectly correlated, that one is completely caused by the other. There is no separate soul or life-force to stick a finger into the brain now and then and make neural cells do what they would not otherwise Actually, of course, this is a working assumption only as long as there are unexplained aspects of behaviour. . . . Our failure to solve a problem so far does not make it insoluble One cannot logically be a determinist in physics and chemistry and biology and a mystic in psychology.[94]

在此，赫布隐含地批评了心理学中从弗洛伊德到荣格的发展轨迹——这一轨迹起初源于年轻弗洛伊德对神经系统实际研究，后来却分岔走向了一条神秘主义神经能量学之路。[95]由于当时弗洛伊德在美国仍极具影响力，赫布在批评时不得不保持委婉，但其核心观点依然清晰明确。

In this, Hebb was implicitly criticizing the Freud-to-Jung trajectory in psychology that starting out from actual research into the nervous system by the young Freud, later branched out along a path of mystical neuro-energetics.[95] Working in a US context in which Freud was still highly influential, Hebb had to be tactful in his criticism, but the thrust is no less clear.

"自我""本我"和"超我"是帮助人们洞察并阐明行为重要事实的概念，但人们也极易把它们视为虚幻实体——仿佛是那些想要这样、反对那样，凭力量或诡计彼此压倒，并施加惩罚或遭受惩罚的人格化主体。弗洛伊德留给我们任务，是把这些暂时理论表述进一步发展，直至这种危险不复存在。当理论变得僵化时，就容易沦为教条；而且，只要它仍有这么多问题尚未解决，心理理论还面临着另一重危险，即可能导致人们重新陷入传统思想中的活力论和非决定论。[96]

Ego, Id, and Superego are conceptions that help one to see and state important facts of behavior, but they are also dangerously easy to treat as ghostly realities, as anthropomorphic agents that want this or disapprove of that, overcoming one another by force or guile, and punishing or being punished. Freud has left us the task of developing these provisional formulations of his to the point where such a danger no longer exists. When theory becomes static it is apt to become dogma; and psychological theory has the further danger, as long as so many of its problems are unresolved, of inviting a relapse into the vitalism and indeterminism of traditional thought.[96]

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遗憾的是，这种对弗洛伊德式非决定论和活力论的倒退，到 20 世纪 60 年代已对法国马克思主义理论家产生了如此深远影响，以至于廷帕纳罗甚至指出，这已成为他们主要特征。[97]

Unfortunately, this lapse into Freudian indeterminism and vitalism had, by the 1960s, so influenced French Marxist theorists that Timpanaro[97] could remark that it was their main distinguishing feature.

新的原子论

The new atomism

化学家们每天发现的那些令人惊叹事实，正是对原子论的一项项新证明。[98]

The remarkable facts that chemists daily discover are as many new proofs of atomism.[98]

牛顿是卢克莱修追随者，也就是一位原子论者。他对声波传播解释明确基于"物体由原子构成"这一思想。由此，他能够正确预测声音应存在衍射现象，[99]这堪称原子假说力量早期杰出证明。但原子论之所以得以具体化，是通过化学而非我们如今所称物理学实现的。

Newton was a follower of Lucretius, that is, an atomist. His account of the propagation of sound waves is explicitly based on the idea that bodies are made of atoms. From this, he was able to correctly predict that diffraction of sound should exist,[99] a remarkable early demonstration of the power of the atomic hypothesis. But it was through chemistry rather than what we would now call physics that atomism became concrete.

道尔顿

Dalton

虽然"原子"一词在 18 世纪期间仍在使用，但主要出现在对伊壁鸠鲁学派的历史叙述中[100]，或作为通俗散文中文学隐喻——其前提是受过教育人都读过卢克莱修作品。直到道尔顿开展研究后，化学家才开始严肃使用这一术语。

While the term 'atom' continued to be used during the eighteenth century, it was mentioned mainly in the context of historical accounts of the Epicureans[100] or as a literary metaphor in popular prose, the assumption being that educated people had read Lucretius. It was not used seriously by chemists until the work of Dalton.

他在研究不同气体在水中可溶解量的实验过程中，[101]逐步形成了关于原子思想。由此，他提出了溶液中气体分压概念，这一概念至今仍在当代化学中沿用。他关于分压理论模型是：当一种气体（例如氧气）溶解于水时，氧气原子会与水"原子"交错排列，形成规则网格。

He developed his thoughts on atoms in the context of experiments he did on the quantities of different gases that could be dissolved in water.[101] From this, he developed the idea of partial pressures of gases in solution, a notion that continues to be used in contemporary chemistry. The theoretical model he had for partial pressure was that when a gas, let us say oxygen dissolves in water, then the oxygen atoms are interspersed with water 'atoms' in a regular grid.

道尔顿发现了溶液中分压概念，这使他得以估算大气中主要气体比例并提出“道尔顿定律”。他还利用分压原理并测量气体密度，提出了原子量概念。他是第一个明确区分所谓“原子的机械混合物”（例如溶解在水中氧气）与化学化合物的人。

Dalton discovered the notion of partial pressure in a solution, which led him to estimate the proportions of the principal gases in the atmosphere and formulate ‘Dalton’s Law’. He also introduced the concept of atomic weights by utilizing the idea of partial pressure and measuring the densities of gases. He was the first to clearly distinguish what he termed mechanical mixtures of atoms, for example, oxygen dissolved in water, from chemical compositions.

本书的一个主要目标，便是阐明确定简单物质和化合物中基本粒子的相对重量、构成一个化合物粒子的简单基本粒子数量，以及形成一个更复杂化合物粒子所需较简单的复合粒子数量的重要性与优势。[102]

Now it is one great object of this work, to shew the importance and advantage of ascertaining the relative weights of the ultimate particles, both of simple and compound bodies, the nuimber of simple elementary particles which constitute one compound particle, and the number of less compound particles which enter into the formation of one more compound particle[102]

他在分子量相对顺序上判断正确。他对氮和氧分子量估算误差至少为两倍；对于化合物气体而言，其误差则小得多。氮分子和氧分子误差较大原因，很可能是他假设这些气体为单原子气体。

He got the relative order of molecular weights right. His estimates of the molecular weights of nitrogen and oxygen are out by a factor of at leas two; for compound gases, his errors are much smaller. The larger errors for Nitrogen and Oxygen molecules are probably due to his assumption that these gases were monatomic.

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在把这一理论应用于求解弹性流体比热之前，我们必须首先确定其基本粒子的相对重量。暂且假设（这一点将在后文证明）：如果一个氢原子质量为 1，则氧为 7，氮为 5，亚硝酸气体为 12，一氧化二氮为 17，碳酸为 19，氨气为 6，碳化氢为 7，烯烃气体为 6，硝酸为 19，一氧化碳为 12。[103]

Before we can apply this doctrine to find the specific heat of elastic fluids, we must first ascertain the relative weights of their ultimate particles. Assuming at present what will be proved hereafter, that if the weight of an atom of hydrogen be 1, that of oxygen will be 7, azote 5, nitrous gas 12, nitrous oxide 17, carbonic acid 19, ammoniacal gas 6, carburetted hydrogen 7, olefiant gas 6, nitric acid 19, carbonic oxide 12,[103]

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他得出估算值汇总于表 5.1 中，该表将其与现代数值进行对比，并把他所命名化合物名称转换为现代化学式。

The estimates that he arrived at are summarized in Table 5.1, which compares them with the modern values, and translates his names for compounds into modern chemical formulae.

在[104]中，他详细阐述了原子存在多种级别假说。基本原子即元素——当时他已能识别出二十种。这些元素可以成对结合形成他所称的二元原子，成三结合形成三元原子，以此类推。这些更高层次原子组合与我们现在所知分子非常接近。如图 5.2 所示，现在已知他的一些化学式是错误的。例如，他认为水由一个氢原子和一个氧原子组成，这也是基于氢气是单原子这一假设得出又一结论。

In[104] he elaborated the hypothesis that atoms came in several orders. The primary atoms were the elements – of which he was already able to identify twenty. Pairs of these could be bound together to form what he called binary atoms, triples to form ternary atoms etc. These higher orders correspond pretty closely to what we now know as molecules. As Figure 5.2 shows, some of his formulae are now known to be wrong. For instance, he believed water was made of one hydrogen and one oxygen atom, another consequence of the assumption that hydrogen gas was monatomic.

他关于水“原子”构想在某些方面竟出人意料地具有现代性。如今我们认为原子由中子和质子组成原子核以及围绕其周围电子云构成。而他则认为，二元、三元等原子具有由基本粒子组成的核心，周围环绕着他所称的“热气层”。

The idea he had for a water‘atom’was in some ways strikingly modern. We now think of atoms as being made of a nucleus of neutrons and protons surrounded by a cloud of electrons. He thought that binary and ternary, etc., atoms had a nucleus of elementary particles surrounded by what he called an atmosphere of heat.

从道尔顿 1808 年模型到 20 世纪中叶模型，转变主要体现在两点：

The shift from Dalton’s model of 1808 to the mid-twentieth-century model came down to two points:

1. 道尔顿把化学元素视为基本粒子。20 世纪中叶理论则把质子和中子视为“基本”粒子。20 世纪末理论甚至把这些粒子进一步分解为夸克和胶子。

1. Dalton took the chemical elements to be elementary particles. Mid-twentieth-century theory took protons and neutrons to be the‘elementary’particles. Late

twentieth-century theory decomposed even these into quarks and gluons.

表 5.1 道尔顿对相对分子量的估算与今知数值的比较

	道尔顿的名称	实际分子量	与氢分子的比值	道尔顿估值
H_2	氢	2.01594	1	1
N_2	氮（Azote）	28.0134	13.89595	5
O_2	氧	31.9988	15.87289	7
N_2O	一氧化二氮	46.0055	22.82087	17
NO	亚硝气	30.0061	14.88442	12
CO_2	碳酸气	44.0098	21.83091	19
NH_3	氨气	17.03061	8.447975	6

Table 5.1 Comparisons of Dalton’s Estimates of Relative Molecular Weights to Those We Know Today

	Dalton’s name	Actual molecular weight	Ratio to hydrogen molecule	Dalton’s estimate
H_2	Hydrogen	2.01594	1	1
N_2	Azote	28.0134	13.89595	5
O_2	Oxygen	31.9988	15.87289	7
N_2O	Nitrous oxide	46.0055	22.82087	17
NO	Nitrous gas	30.0061	14.88442	12
CO_2	Carbonic acid	44.0098	21.83091	19
NH_3	Amoniacal gas	17.03061	8.447975	6

校注：底本 N_2O 行的实际分子量印为 46.0055，相对氢分子的比值为 22.82087；两项与该行化学式之间存在疑点，此处逐项照录，未以推算值替换。Amoniacal 亦为底本拼写。

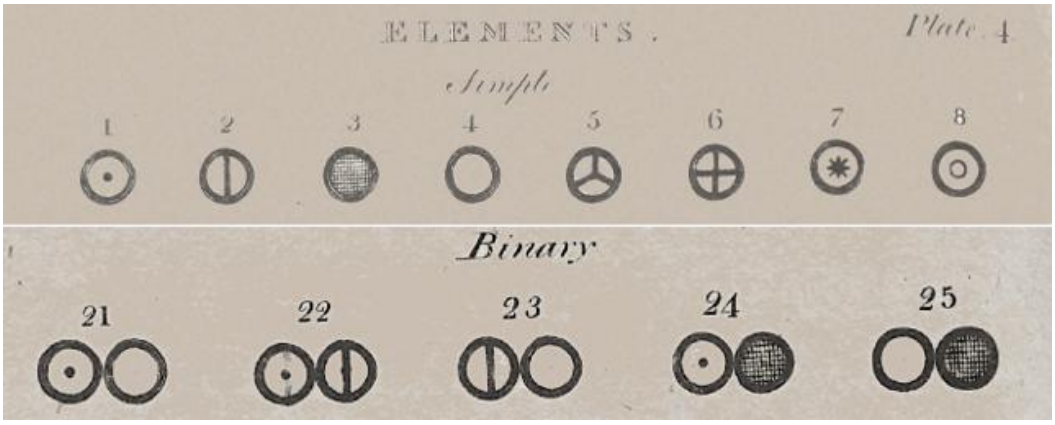


图 5.2 道尔顿表示元素原子的符号：[1] 氢，[2]Azote（氮），[3] 碳，[4] 氧，[5] 磷，[6] 硫，[7] 苦土（Magnesia），[8] 石灰（钙）；以及若干化合物：[21] 水，[22] 氨，[23] 一氧化二氮，[24] 成油气（乙烯 C_2H_2），[25] 碳氧化物（一氧化碳）。注意，他尚未认识到气态氢为 H_2，因此水等物质的组成显得古怪。取自 John Dalton 的《化学哲学新体系》。

Figure 5.2 Dalton’s symbolic notation for the atoms of elements. [1] Hydrogen, [2] Azote (Nitrogen), [3] Carbone, [4] Oxygen, [5] Phosphorous, [6] Sulphur, [7] Magnesia, [8] Lime (Calcium); also for some compounds: [21] Water, [22] Ammonia, [23] Nitrous oxide, [24] Olefiant gas (Ethylene C_2H_2), [25] Carbonic oxide (Carbon monoxide). Note that he did not realize that gaseous hydrogen was H_2, hence the odd composition of water, etc. From John Dalton, *A New System of Chemical Philosophy*.

ELEMENTS：元素；Simple：简单的；Binary：二元的；Plate 4：图版 4。各编号对应图注中的原子及化合物。

校注：底本将 Ethylene（乙烯）的化学式印作 C_2H_2，此处保留原书式并提示，未静默改动；图中历史元素分类及图注 Magnesia/Lime 也按底本保留。

2. 道尔顿把气体的弹性归因于围绕由化学元素构成原子核的热质或热的气层。当气体受热时，热质被添加到原子中，原子因此变大，从而使气体膨胀。这与现代基础教科书中描述的围绕原子核电子云有明显相似之处。我们认为，至少固体的弹性是由电子云之间静电排斥力造成。甚至在玻尔原子模型中，我们也曾有这样一个概念：当向原子输入能量时，其电子会跃迁至更高能级，从而导致原子膨胀，这与

2. Dalton attributed the elasticity of gases to an *atmosphere* of *caloric* or heat around a nucleus made up of chemical elements. As a gas was heated, *caloric* was added to the atoms, which grew larger, thus making the gas expand. This has obvious analogies to the *cloud* of electrons that modern elementary textbooks describe around the nucleus. We accept that it is the electrostatic repulsion of the electron clouds that is responsible for at least the elasticity of solids. We even had the concept in

道尔顿关于“随着热量增加，热质气层会膨胀”的观点相去不远。

the Bohr model of the atom that when you added energy to an atom, its electrons moved to higher levels with a consequent expansion, which is not far from Dalton’s concept that the *atmosphere* of *caloric* expanded with the addition of heat.

唯物主义的认知过程总是如此：当人们发现某种被视为原子或不可分割粒子（用卢克莱修的话说就是“原质”）时，却又发现它们是由更基本粒子组成。目前通用标准模型包含一组基本粒子：夸克、胶子、轻子和中微子。这些粒子之下是否还存在更细微层次，仍有待观察。

There is a repeated process by which materialist understanding discovers what it takes to be atomic or indivisible particles, *primordia* in Lucretius terms, only to find that they are composed of even more basic ones. The Standard Model in use now has a set of elementary particles: quarks, gluons, leptons and neutrinos. It remains to be seen whether a further level of detail will be detected below these.

热、物质与时间

Heat, matter and time

道尔顿思想，就像化合物一样，建立在他人奠基性工作之上。托里切利 1643 年发现并测定大气压，为理解分压奠定基础。随后几年，化学家们鉴定出新元素气体：1766 年氢气、1772 年氮气和 1777 年氧气，这些发现对于理解构成大气层各种分压至关重要。然而，道尔顿研究仍以热是一种称为“热质”（caloric）的独立实体这一观念为基础。这一概念基于布莱克和瓦特开创性工作，他们量化了热量、比热和潜热，从而为蒸汽动力工业资本主义时代奠定基础。

Dalton’s ideas, much like compounds, are built on the foundational work of others. Torricelli’s discovery and measurement of atmospheric pressure in 1643 laid the groundwork for understanding partial pressures. In the years that followed, chemists identified new elemental gases: hydrogen in 1766, nitrogen in 1772 and oxygen in 1777, which were crucial for conceptualizing the various partial pressures that constitute the atmosphere. However, Dalton’s work still operated under the notion of heat as a separate entity, known as ‘caloric’. This concept was based on the pioneering work of Black and Watt, who quantified heat, specific heat and latent heat, thereby setting the stage for the era of steam-powered industrial capitalism.

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我们此前已论述过，这一社会新技术基础如何为理解经济提供关键概念工具。[105]但出于提高工程效率的需要，它同时也为热力学研究奠定基础，并推动了原子论发展下一阶段：把热理解为运动。

We have already commented on how this new technical basis for society prepared key conceptual tools for the understanding of the economy.[105] But through the necessities of engineering efficiency, it also prepared the study of thermodynamics and motivated the next stage in the development of atomism: the understanding of heat as motion.

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如果蒸汽机没有被发明出来，我们无疑会停留在低于当前理论水平位置。热所取得的成就通过蒸汽机，以更加强烈语气向思考者们提出了这个问题：这种能够让我们取代风力、水力、马力和人力的一种媒介究竟是什么？[106]

If the steam-engine had not been invented, we should assuredly stand below the theoretic level which we now occupy. The achievements of Heat, through the steam-engine, have forced, with augmented emphasis, the question upon thinking minds: What is this agent, by means of which we can supersede the force of winds, of rivers, of horses and men ?[106]

在 18 世纪末至 19 世纪初，关于热的哲学存在两大流派。一方面，有被称为唯物主义者的人，如道尔顿和许多其他化学家，他们认为热是物质的一种独特形式。另一方面，则有被称为“机械论者”的学派，他们认为热是原子的机械运动。最终占上风机械论思想，其渊源至少可追溯至培根，他认为热仅在热体的较小粒子中发生膨胀。这种膨胀被其他粒子所阻碍、排斥并击退，从而产生一种来回运动，导致了热与火的狂暴。[107]

In the late eighteenth and early nineteenth centuries there were two schools of philosophy on heat. On the one hand, there were those who were styled *materialists*, like Dalton and many other chemists, who thought that heat was a distinct form of matter. On the other hand, there were what were termed *mechanists* for whom heat was the mechanical motion of atoms. The mechanistic ideas, which eventually triumphed, went back at least as far as Bacon,who believed that heat expanded only in the smaller particles of a hot body. This expansion was checked, repelled and beaten back by other particles, creating a back-and-forth motion that caused the fury of heat and fire.[107]

但“热是机械运动”这一观点，不得不面对一种起初看似同样合理的唯物主义观点。

But the idea that heat was mechanical motion had to contend with what initially seemed an equally plausible materialist view.

热量会流动。这与液体行为极为相似。如果把一块湿布与一块干布接触，湿气就会向干布扩散；如果把一块热金属与一块冷金属接触，热量也会以类似方式扩散。

Heat flows. This seemed very like the behaviour of a liquid. If you bring a wet cloth into contact with a dry one, the dampness spreads into the dry cloth. If you bring a hot piece of metal in contact with a cold one, the heat spreads into a similar way.

当气体被压缩时，它会升温。这被解释为热量被挤出了原子之间空隙。

When a gas is compressed, it heats up. This was interpreted as the heat being squeezed out of the gaps between the atoms.

摩擦被解释为释放了此前被物体过于紧密地封存的热量。如果观察到火石和钢片摩擦产生火花，这种解释似乎合乎情理。敲击似乎把火花从钢片中释放出来，这暗示着热量原本潜藏在其中。

Friction was interpreted as releasing heat that had hitherto been too tightly contained in objects. If one observed a spark being struck from a flint and steel, this seemed a plausible account. The blow appeared to release the spark from the steel, implying that the heat was latent within it.

温度计使科学家能够研究比热，即使不同物质温度升高相同度数所需热量。水每升高一度所能吸收热量是汞的三十倍。化合物“热容量”被认为反映了原子之间可供热量进入空间大小。

Thermometers allowed scientists to study specific heat, which refers to the amount of heat needed to raise the temperature of different substances by the same number of degrees. Water can absorb thirty times more heat than mercury for each degree of increase in temperature. The ‘heat capacity’ of a compound was taken to indicate how much space there was between the atoms for heat to enter.

据廷德尔所述，一项关键观察结果推翻了唯物主义解释。那就是伦福德伯爵在镗削炮管时的经历。伦福德观察到，炮管在镗孔时会变得非常烫。

According to Tyndall, a crucial observation discredited the materialist account. It was Count Rumford’s experience with cannon boring. Rumford observed that cannons got very hot when being bored.

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最近，我在慕尼黑军械库车间里负责监督大炮镗孔工作，我惊讶地发现，黄铜炮在镗孔过程中短时间内就会产生相当大热量；而镗刀从炮身上分离出来的金属屑所产生热量则更为强烈（通过实验我发现，其温度远高于沸水）。

Being engaged, lately, in superintending the boring of cannon, in the workshops of the military arsenal at Munich, I was struck with the very considerable degree of heat which a brass gun acquires, in a short time, in being bored; and with the still more intense heat (much greater than that of boiling water, as I found by experiment,) of the metallic chips separated from it by the borer.

我越是深思这些现象，就越觉得它们奇妙而引人入胜。对它们进行彻底研究，似乎甚至有望让我们更深入地洞察热的隐秘本质；并使我们能够就一种火热流体存在与否提出一些合理推测：关于这一课题，各个时代哲学家们一直意见分歧。[108]

The more I meditated on these phenomena, the more they appeared to me to be curious and interesting. A thorough investigation of them seemed even to bid fair to give a farther insight into the hidden nature of heat; and to enable us to form some reasonable conjectures respecting the existence, or non-existence, of an igneous fluid,: a subject on which the opinions of philosophers have, in all ages, been much divided.[108]

正如我们所言，关于摩擦标准解释是：它释放了被摩擦物体中潜藏热量。唯物主义者秉持“无中不能生有”原则。热量总和保持恒定。如果燧石与钢片摩擦产生火花时释放了热量，那么这热量在燧石撞击钢片之前就必须已经隐藏在钢片之中。同理，被镗削的大炮所产生热量，在被镗床磨成粉末之前，就必须已经潜藏在青铜之中。伦福德对此并不信服，于是进行了实验。

As we have said, the standard account of friction was that it released heat latent in the subject being rubbed. The materialists held a ‘nothing from nothing’ principle. The total amount of heat remained constant. If heat was released in a flint on steel spark, then it must have been hidden in the steel before the flint struck it. Analogously, the heat of a bored cannon must have been latent in the bronze before it was ground to dust by the boring machine. Rumford was not convinced and experimented.

上述机械操作中实际产生热量究竟从何而来？

From whence comes the heat actually produced in the mechanical operation above mentioned ?

难道是镗刀从金属块中分离出来金属碎屑所提供吗？

Is it furnished by the metallic chips which are separated by the borer from the solid mass of metal ?

如果是这样，那么根据现代关于潜热和热质理论，这些被削成碎屑金属部分热容量不仅应该发生变化，而且这种变化还应足够大，足以解释所产生所有热量。

If this were the case, then, according to the modern doctrines of latent heat, and of caloric, the capacity for heat of the parts of the metal, so reduced to chips, ought not only to be changed, but the change undergone by them should be sufficiently great to account for all the heat produced.

但实际上并未发生此类变化；因为我发现，当取等量（按重量计）这些金属屑，以及用细锯从同一金属块上切下薄片，并把它们在相同温度（即沸水温度）下放入等量冷水中（即 59°F）时，从观察结果来看，放入金属屑那部分水，其温度既没有比放入金属薄片另一部分水低，也没有比后者高。[109]

But no such change had taken place; for I found, upon taking equal quantities, by weight, of these chips, and of thin slips of the same block of metal separated by means of a fine saw, and putting them, at the same temperature, (that of boiling water,) into equal quantities of cold water, (that is to say, at the temperature of 59° F .) the portion of water into which the chips were put was not, to all appearance, heated either less or more than the other portion, in which the slips of metal were put.[109]

他得以驳斥了“摩擦热是黄铜中释放出的某种物质”这一假说。其他实验证明，加热或冷却过程中并未伴随质量变化。相反，热量必定来自那匹不幸被套在驱动镗床的马所做的功。他认为马的力量可以产生热量，甚至可以用于烹饪，但产生热量绝不会超过把马的饲料作为燃料所能获得热量。

He was able to reject the hypothesis that the heat of friction was some substance released from the brass. Other experiments proved that there was no weight change associated with heating or cooling. Instead, the heat must have come from the work put into the borer by the unfortunate yoked horse driving it. He held that heat could be produced by a horse’s strength and indeed used for cooking, but never more heat than could be obtained by using the horse’s food as fuel.

在探讨这一课题时，我们绝不能忽视一个最引人注目现象：即这些实验中摩擦产生热源显然是取之不尽的。毋庸赘言，任何隔离的物体或物体系统能够无限持续提供的东西，绝不可能是某种物质；在我看来，若非“运动”（MOTION），要对那些实验中能够被激发并传递任何事物形成清晰认识，似乎极其困难，甚至完全不可能。[110]

In reasoning on this subject we must not forget that *most remarkable circumstance*, that the source of the heat generated by friction in these experiments appeared evidently to be *inexhaustible* It is hardly necessary to add, that anything which any *insulated* body or system of bodies can continue to furnish *without limitation* cannot possibly be a *material substance*; and it appears to me to be extremely difficult, if not quite impossible, to form any distinct idea of anything capable of being excited and communicated m those experiments, except it be MOTION.[110]

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这里明显呼应了马克思关于劳动力和工资购买的生活资料作为生产性劳动必要能量来源论述。在第［5］章中，我们曾提及马克思剥削理论依赖于功与能量物理等价相关概念。上文我们已引用廷德尔关于这些主题讲座内容。事实证明，马克思实际上曾旁听过这些讲座，廷德尔在其中总结了伦福德研究成果。

There are obvious echoes here of Marx discussing labour power and wage goods as the necessary source of energy for productive labour. In Chapter [5], we referred to the dependence of Marx’s theory of exploitation on concepts relating to the physical equivalence of work and energy. We have cited above from Tyndall’s lectures on these topics. It turns out that Marx actually attended these lectures in which Tyndall summarized the work of Rumford.

马克思和恩格斯对许多参与热力学（包括第一定律和第二定律）发展科学家著作有所了解，在某些情况下还进行了深入研究——其中包括赫尔曼·冯·亥姆霍兹、朱利叶斯·罗伯特·迈尔、詹姆斯·普雷斯科特·焦耳、尤斯图斯·冯·李比希、让·巴蒂斯特·约瑟夫·傅里叶、萨迪·卡诺、鲁道夫·克劳修斯、威廉·汤姆森、彼得·格思里·泰特、威廉·格罗夫、詹姆斯·克拉克·麦克斯韦以及路德维希·爱德华·玻尔兹曼。此外，我们知道，在 1867 年《资本论》第一卷出版前后，马克思曾多次参加自然科学公开讲座，其中包括英国物理学家约翰·廷德尔（其著作《热作为运动的一种形式》）所做一系列讲座。廷德尔本身就是当时物理学发展进程中重要人物，他是 J.R. 迈尔思想主要倡导者——迈尔是能量守恒定律（热力学第

Marx and Engels had some familiarity with and in some cases had closely studied the works of many of the scientists involved in the development of thermodynamics (both the first and second laws) –including Hermann von Helmholtz, Julius Robert Mayer, James Prescott Joule, Justus von Liebig, Jean-Baptiste Joseph Fourier, Sadi Carnot, Rudolf Clausius, William Thomson, Peter Guthrie Tait, William Grove, James Clark Maxwell, and Ludwig Eduard Boltzmann. In addition, we knew that Marx had attended numerous public lectures on natural science in the years leading up to and following the publication of Capital, Volume I in 1867, and that among these was a series of lectures by the English physicist John Tyndall, author of Heat Considered as a Mode of Motion. Tyndall, a major figure in the developing physics in his own right, was

一定律）共同发现者之一。马克思关注了廷德尔关于太阳光研究，特别是其与热相关部分。[111]

廷德尔讲座使他了解到，热是原子或分子运动结果；冰的膨胀、熔化和汽化，是由于分子运动克服了内聚力所致；他还能够估算大气中气体分子实际速度。[112] 从《自然辩证法》笔记和片段中可以明显看出，恩格斯对当时科学领域这些主题非常熟悉。[113]

但现在我们要转向两个问题，这些问题似乎超出了马克思和恩格斯对其所处时代力学的认知范围，这些问题是麦克斯韦和玻尔兹曼能够提出但尚未给出充分答案的。[114]

1. 为什么时间似乎具有方向性？

2. 知识或信息与功之间有何关系？

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从历史上看，这两个问题曾以相反顺序被提出，但我们将先讨论“时间之箭”。这与第 5.1 节和第 5.4 节中已提出问题是相关的，因为恩格斯和达尔文不仅都隐含接受了时间具有方向性这一观点，还分别针对这种方向性提出了不同假说。

自古以来，过去与未来之间差异以及时间“流逝”便显而易见。出生、成长、死亡和衰败都是显而易见。人们从经验中知道，春天雪会融化，热锅会冷却等等。这种时间方向性并不值得大惊小怪。时间流逝或许令人惋惜，但似乎是不言而喻。就连牛顿这样严谨哲学家，也把时间流逝视为公理性假设，这一点我们此前已加以批判。[115]拉格朗日力学和哈密顿力学发展——这两者侧重于受守恒定律约束完整轨迹——突显了把时间视为流动观点之不足，并揭示出力学基本定律在时间轴上具有对称性。

在抽象动力学中，若把系统中每个运动粒子的运动瞬间逆转，则该系统将向后运动，其中每个粒子都将沿其原路径运动——当再次处于同一位置时，速度与先前相同——也就是说，用数学术语表述，当 t 变为 $-t$ 时，任何解仍然是解。

因此，如果宇宙中每一个物质粒子的运动在任何瞬间都被精确逆转，自然进程从那时起将永远被彻底逆转。瀑布脚下爆裂泡沫气泡会重新聚合并落回水中；热运动会重新凝聚其能量，把水流以水滴形式抛向瀑布上方，重新形成一股紧密上升水柱。原本由固体摩擦产生、并通过传导、辐射及吸收而消散热量，将重新回到接触点，并使运动物体逆着它先前屈服那股力量反弹回去。巨石将从泥浆中回收重建其原有崎岖形态所需材料，并重新与它们曾经脱落山峰合为一体。[116]

the principal advocate of the ideas of J.R. Mayer –one of the co-discoverers of the conservation of energy (the first law of thermodynamics). Marx followed Tyndall’s research on the Sun’s rays, particularly as it related to heat.[111]

Tyndall’s lectures taught him that heat is a result of atomic or molecular motion, and that the expansion, melting and vaporization of ice occur due to molecules’ motion overcoming cohesion forces, and he could estimate the actual velocities of gas molecules in the atmosphere.[112] Engel’s familiarity with these topics of the science of his time is obvious from the notes and fragments for *Dialectics of Nature*.[113]

But we now move to two questions that pass beyond the boundary of what Marx and Engels seem to have known about the mechanics of their day, questions that Maxwell and Boltzmann could pose but not yet give an adequate answer to.[114]

1. Why does time appear to have a direction?

2. How does knowledge or information relate to work?

Historically, the questions were posed in the opposite order, but we will deal with times arrow first. It relates to issues already raised in Sections 5.1 and 5.4 since both Engels and Darwin implicitly accept not only that time is directional, but also provide distinct hypotheses for that directionality.

The difference between the past and the future, and the ‘flow’ of time has been obvious since antiquity. Birth, growth, death and decay were obvious. People knew from experience that snow melts in the spring, that hot pans cool down, etc. This direction of time was unremarkable. Time’s flow and passage might be regretted but seemed self-evident. Even a careful philosopher like Newton took time-flow as an axiomatic assumption, which we criticized earlier.[115] The development of Lagrangian and Hamiltonian mechanics, which focus on complete trajectories constrained by conservation laws, emphasized the inadequacy of viewing time as a flow, revealing that the basic laws of mechanics were symmetrical with respect to the time axis.

IN abstract dynamics an instantaneous reversal of the motion of every moving particle of a system causes the system to move backwards, each particle of it along its old path,-and at the same speed as before when again in the same position – that is to say, in mathematical language, any solution remains a solution when t is changed into $-t$.

If, then, the motion of every particle of matter in the universe were precisely reversed at any instant, the course of nature would be simply reversed for ever after. The bursting bubble of foam at the foot of a waterfall would reunite and descend into the water : the thermal motions would re-concentrate their energy and throw the mass up the fall in drops reforming into a close column of ascending water. Heat which had been generated by the friction of solids and dissipated by conduction, and radiation with absorption, would come again to the place .of contact and throw the moving body back against the force to

which it had previously yielded. Boulders would recover from the mud the materials required to rebuild them into their previous jagged forms, and would become reunited to the mountain peak from which they had formerly broken away.[116]

追踪两颗相撞台球轨迹时，若把轨迹方向颠倒，情景似乎完全相同。在真实台面上，球体因摩擦而减速；而在真空中，原子则无摩擦地滑行，从而实现了完美可逆性。然而，热力学却揭示了根本上不可逆过程。热量自然从高温物体流向低温物体，封闭系统熵不会减少。虽然理论力学可能不区分过去和未来，但热力学证实了我们对时间方向性的本能感知。

〔校注：底本写 closed systems（封闭系统）的熵不减；此处保留原措辞。该说法没有在此句中明示隔离条件，不将原书措辞静默改为 isolated systems。〕

Tracing the paths of two colliding billiard balls, the scenario appears identical if the paths are reversed. On a real table, balls decelerate due to friction, whereas atoms glide without friction through space, allowing perfect reversibility. However, thermodynamics has uncovered fundamentally irreversible processes. Heat naturally flows from hot to cold bodies, and the entropy of closed systems does not decrease. While theoretical mechanics may not differentiate between past and future, thermodynamics confirms our innate sense of time’s directionality.

热力学最初是一门经验科学，源于生产实践需求。[117]致力于完善蒸汽机工程师们是其最初推动力。用马克思话来说，这是一门由新的物质生产方式驱动科学。作为资本主义工业节省劳动时间冲动产物，热力学断言了时间方向不可抗拒客观性。鉴于此，热原子论面临挑战在于证明，力学定律本身就使时间指针指向未来。

Thermodynamics starts as an empirical science, an outgrowth of the practical needs of production.[117] Engineers perfecting steam engines were its initial impulse. It was, in Marx’s terms, a science driven by the new mode of material production. A child of capitalist industry’s drive to economize on labour time, thermodynamics asserted the inexorable objectivity of time’s direction. Given this, the challenge for the atomic theory of heat was to demonstrate that the laws of mechanics themselves set time’s compass pointing to the future.

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汤姆森——上文曾提及他指出了时间理论可逆性——对此给出如下解释。

Thomson, whose pointing out of the theoretical reversibility of time is cited above, gave an explanation in the following way.

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他没有去探讨时间可逆性更复杂实例——例如瀑布倒流或摩擦力逆向作用——而是选择聚焦于更简单热传导案例。他提出，设想一根金属棒，左侧温度均匀而较热，右侧温度均匀而较冷，且整体被完美隔热层包裹。随着时间推移，热量会扩散到整根金属棒，最终导致从一端到另一端温度保持一致。

Instead of exploring the more complex instances of time reversal, such as waterfalls flowing in reverse or friction working backwards, he chose to focus on the simpler case of heat transfer. Imagine a metal bar, he proposed, uniformly heated on the left side and uniformly cooled on the right, all encased in perfect insulation. Over time, the heat would spread throughout the bar, eventually leading to a consistent temperature from one end to the other.

他承认，尽管当时人们普遍认为金属棒中热量源于其原子运动，但在 1875 年，物理学界对原子在金属内部振动机制尚不完全了解。他设想了一个由带小孔的可透过隔板隔开腔体，腔内分别充有热气和冷气。如果能远程控制这些小孔开闭，左侧高速热分子将穿过隔膜，而右侧低速冷分子则会从右向左通过。这种交换将使两侧分子平均速度趋于平衡。

He acknowledges that while it was accepted that heat in a bar of metal was due to the motion of the bar’s atoms, in 1875, physics had an imperfect understanding of how atoms vibrated within metal. He proposed a cavity with warm and cold gas, separated by a permeable barrier with small holes. If these holes could be opened or closed remotely, hot fast molecules from the left would pass through the barrier, while cold slow ones would pass from right to left. This exchange would equalize the average speeds of the molecules on both sides.

汤姆森明确考虑的是有限系统，其中气体分子数量是固定的。但他有限系统仍涉及数量极其庞大分子：例如，其中一个例子中就包含 8,000,000,000,000 个氮分子。我们将用数量少得多的分子——仅 80 个——来说明这一原理。我们不考虑分子在三维空间中以不同速度自由运动情况，而是采用一个更简单模型。我们将运动限制在二维空间内，以便绘制简单示意图。进一步把分子运动方向限定为上下或左右，并假设速率均为常数。换言之，我们将对模型

Thompson was explicitly considering finite systems, with fixed numbers of gas molecules. But his finite systems still involved very large numbers of molecules: 8,000,000,000,000 molecules of nitrogen in one example. We shall illustrate the principle with much smaller numbers: just eighty molecules. Instead of looking at molecules moving freely in three dimensions at varying velocities, we will take a much simpler model. We will restrict the motion to two dimensions to allow us to draw

自由度进行量子化，或减少其数量，从而得到所谓“格子气模型”。

〔校注：本段底本拼作 Thompson，前后文作 Thomson；中文按同一人物统一为汤姆森，英文保留底本拼写。〕

simple pictures. We will further restrict the directions of motion of the molecules to be either up/down or left/right and assume velocities are constant. In other words, we will quantize or reduce the number of degrees of freedom of the model, obtaining what is termed a lattice gas model.

这些简单格子模型已被证明足以满足许多实际需求——例如空气动力学计算机模拟。[118]平面格子气模型由于涉及镶嵌结构，可以是六边形格子，也可以是我们将要使用直角格子（图 5.4）。在这两种类型中，其行为均由一组规则来规定，这些规则把进入格子粒子与离开格子粒子联系起来，从而满足质量、动量和能量守恒。本例中将使用规则在图 5.3 中以示意图形式展示。

These simple lattice models have been proven to be adequate for many practical purposes –computer simulation of aerodynamics being an example.[118] Lattice gas models for the plane, since they involve tessellation, can either be hexagonal or the right-angle one we will use (Figure 5.4). In both types, the behaviour is specified by a set of rules that relate incoming particles on the lattice to outgoing ones in such a way as to conserve mass, momentum and energy. The rules we will use in this example are displayed diagrammatically in Figure 5.3.

格子气模型便于高效地进行数字模拟。作者们为此专门构建了专用计算机。[119]由于这些模型处理的是以恒定速率运动粒子，我们无法像汤姆森最初例子那样模拟温度差，但可以通过压力差来阐释一个类似、实质上不可逆过程。

Lattice gas models lend themselves to efficient digital implementation. The authors have constructed specialized machines for this purpose.[119] Since they work with particles moving at constant velocities, we cannot model temperature differences as in Thomson’s original example, but we can illustrate a similar effectively irreversible process using pressure differences.

请看图 5.4。你可以看到，最初有序状态最终演变为气体原子以看似随机方式分布在格子区域内（图 5.4.b）。

Look at Figure 5.4. You can see how initial order ends up with gas atoms spread through the area of lattice in what appears to be a random fashion(Figure 5.4.b).

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这两幅图生动地阐释了熵增概念。图（a）描绘了一种高度有序状态，所有八十个原子都位于左侧，并向右或向下运动。图（b）则显示这些原子已分散在整个区域内，在动量方面，它们向任何方向（上、下、左或右）运动概率均等。

The two images effectively illustrate the concept of increasing entropy. Image (a) depicts a highly ordered state with all eighty atoms positioned to the left, moving either right or down. Image (b) shows these atoms spread out across the area and, in terms of momentum, with an equal likelihood of moving in any direction—up, down, left or right.

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一旦原子在格子状态中随机分布，在后续观测中发现它们全都位于同一侧概率就微乎其微。如果我们在过程进行时记录左侧原子数量（图 5.5），会发现左侧原子数起初为 80，但很快演变为在 40 附近不规则地振荡。

Once the atoms are randomly distributed among lattice states, the chances of finding them all on one side upon further observation are exceedingly slim. If we record how many atoms were on the left as the process proceeds (Figure 5.5), we see the left count starts at eighty but rapidly evolves to oscillate noisily around forty.

格子气简单状态结构使其便于进行概率分析。图 5.4 中快照是在原子离开格子节点时拍摄。共有 80 个格子位置可供原子离开，每个格子节点离开原子数在 0 到 4 之间。

The simple state structure of lattice gases lends them to easy probability analysis. The snapshots in Figure 5.4 are taken as atoms leave lattice nodes. There are eighty lattice positions from which atoms can depart, and each lattice node can have between zero and four atoms leaving it.

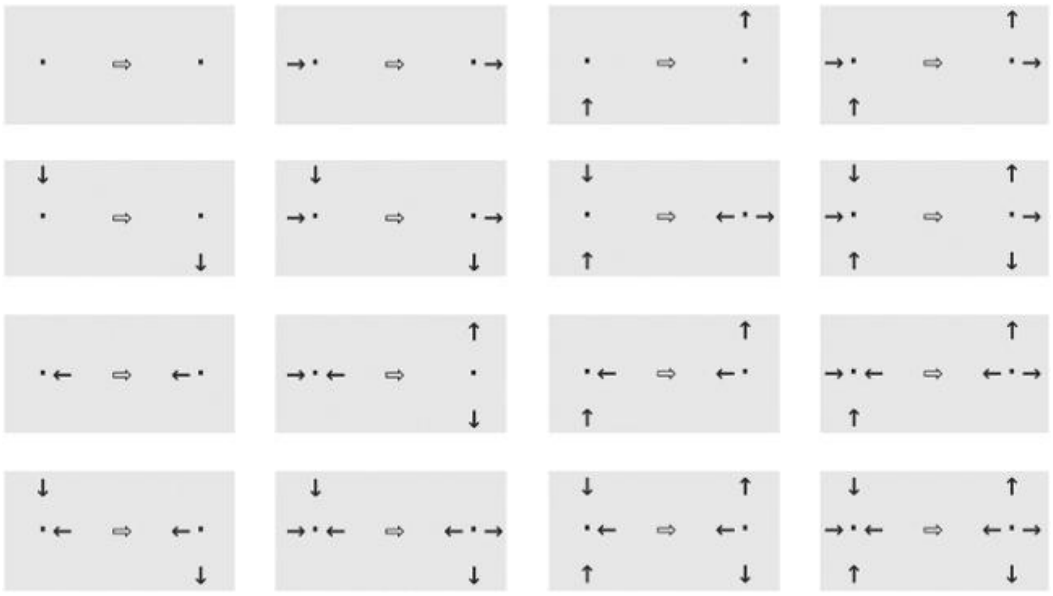


图 5.3 矩形二维格子气系统的演化规则。每个灰色矩形代表一条规则，指示入射与出射分子的配置，并保持物质、能量和动量守恒。作者绘图。

Figure 5.3 The evolution rules of a rectangular 2D lattice gas system are illustrated, with each grey rectangle representing a rule indicating the configuration of incoming and outgoing molecules, conserving matter, energy and momentum. Figure by authors.
每个灰框中双箭头左侧为入射配置，右侧为出射配置；细箭头表示粒子运动方向。

由于气体中有八十个原子，平均每个格子节点有一个原子离开。因此，每条出射路径有 $\frac{1}{4}$ 的概率被离开的原子占据，有 $\frac{3}{4}$ 的概率为空。显然，如果以原子数相对于格点数来衡量的气体总压力更高或更低，这些概率就会不同。

网格右半部有 160 条不同的出射路径。因此，右半部任何位置都没有原子的概率必定为 $\left(\frac{3}{4}\right)^{160}$，约为 1.022×10^{-20}，即 1/100,000,000,000,000,000,000。假设你有我们在 20 世纪 90 年代制造的那种物理装置，每秒能更新格子气两千万次，那么你必须让它运行大约 150,000 年，才有较大的机会使所有气体分子同时回到左半部。这还只是原子数极少的情形。汤姆森的例子假定了多得多的分子——8,000,000,000,000 个，而且这些分子的运动自由度大得多。它们可以向任何方向运动，速率可以连续变化。在这样的例子中，所有分子最终聚集在同一侧的概率如此之小，以至于在我们如今估算的整个宇宙存续时间内也不会发生。[120]如果暗能量确实存在，相空间因而不断膨胀，那么即使宇宙具有无限的年龄，也不会出现回归。

Since there are eighty atoms in the gas, on average there is one atom leaving each lattice node. Each exit path thus has a $\frac{1}{4}$ chance of being occupied by a departing atom and a $\frac{3}{4}$ chance of being empty. Obviously, these chances would be different if the overall pressure of the gas, in terms of atoms to grid positions, was higher or lower.

There are 160 different exit paths on the right-hand side of the grid. The probability of there being no atoms anywhere on the right-hand side must therefore be $\left(\frac{3}{4}\right)^{160}$ which is about 1.022×10^{-20} or 1/100,000,000,000,000,000,000. Suppose you had the physical apparatus we built in the 1990s that could update a lattice gas twenty million times a second, then you would have to run it for around 150,000 years to have a good chance of the gas molecules all returning simultaneously to the left-hand side. This is with a tiny number of atoms. Thomson in his examples assumed vastly greater numbers 8,000,000,000,000 molecules, and these molecules had much more freedom of movement. They could move in any direction and at a continuous range of velocities. In such an example, the chances of all molecules ending up on one side would be so low that it would not have occurred in what we now estimate to be the entire duration of the universe.[120] If dark energy is real, and phase space consequently expands remorselessly, then even the age of an infinite universe is not going to give us recurrence.

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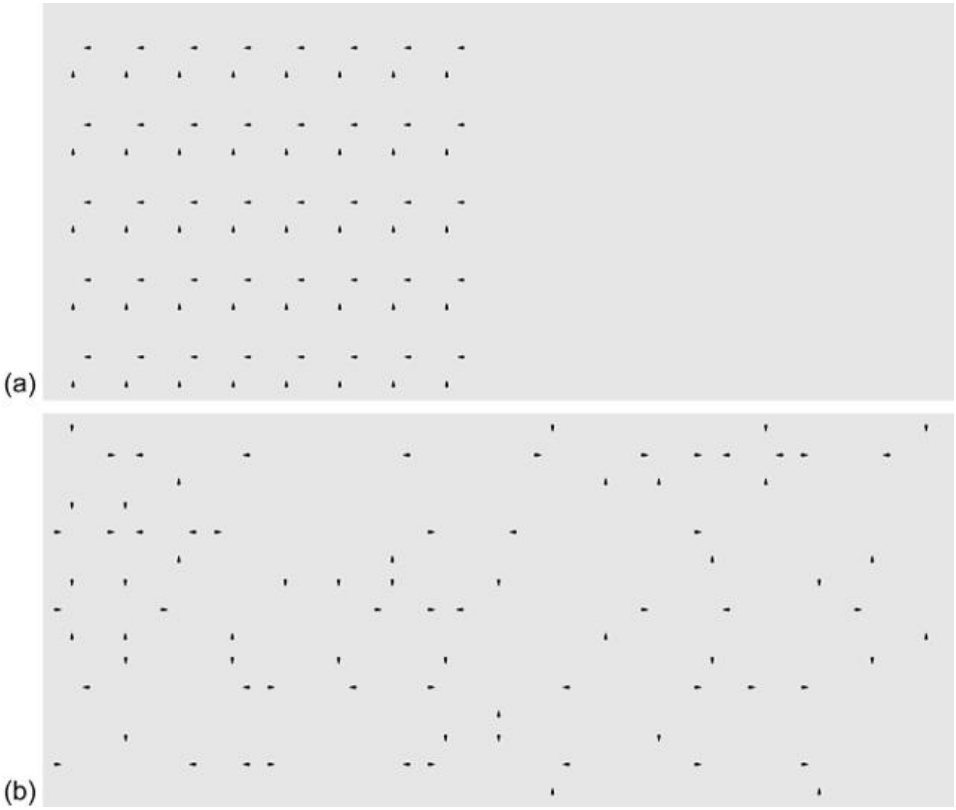


图 5.4 尺寸为 5×165 的气体格子。泪滴形状表示运动方向。格子气最初处于低熵配置：所有原子都在左侧，且只具有四种可能原子动量状态中的两种。经过 1,000 个时间步，原子均匀分布。作者绘图。

Figure 5.4 Gas lattice of dimension 5 × 165. Motion is indicated by the teardrop shape. The lattice gas is initially configured in a low-entropy state, with all atoms on the left side and only two of four possible atomic momentum states. After 1,000 time steps, the atoms are evenly distributed. Figure by authors.

（a）初始状态；（b）经过 1,000 个时间步后的状态。

校注：底本图注明写 5×165，与正文 80 个格点的说明不一致；图注尺寸保留原书，未擅改。

因此，从一团空气极有序状态出发——无论是从温度角度，还是从氧分子和氮分子相对分布角度来看（因为汤姆森同时举了这两个例子）——在允许其自由演化之后，重新回到那种有序状态可能性微乎其微。

因此，诸如热量从高温向低温流动等理论过程，可以以一种与牛顿原子运动相容方式得到解释。然而，汤姆森所描述机械逆转可能性，引发了一个问题：如果原子在达到平衡后突然逆转运动，会发生什么？

So, starting from a very ordered state of a volume of air, whether in terms of temperature or in terms of relative distribution of the oxygen and nitrogen molecules, for Thomson used both examples, a return to that order after it was allowed to freely evolve is vanishing unlikely.

Theoretical processes, such as heat flow from hot to cold, are thus explained in a way compatible with Newtonian atomic motion. However, the possibility of mechanical reversal, as described by Thomson, raises questions about what would happen if atoms’ motion suddenly reversed after reaching equilibrium.

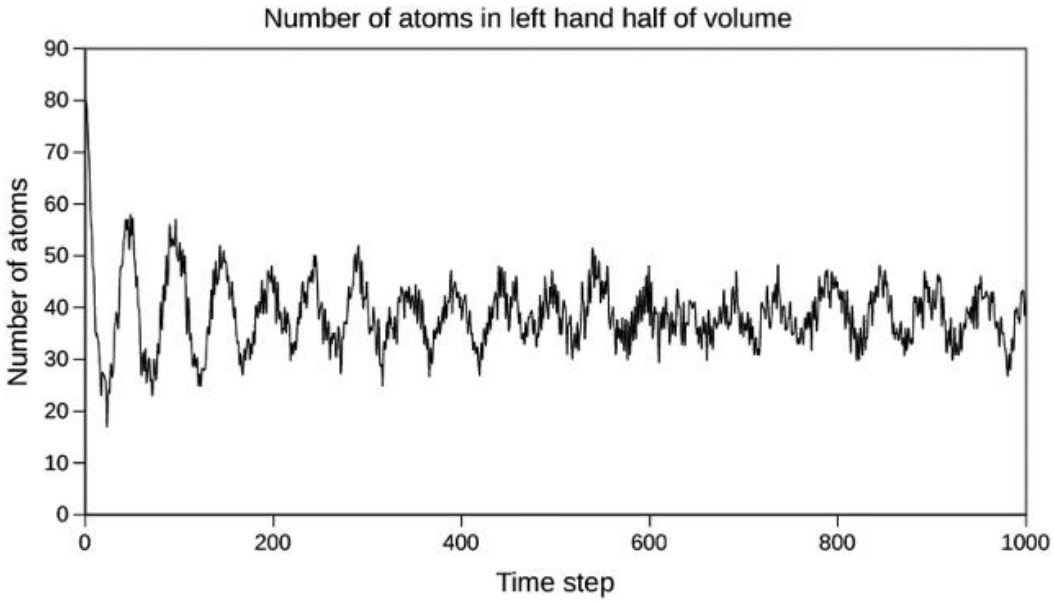


图 5.5 起初，八十个原子全在左侧。系统趋于较高熵的状态，左侧原子数围绕四十这一均值振荡，即约一半原子处于空间的左半部。这种熵涨落是汤姆森论证的一个依据。作者绘图。

Figure 5.5 At the start, all eighty atoms are on the left. The system settles to a higher entropy state with oscillations around a mean of forty, with around half of the atoms in the left half of the space. This entropy fluctuation is a basis for Thomson’s argument. Figure by authors.

图题：空间左半部的原子数；纵轴 Number of atoms：原子数；横轴 Time step：时间步。

此前发生碰撞和扩散现在必须“倒转”进行。经过与最初扩散过程完全相同时间后，系统必须恢复到初始状态：左侧是高温、高速分子，右侧是低温、低速分子，只不过此时所有分子运动方向都必须与之前相反。同样地，如果我们在格子气模拟中于第 1,001 步应用图 5.3 中逆向规则，并重复此操作 1,000 次，那么所有原子最终都会聚集在左侧。但这一次，所有动量都将指向左侧或向上。秩序将从无序中重现。

然而，这种回归“伊甸园”前景极其敏感。只要改变图（b）中一个原子动量，应用反向演化规则就无法重现图（a）状态，而是会产生另一种原子四处散乱、看似无序状态。此外，即使反转操作完美无缺，结果也只是昙花一现：

现在假设在开始后某个时刻，温度已大致达到平衡，并让每个粒子运动瞬间发生逆转。每个分子将沿其原路径回溯，而在与最初时间间隔相等第二个时间间隔结束时，每个分子将处于与起始时相同位置，并以相同速度运动；因此，将再次出现最初给定不均匀温度分布，唯一区别在于每个粒子都朝着与其初始运动方向相反方向运动。这一差异不会妨碍随后立即开始均化过程；尽管各分子运动轨迹完全不同，但平均而言，该过程将遵循与系统最初被置于自发状态后立即发生情况相同规律。[121]

The collisions and diffusion that had previously occurred must now play backwards. After the exact same time that the diffusion had originally taken, the initial starting point must be recovered with hot, fast molecules to the left and cold, slow ones to the right, except that in each case now, the movements must be in the reverse sense. Similarly, were we in our lattice gas simulation to apply the inverse rules of Figure 5.3 at step 1,001 and continue doing this 1,000 times then all our atoms would end up on the left. But this time all momenta would be towards the left or upwards. Order would have been recreated from disorder.

But this prospect of a return to Eden is exquisitely sensitive. Change the momentum of one atom in picture (b), and the application of the reversed evolution rules will fail to recreate picture (a), yielding instead another apparently disordered state with atoms all over the place. Moreover, even should the reversal work perfectly, the result would be evanescent:

Suppose now the temperature to have become thus very approximately equalised at a certain time from the beginning, and let the motion of every particle become instantaneously reversed. Each molecule will retrace its former path, and at the end of a second interval of time, equal to the former, every molecule will be in the same position, and moving with the same velocity, as at the beginning; so that the given initial unequal distribution of temperature will again be found, with only the difference that each particle is moving in the direction reverse to that of its initial motion. This difference will not prevent an instantaneous subsequent commencement of equalisation, which, with entirely different paths for the individual molecules, will go on in the average according to the same law

as that which took place immediately after the system was first left to itself.[121]

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因此，根据汤姆森观点，我们将得到一种时间对称性（图 5.6）。从我们通常时间方向来看，气体从熵低、概率小状态向熵高、概率大状态运动。如果我们把所有运动方向逆转，它将从熵高状态演化回熵低状态，但随后会穿过熵低状态，并再次进入概率更大熵高状态。

So, according to Thomson, we would have a time symmetry(Figure 5.6). Viewed from the perspective of our normal time direction, the gas moves from a low-entropy improbable state to a high entropy probable state. If we reverse all motions, it evolves back from high entropy to low entropy, but would then pass through the low-entropy state and again move into a more probable high entropy state.

既然热力学时间之箭具有这种更深层次时间对称性，那么它究竟意味着什么？

What does the thermodynamic arrow of time then amount to, given that it possesses this deeper time symmetry?

回顾图 5.6。从时间 10 开始，所有原子运动都发生逆转，因此系统以逆向方式重演其历史，直到时间 20，它到达了时间 0 状态的镜像。[122]由于时间 20 这种低熵状态本质上是不太可能出现，因此极有可能，从时间 20 开始，该系统将演化为一种更无序状态。但这一论点隐含了时间 0 之前“前史”。

Look back at Figure 5.6. From time 10, all atomic motions reverse, so the system recapitulates its history in reverse until time 20, it arrives at a mirror image[122] of its state at time 0. Since this low-entropy state at time 20 is inherently improbable, the overwhelming likelihood is that from time 20 on the system evolves into a more disordered state. But this argument implies the prehistory before time 0.

根据力学定律时间对称性，时间-1 状态必须是时间 21 状态镜像。A 时段是 D 时段镜像。因此，可以推断时间-1 和时间 21 熵必须相同，而通过归纳法，在通向时间 0 的 A 时段内，熵本应呈下降趋势！

By the time symmetry of the laws of mechanics, the state at time −1 must be the mirror image of that at time 21. Period A is the reflection of period D. As such, the entropy must be inferred to be the same at time −1 and time 21, and by induction, the entropy for period A leading up to time 0 *should have been falling!*

因此，把时间不对称归因于概率解释并非最终的答案。如果我们假设一个孤立系统当前处于低熵状态，那么它在一小时后处于高熵状态确实很可能。但同样，一小时前它处于比现在更高熵状态也必然很可能。

So the explanation that time asymmetry arises from probability turns out not to be final. If we assume that an isolated system is in a low-entropy state now, then it is indeed probable that it will be in a higher entropy state in an hour’s time. But it must also be probable that an hour ago it was in a higher entropy state than it is now.

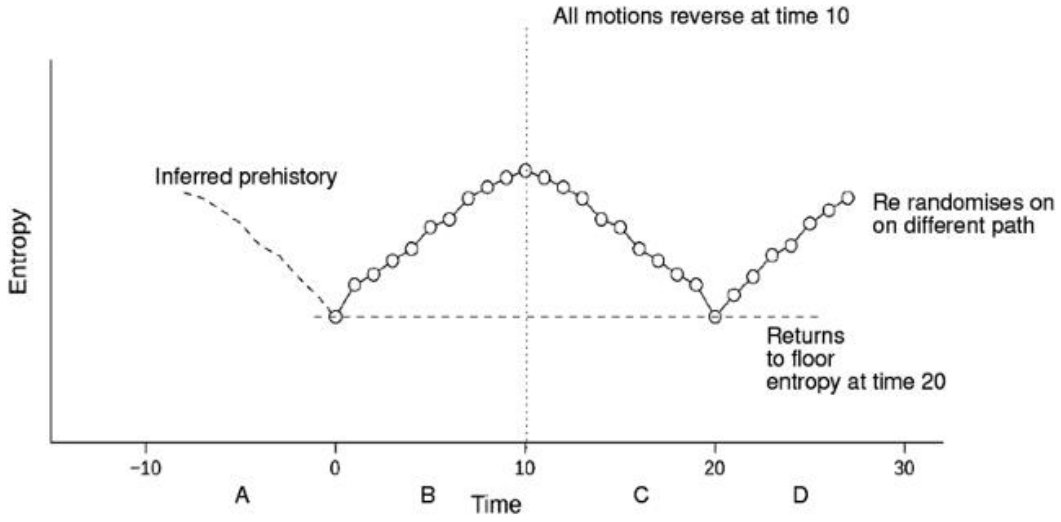


图 5.6 汤姆森所描述的有限系统的时间演化。时间 0 时，系统处于低熵状态，随后向熵增加的方向演化。时间 10 时，所有粒子的运动方向被逆转。系统于是演化回原来的低熵状态，并在时间 20 到达该状态。此后熵再次增加，但在微观层面沿着一条不同的路径演化。作者绘图。

Figure 5.6 The time evolution of a finite system as described by Thomson. At time 0, the system is in a low-entropy state and evolves towards increasing entropy. At time 10, all particle motions are reversed. The system now evolves back to its original low-entropy state, attaining it at time 20. Entropy then starts to increase again but follows, at the microscopic level, a different path. Figure by authors.

纵轴 Entropy：熵；横轴 Time：时间；Inferred prehistory：推断的前史；All motions reverse at time 10：时间 10 时所有运动逆转；Returns to floor entropy at time 20：时间 20 时回到最低熵；Re randomises on different path：沿不同路径重新随机化。A、B、C、D 标示四个时段。

如果你有一个像格子气那样有限模型系统，它遵循对称动力学定律（这些定律具有守恒性质），那么你确实会观察到这样的景象。如果我们将逆演化规则对图 5.4.a 应用 1,000 次，就会得到一幅看似随机的图 c，其中原子分布于整个体积中。对图 c 应用正向规则，最终又会回到图 5.4.a。把这一概念推广到宏观层面时，其蕴含意义令人不安。我们是否应该根据概率推断：一秒钟前世界中那个看似不太可能出现低熵状态，很可能是在两秒钟前一个高熵状态之后才出现？

If you had a finite model system like a lattice gas that evolves under symmetrical dynamical laws, ones which have conservative properties, that is indeed what you would observe. If we applied reverse evolution rules 1,000 times to picture 5.4.a, we would get an apparently random picture *c* with atoms throughout the volume. Apply the forward rules to picture *c* and we would end up again with picture 5.4.a. When applying this concept globally, the implications are unsettling. Should we infer, based on probability, that a seemingly improbable low-entropy state of the world one second ago was likely preceded by a higher entropy state two seconds prior?

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认为当前熵增加源于几秒钟或几天前一瞬间低熵状态——而在那之前时间曾逆向流动——这种观点是否合理？

Is it plausible to believe that the current rise in entropy originates from a fleeting low-entropy condition a few seconds or days earlier, before which time flowed in reverse?

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难道我们对过去记忆仅仅是虚幻？汤姆森关于封闭系统的论点似乎暗示了这种可能性。他说，在熵减阶段（时间从-10 到 0，或从 10 到 20 之间，见图 5.6），意识会发生逆转。

Could it be that our recollections of the past are merely deceptive? Thomson’s argument about closed systems would seem to imply this possibility. He says that during an entropy-reducing phase (times −10 to 0 or times 10 to 20 in Figure 5.6) consciousness would reverse.

如果关于生命唯物主义假说也是正确的，那么生物将向后生长，虽然对未来有清醒认知，却对过去毫无记忆，并会重新回到未出生状态。[123]

if also the materialistic hypothesis of life were true, living creatures would grow backwards, with conscious knowledge of the future, but no memory of the past, and would become again unborn.[123]

我们会知道未来，但因记住了它，便以为那是过去。

We would know the future but, remembering it, think it was the past.

他回避了“这种情况正在发生”这一结论，因为他说自己只是在考虑有限且孤立系统。他谨慎指出，这并不适用于与宇宙其余部分相互作用系统：

He avoids the conclusion that this is happening because he says he was just considering finite and isolated systems. He is careful to state that it does not apply to systems communicating with the rest of the universe:

关于热传导这一众所周知古老定律，有一种奇怪却真实理解：即在 1,000 年时间里，一根铁棒一半自行比另一半升温一度可能性极小；而且，这一现象在 1,000,000 年内发生概率，是 1,000 年内发生概率 1,000 倍，并且在极其漫长时间内肯定会发生。但须记住，我们假设这根铁棒表面覆盖着一层不透热的漆层。若摒弃这一不可能理想假设，并认为宇宙中分子数量是无限；那么我们便可断言，除非有外部热源或冷源作用，否则铁棒一半温度永远不会高于另一半。这一实例足以阐明能量耗散理论所依据哲学基础。[124]

It is a strange but a true conception of the old well-known law of the conduction of heat to say that it is very improbable that in the course of 1,000 years one half the bar of iron shall of itself become warmer by a degree than the other half; and that the probability of this happening before 1,000,000 years pass is 1,000 tines as great as that it will happen in the course of 1,000 years, and that it certainly will happen in the course of some very long time. But let it be remembered that we have supposed the bar to be covered with an impermeable varnish. Do away with this impossible ideal, believe number of molecules in the universe to be infinite ; then we may say one half of the bar will never become warmer than the other, except by the agency of external sources of heat or cold. This one instance suffices to explain the philosophy of the foundation on which the theory of the dissipation of energy rests.[124]

在上述对汤姆森 1875 年论文论述中，我们的评注自由采用了“概率较低状态熵值较低”这一概念。这一结论证明归功于玻尔兹曼 1872 年论文（译为[125]），他在其中通过统计学论证证明了封闭系统中熵必然趋于增大。[126]1876 年，玻尔兹曼挚友洛施密特提出了与汤姆森类似的可逆性论点，[127]他在其中明确指出，如图 5.6 所示，若粒子运动方向发生逆转，则熵必然减小。

In the account of Thomson’s 1875 paper above, our commentary has freely used the notion that less probable states have lower entropy. Proof of this was due to Boltzmann’s paper of 1872 (translated as[125]) in which he proved in terms of statistical arguments that the entropy in a closed system must tend to increase.[126] In 1876, Boltzmann’s close friend Loschmidt made a similar reversibility argument to Thomson[127] where he explicitly pointed out that in the event of particle reversal then entropy must decrease as shown in Figure 5.6.

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玻尔兹曼 1872 年证明中关键细节基于这样一个假设：当气体中两个分子（A 和 B）在位置 x 发生碰撞时，每个分子到达位置 x 的概率在统计上相互独立[128]。但在两个分子刚刚碰撞之后，它们位置和速度并非相互独立。碰撞这一行为使得它们位置产生相关性——它们必须非常接近彼此才会发生碰撞。

The fine detail of Boltzmann’s 1872 proof rested on the assumption that when two molecules (A and B) of a gas collided at location x, the probability that each molecule would arrive at position x was statistically independent.[128] But immediately after two molecules collide, their position and velocities are not independent. The act of collision has correlated both their positions – they must have been very close to one another to collide.

如果把图 5.6 中时间 10 时所有运动方向逆转，那么分子运动就绝对不是相互独立。它们运动已被预先确定为恰好能使系统返回起始位置。在这种情况下，玻尔兹曼用来推导时间箭头假设就被违背了，时间箭头可能会逆转——熵可能会随时间而减小。

If you reverse all motions at time 10 in Figure 5.6, then the motions of the molecules are most definitely not independent. They are predetermined to be exactly such as will return the system to its starting position. In that case, the assumption that Boltzmann had relied upon to deduce an arrow of time is violated, and the time arrow could reverse –entropy could decline with time.

但我们确实把熵增加作为经验事实所经历。如何从宇宙本质及其初始状态角度来解释这一点呢？

But we do experience a rise in entropy as an empirical given. How is one to explain this in terms of the nature of the universe and its initial state?

我们有两种设想可供选择。一种假设是，整个宇宙目前正处于一种极不可能状态。然而，也可以设想，这种不大可能的状态持续漫长岁月，以及从这里到天狼星距离，与宇宙年龄和规模相比都微不足道。因此，在这个整体上处于热平衡状态（因而已“死亡”）宇宙中，必定存在一些大小与我们银河系相当相对较小区域（我们称之为“世界”），它们在相对短暂的漫长年代里会显著偏离热平衡状态。在这些“世界”中，状态概率增加与减少同样频繁。对于宇宙整体而言，时间两个方向是无法区分，正如在空间中没有“上”与“下”之分。然而，正如在地球表面某个地点，我们可以把指向地球中心方向称为“下”，同样，处于某个“世界”中、处于特定时间段生物，可以把时间方向定义为从概率较低状态向概率较高状态发展（前者即“过去”，后者即“未来”），并基于这一定义，它会发现这个与宇宙其余部分隔离小区域，“最初”总是处于一种概率较低状态。在我看来，这种观点似乎是理解热力学第

One has the choice of two kinds of pictures. One can assume that the entire universe finds itself at present in a very improbable state. However, one may suppose that the eons during which this improbable state lasts, and the distance from here to Sirius, are minute compared to the age and size of the universe. There must then be in the universe, which is in thermal equilibrium as a whole and therefore dead, here and there relatively small regions of the size of our galaxy (which we call worlds), which during the relatively short time of eons deviate significantly from thermal equilibrium. Among these worlds the state probability increases as often as it decreases. For the universe as a whole, the two directions of time are indistinguishable, just as in space there is no up or down. However, just as at a certain place on the Earth’s surface we can call ‘down’ the direction towards the centre of the Earth, so a living being that finds itself

二定律及每个独立世界热寂现象有效性唯一途径，而无需假设整个宇宙从一个确定初始状态单向演变为最终状态。有人提出异议，认为为了解释一小部分世界生命状态，而把宇宙如此大一部分设想为“死亡”状态，这种做法既不经济也毫无意义——我认为这一异议站不住脚。我至今仍清楚记得，曾有人坚决不相信太阳可能距离地球 2000 万英里，其理由是：难以想象竟会有如此广阔空间仅充满以太，而生命却如此稀少。[129]

〔校注：底本本引文包含 without without 与 to to 重复，以及太阳距离 20 million miles 的原述；英文照录，中文不重复语词，距离按原述保留。〕

in such a world at a certain period of time can define the time direction as going from less probable to more probable states (the former will be the ‘past’ the latter the ‘future’) and by virtue of this definition he will find that this small region, isolated from the rest of the universe, is ‘initially’ always in an improbable state. This viewpoint seems to me to be the only way one can understand the validity of the second law and the heat death of each individual world without without invoking an unidirectional change of the entire universe from a definite initial state to to a final state. The objection that it is uneconomical and hence senseless to imaging such a large part of the universe as being dead in order to explain why a small part is living –this objection I consider invalid. I remember only too well a person who absolutely refused to believe that the Sun could be 20 million miles from the Earth, on the grounds that it is inconceivable that there could be so much of space filled only with aether and so little with life.[129]

现代物理学通常直接接受这样观点：整个宇宙始于一次熵值较低大爆炸——即玻尔兹曼所提出第一个备选方案。[130]但关于其第二个备选方案变体——即存在多个局部区域或“岛屿宇宙”——至今仍在讨论之中。[131] 现代“热寂”情景涉及恒星坍缩成黑洞，最终蒸发为辐射。[132]

Modern physics generally just accepts that the whole universe just started in a low entropy big bang –the first of Boltzmann’s alternatives.[130] But variants of his second alternative, with multiple local regions or island universes, are still discussed.[131] Modern ‘heat death’ scenarios involve stars collapsing into black holes, which eventually evaporate to radiation.[132]

玻尔兹曼对熵阐释可用于解释结构最初如何形成。卢克莱修把此归因于“clinamen”（即原子的不可预测偏转）。卢克莱修设想原子最初处于“下落”状态，因此他假设它们沿着平行向下轨迹运动。如今我们知道，正如玻尔兹曼所言，“在太空中没有上下之分”。因此，我们不能假设初始状态下存在平行下落。据推测，最初大爆炸没有优先方向，而仅仅是一团由电子、质子、中子和电磁辐射（光子）组成物质。这些物质相当于卢克莱修所写“corpora”（即“第一体”）。

Boltzmann’s account of entropy can be used to explain how structure initially emerges. Lucretius attributed this to the *clinamen* or unpredictable swerve of atoms. Lucretius envisaged the atoms as initially ‘falling’, so he assumed parallel downward paths. We now know that, as Boltzmann said, ‘in space there is no up or down’. So we cannot assume initial parallel falling. The initial big bang is assumed to have had no preferred direction, but just a mass of electrons, protons, neutrons and electromagnetic radiation (photons). These were equivalent to the *corpora* or first bodies of which Lucretius wrote.

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如果我们假设它们处于热平衡状态——宇宙微波背景辐射相对均匀性也支持这一观点——那么卢克莱修面临问题依然存在。是什么让这些“最初实体”聚集在一起？

If we assume these were in thermal equilibrium, which the relative uniformity of the cosmic background radiation suggests, then the problem that Lucretius faced remains. What brought the ‘first bodies’ together?

为什么宇宙没有一直保持为亚原子粒子海洋？这在最基本层面上，正是那个古老哲学问题：为何秩序会从混沌中产生。

Why did the universe not remain a sea of subatomic particles? This, in its most elementary form, is the old philosophical question as to why order emerges from chaos.

弗劳奇[133]依据玻尔兹曼的 S 公式，为此构造了一个简单论证。把公式应用于最初含有 N 个粒子的整个宇宙，就得到 $S = k\log(N$个粒子的状态数$)$。接着他指出，由于对数值即使对于宇宙学数量也小于 100，初始中子—质子气体的熵可以近似为 $S(n, p\,\text{gas}) \approx kN$。如果四个核子结合成一个 α 粒子，即卢克莱修所谓实体的首次聚合，就会释放出一个 7 MeV 的 γ 光子。所得 α 粒子与光子构成的气体或等离子体，其熵可写为：

Frautschi[133] constructs a simple argument based on Boltzmann’s S formula for this. Applying it to the whole universe with N initial particles, we have $S = k\log(\text{number of } N \text{ particle states})$. Next, he shows that since logs, even for cosmological quantities, are less than 100, you can approximate the entropy of the initial neutron-proton gas by $S(n, p\ \text{gas}) \approx kN$. If four nucleons combine to form an alpha particle, a first agglomeration of Lucretian corpora, a 7 MeV (γ) photon is released. The entropy of the resulting alpha and photon gas, or plasma, can be written:

$$S(\alpha, \gamma\ \text{gas}) \approx k(N_\alpha + N_\gamma) \approx k\left(\tfrac{N}{4} + \tfrac{N\,7\text{MeV}}{3kT}\right)$$

$$S(\alpha, \gamma\ \text{gas}) \approx k(N_\alpha + N_\gamma) \approx k\left(\tfrac{N}{4} + \tfrac{N\,7\text{MeV}}{3kT}\right)$$

这里的关键是，光子的熵贡献与温度 T 成反比。如果 $kT > 7$ MeV，解离的中子和质子的熵就高于 α 粒子与 γ 射线的熵之和。随着宇宙膨胀，中子—质子气体的温度降至某一临界值以下，[134]在这一温度下，氦核与 γ 射线的熵之和超过孤立核子的熵，于是形成了原初氦。进一步的膨胀使 γ 光子冷却为我们如今观测到的 2.4 开尔文微波背景。

〔校注：2.4 Kelvin 为底本原述，照录。〕

弗劳奇[135]关于秩序源于混沌的论证，依赖于玻尔兹曼本人尚不知道的几项知识：亚原子粒子、α 粒子的结合能以及爱因斯坦的质能等价关系。这些发现都出现在玻尔兹曼自杀早逝后的二十年内；倘若他没有自杀，我们有理由认为他本可以亲眼见到这些发现。自玻尔兹曼发表最初那篇论文以来的一个半世纪中，他的思想得到了日益广泛的应用。我们对信息、[136]经济[137]乃至引力[138]的理解，均已被证明可以从他的熵统计理论中推导出来。

The important point here is that the entropy contribution of the photons is inversely proportional to temperature *T*. If $kT > 7$ MeV, the entropy of the disassociated neutrons and protons is higher than that of the alpha particles plus gamma rays. As the universe expanded, the temperature of the neutron-proton gas fell below a critical level[134] at which the entropy of the helium nuclei plus gamma rays became greater than that of the isolated nucleons, and primordial helium was formed. Further expansion cooled the γ photons down to the 2.4 Kelvin microwave background we now see.

Frautschi’s[135] demonstration of order from chaos relies upon several things that Boltzmann himself did not know: sub-atomic particles, the binding energy of α particles, Einstein’s mass-energy equivalence. All were discovered within two decades of Boltzmann’s premature death by suicide, in the absence of which he might reasonably be expected to have seen them. In the century and a half since Boltzmann’s original paper, the application of his ideas have spread. Our understanding of information,[136] the economy[137] and even gravity[138] have been shown to be derivable from his statistical theory of entropy.

第 5 章原注 / Chapter 5 Notes

↑1. 弗里德里希 • 恩格斯、Tristram Hunt：《家庭、私有制和国家的起源》（The Origin of the Family, Private Property and the State）（Penguin，2010）。

1 Friedrich Engels and Tristram Hunt, *The Origin of the Family, Private Property and the State* (Penguin, 2010). p. 243

↑2. 顾名思义，这些是他担任格拉斯哥大学哲学教授时所讲授的课程。20 世纪中叶，人们在 John Anderson 的笔记中发现了这些讲义的记录；Anderson 曾是斯密的学生，后来也成为教授。

2 As the name implies, they were the lectures he gave a Professor of Philosophy at the University of Glasgow. In the mid-twentieth century, transcripts of the lectures were discovered in notebooks of John Anderson who had been one of Smith’s students and later became a professor himself. p. 243

↑3. 恩格斯、Hunt：《家庭、私有制和国家的起源》，第 IX 章。

3 . Engels and Hunt, *The Origin of the Family, Private Property and the State*, Chapter IX. p. 243

↑4. 亚当 • 斯密：《法理学讲义》（Lectures on Jurisprudence），《亚当 • 斯密著作与通信格拉斯哥版》，1978，第 I 部分第 2 节。

4 Adam Smith, *Lectures on Jurisprudence. The Glasgow edition of the Works and Correspondence of Adam Smith*, 1978, Part I, section 2. p. 243

↑5. 斯密：《法理学讲义》，《亚当 • 斯密著作与通信格拉斯哥版》，第 I 部分第 2 节。

5 Smith, *Lectures on Jurisprudence. The Glasgow edition of the Works and Correspondence of Adam Smith*, Part I, section 2. p. 243

↑6. “国家是阶级矛盾不可调和的产物和表现。国家是在阶级矛盾客观上不能调和的地方、时候和程度上产生的。反过来说，国家的存在证明阶级矛盾不可调和。”V. I. 列宁：《国家与革命》（State and Revolution）（Haymarket Books，2015），第 1 章第 1 部分。

6 ‘The state is a product and a manifestation of the irreconcilability of class antagonisms. The state arises where, when and insofar as class antagonism objectively cannot be reconciled. And, conversely, the existence of the state proves that the class antagonisms are irreconcilable.’ V. I. Lenin, *State and Revolution* (Haymarket Books, 2015), Chapter 1, Part 1 p. 243

↑7. 斯密：《法理学讲义》，《亚当 • 斯密著作与通信格拉斯哥版》，第 I 部分第 2 节。

7 Smith, *Lectures on Jurisprudence. The Glasgow edition of the Works and Correspondence of Adam Smith*, Part I, section 2. p. 243

↑8. 斯密：《法理学讲义》，《亚当 • 斯密著作与通信格拉斯哥版》，第 I 部分第 3 节。

8 Smith, *Lectures on Jurisprudence. The Glasgow edition of the Works and Correspondence of Adam Smith*, Part I, section 3. p. 243

↑9. 恩格斯、Hunt：《家庭、私有制和国家的起源》，第 V 章。

9 Engels and Hunt, *The Origin of the Family, Private Property and the State*, Chapter V. p. 243

↑10. 斯密：《法理学讲义》，《亚当 • 斯密著作与通信格拉斯哥版》，第 I 部分第 6 节。

10 Smith, *Lectures on jurisprudence. The Glasgow edition of the works and correspondence of Adam Smith*, Part I, section 6. p. 243

↑11. 值得注意的是，同一过程也在美利坚帝国发生。其公民逐渐不再喜欢战争的艰辛，因此从 20 世纪 80 年代开始，它越来越多地雇用阿富汗人和阿拉伯人作战；本 • 拉登后来让他的追随者转而攻击昔日支付报酬的雇主。若能以历史为鉴，类似事情不会是最后一次发生。

11 It is interesting to note that this same process is occurring in the American Empire. Its citizens, having grown less fond of the hardships of war, from the 1980s, it increasingly resorted to hiring Afghans and Arabs to fight; bin Laden in due course turned his followers on his erstwhile paymasters. If history is any judge, that will not be the last time something similar occurs. p. 243

↑12. Ibn Khaldun、Franz Rosenthal、Nessim Joseph Dawood：《历史绪论：历史学导论，三卷本》（The Muqaddimah: An Introduction to History; in Three Volumes）（普林斯顿大学出版社，1969），第 1 卷第 43 页。

12 Ibn Khaldun, Franz Rosenthal, and Nessim Joseph Dawood, *The Muqaddimah: An Introduction to History; in Three Volumes* (Princeton University Press, 1969), 1, 43. p. 243

↑13. Bartkus 和 Hassan（“伊本 • 赫勒敦与亚当 • 斯密：对劳动分工理论及现代经济思想的贡献”，Ibn Khaldun and Adam Smith: Contributions to the Theory of the Division of Labor and Modern Eco-

13 Bartkus and Hassan (‘Ibn Khaldun and Adam Smith: Contributions to the Theory of the Division of Labor and Modern Economic Thought’) speculated that Smith derived his ideas from Khaldūn, that Newton cribbed from subsequently lost works of Archimedes. p. 243

nomic Thought）推测，斯密的思想源自赫勒敦，牛顿则抄用了后来失传的阿基米德著作。

p. 244 ↑14. 参见 Jeremy L. England：“自我复制的统计物理学”（Statistical Physics of Self-replication），The Journal of Chemical Physics，第 139 卷，121923（2013），doi:10.1063/1.4818538。

14 Cf. Jeremy L. England, ‘Statistical Physics of Self-replication’, *The Journal of Chemical Physics* 139, no. 121923 (2013), doi: 10.1063/1.4818538.

p. 244 ↑15. Denis Diderot，载 D. Diderot 与 J. D’Alembert 编：《科学、艺术与工艺百科全书》（Encyclopédie des arts et des métiers），35 卷（Briasson，1751—1780；重印本，Stuttgart/Bad Cannstatt：Frommann，1966），第 V 卷第 270 页。在线有多个版本：Lexilogos、官方 ARTFL 项目及新的 ENCRRE 项目。

15 Denis Diderot in D. Diderot and J. D’Alembert, eds, *Encyclopédie des arts et des métiers*, 35 vols. (Briasson, 1751–1780; Reprint, Stuttgart/Bad Cannstatt: Frommann, 1966), V:270. Different online versions available: Lexilogos, official ARTFL project and the new ENCRRE project.

p. 244 ↑16. Denis Diderot：“达朗贝尔与狄德罗的对话（1769）”（Conversation Between D’Alembert and Diderot），载 Diderot, Interpreter of Nature，Jean Stewart 与 Jonathan Kemp 译（International Publishers，1943），无页码，https://www.marxists.org/reference/archive/diderot/1769/conversation.htm（访问日期：2022 年 11 月 12 日）。

16 Denis Diderot, ‘Conversation Between D’Alembert and Diderot (1769)’, in Diderot, *Interpreter of Nature*, trans. Jean Stewart and Jonathan Kemp (International Publishers, 1943), s.p, https://www.marxists.org/reference/archive/diderot/1769/conversation.htm (accessed on 12 November 2022).

p. 244 ↑17. 狄德罗：“达朗贝尔与狄德罗的对话（1769）”，https://www.marxists.org/reference/archive/diderot/1769/conversation.htm（访问日期：2022 年 12 月 18 日）。

17 Diderot, ‘Conversation Between D’Alembert and Diderot (1769)’, https://www.marxists.org/reference/archive/diderot/1769/conversation.htm (accessed on 18 December 2022).

p. 244 ↑18. 狄德罗：“达朗贝尔与狄德罗的对话（1769）”，https://www.marxists.org/reference/archive/diderot/1769/conversation.htm（访问日期：2022 年 12 月 19 日）。

18 Diderot, ‘Conversation Between D’Alembert and Diderot (1769)’, https://www.marxists.org/reference/archive/diderot/1769/conversation.htm (accessed on 19 December 2022).

p. 244 ↑19. Robyn P. Araujo、Lance A. Liotta：“任意规模网络实现稳健完全适应的拓扑要求”（The Topological Requirements for Robust Perfect Adaptation in Networks of Any Size），Nature Communications，第 9 卷，1757（2018），https://doi.org/10.1038/s41467-018-04151-6。

19 Robyn P. Araujo and Lance A. Liotta, ‘The Topological Requirements for Robust Perfect Adaptation in Networks of Any Size’, *Nature Communications* 9, no. 1757 (2018), https://doi.org/10.1038/s41467-018-04151-6.

p. 244 ↑20. Helen De Cruz：“动物认知、物种不变论与数学实在论”（Animal Cognition, Species Invariantism and Mathematical Realism），载 Advances in Experimental Philosophy，Andrew Aberdein 与 Matthew Inglis 编（Bloomsbury Academic，2019），第 39—62 页。

20 Helen De Cruz, ‘Animal Cognition, Species Invariantism and Mathematical Realism’, in *Advances in Experimental Philosophy,* ed. Andrew Aberdein and Matthew Inglis (Bloomsbury Academic, 2019), 39–62.

p. 244 ↑21. Elisabeth Brannon、Jessica Cantlon：“猴子与大学生的基础数学”（Basic Math in Monkeys and College Students），PLoS Biology，第 5 卷第 12 期（2007），e328，https://doi.org/10.1371/journal.pbio.0050328；Brian Butterworth：《鱼会计数吗？动物揭示我们独特数学心智的奥秘》（Can Fish Count?: What Animals Reveal about our Uniquely Mathematical Mind），Kindle 版（Quercus，2022）。

21 Elisabeth Brannon and Jessica Cantlon, ‘Basic Math in Monkeys and College Students’, *PLoS Biology* 5, no. 12 (2007): e328, https://doi.org/10.1371/journal.pbio.0050328; Brian Butterworth, *Can Fish Count?: What Animals Reveal about our Uniquely Mathematical Mind,* Kindle edn (Quercus, 2022).

p. 244 ↑22. Peter Wolfendale：《理性的复仇》（The Revenge of Reason），Ray Brassier 与 Reza Negarestani 作序（Urbanomic/Mono，2023）。

22 Peter Wolfendale, *The Revenge of Reason*, fore. Ray Brassier and Reza Negarestani (Urbanomic/Mono, 2023).

p. 244 ↑23. 卡尔 • 马克思：《1844 年经济学哲学手稿：异化劳动》，Martin Milligan 译（Progress Publishers，1959），Marxists Internet Archive，https://www.marxists.org/archive/marx/works/1844/manuscripts/labour.htm（访问日期：2022 年 12 月 26 日）。

23 Karl Marx, *Economic and Philosophical Manuscripts of 1844: Estranged Labor*, trans. Martin Milligan (Progress Publishers, 1959), Marxists Internet Archive URL:https://www.marxists.org/archive/marx/works/1844/manuscripts/labour.htm (accessed on 26 December 2022).

†24. 马克思：《1844 年经济学哲学手稿》，无页码。

24 Marx, *Economic and Philosophical Manuscripts of 1844*, s.p. p. 244

†25. “在这个特定问题上，唯物主义与‘马赫主义’的区别在于：唯物主义完全符合自然科学，把物质看作第一性的，把意识、思想、感觉看作第二性的，因为在明确的形式下，感觉只同物质的较高形式（有机物质）相联系，而‘在物质结构的基础中’，只能推测存在着一种类似于感觉的能力。例如，著名德国科学家恩斯特·海克尔、英国生物学家劳埃德·摩根等人就作出了这种假设，更不用说前文提到的狄德罗的猜想了。”（引自 V. I. 列宁：《全集：唯物主义和经验批判主义》，第 14 卷［Progress Publishers，1977］，第 46 页。）

25 ’ The difference between materialism and “Machism” in this particular question thus consists in the following.Materialism, in full agreement with natural science, takes matter as primary and regards consciousness, thought, sensation as secondary, because in its well-defined form sensationist associated only with the higher forms of matter (organic matter), while “in the foundation of the structure of matter” one can only surmise the existence of a faculty akin to sensation. Such, for example, is the supposition of the well-known German scientist Ernst Haeckel, the English biologist Lloyd Morgan and others, not to speak of Diderot’s conjecture mentioned above”’ (A quote from V. I. Lenin, *Collected Works: Materialism and Empirio-criticism,* vol. 14 [Progress Publishers, 1977], 46). p. 244

†26. Katerina Kolozova：“非人性的人工制品：关于意指自动机及其物理支撑在幻想统一体中的唯物主义论述”（The Artifact of Non-Humanity: A Materialist Account of the Signifying Automaton and Its Physical Support in a Fantasized Unity），Philosophy Today，第 65 卷第 2 期（2021 年春），第 359—374 页，doi:10.5840/philtoday2021420402n。

26 Katerina Kolozova, ‘The Artifact of Non-Humanity: A Materialist Account of the Signifying Automaton and Its Physical Support in a Fantasized Unity’, *Philosophy Today* 65, no. 2 (Spring 2021): 359–74, doi:10.5840/philtoday2021420402n. p. 244

†27. 许煜:《递归性与偶然性》（Recursivity and Contingency）（Rowman & Littlefield International，2019），第 41 页。

27 Yuk Hui, *Recursivity and Contingency* (Rowman & Littlefield International, 2019), 41 p. 244

†28. 亚里士多德：《物理学》，载《亚里士多德全集》修订牛津译本，Jonathan Barnes 译编（Oxford University Press，1991），195b31—199a9—199a19。

28 Aristotle, *Physics*, *The Complete Works of Aristotle, The Revised Oxford Translation*, trans. and ed. Jonathan Barnes (Oxford University Press, 1991), 195b31-199a9-199a19. p. 245

†29. 亚里士多德：《形而上学》，W. D. Ross 修订并撰导论与评注（Clarendon Press，1924），第 12 卷，MIT Classics Internet Archive，http://classics.mit.edu/Aristotle/metaphysics.html。

29 Aristotle, *Metaphysics*, a revised text with introduction and commentary from W. D. Ross (Clarendon Press, 1924), Book 12, the MIT Classics Internet Archive http://classics.mit.edu/Aristotle/metaphysics.html. p. 245

†30. 卡尔·马克思：《德谟克利特的自然哲学和伊壁鸠鲁的自然哲学的差别》，载《马克思恩格斯全集》英文版第 1 卷（Progress Publishers，1975），无页码，Marxist Internet Archive，https://www.marxists.org/archive/marx/works/1841/dr-theses/index.htm（访问日期：2021 年 11 月 27 日）。

30 Karl Marx, The *Difference Between the Democritean and Epicurean Philosophy of Nature, in Marx-Engels Collected Works,* vol. 1 (Progress Publishers, 1975), s.p., the Marxist Internet Archive https://www.marxists.org/archive/marx/works/1841/dr-theses/index.htm (accessed on 27 November 2021). p. 245

†31. Julian Hocheld:“阶级利益概念”（The Concept of Class Interest），The Polish Sociological Bulletin，第 16 期（1967），第 5—14 页，http://www.jstor.org/stable/44815383。

31 Julian Hocheld, ‘The Concept of Class Interest’, *The Polish Sociological Bulletin* 16 (1967): 5–14, http://www.jstor.org/stable/44815383. p. 245

†32. Denis Diderot：《狄德罗全集》（OEuvres complètes de Diderot），第 1 卷，J. Assézat 编（巴黎，1875），第 304 页。

32 Denis Diderot, *OEuvres complètes de Diderot*, vol. 1, éd. Par J. Assézat (Paris, 1875), 304. p. 245

†33. 马克思：《1844 年经济学哲学手稿》，https://www.marxists.org/archive/marx/works/download/pdf/Economic-Philosophic-Manuscripts-1844.pdf（访问日期：2023 年 1 月 9 日）。

33 Marx, *Economic and Philosophical Manuscripts of 1844*, https://www.marxists.org/archive/marx/works/download/pdf/Economic-Philosophic-Manuscripts-1844.pdf (accessed on 9 January 2023). p. 245

p. 245 ↑34. Katerina Kolozova：“现代性中什么值得挽救：从非哲学马克思主义到齐泽克普遍主义的实在论视角”（What Is Worth Salvaging In Modernity: A Realist Perspective From Non-Philosophical Marxism to Žižek’s Universalism），载 Understanding Žižek, Understanding Modernism，Zahi Zaloua 与 Jeffrey R. Di Leo 编（Bloomsbury，2022）。

34 Katerina Kolozova, ‘What Is Worth Salvaging In Modernity: A Realist Perspective From Non-Philosophical Marxism to Žižek’s Universalism’, in *Understanding Žižek, Understanding Modernism,* ed. Zahi Zaloua and Jeffrey R. Di Leo (Bloomsbury, 2022).

p. 245 ↑35. 卡尔·马克思：《对黑格尔哲学的总体批判》，无页码，https://www.marxists.org/archive/marx/works/down-load/pdf/Economic-Philosophic-Manuscripts-1844.pdf（访问日期：2023 年 1 月 9 日）。

〔校注：底本网址路径作 down-load，与原注 33 的 download 不同；两者分别按底本保留。〕

35 Karl Marx, *Critique of Hegel’s Philosophy in General*, s.p, https://www.marxists.org/archive/marx/works/down-load/pdf/Economic-Philosophic-Manuscripts-1844.pdf (accessed on 9 January 2023).

p. 245 ↑36. Katerina Kolozova：《迈向社会主义的激进形而上学：马克思与拉鲁埃尔》（Toward Radical Metaphysics of Socialism: Marx and Laruelle）（Punctum Books，2015）。

〔校注：底本此条漏印注号 36；据正文注 36 和前后 35、37 序列恢复。原文第 254 页 block2 第 0 行同时含本注末尾和注 37 号，源行引用因此交叠。〕

36 Katerina Kolozova, *Toward Radical Metaphysics of Socialism: Marx and Laruelle* (Punctum Books, 2015).

p. 245 ↑37. 狄德罗：“达朗贝尔与狄德罗的对话（1769）”，https://www.marxists.org/reference/archive/diderot/1769/conversation.htm（访问日期：2023 年 12 月 12 日）。

〔校注：底本网址前缀误印 fh ttps://；中文恢复 https://，英文保留底本。〕

37 Diderot, ‘Conversation Between D’Alembert and Diderot (1769)’, fh ttps://www.marxists.org/reference/archive/diderot/1769/conversation.htm (accessed on 12 December 2023).

p. 245 ↑38. 卡尔·马克思：《资本论》，第 1 卷，Marxist Internet Archive 在线版（Progress Publishers，1954），第 7 页。

38 Karl Marx, *Capital*, vol. 1, As online at Marxist Internet Archive (Progress Publishers, 1954), 7.

p. 245 ↑39. cwt 是一种英制计量单位，表示 112 磅。1 quarter 为 $\frac{1}{4}$ 英担，即 28 磅。

〔校注：本注按底本所列重量换算照译；正文中的 quarter 统一保留为“夸特”，不与液量单位 quart（夸脱）混同。底本分数已依据原注图像恢复。〕

39 A cwt is an Imperial English measure meaning 112 lbs. A quarter is $\frac{1}{4}$ cwt = 28lb.

p. 245 ↑40. 马克思：《资本论》，第 1 章，第 28 页。

40 Marx, *Capital*, Chap 1, 28.

p. 245 ↑41. 马克思：《资本论》，第 42 页。

41 Marx, *Capital*, 42.

p. 245 ↑42. 马克思：《资本论》，第 43 页。

42 Marx, *Capital*, 43.

p. 245 ↑43. 恩格斯致马克思，1858 年 7 月 14 日。

43 Engels to Marx, 14 July 1858.

p. 245 ↑44. A. M. Shaikh：“劳动价值论的实证力量”（The Empirical Strength of the Labour Theory of Value），载《马克思主义经济学：重新评价》（Marxian Economics: A Reappraisal），R. Bellofiore 编，第 2 卷（Macmillan，1998），第 225—251 页；P. Petrovic：“生产价格对劳动价值的偏离：一些方法论与实证证据”（The Deviation of Production Prices from Labour Values: Some Methodolog and Empirical Evidence），Cambridge Journal of Economics，第 11 卷（1987），第 197—210 页；David Zachariah：《检验瑞典的劳动价值论》（Testing the Labor Theory of Value in Sweden）（2004），http://reality.gn.apc.org/econ/DZ_article1.pdf；W. Paul Cockshott 与 Allin F. Cottrell：“劳动时间与其他价值基础的比较：研究札记”（Labour Time Versus Alternative Value Bases: A Research Note），Cambridge Journal of Economics，第 21 卷第 4 期（1997），第 545—549 页；W. Paul Cockshott 与 Allin Cottrell：“价格

44 A. M. Shaikh, ‘The Empirical Strength of the Labour Theory of Value’, in *Marxian Economics: A Reappraisal*, ed. R. Bellofiore, vol. 2 (Macmillan, 1998), 225–51; P. Petrovic, ‘The Deviation of Production Prices from Labour Values: Some Methodolog and Empirical Evidence’, *Cambridge Journal of Economics* 11 (1987): 197–210; David Zachariah, *Testing the Labor Theory of Value in Sweden* (2004), http://reality.gn.apc.org/econ/DZ_article1.pdf; W. Paul Cockshott and Allin F. Cottrell, ‘Labour Time Versus Alternative Value Bases: A Research Note’, *Cambridge Journal of Economics* 21, no. 4 (1997): 545–9; W. Paul Cockshott and Allin Cottrell, ‘Robust Correlations Between Prices and Labour Values: A Comment’, *Cambridge Journal of Economics* 29, no. 2 (2005): 309–16.

与劳动价值之间的稳健相关性：评论”（Robust Correlations Between Prices and Labour Values: A Comment），Cambridge Journal of Economics，第 29 卷第 2 期（2005），第 309—316 页。

↑45. 该书第十一章包含了跨越多个世纪的大量价格统计数据。

45 Chapter XI of that book contained extensive statistical data on prices over many centuries. p. 245

↑46. 这里使用的是源自 20 世纪物理学的术语。埃米 • 诺特在“变分问题的不变量”（Invariant Variation Problems），Transport Theory and Statistical Physics，第 1 卷第 3 期（1971），第 186—207 页（原作发表于 1918 年）中证明，经典物理学的守恒定律可以视为自然规律在时间和空间中具有对称性的结果。例如，动量守恒源于物理规律在空间中处处相同这一事实，即位移对称性。可以证明，时间的均匀性蕴含能量守恒。相关解释可参见 Jozef Hanc、Slavomir Tuleja 与 Martina Hancova：“对称性与守恒定律：诺特定理的推论”（Symmetries and Conservation Laws: Consequences of Noether’s Theorem），American Journal of Physics，第 72 卷第 4 期（2004），第 428—435 页。她的工作最初适用于受拉格朗日最小作用量原理支配的系统，这些系统体现了经典力学的决定论唯物主义。此后，它已成为物理学其他领域的一种基础方法。我们在 Allin F Cottrell 等人的《经典经济物理学》（Classical Econophysics）（Routledge，2009），第 11 章和第 5 章，以及 Paul Cockshott 的“对称性与交换关系”（Symmetries and Exchange Relations），Política y cultura，第 23 期（2005），第 279—304 页中，探讨了商品交换所定义的度量空间中的对称性与马克思的价值守恒规律之间的关系。

〔校注：底本连续两次出现 Emmy，并将 Chapter 误印为 Chaper；英文照录，中文按语义整理。〕

46 We are here using terminology derived from twentieth-century physics. Emmy Emmy Noether, ‘Invariant Variation Problems’, *Transport Theory and Statistical Physics* 1, no. 3 (1971): 186–207 (1918) established that the conservation laws of classical physics could be considered as the result of the symmetry in time and space of the laws of nature. For example the conservation of momentum arises from the fact that the laws of physics are the same everywhere in space –a displacement symmetry. Homogeneity of time can be shown to imply the conservation of energy. For an explanation consult Jozef Hanc, Slavomir Tuleja, and Martina Hancova, ‘Symmetries and Conservation Laws: Consequences of Noether’s Theorem’, *American Journal of Physics* 72, no. 4 (2004): 428–35. Her work applied originally to systems governed by the Lagrange principle of least action, encoding the deterministic materialism of classical mechanics. It has subsequently become a foundational approach in other fields of physics. We have explored the relationship between symmetries in the metric space defined by commodity exchange and the Marxian law of conserved value in Allin F Cottrell et al., *Classical Econophysics* (Routledge, 2009), Chaper 11 and 5. Paul Cockshott, ‘Symmetries and Exchange Relations’, *Política y cultura*, no. 23 (2005): 279–304. p. 245–246

↑47. 马克思：《资本论》，第 152 页。

47 Marx, *Capital*, 152. p. 246

↑48. 使用国际单位制，所做的功等于发动机施加的力乘以该力作用的距离，因此单位为牛顿 • 米。但力等于质量乘以加速度，所以这换算为（千克 • 米/秒 ^{2}）• 米，与动能 $\frac{1}{2}mv^2$ 具有相同量纲。

48 Using SI units, work done is force applied by the engine times distance it was applied thus Newton Metres. But force is mass times acceleration, so this translates to (Kilogram Meter/Second^2) Metres, which has the same dimensions as kinetic energy $\frac{1}{2}mv^2$. p. 246

↑49. 马克思：《资本论》，第 162 页。

49 Marx, *Capital*, 162. p. 246

↑50. W. Paul Cockshott、A. Cottrell 与 G. J. Michaelson:《用投入产出表检验劳动价值论》（Testing Labour Value Theory with Input/output Tables）（斯特拉斯克莱德大学计算机科学系，1993）。

50 W. Paul Cockshott, A. Cottrell, and G. J. Michaelson, *Testing Labour Value Theory with Input/output Tables* (Department of Computer Science, University of Strathclyde, 1993). p. 246

↑51. Karl Marx、Salomea W. Ryazanskaya 与 Maurice Dobb：《政治经济学批判》（A Contribution to the Critique of Political Economy）（Progress Publishers，1978），序言。

51 Karl Marx, Salomea W. Ryazanskaya, and Maurice Dobb, *A Contribution to the Critique of Political Economy* (Progress Publishers, 1978), Preface. p. 246

↑52. Guido Starosta：“商品形式与辩证方法：论马克思在《资本论》第一章中的叙述结构”（The Commodity-form and the Dialectical Method: On the Structure of Marx’s Exposition in Chapter 1 of Capital），Science & Society，第 72 卷第 3 期（2008），第 295—318 页。

52 Guido Starosta, ‘The Commodity-form and the Dialectical Method: On the Structure of Marx’s Exposition in Chapter 1 of Capital’, *Science & Society* 72, no. 3 (2008): 295–318. p. 246

↑53. Starosta：“商品形式与辩证方法：论马克思在《资本论》第一章中的叙述结构”，第 298 页。

53 Starosta, ‘The Commodity-form and the Dialectical Method: On the Structure of Marx’s Exposition in Chapter 1 of Capital’, 298. p. 246

p. 246 ↑54. 卡尔 • 马克思：“阿道夫 • 瓦格纳边注”（Notes on Adolph Wagner），Texts on Method（1975），第 201 页。

54 Karl Marx, ‘Notes on Adolph Wagner’, *Texts on Method*, 1975, 201.

p. 246 ↑55. Diane Elson：《价值：资本主义中劳动的表现》（Value: The Representation of Labour in Capitalism）（Verso Books，2016），重印本。

55 Diane Elson, *Value: The Representation of Labour in Capitalism* (Verso Books, 2016), reprint.

p. 246 ↑56. 马克思：《资本论》，第 3 章。

56 Marx, *Capital*, Chapter 3.

p. 246 ↑57. S. Meikle：《亚里士多德的经济思想》（Aristotle’s Economic Thought）（Oxford University Press，1997）。

57 S. Meikle, *Aristotle’s Economic Thought* (Oxford University Press, 1997).

p. 246 ↑58. 该理论把一种共同实体——燃素——设定为火的本质。这似乎能够解释各种观察结果，但后来被证明是错误的。

58 This theory posited a common substance, phlogiston, as the essence of fire, which seemed to account for various observations but was later disproved.

p. 246 ↑59. 18 世纪的 fossils 一词源自拉丁语 fossa（沟），指一切从地下挖出的东西。因此，他所谓的易燃矿物，指的是煤和矿物油。

59 In the eighteenth century fossils, from the latin *fossa* or ditch, referred to anything dug out of the earth. By inflammable fossils he thus means coal and mineral oil.

p. 246 ↑60. 即我们今天所谓的 SO_2。

60 What we would now call SO_2.

p. 246 ↑61. 现在称为发烟硝酸；斯特雷奇观察到的现象，是由溶解在酸中的 NO_2 引起的。

61 This is now called fuming nitric acid, and the phenomenon Strachey observed is due to NO_2 in solution in the acid.

p. 246 ↑62. John Strachey：《关于不同土层与矿物层的观察》（Observations on the Different Strata of Earths and Minerals）（1730）。

62 John Strachey, *Observations on the Different Strata of Earths and Minerals* (1730).

p. 246 ↑63. 参见 Emmanuel Farjoun 与 Moshe Machover：《混沌的规律：政治经济学的概率方法》（Laws of Chaos, a Probabilistic Approach to Political Economy）（Verso，1983）；E. Farjoun：“以什么生产商品？”（Production of Commodities by Means of What?），载《李嘉图、马克思、斯拉法》（Ricardo, Marx, Sraffa），E. Mandel 编（Cambridge University Press，1984）。

63 See Emmanuel Farjoun and Moshe Machover, *Laws of Chaos, a Probabilistic Approach to Political Economy* (Verso, 1983); E. Farjoun, ‘Production of Commodities by Means of What?’, in *Ricardo, Marx, Sraffa*, ed. E Mandel (Cambridge University Press, 1984).

p. 247 ↑64. Khaldun、Rosenthal 与 Dawood：《历史绪论》（The Muqaddimah: An Introduction to History; in Three Volumes），第 1 卷。

64 Khaldun, Rosenthal, and Dawood, *The Muqaddimah: An Introduction to History; in Three Volumes*, 1.

p. 247 ↑65. David Ricardo：“政治经济学及赋税原理”（Principles of Political Economy and Taxation），载《李嘉图著作与通信集》（The Works and Correspondence of David Ricardo），P. Sraffa 编，第 1 卷（Cambridge University Press，1951）。

65 David Ricardo, ‘Principles of Political Economy and Taxation’, in *The Works and Correspondence of David Ricardo*, ed. P. Sraffa, vol. 1 (Cambridge University Press, 1951).

p. 247 ↑66. “我看得远，是因为我站在巨人的肩膀上。”——牛顿。

66 ‘I see far because I stand on the shoulders of giants’, Newton

p. 247 ↑67. Friedrich Engels：《自然辩证法》（Dialectics of Nature）（Progress Publishers，1974）。

67 Friedrich Engels, *Dialectics of Nature* (Progress Publishers, 1974).

p. 247 ↑68. 关于这段历史，可阅读 Cardwell 的《从瓦特到克劳修斯：工业时代早期热力学的兴起》（From Watt to Clausius: The Rise of Thermodynamics in the Early Industrial Age）。

68 For an account of the history of this read Cardwell (*From Watt to Clausius: The Rise of Thermodynamics in the Early Industrial Age*).

p. 247 ↑69. 两人当时都受雇于格拉斯哥大学。

69 Both were employed at Glasgow University.

p. 247 ↑70. Karl Marx 与 Friedrich Engels：《马克思恩格斯全集》英文版（Marx-Engels Collected Works），第 40 卷（Progress Publishers），第 441 页。

70 Karl Marx and Friedrich Engels, *Marx-Engels Collected Works*, vol. 40 (Progress Publishers), 441.

p. 247 ↑71. Karl Marx 与 Friedrich Engels：《马克思恩格

斯全集》英文版（Marx-Engels Collected Works），第 41 卷（Progress Publishers），第 246 页。

71 Karl Marx and Friedrich Engels, *Marx-Engels Collected Works*, vol. 41 (Progress Publishers), 246.

↑72. Adrian Desmond 与 James Moore：《达尔文的神圣事业：对奴隶制的憎恨如何塑造了达尔文的人类进化观》（Darwin’s Sacred Cause: How a Hatred of Slavery Shaped Darwin’s Views on Human Evolution）（HMH，2014）。

72 Adrian Desmond and James Moore, *Darwin’s Sacred Cause: How a Hatred of Slavery Shaped Darwin’s Views on Human Evolution* (HMH, 2014). p. 247

↑73. R. Chambers：《创世自然史遗迹》（Vestiges of the Natural History of Creation [by R. Chambers]）（John Churchill，Princes Street，Soho，1853），第 217 页，https://books.google.co.uk/books?id=qDoDAAAAQAAJ。

73 R. Chambers, *Vestiges of the Natural History of Creation [by R. Chambers]* (John Churchill, Princes Street, Soho, 1853), 217, https://books.google.co.uk/books?id=qDoDAAAAQAAJ.. p. 247

↑74. Chambers：《创世自然史遗迹》，第 219 页。

74 Chambers, *Vestiges of the Natural History of Creation*, 219. p. 247

↑75. Sean Sayers：“马克思与目的论”（Marx and Teleology），Science & Society，第 83 卷第 1 期（2019），第 37—63 页，https://doi.org/10.1521/siso.2019.83.1.37；电子预印本：https://doi.org/10.1521/siso.2019.83.1.37；https://doi.org/10.1521/siso.2019.83.1.37。

75 Sean Sayers, ‘Marx and Teleology’, *Science & Society* 83, no. 1 (2019): 37–63, https://doi.org/10.1521/siso.2019.83.1.37, eprint: https://doi.org/10.1521/siso.2019.83.1.37, https://doi.org/10.1521/siso.2019.83.1.37. p. 247

↑76. Charles Darwin：《物种起源》（On the Origin of the Species）（Project Gutenberg 电子书，2009），第 3 章。

76 Charles Darwin, *On the Origin of the Species* (Project Gutenberg EBook, 2009), Chapter 3. p. 247

↑77. Charles Darwin：《人和动物的情感表达》（The Expression of the Emotions in Man and Animals）（Project Gutenberg 电子书，1998）。

77 Charles Darwin, *The Expression of the Emotions in Man and Animals* (Project Gutenberg EBook, 1998). p. 247

↑78. Darwin：《物种起源》。

78 Darwin, *On the Origin of the Species*. p. 247

↑79. Darwin：《物种起源》，第 1 章。

79 Darwin, *On the Origin of the Species*, Chapter 1. p. 247

↑80. Darwin：《物种起源》，第 1 章。

80 Darwin, *On the Origin of the Species*, Chapter 1. p. 247

↑81. Lyudmila N. Trut、I. Z. Plyusnina 与 I. N. Oskina：“狐狸驯化实验及狗演化中的争议问题”（An Experiment on fox Domestication and Debatable Issues of Evolution of the Dog），Russian Journal of Genetics，第 40 卷第 6 期（2004），第 644—655 页。

81 Lyudmila N. Trut, I. Z. Plyusnina, and I. N. Oskina, ‘An Experiment on fox Domestication and Debatable Issues of Evolution of the Dog’, *Russian Journal of Genetics* 40, no. 6 (2004): 644–55. p. 247

↑82. Adam S. Wilkins、Richard W. Wrangham 与 W. Tecumseh Fitch：“哺乳动物的‘驯化综合征’：基于神经嵴细胞行为与遗传学的统一解释”（The “Domestication Syndrome” in Mammals: A Unified Explanation Based on Neural Crest Cell Behavior and Genetics），Genetics，第 197 卷第 3 期（2014），第 795—808 页。

82 Adam S. Wilkins, Richard W. Wrangham, and W. Tecumseh Fitch, ‘The “Domestication Syndrome” in Mammals: A Unified Explanation Based on Neural Crest Cell Behavior and *Genetics*’, *Genetics* 197, no. 3 (2014): 795–808. p. 247

↑83. Darwin：《物种起源》，第 3 章。

83 Darwin, *On the Origin of the Species*, Chapter 3. p. 247

↑84. Darwin：《物种起源》，第 3 章。

84 Darwin, *On the Origin of the Species*, Chapter 3. p. 247

↑85. Darwin：《物种起源》，第 3 章。

85 Darwin, *On the Origin of the Species*, Chapter 3. p. 247

↑86. Darwin：《物种起源》，第 4 章。

86 Darwin, *On the Origin of the Species*, Chapter 4. p. 247

↑87. Darwin：《物种起源》，第 4 章。

87 Darwin, *On the Origin of the Species*, Chapter 4. p. 247

↑88. Darwin：《物种起源》，第 4 章。

88 Darwin, *On the Origin of the Species*, Chapter 4. p. 247

↑89. Stephen Jay Gould：《熊猫的拇指：自然史再思考》（The Panda’s Thumb: More Reflections in Natural History）（W. W. Norton & Company，2010），第 28 章。

89 Stephen Jay Gould, *The Panda’s Thumb: More Reflections in Natural History* (W.W. Norton & Company, 2010), Chapter 28. p. 247

↑90. Lewis H. Morgan：《古代社会》（Ancient Society）（Harvard University Press，2013）。

90 Lewis H. Morgan, *Ancient Society* (Harvard University Press, 2013). p. 247

p. 247 ↑91. Charles Darwin:《人类的由来及与性有关的选择》(The Descent of Man in Relation to Sex)(Murray，1871)，第 III 章。

91 Charles Darwin, *The Descent of Man in Relation to Sex* (Murray, 1871), Chapter III.

p. 247 ↑92. Donald Olding Hebb：“行为的组织：神经心理学理论”(The Organization of Behavior; A Neuropsycholocigal Theory)，A Wiley Book in Clinical Psychology，第 62 卷（1949)，第 78 页。

92 Donald Olding Hebb, ‘The Organization of Behavior; A Neuropsycholocigal Theory’, *A Wiley Book in Clinical Psychology* 62 (1949): 78.

p. 247 ↑93. Darwin：《人和动物的情感表达》。

93 Darwin, *The Expression of the Emotions in Man and Animals*.

p. 248 ↑94. Hebb：“行为的组织：神经心理学理论”(The Organization of Behavior; A Neuropsycholocigal Theory)。

94 Hebb, ‘The Organization of Behavior; A Neuropsycholocigal Theory’.

p. 248 ↑95. 关于这一分歧，参见 Harry K. Wells：《西格蒙德·弗洛伊德：巴甫洛夫学派的批判》(Sigmund Freud, a Pavlovian Critique)(International Publishers，1960)。

95 For an account of this dichotomy see Harry K. Wells, *Sigmund Freud, a Pavlovian Critique* (International Publishers, 1960).

p. 248 ↑96. Hebb：“行为的组织：神经心理学理论”(The Organization of Behavior; A Neuropsycholocigal Theory)。

96 Hebb, ‘The Organization of Behavior; A Neuropsycholocigal Theory’.

p. 248 ↑97. Sebastiano Timpanaro：《论唯物主义》(On Materialism)(Routledge，2003)。

97 Sebastiano Timpanaro, *On Materialism* (Routledge, 2003).

p. 248 ↑98. Ludwig Boltzmann:《理论物理学与哲学问题：文选》(Theoretical Physics and Philosophical Problems: Selected Writings)，第 5 卷(Springer Science & Business Media，2012)，第 14 页。

98 Ludwig Boltzmann, *Theoretical Physics and Philosophical Problems: Selected Writings*, vol. 5 (Springer Science & Business Media, 2012), 14.

p. 248 ↑99. Newton：《自然哲学的数学原理……由 Andrew Motte 翻译……附牛顿的世界体系；W. Emerson 对〈原理〉的简评与辩护；John Machin 依据引力论述的月球运动定律……(先生的序言)》(The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which are Added Newton’s System of the World; A Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . .(The Preface of Mr)，第 8 节，命题 XLI。

〔校注：本注书目在底本中包含截断的 (The Preface of Mr)，按原样保留，不推补缺失信息。〕

99 Newton, *The Mathematical Principles of Natural Philosophy . . . Translated . . . by Andrew Motte. To Which are Added Newton’s System of the World; A Short Comment on, and Defence of the Principia, by W. Emerson. With the Laws of the Moon’s Motion According to Gravity, by John Machin . . .(The Preface of Mr)*, Section 8, Proposition XLI.

p. 248 ↑100. J. J. Brucker 与 W. Enfield:《哲学史：从最早时代到本世纪初；由 William Enfield 根据布鲁克的〈哲学批判史〉编写》(The History of Philosophy, from the Earliest Times to the Beginning of the Present Century; Drawn up from Brucker’s Historia Critica Philosophiæ: by William Enfield)，《哲学史：从最早时代到本世纪初；由 William Enfield 根据布鲁克的〈哲学批判史〉编写》，第 1 卷（P. Wogan，1792)，https://books.google.co.uk/books?id=ximMZcSJY8gC。

〔校注：本注底本重复列出书名，中文同样保留重复书目信息。〕

100 J. J. Brucker and W. Enfield, *The History of Philosophy, from the Earliest Times to the Beginning of the Present Century; Drawn up from Brucker’s Historia Critica Philosophiæ: by William Enfield*, The History of Philosophy, from the earliest times to the beginning of the present century; drawn up from Brucker’s Historia critica philosophiæ: by William Enfield v. 1 (P. Wogan, 1792), https://books.google.co.uk/books?id=ximMZcSJY8gC.

p. 248 ↑101. John Dalton：“III. 论水和其他液体对气体的吸收”(III. On the Absorption of Gases by Water and Other Liquids)，The Philosophical Magazine，第 24 卷第 93 期（1806)，第 15—24 页。

101 John Dalton, ‘III. On the Absorption of Gases by Water and Other Liquids’, *The Philosophical Magazine* 24, no. 93 (1806): 15–24.

p. 248 ↑102. John Dalton：《化学哲学新体系》(A New System of Chemical Philosophy)，第 1 卷(Bickerstaff，1808)，第 213 页。

102 John Dalton, *A New System of Chemical Philosophy*, vol. 1 (Bickerstaff, 1808), 213.

↑103. Dalton：“III. 论水和其他液体对气体的吸收”，第 73 页。

103 Dalton,‘III. On the Absorption of Gases by Water and Other Liquids’, 73. p. 248

↑104. Dalton：“III. 论水和其他液体对气体的吸收”。

104 Dalton,‘III. On the Absorption of Gases by Water and Other Liquids’. p. 248

↑105. 参见 P. Cockshott 等：《经典经济物理学：古典政治经济学、热力学与信息论论文集》（Classical Econophysics: Essays on Classical Political Economy, Thermodynamics and Information Theory）（Routledge，2008）的前几章。

105 See the first chapters of P. Cockshott et al., *Classical Econophysics: Essays on Classical Political Economy, Thermodynamics and Information Theory* (Routledge, 2008). p. 248

↑106. John Tyndall：《热作为运动的一种形式》（Heat Considered as a Mode of Motion）（D. Appleton/Company，1875），第 xiii 页。

106 John Tyndall, *Heat Considered as a Mode of Motion* (D. Appleton/Company, 1875), xiii. p. 248

↑107. Tyndall：《热作为运动的一种形式》，第 xiii 页。

107 Tyndall, *Heat Considered as a Mode of Motion*, xiii. p. 248

↑108. Benjamin Count Rumford：“关于摩擦所激发的热的来源的研究，伦福德伯爵本杰明著，Frsmria”（An Inquiry Concerning the Source of the Heat Which is Excited by Friction. by Benjamin Count of Rumford, Frsmria），Philosophical Transactions of the Royal Society of London（1798），第 80—102 页。

108 Benjamin Count Rumford,‘An Inquiry Concerning the Source of the Heat Which is Excited by Friction. by Benjamin Count of Rumford, Frsmria’, *Philosophical Transactions of the Royal Society of London* (1798): 80–102. p. 248

↑109. Rumford：“关于摩擦所激发的热的来源的研究，伦福德伯爵本杰明著，Frsmria”。

109 Rumford, ‘An Inquiry Concerning the Source of the Heat Which is Excited by Friction. by Benjamin Count of Rumford, Frsmria’. p. 248

↑110. Rumford；斜体及大写均为原文所有。

110 Rumford, italics and capitals in the original. p. 248

↑111. Paul Burkett 与 John Bellamy Foster：“马克思政治经济学批判中的物质变换、能量与熵：超越波多林斯基神话”（Metabolism, Energy, and Entropy in Marx’s Critique of Political Economy: Beyond the Podolinsky Myth），Theory and Society，第 35 卷第 1 期（2006），第 109—156 页，ISSN：03042421、15737853，http://www.jstor.org/stable/4501746。

111 Paul Burkett and John Bellamy Foster, ‘Metabolism, Energy, and Entropy in Marx’s Critique of Political Economy: Beyond the Podolinsky Myth’, *Theory and Society* 35, no. 1 (2006): 109 – 56, issn: 03042421, 15737853, http://www.jstor.org/stable/4501746. p. 248 p. 248

↑112. 廷德尔指出，在 0°C 时，氧分子的运动速度为每秒 1,514 英尺。

112 Tyndall stated that at 0°C oxygen. molecules move at 1,514 feet per second. p. 248

↑113. Engels：《自然辩证法》，第 243—256 页。

113 Engels, *Dialectics of Nature*, 243–56. p. 248

↑114. 关于这两个问题，H. Price 的《时间之箭与阿基米德之点：时间物理学的新方向》（Time’s Arrow and Archimedes’ Point: New Directions for the Physics of Time）（Oxford University Press，USA，1997）给出了深入而出色的唯物主义论述。本章以下内容主要借鉴他的著作，以及 Carlo Cercignani 与 Roger Penrose 的《路德维希·玻尔兹曼：相信原子的人》（Ludwig Boltzmann: The Man Who Trusted Atoms）（Oxford University Press，1998）。

114 For both questions H. Price, *Time’s Arrow and Archimedes’ Point: New Directions for the Physics of Time* (Oxford University Press, USA, 1997) gives a good in-depth materialist account and what we will say in this chapter draws heavily on his work and on Carlo Cercignani and Roger Penrose, *Ludwig Boltzmann: The Man Who Trusted Atoms* (Oxford University Press, 1998). p. 248 p. 248

↑115. 本书第 3.3.3.0.1 节。

115 Chapter 3.3.3.0.1 of this book. p. 248

↑116. William Thomson：“能量耗散的动力学理论”（The Kinetic Theory of the Dissipation of Energy），Proceedings of the Royal Society of Edinburgh，第 8 卷（1875），第 325—334 页。

116 William Thomson, ‘The Kinetic Theory of the Dissipation of Energy’, *Proceedings of the Royal Society of Edinburgh* 8 (1875): 325–34. p. 248

↑117. 相关论述参见 Cardwell：《从瓦特到克劳修斯：工业时代早期热力学的兴起》。

117 For an account see Cardwell, *From Watt to Clausius: The Rise of Thermodynamics in the Early Industrial Age*. p. 248

↑118. U. B. Frisch、B. Hasslacher 与 Y. Pomeau：“纳维–斯托克斯方程的格子气自动机”（Lattice Gas p. 249

Automata for the Navier Stokes Equation），Physical Review Letters，第 56 卷（1986），第 1505—1508 页。

118 U. B. Frisch, B. Hasslacher, and Y. Pomeau, ‘Lattice Gas Automata for the Navier Stokes Equation’, *Physical Review Letters* 56 (1986): 1505–8.

p. 249 ↑119. Peter Barrie 等：“高度并发机器的设计与验证”（Design and Verification of a Highly Concurrent Machine），Microprocessors and Microsystems，第 16 卷第 3 期（1992），第 115—123 页；George J. Milne 与 William P. Cockshott：《物理过程模拟的方法与装置》（Method and Apparatus for Simulation of a Physical Process），美国专利 5,485,599，1996 年。

119 Peter Barrie et al., ‘Design and Verification of a Highly Concurrent Machine’, *Microprocessors and Microsystems* 16, no. 3 (1992): 115–23; George J. Milne and William P. Cockshott, *Method and Apparatus for Simulation of a Physical Process*, US Patent 5,485,599, 1996.

p. 249 ↑120. Jean Bricmont：《理解统计力学》（Making Sense of Statistical Mechanics）（Springer，2022），第 8 章，估计装有 2 克 H_2 的圆柱体达到所有分子都在容器一半中的状态，所需时间为 $2^{10^{23}}$ 量级。这是 2^N，其中 N 为阿伏伽德罗常数，即每摩尔气体的分子数。他没有给出表达式的时间常数。更准确地说，应对某个 T 写成 $2^{10^{23}}T$；但即使令 T 为原子碰撞之间的平均时间，取 $t = 0.2 \times 10^{-9}$ 秒，指数项仍大到使时间常数问题在实践中无关紧要。

120 Jean Bricmont, *Making Sense of Statistical Mechanics* (Springer, 2022), Chapter 8 gives an estimate of a time of order $2^{10^{23}}$ for a cylinder with 2 gm of H_2 to a rrive at a state with all the molecules in one-half of the cylinder. This being 2^N for N= Avogadro’s number: the number of molecules per Mole of gas. He does not give the time constant for the expression. More precisely, one should write $2^{10^{23}}T$ or some T , but even if we let T . be the mean time between atomic collisions with $t = 0.2 \times 10^{-9}$ seconds, the exponential term is so great that the issue of the time constant becomes practically irrelevant.

p. 249 ↑121. Thomson：“能量耗散的动力学理论”。

121 Thomson, ‘The Kinetic Theory of the Dissipation of Energy’.

p. 249 ↑122. 这一状态是镜像，因为此时每个原子都沿着与其原始方向相反的方向运动。

122 This state is a mirror image because every atom is now moving in the reverse of its original direction.

p. 249 ↑123. Thomson：“能量耗散的动力学理论”。

123 Thomson, ‘The Kinetic Theory of the Dissipation of Energy’.

p. 249 ↑124. Thomson：“能量耗散的动力学理论”。

124 Thomson, ‘The Kinetic Theory of the Dissipation of Energy’.

p. 249 ↑125. L. Boltzmann：Sitzungsberichte Akad. Wiss.，维也纳，II，第 66 卷，第 275—370 页；英译：Brush, S. G.（1966），《气体动理论》第 2 卷《不可逆过程》（Kinetic Theory: Vol. 2 Irreversible Processes）（Permagon Press，1872）。

125 L. Boltzmann, *Sitzungsberichte Akad. Wiss., Vienna, II, 66: 275-370; English transl.: Brush, SG (1966) Kinetic Theory: Vol. 2 Irreversible Processes* (Permagon Press, 1872).

p. 249 ↑126. 更准确地说，他的原论文表明，具有函数形式 $H \sim \int p \log p$ 的量，在系统随时间演化时趋于减小。该量为负，因为概率 $p < 1$，$\log p < 0$。现在我们通常使用玻尔兹曼后来的表述 $S = k \log$（N article states 的数目），这是趋于增加的正熵。

〔校注：底本熵公式括号内印作 N article states，疑有缺字；这里保留该字样，不推补 particle。H 公式及不等号按源图恢复。〕

126 More precisely, his orginal paper showed that a quantity with functional form $H \sim \int p \log p$, which is negative since probability $p < 1$, $\log p < 0$ H tends to decrease in the time evolution of a system. We now normally use Boltzmann’s later formulation $S = k \log$(number of N article states) which is positive entropy which tends to increase.

p. 249 ↑127. Loshmidt:“论计及重力时物体系统的热平衡状态”（Ueber den znstand des Warmegleichgewichtes eines Systems von Korpern mit Rucksicht auf die Schwerkraft），Wiener Berichte，第 73 卷（1876），第 135 页。

127 Loshmidt, ‘Ueber den znstand des Warmegleichgewichtes eines Systems von Korpern mit Rucksicht auf die Schwerkraft’, *Wiener Berichte* 73 (1876): 135.

p. 249 ↑128. 形式地，令 $P^1(x,\xi,t)$ 为一个分子在时间 t 位于位置 x、速度为 ξ 的概率函数；令 $P^2(x_1,\xi_1,x_2,\xi_2,t)$ 表示分子 1 的位置和速度坐标为 x_1,ξ_1，分子 2 的位置和速度为 x_2,ξ_2 的概率。玻尔兹曼的统计独立性假设为 $P^2(x_1,\xi_1,x_2,\xi_2,t) = P^1(x_1,\xi_1,t) \times P^1(x_2,\xi_2,t) \times P^1(x_2,\xi_2,t)$。

〔校注：底本独立性公式末尾另重复印有 $\times P^1(x_2, \xi_2, t)$。本译保留此重复项并标明，不将其静默删去。〕

128 Formally let $P^1(x,\xi,t)$ be the probability function for one molecule being in locality x with velocity ξ at time t and let $P^2(x_1,\xi_1,x_2,\xi_2,t)$ denote the probability of molecule 1 having position and velocity coordinates x_1,ξ_1 and molecule 2 have position and velocity x_2,ξ_2. Boltzmann’s statistical independence assumption was that $P^2(x_1,\xi_1,x_2,\xi_2,t) = P^1(x_1,\xi_1,t) \times P^1(x_2,\xi_2,t) \times P^1(x_2,\xi_2,t)$.

↑129. Ludwig Boltzmann：“答策梅洛关于热理论的评论”（Reply to Zermelo’s Remarks on the Theory of Heat），载《气体动理论：经典论文选集及历史评注》（The Kinetic Theory Of Gases: An Anthology of Classic Papers with Historical Commentary）（World Scientific，2003），第 412—419 页。

129 Ludwig Boltzmann, ‘Reply to Zermelo’s Remarks on the Theory of Heat’, in *The Kinetic Theory Of Gases: An Anthology of Classic Papers with Historical Commentary* (World Scientific, 2003), 412–19. p. 249

↑130. 关于回归与不可逆性争论涉及的问题，可参见 Jean Bricmont：“混沌的科学还是科学中的混沌？”（Science of Chaos or Chaos in Science?），arXiv 预印本 chao-dyn/9603009，1996。

130 A good overview of the issues involved in the disputes over recurrence and irreversibility can be found in Jean Bricmont, ‘Science of Chaos or Chaos in Science?’, *arXiv preprint chao-dyn/9603009*, 1996. p. 249

↑131. 相关讨论参见 Rüdiger Vaas：“时间之前的时间：当代宇宙学中的宇宙分类，以及如何避免世界开端与永恒的二律背反”（Time before Time-Classifications of Universes in Contemporary Cosmology, and How to Avoid the Antinomy of the Beginning and Eternity of the World），arXiv 预印本 physics/0408111，2004。

131 See Rüdiger Vaas, ‘Time before Time-Classifications of Universes in Contemporary Cosmology, and How to Avoid the Antinomy of the Beginning and Eternity of the World’, *arXiv preprint physics/0408111*, 2004, for a discussion. p. 249

↑132. Steven Frautschi：“膨胀宇宙中的熵”（Entropy in an Expanding Universe），Science，第 217 卷第 4560 期（1982），第 593—599 页。

132 Steven Frautschi, ‘Entropy in an Expanding Universe’, *Science* 217, no. 4560 (1982): 593–9. p. 249

↑133. Frautschi：“膨胀宇宙中的熵”（Entropy in an Expanding Universe）。

133 Frautschi, ‘Entropy in an Expanding Universe’. p. 249

↑134. 按他的近似计算，此时 $kT \approx 28/9$ MeV。

134 In his approximate calculation when $kT \approx 28/9$ MeV. p. 249

↑135. Frautschi：“膨胀宇宙中的熵”（Entropy in an Expanding Universe）。

135 Frautschi, ‘Entropy in an Expanding Universe’. p. 249

↑136. 参见 C. Shannon：“通信的数学理论”（A Mathematical Theory of Communication），The Bell System Technical Journal，第 27 卷（1948），第 379—423、623—656 页；Rolf Landauer：“计算过程中的不可逆性与热量产生”（Irreversibility and Heat Generation in the Computing Process），IBM Journal of Research and Development，第 5 卷（1961），第 183—191 页。

136 See C. Shannon, ‘A Mathematical Theory of Communication’, *The Bell System Technical Journal* 27 (1948): 379–423 and 623–56; Rolf Landauer, ‘Irreversibility and Heat Generation in the Computing Process’, *IBM Journal of Research and Development* 5 (1961): 183–91. p. 250

↑137. 参见 Farjoun 和 Machover：《混沌的规律：政治经济学的概率方法》（Laws of Chaos, a Probabilistic Approach to Political Economy）；W. Paul Cockshott 等：《经典经济物理学》（Classical Econophysics），第 2 卷（Routledge，2009）；J. Barkley Rosser：“熵与经济物理学”（Entropy and Econophysics），The European Physical Journal Special Topics，第 225 卷第 17 期（2016），第 3091—3104 页。

137 See Farjoun and Machover, *Laws of Chaos, a Probabilistic Approach to Political Economy*; W. Paul Cockshott et al., *Classical Econophysics*, vol. 2 (Routledge, 2009); J. Barkley Rosser, ‘Entropy and Econophysics’, *The European Physical Journal Special Topics* 225, no. 17 (2016): 3091–104. p. 250

↑138. 参见 Erik Verlinde：“论引力的起源与牛顿定律”（On the Origin of Gravity and the Laws of Newton），Journal of High Energy Physics，2011 年第 4 期，第 1—27 页；他从熵推导出牛顿定律。这一推导虽有影响，却存在争议；参见 Shan Gao：“引力是熵力吗？”（Is Gravity an Entropic Force?），Entropy，第 13 卷第 5 期（2011），第 936—948 页。

138 See Erik Verlinde, ‘On the Origin of Gravity and the Laws of Newton’, *Journal of High Energy Physics* 2011, no. 4 (2011): 1–27, who derives Newton’s laws from entropy. This derivation, though influential, is controversial; see Shan Gao, ‘Is Gravity an Entropic Force?’, *Entropy* 13, no. 5 (2011): 936–48. p. 250

第 6 章唯心主义的复兴与回应

Chapter 6: IDEALIST REPRISE AND RESPONSES

对玻尔兹曼的反对

Opposition to Boltzmann

p. 123

玻尔兹曼职业生涯大部分时间作为物理学家度过。然而，他生命最后几年是在维也纳大学担任哲学教授。在这个职位上，他接替了同样身兼物理学与哲学双重身份的马赫。

Boltzmann spent a large part of his career as a physicist. His final years, though, were as the professor of philosophy at the University of Vienna. In this post, he replaced Mach, who also straddled physics and philosophy.

无论在哲学还是物理学领域，两人都是对手，各自主张截然相反立场。作为唯物主义原子论者的玻尔兹曼，与作为唯心主义者的马赫针锋相对。

In terms both of philosophy and physics the two were rivals, proponents of polar opposite positions. Boltzmann, the materialist atomist, against Mach, the idealist.

尽管如今玻尔兹曼的原子论已完全没有争议，但在他生前却远非如此。许多欧洲大陆的著名物理学家都对原子的存在深表怀疑。例如，庞加莱把玻尔兹曼、汤姆森和麦克斯韦提出的关于热的原子论解释称为“英国[1]动理论”，并怀疑它能否解释熵增定律。

Although Boltzmann’s atomism is now completely uncontroversial, that was far from the case in his lifetime. A lot of prominent continental physicists were highly sceptical about the existence of atoms. Poincaré, for example, referred to the atomistic account of heat propounded by Boltzmann, Thomson and Maxwell as the ‘English[1] kinetic theory’ and doubted it as an explanation of the law of increasing entropy.

我不知道是否有人指出过，英国的这些动理论能够摆脱这一矛盾。按照这些理论，世界最初趋向于一种状态，并长久停留其中而不发生明显变化；这与经验相符。但如果不违背上述定理，世界就不会永远保持这种状态；它只会停留极其漫长的时间，分子越多，这段时间就越长。这种状态并非宇宙的最终死亡，而是一种沉睡，宇宙将在百万乘百万个世纪之后从中苏醒。按照这种理论，要看到热从冷物体传向热物体，并不需要麦克斯韦妖那样敏锐的视力、智慧和灵巧；稍有耐心便足够了。[2]

I do not know if it has been remarked that the English kinetic theories can extricate themselves from this contradiction. The world, according to them, tends at first toward a state where it remains for a long time without apparent change; and this is consistent with experience; but it does not remain that way forever, if the theorem cited above is not violated; it merely stays there for an enormously long time, a time which is longer the more numerous are the molecules. This state will not be the final death of the universe, but a sort of slumber, from which it will awake after millions of millions of centuries. According to this theory, to see heat pass from a cold body to a warm one, it will not be necessary to have the acute vision, the intelligence, and dexterity of Maxwell’s demon; it will suffice to have a little patience.[2]

p. 123–124

庞加莱对原子论的反对建立在严格的数学异议之上。这些异议源于抽象的拉格朗日力学或哈密顿力学，而非哲学或本体论。实质上，他利用刘维尔定理，把汤姆森在前引段落中给出的启发式解释转化为形式化的数学论证。由此可以证明：在时刻 t，取 $6N$ 维相空间内的一个微小体积 δ_t，使其沿最小作用量路径演化一个很短的时间 ϵ，它就会映射到相空间中的一个新区域 $\delta_{t+\epsilon}$，而体积保持不变。因此，他认为，若从处于某一低熵状态的封闭系统出发，即从只占相空间中可能位置的一小部分的某个 δ_t 出发，最后总会落到一个等体积的区域。也就是说，所得相空间体积同样不大可能出现。策梅洛也提出了类似论证。这些论证依赖于一个看似必然的结论：任何按确定性规律运行的有限粒子系统都会发生回归。对格子气例子那样简单且规模较小的模型，这一点当然成立；但如果宇宙无限——这实质上是汤姆森的观点——或者即使体积不大的气体，其回归也需要漫长得多的时间，以至于我们不妨把系统视为无限——这是玻尔兹曼的回应——那么这一论证便不能成立。

The opposition to atomism by Poincaré was founded on strictly mathematical objections. They were ones founded on abstract Lagrangian or Hamiltonian mechanics rather than philosophy or ontology. In essence, he translated the heuristic explanation given by Thomson in the passages quoted earlier into formal maths using Liouville theorem. This can be used to show that if you take a small volume δ_t in a 6N dimensional phase space at time t and allow it to evolve along a course of least action for some small time ϵ it maps to a new region in phase space $\delta_{t+\epsilon}$ whose volume is unchanged. Thus he argued that if you start with a closed system in some low-entropy state, that is, in some δ_t that is a small part of the possible phase space positions, it will always end up in an equivalent volume. That is to say, the resultant volume of phase space is equally improbable, while similar arguments were being made by Zermelo. They rested on the apparent necessity for recurrences to occur in any finite system of particles operating by deterministic laws. While this is certainly valid for simple relatively small models like our lattice gas example, the argument fell

if either the universe is infinite –which was essentially Thomsons point –or that such recurrences would take such a long time for even modest volumes of gas, that we might as well treat the system as infinite –Boltzmann’s response.

因此，当策梅洛仅凭“气体中初始状态必然会重现”这一理论事实——却未计算这将耗时多久——便得出“必须摒弃气体理论的假设，否则就必须对其进行根本性修改”结论时，他就像一个掷骰子的人：他计算出连续掷出 1000 个“1”概率不为零，却尚未观察到这样序列，就断定自己骰子必定是做过手脚！[3]

Thus when Zermelo concludes, from the theoretical fact that the initial states in a gas must recur –without having calculated how long a time this will take –, that the hypotheses of gas theory must be rejected or else fundamentally changed, he is just like a dice player who has calculated that the probability of a sequence of 1000 one’s is not zero, and then concludes that his dice must be loaded since he has not yet observed such a sequence![3] p. 124

庞加莱回归论证中还隐含着这样一个假设：在所有时刻，原子位置都是以任意精度客观确定。如果放弃这一假设，如果我们允许原子位置存在任何绝对、客观不确定性，那么碰撞过程就会放大这种初始不确定性程度。

There is also an implicit assumption in the Poincaré reccurence argument that the positions of the atoms are objectively defined to arbitrary precision at all instants. If this is abandoned, if we allow for any absolute, objective uncertainty in the positions of atoms, then the collision process magnifies the scale of that initial uncertainty.

我们此前[4]在讨论弗雷德金假设的可逆计算机[5]时谈过这个问题；在某种意义上，这些计算机是庞加莱论证的一种变体。用我们稍后将讨论的现代信息论来说，假定粒子位置具有任意精度，就意味着系统包含无限的信息，从而在相空间中占据零体积。由于熵以相空间体积的对数来定义，而 log(0) 没有定义，策梅洛 / 庞加莱回归悖论的概念设定本身就使熵的概念失去了意义。

We have discussed this earlier[4] in our treatment of Fredkin’s hypothetical reversible computers[5] which are in a sense a variant of the Poincaré argument. In terms of modern information theory, which we will discuss later, the assumption of arbitrarily precise particle positions implies a system with infinite information content and thus a zero volume in phase space. Since entropy is defined in terms of a logarithm over volumes in phase space, and since log(0) is an undefined function, the very conceptual setup of the Zermelo/Poincaré recurrence paradox renders the concept of entropy meaningless.

人们要么像庞加莱那样得出结论：原子论和“英国动理论”是错误的；要么认为经典分析所假设关于位置任意精度数学是不符合物理实际。随后几十年量子力学发展指向了后一种结论。如果位置不能被任意分割，那么虽然仍可以存在如前所述格子模型等确定性模型，但原则上也可能存在非确定性格子模型。

One either concludes, as Poincaré did, that atomism and the ‘English kinetic theory’ are false or that the arbitrary precision mathematics of position assumed by classical analysis is unphysical. The development of quantum mechanics in subsequent decades pointed to the latter conclusion. If positions are not arbitrarily divisible, then you can still have deterministic models like the lattice models presented earlier, but you could, in principle, also have non-deterministic lattice models.

马赫主义

Machism

玻尔兹曼生命最后几年，都耗费在看来可能是一场徒劳斗争上——试图说服德语科学界接受他的原子理论。除了前文提到批评者外，马赫这一显赫人物也与他针锋相对。马赫在 1901 年之前一直担任维也纳大学哲学教授，他宣扬反原子论和反实在论哲学思想，对一代欧洲科学家产生了深远影响。他的哲学立场在当今关于量子力学诠释争论中依然回响不绝。1901 年马赫退休后，玻尔兹曼接替他担任维也纳大学哲学教授席位，并开始讲授科学哲学。遗憾的是，他的任期很短。

The final years of Boltzmann’s life were consumed by what likely felt like a futile struggle to persuade the German-speaking scientific community to embrace his atomic theories. In addition to previously mentioned critics, the prominent figure of Mach stood against him. Mach, a professor of Philosophy at Vienna until 1901, propagated anti-atomistic and anti-realist philosophies that greatly impacted a generation of European scientists. His philosophical stance still resonates in today’s debates on quantum mechanics interpretations. Following Mach’s retirement in 1901, Boltzmann succeeded him as the Chair of Philosophy at Vienna, where he be- p. 124–125

gan teaching the philosophy of science. Sadly, his tenure was brief.

p. 125

1906 年 9 月 6 日，他与妻子和女儿一同前往的里雅斯特附近杜伊诺湾游玩。当妻子和女儿正在享受游泳之乐时，他上吊自杀了。此前他在莱比锡时曾有过一次自杀未遂。当得知深陷抑郁玻尔兹曼自杀身亡消息时，世界各地物理学家都深感悲痛。许多物理学家难以理解他这一举动。他的自杀似乎更多源于个人生活中因素（抑郁症和健康状况恶化），而非任何学术问题。在抑郁和健康状况不佳情况下，他就在实验即将验证其研究成果之际自杀了。奥斯特瓦尔德在悼念玻尔兹曼时，把他描述为那些为科学真理而奋斗人们所必须付出巨大健康与精力牺牲的牺牲品。一些物理学家则把玻尔兹曼自杀归因于精神不稳定。据钱德拉塞卡所述，奥斯特瓦尔德和马赫对其理论猛烈抨击使他深陷抑郁，这促使他自杀。[6]

With his wife and daughter he visited a place at the Bay of Duino near Trieste on September 6, 1906. When his wife and daughter were enjoying swimming he hanged himself. He had made an earlier suicide attempt when he was at Leipzig. Physicists everywhere were devastated by the news that Boltzmann, in deep depression, had committed suicide. His act was difficult for many physicists to understand. His suicide seems to have been due to factors in his personal life (depressions and decline of health) rather than to any academic matters. Depressed and in bad health, he committed suicide just before experiment verified his work. In a tribute to Boltzmann, Ostwald described Boltzmann as a victim of the immense sacrifices of health and strength demanded of those who struggle for scientific truth. Some physicists ascribed Boltzmann suicide to mental instability. According to Chandrasekar, he was greatly depressed by the violent attacks made on his ideas by Ostwald and Mach. This made him to commit suicide.[6]

就在他去世之际，又有两位著名人物加入了关于唯物主义思想之争：爱因斯坦和列宁。在探讨爱因斯坦如何于 1905 年从科学上为玻尔兹曼正名、列宁如何于 1908 年从哲学上为他正名之前，我们必须先梳理一下马赫对原子理论提出异议。在第五章中，我们曾回顾过从 17 世纪到 19 世纪初，热学领域曾存在两种相互竞争理论。机械论认为热是微小粒子运动；另一种理论则认为热是一种物质，通常被称为“热质”（caloric）。尽管如今看来难以置信，但马赫却是“热质说”晚期倡导者。

He died just as two more famous figures were entering the battle of ideas over materialism: Einstein and Lenin. Before we go into how Einstein vindicated Boltzmann scientifically in 1905 and Lenin vindicated him philosophically in 1908, we must run through the objections raised by Mach to the atomic theory. In Chapter 5, we recounted how, from the seventeenth to the early nineteenth centuries, there were two competing theories of heat. The mechanical theory held that heat was the motion of small particles; the alternative theory held that heat was a substance, often called *caloric*. Incredible as it now seems, Mach was a late proponent of what amounted to the *caloric* theory of heat.

人们本以为，这一切在 19 世纪中叶就已解决，但他在一部著作[7]（初版于 1872 年）中，给出了他认为热的机械理论缺乏依据的理由。

One would have imagined that all this would have been settled by the mid-nineteenth century, but in[7] (originally published 1872), he gave arguments as to why he considered the mechanical theory of heat unjustified.

如果大家还记得，反对热质说主要论据是：实验证明热量并非守恒量。热量是可以被消耗。蒸汽机和卡诺理想热机都能利用热量来做功。正如伦福德所展示那样，热量也可以通过摩擦做功产生。廷德尔、麦克斯韦、汤姆森和玻尔兹曼得出结论是：热量仅仅是机械能一种形式。马赫试图论证这一结论是不成立。

If you recall, the key arguments against the caloric theory had been that experiments proved that heat was not a conserved quantity. Heat could be used up. Steam engines and the ideal heat engines of Carnot were able to use up heat to do work. Heat could also be created from work by friction, as Rumford had shown. The conclusion of Tyndall, Maxwell, Thomson and Boltzmann was that heat was just a form of mechanical energy. Mach tried to argue that this was an illegitimate conclusion.

他以电现象作类比。

The analogy he made was with electricity.

p. 125–126

1785 年，库仑制造了他的扭秤，借此测量带电物体之间的斥力。假设有两个小球 A、B，它们整体上带有相同的电荷。当两个球心相距一定距离 r 时，它们会相互施加某一斥力 p。现在让另一小球 C 与 B 接触，使两者带上相等的电荷，然后在相同距离 r 处分别测量 B 与 A、C 与 A 之间的斥力。这两项斥力的和仍为 p。因此，有某种东西保持了恒定。如果我们把这一效应归于某种物质，自然就会推断出它的恒定性。但这一说明的关键是电力 p 的可分性，而不是物质这一比喻。

In 1785 Coulomb constructed his torsion balance, by which he was enabled to measure the repulsion of electrified bodies. Suppose we have two small balls, A, B, which over their whole extent are similarly electrified. These two balls will exert on one another, at a certain distance *r* of their centres from one another, a certain repulsion *p*. We bring into contact with B, now, a ball C, suffer both to be equally electrified, and then measure the repulsion of B with A and of C from A at the same distance *r* . The sum of these repulsions is again *p*. Accordingly something has remained constant. If we ascribe this effect to a substance, then we infer naturally its constancy. But the es-

sential point of the exposition is the divisibility of the electric force *p* and not the simile of substance.

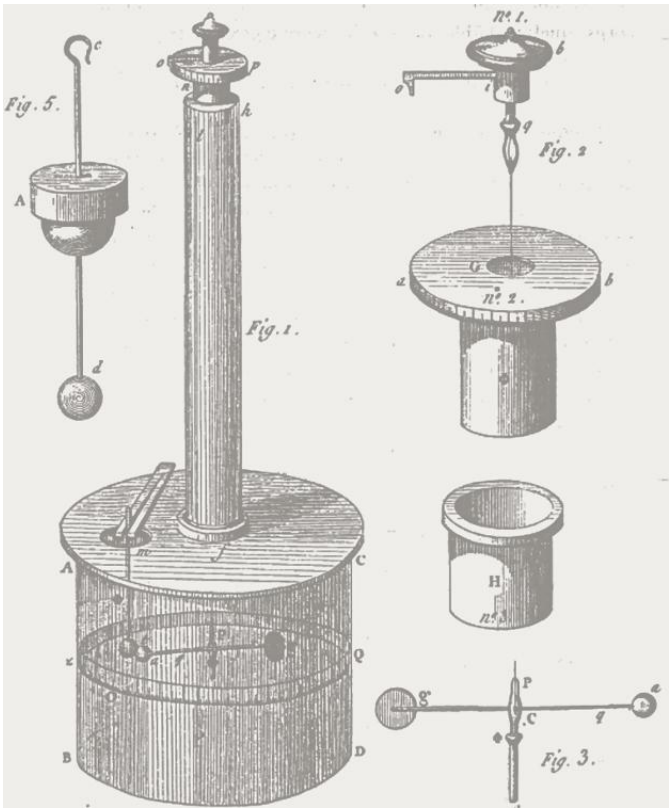


图 6.1 马赫论证中使用的库仑扭秤。出自库仑 1785 年的文献。

Figure 6.1 Coulomb balance as used in Mach’s argument. From Coulomb 1785.
原图字母为部件标号；Fig. 为图号。

1838 年，里斯制造了他的电气空气温度计（热电计）。这一仪器测量莱顿瓶放电产生的热量。该热量并不正比于按库仑的方法测得的瓶中电量；若以 q 表示这一电量、s 表示电容，则热量正比于 q^2/s，或者更简单地说，正比于带电瓶的能量。现在，若让莱顿瓶通过温度计完全放电，就能得到一定的热量 W。但如果让放电通过温度计进入第二个瓶子，得到的热量就少于 W。不过，再使两个瓶子通过空气温度计完全放电，就能得到余下的热量；此时，热量又正比于两个瓶子的能量。因此，在第一次不完全放电时，电的做功能力就损失了一部分。[8]

〔校注：英文底本将 to 印作 tc，照录。〕

In 1838 Riess constructed his electrical air-thermometer (the thermoe lectrometer). This gives a measure of the quantity of heat produced by the discharge of jars. This quantity of heat is not proportional tc the quantity of electricity contained in the jar by Coulomb’s measure, but if *q* be this quantity and *s* be the capacity, is proportional to q^2/s , or, more simply still, to the energy of the charged jar. If, now, we discharge the jar completely through the thermometer, we obtain a certain quantity of heat, W. But if we make the discharge through the thermometer into a second jar, we obtain a quantity less than W, But we may obtain the remainder by completely discharging both jars through the air-thermometer, when it will again be proportional to the energy of the two jars. On the first, incomplete discharge, accordingly, a part of the electricity’s capacity for work was lost.[8]

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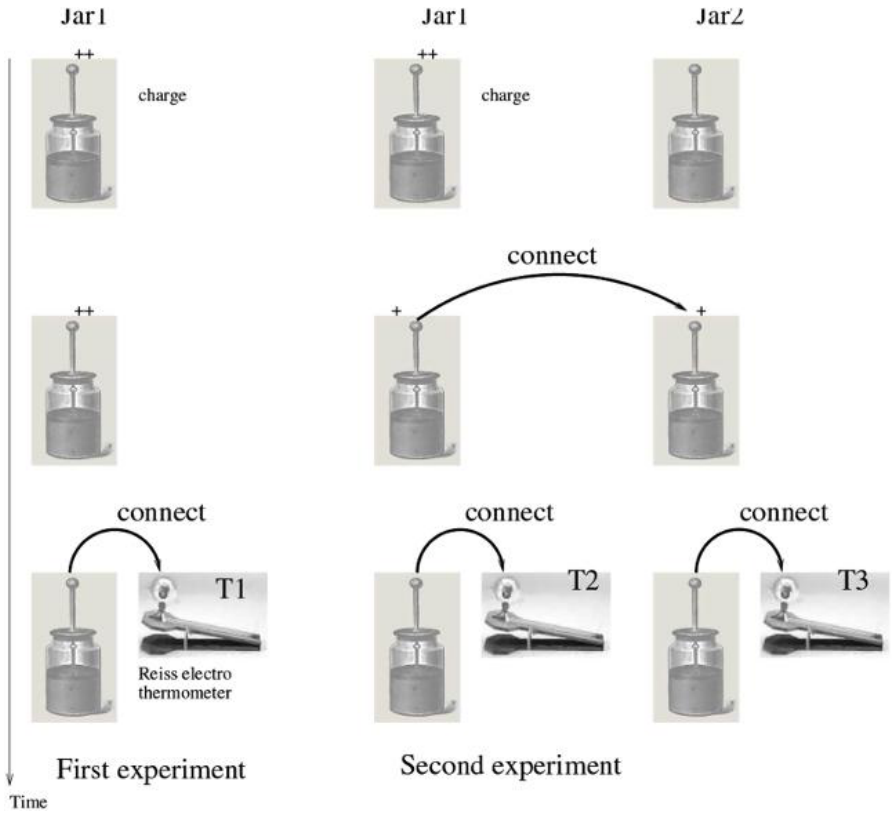


图 6.2 马赫所描述的莱顿瓶实验。作者绘图。

Figure 6.2 The Leiden jar experiments as described by Mach. Figure by authors.
Jar 1 / Jar 2：莱顿瓶 1 / 2；charge：充电；connect：连接；Riess electro thermometer：里斯电气温度计；First experiment / Second experiment：第一次 / 第二次实验；Time：时间；T1—T3 为温度计标记。

他接着论证说，库仑[9]实验先进行，只是历史上的偶然。如果先做的是莱顿瓶实验，那么测量电所用的量就会是电能，而不是电荷。

He then argues that it was just a matter of historical accident that the Coulomb[9] experiment was performed first. If the Leiden jar experiment had been done first, then the way of measuring electricity would have been in terms of quantities of electrical energy, not electrical charge.

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马赫提出，我们对“电”守恒（现被理解为电荷守恒）信念，源于库仑最初实验。倘若我们以能量来定义电，或许会发现电容器之间电转移——库仑球体也起着电容器作用——并不守恒。马赫把这种假想“电”损失与卡诺热机中热量非守恒现象进行了类比。

Mach proposed that our belief in the conservation of electricity, now understood as charge conservation, stems from Coulomb’s initial experiments. Had we defined electricity in terms of energy, we might have observed that electric transfer between capacitors, which Coulomb’s spheres also functioned as, is not conservative. Mach drew parallels between this hypothetical loss of ’electricity’ and the non-conservation of heat in Carnot’s heat engines.

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他认为，把电荷看作守恒量，与把热质看作守恒量，理由同样充分。静电学承认，同样数量的电荷，随其电势不同，可以对应不同的能量。同样，一定数量的热质也可能随其势的不同而包含不同的能量。因此，温度可以看作电压在热学中的对应量：电压表示电荷的势，而温度表示物体内热质的势。

He argued that there is as much reason to consider electric charge conserved as there is to consider caloric conserved. Electrostatics acknowledges that a specific amount of charge can represent varying amounts of energy depending on its potential. Similarly, a certain amount of caloric might contain different energy levels based on its potential. Thus, temperature could be seen as a thermal equivalent to electrical voltage, where voltage represents the potential of a charge, and temperature represents the potential of caloric within a body.

如果科学界愿意接受电荷概念，那么就没有理由拒绝热质。

If science was willing to accept the notion of electric charge, then it had no reason to reject caloric.

我们该如何看待这一论点？

What are we to make of this argument?

对马赫静电类比的批判

Critique of Mach's electrostatic analogy

再看马赫如何结束他的第一个类比，即库仑类比："如果我们把这一效应归于某种物质，自然就会推断出它的恒定性。但这一说明的关键是电力 p 的可分性，而不是物质这一比喻。"注意，他在这里留了退路。他甚至没有直言自己赞同电荷具有实体性。他把电荷概念称为一种比喻，说真实存在的是电力的可分性。这符合他总体上的工具主义科学观：科学研究的是仪器测量结果之间的关系，而非发现究竟是什么东西使仪表指针运动。

Look again at how Mach concludes his first, Coulomb analogy: 'If we ascribe this effect to a substance, then we infer naturally its constancy. But the essential point of the exposition is the divisibility of the electric force *p* and not the simile of substance'. Notice that he is hedging his bets here. He is not saying outright that he even agrees that electric charge is substantial. He calls the notion of charge a simile, saying that what is real is the divisibility of the electric force. This is in keeping with his general instrumentalist approach to science, whereby science is about relations between measurements on instruments rather than about discovering the what the things are that cause dials to move.

但如果我们考察库仑实验中一系列操作，就会发现"力的可分性"并不是对实验过程的准确描述。

But if we examine the sequence of actions performed in the Coulomb experiment we find that 'divisibility of force' is not an accurate description of what is going on.

第一步，实验者给一个小球充电（图 6.1 中标为 d），然后将它放入扭秤。他利用扭秤顶部的刻度盘，读出该球对小球 a 施加的静电力。接着把小球 d 取出，让它与另一个小球 b 接触（库仑图中未画出 b）。然后依次把 d、b 放入扭秤，分别从刻度盘上读出它们造成的偏转。

In the first step, an experimenter charged up *a* ball (shown as *d* in Figure 6.1) and inserted it into the torsion balance. He then read off the electrostatic force it exerted on the ball *a* using the dial at the top of the balance. Ball *d* was then removed from the balance and brought into contact with another ball *b* (which is not shown in Coulomb's diagram). Balls *d*, *b* are then successively placed in the balance, and the deflections they induce read off on the dial.

把这个实验叫作对力的分割的测量，其实并不恰当，因为这里是在三个不同的时刻，分别进行了三次力的测量。真正展示力的分割，需要被分割的力同时作用，如图 6.3 所示。库仑观察到的是：似乎有某种东西能够在小球之间传递，使这些不同小球在不同时刻所能产生的力保持守恒。他据此推断，有某种东西——不妨称为电荷——被分配到了小球之间。

It is not really valid to call this a measurement of division of forces since what we have is three distinct measurements of forces at three different times. An actual division of force demonstration requires that the forces being divided act simultaneously, as in Figure 6.3. What Coulomb had observed was that there appeared to be something which could be transferred between balls in such a way that the forces that these distinct balls could induce at distinct times were conserved. He deduced that there was something, call it electric charge, which was being divided between the balls.

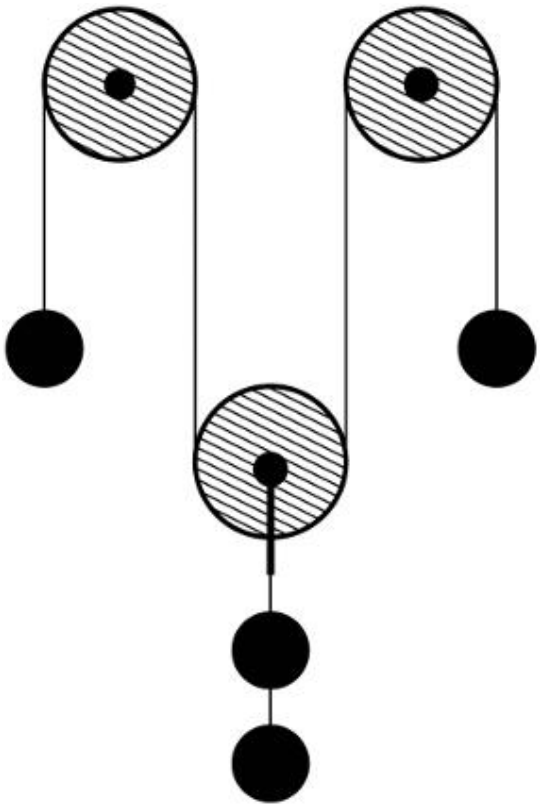


图 6.3 真正的力的分割装置示意图。两侧重物的平衡力与中央重物的力同时作用。作者绘图。

Figure 6.3 Diagram of an actual division of forces setup. The balancing forces of the peripheral weights act simultaneously with the central ones. Figure by authors.

马赫声称库仑目睹了力的分割，这是一种歪曲。库仑实际注意到的是某种能够产生力的东西被分割了。马赫反对认识物自体的康德主义偏见，促使他拒绝这一观念，并错误地声称观察到的是力的分割。在力学中，力是加速度的直接动因。因此，能够积聚在髓球上或莱顿瓶内的并不是力本身，而是有潜能产生力的某种物质。

Mach’s assertion that Coulomb witnessed a division of force is a distortion. Coulomb actually noted a division of something that could potentially cause force. Mach’s Kantian bias against the knowledge of things in themselves prompted him to reject this notion and incorrectly assert that a division of forces was observed. In mechanics, force is an immediate agent of acceleration. Therefore, what can be accumulated on pith balls or in Leiden jars is not force per se, but rather a substance that has the potential to generate force.

如果我们现在审视马赫对莱顿瓶实验描述，会发现它同样存在谬误。绘制该实验电路图（图 6.4）有助于理解。通过该图，我们可以看出他的分析中遗漏了一个关键步骤。当第一个莱顿瓶与第二个连接时，是通过一个具有一定电阻连接（图中 R1）实现。马赫论述中忽略了通过该电阻器散失热量。

If we now look at Mach’s account of the Leiden jar experiment, we find it equally faulty. It helps to draw a circuit diagram of it (Figure 6.4). With this, we can see that a key step is left out of his analysis. When the first leiden jar is connected to the second, it is done via a connection which has some resistance (R1 in the diagram). The heat lost through this resistor is left out of Mach’s account.

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若不测量 h_1，即把一半电荷转移到电容器 C2 时释放的热量，我们得到的实验就类似于库仑实验，但会发现 $h_0 > h_2 + h_3$。这样看来，电似乎就不守恒了。他想要在电荷的电压与假想热质的温度之间建立类比，但所用的思想实验，是使电能（电荷 × 电压平方）通过电阻放电转化为热。在图 6.4 中，他把电阻 R1 上损失的热量比作卡诺热机输出的功；从热力学来看，这个类比荒谬不通。马赫把有用能经电阻损失为热，与热转化为有用能相提并论。但热力学第二定律告诉我们，这两种转化并不对称。电

If we did not measure h1 the heat released in shifting half the charge to capacitor C2, we would have an experiment similar to the Coulomb one, but we would note that $h_0 > h_2 + h_3$. It would thus appear that electricity was not conserved. The analogy he wants to make is between the voltage of a charge, and the temperature of a hypothetical *caloric*. But he does this by a thought experiment in which electric energy (charge × voltage^2) is converted to heat by discharge through a resistor. In Figure 6.4, the heat lost through resistor R1 is supposed to be analogous to the work out of a Carnot heat engine

能可以以 100% 的效率转化为热，而热转化为有用能，无论是电能还是机械功，效率总是低于 100%。

〔校注：底本电能括号中写作 charge × voltage²，量纲可疑；此处保留其平方标记，未静默改为电荷 × 电压。〕

but this an absurd analogy on thermodynamic grounds. Mach is matching a loss of useful energy to heat through a resistor, with a conversion of heat to useful energy. But the second law of thermodynamics tells us that such transfers are not symmetric. You can transform electrical energy into heat with 100 per cent efficiency, but the conversion of heat to useful energy, whether this is electricity or mechanical work, is always less than 100 per cent efficient.

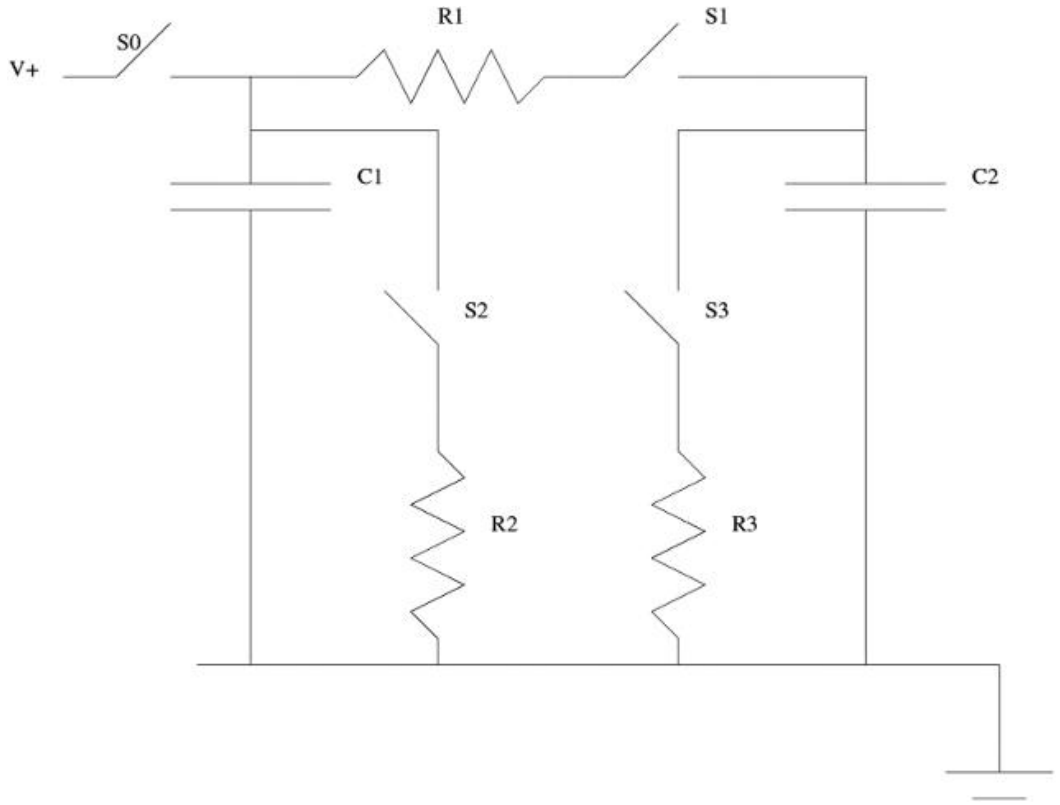


图 6.4 用现代电路图表示马赫所描述的莱顿瓶实验。步骤：（1）闭合开关 S0，给 C1 充电；（2）断开 S0，再闭合 S2，测量 R2 上耗散的热量，记为 h_0；（3）重复步骤（1）；（4）闭合 S1，测量 R1 上耗散的热量，记为 h_1；（5）断开 S1，再同时闭合 S2 和 S3，测量 R2 上的热量（h_2）和 R3 上的热量（h_3）。作者绘图。

Figure 6.4 The Leiden jar experiment described by Mach as a modern circuit diagram. Steps: (1) close switch S0 and charge C1; (2) open switch S0, then close switch S2 and measure heat dissipated in R2. Call this h0; (3) repeat step (1); (4) close Switch S1 and measure heat dissipated in R1, call this h1; (5) open S1 and then simultaneously close S2 and S3, measuring heat in R2 (h2) and R3 (h3). Figure by authors.

S0—S3：开关；C1、C2：电容器；R1—R3：电阻；V+：正电压端；下方符号：接地。

假设有一幅与图 6.4 对应的热学示意图：用热库代替电容器，并在原图电阻所在的位置放置热机。

Suppose we had a thermal equivalent diagram to Figure 6.4 with thermal reservoirs substituting for capacitors and heat engines where the resistors occur in the original diagram.

在原始图中描述运行过程中，所有电能最终都会以热能形式散失。而在热力版本中，只有一部分初始热能被转化为功。

In operations described in the original diagram, all the electrical energy is eventually dissipated as heat. In the thermal version, only a part of the starting thermal energy gets transferred into work out.

对马赫所尝试类比进行仔细考察后，会发现它根本行不通。随着我们在此详细梳理马赫论点，其荒谬之处便显而易见。他试图通过思想实验中类比来反驳原子理论，却实际上忽略了熵增定律。正是麦克斯韦和玻尔兹曼原子理论解释了熵增加。马赫却试图用类比来反驳它，完全无视了熵增加。

A close examination of Mach’s attempted analogy shows that it simply does not work. As we go through Mach’s argument here in detail it’s full absurdity sinks in. He was attempting to refute the atomic theory by making analogies in thought experiments in which he effectively ignored the law of increasing entropy. It is Maxwell and Boltzmann’s atomic theory that explains

increasing entropy. Mach tried to refute it with analogies, just ignoring entropy increase.

p. 130–131 或许可以为他辩解说，这本书初版于 1872 年，时间还相当早，恰逢玻尔兹曼发表那篇关于熵的里程碑式论文。那时尚早于麦克斯韦的《热学理论》一书，[10]但麦克斯韦论述中的关键内容已经发表。[11]即便允许马赫用几年时间消化这些思想，这个借口也很勉强。马赫在此后的岁月里并未撤回异议。1894 年 10 月，他又在 The Monist 第 5 卷第 1 期上以英语发表了这篇文章，只作少量修改，再次使用同样的静电类比来反对热的机械理论。

〔校注：“早于麦克斯韦的《热学理论》”为底本原述；注 10 列 1875 年版本，未据此改写正文。〕

In mitigation, one might say that the first publication date of the book, 1872, was quite early, the same year as Boltzmann’s landmark paper on entropy. It was before Maxwell’s book on the Theory of Heat,[10] but the key elements of Maxwells account had already been published.[11] Even supposing that Mach be allowed a few years to absorb these ideas, the excuse wears thin. Mach did not reverse his objections in subsequent years. He had the text published again in English in Vol. 5, No. I, of The Monist, October, 1894, with minor modifications repeating the same electrostatic analogies against the mechanical theory of heat.

p. 131 可能会有人反驳说，直到那时，证明原子存在证据还只是归纳性。克劳修斯、麦克斯韦和玻尔兹曼曾基于公认力学原理，构建了一种能够解释以下现象理论：

1. 温度和压力。
2. 熵的增加。
3. 气体的黏度。
4. 气体中的“等体积定律”。
5. 与化学的原子理论相一致。

It may be objected that up, until that date, the evidence for the existence of atoms was inductive. Clausius, Maxwell and Boltzmann had, from accepted principles of mechanics, constructed a theory that explained:

1. Temperature and pressure.
2. Entropy increase.
3. Viscosity in gases.
4. The ‘law of equivalent volumes’ in gases.
5. Was consistent with the atomic theory of chemistry.

爱因斯坦驳倒马赫派对原子的反对

Einstein demolishes Machist opposition to atoms

谁能想到，就在 1900 年，人们还在围绕原子是否真实存在争斗，甚至可以说斗得你死我活。维也纳的大哲学家恩斯特·马赫说：不。大化学家威廉·奥斯特瓦尔德也说：不。然而在那个关键的世纪之交，一个人基于根本的理论依据，挺身捍卫原子的实在性。他就是路德维希·玻尔兹曼。……人类的进步在那一刻悬于一架精细的思想天平之上；倘若当时反原子学说真的占了上风，我们的进步肯定会倒退几十年，甚至可能倒退一百年。[12]

Who could think that, only in 1900, peoples were battling, one might say to the death, over the issue whether atoms are real or not. The great philosopher Ernst Mach in Vienna said, NO. The great chemist Wilhelm Ostwald said, NO. And yet one man, at that critical turn of the century, stood up for the reality of atoms on fundamental grounds of theory. He was Ludwig Boltzmann. . . . The ascent of man teetered on a fine intellectual balance at that point, because had the anti-atomic doctrines then really won the day, our advance would certainly have been set back by decades, and perhaps a hundred years.[12]

就在玻尔兹曼自杀前不久，一位名不见经传专利局职员发表了一系列令人惊叹四篇论文，这些论文不仅彻底证明了玻尔兹曼正确性，还把 20 世纪物理学引向了一条革命性新道路。

1. 1905 年 6 月：a.《论热的分子动理论所要求的静止液体中悬浮微粒的运动》。[13]

b.《关于光的产生与转化的一种启发式观点》。[14]

c.《论运动物体的电动力学》。[15]

2. 1905 年 9 月，《物体的惯性是否取决于其能量含量?》。[16]

Shortly before Boltzmann’s suicide, an obscure patent clerk published an astonishing sequence of four papers that both completely vindicated Boltzmann and set twentieth-century physics on a revolutionary new path.

1. June 1905, a. *On the movement of small particles suspended in stationary liquids required by the molecularkinetic theory of heat.*[13]

b. *On a heuristic point of view concerning the production and transformation of light.*[14]

c. *On the electrodynamics of moving bodies*[15]

2. September 1905, *Does the inertial of a body depend on its energy content?*[16]

p. 131–132 论文 [a] 表明，微粒的布朗运动可以根据分子动理论作出定量预测，由此奠定了基础。[17]在题为“悬浮粒子的平均位移公式：测定原子实际大小的新方法”的一节中，他说明：若假定阿伏伽德罗常数为

Paper [a] sets the stage by showing that the Brownian motion of small particles could be quantitatively predicted from the molecular kinetic theory.[17] In a section entitled *‘Formula for the Mean Displacement of*

6×10^{23}，那么直径为 1 微米的小球在水分子随机撞击下，预计每分钟会移动约 6 微米。反过来，若通过实验精确测量这些小球的运动，就能更精确地估计阿伏伽德罗常数。这一预测很快在实验中得到验证，[18]并在这一过程中将阿伏伽德罗常数的估计值修正为 7.15×10^{23}。

〔校注：底本“in experiments by”后没有作者姓名，直接接注 18；该注列 Jean Perrin。中文按句意连接，英文保留原貌。〕

Suspended Particles. A New Method of Determining the Actual Size of Atoms’, he showed that if one assumes Avogadro’s number was 6×10^{23}, then spheres of 1-micron diameter could be expected to wander about 6 microns per minute due to random bombardment by water molecules. Alternatively, if one were by experiment to obtain accurate measurements of the movement of such spheres one could get a more precise estimate of Avogadro’s number. The prediction was quickly validated in experiments by[18] who, in the process, refined the estimate of Avogadro’s number to 7.15×10^{23}.

仅凭这篇论文，就足以彻底驳倒马赫派反对原子、分子及热的动理论的论点。但若考察爱因斯坦在那个夏天接连发表的整组论文，就能看到一个逐渐展开的主题：捍卫关于实在的微粒观，反对与之争论、受到马赫及其支持者青睐的能量论学派。

On its own, this paper would have been enough to permanently refute the Machist arguments against atoms, molecules and the kinetic theory of heat. But if we look at the whole volley of papers that Einstein published over that summer, we see a developing theme of defending a corpuscular view of reality against the contenting energeticist school favoured by Mach and his supporters.

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爱因斯坦[19]论证了为什么应把光看作微粒。他在开篇写道：

Einstein[19] lays out an argument as to why light should be viewed as corpuscular. He introduces it by saying:

光波动理论及其连续空间函数已被证明是纯粹光学现象绝佳模型，预计永远不会被其他理论所取代。尽管如此，我们应当考虑到，光学实验观测到只是时间平均值，而非瞬时值。因此，尽管麦克斯韦理论与实验结果完全吻合，但使用连续空间函数来描述光，可能会导致与实验结果矛盾，特别是在应用于光产生和转化时。

The wave theory of light with its continuous spatial functions has proven to be an excellent model of purely optical phenomena and presumably will never be replaced by another theory. Nevertheless, we should consider that optical experiments observe only time-averaged values, rather than instantaneous values. Hence, despite the perfect agreement of Maxwell’s theory with experiment, the use of continuous spatial functions to describe light may lead to contradictions with experiments, especially when applied to the generation and transformation of light.

特别是，黑体辐射、光致发光、紫外线产生阴极射线以及其他与光产生和转化相关现象，似乎在假设光能量在空间中呈不连续分布情况下，能得到更好建模。根据这一图景，点光源发出光波能量并非连续扩散到越来越大体积中，而是由有限数量能量量子组成，这些量子在空间中特定点上局域化，在运动过程中不会分裂，且只能作为整体被吸收或产生。[20]

In particular, black body radiation, photoluminescence, generation of cathode rays from ultraviolet light and other phenomena associated with the generation and transformation of light seem better modeled by assuming that the energy of light is distributed discontinuously in space. According to this picture, the energy of a light wave emitted from a point source is not spread continuously over ever larger volumes, but consists of a finite number of energy quanta that are spatially localised at points of space, move without dividing and are absorbed or generated only as a whole.[20]

他是如何论证这一观点？

How does he justify his suggestion?

起初，他考察了处于热平衡状态腔内气体，假设其本质上具有微粒性，由原子和电子组成。他提出，受各种局部静电势限制电子可以作为电磁辐射谐振器。运用分子动理论，他推导出了谐振器平均能量公式，表明该能量应与气体温度成正比。

Initially, he examines a gas within a cavity at thermal equilibrium, presuming it to be corpuscular in nature, composed of atoms and electrons. He posits that the electrons, confined by various local electrostatic potentials, can serve as resonators for electromagnetic radiation. Employing molecular kinetic theory, he deduces a formula for the average energy of a resonator, indicating that it should be directly proportional to the gas temperature.

到这里，他阐述的仍是当时公认的光发射理论。然而，他证明这一理论的预测偏离了实验结果。考虑频率越来越高的辐射时，理论所预测的腔内辐射能会趋于无穷大。他推导出一个方程，说明体积为 v_0 的腔内单色光的熵应是多少，以及当光的全部能量集中到腔内某个子体积 v 中时，这个熵会怎样变化。[21]对此，他指出：

Up to this point, he was outlining the accepted theory of light emission. However, he demonstrates that the theory’s predictions deviate from experimental results. Considering increasingly higher frequencies of radiation, the predicted radiant energy within a cavity would theoretically become infinite. He derives an equation for what the entropy of monochromatic light within

a cavity v_0 would be and how this entropy would change if all the energy of the light were concentrated in some sub-volume *v* of the cavity.[21] Of this, he observes:

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该方程表明，密度足够低单色辐射熵随体积变化规律，与理想气体或稀溶液熵变化规律相同。下文将根据玻尔兹曼先生提出原理来解释上述方程，该原理指出：系统熵是其状态概率函数。[22]

This equation shows that the entropy of monochromatic radiation of sufficiently low density varies with volume according to the same law as the entropy of an ideal gas or that of a dilute solution. In the following the equation just found will be interpreted in terms of the principle introduced by Mr. Boltzmann that says that the entropy of a system is a function of the probability of its state.[22]

这立即表明——尽管爱因斯坦并未对此多加论述——光表现出了类似原子集合体微粒性质。

This immediately indicates, though Einstein does not labour the point, that the light is behaving in a corpuscular way like a collection of atoms.

接着，他把玻尔兹曼的熵公式用于体积为 v_0 的气体中的分子，推导出当全部分子都处于子体积 $v < v_0$ 内时的熵变公式；这基本上是上一章半个圆筒例子的推广。结果表明，气体熵变公式与他推导出的单色辐射熵变公式具有相同的函数形式。[23]他由此得出结论：

He then uses Boltzmann’s entropy formula for the molecules in a gas of volume v_0 and derives a formula for what the change in entropy would be if all the molecules were within a sub-volume $v < v_0$ – basically a generalization of the half cylinder examples given in the previous chapter. It turns out that the formula for the change in entropy of the gas is functionally the same[23] as the formula he derived for the change in the entropy of monochromatic radiation. He then concludes:

从热理论来看，低密度单色辐射（在维恩辐射公式的适用范围内）的行为，仿佛它由大小为 $\frac{R\beta\nu}{N}$ 的独立能量量子组成。*

〔校注：底本公式后有“>8”异常字符，英文照录；中文保留句号和星号，不将异常字符解释为公式或注号。〕

In terms of heat theory monochromatic radiation of low density (within the realm of validity of Wien’s radiation formula) behaves as if it consisted of independent energy quanta of the magnitude $\frac{R\beta\nu}{N}$. >8*

〔原书星号页脚注〕* 这一量的现代记法是 $h\nu$，其中 h 为普朗克常数，ν 为频率。他写成 $\frac{R\beta\nu}{N}$，但他所用的常数 R, β, N 保证 $\frac{R\beta}{N} = h$。底本在常数列表后另印有 $\frac{n!}{r!(n-r)!}$。

〔校注：底本在 R、β、N 后混入组合数分式 n!/[r!(n−r)!]，与当前句意不相衔接。英文保留原位置，中文说明其原有位置，不将其改写为新的物理关系；Plank 拼写亦照录英文。〕

* The modern notation for this is $h\nu$ for $h =$ Plank’s constant and ν frequency. He wrote $\frac{R\beta\nu}{N}$ but his R, β, N $\frac{n!}{r!(n-r)!}$ constants ensure $\frac{R\beta}{N} = h$.

请思考他的成就。他两次应用了玻尔兹曼对熵统计学解释：最初是在辐射背景下，随后又在气体分子背景下。他证明了光熵随体积变化方式与气体熵相同。这使他推断出光具有粒子性，由后来被称为“光子”粒子组成。值得注意的是，他在推导腔内光熵时，并未预设光粒子性；而是从经验上得到验证黑体辐射公式中推导出了这一结论。因此，玻尔兹曼统计力学不仅在原子层面正确——爱因斯坦关于布朗运动论文便证明了这一点——而且也适用于光，从而推导出了光粒子性。

Reflect on his achievements. He applied Boltzmann’s statistical interpretation of entropy twice: initially in the context of radiation, and subsequently in the context of gas molecules. He demonstrated that the entropy of light changes with volume in the same manner as the entropy of gas. This led him to deduce that light is particulate, consisting of what would later be termed photons. Notably, he did not presuppose the particulate nature of light in his derivation of light’s entropy within a cavity; rather, he deduced it from the empirically validated formula for black body radiation. Thus, Boltzmann’s statistical mechanics was not only correct regarding atoms, as evidenced by Einstein’s paper on Brownian motion, but it was also applicable to light, inferring its particulate nature.

鉴于当时人们对玻尔兹曼关于气体原子理论仍持极大怀疑态度，这无疑是一个大胆推论。它意味着物理学界不仅必须摒弃对气体理论疑虑，还必须面对一个或许更为艰巨挑战——接受电磁辐射粒子理论，而此前关于电磁辐射连续波理论似乎一直是 19 世纪物理学辉煌成就。

Given that Boltzmann’s atomic account of gases was still being treated with great scepticism, this was an audacious extension. It implied that the physics community must both drop their doubts about the gas theory and face a perhaps harder challenge – to accept a particulate theory for electromagnetic radiation, the continuous wave account of which had up until then seemed a triumph of nineteenth-century physics.

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单就这一点而言，这不过是一个大胆假设，但爱

因斯坦提供了实验证据来支持这一观点。光电效应早已为人所知，这是一种入射光导致真空管中冷阴极释放电子并向阳极移动现象。这种运动会产生可检测到电流。在更现代应用中，太阳能电池板正是基于这一原理工作。通过对阳极施加反向偏压，电子会被排斥，阻止电流流动。所谓“截止电压”——即电流停止流过点——表示电子从阴极发射出来时所具有能量（单位为电子伏特）。

This would merely be a bold hypothesis by itself, but Einstein provided experimental evidence to support it. The photoelectric effect, already known, is a phenomenon where incident light causes electrons to be emitted from a cold cathode in a vacuum tube, moving towards the anode. This movement generates a detectable current. In a more contemporary application, this principle is how solar panels operate. By applying a reverse bias to the anode, electrons are repelled, preventing current flow. The ‘stopping voltage’–the point at which current ceases–indicates the energy, in electron volts, that the electrons possess when emitted from the cathode.

此前关于光波动理论曾预测，截止电压将由入射光强度决定。爱因斯坦理论则指出，它应由光频率决定。高频光量子具有更大能量，因此会把电子激发到更高电压。

The previous wave theory of light predicted that the stopping voltage would be determined by the intensity of the incident light. Einstein’s account said that it should be determined by the frequency of the light. High-frequency light quanta had more energy and would thus excite electrons to higher voltages.

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实验数据支持爱因斯坦量子理论，而非波理论。截止电压取决于频率，而非强度。正是凭借在光电效应方面研究，爱因斯坦获得了诺贝尔物理学奖。

The experimental data aligns with Einstein’s quantum theory over the wave theory. The stopping voltage is dependent on frequency, not intensity. It was for his work on the photoelectric effect that Einstein received the Nobel Prize in Physics.

一个世纪前，光电效应还是一种鲜为人知现象，只有实验物理学家才了解。如今，它已成为我们日常生活一部分。太阳能电池板、LED 灯泡以及智能手机中摄像头都利用了这一效应。事实上，现代数码相机芯片中传感器非常灵敏，以至于击中像素离散光子数量统计波动，会在均匀背景上表现为可见“散粒噪声”。现代技术使我们能够在手机上观察到光的量子化现象。

A century ago, the photoelectric effect was an obscure phenomenon known only to experimental physicists. Now, it’s a part of our daily lives. Solar panels, LED light bulbs and the cameras in our smartphones all utilize it. In fact, the sensors in modern digital camera chips are so sensitive that the statistical fluctuation in the number of discrete photons striking pixels becomes visible ‘shot noise’ against a uniform background. Modern technology has made it possible to observe the quantization of light on our cell phones.

有人可能会反驳说，爱因斯坦对光电效应解释是一种事后推断。当时，光电效应实验证据早已确立。爱因斯坦理论为这些观测结果提供了解释，而连续统理论却无法做到这一点。然而，这并非一种预测。1922 年，人们发现 X 射线可以被电子散射，从而导致电子获得动量，这一发现被证明是对爱因斯坦理论关键性证实。

It could be objected that his photo-electric was a post-diction on his part. The experimental evidence for the photoelectric effect was well-established. Einstein’s theory provided an explanation for these observed results, which the continuum theory could not. However, this was not a prediction. The 1922 discovery that X-rays could be scattered by electrons, resulting in a gain of momentum by the electrons, proved to be a crucial confirmation of Einstein’s theory.

他在 1905 年的最后两篇论文讨论了狭义相对论，[24]并推导出著名的 $e = mc^2$ 公式。[25]他推导相对论时使用的论证与唯物主义问题只有间接联系，所以这里不作讨论。[26]物质与能量彼此等价的结论，既具有哲学意义，也具有巨大的实践意义。在哲学上，它削弱了把“能量论”当作唯物主义替代方案的可能性。[27]在历史上，它开启了原子能及其所带来的一切可能。如果说有什么让原子观念深入公众心中，那就是原子能。

His final two 1905 papers address the theory of special relativity[24] and derive his famous $e = mc^2$ formula.[25] His arguments in deriving the theory of relativity are only tangentially relevant to the issue of materialism, so we will not go into them here.[26] The conclusion that matter and energy were equivalent to one another did have philosophical as well as immense practical relevance. Philosophically, it undermined the ability to appeal to ‘energetism’ as an alternative to materialism.[27] Historically, it opened up the possibility of atomic power with all that entails. If anything established the idea of atoms in the popular mind, it was atomic energy.

贝希托尔德[28]把马赫的原子观概括如下：“马赫的原子观可以归纳为五个主要观点：（a）原子不作为

Bächtold[28] summarizes Mach’s view on atoms as: ‘Mach’s view on atoms can be described as consisting

自在之物存在；（b）原子概念只是感觉复合体的思想符号；（c）原子论是一种没有成果的假说；（d）原子论是一种人为的假说；（e）原子论是一种没有充分依据的假说。”

of five main ideas: (a) atoms do not exist in themselves, (b) the notion of atom is only a thought-symbol for a complex of sensations, (c) atomism is an unfruitful hypothesis, (d) atomism is an artificial hypothesis and (e) atomism is an unjustified hypothesis’.

到了 20 世纪中叶，马赫学派对原子观念反对似乎已显得荒谬且令人反感。广岛毁灭绝非源于某种“感觉复合体”。

By the mid-twentieth century, the Machist objection to the idea of atoms just seemed absurd and obscene. Hiroshima was not destroyed by a‘complex of sensations’.

哲学马赫主义与列宁的干预

Philosophical Machism and Lenin’s intervention

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马赫影响不仅体现在阻碍了原子理论被接受。他的哲学思想塑造了半代科学家思维方式。它确立了工具主义——即科学仅仅是用于把仪器读数与感官印象相关联公式这一学说——作为许多早期量子物理学家思考量子力学所引发概念革命出发点。

Mach’s influence did not just extend to holding back the acceptance of the atomic theory. His philosophical teaching conditioned half of a generation of scientists. It established instrumentalism, the doctrine that science was just about formulae used to correlate instrument readings and sense impressions, as the starting point from which many of the original quantum physicists thought through the conceptual revolution induced by quantum mechanics.

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与牛顿和莱布尼茨一样，[29]马赫强调科学思想中简洁性的优点。

Like Newton and Leibniz,[29] Mach emphasized the virtues of simplicity in scientific thought.

科学知识传播总是涉及描述，即在思想中对事实模仿性再现，其目的是取代新经验并省去相关麻烦。同样，为了节省教学和学习劳力，人们寻求简洁、概括描述。这实际上就是自然法则全部内容。只要知道重力加速度数值以及伽利略下落定律，我们便拥有了简单而简明指引，可以在思想中重现落体一切可能运动。这样公式完全可以替代一份完整下落运动表，因为借助该公式，可以随时轻松推导出该表中数据，而丝毫不给记忆带来负担。

The communication of scientific knowledge always involves description, that is, a mimetic reproduction of facts in thought, the object of which is to replace and save the trouble of new experience. Again, to save the labor of instruction and of acquisition, concise, abridged description is sought. This is really all that natural laws are. Knowing the value of the acceleration of gravity, and Galileo’s laws of descent, we possess simple and compendious directions for reproducing in thought all possible motions of falling bodies. A formula of this kind is a complete substitute for a full table of motions of descent, because by means of the formula the data of such a table can be easily constructed at a moment’s notice without the least burdening of the memory.

……

. . .

在代数中，我们尽可能把所有形式相同数值运算一次性完成，从而仅需针对个别情况进行少量工作。代数和分析中符号使用——这些符号仅仅代表待执行运算——源于这样一种观察：通过把所有机械性运算交由双手完成，我们可以实质性地减轻思维负担，从而把思维能力留给更重要、更困难任务。这种方法一个结果——也印证了其经济性——便是计算机制造。数学家巴贝奇（差分机发明者）可能是第一个清晰认识到这一事实的人，他在其著作《制造业与机械的经济性》中曾提及这一点，尽管只是略为提及。[30]

In algebra we perform, as far as possible, all numerical operations which are identical in form once for all, so that only a remnant of work is left for the individual case. The use of the signs of algebra and analysis, which are merely symbols of operations to be performed, is due to the observation that we can materially disburden the mind in this way and spare its powers for more important and more difficult duties, by imposing all mechanical operations upon the hand. One result of this method, which attests its economical character, is the construction of calculating machines. The mathematician Babbage, the inventor of the difference-engine, was probably the first who clearly perceived this fact, and he touched upon it, although only cursorily, in his work, The Economy of Manufactures and Machinery.[30]

问题在于，这一节约原则与一种实质上属于个人主义或心理主义的观念纠缠在一起：信息的节约究竟发生于何处？这里把个体感知与科学这一社会活动混为一谈。

The problem is that this economic principle was tied up with what amounted to an individualist or psychological conception where the information economy occured

当我们在思想中把一个物体从它运动于其中的可变环境中分离出来时，实际上是在我们的全部感觉之流中，抽出一组思想所关注的、比其他感觉相对更稳定的感觉。这组感觉并非绝对不变。它的成员时而出现，时而消失，或发生改变。它从不以完全相同的样态重现。然而，其恒定要素的总量与可变要素的总量相比，总是占如此大的比重，尤其是考虑到转变的连续性时更是如此，以至于就当前目的而言，前者通常似乎足以确定物体的同一性。但是，由于我们可以从这组感觉中分离出任何一个成员，而物体对我们来说仍不失为同一个，我们便容易相信：把所有成员都抽离之后，还会剩下某种额外的东西。于是，我们形成了一种有别于其属性的实体观念，即物自体观念，而把我们的感觉仅仅视为这个物自体的属性的符号或标志。然而，更妥当的说法应当是：物体或事物是感觉群的简明心理符号——这些符号并不存在于思想之外。[31]

–a confusion of individual perception with the social activity of science.

In mentally separating a body from the changeable environment in which it moves, what we really do is to extricate a group of sensations on which our thoughts are fastened and which is of relatively greater stability than the others, from the stream of all our sensations. Absolutely unalterable this group is not. Now this, now that member of it appears and disappears, or is altered. In its full identity it never recurs. Yet the sum of its constant elements as compared with the sum of its changeable ones, especially if we consider the continuous character of the transition, is always so great that for the purpose in hand the former usually appear sufficient to determine the body’s identity. But because we can separate from the group every single member without the body’s ceasing to be for us the same, we are easily led to believe that after abstracting all the members something additional would remain. It thus comes to pass that we form the notion of a substance distinct from its attributes, of a thing-in-itself, whilst our sensations are regarded merely as symbols or indications of the properties of this thing-in-itself. But it would be much better to say that bodies or things are compendious mental symbols for groups of sensations—symbols that do not exist outside of thought.[31]

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他的意思是，世界上物体并不存在，只有我们对它们思考或感觉才存在。这种唯心主义观点几乎与贝克莱学说如出一辙。我们确实对引发白垩纪—第三纪灭绝事件小行星有一种象征性表征。但杀死恐龙并非这种象征性表征，而是那块真实存在岩石与金属，正是它形成了希克苏鲁伯陨石坑。

He is saying that objects in the world do not exist, only our thoughts or sensations of them. This is almost Berkeleyan in its idealism. We do have a symbolic representation for the asteroid that caused the KT extinction event. But it was not this symbolic representation that killed the dinosaurs. It was a real mass of rock and metal that created the Chicxulub crater.

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为什么说事物只是感觉群的心理符号、而这些符号并不存在于思想之外，会是更好的说法呢？

Why would it be better to say that things are just mental symbols for groups of sensation symbols that do not exist outside of thought?

他如何把这种个人主义与他在其他地方所承认科学集体性相调和？

How does he reconcile this individualism with what he knows about the collective nature of science which he acknowledges elsewhere?

通过交流，许多人最初各自获得经验被汇集为一。知识传播，以及每个人为了以最少思维消耗来管理自己经验储备而感受到必要性，迫使我们把知识归纳为经济形式。但这恰恰为我们提供了一条线索，它剥去了科学一切神秘面纱，并向我们展示了科学真正力量。就具体结果而言，它所提供，是我们即使不借助方法，只要时间足够长，也能达到。整个数学领域中，没有哪个问题不能通过直接计数来解决。但借助现有数学工具，许多计数运算可以在几分钟内完成，而如果没有数学方法，这些运算可能需要一辈子才能完成。正如一个完全依赖自身劳动成果人永远无法积累巨额财富，相反，许多人劳动成果积累集中于一人之手，才是财富与权力基础；同样，任何值得称之为知识东西，都无法被局限于人类寿命范围之内、且仅具备有限能力单个头脑所汇集，除非通过最精妙思维节约，以及对成千上万合作者所积累、经过经济有序整理经验进行汇集。[32]

By communication, the experience of many persons, individually acquired at first, is collected in one. The communication of knowledge and the necessity which every one feels of managing his stock of experience with the least expenditure of thought, compel us to put our knowledge in economical forms. But here we have a clue which strips science of all its mystery, and shows us what its power really is. With respect to specific results it yields us nothing that we could not reach in a sufficiently long time without methods. There is no problem in all mathematics that cannot be solved by direct counting. But with the present implements of mathematics many operations of counting can be performed in a few minutes which without mathematical methods would take a lifetime. Just as a single human being, restricted wholly to the fruits of his own labor, could never amass a fortune, but on the contrary the accumulation of the labor of many men in the hands of one is the foundation of wealth and power, so, also, no knowledge worthy of the name can be gathered up in a single human mind limited to the span of a human life and gifted only with finite powers, except by the most exquisite economy of thought and by the careful amassment of the economically ordered experience of thousands of co-workers.[32]

上述论述很好地认识到了信息的节约以及科学的集体性质。但在其他地方，我们看到的却是个人主义和心理学式的解释。马赫十分愿意承认科学通过集体努力而进步，那么过去就必定有真实的人提出了运动定律。确实存在一个真实的牛顿，马赫使用的正是他的运动定律。如果马赫能够接受其他科学家的实在性，为何又否认牛顿所解释其运动的那些行星天体的实在性？

The above is a good recognition of information economy along with the collective character of science. But in other places, we get individualistic and psychological accounts. Mach was quite willing to recognize that science advanced by collective effort so there must have been real people in the past who came up with the laws of motion. There was a real Newton whose laws of motion Mach used. If Mach could accept the reality of other scientists, why did he reject the reality of the planetary bodies whose motions Newton explained?

他对原子反对，正是源于这种对科学过程理解。

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His opposition to atoms sprung directly from this conception of the scientific process.

当几何学家想理解一条曲线的形状时，会先把它分解成微小的直线元素。但他完全清楚，这些元素只是暂时而任意的手段，用于分部分地理解他无法作为整体把握的对象。一旦找到曲线的规律，他就不再考虑这些元素。同样，物理科学若把自己创造的、可变而节约的工具——分子和原子——当作现象背后的实在，忘却其年长的姊妹哲学最近获得的智慧，以机械神话取代旧的泛灵论或形而上学图式，从而制造无穷无尽的假想问题，那就有失其本分。原子必须始终像数学函数一样，作为表征现象的工具。然而，随着智力在与研究对象接触中逐渐受到训练，物理科学将放弃以石子拼成马赛克的游戏，转而探寻现象的生命之流所流经的河床的边界和形态。它所追求的目标，是对事实最简单、最节约的抽象表达。

When a geometer wishes to understand the form of a curve, he first resolves it into small rectilinear elements. In doing this, however, he is fully aware that these elements are only provisional and arbitrary devices for comprehending in parts what he cannot comprehend as a whole. When the law of the curve is found he no longer thinks of the elements. Similarly, it would not become physical science to see in its self-created, changeable, economical tools, molecules and atoms, realities behind phenomena, forgetful of the lately acquired sapience of her older sister, philosophy, in substituting a mechanical mythology for the old animistic or metaphysical scheme, and thus creating no end of suppositious problems. The atom must remain a tool for representing phenomena, like the functions of mathematics. Gradually, however, as the intellect, by contact with its subject-matter, grows in discipline, physical science will give up its mosaic play with stones and will seek out the boundaries and forms of the bed in which the living stream of phenomena flows. The goal which it has set itself is the simplest and most economical abstract expression of facts.

这种放弃探究现象背后真理观点，无法解释爱因斯坦如何能基于原子理论预测未被观测到现象。如果原子并不存在，爱因斯坦又如何能准确预测花粉粒扩散速率？虽然一个能经济地解释过去观测结果公式很有用，但如果没有一个能准确捕捉现实理论，预测此前未被观测到现象将是不可能。这一点既适用于爱因斯坦对布朗运动和光电效应预测，也适用于更宏大天文预测。勒维耶利用牛顿万有引力定律预测海王星位置，其依据正是需要解释其他行星轨道上摄动。该定律指明了搜索方向，而当人们在该处观测时，海王星确实被发现了。若仅把海王星视为表征这些摄动一种工具，那便是毫无意义。同样，广义相对论并非对既定事实简单表述；它是一套描述真实宇宙几何性质理论。该理论预言了 1915 年尚不为人知、且直到很久以后才被观测到众多现象。爱因斯坦理论预测了 1919 年日食期间埃丁顿观测结果：恒星视位置会因太阳引力而发生偏移。近年来，该理论预测引力波和黑洞也已得到观测证实。

This viewpoint, which abandons efforts to comprehend the truths behind phenomena, fails to explain how Einstein could predict unobserved phenomena based on atomic theory. If atoms are non-existent, how could Einstein accurately forecast the diffusion rate of pollen grains? While a formula that economically accounts for past observations is useful, without a theory that accurately captures reality, predicting previously unobserved phenomena would be unfeasible. This applies as much to Einsteins Brownian motion and photoelectric predictions as it does to grander astronomical ones. Le Verrier’s prediction of Neptune’s location, using Newton’s law of gravity, was based on the need to explain the perturbations in other planets’ orbits. This law indicated where to search, and upon looking there, Neptune was indeed observed. To consider Neptune merely as a tool to represent these perturbations would be non-sensical. Similarly, general relativity was not just a simple representation of pre-existing facts; it was a theory describing the real universe’s geometric properties. It forecasted numerous phenomena that were unknown in 1915 and only observed much later. Einstein’s theory predicted Edingtons’s observations during the 1919 solar eclipse that the apparent positions of stars would be shifted by the Sun’s gravity. In more recent years, the gravitational waves and black holes predicted by the theory have been observed.

在机器学习领域，人们普遍认为，一个在训练集上拟合良好模型在新数据上可能表现不佳。这一现象已广为认可，因此在研究中，使用与训练集不同数据集来评估模型已成为标准做法。只有在未见过示例上表现良好模型才会被视为值得信赖。同样，关于熵的原子理论、关于光的光子理论以及广义相对论都被视为稳健模型。它们优势不仅在于拟合现有数据，更在于把握了客观的因果机制。

In machine learning, it’s a common understanding that a model well-fitted to the training set might perform poorly on new data. This phenomenon is so recognized that evaluating a model on a separate dataset from the training set is a standard practice in research. Only models that perform well on previously unseen examples are deemed trustworthy. Similarly, the atomic theory of entropy, the photon theory of light and general relativity are considered robust models. Their strength lies not just in fitting existing data but in encapsulating objective causal mechanisms.

就在爱因斯坦论文发表之后，但在这些论文尚未产生太大影响之前，一位笔名为列宁俄罗斯社会主义者出版了一本书，对马赫主义进行了批判。[33] 事实证明，这位社会主义者后来成为了苏联首任总理，[34]因此，他对马赫批判被反复转载，并对苏联哲学产生了重大影响。

〔校注：底本 attaching Machism 按上下文应为 attacking；英文照录，中文按批判译。〕

Just after Einstein’s papers came out, but before they had yet made much impact, a Russian socialist with the pen-name Lenin, published a book attaching Machism.[33] As it transpired, this socialist later went on to become the first Soviet Premier,[34] and as a result, his criticism of Mach was repeatedly reprinted and had a big influence on Soviet philosophy.

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作为政治家和政治理论家，列宁对马赫批判与爱因斯坦相比，技术性要弱得多。爱因斯坦通过质疑马赫关于原子物理主张来间接反驳他，而列宁则直接挑战了马赫理论哲学基础。

As a politician and political theorist, Lenin’s critiques of Mach were significantly less technical compared to Einstein’s. While Einstein indirectly contested Mach by disputing his physical claims about atoms, Lenin directly confronted the philosophical bases of Mach’s theories.

马赫对原子存在深刻怀疑——这似乎与 19 世纪科学进步相矛盾——是更广泛反唯物主义立场一部分。源于古希腊唯物主义原子论，传统上一直与这种观点相对立。马赫不仅否认原子存在，更否认所有物质对象存在：“更恰当说法是，即物体或事物是感觉群的简明心理符号——这些符号在思想之外并不存在。”[35]

Mach’s profound scepticism regarding the existence of atoms, which appears contradictory to the progress of nineteenth-century science, was an element of a broader anti-materialist stance. Atomism, originating from ancient Greek materialism, traditionally countered this perspective. Mach did not just deny the existence of atoms but of all material objects: ‘it would be much better to say that bodies or things are compendious mental symbols for groups of sensations –symbols that do not exist outside of thought’.[35]

他断言，感觉才是根本，而物质实体仅仅是感觉群的心理或概念符号。因此，如果一个人始终对物质世界持怀疑态度，那么从逻辑上讲，这种怀疑自然会延伸到对物质最小单元——原子——存在产生怀疑。

He asserts that sensations are fundamental, and material bodies merely represent mental or conceptual symbols for clusters of sensations. Consequently, if one consistently doubts the material world, it logically extends to doubting the existence of matter’s minutest elements –atoms.

马赫把物理科学视为心理学一个分支，把心理现象和个人经验置于现实核心。对他而言，其他一切都是幻象，尽管是一个连贯幻象。列宁曾就这段话评论道：

Mach considered physical science a branch of psychology, placing psychological phenomena and personal experience at the core of reality. To him, everything else was an illusion, albeit a coherent one. Lenin remarked on this passage:

老调重弹了，最可敬的教授！这完全是在重复贝克莱的话：物质是一个赤裸的抽象符号。但实际上，赤裸的正是恩斯特·马赫，因为，如果他不承认“可感内容”是独立于我们而存在的客观实在，那么剩下的就只有一个“赤裸的抽象的”我，一个必定要大写并用斜体标出的我，等同于“那架发疯的钢琴，它以为自己是世上唯一存在的东西”[36]。如果我们的感觉的“可感内容”不是外部世界，那么就只剩下这个沉湎于空洞“哲学”幻想的赤裸的我。这是一种愚蠢而徒劳的消遣！

An old song, most worthy Professor! This is a literal repetition of Berkeley who said that matter is a naked abstract symbol. But it is Ernst Mach, in fact, who goes naked, for if he does not admit that the ‘sensible content’ is an objective reality, existing independently of us, there re- mains only a ‘naked abstract’ I, an I infallibly written with a capital letter and italicised, equal to ‘the insane piano, which imagined that it was the sole existing thing in this world’[36]. If the ‘sensible content’ of our sensations is not the external world, then nothing exists save this naked I engaged in empty ‘philosophical’ fancies. A stupid and fruitless occupation!

马赫把“事物”仅仅是感觉符号这一比喻一直推演到原子层面，并以钠为例加以说明。

Mach pursued the metaphor that ‘things’ were just symbols for sensations all the way down to atoms, giving as an example sodium.

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物理学家看到一个物体被弯曲、拉伸、熔化并汽化，便把它切割成更小永久性部分；化学家则把它分解为元素。然而，即便是元素也并非一成不变。以钠为例。当受热时，这种白色、银色块状物会变成液体；若继续加热并隔绝空气，它便转化为紫色蒸汽；若再进一步加热，则会发出黄光。之所以仍保留“钠”这一名称，既是因为这些转变具有连续性，也是出于一种必要的节约本能。通过冷凝蒸汽，可以使白色金属重新显现。事实上，即使金属投入水中并转化为氢氧化钠，那些消失性质通过巧妙处理仍可重新显现；正如一个运动物体经过柱子后方、暂时消失在视线中，过一会儿又会重新出现。毫无疑问，只要某一组性质有可能出现，随时准备好相应名称和概念是非常方便。但这个名称和概念，不过是这些现象一个简明而节约的符号罢了。对于那些无法借此唤起一大群井然有序感官印象人来说，它不过是一个空洞词语。对于化学元素被进一步分解成分子和原子，情况也是如此。[37]

The physicist who sees a body flexed, stretched, melted, and vaporised, cuts up this body into smaller permanent parts; the chemist splits it up into elements. Yet even an element is not unalterable. Take sodium. When warmed, the white, silvery mass becomes a liquid, which, when the heat is increased and the air shut out, is transformed into a violet vapor, and on the heat being still more increased glows with a yellow light. If the name sodium is still retained, it is because of the continuous character of the transitions and from a necessary instinct of economy. By condensing the vapor, the white metal may be made to reappear. Indeed, even after the metal is thrown into water and has passed into sodium hydroxide, the vanished properties may by skilful treatment still be made to appear; just as a moving body which has passed behind a column and is lost to view for a moment may make its appearance after a time. It is unquestionably very convenient always to have ready the name and thought for a group of properties wherever that group by any possibility can appear. But more than a compendious economical symbol for these phenomena, that name and thought is not. It would be a mere empty word for one in whom it did not awaken a large group of well-ordered sense-impressions. And the same is true of the molecules and atoms into which the chemical element is still further analysed.[37]

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因此，在马赫看来，钠原子只是一组感觉印象的符号。归根结底，既不存在钠原子，也不存在任何原子；它们只是我们用来赋予感觉印象以一致性的观念。

So, for Mach the sodium atom was just a symbol for a set of sense impressions. Ultimately, there were no sodium atoms, no atoms in general; these were just ideas we used to impose consistency on our impressions.

对列宁而言，唯物主义核心在于它主张物质世界独立于我们感知。这一假设是所有务实的人所共有，只有唯心主义哲学家对此提出质疑。列宁指出，马赫颠覆了这一观点，把人类经验置于一切之上。尽管马赫拥有物理学背景并为科学哲学做出了贡献，但列宁认为，他哲学不过是贝克莱主教思想的回声。[38]作为一名社会主义政治家，列宁深刻认识到哲学与政治之间密切联系。贝克莱曾明确表示，他唯心主义旨在对抗当时无神论者。《百科全书》编纂者、法国大革命的哲学家狄德罗曾严厉批评贝克莱，视其为教会和王权捍卫者。在列宁所处时代，中欧和东欧君主制正面临革命运动威胁，这些运动最终在 1917 年至 1918 年间推翻了这些政权。在列宁看来，马赫对贝克莱思想复兴，是又一次旨在强化教会和王权、以对抗无神论社会主义运动哲学干预。值得注意的是，马赫作为维也纳大学哲学教授职位，必须得到奥匈帝国皇帝弗朗茨·约瑟夫亲自批准。

For Lenin, the crux of materialism lay in its claim that the material world is independent of our perception. This assumption is one shared by all pragmatic individuals, with only idealist philosophers questioning it. Lenin argued that Mach subverted this idea, elevating human experience above all else. Despite Mach’s background in physics and his contributions to the philosophy of science, Lenin contended that his philosophy merely echoed that of Bishop Berkeley.[38] As a socialist politician, Lenin was keenly aware of the deep connection between philosophy and politics. Berkeley had made it clear that his idealism aimed to counter the atheists of his time. Diderot, the Encyclopedist and philosopher of the French Revolution, had sharply criticized Berkeley, viewing him as a defender of the Church and Crown. In Lenin’s era, the monarchies of Central and Eastern Europe were under threat from revolutionary movements that would, between 1917 and 1918, ultimately topple them. To Lenin, the revival of Berkeleyian ideas by Mach was another philosophical intervention into politics designed to reinforce Church and Crown against the atheistic socialist movement. It is perhaps worth noting that Mach’s position, professor of philosophy in Vienna, was one that had to be personally approved by the Austro-Hungarian Emperor Franz Joseph.

列宁指出，马赫唯心主义并不彻底。他援引马赫话说，随着科学发展，原则上应该可以通过检查大脑，来判断其正在经历什么样体验。这如今确实已

Mach, Lenin said, was only halfhearted in his idealism. He cites Mach as saying that with the advance of science, it should in principle be possible, by examin-

经可以实现，Shen 等人[39]表明，通过结合功能性磁共振成像和对抗性神经网络，端到端模型可以学习大脑活动与感知之间直接映射关系。当受试者注视一幅图片时，神经网络以功能性磁共振成像数据为输入，便能在屏幕上生成对该人所见景象尚可再现。具体示例见图 6.5。但列宁质疑：这如何能与马赫把感觉视为第一性相一致？因为这实际上承认了感觉是组织化物质——即大脑——结果。

ing a brain, to say what experiences it was undergoing. This is indeed now possible, Shen et al.[39] show that using a combination of functional magnetic resonance and adversarial neural nets the end-to-end model can learn a direct mapping between brain activity and perception. When a subject is looking at a picture, the neural net, using as inputs fMR data, can generate on a screen a passable representation of what the person is seeing. For examples see Figure 6.5. But Lenin asks how this can be consistent with Mach making sensations primary, since it is effectively an admission that sensations are a result of organized matter –the brain.

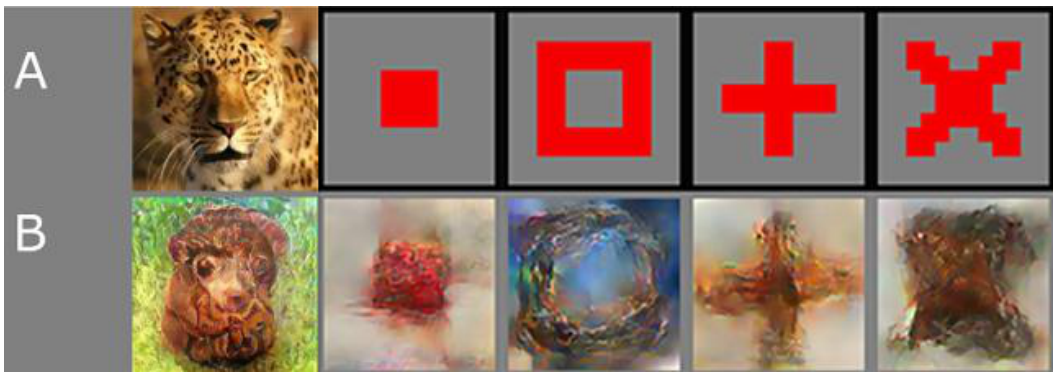


图 6.5 利用脑扫描，可以重建一个人正在看到的景象。A 行：呈现给受试者的图像；B 行：利用关于其脑活动的实时信息，由计算机重建的感知内容。取自 Shen 等人的研究。

Figure 6.5 It is possible, using brain scanning to reconstruct what a person is seeing. Row A: images presented to a subject; row B: computer reconstruction of what they are perceiving using real-time information about the activity of their brain. From the work of Shen et al.

A：呈现图像；B：重建图像。

这意味着，在我们之外，独立于我们和我们的意识，存在着物质的运动，例如具有一定波长和速度的以太波；它们作用于视网膜，使人产生某种颜色的感觉。自然科学正是这样看待这个问题的。它用存在于人类视网膜之外、人体之外并独立于人的不同波长的光波，解释各种颜色的感觉。这就是唯物主义：物质作用于我们的感觉器官，产生感觉。感觉依赖于大脑、神经、视网膜等，即依赖于以一定方式组织起来的物质。物质的存在并不依赖于感觉。[40]

This means that outside us, independently of us and of our minds, there exists a movement of matter, let us say of ether waves of a definite length and of a definite velocity, which, acting upon the retina, produce in man the sensation of a particular colour. This is precisely how natural science regards it. It explains the sensations of various colours by the various lengths of light-waves existing outside the human retina, outside man and independently of him. This is materialism: matter acting upon our sense-organs produces sensation. Sensation depends on the brain, nerves, retina etc, ie on matter organised in a definite way. The existence of matter does not depend on sensation.[40]

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量子理论中的唯心主义与唯物主义

Idealism and materialism in the quantum theory

列宁写作时，物质的原子理论正处于革命性变革之中。同时，爱因斯坦提出光由量子化的光子组成，给经典物理学带来了矛盾：这似乎与有大量经验证据支持、已经确立的光的波动理论相冲突。随后，卢瑟福[41]通过薄金箔对 β 辐射的散射表明，此前被认为是实心的原子，实际上大部分是空隙，只有一个很小的原子核。同年，尼科尔森[42]提出，原子的大部分体积由电子轨道占据，而电子允许的角动量是普朗克常数 h 的倍数。

〔校注：底本确实写 beta radiation，照译 β 辐射；未静默改成 α 粒子。〕

Lenin was writing during a revolutionary period in the atomic theory of matter. Meanwhile, Einstein had introduced a contradiction to classical physics by proposing that light consists of quantized photons, seemingly at odds with the well-established wave theory of light, which was supported by considerable empirical evidence. Then Rutherford[41] showed, by the scattering of beta radiation from thin gold foil that atoms, which had previously been thought of as solid, were in fact mostly void with only a tiny nucleus. In the same year Nicholson,[42] proposed that the bulk of the volume of atoms was

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occupied by electron orbits, with the allowable angular momenta of the electrons being multiples of Plank’s constant *h*.

因此，如果如索末菲所提出的那样，常数 h 在原子层面具有意义，那么当电子离开或返回时，原子的角动量或许只能以离散的量增加或减少。不难看出，这种观点比通常的解释更容易理解；后者被认为意味着能量本身具有原子式的构成。[43]

If, therefore, the constant *h* has, as Sommerfeld has suggested, an atomic significance, it may mean that the angular momentum of an atom can only rise or fall by discrete amounts when electrons leave or return. It is readily seen that this view presents less difficulty to the mind than the more usual interpretation, which is believed to involve an atomic constitution of energy itsef.[43]

p. 140–141

到第二年，玻尔[44]已把这一观点整合进原子理论中，该理论能够解释所观测到元素发射光谱。在他模型中，原子类似于一个微型太阳系，电子围绕带正电原子核运行。这是一种对原子结构半量子化处理。虽然轨道是按经典理论分析，但它们受普朗克常数限制，该常数把电子可能角动量进行了量子化。然而，玻尔对爱因斯坦关于光具有粒子性观点并不满意，认为这与已知光干涉能力相矛盾。他与其他人一起，试图构建一种辐射理论[45]，把他原子模型与经典电磁学相结合，从而消除对光子需求。尽管他 1924 年提出理论细节很快就被实验数据所推翻，但他在这篇论文中哲学论述，间接地为后来马赫学派物理学哲学观点复兴铺平了道路。

By the following year, Bohr[44] had integrated this into a theory of the atom, which could explain the observed emission spectra of the elements. In his model, the atom resembled a miniature solar system with electrons orbiting a positively charged nucleus. It was a semi-quantum approach to atomic structure. While orbits were analysed classically, they were restricted by Planck’s constant, which quantized the possible angular momenta for the electrons. Bohr, however, was dissatisfied with Einstein’s idea that light was particulate, finding it contradictory to the known interference capabilities of light. Along with others, he attempted to construct a theory[45] of radiation that would integrate his model of the atom with classical electrodynamics, obviating the need for the photon. While the details of his 1924 theory were soon contradicted by experimental data, his philosophical remarks in the paper opened up what later, and indirectly, became a recurrence of essentially Machian philosophical views of physics.

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尽管对应原理使得通过估算跃迁概率来推断原子在给定定态中停留平均时间成为可能，但在确定与跃迁相关辐射发射发生时间间隔这一问题上，却存在巨大困难。事实上，这一难题连同量子理论中其他著名悖论，加剧了各方表达疑虑：即关于物质与辐射之间相互作用详细解释，是否能够完全基于迄今用于解释自然现象那种时空因果描述来给出。[46]

Although the correspondence principle makes it possible through the estimation of probabilities of transition to draw conclusions about the mean time which an atom remains in a given stationary state, great difficulties have been involved in the problem of the time-interval in which emission of radiation connected with the transition lakes place. In fact, together with other well-known paradoxes of the quantum theory, the latter difficulty has strengthened the doubt, expressed from various sides whether the detailed interpretation of the interaction between matter and radiation can be given at all in terms of a causal description in space and time of the kind hitherto used for the interpretation of natural phenomena[46]

玻尔因果描述暗示了能量和动量守恒。在他 1924 年论文中，他试图通过提出能量和动量守恒在统计平均意义上成立，来弥合经典力学与量子力学之间鸿沟。他推测，虽然动量在许多电子能级跃迁中会守恒，但在每个单独电子与假设虚拟电磁场相互作用中，动量可能并不守恒。然而，这一观点很快就被推翻了——电子散射光子实验证实，在每次相互作用中动量都是守恒，这导致爱因斯坦光子假说被广泛接受。

Bohr’s causal description implied that energy and momentum conservation were maintained. In his 1924 paper, he tried to bridge the gap between classical and quantum mechanics by proposing that energy and momentum conservation applied on average, statistically. He theorized that while momentum would be conserved across many electron energy transitions, it might not be in each separate electron interaction with a hypothesized virtual electromagnetic field. However, he was quickly disproven as experiments with photon scattering by electrons confirmed that momentum was conserved in each interaction, leading to the widespread acceptance of Einstein’s photon hypothesis.

不过，关于量子系统中因果关系哲学问题仍由玻尔不断提出。

However, the philosophical points about causality in quantum systems continued to be raised by Bohr.

尽管由此给量子理论构建带来了困难，但正如我们将看到，其本质似乎可以表述为所谓“量子公设”——该公设赋予任何原子过程一种本质上不连续性，或者更确切地说，一种与经典理论完全相悖“个体性”，并以普朗克“作用量子”为象征。

Notwithstanding the difficulties which hence are involved in the formulation of the quantum theory, it seems, as we shall see, that its essence may be expressed in the so-called quantum postulate, which attributes to any atomic process an essential discontinuity, or rather individuality, completely foreign to the classical theories and symbolised by Planck’s quantum of action.

这一公设意味着放弃对原子过程作因果性的时空协调描述。事实上，我们通常对物理现象的描述，完全建立在这样一种观念上：可以观察相关现象而不对其造成可察觉的扰动。

This postulate implies a renunciation as regards the causal space-time co-ordination of atomic processes. Indeed, our usual description of physical phenomena is based entirely on the idea that the phenomena concerned may be observed without disturbing them appreciably.

……

. . .

而量子公设意味着，任何对原子现象的观察，都涉及与观测装置之间不可忽略的相互作用。因此，既不能赋予现象、也不能赋予观测装置以通常物理意义上的独立实在性。毕竟，观测概念在某种意义上是任意的，因为它取决于哪些对象被纳入待观测的系统。当然，最终每一次观测都可以归结为我们的感觉知觉。[47]

Now the quantum postulate implies that any observation of atomic phenomena will involve an interaction with the agency of observation not to be neglected. Accordingly, an independent reality in the ordinary physical sense can neither be ascribed to the phenomena nor to the agencies of observation. After all, the concept of observation is in so far arbitrary as it depends upon which objects are included in the system to be observed. Ultimately every observation can of course be reduced to our sense perceptions.[47]

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虽然最后一句颇有马赫主义认识论意味，但玻尔通过此前提及“观测装置”这一表述，在某种程度上为其论断规避了主观主义指责。例如，光子观测装置可以是一块照相底片。鉴于后来被称为“波函数坍缩”现象，碘化银晶体与视紫红质分子在视网膜中作用同样有效。他接着说道：

While the last sentence is redolent of the Machist theory of knowledge, Bohr hedged his statement against subjectivism somewhat with the previous reference to ‘agencies of observation’. An agency of observation of photons could, for instance, be a photographic plate. In the light of what would latter be called the collapse of the wave function, a crystal of silver iodide would be as effective as a molecule of visual purple in the retina. He goes on:

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就光而言，其在时空中传播可由电磁理论充分阐明。特别是真空中发生干涉现象以及物质介质光学性质，完全受波理论叠加原理支配。然而，在辐射与物质相互作用过程中能量和动量守恒——如光电效应和康普顿效应所示——恰恰通过爱因斯坦提出光量子概念得到了恰当阐释。

As regards light, its propagation in space and time is adequately expressed by the electromagnetic theory. Especially the interference phenomena in vacuo and the optical properties of material media are completely governed by the wave theory superposition principle. Nevertheless, the conservation of energy and momentum during the interaction between radiation and matter, as evident in the photoelectric and Compton effect, finds its adequate expression just in the light quantum idea put forward by Einstein.

众所周知，这一表面矛盾一方面引起对叠加原理有效性的怀疑，另一方面引起对守恒定律有效性的怀疑；这些怀疑已被直接实验明确排除。这种情况似乎清楚地表明，不可能对光现象作因果性的时空描述。一方面，当我们试图根据量子公设追踪光的时空传播规律时，只能进行统计考量；另一方面，要使以作用量子为特征的单个光过程满足因果性要求，就必须放弃时空描述。[48]

As is well known, the doubts regarding the validity of the superposition principle on one hand and of the conservation laws on the other, which were suggested by this apparent contradiction, have been definitely disproved through direct experiments. This situation would seem clearly to indicate the impossibility of a causal space-time description of the light phenomena. On one hand, in attempting to trace the laws of the time-spatial propagation of light according to the quantum postulate, we are confined to statistical considerations. On the other hand, the fulfilment of the claim of causality for the individual light processes, characterised by the quantum of action, entails a renunciation as regards the space-time description.[48]

玻尔因其表述晦涩而受到批评，因此让我们更仔细地看看他似乎想要表达什么。[49]这些措辞中蕴含着许多隐含含义。

Bohr was criticized for the obscurity of his statements, so let’s look a bit more closely at what he seems to have meant.[49] There are a lot of implied meanings in the language.

首先，他所说的叠加原理指什么？在当代关于量子计算等问题的讨论中，人们谈的是基矢态的叠加；但在 20 世纪 20 年代中期，物理学用这个术语指产生干涉的波的叠加。因此，他所说的叠加原理就是光学干涉。他提到的疑虑，即如何把干涉现象与爱因斯坦处理光子时涉及的守恒定律结合起来，正是他本人在玻尔、克拉默斯和斯莱特合著论文中表达的疑虑。[50]

First, what does he mean by the superposition principle? In contemporary discussions about, for example, quantum computing, one speaks of superposition of base states, but in the mid-1920s, the term was used in physics to refer to the superposition of waves producing interference, so by the superposition principle, he means optical interference. The doubts to which he refers, about combining interference phenomena with the conservation laws involved in Einstein’s treatment of the photon, are therefore his own doubts expressed in Bohr, Kramers and Slater.[50]

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当他谈到试图追踪光的时空传播时，指的是求解麦克斯韦方程组，从而得到光波的振幅。但这个解只给出到达某处——例如照相底片上某一点——的光的振幅。电磁场振幅的平方给出到达该点的光子通量，但单个光子随机到达，服从泊松统计。因此，他说“我们只能进行统计考量”。[51]

When he talks about attempting to trace the time-space propagation of light, he is talking about the solution to Maxwell’s equations, which give the amplitude of the light wave. But this solution only gives an amplitude for the light arriving at, for example, a given point on a photographic plate. The square of this electromagnetic field amplitude gives the flux of photons arriving at that point, but the individual photons arrive randomly with Poisson statistics. Thus, his statement that ‘we are confined to statistical considerations’[51].

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当他接着将这一点与“满足因果性要求”相对照时，指的仍是守恒定律，正如他 1924 年的论文中所说。经典物理学的整个因果结构都围绕能量和动量守恒建立。行星及其卫星遵循的时空路径，是使能量和动量守恒的路径。但他说，在量子力学中，如果要讨论能量守恒，就必须从麦克斯韦的光模型转向爱因斯坦的模型：我们所能说的，只是一个原子在时刻 t_0 发射一个光子，另一个原子在时刻 t_1 吸收这个光子。在这两个事件之间，无法为光子指定一条确定的路径。

When he then contrasts this with ‘the fulfilment of the claim of causality’, he is referring, as he did in his 1924 paper, to the conservation laws. The whole causal structure of classical physics was structured around the conservation of energy and momentum. The space-time paths followed by the planets and their satellites were those which conserved energy and momentum. But in quantum mechanics, he said that if we want to talk about energy conservation, we are forced to move from Maxwell’s model of light to Einstein’s and all you have is an emission of a photon from one atom at time t0 and an absorption of the photon by another atom at time t1. In between those to events we can assign no definite path to the photon.

如果你搭建一个使用衍射光栅光谱仪，使其暴露在极低单色光通量下——平均每次只有一个光子通过光栅——在长时间曝光后，你仍然会得到一条清晰光谱线。[52] 为了产生这种效应，每个光子必须同时穿过多条光栅条纹，因此它们在发射与吸收之间位置是未定义。

If you set up a spectrometer using a diffraction grating, expose it to a very low flux of monochromatic light such that on average only one photon at a time is passing through the grating, you still get a clear spectral line on long exposure.[52] The photons must each have passed through multiple grating lines at once to produce this effect, so their positions between emission and absorption are undefined.

此外，通过关系式（2）[53] 表达经典概念中限制，与经典力学有限有效性密切相关，在物质波理论中，经典力学对应于几何光学，其中波传播通过“光线”来描述。只有在这个极限下，才能基于时空图景对能量和动量进行明确定义。[54]

The limitation in the classical concepts expressed through relation (2)[53] is, besides, closely connected with the limited validity of classical mechanics, which in the wave theory of matter corresponds to the geometrical optics, in which the propagation of waves is depicted through ‘rays’. Only in this limit can energy and momentum be unambiguously defined on the basis of space-time pictures.[54]

玻尔关注是，如果采用波模型，在多大程度上可以对光子或其他粒子进行定位。

Bohr is concerned with the extent to which you can localize a photon or other particle if you use the wave model.

如果我们从波论角度观察粒子在空间中运动，那么只有把它视为“波包”时，才能在其轨迹上对其进行定位。这个波包可以被视为不同频率波叠加。我

If we view the passage of a particle through space from the standpoint of the wave account, then it can only be localized on its trajectory to the extent that we view it as a ‘wave packet’. This wave packet can be viewed as

们想要实现定位越精确，就越需要考虑更多不同频率分量。

the sum of waves of different frequencies. The sharper the localization we want to achieve, the more different frequency components we need to take into account.

上文关系式（2）指是他方程：

His relation (2) above refers to his equation

$$\Delta t \Delta E = \Delta x \Delta I_x = \Delta y \Delta I_y = \Delta z \Delta I_z = h$$

其中 Δt 表示波包通过某一点的时间间隔。这给出了光子到达时刻的不确定性。

where Δt refers to the time interval for a wave packet to pass a point. This gives the uncertainty about when a photon will arrive.

关系式 $\Delta t \Delta E = h$ 来自爱因斯坦的结论：光子的能量与频率通过普朗克常数 h 成反比，使得 $h = E\tau$，其中 τ 是光波的振动周期。

〔校注：底本称能量与 frequency 成反比；与同句 h=Eτ 相应的反比变量应为周期 τ，频率与能量为正比关系。正文保留底本表述。〕

The relation $\Delta t \Delta E = h$ comes from Einstein's result that the energy and frequency of photons are inversely related by Plank's constant *h* such that $h = E\tau$ where τ is the period of vibration of the light wave.

如果我们希望更准确地确定波包何时到达，就必须选择一个傅里叶和包含更多分量频率波包。由于每种频率都对应不同能量，对到达时间确定性越高，对光子能量确定性就越低。这就是玻尔在论述时空图景与能量守恒时所探讨互补关系。

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If we want to have more certainty about precisely when a wave packet will arrive, we have to have a wave packet whose Fourier sum has more component frequencies. Because each frequency has a different associated energy, greater certainty about arrival time means less certainty about photon energy. This is the complementary relationship that Bohr is addressing with respect to the space time picture and the conservation of energy.

$\Delta x, \Delta y$ 分别是波包在 x、y 方向上的长度，其他方向依此类推。这给出了波包到达位置的空间不确定性。

$\Delta x, \Delta y$ is the length of the wave packed in the x, y directions, etc. This gives the spatial uncertainty about where it will arrive.

ΔI_x 是 x 方向的动量范围，其他方向依此类推。因此，$\Delta x \Delta I_x = h$ 表达的就是后来所谓的海森堡不确定性原理。[55]玻尔从爱因斯坦的光子动量公式 $I\lambda = h$ 推导出这一关系；这里的符号 I 表示动量。

ΔI_x is the range of momenta in the x direction, etc. So $\Delta x \Delta I_x = h$ expresses what was later known as Heisenberg's uncertainty principle.[55] Bohr has derived this from Einstein's formula for the momentum of photons $I\lambda = h$, where *I* in this notation means momentum.

这就是玻尔所称互补原理。人们可以把一个过程视为具有确定时空演化，也可以把它视为受守恒定律支配，但不能同时兼顾两者。

This is what Bohr termed the complementary principle. You could either view a process as having a definite space time evolution or as being one governed by conservation laws, but not both at the same time.

玻尔和海森堡后来被视为“哥本哈根诠释”代表人物，尽管有观点认为两人观点存在实质性差异，且关于哥本哈根诠释流行说法在某种程度上是一种神话。[56]如前所述，对玻尔而言，因果诠释是指基于守恒定律诠释；而对海森堡而言，因果诠释则是指波函数确定性演化。波函数随时间以确定性方式演化。该函数为某个态空间正交基分配振幅，使得这些振幅平方即为在相应基矢态下观测到该系统概率。这在形式上与光学情况相同，即场振幅平方即为光子通量。对海森堡而言，对粒子实际探测，或者更普遍地说，对系统基矢态探测，构成了由波函数演化所支配量子世界与观察结果经典世界之间一个“切口”（Schnitt）。

Bohr and Heisenberg were later to be associated with what became known as the Copenhagen interpretation of quantum theory, though it is arguable that they differed substantially in their views and the popular account of the Copenhagen theory is somewhat of a myth.[56] As we have said, for Bohr a *causal* interpretation meant one in terms of the conservation laws. For Heisenberg a *causal* interpretation was in terms of the deterministic operation of wave functions. The wave function evolved in a deterministic fashion with respect to time. The function assigns amplitudes to the orthogonal bases of some state space such that the square of these amplitudes gives the probability of observing the system in the corresponding basis state. This is formally the same as the optical case, where the square of the field amplitude gives the photon flux. For Heisenberg, the actual detection of a particle, or more generally, the detection of a basis state of the system, constituted a cut or *schnit* between the quantum world governed by the evolution of the wave function and the classical world in which the result was observed.

直到最近，这似乎还是一个相当深奥话题，但随着技术进步，它现在正被应用于量子密码学这一实际领域。例如，廖等人[57]成功演示了利用量子密码学向轨道卫星传输数据。

Until recently, this would have seemed a rather arcane topic, but with the advance of technology, it is now being harnessed for practical purposes in quantum cryptography. For instance, Liao et al.[57] were able to demonstrate that they were able to use quantum cryptography to transmit data to an orbiting satellite.

如今，量子密码学已能为普通数字密码学分发超长密钥，其长度之大，使得有效暴力破解成为不可能。我们将解释该技术工作原理，并利用其运作过程来阐释量子力学中不同唯心论和唯物论解释。

It is now possible to use quantum cryptography to distribute long keys for ordinary digital cryptography, so long that effective brute force decryption is ruled out. We will give an explanation of how this technology works and then use its operation to illustrate different idealist and materialist accounts of QM.

请看图 6.6 所示实验装置。它展示了如何利用初始偏振片制备一束垂直偏振光。若在此光束中放置一个与第一偏振片成 45° 角第二偏振片，则最初偏振光束中光子只有一半能通过第二偏振片。

Consider the setup in Figure 6.6. It shows how an initial polarizer can be used to prepare a vertically polarized light beam. If you place a second polarizer at 45° in this beam, then only half of the photons in the first polarized beam will get through the second one.

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图 6.6 展示是简单偏振片，但实际上还存在更复杂偏振装置，它们能把初始偏振光子流分为两路：一路为垂直偏振光，另一路为水平偏振光。这为信息传输提供了一种潜在途径。假设我们把垂直偏振光子记为 1，水平偏振光子记为 0，则可将其用作传输编码。

Figure 6.6 shows simple polarizing sheets, but more sophisticated polarizing devices exist which will sort an initially polarized stream of photons into two, one of vertically polarized and one of horizontally polarized light. This provides a potential way to transmit information. Suppose we treat vertically polarized photons as 1 and horizontally polarized ones as 0, then this can be used as a transmission code.

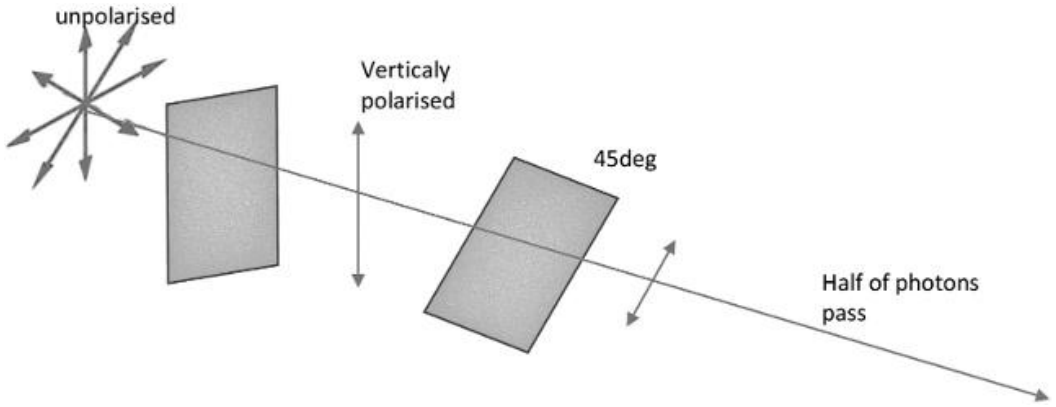


图 6.6 光通过偏振片后，可以形成垂直偏振光束。若再通过相对第一片旋转 45° 的第二片偏振片，就得到偏振方向为 45°、光子数减半的光束。作者绘图。

Figure 6.6 If light is passed through a polarizer a vertically polarized beam can be created. If this is then passed through a second polariser at 45° to the first, then a beam with 45° polarization and half the number of photons results. Figure by authors.
unpolarised：非偏振光；Vertically polarised：垂直偏振；45deg：45°；Half of photons pass：一半光子通过。

我们还可以进一步增加复杂性，允许发送方使用两片相互成 45° 角偏振片中任意一片来编码信息。

We can complicate this further by allowing the sender to encode their message with either of two polarizers at 45° to one another.

假设爱丽丝想向鲍勃发送比特序列 1001011001，并且她随机选择使用垂直偏振片还是对角线偏振片来发送各个比特。在下文中，我们把垂直偏振片表示为 +，对角线偏振片表示为 x。

Suppose Alice wants to send to Bob the bit sequence 1001011001 and she randomly selects whether to send individual bits with vertical or diagonal polarizers. In what follows, we indicate a vertical polarizer as + and a diagonal one as x.

序列	含义
1001011001	爱丽丝尝试发送的比特流
xx+x++x++x	她使用的编码偏振片

Sequence	Meaning
1001011001	Bit stream Alice tries to send
xx+x++x++x	Encoding polarizer she used

鲍勃既不知道初始比特流，也不知道偏振片设置顺序。他可以猜测使用了哪种偏振片设置。如果对于某个给定比特，他做出了正确选择，那么通过使用分束偏振片，他就能正确读取爱丽丝在该序列中发送 1 或 0。如果他做出了错误选择，那么他将以 50% 概率读取 0 或 1，无论爱丽丝实际发送是什么。例如，我们可能会遇到以下情况：

Bob does not know either the initial bit-stream or the sequence of polarizer settings. He can guess which polarizer setting was used. If, for a given bit, he makes the correct choice, then by using a beam splitting polarizer he can correctly read the 1 or the 0 that Alice sent for that bit in the sequence. If he makes the wrong choice, then he will read either a 0 or a 1 with a 50 per cent probability of whatever Alice sent. For example, we might have:

序列	含义
1001011001	爱丽丝尝试发送的比特流
xx+x++x++x	她使用的编码偏振片
xx++x+++xx	鲍勃的偏振片设置
101**1*0*1	鲍勃收到的结果；* 表示随机的 1 或 0

Sequence	Meaning
1001011001	Bit stream Alice tries to send
xx+x++x++x	Encoding polarizer she used
xx++x+++xx	Bob’s polarizer settings
101**1*0*1	What Bob recieves, * indicates random 1 or 0

校注：底本接收串第三位为 1，而该位发送比特为 0、双方偏振设置均为 +，与正文匹配时正确读取的规则不符。保留原串，未静默改成 100**1*0*1。

量子传输完成后，鲍勃通过普通数字信道告知爱丽丝他使用偏振器设置，爱丽丝则回复一个比特流，告知他原始消息中哪些比特是通过正确偏振器接收到。鲍勃会丢弃接收到消息中那些使用错误偏振器比特。该过程结束后，鲍勃和爱丽丝共享一个已知比特串，该比特串可作为加密密钥使用。

After the quantum transmission is finished, Bob uses an ordinary digital channel to tell Alice which polarizes he used, she replies with a bit stream telling him which bits in the original message were received with the correct polarizer. Bob discards the bits in his received message that used the wrong polarizer. At the end of the procedure, Bob and Alice share a known bit-string that can be used as a cryptography key.

这里关键点在于，每次错误地选择偏振片都会产生一个与发送信号无关随机读数。

The important step here is that each incorrect selection of a polarizer gives a random reading unrelated to what was sent.

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以偏振光为例的好处，是可以借助一个只需经典欧几里得几何便能理解的简单情境，引入冯 • 诺伊曼用于奠定量子理论数学基础的线性代数。[58]请看图 6.7。图中有两个参考系：H、V 轴，即水平轴和垂直轴；另一个参考系的轴标为 L、R，意为左与右。L、R 轴相对 H、V 轴旋转了 45°。我们假定 L、R、H、V 都是等长向量，并且不失一般性地假定长度为 1。

The nice thing about using polarized light as an example is that it introduces the linear algebra that provided the mathematical foundation of quantum theory from Neumann[58] in a simple context that can be understood with no more than classical Euclidean geometry. Consider Figure 6.7. The images show two frames of reference, the H, V or Horizontal and Vertical axes, and another frame of reference, whose axes we have labelled L, R, meaning Left and Right. The L, R axes are rotated 45° relative to the H, V axes. We assume that L, R, H, V are all vectors of the same length. Without loss of generality, we assume length 1.

显然，对于像 [H,V] 或 [L,R] 这样成直角坐标轴对，我们可以使用任一坐标系来表示单位圆内任何位置。这包括坐标轴本身端点。

Clearly, with pairs of axes at right angles like [H,V] or [L,R] we can express any position within the unit circle using either coordinate system. That includes the endpoints of the axes themselves.

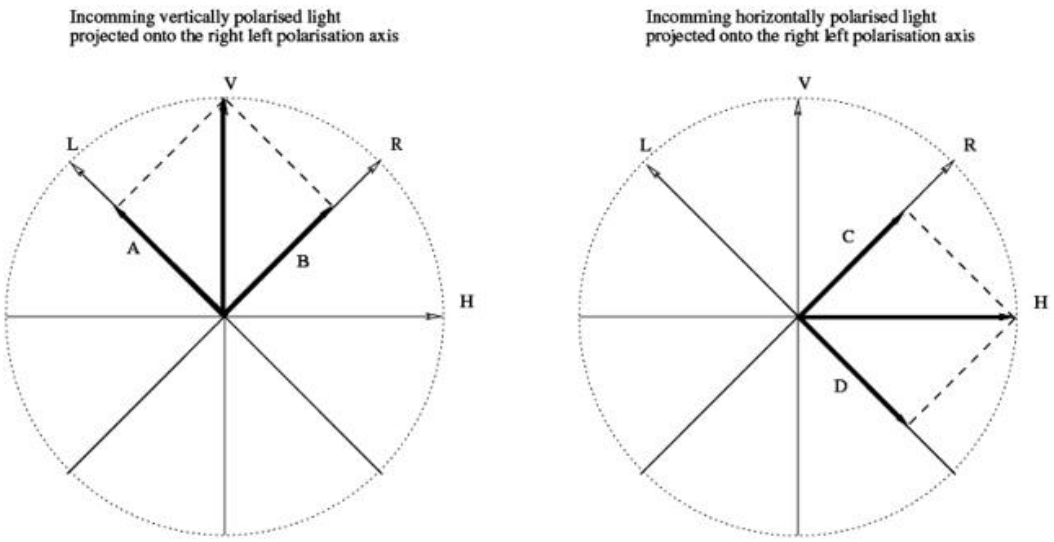


图 6.7 左图表示垂直偏振光的振幅如何投影到一个相对原偏振轴旋转 45° 的偏振器的左、右两轴。表示初始振幅的垂直向量 V，在新的参考系中投影为两个等长的分量向量 A、B。右图为水平偏振光的相应情形。作者绘图。

Figure 6.7 Left picture shows how the amplitude of vertically polarized light gets projected onto the Left and Right axes of a polarizer at 45° to the original axis of polarization. The vertical vector V representing the initial amplitude projects down two equal length component vectors A,B on the new frame of reference. The right picture shows the corresponding situation for horizontally polarized light. Figure by authors.

左图标题：入射垂直偏振光在左右偏振轴上的投影；右图标题：入射水平偏振光在左右偏振轴上的投影。V/H：垂直/水平；L/R：左/右；A、B、C、D：图示分量。

因此，图中标记为 V 的点，在水平—垂直坐标系中表示为 $V = [0, 1]_{HV}$，在左—右坐标系中的坐标为 $[A, B]_{LR}$。同样，$H = [1, 0]_{HV} = [D, C]_{LR}$。利用勾股定理，可以容易地求出 A、B、C、D：

So, the point labelled V in the diagram has coordinates $V = [0, 1]_{HV}$ in Horizontal Vertical system and coordinates $[A, B]_{LR}$ in the Left, Right system. Similarly, $H = [1, 0]_{HV} = [D, C]_{LR}$. By Pythagoras theorem you can readily solve for A, B, C, D as

$$V = [0, 1]_{HV} = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]_{LR}$$

且

and

$$H = [1, 0]_{HV} = \left[\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]_{LR}$$

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到目前为止，这仅仅是一个在笛卡尔几何中改变参考系例子，所采用论证方式对任何 17 世纪科学家来说都很熟悉。我们可以分两步把它扩展为关于量子测量讨论。

Up till now this has just been an example of changing the frame of reference in Cartesian geometry, using a style of argument that would have been familiar to any seventeenth-century scientist. We can evolve it to one about quantum measurement in two steps.

1. 把这些参考系看作实际的物理偏振器，它们能把初始光束分成电场相互垂直的两束分量光。光通过第一个分束器后，垂直偏振光束 V 以 45° 进入第二个分束器，分为 L、R 两束。若 V 的振幅为 v，则 L、R 两束的振幅均为 $v/\sqrt{2}$。光束亮度正比于电场振幅的平方，因此 L、R 两束的亮度各为 V 的一半。这些结论都来自前述基本几何论证。

1. Consider the frames of reference to be actual physical polariers capable of splitting an initial beam into two component beams whose electric fields are at right angles to one another. The light passes through a first beam splitter and the vertically polarized beam (V) enters a second one at 45° which splits it into L and R component beams. If the amplitude of the beam V is *v* then the amplitude of the L and R beams will be $v/\sqrt{2}$. Because the brightness of a beam is proportional to the square of the field amplitude, then L and R beams will be half as bright as the V beam. All of this follows from the basic geometric arguments given earlier.

2. Now consider what is happening at the level of individual photons. If we are sending information on sin-

2. 现在考虑在单个光子层面上发生了什么。如果我们传输是具有两种可能（偏振）状态单粒子信息，那么我们所处理就不是比特，而是量子比特。

描述量子比特和其他量子态时，物理学家用一种专门记号——狄拉克记号——表达前述几何关系。不熟悉它的人可能觉得有些奇怪，但基本原理很简单。在初始参考系中，我们用标准笛卡尔记号把垂直轴向量写成 $[0,1]_{HV}$；他们则写成 $0|H\rangle + 1|V\rangle$。这依然只是把参考系中的坐标明确写出来的一种方式。$|\rangle$ 这种括号形式称为 ket，即右矢。

按量子力学的标准解释，一个量子比特以两个正交纯态之一发送，即 $1|V\rangle+0|H\rangle$ 或 $0|V\rangle+1|H\rangle$。记号 $|H\rangle$ 表示一个带标签的态，这里的 H 代表水平偏振。如果旋转 45°，并把这些纯态投影到由新偏振器形成的基矢态[59]上，把这些基矢态标为 L、R，就得到：

gle particles with two possible (polarization) states, then instead of bits we have qubits.

In describing qubits and other quantum states, the physicists use a specialized notation, the Dirac notation, for the geometry that we described earlier. This can look a bit strange if you are unfamiliar with it, but the basic principle is simple. Where we used standard Cartesian notation for the initial frame of reference, writing the vertical axis vector as $[0,1]_{HV}$ they would write it as $0|H\rangle + 1|V\rangle$, but this is still just an explicit way of writing down coordinates in a frame of reference. The bracketing construct $|\rangle$ is referred to as a *ket*.

In the standard account of QM what is happening when a qubit is sent in one of two orthogonal pure states $1|V\rangle+0|H\rangle$ or $0|V\rangle+1|H\rangle$. The notation $|H\rangle$ indicates a labelled state, *H* standing for horizontal polarization in this case. If we perform a 45° rotation and project these pure states onto the basis states[59] formed by the new polarizer which we will label L, R we get

$$0|H\rangle + 1|V\rangle \rightarrow \frac{1}{\sqrt{2}}|L\rangle + \frac{1}{\sqrt{2}}|R\rangle$$

这显然只是方程（6.1）符号转换，但在实际中这意味着什么？

当然，无论我们采纳哪种观点，实验证据都不会因此而改变。我们仍然观察到，沿 L 和 R 方向亮度仅为移除第二个偏振片并测量光束 V 亮度时所得到值一半。经典理论通过把电场在新 45° 坐标系中分解为等大且相互独立分量，对此给出了通俗易懂解释。

This is evidently just a notational translation of equation (6.1), but what does it mean in practice?

The experimental evidence does not of course change as a result of which view we take. We still observe that the brightness going down L and R is half of what we would have gotten if we removed the second polarizer and measured the brightness of the beam V. The classical picture gives an intelligible account of this in terms of the geometric decomposition of the electric field into equal independent components in the new 45° coordinate system.

玻尔的解释是，在表达式 $\frac{1}{\sqrt{2}}|L\rangle+\frac{1}{\sqrt{2}}|R\rangle$ 中，右矢前的振幅系数 $\frac{1}{\sqrt{2}}$ 表示在这条路径上发现光子的概率的平方根。这显然与经典解释一样符合实验。无论为 $\frac{1}{\sqrt{2}}$ 赋予何种本体论含义，$\left(\frac{1}{\sqrt{2}}\right)^2$ 仍意味着亮度是原来的 $\frac{1}{2}$。

The Bohr interpretation is that the $\frac{1}{\sqrt{2}}$ amplitude coefficients in front of the *kets* in the expression $\frac{1}{\sqrt{2}}|L\rangle + \frac{1}{\sqrt{2}}|R\rangle$ express the square root of the probability that the photon will be found going down this path. This clearly fits the experiment just as well as the classical account. Whatever ontological semantics you associate it with the $\frac{1}{\sqrt{2}}$, $\left(\frac{1}{\sqrt{2}}\right)^2$ is still $\frac{1}{2}$ as bright.

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这套计算无疑是正确。玻尔互补观点最初是通过波包模型中波包在傅里叶意义上位置不确定性来阐述，而在分束器等实验装置中，这一特性表现得尤为明显。在实验装置中两个探测器之一触发之前，光子可能位于任一通道中。由于两束光分支可以任意延长，光子空间位置也变得任意不确定。

The maths certainly works. Bohr’s complementary view, initially expressed in terms of how the wave packet view involved an ambiguous position to the packet in Fourier terms, becomes much more pronounced with a setup like a beam splitter. Until one of the two detectors in the setup fires, the photon could be in either channel. Since the arms of the two beams can be arbitrarily extended, the spatial position of the photon also becomes arbitrarily indeterminate.

玻尔互补方法实质上指出：数学推导是明确且经验上正确，但在光子被探测之前，无法对其位置进行有意义描述。我们只能基于把光视为波假设进行计算，以预测最终位置相对概率。光子抽象概念本质上与特定位置能量量子发射或吸收过程紧密相连。

Bohr’s complementary approach effectively says that the maths is well defined and empirically correct, but that you cannot meaningfully speak of the position of the photon before it is detected. You just have to use the calculations based on treating light as a wave to predict

它使我们能够把发生事情理解为一种能量守恒过程，但禁止我们同时讨论连接这两个事件时空过程。这些必须通过光波来处理，光波告诉我们不同位置预期亮度之间关系。

the relative probabilities of final positions. The photon abstraction is inherently tied to the process of emission or absorption of quanta of energy at particular places. It allows us to understand what happens as something that conserves energy, but it prohibits us from simultaneously talking about the space-time process that connects the two events. These have to be dealt with in terms of light waves, which tell us the relationship between expected brightness in different places.

海森堡方法则略有不同。波（或波函数）随时间沿所有可能路径演化，随后，当进行测量且其中一个探测器触发时，量子世界与经典世界之间便发生了一次“切割”。这有时被称为波函数坍缩。

Heisenberg’s approach was subtly different. The wave (or wave function), evolves through time along all possible paths and then, when a measurement is performed and one of the detectors fires, then there is a‘cut’ between the quantum world and the classical world. This is sometimes spoken of as there being a collapse of the wave.

玻尔的这一表述并没有内在的唯心主义性质。如果一个人是严格的唯物主义者，承认存在的一切就是原子以及原子借以交换力和能量的光子，[60]那么谈论光子在两次影响原子之间的位置，就成了没有意义的唯心主义。我们可能愿意想象光子在发射与吸收之间具有一个位置，但那只是想象，只是在头脑中构造一幅观念图景。既然光子和原子是实在的基本组成部分，那么，独立于原子相互作用的、观念中的光子位置，按定义就是不真实的。

There is nothing inherently idealistic about the Bohr statement of things. If one is a strict materialist and accepts that all that exists are the atoms and the photons[60] by which they exchange force and energy, then speaking of where the photons are between when they influence atoms is meaningless idealism. We may like to imagine the photon as having a position in between emission and absorption, but that is just imagination. It is just an attempt to form an ideal picture in our minds. Since the photons and the atoms are the fundamental components of reality, an ideal position of a photon independent of atomic interaction is by definition unreal.

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但海森堡的解释确实可能引向唯心主义的解释，如今常称为哥本哈根诠释。[61]按照这种解释，正是认识主体即观察者的参与，使波函数坍缩并出现一个确定的结果。海森堡[62]声称：“当‘客观实在’这样的概念用于原子物理学中的情形时，它们并没有不言自明的含义。”赋予观察者的心理体验以特权地位，这看起来很像回到马赫式实证主义；在他受训练的年代，这种思潮曾深刻影响德国物理学。

Heisenberg’s account does lead to potentially idealist accounts though, now often known as the Copenhagen interpretation,[61] in which it is the participation of the knowing subject, the observer, that causes the wave function to collapse and a definite result to be seen. Heisenberg[62] claims: ‘such concepts as “objective reality” have no immediately evident meaning, when they are applied to the situation which one finds in atomic physics’. This process of giving a privileged position to the psychological experience of the observer looks very like a reversion to the Machist positivism that had been so influential in German physics back when he had been trained.

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如果我们把视角从 20 世纪 30 年代物理实验室转向现代通信技术，观察者特殊角色便完全失去了可信度。当廖等人[63]架设激光器和望远镜，向无人卫星“墨子号”发送偏振光子时，观察者又在哪里呢？

If we move from the perspective of the 1930s physics lab to that of modern communications technology, the special role of the observer loses all credibility. When Liao et al.[63] set up their lasers and telescopes to send polarized photons to the unmanned satellite *Micius*, where was the observer?

“墨子”号上自动设备记录了偏振光子，机载处理器利用这些光子把加密密钥存储在随机存取存储器（RAM）中，随后用该密钥向地面发送无线电信息。那里肯定没有人。如果波包发生了坍缩，那么那是那里设备在无人协助情况下完成。难道我们要说计算机是主体吗？

The automatic equipment in *Micius* recorded the polarized photons and the on-board processor used them to store in RAM the encryption key that it used to send subsequent radio messages to the ground. There was certainly no person up there. If wave packets collapsed, then the equipment up there did it unaided. Are we to say computers are subjects?

抑或我们只应承认，哲学意义上的主体[64]与理解这类工程技术无关？

Or should we just recognize that the philosophical subject[64] is irrelevant to understanding this sort of engineering?

海森堡学派潜藏的主观主义引起了反对，最初的反对见相关研究，[65]后来则有玻姆[66]、[67]和埃弗雷特[68]、[69]提出的其他实在论解释。

The latent subjectivism of the Heisenberg school gave rise to opposition, initially from[65] and later from the alternative realist interpretations of Bohm[66],[67] and Everett.[68],[69]

这两种替代性实在论方法，分别优先考虑玻姆粒子诠释和埃弗雷特波诠释。

The two alternative realist approaches prioritize either the particle interpretation for Bohm or the wave interpretation for Everett.

玻姆认为，微观层面上看似存在非决定论与宏观层面上存在统计可预测性，不应阻止我们寻找更深层次决定论机制。

Bohm argued that the existence of apparent indeterminism at a microscopic level and statistical predictability at a macro level should not prevent us from looking for deeper deterministic mechanisms.

在给定统计规律的范围内，个体行为的无规律性，一般而言，与更广泛范围内起作用的更详细的个体规律并不矛盾。[70]

lawlessness of individual behaviour in the context of a given statistical law is, in general, consistent with more detailed individual laws operating in a broader context.[70]

他举例说明了一家寿险公司运作：该公司能够预测一名男子在特定年份死亡可能性。他们无法确切预测该男子是否会死亡，但可以针对其承保数百万客户群体做出准确预测。某人死亡仅在统计上可预测这一事实，并不意味着该个体死亡没有明确物质原因。也许是由于飞行控制机制故障导致客机空难，也许是肺癌。如果我们仅满足于统计死亡率，我们就不会去研究癌症成因、737MAX 飞行计算机可靠性等问题。玻姆借鉴了德布罗意早期研究[71]，后者最先证明物质粒子与光一样，也具有波特性。

He gave as an example the operation of a life insurance company that can predict the likelihood of a man dying in a particular year. They cannot predict with certainty if he will die, but they can make an accurate prediction over the ensemble of millions of customers they assure. The fact that the death of one man was only statistically predictable does not mean that there was not a definite material cause of this individual death. Perhaps it was an airliner crash due to a faulty flight control meachanism or perhaps lung cancer. Were we to just satisfy ourselves with statistical death rates we would not research into causes of cancer, the reliability of 737MAX flight computers, etc. Bohm took up the early work of De Broglie,[71] who was the first to show that material particles, like light, also show wave-like properties.

量子关系式“能量 =h× 频率”，使我们把一种周期现象与任何孤立的一份物质或能量联系起来。与这份物质绑定的观察者，会把它与一个由其内部能量、即“静止质量”确定的频率联系起来。对于一位观察者，若这份物质以速度 βc 匀速运动，他就会看到这个频率因洛伦兹—爱因斯坦时间变换而降低。我已经证明（巴黎科学院《报告》，9 月 10 日及 24 日），只要假定波以速度 c/β 传播，静止观察者就会始终看到这种内部周期现象与一个波同相；该波的频率由运动物体全部能量所对应的量子关系确定。这种速度大于 c 的波不能携带能量。[72]

THE quantum relation, energy = h × frequency, leads one to associate a periodical phenomenon with any isolated portion of matter or energy. An observer bound to the portion of matter will associate with it a frequency determined by its internal energy, namely, by its ‘mass at rest.’ An observer for whom a portion of matter is in steady motion with velocity βc, will see this frequency lower in consequence of the Lorentz-Einstein time transformation. I have been able to show (Comptes rendus, September 10 and 24, of the Paris Academy of Sciences) that the fixed observer will constantly see the internal periodical phenomenon in phase with a wave the frequency of which is determined by the quantum relation using the whole energy of the moving body provided it is assumed that the wave spreads with the velocity c/β. This wave, the velocity of which is greater than c, cannot carry energy.[72]

p. 149–150

这种物质波特性，继爱因斯坦光电效应和相对论之后，将要求人们重新思考动力学定律在微小运动粒子上应用。

This wave property of matter, following on from Einstein’s photo-electric effect and relativity theories, would entail a rethinking of the laws of dynamics when applied to small moving particles.

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“相位波”在确定任何运动物体运动方面具有极其重要意义，我已经能够证明，玻尔原子中轨道稳定性条件表明，该波与闭合路径长度相协调。

The ‘phase wave’ has a very great importance in determining the motion of any moving body, and I have been able to show that the stability conditions of the trajectories in Bohr’s atom express that the wave is tuned with the length of the closed path.

当发光原子穿过狭窄孔隙时，其路径不再是直线；即发生衍射。此时必须放弃惯性原理，并假设任何运动物体始终沿着其“相位波”光线运动；这样，当它通过足够小孔

The path of a luminous atom is no longer straight when this atom crosses a narrow opening; that is, diffraction. It is then necessary to give up the inertia principle, and we must sup-

隙时，其路径就会发生弯曲。动力学必须经历与光学相同演变过程——正如当波动理论取代纯几何光学时，光学所经历那样。基于波理论假设使我们能够解释干涉现象和衍射条纹。借助这些新思想，或许能够把扩散和色散现象与光离散性相协调，并解决量子理论提出几乎所有问题。[73]

pose that any moving body follows always the ray of its‘phase wave’; its path will then bend by passing through a.sufficiently small aperture. Dynamics must undergo the same evolution that optics has undergone when undulations took the place of purely geometrical optics. Hypotheses based upon those of the wave theory allowed us to explain interferences and diffraction fringes. By means of these new ideas, it will probably be possible to reconcile also diffusion. and dispersion with the discontinuity of light, and to solve almost all the problems brought up by quanta.[73]

在德布罗意—玻姆理论中，波方程被视为建立了一个势，其空间导数对粒子施加了一种引导力。由此产生轨迹可能以相当复杂方式弯曲。该理论数学结构非常经典，是哈密顿轨迹推广。它也是确定性。最终结果随机性源于初始粒子位置随机分布。在双缝实验中，粒子最终位置对其穿过缝隙确切位置极为敏感。引导波势会根据粒子起始点，把它们引导至截然不同最终位置。因此，在玻姆理论中，量子随机性与玻尔兹曼统计力学中原本就存在随机性并无二致。[74]这两种情况都假设系综中粒子初始位置是随机。

In the De Broglie/Bohm theory the wave equation is seen as establishing a potential, the spatial derivative of which imposes a guiding force on particles. The resultant trajectories may be curved in rather complicated ways. The mathematical structure of the theory is very classical, a generalization of Hamiltonian trajectories. It is also deterministic. Randomness of final results arises from the random spread of initial particle positions. In the two slit example, the final ending position of a particle is very sensitive to just where it passes through the aperture. The guiding wave potential steers particles to widely different final positions depending on their starting points. So, in Bohm’s account, quantum randomness is no different from the randomness that was already present in Boltzmann’s statistical mechanics.[74] In both cases it is assumed that the initial positions of particles in ensembles are random.

p. 150–151

SQM 中的客观概率[75]意味着，实验结果会呈现正确的频率。然而，反过来显然并不成立。测量结果的统计分布，并不意味着存在客观概率。例如，系统系综的初始状态具有适当的统计分布，再加上适当的演化规律，也能得到这些频率。我们无法确定地预测某个光子会在屏幕的什么位置落下，只是因为不知道单个系统的初始状态。PWT 采用的就是这种方法。[76]、[77]

Objective probability in SQM[75] implies that the correct frequencies of experimental outcomes are observed. However, the reverse is obviously not true. A statistical spread of measurement outcomes does not imply the existence of objective probability. For example, these frequencies also follow from the right statistical spread in the initial state across the ensemble of systems together with the right kind of laws of evolution. The fact that we cannot predict with certainty where a particular photon will hit the screen is only due to our ignorance of the initial state of the individual system. This is the approach taken by PWT[76],[77]

p. 151

玻姆把粒子观视为基本；而埃弗雷特或“多世界”实在论则把波函数视为基本。一切都被视为具有振幅叠加波，或者在有限模型中，被视为具有振幅状态矢量，这些振幅对应于每一个可能正交态。波函数不会发生坍缩，因为万物始终处于状态叠加之中。宇宙演化是一系列对该态矢量进行酉变换，即保持长度不变旋转。观测结果会给出确定答案；光子偏振方向会向左或向右，因为多重宇宙中与记录左偏振和右偏振相关子流形彼此正交。

Bohm took the particle view as fundamental; the Everett or‘many worlds’form of realism takes the wave function as fundamental. Everything is treated in terms of superposed waves with amplitudes, or in a finite model state vectors with amplitudes for every possible orthogonal state. There is no reduction of the wave function because everything is always in a superposition of states. The evolution of the universe is a series of unitary, that is to say, length preserving, rotations of this state vector. Observations yield definite answers; polarizations of photons go left or right, because the submanifolds of the multiverse associated with recording left and right polarizations are orthogonal to one another.

未来是否会有任何实验能够分出埃弗雷特和玻姆谁是正确的，我们无从知晓。目前，玻尔、海森堡、玻姆和埃弗雷特理论都与现有证据同样相容。但玻姆和埃弗雷特提出理论在解释现象方面与海森堡实证主义同样有效，这一事实表明，科学并不要求采用主观主义理论。正如我们之前所说，玻尔本人理论在原则上也与严格唯物主义相容。

Whether any experiments will, in the future, be able to resolve whether Everett or Bohm was right we cannot know. At present, the accounts of Bohr, Heisenberg, Bohm and Everett are all equally compatible with the evidence. But the fact that Bohm and Everett have produced theories that explain things just as well as Heisenberg’s positivism shows that science does not demand

a subjectivist theory. Bohr’s own account was, as we said earlier, in principle also compatible with strict materialism.

第 6 章原注 / Chapter 6 Notes

p. 250 ↑1. 汤姆森和麦克斯韦是苏格兰人，玻尔兹曼是奥地利人。

1 Thomson and Maxwell were Scottish, Boltzmann Austrian.

p. 250 ↑2. Henri Poincaré：“机制与经验”（Mechanism and Experience），载《气体动理论：经典论文选集及历史评注》（The Kinetic Theory Of Gases: An Anthology of Classic Papers with Historical Commentary）（World Scientific，2003），第 377—381 页。

2 Henri Poincaré, ‘Mechanism and Experience’, in *The Kinetic Theory Of Gases: An Anthology of Classic Papers with Historical Commentary* (World Scientific, 2003), 377–81.

p. 250 ↑3. Boltzmann：“答策梅洛关于热理论的评论”（Reply to Zermelo’s Remarks on the Theory of Heat）。

3 Boltzmann, ‘Reply to Zermelo’s Remarks on the Theory of Heat’.

p. 250 ↑4. Paul Cockshott、Lewis M. Mackenzie 与 Gregory Michaelson：《计算及其极限》（Computation and its Limits）（Oxford University Press，2012），第 117—120 页。

4 Paul Cockshott, Lewis M. Mackenzie, and Gregory Michaelson, *Computation and its Limits* (Oxford University Press, 2012), 117–20.

p. 250 ↑5. E. Fredkin 与 T. Toffoli：“保守逻辑”（Conservative Logic），International Journal of Theoretical Physics，第 21 卷第 3 期（1982），第 219—253 页。

5 E. Fredkin and T. Toffoli, ‘Conservative Logic’, *International Journal of Theoretical Physics* 21, no. 3 (1982): 219–53.

p. 250 ↑6. S. Rajasekar 与 N. Athavan：“路德维希 • 爱德华 • 玻尔兹曼”（Ludwig Edward Boltzmann），arXiv 预印本 physics/0609047，2006。

6 S. Rajasekar and N. Athavan, ‘Ludwig Edward Boltzmann’, *arXiv preprint physics/0609047*, 2006,

p. 250 ↑7. Ernst Mach：《能量守恒原理的历史与根源》（History and Root of the Principle of the Conservation of Energy）（Open Court Publishing Company，1910）。

7 Ernst Mach, *History and Root of the Principle of the Conservation of Energy* (Open Court Publishing Company, 1910).

p. 250 ↑8. Mach：《能量守恒原理的历史与根源》（History and Root of the Principle of the Conservation of Energy），第 45 页。

8 Mach, *History and Root of the Principle of the Conservation of Energy*, 45.

p. 250 ↑9. Charles Augustin Coulomb：“关于扭转力及金属丝弹性的理论和实验研究：此理论在工艺和各种物理实验中使用金属方面的应用；用于测量最微小力的各种扭秤的制造；关于弹性与内聚力定律的观察”（Recherches théoriques et expérimentales sur la force de torsion: & sur l’élasticité des fils de métal: application de cette théorie à l’emploi des métaux dans les arts & dans différentes expériences de physique: construction de différentes balances de torsion, pour mesurer les plus petits degrés de force: observations sur les loix de l’élasticité & de la coherence），1784。

9 Charles Augustin Coulomb, ‘Recherches théoriques et expérimentales sur la force de torsion: & sur l’élasticité des fils de métal: application de cette théorie à l’emploi des métaux dans les arts & dans différentes expériences de physique: construction de différentes balances de torsion, pour mesurer les plus petits degrés de force: observations sur les loix de l’élasticité & de la coherence’, 1784.

p. 250 ↑10. James Clerk Maxwell:《热学理论》（The Theory of Heat）（Longmans, Green, and Co，1875）。

10 James Clerk Maxwell, *The Theory of Heat* (Longmans, Green, and Co, 1875).

p. 250 ↑11. James Clerk Maxwell：“IV. 论气体的动力学理论”（IV. On the Dynamical Theory of Gases），Philosophical Transactions of the Royal Society of London，第 157 期（1867），第 49—88 页。

11 James Clerk Maxwell, ‘IV. On the Dynamical Theory of Gases’, *Philosophical Transactions of the Royal Society of London*, no. 157 (1867): 49–88.

p. 250 ↑12. Jacob Bronowski：《人类的进步》（The Ascent of Man）（Random House，2011）。

12 Jacob Bronowski, *The Ascent of Man* (Random House, 2011).

p. 250 ↑13. A. Einstein：“论热的分子动理论所要求的静止液体中悬浮微粒的运动”（On the Movement of Small Particles Suspended in Stationary Liquids Required by the Molecular-kinetic Theory of Heat），Annals of Physics，第 17 卷（1905），第 549 页。

13 A. Einstein, ‘On the Movement of Small Particles Suspended in Stationary Liquids Required by the Molecular-kinetic Theory of Heat’, *Annals of Physics* 17 (1905): 549.

↑14. A. Einstein：“关于光的发射和转化的一种启发式观点”（Concerning an Heuristic Point of View Toward the Emission and Transformation of Light），American Journal of Physics，第 33 卷第 5 期（1965），第 367 页。

14 A. Einstein, ‘Concerning an Heuristic Point of View Toward the Emission and Transformation of Light’, *American Journal of Physics* 33, no. 5 (1965): 367. p. 250

↑15. Albert Einstein:“论运动物体的电动力学”（On the Electrodynamics of Moving Bodies），Annalen der physik，第 17 卷第 10 期（1905），第 891—921 页。

15 Albert Einstein, ‘On the Electrodynamics of Moving Bodies’, *Annalen der physik* 17, no. 10 (1905): 891–921. p. 250

↑16. Albert Einstein：“物体的惯性是否取决于其能量含量”（Does the Inertia of a Body Depend Upon its Energy-content），Annals of Physics，第 18 卷（1905），第 639—641 页。

16 Albert Einstein, ‘Does the Inertia of a Body Depend Upon its Energy-content’, *Annals of Physics* 18 (1905): 639–41. p. 250

↑17. 关于该论文及其意义的概述，参见 Debashish Chowdhury：“爱因斯坦布朗运动理论一百年：从花粉粒到蛋白质列车——1”（100 Years of Einstein’s Theory of Brownian Motion: From Pollen Grains to Protein Trains—1），Resonance，第 10 卷第 9 期（2005），第 63—78 页。

17 For a summary of the paper and its implications see Debashish Chowdhury, ‘100 Years of Einstein’s Theory of Brownian Motion: From Pollen Grains to Protein Trains—1’, *Resonance* 10, no. 9 (2005): 63–78. p. 251

↑18. Jean Perrin：“布朗运动与分子实在性”（Mouvement brownien et réalité moléculaire），1909。

18 Jean Perrin, ‘Mouvement brownien et réalité moléculaire’, 1909. p. 251

↑19. Einstein:“关于光的发射和转化的一种启发式观点”（Concerning an Heuristic Point of View Toward the Emission and Transformation of Light）。

19 Einstein, ‘Concerning an Heuristic Point of View Toward the Emission and Transformation of Light’. p. 251

↑20. A. Einstein：“关于光的发射和转化的一种启发式观点”（Concerning an Heuristic Point of View Toward the Emission and Transformation of Light），American Journal of Physics，第 33 卷第 5 期（1965），第 367 页。

20 A. Einstein, ‘Concerning an Heuristic Point of View Toward the Emission and Transformation of Light’, *American Journal of Physics* 33, no. 5 (1965): 367. p. 251

↑21. 例如，全部电磁能恰好集中于腔的一半。他的方程为 $S - S_0 = k \ln \left(\frac{v}{v_0}\right)^{\frac{E}{h\nu}}$，其中 S, S_0 分别为最终和初始的熵，v, v_0 分别为最终和初始的体积，k, h 分别为玻尔兹曼常数和普朗克常数，ν 为频率，E 为能量。

〔校注：公式据源图恢复。底本 Plank 及 requency 等拼写和多余句点在英文中照录，中文按符号含义翻译。〕

21 An example would be if all the electromagnetic energy happened to be concentrated in one half of the cavity. His equation says $S - S_0 = k \ln \left(\frac{v}{v_0}\right)^{\frac{E}{h\nu}}$ where S, S_0 are the. final and initial entropies, v, v_0 final and. initial volumes, k, h the. Boltzmann and Plank constants, ν . requency and E . energy. p. 251

↑22. Einstein:“关于光的发射和转化的一种启发式观点”（Concerning an Heuristic Point of View Toward the Emission and Transformation of Light）。

22 Einstein, ‘Concerning an Heuristic Point of View Toward the Emission and Transformation of Light’. p. 251

↑23. $S - S_0 = k \ln \left(\frac{v}{v_0}\right)^n$，其中 n 为气体分子的数量。这显然与他先前给出的电磁熵变方程具有相同的函数形式。

23 $S - S_0 = k \ln \left(\frac{v}{v_0}\right)^n$ where *n* is the number of gas molecules. This was clearly the same functional form as the equation for change in electromagnetic entropy he gave earlier. p. 251

↑24. Einstein：“论运动物体的电动力学”（On the Electrodynamics of Moving Bodies）。

24 Einstein, ‘On the Electrodynamics of Moving Bodies’. p. 251

↑25. Einstein：“物体的惯性是否取决于其能量含量”（Does the Inertia of a Body Depend Upon its Energy-content）。

25 Einstein, ‘Does the Inertia of a Body Depend Upon its Energy-content’. p. 251

↑26. 尚未读过的读者或许会喜欢 A. Einstein 的《相对论》（Relativity）（Methuen/Company，1920）；他在书中为非专业读者作了说明。

26 Readers who have not done so may well enjoy A. Einstein, *Relativity* (Methuen/ Company, 1920) where he gave an account for the non-specialist readership. p. 251

↑27. 在非技术性论述中，能量概念长期带有神秘主义意味：vis viva（活力）呼应 élan vital（生命冲力），又逐渐滑向神秘能量——参见 Adrian Tait：“[一种]

27 The notion of energy has long, in non technical accounts had mystical overtones, *vis viva* echoes *elan vital* and shades off into mystic energy Adrian Tait, ‘[A] p. 251

神秘而不可言说的力与能量”（[A] Mystic, Ineffable Force and Energy），Arthur Machen: Critical Essays，2021，第 193 页——或昆达里尼能量——参见 Lilian Silburn：《昆达里尼：深处的能量》（Kundalini: The Energy of the Depths）（SUNY Press，1988）。不知为何，原子似乎并未对神秘主义边缘群体产生同样的吸引力。

Mystic, Ineffable Force and Energy’, *Arthur Machen: Critical Essays*, 2021, 193 or *kundalini energy*, Lilian Silburn, *Kundalini: The Energy of the Depths* (SUNY Press, 1988). For some reason atoms don’t seem to have had the same appeal to the esoteric fringe.

p. 251 ↑28. Manuel Bächtold：“为马赫的原子观辩护”（Saving Mach’s View on Atoms），Journal for General Philosophy of Science / Zeitschrift für allgemeine Wissenschaftstheorie，第 41 卷第 1 期（2010），第 1—19 页，ISSN：09254560、15728587，`http://www.jstor.org/stable/20722524`。

28 Manuel Bächtold, ‘Saving Mach’s View on Atoms’, *Journal for General Philosophy of Science / Zeitschrift für allgemeine Wissenschaftstheorie* 41, no. 1 (2010): 1 – 19, issn: 09254560, 15728587, `http://www.jstor.org/stable/20722524`.

p. 251 ↑29. 见第 [3.3.1] 节。

29 See chap. [3.3.1].

p. 251 ↑30. Ernst Mach：“通俗科学讲演”（Popular Scientific Lectures），“物理研究的节约性质”（The economical nature of physical inquiry）章（Open Court，1898）。

30 Ernst Mach, ‘Popular Scientific Lectures’, chap. The economical nature of physical inquiry (Open Court, 1898).

p. 251 ↑31. Mach：“通俗科学讲演”（Popular Scientific Lectures）。

31 Mach, ‘Popular Scientific Lectures’.

p. 251 ↑32. Mach：“通俗科学讲演”（Popular Scientific Lectures）。

32 Mach, ‘Popular Scientific Lectures’.

p. 251 ↑33. V. I. Lenin：《唯物主义和经验批判主义》（Materialism and Empirio-criticism）（International Publishers，1908）。

33 V. I. Lenin, *Materialism and Empirio-criticism* (International Publishers, 1908).

p. 251 ↑34. 正式职称为苏联人民委员会主席。

34 Officially Chairman of the Council of People’s Commissars of the Soviet Union.

p. 251 ↑35. Mach：“通俗科学讲演”（Popular Scientific Lectures）。

35 Mach, ‘Popular Scientific Lectures’.

p. 251 ↑36. 这里列宁引用的是 18 世纪唯物主义者狄德罗：为了让我的体系具备充分的说服力，还请注意，它同样面临贝克莱针对物体的存在所提出的那个不可克服的困难。有一刻，有感觉的羽管键琴陷入狂乱，以为它是世界上唯一的羽管键琴，宇宙的一切和声都发生在它之中。Denis Diderot 等：《达朗贝尔与狄德罗的对话》（Entretien entre d’Alembert et Diderot）（Garnier-Flammarion，1965）。

36 Here Lenin is quoting the eighteenth-century materialist Diderot: Et pour donner à mon système toute force, remarquez encore qu’il est sujet à la même difficulté insurmontable que Berkeley a proposée contre l’existence des corps. Il y a un moment de délire où le clavecin sensible a pensé qu’il était le seul clavecin qu’il y eût au monde, et que toute l’harmonie de l’univers se passait en lui. Denis Diderot et al., *Entretien entre d’Alembert et Diderot* (Garnier-Flammarion, 1965).

p. 251 ↑37. Mach：“通俗科学讲演”（Popular Scientific Lectures）。

37 Mach, ‘Popular Scientific Lectures’.

p. 252 ↑38. 据 Cercignani 与 Penrose 的《路德维希·玻尔兹曼：相信原子的人》（Ludwig Boltzmann: The Man Who Trusted Atoms），玻尔兹曼采取了非常相近的立场，把对手视为贝克莱式唯心主义者：在玻尔兹曼看来，哲学的中心问题是存在与知识之间的关系，即 gnoseology（英语译本 [2] 通常把这个词译为 epistemology，尽管这在某种程度上缩小了其范围）。最能说明他在这一问题上态度的，是他在哲学意味最少、也最有趣的一篇文章中近乎随口说出的一句话；那篇文章记述了他 1905 年的“黄金国”之旅。这句话常被引用，出现在他描写加利福尼亚大学伯克利分校校园之后：“伯克利这个名字属于一位备受推崇的英国哲学家；人们甚至把人脑所孵化出的最大蠢事——否认物质世界存在的哲学唯心主义——

38 According to Cercignani and Penrose, *Ludwig Boltzmann: The Man Who Trusted Atoms* Boltzmann took a very similar standpoint, seeing his opponents as Berkeleyian idealists: The central point of philosophy is, according to Boltzmann, the problem of the relationship between existence and knowledge, gnoseology (this term is usually rendered as epistemology in the English translation [2], although this somehow restricts its scope). To characterise his attitude in this respect, nothing serves better than a sentence thrown almost casually into the least philosophical and most entertaining of his essays, describing his trip to ‘Eldorado’ in 1905. This sentence is frequently quoted and follows his description of the campus of the University of California at Berkeley: ‘The name of Berkeley is that of a highly esteemed

的发明归功于他。”Cercignani 与 Penrose：《路德维希·玻尔兹曼：相信原子的人》，第 170 页。

English philosopher, who is even credited with being the inventor of the greatest folly ever hatched by a human brain, of philosophical idealism that denies the existence of the material world’. Cercignani and Penrose, *Ludwig Boltzmann: The Man Who Trusted Atoms*, 170

↑39. Guohua Shen 等：“从人脑活动进行端到端深度图像重建”（End-to-End Deep Image Reconstruction From Human Brain Activity），Frontiers in Computational Neuroscience，第 13 卷（2019），21，ISSN：1662-5188，https://doi.org/10.3389/fncom.2019.00021，https://www.frontiersin.org/article/10.3389/fncom.2019.00021。

39 Guohua Shen et al.,‘End-to-End Deep Image Reconstruction From Human Brain Activity’, *Frontiers in Computational Neuroscience* 13 (2019): 21, issn: 1662-5188, https://doi.org/10.3389/fncom.2019.00021, https://www.frontiersin.org/ article/10.3389/fncom.2019.00021. p. 252

↑40. Lenin：《唯物主义和经验批判主义》（Materialism and Empirio-criticism），第 55 页。

40 Lenin, *Materialism and Empirio-criticism*, 55. p. 252

↑41. Ernest Rutherford：“物质对 α 和 β 粒子的散射与原子结构”（The Scattering of α and β Particles by Matter and the Structure of the Atom），Philosophical Magazine，第 92 卷第 4 期（2012），第 379—398 页。

41 Ernest Rutherford,‘The Scattering of *α* and *β* Particles by Matter and the Structure of the Atom’, *Philosophical Magazine* 92, no. 4 (2012): 379–98. p. 252

↑42. John William Nicholson：“太阳日冕的构成。IL”（The Constitution of the Solar Corona. IL），Monthly Notices of the Royal Astronomical Society，第 72 卷第 8 期（1912），第 677—693 页。

42 John William Nicholson,‘The Constitution of the Solar Corona. IL’, *Monthly Notices of the Royal Astronomical Society* 72, no. 8 (1912): 677–93. p. 252

↑43. Nicholson：“太阳日冕的构成。IL”（The Constitution of the Solar Corona. IL）。

43 Nicholson,‘The Constitution of the Solar Corona. IL’. p. 252

↑44. Niels Bohr：“I. 论原子和分子的构成”（I. On the Constitution of Atoms and Molecules），The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science，第 26 卷第 151 期（1913），第 1—25 页。

44 Niels Bohr, ‘I. On the Constitution of Atoms and Molecules’, *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* 26, no. 151 (1913): 1–25. p. 252

↑45. Niels Bohr、Hendrik Anton Kramers 与 John Clarke Slater：“LXXVI. 辐射的量子理论”（LXXVI. The Quantum Theory of Radiation），The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science，第 47 卷第 281 期（1924），第 785—802 页。

45 Niels Bohr, Hendrik Anton Kramers, and John Clarke Slater, ‘LXXVI. The Quantum Theory of Radiation’, *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* 47, no. 281 (1924): 785–802. p. 252

↑46. Bohr、Kramers 与 Slater：“LXXVI. 辐射的量子理论”（LXXVI. The Quantum Theory of Radiation）。

46 Bohr, Kramers, and Slater,‘LXXVI. The Quantum Theory of Radiation’. p. 252

↑47. Niels Bohr 等：《量子公设与原子理论的近期发展》（The Quantum Postulate and the Recent Development of Atomic Theory），第 3 卷（英国：R. & R. Clarke, Limited 印刷，1928）。

47 Niels Bohr et al., *The Quantum Postulate and the Recent Development of Atomic Theory*, vol. 3 (Printed in Great Britain by R. & R. Clarke, Limited, 1928). p. 252

↑48. Bohr 等：《量子公设与原子理论的近期发展》（The Quantum Postulate and the Recent Development of Atomic Theory）。

48 Bohr et al., *The Quantum Postulate and the Recent Development of Atomic Theory*. p. 252

↑49. 这里对他 1928 年论文的阐述参考了 Howard 和 Camilleri 的研究：“谁发明了‘哥本哈根诠释’？一项神话研究”（Who Invented the ‘Copenhagen Interpretation’? A Study in Mythology），Philosophy of Science，第 71 卷第 5 期（2004），第 669—682 页；“玻尔、海森堡及其对互补性的不同看法”（Bohr, Heisenberg and the Divergent Views of Complementarity），Studies In History and Philosophy of Science Part B: Studies In History and Philosophy of Modern Physics，第 38 卷第 3 期（2007），第 514—528 页。

49 This account of his 1928 paper draws on Howard and Camilleri (’ Who Invented the ‘Copenhagen Interpretation’? A Study in Mythology’, *Philosophy of Science* 71, no. 5 (2004): 669–82; ‘Bohr, Heisenberg and the Divergent Views of Complementarity’, *Studies In History and Philosophy of Science Part B: Studies In History and Philosophy of Modern Physics* 38, no. 3 (2007): 514–28). p. 252

↑50. Bohr、Kramers 与 Slater：“LXXVI. 辐射的量 p. 252

子理论”（LXXVI. The Quantum Theory of Radiation）。

50 Bohr, Kramers, and Slater, ‘LXXVI. The Quantum Theory of Radiation’.

p. 252 ↑51. 现代数码相机十分灵敏，在弱光下拍照时，可以看到光子随机到达产生的“散粒噪声”，表现为各像素之间不相关的斑点。

51 Modern digital cameras are so sensitive that when taking photos in low light intensities you can see the ‘shot noise’ of the random arrival of photons as an uncorrelated speckle between individual pixels.

p. 252 ↑52. 玻尔似乎直到后来与爱因斯坦讨论时才考虑过这类例子。参见 Jan Faye：“量子力学的哥本哈根诠释”（Copenhagen Interpretation of Quantum Mechanics），载《斯坦福哲学百科全书》（The Stanford Encyclopedia of Philosophy），2019 年冬季版，Edward N. Zalta 编（斯坦福大学形而上学研究实验室，2019）。

52 Bohr had apparently not considered this sort of example until later discussions with Einstein, Jan Faye, ’Copenhagen Interpretation of Quantum Mechanics’, in *The Stanford Encyclopedia of Philosophy*, Winter 2019, ed. Edward N. Zalta (Metaphysics Research Lab, Stanford University, 2019).

p. 252 ↑53. $\Delta t \Delta E = \Delta x \Delta I_x = \Delta y \Delta I_y = \Delta z \Delta I_z = h$

53 $\Delta t \Delta E = \Delta x \Delta I_x = \Delta y \Delta I_y = \Delta z \Delta I_z = h$

p. 253 ↑54. Bohr 等：《量子公设与原子理论的近期发展》（The Quantum Postulate and the Recent Development of Atomic Theory）。

54 Bohr et al., *The Quantum Postulate and the Recent Development of Atomic Theory*.

p. 253 ↑55. 海森堡在前一年用另一种方法推导出了这一关系。他设想一个思想实验：用高频辐射照射电子，试图确定它的位置。要越精确地确定位置，所需光频率就越高。频率更高的光会扰动电子的运动，由此得到位置不确定性与动量不确定性之间的反比关系。玻尔直接从波的性质出发的推导更具一般性。

55 Heisenberg had derived this the year earlier by a different method involving a thought experiment in which you attempt to fix the position of an electron by illuminating it with high-frequency radiation. The more precisely you want to fix its position, the higher the light frequency you need. This higher-frequency light perturbs the motion of the electron, giving the inverse relationship between positional and momentum uncertainty. Bohr’s direct derivation from the properties of waves is more general.

p. 253 ↑56. Camilleri：“玻尔、海森堡及其对互补性的不同看法”（Bohr, Heisenberg and the Divergent Views of Complementarity）；Faye：“量子力学的哥本哈根诠释”（Copenhagen Interpretation of Quantum Mechanics）；Howard：“谁发明了‘哥本哈根诠释’？一项神话研究”（Who Invented the‘Copenhagen Interpretation’？A Study in Mythology）。

56 Camilleri, ‘Bohr, Heisenberg and the Divergent Views of Complementarity’; Faye, ‘Copenhagen Interpretation of Quantum Mechanics’; Howard, ‘Who Invented the ‘Copenhagen Interpretation’？A Study in Mythology’.

p. 253 ↑57. Sheng-Kai Liao 等：“卫星到地面的量子密钥分发”（Satellite-to-ground Quantum Key Distribution），Nature，第 549 卷第 7670 期（2017），第 43—47 页。

57 Sheng-Kai Liao et al., ‘Satellite-to-ground Quantum Key Distribution’, *Nature* 549, no. 7670 (2017): 43–7.

p. 253 ↑58. J. von Neumann：《量子力学的数学基础》（Mathematical Foundations of Quantum Mechanics），R. T. Beyer 据 1931 年德文版译为英文（Princeton University Press，1955）。

58 J. von Neumann, *Mathematical Foundations of Quantum Mechanics*, Engl. transl. of the 1931 German edition by R. T. Beyer (Princeton University Press, 1955).

p. 253 ↑59. 物理学使用“基矢态”（basis states）一词，指普通人所说的坐标系的轴。

59 Physics uses the term basis states to mean what lay people call the axes of a coordinate system.

p. 253 ↑60. 或者用更符合当代的说法：基本粒子和玻色子。

60 Or to be more up to date, elementary particles and Bosons.

p. 253 ↑61. Howard（“谁发明了‘哥本哈根诠释’？一项神话研究”）认为，哥本哈根诠释这个术语是海森堡在 1955 年的一篇论文中发明的：Werner Heisenberg，“量子理论诠释的发展”（The Development of the Interpretation of the Quantum Theory），Niels Bohr and the Development of Physics，1955，第 12 页。在文中，他试图把自己的观点投射到丹麦人玻尔身上：1955 年的新情况，是海森堡把他混合起来的一套思想命名为“哥本哈根诠释”；一经这样命名，他便经常强化这种统一的哥本哈根观点的发明，并以其主要代言人自居（例如见 Heisenberg [1955]）。为什么？

61 Howard (‘Who invented the “Copenhagen Interpretation”？A Study in Mythology’) claims that the term Copenhagen interpretation was invented by Heisenberg in a 1955 paper Werner Heisenberg, ‘The Development of the Interpretation of the Quantum Theory’, *Niels Bohr and the Development of Physics*, 1955, 12 in which he attempted to project his own opinions on to the Danish Bohr: What was new in 1955 was Heisenberg’s dubbing his amalgam of ideas the ‘Copenhagen Interpretation’, but having so dubbed it, Heisenberg regularly reinforced the invention of a unitary Copenhagen

回想海森堡 1955 年的处境会有所帮助：这位 20 世纪 20 年代曾深受玻尔喜爱的人，在战后却成为被哥本哈根核心圈从道义上排斥的人，主要因为 1941 年 9 月他接掌德国原子弹计划之后访问哥本哈根，那次不幸的访问使他与玻尔的关系严重破裂。对于骄傲而曾雄心勃勃的海森堡来说，还有什么办法比让自己成为哥本哈根诠释的声音更适合重获这个家族的成员身份呢？Howard：“谁发明了‘哥本哈根诠释’？一项神话研究”（Who Invented the“Copenhagen Interpretation”？A Study in Mythology）。

point of view and posed as its chief spokesperson (see eg Heisenberg [1955]). Why? It helps to recall Heisenberg’s situation in 1955, especially the fact that the person who was Bohr’s favorite in the 1920s had become a moral exile from the Copenhagen inner circle in the postwar period, mainly because of the bitter rupture in Heisenberg’s relationship with Bohr during his ill-fated visit to Copenhagen in September 1941 after taking over the leadership of the German atomic bomb project. What better way for a proud and once ambitious Heisenberg to reclaim membership in the Copenhagen family than by making himself the voice of the Copenhagen interpretation? Howard, ‘Who Invented the “Copenhagen Interpretation”？A Study in Mythology’

↑62. Heisenberg：“量子理论诠释的发展”（The Development of the Interpretation of the Quantum Theory），第 23 页。

62 Heisenberg, ‘The Development of the Interpretation of the Quantum Theory’, 23. p. 253

↑63. Liao 等：“卫星到地面的量子密钥分发”（Satellite-to-ground Quantum Key Distribution）。

63 Liao et al., ‘Satellite-to-ground Quantum Key Distribution’. p. 253

↑64. 进一步讨论见第 [sec:altsub]?? 节。

〔校注：底本交叉引用未解析，显示为 [sec:altsub]??；保留，不猜填节号。〕

64 See further discussion in Section [sec:altsub]??. p. 253

↑65. Albert Einstein、Boris Podolsky 与 Nathan Rosen：“量子力学对物理实在的描述能否被认为是完备的？”（Can Quantum-mechanical Description of Physical Reality be Considered Complete?），Physical Review，第 47 卷第 10 期（1935），第 777 页。

65 Albert Einstein, Boris Podolsky, and Nathan Rosen, ‘Can Quantum-mechanical Description of Physical Reality be Considered Complete?’, *Physical Review* 47, no. 10 (1935): 777. p. 253

↑66. David Bohm：《现代物理学中的因果性与机遇》（Causality and Chance in Modern Physics）（Routledge，2004）。

66 David Bohm, *Causality and Chance in Modern Physics* (Routledge, 2004). p. 253

↑67. 玻姆本人的论述见《现代物理学中的因果性与机遇》。详细的现代阐述还可参见 Holland：《运动的量子理论：德布罗意—玻姆量子力学因果解释》（The Quantum Theory of Motion: An Account of the de Broglie-Bohm Causal Interpretation of Quantum Mechanics）；以及 Bricmont：《理解量子力学》（Making Sense of Quantum Mechanics）和“量子力学史，或错误的喜剧”（History of Quantum Mechanics or the Comedy of Errors），arXiv 预印本 arXiv:1703.00294，2017。

67 His own account is given in Bohm (*Causality and Chance in Modern Physics*) and detailed modern accounts in the work of of Holland (*The Quantum Theory of Motion: An Account of the de Broglie-Bohm Causal Interpretation of Quantum Mechanics*) and Bricmont (*Making Sense of Quantum Mechanics*; ‘History of Quantum Mechanics or the Comedy of Errors’, *arXiv preprint arXiv:1703.00294*, 2017) can also be consulted. p. 253

↑68. Hugh Everett：“量子力学的‘相对态’表述”（“Relative State”Formulation of Quantum Mechanics），The Many Worlds Interpretation of Quantum Mechanics，2015，第 141—150 页。

68 Hugh Everett, ‘“Relative State” Formulation of Quantum Mechanics’, *The Many Worlds Interpretation of Quantum Mechanics*, 2015, 141–50. p. 254

↑69. 随着量子计算的发展，埃弗雷特的解释日益流行。这个研究领域由支持埃弗雷特的 Deutsch 开创：“量子理论、丘奇—图灵原理与通用量子计算机”（Quantum Theory, the Church-Turing Principle and the Universal Quantum Computer），Proceedings of the Royal Society of London A，1985，第 97—117 页。他此后仍不断为这一解释辩护，参见 D. Deutsch：《实在的结构》（The Fabric of Reality）（Penguin，2011）；David Deutsch：“实验检验的逻辑，特别是对埃弗雷特量子理论的检验”（The Logic of Experimental Tests, Particularly of Everettian Quantum Theory），Studies in History and Philosophy of Science Part B: Studies in

69 Everett’s account has become increasingly popular with the development of quantum computing. This field of research was initiated by the Everettian Deutsch (’Quantum Theory, the Church-Turing Principle and the Universal Quantum Computer’, *Proceedings of the Royal Society of London A*, 1985, 97–117.) who has continued to argue for it D. Deutsch, *The Fabric of Reality* (Penguin, 2011); David Deutsch, ‘The Logic of Experimental Tests, Particularly of Everettian Quantum Theory’, *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics* 55 (2016): 24–33. p. 254

History and Philosophy of Modern Physics，第 55 卷（2016），第 24—33 页。

p. 254 ↑70. David Bohm：《整体性与隐缠序》（Wholeness and the Implicate Order）（Routledge，2002），第 87 页。

70 David Bohm, *Wholeness and the Implicate Order* (Routledge, 2002), 87.

p. 254 ↑71. Louis De Broglie：“波与量子”（Waves and Quanta），Nature，第 112 卷第 2815 期（1923），第 540 页。

71 Louis De Broglie, ‘Waves and Quanta’, *Nature* 112, no. 2815 (1923): 540.

p. 254 ↑72. De Broglie：“波与量子”（Waves and Quanta）。

72 De Broglie, ‘Waves and Quanta’.

p. 254 ↑73. De Broglie：“波与量子”（Waves and Quanta）。

73 De Broglie, ‘Waves and Quanta’.

p. 254 ↑74. Antony Valentini：“信号局域性、不确定性与亚量子 H 定理。I”（Signal-locality, Uncertainty, and the Subquantum H-theorem. I），Physics Letters A，第 156 卷第 1—2 期（1991），第 5—11 页。

74 Antony Valentini, ‘Signal-locality, Uncertainty, and the Subquantum H-theorem. I’, *Physics Letters* A 156, no. 1–2 (1991): 5–11.

p. 254 ↑75. 标准量子力学（Standard Quantum Mechanics），即海森堡的观点。

75 Standard Quantum Mechanics, that is, the Heisenberg view.

p. 254 ↑76. 导引波理论（Pilot Wave Theory），即玻姆的观点。

76 Pilot Wave Theory, the Bohm view.

p. 254 ↑77. Philipp Roser：“德布罗意—玻姆导引波图景中的量子计算”（Quantum Computing in the de Broglie-Bohm Pilot-wave Picture），arXiv 预印本 arXiv:1012.4843，2010。

77 Philipp Roser, ‘Quantum Computing in the de Broglie-Bohm Pilot-wave Picture’, *arXiv preprint arXiv:1012.4843*, 2010,

第 7 章逻辑与唯物主义

Chapter 7: LOGIC AND MATERIALISM

引言

Introduction

显然，哲学家——即研究现实本质全貌的人——的职责，还包括探究三段论推理的原则。[1]

Clearly then it is the function of the philosopher, ie the student of the whole of reality in its essential nature, to investigate also the principles of syllogistic reasoning.[1]

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自古以来，许多长盛不衰的跨国国家体系都明确以某种有组织宗教作为其主导意识形态。在当今时代，例子包括：神圣罗马帝国、西班牙帝国、葡萄牙帝国和法兰西帝国的天主教；奥斯曼帝国的伊斯兰教；以及大英帝国的圣公会。除了一项重大例外，从未出现过基于世俗哲学的此类体系；即便美国帝国，也公开奉行自然神论，[2] 即使不是彻头彻尾的基督教。

Since antiquity, many long-lived transnational state systems have been explicitly founded on some organized religion as their explicit dominant ideology. In the current era, examples include Roman Catholicism for the Holy Roman, Spanish, Portuguese and French Empires; Islam for the Ottoman Empire; and Anglicanism for the British Empire. With one major exception, there have been no such systems based on secular philosophies; even the United States empire is openly deist,[2] if not outright Christian.

这一唯一的例外就是辩证唯物主义，即 19 世纪末以来世界共产主义运动的哲学。正如采纳主流宗教的帝国曾向数十亿人传授这些宗教一样，直到 1989 年，前苏联和欧洲社会主义国家也曾向数十亿人教授辩证唯物主义；中国和现存社会主义国家则延续至今。辩证唯物主义的影响范围不容低估。例如，2021 年圣公会有 8500 万领圣餐者，[3]明显少于拥有 9500 万党员的中国共产党。[4]

That one exception is *dialectical materialism*, the philosophy of the world communist movement since the late nineteenth century. Just as billions of people were taught the dominant religions in the empires that embraced them, so billions of people have been taught *dialectical materialism* in the former USSR and European socialist states until 1989, and in China, and extant socialist states, to the present. The reach of *dialectical materialism* should not be underestimated. Thus, in 2021, the Anglican Church had 85 million communicants[3]: markedly less than the Chinese Communist Party with 95 million members.[4]

就其唯物主义成分而言，辩证唯物主义具有深刻进步性，它摒弃一切形式的唯心主义，并积极倡导科学与理性作为人类理解和改造世界的手段。然而，正如我们将要探讨的，其辩证成分在哲学和意识形态层面存在更多问题。

In its materialist component, dialectical materialism is profoundly progressive, rejecting all forms of idealism and actively promoting science and rationality as the means for humanity to understand and transform the world. However, as we shall explore, the dialectical component is much more problematic, philosophically and ideologically.

特别是在苏联，辩证唯物主义被用于党派目的，既用来压制反对派，也用来限制被认为适合进行科学研究的领域。这对苏联遗传学的影响——即实际上拉马克主义被推崇而孟德尔主义被边缘化——已有充分记载。[5]本文将重点探讨其对形式逻辑研究这一较少为人所知的影响；尽管如此，这一影响对苏联数学的发展，特别是计算机科学和实用计算领域，却具有深远意义。

In particular, in the USSR, dialectical materialism was deployed for partisan purposes, both to suppress opposition, and to constrain what were deemed appropriate areas for scientific investigation. The effects on Soviet genetics, with what was effectively Lamarckianism promoted over Mendelism, are well documented.[5] Here, we will focus on the less well-known impact on the study of formal logic, which, nonetheless, had profound implications for the development of Soviet mathematics, in particular computer science and practical computing.

逻辑概述

Logic overview

逻辑[6]是一种形式化推理系统。需要特别注意的是，逻辑关注的是真值（即真与假），而非真理本身。我们所推理的事物被假定为独立于逻辑而存在，其真

A logic[6] is a formalized system for reasoning. It is important to note that logic is about *truth values* (i.e. true and false) rather than *the truth*. The things we reason

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伪已然确定。这些事物可能是事实、假设、推测或信念。逻辑本身无法确定某事是否真实；它关注的是推理的正确性。

当我们说逻辑是形式化的，是指存在以下规则：

• 构造用于推理的陈述，即文法或句法；

• 为命题赋予意义，即语义学；

• 操作陈述，以确定其性质，通常是判断它们总为真或假、还是部分情形下为真或假，即通过证明或求值。

我们将在下文进一步探讨这些内容，但在此不会进行详尽或形式化论述。

逻辑论证是从前提（也称为假设）推导出结论的过程。推理步骤是通过推理规则进行的，这是古代更为复杂的三段论的一种现代形式，将在后续章节中讨论。

三段论一般采用如下三项形式：

前提 $_1$

前提 $_2$

结论

读作：给定前提 $_1$ 和前提 $_2$，可以得出结论。

例如，假设 A 和 B 是两个命题，且我们接受“如果 A 成立，则 B 也成立”这一前提。那么，如果我们认为 A 成立，我们就会合理地推断 B 也成立，即 B 在逻辑上由 A 推导而出。用三段论形式表示，即

1. A

2. 如果 A，那么 B

3. B

当代逻辑的这一基本规则称为肯定前件式（Modus Ponens），即肯定的方法。例如：

〔校注：底本拉丁词误印 Modus Ponems，英文照录，中文用通行拼写 Ponens。〕

1. 巴斯特是一只猫

2. 如果巴斯特是一只猫，那么巴斯特喜欢鱼

3. 所以巴斯特喜欢鱼

这看起来完全合乎情理。

p. 155 在论辩中，人们可能会激烈质疑 A 是否成立，或者 B 是否确实能从 A 推导出来。例如，考虑以下情况：

1. 巴斯特是一位战斗女神

2. 如果巴斯特是一位战斗女神，那么巴斯特保卫国王

3. 所以巴斯特保卫国王

about are *assumed* to be true or false independently of logic. They may be factual, or hypothetical, or speculations or beliefs. Logic cannot in itself establish whether or not something is truthful; rather, it is concerned with *correct reasoning*.

When we say logic is *formalized*, we mean that there are rules for:

• constructing statements to reason about, i.e.grammar or syntax;

• giving meaning to statements, i.e. semantics;

• manipulating statements to establish their properties, commonly whether they are always, or partially, true or false, i.e. through proof or evaluation.

We will further explore these below, but we won’t exhaustively or formally treat them here.

A logical argument proceeds from *premises*, also known as *assumptions*, to *conclusions*. Reasoning steps are through *rules of inference*, a modern form of the more convoluted *syllogism* of antiquity, discussed in subsequent sections.

A syllogism has the general three term form:

$premise_1$

$premise_2$

conclusion

and reads as: given $premise_1$ and $premise_2$, we can conclude *conclusion*.

For example, suppose *A* and *B* are statements, and we accept that if *A* is the case then *B* is the case. Then, if we take it that *A* is the case, we will reasonably infer that *B* is the case as well, that is *B* logically follows from *A*. In syllogistic form, this is

1. A

2. if *A* then *B*

3. *B*

This fundamental rule of contemporary logic is known as *Modus Ponems*, or the method of affirming. For example, consider:

1. Bastet is a cat

2. if Bastet is a cat then Bastet likes fish

3. so Bastet likes fish

This seems entirely reasonable.

In arguments, people may vociferously question whether or not *A* is the case, or whether or not *B* actually follows from *A*. For example, consider:

1. Bastet is a warrior goddess

2. if Bastet is a warrior goddess then Bastet defends the king

3. so Bastet defends the king

我们可能会反驳道：

• 没有女神；

• 有女神，但没有战斗女神；

• 有战斗女神，但巴斯特不是其中之一；

• 巴斯特是一位战斗女神，但没有国王；

• 巴斯特是一位战斗女神，但她不保卫国王；

• 等等。

但如果我们承认“巴斯特是一位战斗女神”为真，并承认“如果巴斯特是一位战斗女神，那么她保卫国王”为真，那么显然，“巴斯特保卫国王”就是一个无可置疑的结论。

也就是说，一般而言，只要假定 A 以及“A 蕴含 B”都成立，就没有人会质疑推导出 B。任何争论都针对前提，而非推导；前提的真或假最终是在逻辑之外确定的。[7]

在当代逻辑中，我们现在把命题逻辑与谓词逻辑区分开来。通常把二者统称为“演算”，这源于乔治·布尔——他把自己开创性的体系（下文将讨论）命名为“逻辑演算”。[8]

命题逻辑涉及命题（即简单陈述）的真值判断。命题由真值及其抽象变量构成。命题可以被否定（非），并通过析取（或）、合取（且）和蕴含（如果……那么……）进行组合。

谓词逻辑则涉及关于事物集合及其成员的命题的真值问题。谓词逻辑在命题逻辑的真值和变量基础上引入了谓词——即返回真值的函数。此外，在谓词逻辑中，命题可以进行量化，以表达全称性质（即所有事物都具有某种性质）和存在性性质（即某些事物，即至少一个，具有某种性质）。

我们还将不涉及特定对象的“纯粹逻辑”与涉及特定事物领域的“应用逻辑”区分开来。正如我们将要讨论的，从古代到 19 世纪末，逻辑的这些方面一直被混为一谈。

一个关键的哲学问题涉及真值和推理规则的地位，这是关于数字和函数等数学实体的地位这一更广泛问题的一部分。它们仅仅是纸上的符号吗？它们是否具有某种更深层的观念实在？或者，正如我们将论证的，它们是物质化数学系统从物质现实中抽象出来的、并与之具有强烈对应关系的组成部分？

We might argue:

• there are no goddesses;

• there are goddesses but no warrior goddesses;

• there are warrior goddesses but Bastet isn’t one;

• Bastet is a warrior goddess but there is no king;

• Bastet is a warrior goddess but she doesn’t defend the king;

• etc.

But if we accept the truth of ‘Bastet is a warrior goddess’, and of ‘if Bastet is a warrior goddess then she defends the king’, then clearly ‘Bastet defends the king’ is an unimpeachable conclusion.

That is, in the general case, nobody questions the deduction of *B*, assuming that both *A*, and *A* implies *B*, are the case. Any argument is about the premises, whose truth or falsity is ultimately determined outside of logic,[7] not the deduction.

In contemporary logic, we now distinguish *propositional* from *predicate* logic. It is common to refer to both as *calculi*, after George Boole, who called his pioneering system, discussed below, the *Calculus of Logic*.[8]

Propositional logic is to do with propositions (i.e. simple statements) being true or false. Propositions are built from truth values, and variables that abstract over them. Propositions may be *negated* (not) and combined through *disjunction* (or), *conjunction* (and), and *implication* (if . . . then . . .).

Predicate logic is then to do with propositions about collections of things, and their members, being true or false. Predicate logic extends propositional logic truth values and variables with *predicates*, which are functions that return truth values. Further, in predicate logic, propositions may be *quantified* to express *universal* properties, that is, all things having some property, and *existential* properties, that is, some things (i.e. at least one) having some property.

We also distinguish *pure* logics, which are not about anything in particular, from *applied* logics, which are about specific domains of things. As we shall discuss, from antiquity until the late nineteenth century, all of these aspects of logic were conflated.

A key philosophical question concerns the status of truth values and rules of inference, as part of a wider question about the status of mathematical entities like numbers and functions. Are they just marks on paper? Do they have some deeper ideal reality? Or are they, as we shall argue, components of materialized mathematical systems, abstracted from and with strong correspondences, to material reality?

p. 155–156

逻辑与辩证法

Logic and dialectics

p. 156

逻辑与辩证法是唯物辩证法的核心组成部分。[9]然而，厘清逻辑与辩证法的关系却是一件出人意料地困难的事。

Logic and dialectics are core components of the *materialist dialectic.*[9] However, untangling logic and dialectics is a curiously difficult business.

亚里士多德实际上所指的正是辩证法，而非逻辑学。在《修辞学》中，[10]他把关注非正式说服的修辞学，与关注通过三段论进行形式推理的辩证法区分开来（第 13 页）。这暗示着逻辑学是辩证法的一部分。

Aristotle actually refers to dialectics rather than logic. In *The 'Art' of Rhetoric*,[10] he distinguishes rhetoric, concerned with informal persuasion, from dialectics, concerned with formal reasoning through the syllogism (p13). This implies that logic is part of dialectics.

亚里士多德指出，修辞学和辩证法都使用三段论推理和归纳推理。基于辩证三段论的推理，必须把所有步骤明确表述出来；但修辞三段论，即省略三段论，可以省略步骤。此外，辩证归纳以发现模式为基础，修辞归纳则以具体例子为基础。总的来说：

Aristotle notes that both rhetoric and dialectics employ syllogistic and inductive reasoning. For reasoning based on the dialectical syllogism, all steps must be made explicit. However, for the rhetorical syllogism, the *enthymeme*, steps may be elided. Further, dialectical induction is based on finding patterns, whereas rhetorical induction is based on concrete examples. Overall:

> 因此，修辞学的功能是处理那些我们加以审议、但尚无系统规则的事物；（第 23 页）

> The function of Rhetoric, then, is to deal with things about which we deliberate, but for which we have no systematic rules; (p23).

这暗示着辩证法是系统化的。

The implication is that dialectics is systematic.

亚里士多德还把关注特定领域的科学，与作为普遍适用话语模式的修辞学和辩证法区分开来。这一点体现在随后的“三艺”与“四艺”的区分中，“三艺”为针对“四艺”应用领域的推理和论证提供了纯粹的工具。

Aristotle also distinguishes sciences, concerned with particular domains, from both rhetoric and dialectic as universally applicable modes of discourse. This is reflected in the subsequent Trivium/Quadrivium distinction, with the Trivium providing the pure tools for reasoning and arguing about the Quadrivium applied domains.

他进一步指出，当一个人对某个领域形成更深刻的理解时：

He further says that, as someone develops richer understandings of a domain:

> 他就越会在不知不觉中形成一门与辩证法和修辞学完全不同的科学。因为一旦找到第一原理，那就不再是辩证法或修辞学，而是他已经掌握其原理的那门科学。（第 31 页）

> the more he will unconsciously produce a science quite different from Dialectic and Rhetoric. For if once he hits upon first principles, it will no longer be Dialectic or Rhetoric, but that science whose principles he has arrived at. (p31)

显然，这一点同样适用于辩证法与逻辑之间的动态关系本身。正如我们将看到的，随着逻辑日益建立在数学基础上，辩证法的空间便随之缩小，这与前几章所讨论的、更广泛的科学进步压缩了宗教和哲学的空间的情况如出一辙。

Clearly, this applies to the dialectics/logic dynamic itself. As we shall see, as logic became more mathematically grounded, so the space for dialectics shrank, much as wider scientific advances shrank the space for both religion and philosophy, as discussed in earlier chapters.

在《工具论》（《前分析篇》）[11] 中，亚里士多德在讨论三段论的前提类型时，进一步将逻辑从辩证法和科学中抽象出来（第 200 页和第 202 页）。一个证明性（即科学性）的前提是真实的，并基于“基本公设”；而对于辩证法的前提，则可能在两个“相互矛盾的陈述”之间进行选择。因此，对于三段论而言，无论前提源自何处，它都仅是真或假。也就是说，无论是科学还是辩证法，一旦某个前提确立，便可运用三段论来得出结论。

In *The Organon (Prior Analytics)*,[11] Aristotle further abstracts logic from both dialectics and science in discussing types of premises for syllogisms (p200 & 202). A demonstrative (i.e. scientific) premise is true and based on 'fundamental postulates', whereas, for a dialectical premise, a choice may be made between two 'contradictory statements'. Then, for a syllogism, a premise is simply true or false, regardless of origin. That is, for both science and dialectics, once some premise is established, a syllogism may be applied to draw a conclusion.

数学形式

Mathematical forms

上文已提到，“三艺”中的逻辑有别于“四艺”中的算术、几何学和天文学这些数学科学。此外，亚里士多德在《形而上学》[12]中指出，数学具有层级：

数学也有其分支——在数学领域中，存在一门主要科学和一门次要科学，以及其他依次排列的学科。（第 151 页）

此外，在探讨哲学的层次结构时，他把普遍（即纯粹）数学与具体（即应用）数学科学区分开来：

人们确实可能会提出这样一个问题：第一哲学是普遍的，还是只涉及某一类或某一实体；因为就连数学科学在这方面也存在差异——几何学和天文学涉及特定类型的实体，而普遍数学则同样适用于所有类型。（第 297 页）

亚里士多德虽受过柏拉图主义训练，但在数学的形而上学地位问题上似乎并未明确表态，但他指出，对于他的导师而言，数学介于物质现实与纯粹理念之间：

此外，他［柏拉图］指出，除了可感之物和理念之外，还存在一个中间类别，即数学对象［脚注：即算术数和几何图形］，它们与可感之物的区别在于永恒且不变，与理念的区别则在于：数学对象中存在许多相似的对象，而每个理念本身都是独一无二的。（第 45 页）

对柏拉图主义者而言，形式是抽象的理念，但仍具有实在性。[13]我们认为，数学对象通过其在符号系统中的物理表征而具有物质存在。例如，理念中的立方体无非就是数学立方体，而数学立方体本身是一种表征物理立方体的物质化构造。

We saw above that Logic in the Trivium was distinguished from the mathematical sciences of Arithmetic, Geometry and Astronomy in the Quadrivium. Further, in *The Metaphysics*,[12] Aristotle says that there is hierarchy in mathematics: p. 157

mathematics too has divisions, -there is a primary and a secondary science, and others successively, in the realm of mathematics. (p151)

and, in considering how philosophy is layered, distinguishes universal (i.e. pure) mathematics, from specific (i.e. applied) mathematical sciences:

One might indeed raise the question whether the primary philosophy is universal or deals with some one genus or entity; because even the mathematical sciences differ in this respect - geometry and astronomy deal with a particular kind of entity, whereas universal mathematics applies to all kinds alike. (p297)

Aristotle, a Platonist by training, nonetheless appears uncommitted as to the metaphysical status of mathematics, but notes that, for his master, mathematics lies between material reality and pure idea:

Further, he [Plato] states that besides sensible things and the Forms there exists an intermediate class, *the objects of mathematics* [footnote: ie arithmetical numbers and geometric figures], which differ from sensible things in being eternal and immutable, and from the the Forms in that there are many similar objects of mathematics, whereas each Form is itself unique. (p45)

For Platonists, the *forms* are abstract ideals which nonetheless are real.[13] We argue that mathematical objects have material beings in their physical representations within symbol systems. So, for example, the ideal cube is no more than the mathematical cube, itself a materialized construct that characterizes physical cubes.

三段论

Syllogisms

亚里士多德对三段论的阐述是前现代逻辑的关键。接下来我们将稍作深入探讨，但不会进行形式化论证。简明扼要的论述可参见史密斯。[14]

亚里士多德在《工具论》中探讨了三段论，涉及对个体及事物集合的属性的推理，以及对特定领域和一般领域事物的推理。他大量运用具体实例，正如我们上面所见，他称之为修辞归纳。

首先，在《解释篇》[15]中，亚里士多德定义了命题的句法，但没有使用任何符号。单句由名词及作用于名词的动词构成，还可以进一步连接。命题则是作出肯定或否定的句子，主项可以是全称的，也可

Aristotle’s formulations of syllogisms are key to premodern logic. We will now look in slightly more depth at these, but we won’t give a formal treatment. For a succinct account, see Smith.[14]

Aristotle explores syllogisms in *The Organon*, considering reasoning about properties of individuals and collections of things, and about things from particular and general domains. He makes considerable use of concrete examples, which, as we saw above, he called rhetorical induction.

First of all, in *On Interpretation*,[15] Aristotle defines the syntax of propositions, but without using any notation. Single sentences are composed of nouns, and verbs that act on them, and may be further conjoined. Propo- p. 157–158

以是单称的。《解释篇》还引入了矛盾命题这一关键概念。

sitions are then sentences which are affirmations or denials, and subjects may be universal or singular. *On Interpretation* also introduces the key notion of contradictory propositions.

p. 158 因此，亚里士多德提出了四个基本命题模式，通常表述为：

- A：全称肯定命题，例如“所有 X 都是 Y”；
- I：特称肯定命题，例如“有些 X 是 Y”；
- E：全称否定命题，例如“没有 X 是 Y”；
- O：特称否定命题，例如“有些 X 不是 Y”。

Thus, Aristotle enunciates four fundamental schemes for propositions, commonly expressed as:

- A: universal affirmative, for example, all *X* are *Y*;
- I: particular affirmative, for example, some *X* is *Y*;
- E: universal negative, for example, no *X* is *Y*;
- O: particular negative, for example, some *X* is not *Y*;

这些模式都具有“主项是谓项”的结构。这里的 X、Y、Z 可以一致地替换为具体名词。开头的字母 A、I、E、O 是传统的标识。

These schemes all have the structure: *subject* is/are *predicate*. Here, *X* , *Y* and *Z* may be replaced consistently by concrete nouns. The initial letters A, I, E and O are the classical identifiers.

于是，《前分析篇》[16]探讨的三种三段论格可以表述为：

1. P 是 Q；Q 是 R；P 是 R
2. P 是 Q；P 是 R；Q 是 R
3. P 是 R；Q 是 R；P 是 Q

Then, the three syllogistic *figures* explored in *Prior Analytics*[16] may be expressed as:

1. *P* is *Q*; *Q* is *R*; *P* is *R*
2. *P* is *Q*; *P* is *R*; *Q* is *R*
3. *P* is *R*; *Q* is *R*; *P* is *Q*

需注意，这些属于第二级模式，其中 P、Q 和 R 可分别替换为 A、I、E 和 O 这四种命题形式之一，从而形成第一级模式。

Note that these are second level schema, where *P*, *Q* and *R* may be replaced by one of the four proposition forms A, I, E and O, to give a first level schema.

这些格的有效性取决于我们代入 P、Q 和 R 的项是全称还是特称，以及哪些主项和谓项在哪些项中是共有的。

The validity of these figures then depends on whether the terms we substitute for *P*, *Q* and *R* are universal or particular, and which subjects and predicates are common to which terms.

亚里士多德穷尽考察了这三种格中全称 / 特称与主项 / 谓项的所有可能组合，通过推导矛盾排除了其中许多组合。他还说明了如何把第二格和第三格归约为第一格。

Aristotle exhaustively considers all the universal/particular and subject/ predicate possibilities for the three figures, dismissing many by deriving contradictions. He also shows how the second and third figures may be reduced to the first figure.

总体而言，亚里士多德建立了一个既用于构建又用于分析论证的框架，该框架一直被广泛使用，直至 19 世纪中叶逻辑学和数学发生革命性变革。

Overall, Aristotle developed a framework both for constructing and analysing arguments that was in widespread use until the revolutions in logic and mathematics in the mid-nineteenth century.

亚里士多德之后

After Aristotle

亚里士多德的著作由波爱修斯[17]译成拉丁文；他约生活于公元 475 年至 526 年。此后，这些著作实际上从欧洲思想中失落了数百年，直到被阿拉伯学者“重新发现”。后来，亚里士多德哲学在中世纪天主教学者的一场重要思潮中稳步取得影响，这场思潮如今称为经院哲学。七门自由学艺的划分及其分为“三艺”“四艺”，构成欧洲教育的核心；尤其是 11 世纪末起，第一批大学得到梵蒂冈认可而建立之后，更是如此。

Aristotle’s work was translated into Latin by Boethius,[17] who lived around 475 to 526 CE. Subsequently, it was effectively lost to European thought for several hundred years, until it was ‘rediscovered’ by Arab scholars. Aristotelian philosophy then gained steady traction among a significant movement of medieval Catholic scholars, now known as Scholasticism. The articulation of the Seven Liberal Arts, and their division into the Trivium and Quadrivium, was core to European education, especially after the establishment of the first universities from the late eleventh century onwards under the imprimatur of the Vatican.

法国是三段论推理的一个中心，巴黎大学尤其如此。17 世纪中叶，《波尔—罗亚尔逻辑》[18]由曾就读索邦大学的阿尔诺和尼科尔发展而来。他们虽是神职人员，却信奉与主流天主教存在教义冲突的詹森主义，并深受笛卡尔影响。

France was a centre of syllogistic reasoning, particularly at the University of Paris. In the mid-seventeenth century, the *Port-Royal Logic*[18] was developed by Arnauld and Nicole, who had studied at the Sorbonne. Though they were clerics, they adhered to Jansenism, a sect in doctrinal conflict with mainstream Catholicism, and were heavily influenced by Descartes.

1662 年出版的《波尔—罗亚尔逻辑》保留了 A、I、E 和 O 四种三段论形式，但其术语结构更为精妙，允许使用从属命题。这使得语义和推理都变得更加复杂。该书一直是逻辑学的重要教材，一直延续到 19 世纪相当长的一段时期。该书共出版了 63 版，其中包括 10 版英文版，并在牛津大学和剑桥大学广泛使用。

The *Port-Royal Logic*, published in 1662, retained the A, I, E and O syllogistic forms, but it had a more sophisticated term structure that allowed subordinate propositions. This made both semantics and reasoning more complex. The book became a key text on logic until well into the nineteenth century. It went into sixty-three editions, including ten in English, and was in use at Oxford and Cambridge Universities.

尽管如此，在宗教改革之后，亚里士多德逻辑在英格兰的处境并不乐观——那里，作为国家意识形态的天主教被圣公会取代，后者是英国与新教达成的一种折中方案。弗朗西斯·培根（1561—1626），被许多人视为英国那种令人沮丧的经验主义传统的开创者，完全否定了形式三段论。在《新工具》[19],[20]（1620 年出版）中，他主张逻辑是说服的工具，而非推理的工具，并提倡通过观察进行归纳以求理解。培根的批判基于他对统一推理体系的要求。他认为，逻辑——其中他把三段论与辩证法混为一谈——无法满足这一要求，因为它无法纠正毫无根据的前提。

Nonetheless, Aritotelian logic fared less well in England after the Reformation, where the state ideology of Catholicism was displaced by Anglicanism, the English compromise with Protestantism. Francis Bacon (1561–1626), progenitor of what many view as the doleful British tradition of empiricism, rejected formal syllogism entirely. In *Novum Organon*,[19],[20] from 1620, he argues that logic is a tool of persuasion, not reason, and promotes induction from observation as the means to understanding. Bacon’s critique rested on his requirement for a unitary system of reasoning. He argued that logic, in which he conflates syllogistics and dialectics, cannot furnish this as it is unable to correct for unfounded premises.

托马斯·霍布斯（1588—1679）的哲学可被归类为唯物主义，其观点比培根更为审慎。1655 年，他在《论物体》（De Corpore）[21] 中对亚里士多德逻辑学进行了简明扼要的阐述。尽管他拒绝了亚里士多德的形而上学，但他通过与算术的强烈类比，把三段论推理视为一种计算过程。此后，尽管牛津和剑桥大学仍在教授逻辑学，但学术界对三段论的兴趣寥寥，直至 19 世纪初，三段论才得以复兴，且其复兴与辩证法完全分离。

Thomas Hobbes (1588–1679), whose philosophy may be characterized as materialist, was more measured than Bacon. In 1655, he gave a succinct account of Aristotle’s logic in *De Corpore.*[21] While he rejected Aristotelian metaphysics, he viewed syllogistic reasoning as computational, by strong analogy with arithmetic. Thereafter, while logic continued to be taught at Oxford and Cambridge, there was little academic interest in syllogistics until the early nineteenth century, when its revival was wholly separated from dialectics.

爱尔兰圣公会都柏林大主教理查德·惠特利（Richard Whately，1787—1863）的著作影响尤为深远。在 1826 年撰写的《逻辑要素》（Elements of Logic）[22] 中，他有力地捍卫了逻辑学，强调其普遍适用性，以及将其与研究对象相分离的必要性。通过一个引人入胜的算术类比，他断言逻辑的力量恰恰在于抽象性。

The work of Richard Whately (1787–1863), the Archbishop of Dublin in the Anglican Church of Ireland, was particularly influential. In *Elements of Logic*,[22] written in 1826, he robustly defended logic, emphasizing its universal applicability, and the need to separate it from its subject matter. In an engaging analogy with arithmetic, he asserted that the strength of logic lay precisely in abstraction.

思维规律

The laws of thought

同一时期，逻辑的数学化取得了稳步进展，最终以乔治·布尔 1854 年在《思维规律》[23] 中开创性的代数处理为标志。该书的标题源于这样一种观念：思维以不可改变的定律为基础。这些定律直指现实的基本属性，因此也能解释为什么三段论能够体现无可指摘的推理。

The same period saw steady progress in the mathematization of logic, culminating in George Boole’s seminal algebraic treatment in *The Laws of Thought*[23] in 1854. The book’s title derived from the notion that thinking is underpinned by immutable laws. These go to the heart

关于思维定律的大多数论述都描述了斯坦利·杰文斯（1835—1882）所称的三大基本定律：[24]

1. 同一律。凡是存在的，就是存在的。

2. 矛盾律。任何事物不可能既存在又不存在。

3. 排中律。任何事物必定或者存在，或者不存在。（第 117 页）

p. 160 苏格兰人威廉·汉密尔顿[25]（1788—1856），作为惠特利和布尔的同时代人，详细探讨了这些法则在亚里士多德著作中的起源。在他于 1860 年出版的遗著《形而上学与逻辑讲义》[26]中，汉密尔顿重申了逻辑与形而上学的分离，以及逻辑的普遍适用性。对汉密尔顿而言，思维规律具有某种本质性，甚至对神明也具有约束力。

关于每条定律的地位，尤其是“排中律”（LEM），以及是否还有其他定律，一直存在相当大的争议。

of the fundamental properties of reality and, hence, why the syllogism can capture unimpeachable reasoning.

Most accounts of the laws of thought describe what Stanley Jevons (1835–82) termed the *three primary laws*.[24]

1. The Law of Identity. Whatever is, is.

2. The Law of Contradiction. Nothing can both be and not be.

3. The Law of Excluded Middle. Everything must either be or not be. (p117)

The Scot William Hamilton[25] (1788–1856), a contemporary of Whately and Boole, discussed in detail the laws’ origins in Aristotle’s writings. In his posthumous *Lectures on Metaphysics and Logic*,[26] published in 1860, Hamilton reiterated the separation of logic from metaphysics, and logic’s universal applicability. For Hamilton, there was something essential about the laws of thought, circumscribing even the deity.

There has been considerable disputation over the status of each law, especially the Law of the Excluded Middle (LEM), and whether or not there are further laws.

逻辑运算与真值表

Logical operations and truth tables

亚里士多德及其后继者把语法视为构建正确论证的核心，并依赖日常语言中表达的意义和推理。因此，在布尔之前乃至之后，关于逻辑的大部分讨论都围绕着逻辑运算的确切含义展开。

如今，我们按照路德维希·维特根斯坦（1889—1951）所倡导的方法，用真值表赋予逻辑运算符精确的含义。[27]我们假定基本真值为真和假。核心运算包括：否定（非），把前提的真和假反转；合取（且），要求两个前提都为真；析取（或），要求至少一个前提为真，也允许两个都为真。见表 7.1。

请注意，这种形式的析取称为“相容析取”。我们还可以定义“不相容析取”（xor），当任一前提为真时，它为真；但若两个前提同时为真，则为假。

在逻辑推理中，实质蕴含，即“蕴含”或“如果……那么……”，是构建推理规则的核心。例如：

X 蕴含 Y

我们不希望在 Y 为假的同时 X 为真。也就是说，我们要求其效果为：

非（X 且非 Y）

表 7.2 给出了相应的真值表。

这似乎有悖直觉，因为当 X 为假时，“X 蕴含 Y”总是为真。这里，逻辑上的“真”与现实世界中的“真”之间存在着鲜明区别：一个评估结果为真的逻辑蕴含，除非我们知道第一个前提为真，否则绝

Aristotle, and his successors, focused on grammar as central to constructing correct arguments, and depended on meanings and reasoning expressed in everyday language. Thus, much discussion of logic prior to, and indeed after, Boole was about what exactly logical operations meant.

Today, we use *truth tables* to give logical operators precise meanings, as promoted by Ludwig Wittgenstein (1889–1951).[27] We assume the basic truth values of true and false. Core operations are: negation (not) as reversing true and false premises; conjunction (and) as requiring both premises to be true; and disjunction (or) as requiring either or both premises to be true. See Table 7.1.

Note that this form of disjunction is called *inclusive*. We can also define *exclusive* disjunction (xor), which is true if either premise is true, but not both.

In logical reasoning, *material implication*, that is, ‘implies’ or ‘if . . . then . . .’, is central to forming rules of inference. For:

X implies *Y*

we wouldn’t like *X* to be true at the same time that *Y* is false. That is, we require the effect of:

not (*X* and not *Y*)

Table 7.2 shows the corresponding truth table.

This feels counterintuitive, as ‘*X* implies *Y*’ is true whenever *X* is false. Here the distinction between logical and real-world notions of truth is stark: a logical implication which evaluates to true certainly does not per-

不能据此推导出关于现实世界的任何结论。事实上，基于一个假的前提而得出的真蕴含被称为“空真”。

mit us to deduce anything about the real world unless we know that the first premise is true. Indeed, a true implication resting on a false first premise is termed *vacuously* true.

表 7.1 逻辑运算：否定、合取和相容析取

X	非 X	X	Y	X 且 Y	X	Y	X 或 Y
真	假	假	假	假	假	假	假
假	真	假	真	假	假	真	真
		真	假	假	真	假	真
		真	真	真	真	真	真

Table 7.1 Logical Operations: Negation, Conjunction and Inclusive Disjunction

X	not X	X	Y	X and Y	X	Y	X or Y
true	false	false	false	false	false	false	false
false	true	false	true	false	false	true	true
		true	false	false	true	false	true
		true	true	true	true	true	true

表 7.2 蕴含

p. 161

X	Y	非 Y	X 且非 Y	非（X 且非 Y）
假	假	真	假	真
假	真	假	假	真
真	假	真	真	假
真	真	假	假	真

Table 7.2 Implication

X	Y	not Y	X and not Y	not (X and not Y)
false	false	true	false	true
false	true	false	false	true
true	false	true	true	false
true	true	false	false	true

表 7.3 以算术表示的逻辑运算：否定、合取和相容析取

x	1 − x	x	y	x × y	x	y	x + y
0	1	0	0	0	0	0	0
1	0	0	1	0	0	1	1
		1	0	0	1	0	1
		1	1	1	1	1	1

Table 7.3 Logical Operations as Arithmetic: Negation, Conjunction and Inclusive Disjunction

x	1 − x	x	y	x × y	x	y	x + y
0	1	0	0	0	0	0	0
1	0	0	1	0	0	1	1
		1	0	0	1	0	1
		1	1	1	1	1	1

逻辑的数学化

Mathematizing logic

尽管莱布尼茨（1646—1716）早在布尔之前就已探索过逻辑的形式化，但他的著作直到 20 世纪初才重见天日。[28]因此，布尔（1815—1864）如今被公认为现代逻辑的奠基人，特别是命题演算的奠基人。在《思维规律》[29]一书中，布尔的意图是：

探究思维进行推理时所进行的那些操作的基本规律；用演算的符号语言来表达这些规律，并以此为基础建立逻辑科学并构建其方法；（第 1 页）

布尔认为，人类思维的规律确实是数学规律：

一般推理中的思维运作与代数这一特定科学中的运算之间，不仅存在密切的类比，而且在指导这两类运算的规律上，在相当大程度上也完全一致。（第 6 页）

布尔的关键洞见在于：如果把“论域”表示为“1”，把“无”表示为“0”，那么针对事物类别的逻辑运算便具有算术等价形式。因此，可以运用代数技术来对它们进行操作。

从布尔的系统中推导出命题逻辑是显而易见的，其中 1 代表真，0 代表假。参见表 7.3，可将其与表 7.1 进行对比。[30]

信息理论学家克劳德 • 香农于 1938 年指出，布尔演算与逻辑之间的对应关系可应用于开关电路。[31]

p. 161–162 布尔在发展其逻辑时，举例取材于包括当代经济学和神学在内的多个领域。他直截了当地陈述前提，不作讨论，而是专注于从中能得出哪些逻辑结论。通过这种方式，他展示了如何把日常语言中的逻辑论证形式化。他还进一步证明，所有亚里士多德的三段论形式都包含在他的方法之中。

p. 162 尽管书名如此，布尔几乎没有讨论上文所说的三条思维规律。不过，他把“x 且 x=x”称为“思维的基本规律”（第 49 页），并由此提出矛盾律，即任何事物都不可能既是 x 又不是 x。

虽然布尔对经院逻辑持批判态度，但他承认其重要性。他的概念化仍把关于事物的推理与关于事物集合的推理混为一谈，而且他的符号体系虽巧妙，却显得笨拙。尽管如此，他使推理链中实现了算术上的确定性：也就是说，在布尔那里，逻辑真正成为了一门计算的学科，这正是霍布斯所追求的。

布尔的新颖方法遭到了反对，而基于三段论格式的亚里士多德式逻辑与数学的分离，一直延续到 20 世纪相当长的一段时期。尽管如此，尽管布尔的体系存在不足，但他的工作已被证明具有奠基性意义，并标志着与长期存在的亚里士多德传统发生了根本性的决裂。

While Leibniz (1646–1716) had explored the formalization of logic well before Boole, his work was lost until the early twentieth century.[28] Thus, Boole (1815 – 64) is now recognized as the progenitor of modern logic, particularly propositional calculus. In *The Laws of Thought*,[29] Boole’s intention was

to investigate the fundamental laws of those operations of the mind by which reasoning is performed; to give expression to them in the symbolical language of a Calculus, and upon this foundation to establish the science of Logic and construct its method; (p1)

Boole thought that the laws of human thought were quite literally mathematical:

There is not only a close analogy between the operations of the mind in general reasoning and its operations in the particular science of Algebra, but there is to a considerable extent an exact agreement in the laws by which the two classes of operations are conducted. (p6)

Boole’s key insight was that if a Universe of discourse is represented as ‘1’, and Nothing as ‘0’, then logical operations on classes of things have arithmetical equivalents. Hence, algebraic techniques may be deployed to manipulate them.

It is straightforward to read off propositional logic from Boole’s system, where 1 is true and 0 is false. See Table 7.3, which may be contrasted with Table 7.1.[30]

The information theorist Claude Shannon noted in 1938 that the correspond ence between Boole’s calculus and logic could be applied to switching circuits.[31]

Boole developed his logic with examples drawn from a range of domains, including contemporary economics and theology. He stated premises baldly, without discussion, and focused on what might be logically concluded from them. In so doing, he showed how logical arguments in everyday language might be formalized. He further demonstrated that all the Aristotelian syllogistic figures were subsumed by his approach.

Despite his book’s title, Boole barely discussed the three laws of thought considered above. However, he called *x and x = x* the ‘fundamental law of thought’ (p49), *and* used it to develop the *principle of contradiction*, that is, nothing can be both *x and* not *x*.

While Boole was critical of scholastic logic, he acknowledged its importance. His conceptualization still conflated reasoning about things and about collections of things, and his notation, though cunning, is clumsy. Still, he enabled arithmetic certainty in chains of reasoning: that is, with Boole, logic truly became a matter of computation, as Hobbes had sought.

Boole’s novel approach met with opposition, and the Aristotelian separation of logic based on syllogistic figures from mathematics persisted well into the twentieth century. Nonetheless, despite infelicities in Boole’s system, his work has proved foundational, and marked

a fundamental break with the longstanding Aristotelian tradition.

弗雷格与数学的基础

Frege and the foundations of mathematics

德国数学家戈特洛布·弗雷格（1848—1925）把布尔的进路倒转过来，试图把算术、继而把数学建立在逻辑之上。支撑计算的数理逻辑，最终源自弗雷格的工作。

The German matematician Gottlob Frege (1848–1925) turned Boole's work on its head, and sought to found arithmetic, and then mathematics, on logic. The mathematical logic that underpins computing ultimately flows from Frege's.

弗雷格的《概念文字》[32]、[33]出版于 1879 年，其副标题是：

> 一种以算术为范本的、用于纯粹思维的公式语言

Frege's *Begriffsschrift*,[32],[33] published in 1879, was subtitled:

> a formula language, modeled upon that of arithmetic, for pure thought

对“纯粹思维”的强调，完全承袭了亚里士多德把推理与被推理对象相分离的传统。

The emphasis on 'pure thought' is very much in the Aristotelian tradition of separating reasoning from that which is reasoned about.

在此，弗雷格提出了一系列根本性的创新。虽然他像布尔一样使用了变量和算术运算符，但他用“论元”和“函数”的概念取代了亚里士多德的“主项”和“谓项”概念。此外，他把所有规则建立在实质蕴含和否定之上，展示了如何由此推导出其他逻辑运算符以及所有三段论形式。最后，他引入了一个全称量词，即用于表述某个类中所有成员都具有某种性质的运算符。

Here, Frege introduced a number of fundamental innovations. While, like Boole, he used variables and arithmetic operators, he replaced the Aristotelian notions of subject and predicate with those of *argument* and *function*. Further, he based all rules on material implication and negation, showing how to derive the other logical operators, and all syllogistic forms, from them. Finally, he introduced an operator for universal quantification, that is, for talking about all members of a class having some property.

在 1884 年出版的《算术基础》[34]中，弗雷格详细阐述了其逻辑主义数学哲学。他明确试图把逻辑与主观哲学（他称之为心理学）区分开来，并主张逻辑与数学之间存在紧密联系。然而，他也批评数学接受不完整的证明，要求每一步都应明确说明。如今，这已成为基于计算机的证明的一个特征。

In *The Foundations of Arithmetic*,[34] from 1884, Frege elaborated his *logicist* philosophy of mathematics. He explicitly sought to disassociate logic from subjective philosophy, which he called psychology, and asserted a strong connection with mathematics. However, he also criticized mathematics for accepting incomplete proofs, requiring that every step should be made explicit. Today, this is a characteristic of computer-based proofs.

在 1893 年出版的《算术基本定律》[35]中，弗雷格坦率地指出，逻辑无法通过诉诸外部依据来获得正当性：

In *Basic Laws of Arithmetic*,[35] published in 1893, Frege was frank that logic could not be justified by external appeal: p. 163

> 我们为何以及凭什么权利承认一条逻辑定律为真，逻辑只能通过把它归结为另一条逻辑定律来回答。若无法做到这一点，逻辑便无法给出答案。（第 15 页）

> The question why and with what right we acknowledge a logical law to be true, logic can answer only by reducing it to another law of logic. Where that is not possible, logic can give no answer. (p15)

数与归纳

Numbers and induction

亚里士多德把三段论推理即演绎推理，与归纳推理区分开来；后者是在事物集合中识别模式。朱塞佩·皮亚诺（1858—1932）把它置于形式化基础之上，统一了数、集合和归纳的概念。

Aristotle distinguished syllogistic, or *deductive* reasoning from *inductive* reasoning, that is, identifying patterns in collections of things. This was placed on a formal footing by Giuseppe Peano (1858–1932), who unified notions of number, set and induction.

皮亚诺在 1889 年提出的方法，[36]基于一个很简单的相继数概念：从一开始，[37]反复加一。接着，与布

Peano's approach,[36] from 1889, was based on a very simple conception of *successive* numbers, starting with

尔和弗雷格相似，他把具有某种性质看作属于一个有明确特征的集合。然后，通过把数集合的成员看作一个序列并考察其共同性质，他定义了数集合上的归纳。

one[37] and repeatedly adding one. Peano next presented having a property as akin to being in a collection with well-defined characteristics, much like Boole and Frege. He then defined induction over collections of numbers by considering common properties of their members regarded as a sequence.

一般而言，对于一个归纳证明，我们假设某个性质对 1 成立。然后，设 N 为任意一个数。如果假设该性质对 N 成立，且我们能证明它对 N+1 也成立，那么该性质必然对所有数都成立，因为我们可以从 1 开始逐步推导到任意一个数。也就是说，该性质是数列的一种归纳模式。归纳法在数学的许多领域中都占据核心地位，与传统的反证法相辅相成。它也是递归这一基本编程技术的基础，在递归中，对集合的计算是通过针对子集合的计算来定义的，一直推导到空集合为止。

In general, for an inductive proof, we assume that some property holds for 1. Then, suppose *N* is an arbitrary number. If assuming that the property holds for *N*, we can prove that it holds for *N* +1, then the property must hold for all numbers, as we can work our way forward from 1 to any number. That is, the property is an inductive pattern for the number sequence. Inductive proof is central to many areas of mathematics, complementing the traditional proof by contradiction. It is also the basis of the fundamental programming technique of *recursion*, where computations over collections are defined in terms of computations over sub-collections, down to an empty collection.

与演绎法一样，归纳法是在我们已有一个待推理的性质时，将其形式化的过程。首先提出一个归纳性质——即识别假设模式——则与数学实践密切相关。

As with deduction, induction is a process of formalizing a property once we have one to reason about. Coming up with an inductive property in the first place, that is, identifying hypothetical patterns, is to do with the practice of mathematics.

无限与无穷小

Infinity and infinitesimals

归纳似乎为无限制地计数提供了依据。那么，应如何刻画这种看似无限的情形？亚里士多德在《形而上学》[38]中区分了潜无限与实无限：

Induction appears to justify counting without limit. How then might we characterize such seeming infinity? In the *Metaphysics*,[38] Aristotle distinguishes between *potential* and *actualized* infinities:

p. 163–164

无限、虚空和其他这类概念，被说成潜在地或现实地“存在”，其意义不同于大多数存在者，例如看见者、行走者或被看见者。在后面这些情形中，这种断言有时可以不加限定地为真，因为“被看见者有时因正在被看见而得名，有时因能够被看见而得名；但无限的潜在存在，并不意味着它将来会在现实中独立存在；它只有在认识中才可分离。”（第 IX 卷，第 6 部分，第 447 页）

〔校注：英文底本 sometimes be truly made have 含多余 have，英文照录，中文按可理解的句意连接。〕

Infinity and void and other concepts of this kind are said to ‘be’ potentially or actually in a different sense from the majority of existing things. e.g. that which sees, or walks, or is seen. For in these latter cases the predication may sometimes be truly made have without qualification, since ‘that which is seen is so called sometimes because it is seen and sometimes because it is capable of being seen: but the Infinite does not exist potentially in the sense that it will ever exist separately in actuality it is separable only in knowledge.’ (Book IX, Part 6, p447)

p. 164

埃利亚的芝诺（公元前 495—430 年）提出了一些悖论，围绕事物可以无限地分割、因而分成无限多个部分这一可能性展开。[39]亚里士多德在《物理学》中详细讨论了这类悖论。在《形而上学》中，他用潜无限与实无限的区别质疑无限分割：

Zeno of Elea (495–430 BC) had constructed paradoxes that revolve around being able to divide things indefinitely and hence into an infinite number of components.[39] Aristotle dealt with such paradoxes at length in *Physics*. In *Metaphysics*, he deployed the distinction between potential and actual infinities to contest indefinite division:

因为分割过程永不停止这一事实，使这种现实性以潜在方式存在，却不独立存在。（第 IX 卷，第 6 部分，第 447 页）

For the fact that the process of division never ceases makes this actuality exist potentially, but not separately. (Book IX, Part 6, p447)

牛顿和莱布尼茨的微积分提出了尖锐的可分性问题。[40]具体而言，积分涉及在相邻取值的差异趋于零时，对函数值进行求和。请看图 7.1，其中画出了某个函数 y=f(x) 的曲线。

The calculus of Newton and Leibniz raised acute problems of divisabilty.[40] In particular, integration involves summing the values of a function for successive values

可以把曲线下的区域分成宽度为 dx、高度为 $f(x_i)$ 的矩形，再将面积相加，以近似曲线下的面积：

of vanishingly small differences. Consider Figure 7.1, which shows the curve for some function y = f(x).

We can approximate the area under the curve by dividing it into rectangles of width dx and height f(xi), and summing the areas:

$$f(0) * dx + f(x_1) * dx + f(x_2) * dx + \cdots + f(x_i) * dx \cdots$$

当然，对于每个矩形，其与曲线之间仍存在一个近似三角形的区域，该区域未被计入。这些区域累积起来便形成了一个误差值。

Of course, for each rectangle, there remains a vaguely triangular shape between it and the curve, which is not accounted for. These accumulate as an error value.

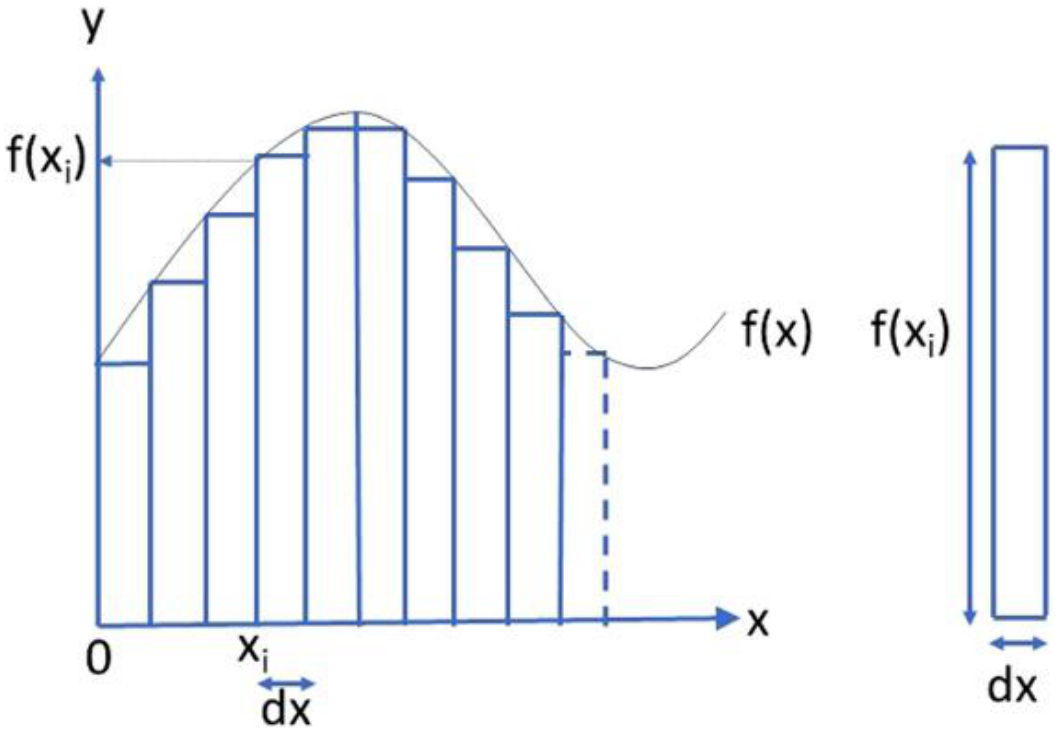


图 7.1 通过面积求和进行积分。作者绘图。

Figure 7.1 Integration by summing areas. Figure by authors.
x、y：坐标轴；dx：矩形宽度；f(xᵢ)：矩形高度；f(x)：函数曲线。

如果我们能使 dx 变小，那么误差就会变小。dx 能变得多小？

If we can make dx smaller, then the error becomes smaller. How small can dx become? p. 165

这归根结底取决于积分起始值与终值之间有多少个值。如果值有无限多个，那么它们之间的差就是零。但这样就会出现一个悖论：把无限多个零相加却得到一个非零的结果——这相当于芝诺所探讨的关于时间和空间的无限分割悖论。

〔校注：“有无限多个值，则它们之间的差为零”为底本论述，照译；并非据此将任意两个不同实数的差视为零。〕

This boils down to how many values there are in between the start and end values of the integration. If there are an infinite number, then the difference between them is zero. But then we have a paradox of summing an infinity of zeros giving a non-zero result: equivalent to the paradoxes of repeatedly dividing time and space that Zeno explored.

实际的解决方法是，把函数对越来越小的 dx 值进行求和，直到相邻两次求和结果之间的差值小到可以忽略不计为止。也就是说，积分会收敛到某个可接受的值。对于某些类别的函数，存在精确的解析解，所得值被解释为“在等价差之和的极限处”的值。然而，该极限通常被视为位于无穷远处，即仿佛积分范围是一个实无限。

The practical solution is to sum the function for smaller and smaller values of dx until the difference between successive sums is small enough not to be concerned about. That is, the integration converges towards some acceptable value. There are precise analytic solutions for some classes of function, and the resulting values are explained as ‘at the limit’ of the equivalent sums of differences. However, the limit is often treated as if it is at infinity, that is, as if the range of integration is an actualized infinity.

通向无限，乃至更远

To infinity, and beyond

对无穷小量和无穷大的表征是格奥尔格·康托尔（1845—1918）工作的核心，该工作为 19 世纪末至 20 世纪初数学逻辑危机奠定了基础。康托尔的研究历经数十年不断完善。泰尔斯对此进行了详尽的阐述。[41]

The characterization of infinitesimals and infinity was central to the work of Georg Cantor (1845–1918), which underpinned the crisis in mathematical logic in the late nineteenth and early twentieth centuries. Cantor's work was refined over several decades.Tiles provides a thorough account.[41]

首先，我们把整数（可用有限位数列表示）与有理数区分开来。有理数虽可表示为两个整数的比值，但若尝试把分子除以分母，其结果可能呈现无限展开。例如：

First of all, we will distinguish the integers, which have finite representations as sequences of digits, from the *rational* numbers, which have finite representations as the ratios of two integers, but may have infinite expansions if an attempt is made to divide the numerator by the denominator. For example:

$$10/3 \Longrightarrow 3.3333333333\ldots$$

不能表示成两个整数之比的数称为实数。例子有 π 和 e。这些数也有无限展开式，但以计算它们的公式来表示，例如对有理数级数求和。

〔校注：底本将不能表示成两个整数之比的数称为 real numbers（实数），通常数学术语应为 irrational numbers（无理数）；正文照录原述。〕

Those numbers which cannot be represented as the ratio of two integers are called the *real* numbers. Examples are π and *e*. Such numbers also have infinite expansions, but are expressed as formulae for calculating them, for example, by summing series of rational numbers.

现在，如果允许无界除法，那么两个有理数之间似乎存在无穷多个实数。康托尔试图利用集合论来刻画这一连续统。

Now, if unbounded division is acceptable, then there appears to be an infinity of real numbers between two rational numbers. Cantor sought to characterize this *continuum* using set theory.

我们把集合写在花括号 { 和} 之间，用逗号分隔元素，例如，偶数集：

We will write sets as between the braces { and }, with elements separated by commas, for example, a set of even numbers:

$$\{2, 4, 6, 8, 10\}$$

空集为 {}。

The empty set is {}.

我们区分序数和基数。序数可作为有序集合中的下标。因此，对于偶数集，第一个元素是 2，第二个元素是 4，第三个元素是 6，依此类推。这里我们使用序数 1、2、3……按顺序选取集合中的元素。基数用于表示集合的大小。因此，上述偶数集合的基数为 5。

We distinguish *ordinal* and *cardinal* numbers. Ordinal numbers may be used as indices in ordered sets. So, for the set of even numbers, the first element is 2, the second element is 4, the third element is 6 and so on. Here we use the ordinals 1, 2, 3 . . . to select elements of the set in order. Cardinal numbers are used for sizes of set. So the cardinality of the set of even numbers above is 5.

p. 166

我们进一步把集合分为有限集和无限集。对于有限集（如上文的偶数集），其基数即为最后一个元素的序数。但对于无限集，其基数不能是序数，因为根据定义，无限集不存在最后一个元素。

〔校注：底本称无限集的基数不能是序数，照译；在通常集合论中基数可用初始序数表示，原文论述不应当作这一点的否定证明。〕

We further distinguish between *finite* and *infinite* sets. For *finite* sets, like that of even numbers above, the cardinality is also the ordinal number for the last element. But, for an *infinite* set, the cardinality can't be an ordinal number, as there is no last element by definition.

虽然显然无法在有限时间内构造出一个无限集合，但某些无限集合可以给出有限的特征描述。特别是，我们可以指定算法来枚举某些无限集合，即生成其相继的成员。而算法本身就是一种有限的、物质化的描述。

While an infinite set clearly cannot be constructed in finite time, some infinite sets can be given finite characterizations. In particular, we can specify *algorithms* to *enumerate* some infinite sets, that is, to generate successive members. And an algorithm is a finite materialized description.

例如，数的无限集合可以通过从 1 开始并反复加 1 来枚举。请注意，前面的句子是一个有限的描述。然而，它指定了一种潜在的无限计算，这种计算无法由一个必然是有限的物质化系统完成。

For example, the infinite set of numbers can be enumerated by starting at 1 and repeatedly adding 1. Note that the previous sentences is a finite description. However, it specifies a potentially infinite computation,

which cannot be completed by a materialized system, which must necessarily be finite.

如果一个集合的元素能与某个整数集建立一一对应关系，则该集合被称为可数集。例如，对于我们这个由五个偶数组成的集合：

A set is said to be *countable*, or *denumerable*, if its members can be put into one-to-one correspondence with some set of integers. For example, for our set of five even numbers:

$$1 \leftrightarrow 2$$
$$2 \leftrightarrow 4$$
$$3 \leftrightarrow 6$$
$$4 \leftrightarrow 8$$
$$5 \leftrightarrow 10$$

因此，从这个意义上说，一个可枚举的集合是可数的，尽管它可能是无限的，因为其元素可以与无限的整数集建立一一对应关系。例如，要生成偶数集，只需把整数依次翻倍。因此，所有偶数的集合可以与所有整数的集合建立一一对应关系。这似乎有悖直觉，因为在有限的整数集中，只有一半是偶数。

Thus, an enumerated set is in this sense countable, even though it may be infinite, as its members can be put into one-to-one correspondence with the infinite set of integers. For example, to generate the set of even numbers, successively double the integers. So the set of all even numbers can be put into one to one correspondence with the set of all integers. This may seem counterintuitive, as only half of a finite set of integers are even.

许多其他由整数派生出来的无限集合都可以与整数建立一一对应关系。这包括有理数，我们可以把它们系统地列举为相继的整数比值。

Many other infinite sets derived from the integers can be put into one-to-one correspondence with them. This includes the rational numbers, which we can systematically enumerate as successive ratios of integers.

为了比较无限集合的性质，康托尔使用希伯来字母 ℵ（aleph），配上相继的下标，以超限数表示这些集合的基数。因此，整数集的基数为 $\aleph_0$（aleph 零），所有可数集也是如此。

〔校注：“所有可数集”均为 ℵ₀ 是底本原述；通常术语下有限集也可数，此处实际讨论可数无限集。〕

In order to compare properties of infinite sets, Cantor used the Hebrew symbol ℵ (*aleph*) with successive subscripts to denote their cardinalities as *transfinite* numbers. Thus, the set of integers has cardinality $\aleph_0$ (*aleph* nought), as do all countable sets.

然而，康托尔借助对角线论证，试图证明实数不可枚举，因而不可数。首先，实数不能表示为两个整数之比，因此必定有无限多个小数位。每一个小数位都可以与整数一一对应，所以可用序数为其编号。假设我们能枚举实数，第 i 个实数 d^i 的各位数字便是 d^i_1, d^i_2, d^i_3，依此类推，如表 7.4 所示。

〔校注：这里再次沿用底本把实数与无理数混同的说法，未静默更改。〕

However, using a *diagonal* argument, Cantor sought to demonstrate that the real numbers are not enumerable and, hence, not countable. First of all, a real number can’t be expressed as a ratio of two integers, so it must have an infinity of decimal digits. Each of those decimal digits can be put into one-to-one correspondence with the integers, so we can index them with ordinal numbers. Suppose we could enumerate the real numbers, so the digits of the ith real number d^i were d^i_1, d^i_2, d^i_3 and so on, as in Table 7.4.

可以构造一个新的实数 $0.n^1n^2n^3n^4\ldots$，使其第一位 n^1 不同于第一个实数的第一位 d^1_1，第二位 n^2 不同于第二个实数的第二位 d^2_2，第三位 n^3 不同于第三个实数的第三位 d^3_3，依此类推。这个新实数不可能在所枚举的序列中，所以实数必定多于 $\aleph_0$，因而不可枚举。

We can make a new real number $0.n^1n^2n^3n^4\ldots$ where the first digit n^1 is different from the first digit of the first real d^1_1, the second digit n^2 is different from the second digit of the second real d^2_2, the third digit n^3 is different from the third digit of the third real d^3_3 and so on. This new real can’t be in the enumerated sequence, so there must be more real numbers than $\aleph_0$, so the reals are not enumerable.

p. 166–167

表 7.4 对角线法

p. 167

1.	0.	d^1_1	d^1_2	d^1_3	d^1_4	⋯
2.	0.	d^2_1	d^2_2	d^2_3	d^2_4	⋯
3.	0.	d^3_1	d^3_2	d^3_3	d^3_4	⋯
4.	0.	d^4_1	d^4_2	d^4_3	d^4_4	⋯
⋯						

Table 7.4 Diagonalization

1.	0.	d_1^1	d_2^1	d_3^1	d_4^1	⋯
2.	0.	d_1^2	d_2^2	d_3^2	d_4^2	⋯
3.	0.	d_1^3	d_2^3	d_4^3	d_4^3	⋯
4.	0.	d_1^4	d_2^4	d_3^4	d_4^4	⋯
⋯						

校注：底本第三行第三个数字位置印为 d^3_4，与第四个数字重复，按原图保留；对角线论证正文仍为 d^3_3。

现在，除了整数集之外，构建集合的集合似乎也是合理的。特别是，给定一个集合，我们可以构造一个包含其所有子集的集合，称为幂集。我们从空集开始，然后依次添加所有单个元素、元素对、元素三元组等等，最终包含整个集合。参见表 7.5。

Now, as well as sets of integers, it seems legitimate to make sets of sets. In particular, given a set, we can construct a set of all of its subsets, known as the *power set*. We start with the empty set {}, and then add all the single elements, pairs of elements, triples of elements and so on, ending with the whole set. See Table 7.5.

表 7.5 幂集

集合	序数	幂集	基数
{1}	1	{{},{1}}	2
{1,2}	2	{{},{1},{2},{1,2}}	4
{1,2,3}	3	{{},{1},{2},{3},{1,2},{1,3},{2,3},{1,2,3}}	8
⋯			

Table 7.5 Power Sets

set	ordinality	power set	cardinality
{1}	1	{{},{1}}	2
{1,2}	2	{{},{1},{2},{1,2}}	4
{1,2,3}	3	{{},{1},{2},{3},{1,2},{1,3},{2,3},{1,2,3}}	8
⋯			

幂集的基数总大于原集合，即成员更多：N 个元素的集合，其幂集有 2^N 个元素。特别地，若整数集基数为 $\aleph_0$，则整数幂集基数为 $2^{\aleph_0}$，记作 $\aleph_1$（aleph 一）。因此，整数的幂集不可数，因为 $\aleph_1$ 必然大于 $\aleph_0$。康托尔随后证明，实数集也有基数 $\aleph_1$。

〔校注：底本将 $2^{\{\aleph_0\}}$ 与 $\aleph_1$ 直接等同并称康托尔证明实数基数为 $\aleph_1$；这涉及连续统假设，不能作为一般集合论中已证明的结论。正文保留作者说法。〕

A power set always has greater cardinality, that is, more members, than the original set: for a set of N elements, the power set has 2^N elements. In particular, if the set of integers has cardinality $\aleph_0$, then the power set of integers has cardinality $2^{\aleph_0}$, denoted $\aleph_1$ (*aleph one*). Thus, the powerset of integers is uncountable, as $\aleph_1$ is necessarily bigger than $\aleph_0$. Cantor then demonstrated that the set of real numbers also has cardinality $\aleph_1$.

从亚里士多德的视角来看，康托尔的对角线论证是不成立的，因为它预设了任意精度下已实现的实数无限集。对于一个以固定精度表示的有限“实数”集，对角线论证所产生的数本就已存在。

From an Aristotelian perspective, Cantor’s diagonalization argument is illegitimate because it presupposes an actualized infinity of real numbers to arbitrary precision. For a finite set of ‘real’ numbers, represented to fixed precision, diagonalization produces a number which is already present.

第 7 章原注 / Chapter 7 Notes

↑1. Aristotle:《形而上学》,第 I—IX 卷,H. Tredennick 英译,洛布古典丛书(William Heinemann,1933),第 161 页。

1 Aristotle, *The Metaphysics, Books I-IX*, trans. H. Tredennick, Loeb Classical Library (William Heinemann, 1933), 161. p. 254

↑2. 正如其官方格言所宣示的:“我们信仰上帝”(In God We Trust)。

2 *In God We Trust*, as their official motto proclaims. p. 254

↑3. Wikipedia:“圣公宗”(Anglican Communion),载 Wikipedia(Wikipedia,2021),https://en.wikipedia.org/wiki/Anglican_Communion 。

3 *Wikipedia*, ‘Anglican Communion’, in *Wikipedia* (*Wikipedia*, 2021), https://en.wikipedia.org/wiki/Anglican_Communion. p. 254

↑4. Statista:“中国共产党”(Chinese Communist Party),载 Topics(Statista,2021),https://www.statista.com/topics/1247/chinesecommunist-party/ 。

4 Statista, ‘Chinese Communist Party’, in *Topics* (Statista, 2021), https://www.statista.com/topics/1247/chinesecommunist-party/. p. 254

↑5. D. Lecourt:《无产阶级科学?李森科事件》(Proletarian Science? The Case of Lysenko)(Verso,1977)。

5 D. Lecourt, *Proletarian Science? The Case of Lysenko* (Verso, 1977). p. 254

↑6. Cockshott、Mackenzie 与 Michaelson:《计算及其极限》(Computation and its Limits)。

6 Cockshott, Mackenzie, and Michaelson, *Computation and its Limits*. p. 254

↑7. 不过,也可能通过中间的逻辑论证来确定。

7 Though perhaps through intermediate logical arguments. p. 254

↑8. G. Boole:《思维规律研究》(An Investigation into the Laws of Thought)(Walton & Maberly,1854)。

8 G. Boole, *An Investigation into the Laws of Thought* (Walton & Maberly, 1854). p. 254

↑9. 后文将进一步探讨这与辩证唯物主义的关系;但请注意,把同一词语的位置互换、分别用作形容词或名词,是有意义的。

9 We’ll later explore how this is related to dialectical materialism, but note that the transposed word use as adjective or noun is significant. p. 254

↑10. Aristotle:《修辞学》(The Art of Rhetoric),J. H. Freese 英译,洛布古典丛书(William Heinemann,1926)。

10 Aristotle, *The Art of Rhetoric*, trans. J. H. Freese, Loeb Classical Library (William Heinemann, 1926). p. 254

↑11. Aristotle:《前分析篇》(Prior Analytics),H. Tredennick 英译,洛布古典丛书(William Heinemann,1962)。

11 Aristotle, *Prior Analytics*, trans. H. Tredennick, Loeb Classical Library (William Heinemann, 1962). p. 254

↑12. Aristotle:《形而上学》,第 I—IX 卷(The Metaphysics, Books I–IX)。

12 Aristotle, *The Metaphysics, Books I–IX*. p. 254

↑13. Plato:《理想国》(The Republic),P. Shorey 英译,第 1 卷,洛布古典丛书(Harvard,1937);Plato:《理想国》,P. Shorey 英译,第 2 卷,洛布古典丛书(Harvard,1962)。

13 Plato, *The Republic*, trans. P. Shorey, vol. 1, Loeb Classical Library (Harvard, 1937); Plato, *The Republic*, trans. P. Shorey, vol. 2, Loeb Classical Library (Harvard, 1962). p. 254

↑14. R. Smith:“亚里士多德的逻辑”(Aristotle’s Logic),载 Plato(Stanford Encyclopedia of Philosophy,2017),https://plato.stanford.edu/entries/aristotle-logic/#AriLogWorOrg 。

14 R. Smith, ‘Aristotle’s Logic’, in *Plato* (Stanford Encyclopedia of Philosophy, 2017), https://plato.stanford.edu/entries/aristotle-logic/#AriLogWorOrg. p. 254

↑15. Aristotle:《解释篇》(On Interpretation),H. P. Cook 英译,洛布古典丛书(William Heinemann,1962)。

15 Aristotle, *On Interpretation*, trans. H. P. Cook, Loeb Classical Library (William Heinemann, 1962). p. 254

↑16. Aristotle:《前分析篇》(Prior Analytics)。

16 Aristotle, *Prior Analytics*. p. 255

↑17. J. Marenbon 编:《剑桥波爱修斯指南》(The Cambridge Companion to Boethius)(Cambridge University Press,2009)。

17 J. Marenbon, ed., *The Cambridge Companion to Boethius* (Cambridge University Press, 2009). p. 255

↑18. J. Buroker:“波尔—罗亚尔逻辑”(Port Royal Logic),载 Stanford Encyclopedia of Philosophy(University of Stanford,2014),https://plato.stanford.edu/entries/port-royal-logic/ 。

18 J. Buroker, ‘Port Royal Logic’, in *Stanford Encyclopedia of Philosophy* (University of Stanford, 2014), https://plato. stanford.edu/entries/port-royal-logic/. p. 255

p. 255 ↑19. F. Bacon：《新工具》（Novum Organon），J. Devey 编（P. F. Collier & Son，1901）。

19 F. Bacon, *Novum Organon,* ed. J. Devey (P. F. Collier & Son, 1901).

p. 255 ↑20. 这一标题呼应亚里士多德的《工具论》（Organon）。

20 Riffing on Aristotle’s *Organon*.

p. 255 ↑21. T. Hobbes：“论物体”（De Corpore），载《托马斯·霍布斯英文著作集》（The English Works of Thomas Hobbes），W. Molesworth 编（John Bohn，1839）。

21 T. Hobbes, ‘De Corpore’, in *The English Works of Thomas Hobbes*, ed. W. Molesworth (John Bohn, 1839).

p. 255 ↑22. R. Whately：《逻辑要素》（Elements of Logic）（James Munroe & Company，1845）。

22 R. Whately, *Elements of Logic* (James Munroe & Company, 1845).

p. 255 ↑23. Boole：《思维规律研究》（An Investigation into the Laws of Thought）。

23 Boole, *An Investigation into the Laws of Thought*.

p. 255 ↑24. W. S. Jevons：《逻辑学初级教程》（Elementary Lessons in Logic）（Macmillan & Co.，1903）。

24 W. S. Jevons, *Elementary Lessons in Logic* (Macmillan & Co., 1903).

p. 255 ↑25. 不要与爱尔兰数学家 William Rowan Hamilton 混淆。

25 Not to be confused with the Irish mathematician William Rowan Hamilton.

p. 255 ↑26. W. Hamilton：《形而上学与逻辑讲义》（Lectures on Metaphysics and Logic），第 1 卷，H. L. Mansel 与 J. Veitch 编（William Blackwood and Sons，1866）。

26 W. Hamilton, *Lectures on Metaphysics and Logic*, vol. 1, ed. H. L. Mansel and J. Veitch (William Blackwood and Sons, 1866).

p. 255 ↑27. L. Wittgenstein：《逻辑哲学论》（Tractatus Logico-Philosophicus），D. F. Pears 与 B. F. McGuinness 英译（Routledge and Kegan Paul，1961），第 32—33 页。

27 L. Wittgenstein, *Tractatus Logico-Philosophicus*, trans. D. F. Pears and B. F. McGuinness (Routledge and Kegan Paul, 1961), 32–3.

p. 255 ↑28. W. Lenzen：“莱布尼茨：逻辑”（Leibniz: Logic），载 Internet Encyclopedia of Philosophy（University of Tennessee at Martin，2017），`https://iep.utm.edu/leib-log/`。

28 W. Lenzen, ‘Leibniz: Logic’, in *Internet Encyclopedia of Philosophy* (University of Tennessee at Martin, 2017), `https://iep.utm.edu/leib-log/`.

p. 255 ↑29. Boole：《思维规律研究》（An Investigation into the Laws of Thought）。

29 Boole, *An Investigation into the Laws of Thought*.

p. 255 ↑30. 请注意，布尔没有使用真值表。不过，他在第 76 页实际上给出了否定的真值表。

30 Note that Boole did not use truth tables. However, on p76 he effectively presents the truth table for negation.

p. 255 ↑31. C. E. Shannon：“继电器与开关电路的符号分析”（A Symbolic Analysis of Relay and Switching Circuits），American Institute of Electrical Engineers Transactions，第 57 卷（1938），第 713—723 页。

31 C. E. Shannon, ‘A Symbolic Analysis of Relay and Switching Circuits’, *American Institute of Electrical Engineers Transactions*, 57 (1938): 713–23.

p. 255 ↑32. 字面意思是“概念文字”。

32 Literally ‘conceptual writing’.

p. 255 ↑33. G. Frege：“概念文字：一种仿照算术语言构造的纯思维公式语言”（Begriffsschrift, a Formula Language, Modeled Upon that of Arithmetic, for Pure Thought），载《从弗雷格到哥德尔：数理逻辑文献集，1879—1931》（From Frege to Gödel: A Source Book in Mathematical Logic, 1879-1931），J. van Heijenoort 编，S. Bauer-Mengelberg 英译（Harvard University Press，1967）。

33 G. Frege, ‘Begriffsschrift, a Formula Language, Modeled Upon that of Arithmetic, for Pure Thought’, in *From Frege to Gödel: A Source Book in Mathematical Logic, 1879-1931*, ed. J. van Heijenoort, trans. S. Bauer-Mengelberg (Harvard University Press, 1967).

p. 255 ↑34. G. Frege：《算术基础》（The Foundations of Arithemtic），J. L. Austin 英译（Harper，1953）。

34 G. Frege, *The Foundations of Arithemtic*, trans. J. L. Austin (Harper, 1953).

p. 255 ↑35. G. Frege：《算术基本规律：体系的阐述》（The Basic Laws of Arithmetic: Exposition of the System），M. Furth 英译（University of California Press，1964）。

35 G. Frege, *The Basic Laws of Arithmetic: Exposition of the System*, trans. M. Furth (University of California Press, 1964).

p. 255 ↑36. G. Peano：“用新方法表述的算术原理”（The Principles of Arithemtic, Presented by a New Method），载《从弗雷格到哥德尔：数理逻辑文献集，1879—

36 G. Peano, ‘The Principles of Arithemtic, Presented by a New Method’, in *From Frege to Gödel: A Source*

1931》（From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931），J. van Heijenoort 编译（Harvard University Press，1967）。

↑37. 如今通常从零开始。

↑38. Aristotle：《形而上学》，第 I—IX 卷（The Metaphysics, Books I-IX）。

↑39. B. Dowden：“芝诺悖论”（Zeno’s Paradoxes），Internet Encyclopedia of Philosophy，`https://iep.utm.edu/zenos-paradoxes/` 。

↑40. M. Tiles：《集合论哲学：康托尔乐园的历史导论》（The Philosophy of Set Theory: An Historic Introduction to Cantor’s Paradise）（Dover，1989）。

↑41. Tiles：《集合论哲学》（The Philosophy of Set Theory）。

Book in Mathematical Logic, 1879–1931, ed. and trans. J. van Heijenoort (Harvard University Press, 1967).

37 It is now usual to start at zero.

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38 Aristotle, *The Metaphysics, Books I-IX*.

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39 B. Dowden, ‘Zeno’s Paradoxes’, *Internet Encyclopedia of Philosophy*, `https://iep.utm` .edu/zenos-paradoxes/.

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40 M. Tiles, *The Philosophy of Set Theory: An Historic Introduction to Cantor’s Paradise* (Dover, 1989).

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41 Tiles, *The Philosophy of Set Theory*.

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第 8 章逻辑与辩证唯物主义

Chapter 8: LOGIC AND DIALECTICAL MATERIALISM

黑格尔与逻辑

Hegel and logic

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……把三段论仅仅看作由三个判断构成，是一种形式主义观点；它忽视了各个规定之间的关系，而推理所涉及的恰恰只是这种关系。把各项之间的联系拆分为孤立的前提和与这些前提相区别的结论，完全是一种纯粹主观的反思：

所有人都会死。盖乌斯是人。因此盖乌斯会死。

一听到这种推理被搬出来，人立刻就会感到厌倦。这种厌倦来自一种形式的徒劳：它通过彼此分离的命题制造出多样性的幻象，而这种幻象在事实本身中立即消解了。G. 黑格尔，《逻辑学》[1]

格奥尔格·黑格尔（1770—1831）的唯心主义在欧洲哲学中占据核心地位。英国传统在复兴逻辑之前先与亚里士多德决裂；与之不同，辩证法处于黑格尔认识论的核心。

从亚里士多德的视角看，黑格尔著作《逻辑学》[2]的书名似乎有些奇怪。亚里士多德把辩证法与科学相对置，认为它是一种说服形式，而非自主理解世界如何运行的途径；他也把辩证法与三段论逻辑相对置，认为它是前提的来源，而非对这些前提进行推理的方法。尽管如此，黑格尔确实认为逻辑可以得到科学阐释，这一点仍有益处。

黑格尔对把三段论的形式与其应用语境相分离的价值深表怀疑。也就是说，三段论本身并无意趣。其重要性在于它如何将特定前提（规定）与特定结论联系起来，而这些联系又依赖先前的推理来确立前提之间的关系。正如我们将看到的，这种态度是辩证唯物主义中反复出现的特征。

尽管如此，黑格尔确实系统地探讨了三段论的各种形式，[3]并借助辩证转化来考察它们之间的关系。不过，他除以普遍（U）、单称（S）或特称（P）来标记三段论中的项之外，并未采用其他形式记号；这比早先的做法更为简化。与前人一样，黑格尔运用了简洁的论证和例子。

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《逻辑学》的序言进一步阐明了黑格尔对三段论逻辑的态度。首先，他指出科学和常识已经取代形而上学，也取代了“逻辑教人如何思考”这一观念。尽管如此，逻辑仍被保留在诸科学之中，“大概是为了某种形式上的效用”，虽然“它的形式和内容在漫长的传承中始终未变”（第 8 页）。

. . . to regard the syllogism as merely consisting *of three judgments* is a formalistic view that ignores the relation of the determinations which alone is at issue in the inference. It is altogether a merely subjective reflection that splits the connection of the terms into isolated premises and a conclusion distinct from them:

All humans are mortal. Gaius is a human. Therefore Gaius is mortal.

One is immediately seized by boredom the moment one hears this inference being trotted out, a boredom brought on by the futility of a form that, by means of separate propositions, gives the illusion of a diversity which is immediately dissolved in the fact itself. G. Hegel, *The Science of Logic*[1]

The idealism of Georg Hegel (1770–1831) is central to European philosophy. Unlike the British tradition, which broke with Aristotle before reviving logic, dialectics is at the heart of Hegel’s epistemology.

The title of Hegel’s book *The Science of Logic*[2] seems strange from an Aristotelian perspective. Aristotle counterpoises dialectics to science, as a form of persuasion rather than an autonomous way of understanding how the world works, and to syllogistic logic, as a source of premises rather than a method for reasoning about them. Still, it is salutary that Hegel does view logic as being amenable to scientific exegesis.

Hegel was highly sceptical about the value of separating the form of a syllogism from its context of application. That is, the syllogism itself is of no interest. Rather, its importance lies in how it connects particular premises (determinations) to particular conclusions, which in turn depend on prior reasoning to establish relationships among premises. As we shall see, this attitude is a recurrent feature of dialectical materialism.

Nonetheless, Hegel does systematically explore the syllogistic forms,[3] and considers how they relate to each other using dialectical transformations. But he deploys no formal notation other than denoting the terms in syllogisms as universal (U), singular (S) or particular (P), a simplification of the earlier practice. Like his predecessors, Hegel uses terse arguments and examples.

Hegel’s attitude towards syllogistic logic is further illuminated in the *Prologue* to *The Science of Logic*. First of all, he noted that science and commonsense had displaced metaphysics, as well as the notion that logic taught one how to think. Nonetheless, logic was retained among the sciences ‘probably for the sake of a certain formal utility’, though ‘its shape and content have remained the same throughout a long inherited tradition’ (p8).

对黑格尔而言，思想蕴含于语言之中，语言的形式决定思想；在这方面，德语显然更胜一筹（!）（《第二版序言》，第 12 页）。而且，逻辑在人类境况中占据核心地位：

逻辑对人来说是如此自然，甚至就是人的本性。但是，如果我们将自然本身，即物理领域，与精神领域相对置，那么就必须说：逻辑是贯穿人一切自然行为——感知、直观、欲求的方式，以及需要和冲动——的超自然要素；它由此使这些行为成为真正属人的东西，尽管还只是形式上属人的东西，使之成为表象和目的。（第 12 页）

在这里，他明确反对亚里士多德的立场：

亚里士多德在同一语境中说："人的本性在如此多的方面都处于束缚之中；但这门并非为任何效用而追求的科学，独自是自在自为地自由的，因而似乎并非人所能拥有。"（第 14 页，引用亚里士多德《形而上学》982b）

黑格尔则认为，恰恰相反，逻辑关乎抽象思维，所以先教给年轻人，具体事务则留待以后。且不论黑格尔未能意识到这种教育是何等的特权，他在这里反映了"三艺"与"四艺"的区分，这种区分当时想必仍然通行。

对黑格尔而言，逻辑的一大益处，是其普遍性使它能够发挥简约表达的作用。然而，脱离内容的逻辑无法达到真理，因此常识将其视为贫瘠无物而加以拒斥，也就不足为奇了：

对于规定推理规则的公式而言——推理实际上是知性的一项主要功能——健全的常识忽视了这些公式在认识中有其位置、必须得到遵守，并且是理性思维必不可少的材料；无论这种忽视多么错误，常识同样正确地认识到：这些公式不加区分地既为谬误服务，也为诡辩服务；无论如何定义真理，就更高的真理，例如宗教真理而言，它们都毫无用处。总的说来，它们只涉及知识的正确性，而不涉及知识的真理性。（第 18 页）

在这里，黑格尔的立场令人想起霍布斯。

黑格尔认为，形式与内容的分离是一种幻象。他在《导言》中进一步详述了这一立场。辩证唯物主义拒斥形式主义，即脱离内容独立研究逻辑的根源，便在于此。

For Hegel, thought is held in language and the form of language determines thought, for which German is clearly superior(!) (*Preface to Second Edition*, p12). And logic is central to the human condition:

So much is logic natural to the human being, is indeed his very *nature*. If we however contrast *nature* as such, as the realm of the physical, with the realm of the spiritual, then we must say that logic is the supernatural element that permeates all his natural behavior, his ways of sensing, intuiting, desiring, his needs and impulses; and it thereby makes them into something truly human, even though only formally human –makes them into representations and purposes. (p12)

Here, he explicitly disagrees with Aristotle's position:

'In so many respects', says Aristotle in the same context, 'is human nature in bondage; but this science, which is not pursued for any utility, is alone free in and for itself, and for this reason it appears not to be a human possession'. (p14, citing Aristotle, *Metaphysics*, 982b)

Hegel argues that, on the contrary, logic is about abstract thought which is why it is taught to young people, with concrete matters coming later. Here, quite apart from Hegel's unawareness of how privileged such education was, he reflects the distinction between the Trivium and Quadrivium, which must have still been current.

For Hegel, a major benefit of logic lies in its utility as an abbreviation because of its universality. However, because logic shorn of content cannot attain truth, it is hardly surprising that it has been rejected by common sense as barren:

Regarding the formulas that define the rules of inference which in fact is a principal function of the understanding, however mistaken healthy common sense might be in ignoring that they have their place in cognition where they must be obeyed, and also that they are essential material for rational thought, it has nonetheless come to the equally correct realisation that such formulas are indifferently at the service just as much of error as of sophistry, and that, however truth may be defined, so far as higher truth is concerned, for instance religious truth, they are useless –that in general they have to do only with the correctness of knowledge, not its truth. (p18)

Here, Hegel's position is reminiscent of Hobbes'.

Hegel argues that the separation of form and content is illusory. He goes on to considerably expand on this position in the *Introduction*. Herein lie the roots of the rejection of formalism, that is, the study of logic independently of content, in dialectical materialism.

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恩格斯与逻辑

Engels and logic

卡尔·马克思（1818—1883）和弗里德里希·恩格斯（1820—1895）的革命世界观以科学唯物主义为前提。马克思几乎没有论述辩证法或三段论逻辑。相比之下，恩格斯在晚期著作中多次提及逻辑。在 1877 年的《反杜林论》[4]中，他有力地断言：逻辑在

The revolutionary world view of Karl Marx (1818–83) and Frederic Engels (1820–95) is premised on scientific materialism. Marx wrote practically nothing about dialectics or syllogistic logic. Engels, in contrast, made a number of references to logic in his later work. In *Anti-*

与辩证法的结合中处于从属地位。首先，他重申亚里士多德的立场：一门科学一旦充分专门化，就会独立于任何范围更广的认识论。三段论逻辑——现在称为形式逻辑——以及辩证法，作为思维规律，独立于哲学：

在以往全部哲学中，还独立存在着的，就是关于思维及其规律的科学——形式逻辑和辩证法。其余一切都归入关于自然和历史的实证科学。（第 40 页）

这一表述在 1880 年的《社会主义从空想到科学的发展》[5]中得到重复，并在 1886 年的《路德维希 • 费尔巴哈和德国古典哲学的终结》[6]中再次重申。

回到《反杜林论》，恩格斯指出，数学必须从逻辑中借用公理，而这些公理是“最贫乏的思想内容的表达”（第 60 页）。他沿袭黑格尔的观点说，仅靠逻辑公理“收效甚微”，要取得进展就必须借助几何学（第 61 页）。他的主要批评是，杜林强调纯数学独立于经验，却不承认它是从现实中抽象出来的（第 61 页）。

恩格斯说，辩证法比形式逻辑更有力：

就连形式逻辑也首先是一种获得新结果、从已知推进到未知的方法；辩证法也是如此，而且更为显著。此外，由于辩证法突破了形式逻辑的狭隘视野，它包含着一种更全面的世界观的萌芽。（第 186 页）

此外，形式逻辑只对初等数学具有意义。在高等数学中，乃至一切新科学中，都需要辩证法才能取得进展：

p. 171–172

初等数学，即常量的数学，至少总体上是在形式逻辑的范围内活动的；变量的数学，其最重要的部分是微积分，本质上无非是辩证法在数学关系中的应用。在这里，与这一方法在新研究领域中的多种应用相比，单纯的证明问题显然退居次要地位。但是，几乎所有高等数学的证明，从微分学最初的证明起，从初等数学的立场看，严格说来都是错误的。当人们像这里一样，试图用形式逻辑来证明在辩证法领域中获得的结果时，这就是必然的。（第 186—187 页）

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然而，他把形式逻辑描述为涉及常量而非变量，这表明他存在根本性的误解。马克思对数学也持这种看法，下文将会谈到。恩格斯还流露出一种奇特的对数学的蔑视：

然而，数学家的抽象要求远不是现实世界必须遵守的规律。（第 75 页）

Dühring,[4] from 1877, he strongly asserted the subordination of logic in its union with dialectics. First of all, he reasserts Aristotle’s position that, once a science is sufficiently specialized, it becomes independent of any wider epistemology. Syllogistic logic, now termed formal, and dialectics, as the laws of thought, are independent of philosophy:

That which still survives, independently, of all earlier philosophy is the science of thought and its laws —formal logic and dialectics. Everything else is subsumed in the positive science of nature and history. (p40)

This formulation was repeated in the 1880 *Socialism: Utopian and Scientific*,[5] and reiterated in 1886 in *Ludwig Feuerbach and the End of Classical German Philosophy.*[6]

Returning to *Anti-Dühring*, Engels argues that mathematics has to borrow axioms from logic, which are‘expressions of the scantiest thought-content’(p60). Following Hegel, he says that logical axioms alone ‘do not cut much ice’, and to make progress it is necessary to draw on geometry (p61). His main plaint is that Dühring emphasizes the independence of pure mathematics from experience rather than acknowledging its abstraction from reality (p61).

Engels says that dialectics is more powerful than formal logic:

Even formal logic is primarily a method of arriving at new results, of advancing from the known to the unknown –and dialectics is the same, only much more eminently so; moreover, since it forces its way beyond the narrow horizon of formal logic, it contains the germ of a more comprehensive view of the world. (p186).

Further, formal logic is only significant for elementary mathematics. In higher mathematics, indeed, in all new science, dialectics is needed to advance:

Elementary mathematics, the mathematics of constant quantities, moves within the confines of formal logic, at any rate on the whole; the mathematics of variables, whose most important part is the infinitesimal calculus, is in essence nothing other than the application of dialectics to mathematical relations. In it, the simple question of proof is definitely pushed into the background, as compared with the manifold application of the method to new spheres of research. But almost all the proofs of higher mathematics, from the first proofs of the differential calculus on, are from the standpoint of elementary mathematics strictly speaking, wrong. And this is necessarily so, when, as happens in this case, an attempt is made to prove by formal logic results obtained in the field of dialectics. (p186–7)

Nonetheless, his characterization of formal logic involving constants rather than variables suggests a fundamental misunderstanding. This was also Marx’s view of mathematics, as shall see below. And Engels betrayed a curious contempt for mathematics:

The abstract requirement of a mathematician is, however, far from being a compulsory law for the world of reality. (p75)

最后，在他身后出版的《自然辩证法》[7]中，恩格斯重谈这些主题。他再次认为，辩证逻辑优于形式逻辑，因为它把分析与推理结合起来：

辩证逻辑与旧的、纯形式的逻辑不同：它不像后者那样，满足于列举思维的运动形式，即各种判断形式和结论形式，并把它们毫无联系地并列起来。相反，它使这些形式彼此推导出来，使一种形式从属于另一种形式，而不是把它们置于同等地位；它从较低的形式发展出较高的形式。（第 296 页）

恩格斯承认，尽管自然科学和历史科学不断发展，形式逻辑仍有一种本质性的、可以认识的特性。尽管如此，他又一次坚决拒斥数学抽象：

他们［数学家和自然科学家］指责黑格尔所做的事，即把抽象推到极端，他们自己却在大得多的规模上这样做。他们忘记了，全部所谓纯数学都涉及抽象，严格说来，它的一切量都是想象的；而一切抽象一旦推到极端，就会转化为无稽之谈，或转化为自身的对立面。（第 359 页）

在《自然辩证法》的笔记中，恩格斯区分了可以通过“物质直观”来证明的数学运算，以及只能通过演绎来证明的逻辑推演：

计算的理性——计算机器！——一种奇特的混淆：把数学运算与纯粹逻辑运算混在一起。数学运算可以作物质性的演示和证明，因为它们以直接的、即使是抽象的物质直观为基础；纯粹逻辑运算则只能通过演绎来证明，因而不可能具有数学运算所具有的确实性——而其中又有多少是错误的！积分机；参见安德鲁斯的演讲，《自然》，9 月 7 日，76.317。[8,9]

无限的本性也曾是恩格斯和杜林共同关心的问题。康托尔曾颇不耐烦地驳斥杜林关于无限的论述。[10]恩格斯也长篇批评了杜林的无限概念。不过，恩格斯把可数性与计数混淆了：

可是，“已经数过的无限数列”这个矛盾又该如何解释呢？只要杜林先生为我们表演一下把它数完的巧妙把戏，我们就能够更仔细地考察这个问题了。[11]

Finally, in the posthumously published *Dialectics of Nature*,[7] Engels revisited these themes. Once again, dialectical logic is superior to formal logic, because it integrates both analysis and reasoning:

Dialectical logic, in contrast to the old, merely formal logic, is not, like the latter, content with enumerating the forms of motion of thought, ie the various forms of judgment and conclusion, and placing them side by side without any connection. On the contrary, it derives these forms out of one another, it makes one subordinate to another instead of putting them on an equal level, it develops the higher forms out of the lower. (p296)

Engels acknowledges that, despite developments in natural and historical sciences, formal logic has an essential and knowable quality. Nonetheless, once again, Engels firmly rejects mathematical abstraction:

What they [mathematicians and natural scientists] charge Hegel with doing, viz., pushing abstractions to the extreme limit, they do themselves on a far greater scale. They forget that the whole of so-called pure mathematics is concerned with abstractions, that all its magnitudes, strictly speaking, are imaginary, and that all abstractions when pushed to extremes are transformed into nonsense or into their opposite. (p359)

In his notes for *Dialectics of Nature*, Engels distinguishes between mathematical operations, which could be proved by ‘material contemplation’, and logical deductions, which could only be proved by deduction:

Calculative reason –calculating machine! –Curious confusion of mathematical operations, which are capable of material demonstration, of proof because they are based on direct, even if abstract, material contemplation, with purely logical ones, which are capable only of proof by deduction, hence are incapable of the positive certainty possessed by mathematical operations—and how many of them wrong! Machine for integration; cf. Andrews’ speech, Nature, Sept. 7, 76.317[8,9] p. 172–173

The nature of infinity had also concerned both Engels and Dühring. Dühring’s writings about infinity, were rather tetchily dismissed by Cantor.[10] Engels, in turn, criticized at length Dühring’s notion of infinity. However, Engels confuses countabilty with counting: p. 173

But what of the contradiction of ‘the counted infinite numerical series’? We shall be in a position to examine this more closely as soon as Herr Dühring has performed for us the clever trick of counting it.[11]

狄慈根、辩证唯物主义与逻辑

Dietzgen, dialectical materialism and logic

约瑟夫·狄慈根（1828—1888）在 1848 年德国革命期间结识马克思，后来成为他的朋友。狄慈根自学成才，恩格斯在《路德维希·费尔巴哈和德国古典哲学的终结》中肯定他独立发现了唯物辩证法。请留意“唯物辩证法”这一表述。它听起来像是一种以唯物主义为基础的辩证法，但这不排除唯物主义

Joseph Dietzgen (1828 – 88) met Marx during the 1848 German Revolution, and subsequently became his friend. Self-educated, Engels credited him, in *Ludwig Fuerbach and the end of Classical German Philosophy*, with independently discovering the materialist dialectic. Note the formulation ‘materialist dialectic’. This sounds like a form of dialectic grounded in materialism.

也能为其他分析方式提供基础。因此，我们可以承认一种非辩证的唯物主义科学。

But this does not exclude materialism grounding other modes of analysis. So, we may admit a non-dialectical materialist science.

伯恩斯[12]认为，狄慈根阅读恩格斯关于费尔巴哈的论述后，在 1887 年的《一个社会主义者在认识论领域的漫游》[13]中创用了“辩证唯物主义”一词。[14]这个看似琐碎的细节，凸显了这些概念的含义是多么游移不定。

Burns[12] argues that Dietzgen coined the expression ‘dialectical materialism’, having read Engels on Feurbach in his *Excursions of a Socialist into the Domain of Epistemology*[13] from 1887.[14] This apparent point of trivia highlights how slippery these notions are.

“辩证唯物主义”听起来像是一种以辩证法为基础的唯物主义，也就是不承认其他分析形式的唯物主义。正如我们将看到的，在后来的马克思主义哲学中，辩证法与科学越来越被混为一谈，二者能够彼此分离的说法也遭到争辩。不明确宣称自己具有辩证性质的科学唯物主义，则被称为机械唯物主义。

‘Dialectical materialism’ sounds like a variety of materialism that is grounded in dialectics, that is, one which admits no other forms of analysis. As we shall see, dialectics and science are increasingly conflated in subsequent Marxist philosophy, and it was disputed that one could be separated from the other. Scientific materialism that was not avowedly dialectical was termed *mechanical*.

狄慈根在最成熟的著作《哲学的积极成果》[15]中，把形式逻辑的前提与从这些前提作出的推演混淆了。他又把批评前提与批评推演混淆起来：

In his most mature work, *The Positive Outcome of Philosophy*,[15] Dietzgen conflated the premises of formal logic with the deductions made from them. He further confuses critiquing premises with critiquing the deduction:

他据以推断狗一般具有警觉性的前提，是由传统传下来的，并凭信念得到接受。但它有事实根据吗？难道不会有一些狗缺少警觉性吗？即便所有推演都很严密，我们的巴哥犬难道不可能很不可靠吗？（XIII）

The premise from which he deducted the watchfulness of dogs in general, was handed down by tradition and had been accepted on faith. But was it founded on fact? Could there not be some dogs who lacked the quality of watchfulness, and might not our pug-dog be very unreliable, in spite of all exact deductions? (XIII)

狄慈根接着引用培根（以及笛卡尔）对三段论逻辑的拒斥。与黑格尔一样，这里需要的是逻辑与辩证法的一种新综合。

Dietzgen goes on to cite Bacon’s (and Descartes’) rejection of syllogistic logic. As with Hegel, what is needed is a new synthesis of logic and dialectics.

p. 174

狄慈根的长处在于明确揭示了辩证法的阶级视角，尤其是在较早写给儿子的书信集《逻辑书简，特别是民主的无产阶级的逻辑》[16]中。在这里，他批评阶级（即资产阶级）逻辑通过强调差异，将剥削合理化为自然现象。这使狄慈根把形式逻辑与阶级逻辑联系起来，因为形式逻辑把前提当作事物彼此分离的方面，并据此推理。不过，他以一种明确的先验论证，拒斥了第二条思维规律，即矛盾表明前提无效。

Dietzgen’s great strength lies in making explicit the class perspective of dialectics, in particular in an earlier collection of letters to his son, *Letters on Logic Especially Democratic-Proletarian Logic*.[16] Here, he criticizes class (i.e. bourgeoisie) logic for rationalizing exploitation as natural, by emphasizing difference. This leads Dietzgen to associate formal logic with class logic because it is based on reasoning from premises as separate aspects of things. However, he rejects the second law of thought, that contradictions demonstrate invalid premises, in an argument that is explicitly transcendental.

狄慈根呼应黑格尔的观点，对逻辑学家不屑一顾，因为他们将逻辑与其内容割裂开来：

Echoing Hegel, Dietzgen has no time for logicians, because they separate logic from its content:

形式逻辑学家既无知又狡黠。他们坚持用传统方式讨论知性或思维，仿佛它们是孤立的东西，却无视逻辑研究的对象与经验世界之间的必然联系。这种相互联系会引向对真理与谬误、意义与无意义、神与偶像的说明，而这对教授们来说很不合时宜。因此，这个不受欢迎的问题被移交给神秘的部门——形而上学和宗教——好让这些官方智慧的尊贵支柱继续为统治阶级效劳。（第四封信）

The formal logicians are as ignorant as they are roguish, when they persist in discussing the intellect or thought in the traditional manner as if they were isolated things, while ignoring the necessary connection of the object of the logical study with the world of experiences. This interconnection leads to an explanation of truth and error, of sense and nonsense, of god and idols, and this is very inopportune for the professors. For this reason this unwelcome problem is handed over to the mystical departments, to metaphysics and religion, so that these venerable pillars of official wisdom may continue their services to the ruling classes. (Fourth Letter)

在《逻辑书简》的开头部分，狄慈根用陶工的类比来拒斥形式逻辑，认为思想不能与行动分离，形式也不能与内容分离。在这里，他似乎忘记了马克思在《资本论》第一卷中的著名论断。[17]

Early on in *Letters*, Dietzgen deploys an analogy of a potter to reject formal logic, saying that thoughts cannot be separated from actions, nor form from content. Here, Dietzgen seems to forget Marx's famous observation in *Capital* Volume 1.[17]

但是，最蹩脚的建筑师与最灵巧的蜜蜂之间的区别在于：建筑师在现实中建造之前，已经在想象中建成了他的建筑。（第 174 页）

But what distinguishes the worst architect from the best of bees is this, that the architect raises his structure in imagination before he erects it in reality. (p174)

狄慈根熟悉亚里士多德的三段论逻辑，合理地批评它无法充分把握事物多方面的属性。然而，他援引自己的革命原则，在后来的《哲学的积极成果》中说：

Familiar with Aristotelian syllogistic logic, Dietzgen makes the reasonable criticism that it could not adequately capture multifaceted properties of things. However, invoking his revolutionary principles, he says in the later *The Positive Outcome of Philosophy*

哲学家们应当放弃试图用三段论证明任何事物的老嗜好。如今，一个论点并非靠言辞，而是靠事实、靠行动来证实。各门科学已充分具备所需的条件，因此“理解的可能性”得到了无可置疑的证明。（XIII）

The philosophers should abandon their old hobby of trying to prove anything by syllogisms. Nowadays, a case is not substantiated by words, but by facts, by deeds. The sciences are sufficiently equipped, and thus the‘possibility of understanding’is demonstrated beyond a doubt. (XIII)

虽然狄慈根已不再属于马克思主义主流经典，但在布尔什维克革命之前，他曾有很大影响。安东·潘涅库克在 1902 年《脑力劳动的本质》《逻辑书简》和《哲学的积极成果》合订本的导言中写道：

While Dietzgen is no longer part of the mainstream Marxist canon, he was highly influential before the Bolshevik revolution. In the introduction to the 1902 combined edition of *The Nature of Brain Work*, *Letters on Logic* and *Positive Outcome of Philosophy*, Anton Pannekoek wrote:

深入研究狄慈根的哲学著作，是理解马克思和恩格斯基本著作的重要而不可或缺的辅助。[18]

a thorough study of Dietzgen's philosophical writings is an important and indispensable auxiliary for the understanding of the fundamental works of Marx and Engels.[18]

我们将考察布尔什维克革命后马克思主义对逻辑的回应，但首先要回过头来看看 20 世纪初形式逻辑的关键发展。

We will explore Marxist responses to logic after the Bolshevik Revolution, but first, we shall backtrack to key developments in formal logic at the start of the twentieth century.

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罗素悖论与《数学原理》

Russell's paradox and Principia Mathematica

19 世纪末，弗雷格的体系被视为形式逻辑的顶峰，正走在数学形式化的康庄大道上。可惜好景不长。1902 年，伯特兰·罗素（1872—1970）致信弗雷格，[19]指出其体系核心存在一个根本矛盾。

At the end of the nineteenth century, Frege's system was seen as the pinnacle of formal logic, on the high road to formalizing mathematics. Alas, this was short lived. In 1902, Bertrand Russell (1872–1970) wrote to Frege,[19] identifying a fundamental contradiction at the heart of his system.

在弗雷格的体系中，谓词用满足某种性质的对象所组成的集合来刻画。罗素悖论涉及这样一个观察：谓词可能适用于自身，也可能不适用于自身。等价地说，集合可能以自身为成员，也可能不以自身为成员。

In Frege's system, predicates are characterized by sets of items that satisfy some property. Russell's paradox involves observing that predicates may or may not apply to themselves. Equivalently, sets may or may not have themselves as members.

一个适用于自身的谓词，若表示为包含自身的集合，听起来似乎会导致无限展开。然而，使用它的名称而非它的内容，可以避免这种情况。也就是说，看似无限的构造可以有有限的表示。

Now, a predicate applying to itself, represented as a set containing itself, sounds like it should lead to an infinite expansion. However, this may be avoided through the use of its name instead of its contents. That is, there are finite representations of apparently infinite constructs.

考虑那些不包含自身的集合。我们可以构造一个由不包含自身的集合组成的集合。这个集合又不包含自身，因此或许它应当包含自身。但若它被包含

Consider sets that don't contain themselves. We might make a set of sets that do not include themselves. In turn, this set does not include itself, so maybe it

在自身之中，它就确实包含自身，于是它又不应当包含自身。

这不只是一个学术练习：在任何具有文件夹系统的计算机上都能演示它。从文件夹创建指向其他文件夹的链接，是常见的做法。那么，要更方便地从一个大文件夹的底部回到顶部，可以把最后一个条目设为指向该文件夹自身的链接。因此，我们可以新建一个文件夹，其中包含指向那些没有自身链接的文件夹的链接。这个新文件夹应不应该链接到自身呢？

正如弗雷格所承认的，[20]罗素悖论使在逻辑中将数学形式化的整个前景受到质疑。尽管如此，罗素和他过去的老师阿尔弗雷德·诺思·怀特海（1861—1947）仍着手尝试，将数学系统地建立在一种与弗雷格或康托尔的体系大致等价的形式逻辑之上。三卷本《数学原理》（PM）[21]于 1910 年至 1913 年间出版，重构了数学的重要部分。如今《数学原理》虽很少有人阅读，尤其因为它采用非标准的记号，但它长期以来一直被誉为奠基之作。

should. But if it is included in itself, then it does include itself and so it shouldn’t.

This is not merely an academic exercise; we can demonstrate this on any computer with a folder system. It is commonplace to make links from folders to other folders. Then, one way to make it easier to get from the bottom of a large folder back to the top is to make the last entry a link to the folder itself. Thus, we can make a new folder with links to folders that don’t have links to themselves. Should that folder link to itself or not?

Russell’s paradox drew into question the whole prospect of formalizing mathematics in logic, as Frege acknowledged.[20] Nonetheless, Russell, and his former teacher Alfred North Whitehead (1861–1947), embarked on trying to systematically ground mathematics in a formal logic that was broadly equivalent to Frege’s or Cantor’s. The three volumes of *Principia Mathematica (PM)*[21] appeared between 1910 and 1913 and reconstructed a significant portion of mathematics. While *PM* is now little read, not least because of its non-standard notation, it has long been heralded as foundational.

希尔伯特纲领

Hilbert’s programme

由大卫·希尔伯特（1862—1943）提出的希尔伯特纲领，规定了 20 世纪乃至今日形式逻辑研究的框架。这一“纲领”并非像宣言那样，是一份固定的宗旨声明；它是从希尔伯特不断发展的构想中整理出来的。

Hilbert’s programme, formulated by David Hilbert (1862–1943), framed the conduct of formal logic in the twentieth century, and to this day. The ‘programme’ was not a settled statement of purpose like a manifesto: rather, it was codified from Hilbert’s evolving conceptions.

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在希尔伯特 1904 年回应罗素悖论的《论逻辑与算术的基础》[22]中，可以看到这一纲领的根源。他首先概述主要数学家对数学基础的看法。利奥波德·克罗内克（1823—1891）不诉诸基础，就默认整数存在，被称为教条主义者。赫尔曼·冯·亥姆霍兹（1821—1894）只承认从经验而来的存在，因此排除以思想实验作为理论基础，被称为经验主义者。埃尔温·克里斯托费尔（1829—1900）被称为机会主义者。他虽然反对克罗内克拒斥无理数，但仍寻求接受无理数的肯定性理由。弗雷格的工作被承认为基础性的，却因体系不够严密、导致自指悖论而受到批评。希尔伯特在这里写道：

We can see the roots of Hilbert’s programme in his 1904 response to Russell’s paradox, *On the foundations of logic and arithmetic*.[22] He began by characterizing leading mathematicians’ views on the foundations of mathematics. Leopold Kronecker (1823–91) was called a *dogmatist* for accepting integers as implicitly existent without recourse to foundations. Herman van Helmholtz (1821–94) was termed an *empiricist* for only accepting existence derived from experience, thus ruling out thought experiments as the basis for theories. Elwin Christoffel (1829–1900) was termed an *opportunist*. While he had opposed Kronecker’s rejection of irrational numbers, he sought positive reasons for accepting them. Frege’s work was acknowledged as foundational, but his system was criticized for a lack of rigour, giving rise to paradoxes of self-reference. Here, Hilbert writes:

> 相反，从一开始，研究数的概念就应当以避免此类矛盾、澄清这些悖论为主要目标之一。[23]（第 130 页）

> *Rather, from the very beginning a major goal of the investigations into the notion of number should be to avoid such contradictions and to clarify these paradoxes.*[23] (p130)

希尔伯特接着将理查德·戴德金（1831—1916）的方法描述为先验方法，因为他假定存在由无限多个对象构成的实无限。我们还会回到这一曾受亚里士多德批评的立场。最后，希尔伯特说，康托尔虽然区分了相容的集合与不相容的集合，却没有给出区分标准，因此不得不作主观主义的假定。[24]

Hilbert went on to characterize Richard Dedekind (1831–1916) as following a *transcendental* method, as he assumed actualized infinities of objects. We will return to this position, which Aristotle criticized. Finally, Hilbert said that Cantor, while distinguishing consistent

and inconsistent sets, gave no criteria for distinguishing them, necessitating *subjectivist* assumptions.[24]

希尔伯特在总结这些评论时说，所有这些困难都可以通过他所谓的公理化方法克服。此外，承认逻辑与算术相互依存，并同时发展二者的规律，或许可以避免这些悖论。

Hilbert concludes these remarks by saying that all these difficulties could be overcome by what he called the *axiomatic method*. Further, the paradoxes might be avoided by acknowledging the co-dependence of logic and arithmetic and concurrently developing their laws.

形式系统以公理和推理规则为基础。公理是对所有论元都为真的基本公式；推理规则用于从公理和其他定理出发，构造或证明新的真公式，即定理。源于亚里士多德的思维规律，可视为公理的先驱。因此，公理化方法就是寻找一组相互独立的公理，使之连同适当的规则，足以展开某一领域的全部内容。

Formal systems are based on *axioms*, elementary formulas which are true for all arguments, and *rules of inference*, for constructing or proving *theorems*, additional true formula, from *axioms* and other *theorems*. The Laws of Thought, derived from Aristotle, might be seen as progenitors of *axioms*. Then, the axiomatic method involves finding sets of independent *axioms* that, together with appropriate rules, are adequate for elaborating all of some domain.

希尔伯特高度重视公理系统的相容性，即不可能利用这些系统推导出矛盾。不能确立相容性，后来成为数学逻辑危机的关键。

Hilbert placed great stress on the *consistency* of axiomatic systems, that is, that it should not be possible to use them to derive contradictions. The impossibility of establishing *consistency* was to prove key to the later crisis in mathematical logic.

在后来的一篇论文《公理化思维》[25]中，希尔伯特首次提到一个“纲领”，[26]并更详细地阐述自己的目标。除了相容性，他还希望确定任意数学问题在原则上是否可解，能够检验数学活动的结果，并确定是否可能存在更简单的证明。在这里，他也回到那个曾使马克思主义者深感困扰的古老的亚里士多德式问题，即“数学中内容与形式主义的关系”（第1113 页）。

In a subsequent paper, *Axiomatic Thought*,[25] Hilbert first referred to a ‘programme’,[26] developing his objectives in greater detail. As well as consistency, he wished to determine the solvability in principle of arbitrary mathematical questions, to be able to check the results of mathematical activity, and to determine whether or not there might be simpler proofs. Here, Hilbert also returned to the old Aristotelian problem that so exercised Marxists, that of ‘the relationship between *content* and *formalism* in mathematics’ (p1113).

尤其值得注意的是，能够判定一个数学问题是否可以在有限步骤内求解，

……触及了数学思维的本质。（第 1113 页）

Notably, the ability to decide whether or not a mathematical question was solvable in a finite number of steps

. . . goes to the essence of mathematical thought. (p1113)

这至今仍通常被称为 Entscheidungsproblem，即判定问题。还有许多其他判定问题，例如两个公式是否等价；但能在有限步骤内证明性质，被视为关键。

This is still commonly referred to as the *Entscheidungsproblem* – the decision problem. There are lots of other decision problems, for example, whether or not two formulae are equivalent, but proof of properties in a finite number of steps was seen as key.

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元理论与逻辑学派

Meta-theory and logical schools

希尔伯特认为，数学抽象应当、也能够独立于任何应用内容而得到研究。但希尔伯特纲领要求数学本身也接受数学推理。也就是说，应当有以数学为内容的元数学，但目的仅在于确立公理的相容性。[27]

Hilbert thought that mathematical abstractions should, and could, be explored independently of any content of application. But Hilbert’s programme required that mathematics itself should be subject to mathematical reasoning. That is, there should be *metamathematics*, with mathematics as its contents, but only to establish the consistency of axioms.[27]

数学用语言[28]来表达，其中有符号、句法和语义。因此，数学将成为自身的元语言，即一种谈论语言的语言。

Now, mathematics is expressed in a language,[28] with symbols, syntax and semantics. Thus, mathematics was to become its own *metalanguage*, a language for talking about language.

在罗素悖论出现后的数十年中，形式逻辑形成了三个学派，反映出不同的回应。可以根据各派允许哪些数学实体，以及允许针对这些实体进行哪些数学推理，来刻画它们的特点；也就是说，要看它们认为元数学应当如何进行。[29]这归根结底取决于它们对无限的态度。

以弗雷格和罗素为代表的逻辑主义者，既接受无限的数学实体，也接受康托尔式的无限推理。他们承认观念性的数学实体具有存在，因而是柏拉图主义者，也因此是唯心主义者。

其次，以希尔伯特为代表的形式主义者，接受无限的数学实体作为研究对象，但力求只使用有限推理。他们不关心数学实体的存在地位。

最后，直觉主义者采取了最激进的立场。他们以克罗内克为基础，诉诸数学直觉，在数学和元数学中都只接受有限构造。他们还拒斥排中律（LEM），即某事及其否定不可能同时为真。

〔校注：底本这里将排中律写成“某事及其否定不可能同时为真”；这实际上是矛盾律的表述。第 7 章列出的排中律是“任何事物必定或者存在，或者不存在”。正文保留底本措辞。〕

这些学派的拥护者之间存在深刻的分歧，有时甚至相互谩骂。尽管如此，它们的形式系统实际上联系十分紧密。

In the decades following Russell’s paradox, three schools of formal logic emerged, reflecting different responses. We may characterize them according to what they accepted as admissible mathematical entities, and what forms of mathematical reasoning about them were admissible, that is, how meta-mathematics might be conducted.[29] This boiled down to their attitude to infinity.

The *logicists*, exemplified by Frege and Russell, accepted both infinite mathematical entities and infinitory reasoning, like Cantor’s. They were Platonists, in that they accorded existence to ideal mathematical entities. Hence, they were idealists.

Next, the *formalists*, like Hilbert, accepted infinite mathematical entities as objects of study, but sought to only use finite reasoning. The existential status of mathematical entities was not of concern.

Finally, the *intuitionists* took the most radical stance. Building on Kronicker, they would only accept finite constructions in both mathematics and meta-mathematics, appealing to mathematical intuition. They also rejected the Law of the Excluded Middle (LEM), that is, that it is not possible for something and its negation to be simultaneously true.

There were profound, and sometimes vituperative, disagreements between proponents of these different schools. Nonetheless, their formal systems were actually very closely related.

直觉主义

Intuitionism

对康托尔的一种回应是彻底拒斥非有限性方法，实际上也拒斥实数。因此，克罗内克在 1886 年主张，通过操作无限序列得到的特定结果，只有能够用整数算术重建、而“不超出有限序列的概念”时，才可以接受。[30]

p. 178

随后，吕伊岑 • 布劳威尔（1881—1966）的直觉主义致力于利用构造性的有限方法，从少数几个“直觉”概念出发重建形式逻辑。1908 年，布劳威尔发表了对经典逻辑的批评，其中的主题在他后来的全部著作中反复出现。[31]首先，他认为，脱离数学的逻辑会导致无根据的结论。与形式主义者相反，布劳威尔认为逻辑应当以数学为基础，而数学本身则以观察和对基本真理的原初直觉为基础。不过，他也认为，即使独立于数学，三段论推理和归谬论证都是可接受的（第 27、29 页）。

布劳威尔拒斥排中律（LEM），[32]即某事及其否定不可能同时为真（第 29 页及以下）。这意味着二者必有其一为真，不允许任何一方的地位不确定。同样，布劳威尔拒斥双重否定可以相互抵消，因为某事“并非不真”仍可能使其地位悬而未定。

One response to Cantor was to entirely reject non-finitary methods, and, in effect, real numbers. Thus, in 1886, Kronecker argued that particular results found by manipulating infinite sequences were only admissible if they could be reconstructed without going ‘beyond the concept of a *finite* series’, using arithmetic over integers.[30]

Subsequently, Luitzen Brouwer’s (1881–1966) intuitionism was concerned with the reconstruction of formal logic from a small number of ‘intuitive’ concepts, using constructive, finite techniques. In 1908, Brouwer published a critique of classical logic, whose themes recur in all his later writing.[31] First of all, he thought that logic separate from mathematics led to unfounded conclusions. Against the formalists, Brouwer thought that logic should be grounded in mathemetics, itself grounded in both observation and primordial intuitions of basic truth. Nonetheless, he suggests that, independently of mathematics, argument by syllogism and contradiction are both acceptable (p27,29).

Brouwer rejects the LEM,[32] that it is not possible for both something and its negation to be true (p29ff). This implies that one or other must be true, not allowing for the status of either to be uncertain. Similarly, Brouwer

〔校注：本段沿用上一处对排中律的误述，并由“不同时为真”转到“必有其一为真”。这两种表述不能等同；译文保留作者原有论述。〕

布劳威尔进一步指出，排中律

要求每个假设或者正确、或者不正确。用数学语言说，对于假定的系统之间以某种方式相互嵌合的每一种情形，要么能构造出其完成，要么能构造出因不可能而受阻的情形。因此，principium tertii exclusi［即排中律］是否有效的问题，等同于是否可能存在不可解数学问题的问题。（第 42 页）

在有限情形中，他接受排中律，因为可以通过穷举来检验。只要无限情形能通过归纳构造出来，他也接受排中律在其中的应用。但是，布劳威尔反对从那些无法证明存在、却被假定存在的实体出发作论证。他尤其拒斥已完成的无限这一假定。

布劳威尔的“原初直觉”（ur-intuition）概念显然是唯心主义的；他在后来的著作中，例如 1929 年的《数学、科学与语言》[33]中，明确表现出了主观主义。

rejects double negation as cancelling, because something not being not true may still leave its status indeterminate.

Brouwer further argues that LEM

demands that every supposition is either correct or incorrect, mathematically: that of every supposed fitting in a certain way of systems in one another, either the termination or the blockage by impossibility, can be constructed. The question of the validity of the principium tertii exclusi [ie LEM] is thus equivalent to the question concerning the *possibility of unsolvable mathematical problems*. (p42)

He was happy with LEM in finite cases, as it may be checked exhaustively. He was also happy with its application to infinite cases, so long as they may be constructed by induction. However, Brouwer objected to arguments from entities which are assumed to exist if their existence cannot be demonstrated. In particular, he rejected assumptions of completed infinities.

Bouwer’s notion of ‘ur-intuition’ is plainly idealist, and his subjectivism is made explicit in his later work, for example, in *Mathematics, Science and Language*[33] from 1929.

从布尔什维克革命到孟什维克化的唯心主义

From the Bolshevik revolution to Menshevizing Idealism

20 世纪头 25 年的灾难性事件夺去了数百万人的生命，又摧毁了数百万人的生活；要充分呈现这些事件而不显得轻描淡写，是不可能的。不过，从我们当前的视角看，1914 年至 1919 年世界大战的关键结果，是苏联的建立。[34]这是第一个由明确奉行唯物主义的群众政党执政的国家。

内战（1917—1923）之后，苏联把重心放在重建和立即改善人民生活上。一旦更广泛的国际革命已陷入停滞这一点变得明确，压倒一切的优先任务便是稳定和加强苏联。为了这一目标，也为了向共产主义未来迈进，科学被视为核心。

战争和移民造成专家短缺，因此起初只要能确保忠诚，国家就无法对剩余的非共产党知识界的意识形态正确性要求过高。与此同时，在共产党领导下不断扩大的教育体系，持续培养出新一代“红色专家”——科学家和工程师。他们明确致力于苏联的目标，却必然要与革命前的一辈人共同工作。

数学发挥了核心作用；这一时期，莫斯科和列宁格勒发展出两个世界级数学中心。纯数学研究虽在继续，但重点在于应用数学。总体上，形式逻辑并不突出。

It is not possible to do justice to the catastrophic events of the first quarter of the twenteith century, killing millions of people and devastating the lives of millions more, without appearing to trivialize them. Nonetheless, from our current perspective, the key outcome of the 1914 to 1919 world war was the establishment of the Soviet Union,[34] the first state governed by a mass party explicitly committed to materialism.

Following the Civil War (1917–23), the Soviet focus was on reconstruction, and immediately bettering people’s lives. Once it was clear that wider international revolution had stalled, the overwhelming priority was to stabilize and strengthen the Union. For this, and for building towards a Communist future, science was deemed central.

With a shortage of experts, lost through war or emigration, the state could not initially afford to place too much premium on the ideological rectitude of the remaining non-Communist intelligentsia, provided their loyalty was assured. At the same time, the expanding education system, under Communist direction, steadily produced a new generation of ‘Red expert’ scientists and engineers, who were explicitly committed to Soviet objectives, but who necessarily worked alongside the pre-Revolutionary cohorts.

p. 179

Mathematics had a central role, and this period saw the growth of two world class mathematical centres, in Moscow and Leningrad. While pure mathematics research continued, the emphasis was on applied mathematics. Overall, formal logic was not prominent.

然而，形式逻辑的地位依然存在争议，甚至在布尔什维克内部围绕苏联发展方向的两极分化中也是如此。例如，1921 年，列宁[35]在讨论托洛茨基的工会提案时，批评布哈林忽视辩证逻辑。[36]

Nonetheless, the status of formal logic was still contested, even among polarizations within the Bolsheviks over the direction that the Soviet Union should take. For example, in 1921, Lenin,[35] in discussing Trotsky's Trade Unions proposal, attacked Bukharin for his neglect of dialectical logic.[36]

1924 年列宁去世后，布尔什维克不同派别之间的斗争变得尖锐。这些斗争以新经济政策（NEP）为背景：新经济政策违背社会主义的愿望，引入相当大的市场成分，试图在战时共产主义这一务实的指令经济所带来的困境之后，加快经济恢复。过度简化地说，“左”派主张迅速从新经济政策转向计划经济，“右”派则主张较慢的转变。这些斗争还与争夺地位、清算旧账深深交织在一起。

After Lenin's death in 1924, the struggles between different Bolshevik factions became acute. These took place against the background of the New Economic Policy (NEP), which, contrary to socialist aspirations, had introduced a substantial market component to try to accelerate recovery following the exigences of the pragmatic command economy of War Communism. To over simplify, the 'left' faction promoted a speedy transition from the NEP to a planned economy, whereas the 'right' faction sought a slower change. And these struggles were deeply entangled with jockeying for position and settling of scores.

正如海伦娜·希恩[37]系统考察的那样，科学与哲学的关系在意识形态和实践两个层面，都成为这些争论的重要组成部分。由此形成两种立场。一方质疑辩证法与科学的相关性，符合亚里士多德传统：成熟的科学独立于其辩证法根源而自主发展。反对者把这一立场称为机械论，认为它源于马克思和恩格斯曾反对的机械唯物主义。另一方则重新强调辩证法，似乎符合汲取黑格尔思想的马克思主义主流传统。

As Helena Sheehan[37] systematically explores, the relationship of science to philosophy became an important component of these disputes, at both ideological and practical levels. Two positions developed. On the one side, the relevance of dialectics to science was questioned. This was in keeping with the Aristotelian tradition that mature sciences developed autonomously of their dialectical roots. This position was characterized by opponents as mechanist, descended from the mechanical materialism that Marx and Engels had opposed. On the other side, was a renewed emphasis on dialectics. This appeared to be in keeping with the mainstream Marxist tradition, drawing on Hegel.

起初，科学话语内部的意识形态斗争针对的是“右”倾，它在论战措辞中被称为孟什维克倾向。在这里，德波林领导的辩证法派在与机械论派的斗争中占了上风。在数学领域，这导致形式逻辑受到越来越强烈的拒斥。辩证法派的优势没有维持多久。在随后反对“左”倾的斗争中，对辩证法的过度依赖被称为唯心主义。正如弗里德曼后来在那篇标题颇能说明问题的《苏联反对形式主义的反叛》[38]中所写：

Initially, ideological struggles within scientific discourse were against the 'right' tendency, characterized rhetorically as Menshevik. Here, the dialecticians under Deborin gained the upper hand against the mechanists. Within mathematics, this led to an increased repudiation of formal logic. The dialecticians' ascendancy was short lived. During subsequent struggles against the 'left' tendency, an over dependence on dialectics was characterized as idealist. As Friedmann subsequently wrote in the telling titled *Revolt Against Formalism in the Soviet Union*[38]:

> 但是，机械论的批评者被热情冲昏头脑，走向了另一个极端——唯心主义。这就需要党的理论家所说的“两条战线上的斗争”。显然，在这里也是实践中的失误引起了对理论问题的注意。（第 307 页）

> But the critics of mechanism, carried away by their zeal, fell into the opposite extreme of idealism. This required a 'struggle on two fronts' as the theoreticians of the Party call it. Apparently it was faults in practice which here too called attention to the theoretical problems. (p307)

p. 180

然而，这一倾向的失败并没有使逻辑得到恢复。相反，两种立场都被混同为孟什维克化的唯心主义，苏联的形式逻辑日益停滞，直到 20 世纪 50 年代初。

However, the defeat of this tendency did not result in the rehabilitation of logic. Rather, both positions were conflated as *Menshevizing idealism*, and formal logic in the Soviet Union increasingly stalled until the early 1950s.

孟什维克化的唯心主义与逻辑

Menshevizing idealism and logic

杰出的数学家安德烈 • 柯尔莫哥洛夫（1903—1987）是新一代红色专家的代表。他 1925 年的论文《论排中律》[39]批评了形式主义和直觉主义，但也表明，经典逻辑与直觉主义逻辑可以在直觉主义框架内得到调和。

The outstanding mathematician Andrei Kolmogorov (1903–87) exemplified the new generation of Red experts. His 1925 paper *On the principle of the excluded middle*[39] offered a critique of formalism and intuitionism, but also showed how classical and intuitionist logic might be reconciled, within an intuitionist framework.

布劳威尔曾主张，用排中律和超限前提来确立有限性结果是不正当的。柯尔莫哥洛夫证明，即使不诉诸这二者，这类有限性结果仍然成立。

Brouwer had argued that it was illegitimate to use both the LEM and transfinite premises to establish finitary results. Kolmogorov demonstrated that such finitary results still stood without recourse to either.

柯尔莫哥洛夫与直觉主义者一样，认为在没有其他证据时，相互矛盾的项应被视为未定。他也与直觉主义者一样，质疑超限前提是否有意义，即使它们能够用来得到有限性结果。

With the intuitionists, Kolmogorov accepted that, in the absence of other evidence, contradictory terms should be regarded as indeterminate. Further, with the intuitionists, he questioned whether transfinite premises had any meaning, even if they might be used to reach finitary results.

柯尔莫哥洛夫把形式主义描述为不对前提的选择作出承诺，听起来与亚里士多德的辩证法相近。他指出，可以把对立项之一增加为公理来解决矛盾，并观察到，从形式主义的角度看：

Kolmogorov’s characterization of formalism, as uncommitted to choice of premises, sounds akin to Aristotelian dialectics. Noting that a contradiction may be resolved by adding one of the opposed terms as an axiom, he observed that, from a formalist perspective:

> 因此，从每一对相互矛盾的公式中选出一个作为新公理，只需考虑是否方便。（第 417 页）
>
> The selection of the formula taken as the new axiom, from each pair of contradictory formulas, is thus subject only to considerations of convenience. (p417)

此外，形式主义逻辑不赋予公理任何意义，而直觉主义则以“表达给予我们的事实”的公理为基础（第 417 页）。这是黑格尔的、也是辩证唯物主义的立场，即一种统一的、有内容的逻辑。然而，柯尔莫哥洛夫仍将“数理逻辑”视为数学中一个独立的组成部分。在这里，他并未采取一种明确带有派别立场的哲学姿态，而是采取数学的立场。

Further, while formalist logic attributed no meaning to axioms, intutionism was based on axioms that ‘express facts given to us’ (p417). This is the Hegelian, and also dialectical materialist, position of an integrated, content-full logic. However, Kolmogorov still identified ‘mathematical logic’ as a distinct component of mathematics. Here, Kolmogorov did not take an explicitly partisan philosophical, as opposed to mathematical, stance.

然而，正如武西尼奇[40]所考察的，就在同一年，即 1925 年，苏联的意识形态家在哲学期刊《在马克思主义旗帜下》明确攻击支撑康托尔成果的唯心主义。武西尼奇指出，当时没有人试图寻找一种唯物主义的替代方案来取代康托尔的理论（第 117—118 页）；虽然希尔伯特受到的关注较少，辩证逻辑却被置于形式逻辑的对立面（第 118—119 页）。不过，这些攻击针对康托尔的集合论，而非苏联数学家（第 122 页）。曾属列宁格勒学派的数学家 G. G. 洛伦兹[41]指出，同一时期，基辅的一个代数学派在乌克兰共产党指示下被关闭，其学者分散到其他中心。

However, as Vucinich[40] explores, in that same year of 1925, Soviet ideologues explicitly attacked the idealism underpinning Cantor’s results in the philosophical journal *Under the Banner of Marxism*. Vucinich notes that no attempt was made to find a materialist alternative to Cantor (p117–8), and that, while less attention was paid to Hilbert, dialectical logic was counterpoised to formal logic (p118–9). Nonetheless, the attacks were against Cantor’s set theory rather than Soviet mathematicians (p122). The former Leningrad School mathematician G. G. Lorentz[41] notes that, in the same period, an algebraic school in Kiev was closed, under the direction of the Ukrainian Communist Party, and its scholars dispersed to other centres.

1931 年，红色专家恩斯特 • 科尔曼（1892—1979）和索菲娅 • 亚诺夫斯卡娅（1896—1966）在同样发表于《在马克思主义旗帜下》的《黑格尔与数学》中，明确阐述了辩证唯物主义的立场。[42]他们反复引用《反杜林论》和不久前问世的《自然辩证法》，在批评黑格尔辩证法的同时，强调辩证唯物主义根植于黑格尔。

The Red experts Ernst Kolman (1892–1979) and Sofya Yanovskaya (1896–1966) provided a clear statement of the dialectical materialist position in 1931, in *Hegel and Mathematics*, also published in *Under the Banner of Marxism.*[42] Repeatedly citing *Anti During*, and the recently available *Dialectics of Nature*, they emphasized the roots of dialectical materialism in Hegel, while criticizing his dialectics. p. 181

科尔曼和亚诺夫斯卡娅概述并拒斥了上述逻辑学派：直觉主义依赖先验假定；逻辑主义把逻辑与数

Kolman and Yanovskaya summarized and rejected the schools of logic we identified above: intuitionism for de-

学统一起来，将理性规律等同于公理和定理；形式主义则脱离内容处理逻辑。他们还拒斥把数学视为物理学一部分的“机械论经验主义者”，并反对马赫将问题心理学化。总的说来：

pending on *a priori* assumptions; logicism for unifying logic and mathematics, and identifying the laws of reason with axioms and theorems; and formalism for treating logic independetly of content. They further reject ‘mechanistic empiricists’ who see mathematics as part of physics, and Mach for psychologizing. Overall:

因此，这些哲学学派都只抓住现实的一个方面，没有任何一个能够理解数学与实践的联系及其发展规律。只有黑格尔给数学下了一个把握事物本质的定义；这个定义完全独立于黑格尔的观点，实际上是深刻唯物主义的。

Thus none of these philosophical schools, which all grasp one and only one side of reality, is in a position to understand the link between mathematics and practice and its laws of development. Hegel alone gave mathematics a definition such as grasped the essence of the matter, a definition which, quite independently of Hegel’s views, is actually profoundly materialist.

不过，从阶级立场看待数学，并不意味着拒斥数学：

Nonetheless, taking a class standpoint on mathematics does not involve rejecting it:

恰恰相反，它［资产阶级数学］必须经过重构，因为它虽然片面而歪曲，却仍客观地表现了物质世界。

on the contrary it [bourgeois mathematics] must be subjected to a reconstruction, since it represents the material world, albeit one-sidedly and distortedly, nevertheless objectively.

科尔曼和亚诺夫斯卡娅指出，试图把分析还原为算术，最终导致了

Kolman and Yanovskaya identify attempts to reduce analysis to arithmetic as ultimately leading to

集合论中那些众所周知的悖论，它们摧毁了专为这一目的建立起来的整个结构，不仅是数学的，而且是逻辑的（原文如此）。

the well-known paradoxes of set theory which destroyed the whole structure, not only of mathematical but also logical (sic), which had been specially erected for that purpose.

因此，尽管理由截然不同，他们与直觉主义者一样，对康托尔的无限持怀疑态度。

Thus, though for very different reasons, they shared the intuitionist suspicion of Cantor’s infinities.

当时，亚诺夫斯卡娅正在翻译和编辑新近重新发现的马克思数学手稿；这些手稿于 1933 年出版。[43] 马克思在讨论泰勒定理时写道：

At the time, Yanovskaya was translating and editing Marx’s recently rediscovered manuscripts on mathematics, which appeared in 1933.[43] In discussing Taylor’s theorem, Marx wrote:

p. 181–182

这种从普通代数出发、而且借助普通代数，跃入变量代数的飞跃，被当作既成事实；它没有得到证明，乍看之下还与通常代数的一切规律相矛盾。……换句话说，起始方程……不仅没有得到证明，而且实际上有意无意地假定了以变量替代常量，这公然违背代数的一切规律——因为代数，因而也包括代数二项式，只容许常量，而且只有两种常量：已知的和未知的。因此，从代数推导这个方程，似乎建立在欺骗之上。（第 117 页）

This leap from *ordinary algebra*, and besides *by means of ordinary algebra*, into the *algebra of variables* is assumed as *un fait accompli*, it is *not proved* and is prima facie *in contradiction to all the laws* of conventional algebra. . . . In other words, the starting equation . . . is not only *not proved* but indeed knowingly or unknowingly assumes a substitution of *variables* for *constants*, which flies in the face of all the laws of algebra –for algebra, and thus the algebraic binomial, only admits of *constants*, indeed only two sorts of *constants*, *known* and *unknown*. The derivation of this equation from algebra therefore appears to rest on a deception. (p117)

p. 182

这仿佛表明，马克思只能接受变量作为实际存在的值的占位符，而不能接受它作为一般的值的占位符。这种思想混乱触及黑格尔对逻辑形式主义批评的核心：把逻辑的内容——这里是常量——抽象成变量，再以任意值替换这些变量，被视为不正当。上面提到的恩格斯关于变量和抽象的断言，很可能源自他与马克思的讨论。

It is as if Marx can only accept variables as place holders for actually existing values, rather than for values in general. This muddleheadedness gets to the heart of the Hegelian critique of formalism in logic: that it is illegitimate to remove content from logic, here constants, by abstraction to variables, which may in turn be replaced by arbitrary values. Engels’ assertions about variables and abstraction, noted above, may well derive from discussions with Marx.

当然，只要明确说明这种抽象，对方程中的任意常量乃至公式作抽象，都是完全正当的。[44] 然而，科尔曼和亚诺夫斯卡娅引用马克思时，并未进一步评论。

Of course it is perfectly legitimate to abstract over any constants, or indeed formulae, in an equation, provided such abstraction is made explicit.[44] However, Kolman and Yanovskaya quote Marx without further comment.

1932 年，柯尔莫哥洛夫探索了一条辩证唯物主义能够接受的直觉主义进路。在《论直觉主义逻辑的解释》[45]中，他将直觉主义逻辑重新表述为问题演算，并证明它在形式上等价于海廷对布劳威尔逻辑的形式化。柯尔莫哥洛夫巧妙地把焦点放在植根于现实的问题上，从而避开了对无内容形式的批评，以及对用变量概括常量的批评：

In 1932, Kolmogorov explored an approach to intuitionism that was acceptable to dialectical materialism. In *On the Interpretation of Intuitionist Logic*,[45] he reformulated intuitionistic logic as a *calculus of problems* and showed that it is formally equivalent to Heyting’s formalization of Brouwer’s logic. Cunningly, Kolmogorov avoided the critique of form without content and that of variables generalizing constants, by focusing on problems, which are grounded in reality:

直觉主义逻辑应由问题演算来取代，因为它的对象实际上不是理论命题，而是问题。（第 328 页）

intuitionistic logic should be replaced by the calculus of problems, for its objects are in reality not theoretical propositions but rather problems. (p328)

也就是说：

That is:

在数学各个具体领域出现的一切情形中，都能使用“问题”和“问题的解”这两个概念，而不会引起误解。（第 329 页）

the concepts ‘problem’ and ‘solution of a problem’ can be employed without misunderstanding in all cases that occur in the concrete areas of mathematics. (p329)

这一障眼法，用对具体问题的抽象替代了对前提的抽象。

This slight of hand substitutes abstraction over concrete problems for abstraction over premises.

有趣的是，柯尔莫哥洛夫把其逻辑的规则称为“计算性的”（第 334 页）。把数学视为计算，是我们的唯物主义观念的核心。

Interestingly, Kolmogorov refers to the rules of his logic as ‘computational’ (p334). The idea of mathematics as computation is central to our conception of materialism.

因此，直觉主义成为苏联构造逻辑学派的基础。具有讽刺意味的是，构造主义如今已成为编程语言理论与实践的一种核心进路。

Thus, intuitionism was the foundation of the Soviet school of constructive logic. Ironically, constructivism is now a core approach for the theory and practice of programming languages.

孟什维克化的唯心主义与英国马克思主义

Menshevizing Idealism and British Marxism

英国共产党（CPGB）成立于 1920 年。作为一个马克思列宁主义政党，该党明确与苏联共产党保持一致，并倡导辩证唯物主义世界观。它虽从未像其他西欧共产党那样拥有众多党员或取得选举成功，却在工会中有相当大的影响力。

〔校注：本段英文第二次出现的 CPGM，依上下文应为前述 CPGB；英文保留底本。〕

The Communist Party of Great Britain (CPGB) was formed in 1920. A Marxist-Leninist party, the CPGM was explicitly aligned with the Communist Party of the Soviet Union and promoted the dialectical materialist world view. While it never enjoyed the mass membership, or electoral success, of other Western European communist parties, it had considerable influence in the trades unions.

英国共产党常被视为一个相对缺乏文化修养的组织，比起理论，它更注重日常斗争。不过，该党拥有相当多今天所谓的公共知识分子。他们经常在广播和大众印刷媒体上出现，著作既由党的出版社出版，也由主流的非党派出版社出版。韦尔斯基的《可见的学院》[46]详细叙述了 20 世纪 30 年代的这一环境。我们尤其关注其中的数学家海曼·莱维（1889—1975）和大卫·盖斯特（1911—1938）。

The CPGB is often seen as a relatively philistine organization, more focused on day to day struggles than theory. Nonetheless, a significant number of what are now known as public intellectuals were members. These regularly appeared on the radio, and in popular print media, and their books were produced by mainstream, non-aligned publishers, as well as the Party press. Werskey’s *The Visible College*[46] provides a through account of this milieu in the 1930s. Of particular interest to us are the mathematicians Hyman Levy (1889–1975) and David Guest (1911–1938).

p. 183

盖斯特于 1938 年在西班牙随国际纵队作战时牺牲，他似乎是对数学基础论争研究最深入的英国马克思主义者。1929 年，他在《数学形式主义》[47]中指出，波兰逻辑学家已在罗素和怀特海的《数学原理》中发现矛盾。他还认为，希尔伯特希望从形式上证明数学相容性的愿望受到挫败，因为唯一的办

Guest, who died fighting with the International Brigades in Spain in 1938, seems to have been the British Marxist who most closely studied the debates around foundations of mathematics. In 1929, in *Mathematical Formalism*,[47] he observed that Polish logicians had found contradictions in Russell and Whiteheads’s

法是“构造一组满足它们的思维对象”（第 210 页）。这归根结底取决于能否对有限整数作出最小限度的刻画；对希尔伯特和直觉主义者来说，这都是一个深刻的问题。

Principia Mathematica. He also suggested that Hilbert’s desire to formally demonstrate the consistency of mathematics was floored because the only way to do so was to ‘produce a set of mental objects satisfying them’ (p210). Ultimately, this depended on being able to minimally characterize the finite integers, a deep problem for Hilbert and the intuitionists.

在 1930 年的《对数学命题的“理解”》[48]中，盖斯特以一个熟悉的理由批评排中律：同时考虑一个命题及其否定是没有意义的。他还说，数学命题与经验命题一样，可能被新证据推翻。相比之下，具体实例的性质可以通过执行过程来确立；他以尝试确定某个特定数是否为质数为例。这似乎与柯尔莫哥洛夫把逻辑规则视为计算性的观点相似。

In *The ‘Understanding’ of the Propositions of Mathematics*,[48] from 1930, Guest critiqued the LEM on the familiar grounds that it is meaningless to simultaneously consider a proposition and its negation. Further, he said that mathematical propositions are like empirical propositions, in that they may be overturned by new evidence. In contrast, properties of concrete instances may be established by carrying out processes, giving as an example trying to establish whether or not a specific number is prime. This seems similar to Kolmogorov’s view of logical rules as computational.

去世前不久，盖斯特似乎同情直觉主义。在评论 E. T. 贝尔的《数学家传》[49]时，他把《数学原理》的改良主义与“直觉主义者的革命性挑战”[50]相对照：

Shortly before his death, Guest appeared sympathetic to intuitionism. In a review of E. T. Bell’s *Men of Mathematics*,[49] he contrasted the reformism of *Principia Mathenmatica* with the ‘revolutionary challenge of the “Intuitionists”’ [50]:

这不正是辩证法的精神突破形式逻辑的坚硬外壳吗？过去，如此多的科学思想曾被囚禁在这层外壳之中！（第 256 页）

But what is this but the spirit of dialectics breaking through the hard shell of formal logic, within which so much scientific thought has been imprisoned in the past! (p256)

在 1931 年伦敦国际科学技术史大会上，科尔曼阐述了关于逻辑的辩证唯物主义立场。他在《数学科学的当前危机及其重建的总纲》[51]中，考察了他所认为的数学当前存在的矛盾：

At the International Congress of the History of Science and Technology, held in London in 1931, the dialectical materialist position on logic was presented by Kolman. In his paper *The Present Crisis in the Mathematical Sciences and General Outlines for Their Reconstruction*,[51] Kolman surveyed what he saw as the current contradictions of mathematics:

数学的一切深刻矛盾——单一与多样之间、有限与无限之间、离散与连续之间、偶然与必然之间、抽象与具体之间、历史与逻辑之间的矛盾，以及理论与实践之间、数学本身与其逻辑基础之间的矛盾——实际上都是辩证矛盾。

All the profound contradictions of mathematics – the contradiction between the singular and the manifold, between the finite and the infinite, the discrete and the continuous, the accidental and the necessary, the abstract and the concrete, the historical and the logical, the contradiction between theory and practice, between mathematics itself and its logical foundation–all are in reality dialectical contradictions.

p. 184

科尔曼虽然承认希尔伯特的进路有一定价值，却批评形式逻辑忽视了数的概念的历史必然性：

While acknowledging some value in Hilbert’s approach, Kolman criticized formal logic for ignoring the historical necessity of the concept of number:

至于希尔伯特的公理化方法，它确实有助于说明各个数学概念之间的逻辑联系，但由于它是事后的构造，同样无法正确描绘发展过程。

As for Hilbert’s axiomatics, it is true that it is of use in explaining the logical connections between individual mathematical concepts, but, since it represents a construction post factum it, too, is unable to give a correct picture of development.

而且，与前面一样，科尔曼把逻辑原子论和直觉主义都斥为唯心主义：

And, as before, Kolman dismisses both logical atomism and intuitionism as idealist:

无论像逻辑主义者那样，把数学概念的世界视为僵化不动的共相的世界，还是像直觉主义者那样，把它视为自由生成的活动领域，都没有什么分别。……有限主义、元逻辑和数学原子论中最精巧的手法，无非表达了资产阶级数学家企图借形式逻辑的帷幕，把自己与物质及辩证法隔开的焦虑；这种做法将他们直接引向经院哲学的荒漠。

It is a matter of indifference whether the world of mathematical concepts is regarded as a world of rigid immovable universals, as it is by the logists, or whether it is looked upon as the sphere of action of the free becoming as it is by the intuitionists. . . . The most refined finesses of finitism, of metalogic, of mathematical atomistics, merely express the anxiety of bour-

geois mathematicians to separate themselves from matter and dialectics by the veil of formal logic, guiding them directly into the desert of scholasticism.

莱维和盖斯特都参加了苏联代表的会议。鲍里斯·赫森（1893—1936）在一篇关于牛顿的论文中，阐述了辩证唯物主义对更广范围的科学分析所持的立场。在为盖斯特所作的悼文《斗争中的数学家》[52]中，莱维说，听众似乎不知所措，但盖斯特发言支持赫森。在同一篇文章中，莱维明确指出，马克思主义内部有关政治和经济的争论，与以下斗争之间存在联系：

Levy and Guest were both present at the Soviet sessions. The dialectical materialist position on analysing science more widely was presented by Boris Hessen (1893–1936) in a paper on Newton. In his eulogy for Guest, *The Mathematician in the Struggle*,[52] Levy reported that the audience seemed nonplussed, but Guest spoke in support of Hessen. In the same article, Levy made explicit the link between the disputes in Marxism over politics and economics, and

高度理论化和抽象性的斗争与困惑。在这方面，大卫能够发挥其十分宝贵的数理逻辑知识的作用。（第 157 页）

struggles and confusions of a highly theoretical and abstract nature. In this, David could bring to bear his very valuable knowledge of mathematical logic. (p157)

此后，盖斯特似乎基本放弃了数学研究，转向更广泛的教育和马克思主义活动。他关于马克思主义哲学的授课笔记在他去世后以《辩证唯物主义教科书》[53]为题出版。在“辩证法与形式逻辑”这一短节中，盖斯特重申，形式逻辑——他称之为“常识的逻辑”（第 68 页）——建立在绝对的抽象之上，因而

Subsequently, Guest seems to have largely abandoned mathematics research for wider educational and Marxist activity. His teaching notes on Marxist philosophy were published posthumously as *A Text Book On Dialectical Materialism*.[53] In a short section on *Dialectics and Formal Logic*, Guest reiterated the line that formal logic, which he characterized as the ‘logic of common sense’ (p68), was based on absolute abstractions and hence was

无法把握变化的内在过程，无法揭示其辩证性质。（第 68 页）

unable to grasp the inner process of change, to show its dialectical character. (p68)

要超越这一点，就需要辩证逻辑：试图进一步发展形式逻辑，会导致形而上学的思维。在这里，盖斯特援引恩格斯在《反杜林论》中将形式逻辑与初等数学、辩证逻辑与高等数学相联系的说法。正如我们将看到的，这一时期图灵的工作，恰恰为刻画“变化的内在过程”奠定了基础，进一步压缩了辩证法的空间。

To go beyond this, dialectical logic was required: attempts to further develop formal logic lead to metaphysical thinking. Here, Guest cited Engels’ association of formal logic with lower mathematics and dialectical logic with higher mathematics, from *Anti-Dühring*. As we shall see, Turing’s work in this period laid the basis precisely for characterizing ‘the inner process of change’, further shrinking the space for dialectics.

1934 年，几位著名的英国共产党成员为《辩证唯物主义的各个方面》[54]撰稿。莱维在《一位科学工作者眼中的辩证唯物主义》[55]中概述了正统观点，却在结尾以亚里士多德式的方式限定了辩证法的范围：

In 1934, several prominent CPGB members contributed to the collection *Aspects of Dialectical Materialism*.[54] In his paper *A Scientific Worker Looks at Dialectical Materialism*,[55] Levy summarized the orthodox account but ended it with an Aristotelian circumscription of dialectics:

p. 185

所谓辩证法的规律……似乎对于科学工作者已经提出的细致分析方法，几乎没有或完全没有增益。……在某种意义上，不能指望它们增添什么，因为它们自称居于科学之上。……因此，对科学而言，它［辩证唯物主义］是一种解释的方法，而非研究的方法。（第 30 页）

the so-called laws of the dialectic . . . appear to add little or nothing to the detailed methods of analysis that scientific workers have produced. . . . In a sense, they cannot be expected to add anything to these, for they profess to stand above science. . . . For science, therefore, it [dialectical materialism] is an *interpretative* method rather than a method of investigation. (p30)

在同一文集中，X 射线晶体学家 J. D. 伯纳尔（1901—1971）也试图将辩证法与科学区分开来，在《辩证唯物主义》[56]一文中呼应了莱维：

In the same collection, the X-ray chrystallographer J. D. Bernal (1901–71) also sought to distance dialectics from science, echoing Levy in his paper *Dialectical Materialism*:[56]

它［辩证唯物主义］并非对科学的批判，也不声称能取代实验方法，或取代对定律和理论的逻辑证明；但它确实以一种十分重要的方式补充科学：提供明确的方法，协调较大范围内的各专门科学，并为新的实验与发现指明方向。（第 98 页）

It [dialectical materialism] is not a critique of science; it does not claim to be a substitute either for experimental method or for the logical proof of laws or theories, but it does in a very important way supplement science by providing a definite method of coordinating the larger groups of special sciences and in pointing the way to new experiment and discovery. (p98)

第 8 章原注 / Chapter 8 Notes

p. 255 ↑1. G. Hegel：《逻辑学》，G. Di Giovanni 英译（Cambridge University Press，2010），第 592—593 页。

1 G. Hegel, *The Science of Logic*, trans. G. Di Giovanni (Cambridge University Press, 2010), 592–3.

p. 255 ↑2. Hegel：《逻辑学》。

2 Hegel, *The Science of Logic*.

p. 255 ↑3. 在所引英译本中占 30 余页。

3 For over 30 pages in the cited translation.

p. 256 ↑4. F. Engels：《反杜林论》（莫斯科：Foreign Languages Publishing House，1954）。

4 F. Engels, *Anti-Dühring* (Foreign Languages Publishing House, Moscow, 1954).

p. 256 ↑5. F. Engels：“社会主义从空想到科学的发展”，载《马克思恩格斯选集》第 II 卷（莫斯科：Foreign Languages Publishing House，1951）。

5 F. Engels, ‘Socialism: Utopian and Scientific’, in *Marx Engels Selected Works Volume II* (Foreign Languages Publishing House, Moscow, 1951).

p. 256 ↑6. F. Engels：“路德维希 • 费尔巴哈和德国古典哲学的终结”，载《马克思恩格斯选集》第 II 卷（莫斯科：Foreign Languages Publishing House，1951）。

6 F. Engels, ‘Ludwig Feuerbach and the End of Classical German Philosophy’, in *Marx Engels Selected Works Volume II* (Foreign Languages Publishing House, Moscow, 1951).

p. 256 ↑7. F. Engels：《自然辩证法》（莫斯科：Foreign Languages Publishing House，1954）。

7 F. Engels, *Dialectics of Nature* (Foreign Languages Publishing House, Moscow, 1954).

p. 256 ↑8. 感谢 Slava Gerovitch 提供这一文献；间接引自 S. Gerovitch《从新话到控制论话语：苏联控制论史》（MIT，2002），第 48 页，该处据 Yanovskaya。

8 Thanks to Slava Gerovitch for this reference, indirectly from S. Gerovitch, *From Newspeak to Cyberspeak: A History of Soviet Cybernetics* (MIT, 2002), 48 after Yanovskaya.

p. 256 ↑9. F. Engels：“《反杜林论》的准备材料”，载《马克思恩格斯全集》第 25 卷（Lawrence & Wishart，1987）。

9 F. Engels, ‘Preparatory Materials for Anti-Dühring’, in *Marx and Engels Collected Works*, vol. 25 (Lawrence & Wishart, 1987).

p. 256 ↑10. G. Cantor：“一般流形论的基础：关于无限理论的数学—哲学研究”，载 W. Ewald 编《从康德到希尔伯特：数学基础文献集》第 II 卷（Clarendon Press，1996），第 878—920 页。

10 G. Cantor, ‘Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation into the Theory of the Infinite’, in *From Kant to Hilbert. A Source Book in the Foundations of Mathematics, Volume II*, ed. W. Ewald (Clarendon Press, 1996), 878–920.

p. 256 ↑11. Engels：《反杜林论》，第 74 页。

11 Engels, *Anti-Dühring*, 74.

p. 256 ↑12. T. Burns：“约瑟夫 • 狄慈根与马克思主义史”，Science and Society 66，第 2 期（2002）：202—227。

12 T. Burns, ‘Joseph Dietzgen and the History of Marxism’, *Science and Society* 66, no. 2 (2002): 202–27.

p. 256 ↑13. J. Dietzgen：《一个社会主义者在认识论领域的漫游》，M. Beer、T. Rothstein 英译（Hottingen-Zurich：Social-Democratic Library，1887），https://www.marxists.org/archive/dietzgen/1887/epistemology.htm 。

13 J. Dietzgen, *Excursions of a Socialist into the Domain of Epistemology*, trans. M. Beer and T. Rothstein (Social-Democratic Library, Hottingen-Zurich, 1887), https://www.marxists.org/archive/dietzgen/1887/epistemology.htm.

p. 256 ↑14. 感谢 Ken Macleod 向我推荐伯恩斯的研究。

14 Thanks to Ken Macleod for referring me to Burns.

p. 256 ↑15. J. Dietzgen：《哲学的积极成果》，E. Untermann 英译（Chicago：Charles H. Kerr and Co，1906），https://www.marxists.org/archive/dietzgen/1887/positive-outcome/index.htm 。

15 J. Dietzgen, *The Positive Outcome of Philosophy*, trans. E. Untermann (Charles H. Kerr and Co, Chicago, 1906), https://www.marxists.org/archive/dietzgen/1887/positive-outcome/index.htm.

p. 256 ↑16. J. Dietzgen：《逻辑书简，特别是民主的无产阶级的逻辑》，E. Untermann 英译（Chicago：Charles H. Kerr and Co，1906），https://www.marxists.org/archive/dietzgen/1870s/letters-on-logic.htm 。

16 J. Dietzgen, *Letters on Logic Especially Democratic-Proletarian Logic*, trans. E. Untermann (Charles H. Kerr and Co, Chicago, 1906), https://www.marxists.org/archive/dietzgen/1870s/letters-on-logic.htm.

p. 256 ↑17. K. Marx：《资本论》第 I 卷，S. Moore、E. Aveling 英译（Lawrence & Wishart，1970）。

17 K. Marx, *Capital Volume I*, trans. S. Moore and E. Aveling (Lawrence & Wishart, 1970).

↑18. Dietzgen：《哲学的积极成果》，第 7 页及以下。

18 Dietzgen, *The Positive Outcome of Philosophy*, 7ff. p. 256

↑19. B. Russell：“致弗雷格的信”，载 van Heijenoort 编《从弗雷格到哥德尔》，第 124—125 页。

19 B. Russell, ‘Letter to Frege’, in van Heijenoort, *From Frege to Gödel*, 124–5. p. 256

↑20. G. Frege：“致罗素的信”，载 van Heijenoort 编《从弗雷格到哥德尔》，第 126—128 页。

20 G. Frege, ‘Letter to Russell’, in van Heijenoort, *From Frege to Gödel*, 126–8. p. 256

↑21. A. N. Whitehead、B. Russell：《数学原理》（Cambridge，1910、1912、1913）。

21 A. N. Whitehead and B. Russell, *Principia Mathematica* (Cambridge, 1910, 1912, 1913). p. 256

↑22. D. Hilbert：“论逻辑与算术的基础”，载 van Heijenoort 编《从弗雷格到哥德尔》，第 129—138 页。

22 D. Hilbert, ‘On the Foundations of Logic and Arithmetic’, in van Heijenoort, *From Frege to Gödel*, 129–38. p. 256

↑23. 原文为斜体。

23 Italics in original. p. 256

↑24. 可以说，希尔伯特这些简略概括为反对形式逻辑的马克思主义者提供了论据；对他们而言，教条主义者、经验主义者和主观主义者等词，具有强烈的哲学和政治意味。

24 Arguably, Hilbert’s potted summaries gave fuel to the Marxist opponents of formal logic, for whom terms like *dogmatist*, *empiricist* and *subjectivist* had strong philosophical and political resonances. p. 256

↑25. D. Hilbert：“公理化思维”，载 W. Ewald 编《从康德到希尔伯特：数学基础文献集》第 2 卷（Oxford University Press，1996），第 1106—1115 页。

25 D. Hilbert, ‘Axiomatic Thought’, in *From Kant to Hilbert: A Source Book in the Foundations of Mathematics, Volume 2*, ed. W. Ewald (Oxford University Press, 1996), 1106–15. p. 256

↑26. [55] 诚然，这一纲领的实行目前仍是一项尚未解决的任务。（第 1115 页）

26 *[55] To be sure, the execution of this programme is at present still an unsolved task.* (1115) p. 256

↑27. D. Hilbert：“初等数论的奠基”，载 W. Ewald 编《从康德到希尔伯特：数学基础文献集》第 2 卷（Oxford University Press，1996），第 1148—1157 页。

27 D. Hilbert, ‘The Grounding of Elementary Number Theory’, in *From Kant to Hilbert: A Source Book in the Foundations of Mathematics, Volume 2*, ed. W. Ewald (Oxford University Press, 1996), 1148–57. p. 256

↑28. 或者，也许是许多略有差别但联系紧密的语言。

28 Or, perhaps, many slightly different, but heavily connected, languages. p. 256

↑29. S. Kleene：《元数学导论》（North-Holland，1952）。

29 S. Kleene, *Introduction to Meta-Mathematics* (North-Holland, 1952). p. 256

↑30. L. Kronecker：“克罗内克 1886 年文献的脚注”，载 W. Ewald 编《从康德到希尔伯特：数学基础文献集》第 2 卷（Oxford University Press，1996），第 947 页。

30 L. Kronecker, ‘Footnote to Kronecker 1886’, in *From Kant to Hilbert: A Source Book in the Foundations of Mathematics, Volume 2*, ed. W. Ewald (Oxford University Press, 1996), 947. p. 256

↑31. M. van Atten、G. Sundholm：“L. E. J. 布劳威尔《逻辑原则的不可靠性》：新译与导论”，History and Philosophy of Logic 38，第 1 期（2017），https://doi.org/10.1080/01445340.2016.1210986 。

31 M. van Atten and G. Sundholm, ‘L.E.J. Brouwer’s “Unreliability of the Logical Principles”. A New Translation, with an Introduction’, *History and Philosophy of Logic* 38, no. 1 (2017), https://doi.org/10.1080/01445340.2016.1210986. p. 257

↑32. tertii exclusi［拉丁文，排除第三种情况］。

32 *tertii exclusi.* p. 257

↑33. L. Brouwer：“数学、科学与语言”，载 P. Mancosu 编《从布劳威尔到希尔伯特：20 世纪 20 年代的数学基础之争》（Oxford University Press，1998），第 45—53 页。

33 L. Brouwer, ‘Mathematics, Science and Language’, in *From Brouwer to Hilbert: The Debate on the Foundations of Mathematics in the 1920s*, ed. P. Mancosu (Oxford University Press, 1998), 45–53. p. 257

↑34. E. H. Carr：《苏俄史》（Macmillan，195078［底本年份如此］）。

34 E. H. Carr, *A History of Soviet Russia* (Macmillan, 195078). p. 257

↑35. V. I. Lenin：“再论工会、目前局势及托洛茨基和布哈林的错误”，载《列宁全集》英文第 1 版，Y. Sdobnikov 译，第 32 卷（莫斯科：Progress Publishers，1965），第 70—107 页，https://www.marxists.org/archive/lenin/works/1921/jan/25.htm 。

35 V. I. Lenin, ‘Once Again On The Trade Unions: The Current Situation and the Mistakes of Trotsky and Buhkarin’, in *Lenin’s Collected Works, 1st English Edition*, trans. Y. Sdobnikov, vol. 32 (Progress Publishers, Moscow, 1965), 70–107, https://www.marxists.org/archive/lenin/works/1921/jan/25.htm. p. 257

p. 257 ↑36. 很难想象当今的国家元首会在工会问题的辩论中讨论逻辑的地位，更不用说是在内战期间了。

36 It is hard to imagine a contemporary head of state arguing about the status of logic in a debate about Trade Unions, let alone during a Civil War.

p. 257 ↑37. H. Sheehan：《马克思主义与科学哲学：批判读本》（Humanity Books，1985）。

37 H. Sheehan, *Marxism and the Philosophy of Science: A Critical Reader* (Humanity Books, 1985).

p. 257 ↑38. G. Friedmann：“苏联反对形式主义的反叛”，Science & Society 2，第 3 期（1938）：300—321。

38 G. Friedmann, ‘Revolt Against Formalism in the Soviet Union’, *Science & Society* 2, no. 3 (1938): 300–21.

p. 257 ↑39. A. Kolmogorov：“论排中律”，载 van Heijenoort 编《从弗雷格到哥德尔》，第 414—437 页。

39 A. Kolmogorov, ‘On the Principle of Excluded Middle’, in van Heijenoort, *From Frege to Gödel*, 414–37.

p. 257 ↑40. A. Vucinich：“苏联的数学与辩证法：斯大林以前的时期”，Historia Mathematics 26（1999）：107—124。

40 A. Vucinich, ‘Mathematics and Dialectics in the Soviet Union: The Pre-Stalin Period’, *Historia Mathematics* 26 (1999): 107–24.

p. 257 ↑41. G. G. Lorentz：“1928 至 1953 年苏联的数学与政治”，Journal of Approximation Theory 116（2002）：169—223。

41 G. G. Lorentz, ‘Mathematics and Politics in the Soviet Union from 1928 to 1953’, *Journal of Approximation Theory* 116 (2002): 169–223.

p. 257 ↑42. E. Kolman、S. Yanovskaya：“黑格尔与数学”，《在马克思主义旗帜下》，1931，https://www.marxists.org/reference/subject/philosophy/works/ru/kolman.htm 。

42 E. Kolman and S. Yanovskaya, ‘Hegel and Mathematics’, *Under the Banner of Marxism*, 1931, https://www.marxists.org/reference/subject/philosophy/works/ru/kolman.htm.

p. 257 ↑43. K. Marx：《马克思数学手稿》（New Park Publications，1983）。

43 K. Marx, *Mathematical Manuscripts of Karl Marx* (New Park Publications, 1983).

p. 257 ↑44. 这后来成为数理逻辑的一个关键特征，如今也是编程中广泛使用的技术。

44 This proved a key feature of subsequent mathematical logic, and is now a widely used technique in programming.

p. 257 ↑45. A. Kolmogorv［底本拼法］：“论直觉主义逻辑的解释”，载 P. Mancosu 编《从布劳威尔到希尔伯特：20 世纪 20 年代的数学基础之争》（Oxford University Press，1998），第 328—334 页。

45 A. Kolmogorv, ‘On the Interpretation of Intuitionistic Logic’, in *From Brouwer to Hilbert: The Debate on the Foundations of Mathematics in the 1920s*, ed. P. Mancosu (Oxford University Press, 1998), 328–34.

p. 257 ↑46. G. Werskey：《可见的学院》（Allan Lane，1978）。

46 G. Werskey, *The Visible College* (Allan Lane, 1978).

p. 257 ↑47. D. Guest：“数学形式主义：1929”，载 C. H. Guest 编《大卫·盖斯特：一位为自由而战的科学家》（Lawrence and Wishart，1939），第 207—210 页。

47 D. Guest, ‘Mathematical Formalism: 1929’, in *David Guest: A Scientist Fights for Freedom*, ed. C. H. Guest (Lawrence and Wishart, 1939), 207–10.

p. 257 ↑48. D. Guest：“对数学命题的‘理解’：1930”，载 C. H. Guest 编《大卫·盖斯特：一位为自由而战的科学家》（Lawrence and Wishart，1939），第 211—213 页。

48 D. Guest, ‘The “Understanding” of the Propositions of Mathematics: 1930’, in *David Guest: A Scientist Fights for Freedom*, ed. C. H. Guest (Lawrence and Wishart, 1939), 211–13.

p. 257 ↑49. D. Guest：“评 E. T. 贝尔《数学家传》”，Labour Monthly，1937 年 10 月。

49 D. Guest, ‘Review of E. T. Bell, Men of Mathematics’, *Labour Monthly*, October 1937.

p. 257 ↑50. 对马克思主义者而言，“改良主义的”和“革命的”同样分别带有强烈的贬义和褒义。

50 ‘Reformist’ and ‘revolutionary’ again have strong negative and positive resonances, respectively, for Marxists.

p. 257 ↑51. E. Kolman：“数学科学的当前危机及其重建的总纲”，载《十字路口的科学：苏联代表于 1931 年 6 月 29 日至 7 月 3 日伦敦国际科学技术史大会提交的论文》，1931，Frank Cass and Co.，https://www.marxists.org/subject/science/essays/colman3.htm 。

51 E. Kolman, ‘The Present Crisis in the Mathematical Sciences and General Outlines for Their Reconstruction’, in *Science at the Crossroads: Papers Presented to the International Congress of the History of Science and technology Held in London from June 29th to July 3rd, 1931 by the delegates of the U.S.S.R, 1931*, Frank Cass and Co., https://www.marxists.org/subject/science/essays/colman3.htm.

↑52. H. Levy：“斗争中的数学家”，载 C. H. Guest 编《大卫·盖斯特：一位为自由而战的科学家》（Lawrence and Wishart，1939），第 150—160 页。

52 H. Levy, ‘The Mathematician in the Struggle’, in *David Guest: A Scientist Fights for Freedom*, ed. C. H. Guest (Lawrence and Wishart, 1939), 150–60. p. 257

↑53. D. Guest：《辩证唯物主义教科书》（Lawrence and Wishart，1939）。

53 D. Guest, *A Text Book of Dialectical Materialism* (Lawrence and Wishart, 1939). p. 257

↑54. H. Levy、J. MacMurray、R. Fox、R. Page、Arnott J. D. Bernal、E. F. Carritt［作者列表标点按底本］：《辩证唯物主义的各个方面》（Watts and Co.，1934）。

54 H. Levy, J. MacMurray, R. Fox, R. Page, Arnott J. D. Bernal, and E. F. Carritt, *Aspects of Dialectical Materialism* (Watts and Co., 1934). p. 258

↑55. H. Levy：“一位科学工作者眼中的辩证唯物主义”，载 H. Levy、J. MacMurray、R. Fox、R. Page Arnott、J. D. Bernal、E. F. Carritt 编《辩证唯物主义的各个方面》（Watts and Co.，1934），第 1—30 页。

55 H. Levy, ‘A Scientific Worker Looks At Dialectical Materialism’, in *Aspects of Dialectical Materialism*, ed. H. Levy, J. MacMurray, R. Fox, R. Page Arnott, J. D. Bernal, and E. F. Carritt (Watts and Co., 1934), 1–30. p. 258

↑56. J. D. Bernal：“辩证唯物主义”，载 H. Levy、J. MacMurray、R. Fox、R. Page Arnott、J. D. Bernal、E. F. Carritt 编《辩证唯物主义的各个方面》（Watts and Co.，1934），第 89—122 页。

56 J. D. Bernal, Dialectical Materialism, in *Aspects of Dialectical Materialism*, ed. H. Levy, J. MacMurray, R. Fox, R. Page Arnott, J. D. Bernal, and E. F. Carritt (Watts and Co., 1934), 89–122. p. 258

第 9 章逻辑危机与反形式主义的顶峰

Chapter 9: THE CRISIS IN LOGIC AND THE APOTHEOSIS OF ANTI-FORMALISM

引言

Introduction

p. 187 希尔伯特纲领可以简要概括为：试图确定罗素和怀特海在《数学原理》中使用的算术形式化是否

Hilbert’s programme may be conveniently summarized as seeking to establish whether or not the formalization of arithmetic used in Russell and Whitehead’s *Principia Mathematica*:

• 相容，即不可能证明一个公式及其否定都是定理；

• is *consistent*, i.e., it is not possible to prove that both a formula and its negation are theorems;

• 完备，即没有不能被证明为定理的定理；

〔校注：底本以“不能被证明的定理”说明不完备性；此处保留作者的用词，不将其静默改为“不可证的真命题”。〕

• is *complete*, i.e., there are no theorems which cannot be proved to be so;

• 具有判定程序，即一种会终止的机械程序或算法，能够确定任意公式是否为定理。

• has a *decision procedure*, i.e., a terminating mechanical procedure or algorithm, to establish whether or not an arbitrary formula is a theorem.

罗素悖论已经强烈暗示，数学不可能相容。随后，正如下文将要探讨的，哥德尔确立了数学不可能完备，而图灵确立了数学不可能拥有判定公式是否为定理的程序。

Russell’s paradox had already strongly suggested that mathematics could not be consistent. Subsequently, as we shall next explore, Gödel established that mathematics could not be complete, and Turing that mathematics could not have a decision procedure for theoremhood.

哥德尔与完备性

Gödel and completeness

元数学用数学来真正谈论数学自身；其关键在于找到公式的数学表示或编码，以便用其他公式来处理它们。第 7 章已说明，如何依照符号和句法来定义逻辑。如果给符号赋予数值，就能把一个公式转化为一个合成的数，再用算术将其分解回符号。

The key to meta-mathematics, where mathematics is used to quite literally talk about itself, lies in finding a mathematical representation or *encoding*, of formulae, for manipulation by other formulae. In Chapter 7, we saw how logic may be defined in terms of its symbols and syntax. If we assign numbers to symbols, we can turn a formula into a composite number, and then decompose it back into its symbols, using arithmetic.

在 1931 年一篇开创性的论文中，库尔特 • 哥德尔（1906—1978）用数论谓词演算的编码，证明了它是不完备的，即存在不能被证明的定理。[1]克莱尼[2]作了详尽的论述。

In a seminal 1931 paper, Kurt Gödel (1906–78) used an encoding of number theoretic predicate calculus to demonstrate that it was incomplete, that is, there are theorems which cannot be proved[1]. Kleene[2] gives a thorough account.

p. 187–188 哥德尔没有采用简单的乘法技巧，而是给符号序列中的每个位置分配一个质数。然后，把每个质数提升到该位置上符号编码所指定的幂，再将这些质数的幂相乘，从而汇合各符号的编码。

Rather than using a simple multiplication technique, Gödel assigned a prime number to each position in a symbol sequence. He then accumulated the codes for symbols by multiplying together the prime numbers, with each raised to the power of the code at its position.

p. 188 例如，假定按表 9.1 给符号赋予编码。那么，肯定前件式三段论的公式为：

〔校注：底本将 Modus Ponens 误拼为 Modus Ponems；英文照录。〕

For example, suppose we give symbols for the codes shown in Table 9.1. Then the formula for the syllogism Modus Ponems is:

$$(a \wedge (a \Rightarrow b)) \Rightarrow b$$

也就是“a，以及 a 蕴含 b，蕴含 b”，转化为：

that is ‘a’, and ‘a’ implies ‘b’, implies ‘b’, becomes:

$$8 \quad 10 \quad 4 \quad 8 \quad 10 \quad 7 \quad 11 \quad 9 \quad 9 \quad 7 \quad 11$$

这可以编码为：

This could be encoded as:

$$2^8 * 3^{10} * 5^4 * 7^8 * 11^{10} * 13^7 * 17^{11} * 19^9 * 23^9 * 29^7 * 31^{11}$$

这一方法适用于任意数量的符号和任意长度的序列。解码要用到一种称为质因数分解的技术，它可以求出一个数中所有质因数的指数。当然，这种编码会得到难以想象的大数，但哥德尔本来就不打算直接处理这些数。

This approach works for an arbitrary number of symbols and sequences of arbitrary lengths. Decoding then involves a technique called *prime factorization*, which enables the exponents of all the prime factors in a number to be found. Of course, this encoding gives unimaginably large numbers, but it was not Gödel’s intention to work directly with them.

表 9.1 符号编码

T（真）	1
F（假）	2
=	3
∧（且）	4
∨（或）	5
¬（非）	6
⇒（蕴含）	7
(	8
)	9
a	10
b	11
c	12
…	

Table 9.1 Codes for Symbols

T (true)	1
F (false)	2
=	3
∧ (and)	4
∨ (or)	5
¬ (not)	6
⇒ (implies)	7
(	8
)	9
a	10
b	11
c	12
…	

哥德尔想要编码的是证明，即一系列公式，其中每个公式要么是公理或定理，要么由公理或定理应用推理规则而得。他展示了如何构造函数来拆解这些哥德尔数，不仅检查其形式是否合乎规则，也检查它们是否对应于有效证明的公式。这可以用来确定一个公式序列是否构成证明，但与检查任意公式是否存在证明，是两回事。

Gödel wanted to encode proofs, that is, sequences of formulae where each is an axiom or a theorem, or follows from an axiom or theorem by application of a rule of inference. He showed how to construct functions that would pull these *Gödel numbers* apart to check, not just that they were well-formed, but that they corresponded to formulae for valid proofs. This works for establishing whether a sequence of formulas is or is not a proof, but it is quite different from checking whether a proof exists for an arbitrary formula.

在假定可以写出一个函数来判断公式是否为定理

p. 188–189

的前提下，哥德尔利用自指构造了一个令人想到罗素悖论的悖论。他展示了如何构造一个公式，设其哥德尔数为 N，它实际上说：

On the assumption that it was possible to write a function to tell whether or not a formula was a theorem, Gödel constructed a paradox using self-reference, reminiscent of Russell’s paradox. He showed how to make a formula with, say, Gödel number *N* that said, in effect:

哥德尔数为 N 的公式不是定理。

p. 189

the formula with Gödel number N is not a theorem.

现在，如果这个公式是定理，它所断言的便是真的，因此它不可能是定理。而如果它不是定理，它所断言的便是假的，因此它必须是定理。也就是说，假定存在一个函数能够检查公式是否为定理，就会使系统不相容；去掉这一假定，则使系统不完备。

〔校注：底本第二个条件句把“不是定理”直接推为“所断言的为假”，但前一引文恰说该公式不是定理。此处按原文保留这一推论，不能把这段概述视为严格证明。〕

Now, if this formula is a theorem, then what it asserts is true, so it can’t be a theorem. And if it isn’t a theorem, then what it asserts is false, so it must be a theorem. That is, assuming that there could be a function to check whether or not a formula was a theorem rendered the system inconsistent. And omitting the assumption rendered it incomplete.

接受排中律（LEM）的逻辑学家一致认为，宁取相容性而非完备性，是最稳妥的做法。对于不完备的系统，会有一些公式，我们无法保证判定它们是否为定理。如果简单地将它们加入系统作为公理，就可能利用它们证明矛盾。然而，对于相容的系统，一旦证明一个公式是定理，就知道无法证明它的否定。

Logicians who accept the LEM are agreed that preferring consistency to completeness is the safest course. For an incomplete system, there will be arbitrary formula whose status as theorems we cannot guarantee to determine. If they are simply added to the system as axioms, then it might be possible to use them to prove contradictions. However, for a consistent system, once we prove that a formula is a theorem, we know that we cannot prove its contrary.

图灵与终止性

Turing and termination

五年后，艾伦·图灵（1912—1954）证明，希尔伯特纲领的第三项要求无法得到满足。[3]也就是说，不可能构造一种会终止的机械程序，来判定一个公式是否为定理。

Five years later, Alan Turing (1912–54) showed that the third requirement of Hilbert’s programme could not be satisfied.[3] That is, it is not possible to construct a terminating mechanical procedure for deciding whether or not a formula is a theorem.

图灵的进路与上文讨论的数理逻辑学家很不一样。他没有使用诸如把谓词演算应用于集合论或数论而得到的形式系统，而是考察人们怎样用手工解决问题。他设想，一个人用铅笔和方格纸，在格子里逐个字母写下问题，然后来回处理，修改已有格子的内容，并写入新的格子。

Turing’s approach was very different from that of the mathematical logicians we have considered above. Rather than using a formal system derived, say, from predicate calculus applied to set or number theory, Turing considered how people solve problems by hand. He speculated about someone using a pencil and squared paper, writing down a problem in the squares, letter by letter, and then working backwards and forwards, changing squares and writing in new ones.

图灵把这一构想推广为现在所谓的图灵机（TM）。图灵机有一条由单元格组成的有限线性纸带，每个格子可以存放一个符号。有一个读写头，能够检查和修改格子；它位于“当前”格子的上方。例如，图 9.1 描绘了一台示意性的图灵机，纸带上逐个符号写着肯定前件式。

Turing generalized this conception to what is now known as a *Turing machine (TM)*. A TM has a finite linear tape of cells. Each cell may hold a symbol. There is a reading head that can inspect and change cells. The head is positioned over the ‘current’ cell. For example, Figure 9.1 shows a stylized TM with Modus Ponems on the tape, symbol by symbol.

纸带可以向左或向右移动，每次移动一格。到达纸带任一端时，便添加新的空格。因此，纸带可以增长到任意长度，但在任何阶段，它总是有界的，即长度始终有限。

The tape may be moved to the left or the right, one cell at a time. New empty cells are added when either end of the tape is reached. Thus, the tape may grow to be arbitrarily long, but, at any stage, it is always bounded, that is, it is always of finite length.

图灵机通过反复检查和修改纸带来执行操作，每次处理一个格子。机器的行为倾向，即当前符号如

A TM executes by repeatedly inspecting and modifying the tape, one cell at a time. The propensity of the machine, that is, how the current symbol determines what it

何结合它此前的操作来决定下一步做什么，称为当前状态。

图灵机由一组各含五个组成部分的指令控制，称为五元组。每条指令规定：

• $state_{old}$：如果机器处于这一旧状态，

• $symbol_{old}$：且读写头下是这一旧符号，

• $state_{new}$：则切换到这一新状态，

• $symbol_{new}$：把读写头下格子的内容改为这一新符号，

• direction：沿这一方向移动纸带一格，即向左或向右，或者停机。

机器的指令设置在控制单元中。控制单元反复寻找一条指令，其 $state_{old}$ 和 $symbol_{old}$ 与当前状态及读写头下的符号相匹配，然后执行它。若不存在这样的指令，计算就失败。利用查找、修改和复制符号序列这些基本操作，可以构造复杂的图灵机。

图灵的关键洞见是，图灵机的纸带可以存放任何符号序列，包括一组图灵机指令的符号序列。事实上，可以构造一台通用图灵机（UTM），配以适当的数据，逐个符号执行存放在其纸带上的任意图灵机。

面对任意图灵机究竟是否会终止这一问题，图灵构造了如下悖论。假设纸带上存放一台“待测试”图灵机的符号及其数据。再假设我们能够构造一台“停机判定”图灵机，它检查纸带：如果待测试机在其数据上停机，就以一个状态停下；如果待测试机在其数据上不停机，就以另一个状态停下。

写出一台不停机的图灵机很容易。它从一个空格开始，写入 0，然后向右移动一格。若右边没有格子，就增加一个。实际上，这台图灵机会永远“循环”，使纸带向无限延伸，却永远达不到无限。

我们可以修改这台停机判定机：如果待测试机在其数据上不停机，它就以一个状态停下；如果待测试机在其数据上停机，它就执行那条循环指令。

假设把修改后的停机判定机应用于它自身及某条纸带。如果它处理自身时不停机，那么它就会停机；如果它处理自身时停机，那么它就进入循环，不再停机。图灵得出结论：不可能构造一台图灵机来判断任意图灵机是否停机；也就是说，图灵机的停机问题不可判定。

will do next, in the light of what it has done previously, is called the current *state*.

A TM is controlled by a set of instructions with five components, known as quintuplets. Each instruction says:

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• $state_{old}$: if the machine is in this old state,

• $symbol_{old}$: and this old symbol is under the head,

• $state_{new}$: then change to this new state,

• $symbol_{new}$: change the cell under the head to this new symbol,

• *direction*: and move the tape one cell in this *direction*, i.e., left or right, or halt.

The machine is set up with the instructions in a control unit. This repeatedly looks for an instruction whose $state_{old}$ and $symbol_{old}$ match the current state and symbol under the head, which is then carried out. If there is no such instruction, the computation fails. Elaborate TMs may be constructed from the basic operations of searching for, changing and copying sequences of symbols.

Turing’s key insight was that a TM tape could hold any symbol sequence, including that for a set of TM instructions. Indeed, it is possible to construct a *Universal* TM (UTM) that will execute an arbitrary TM, held on its tape, symbol by symbol, with appropriate data.

Asking whether or not an arbitrary TM will ever terminate, Turing constructed a paradox as follows: Suppose we have the symbols for a ‘test’ TM and its data on a tape. Let’s assume that we can build a ‘halting’ TM that will inspect the tape and stop in one state if the ‘test’ TM halts on its data and in another state if the ‘test’ TM doesn’t halt on its data.

It is easy to write a TM that doesn’t halt. It starts on a blank cell, writes a 0 and moves one cell to the right. If there isn’t a cell to the right, a new one is added. In effect, this TM will ‘loop’ forever, extending the tape to infinity, without ever reaching it.

We can modify the ‘halting’ TM to halt in one state if the ‘test’ TM doesn’t halt on its data and to execute the loop instruction if the ‘test’ TM does halt on its data.

Suppose we apply the modified ‘halting’ TM to itself and some tape. If it doesn’t halt on itself, then it halts, and if it does halt on itself, then it goes into the loop, and doesn’t halt. Turing concluded that it wasn’t possible to construct a TM to tell if an arbitrary TM halts; that is, the TM *halting problem* is *undecidable*.

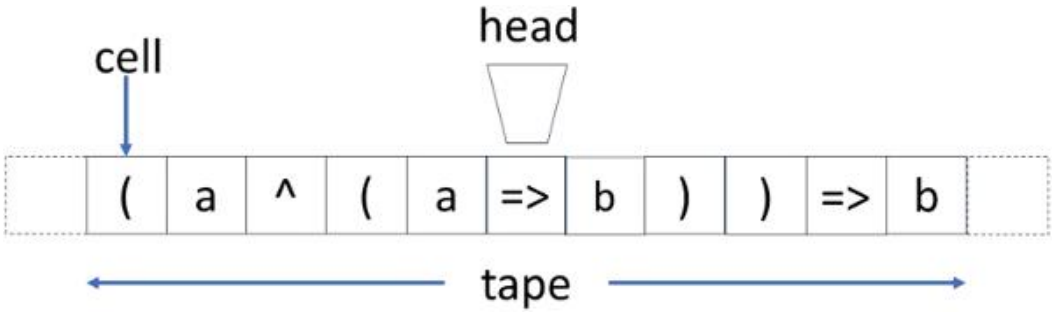


图 9.1 图灵机。作者绘图。

Figure 9.1 Turing machine. Figure by authors.
cell：单元格；head：读写头；tape：纸带。

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这一结果来自图灵在考察生成实数的图灵机时，使用康托尔对角线法作出的更详尽论证。虽然上文批评了对角线法这一整个观念，认为它涉及已完成的无限，但这并不使停机问题核心的悖论失效。

This result falls out of Turing's more elaborate argument using Cantor diagonalization in considering TMs that generate real numbers. While above we critiqued the whole notion of diagonalization as involving completed infinities, this does not invalidate the paradox at the heart of the halting problem..

丘奇—图灵论题

The Church-Turing thesis

在哥德尔和图灵开展工作的同时，人们还探索了其他几种元数学形式体系。斯蒂芬·克莱尼（1909—1994）沿着传统路线发展出递归函数论，将皮亚诺归纳法反转为一种定义数或集合上的函数的形式。[4]

Several other formalisms for meta-mathematics were explored in the same period that Gödel and Turing were working. Along traditional lines, Stephen Kleene (1909–94) developed recursive function theory, which inverted Peano's induction into a form for defining functions over number or sets.[4]

相比之下，摩西·舍恩芬克尔（1888—1942）早在 1924 年就发展了组合逻辑。[5]哈斯克尔·柯里（1900—1982）于 1929 年加以完善，[6]由此引出了阿隆佐·丘奇（1903—1995）自 1932 年起关于 λ 演算的极有影响力的工作。[7]

In contrast, Moses Schönfinkel (1888–1942) had developed combinatory logic in 1924.[5] This was refined by Haskell Curry (1900–82) in 1929,[6] and led to the highly influential work by Alonzo Church (1903–95) on λ calculus, from 1932.[7]

对辩证唯物主义的拥护者而言，这些系统似乎尤其成问题。舍恩芬克尔和柯里的组合逻辑由算子构成；算子的性质通过它们如何消去或复制其他算子、从而与之相互作用来定义。丘奇的 λ 演算则是纯粹抽象的，其规则通过代入来组合抽象。二者都不涉及具体的值。

These systems seem particularly problematic for adherents of dialectical materialism. Schönfinkel's and Curry's combinatory logics are built from operators whose properties are defined by how they interact with other operators by eliminating or duplicating them. And Church's λ calculus is one of pure abstraction, with rules for combining abstractions through substitution. Neither makes any reference to concrete values.

不过，这些系统有一个关键性质：它们都能够表示逻辑和算术，因此都能把握希尔伯特纲领核心的定理与证明概念。这就是丘奇—图灵论题的根源：所有对数进行计算的系统都在如下意义上等价，即任一系统都能翻译成其他任一系统。换言之，没有什么能在一个系统中表达，却不能在其他系统中表达。

Nonetheless, a key property of these systems is that they are all capable of representing logic and arithmetic, so they can all capture the notions of theorem and proof at the heart of Hilbert's programme. This is the root of the Church-Turing thesis: that all systems for performing calculations on numbers are equivalent in the sense that any may be translated into any other. That is, there is nothing that can be expressed in one system that cannot be expressed in any other system.

递归函数论、λ 演算和图灵机在提出后不久，就被证明相互等价。[8]后来，人们证明它们与冯·诺伊曼机，即一般化的数字计算机等价，因而与任意编程语言等价。模拟计算和量子计算[9]的形式化也被证明满足丘奇—图灵论题。

Recursive function theory, λ calculus and TMs were all demonstrated to be equivalent soon after they were developed.[8] Subsequently, they were shown to be equivalent to von Neuman machines, that is, generalized digital computers, and, hence, to arbitrary programming lan-

guages. Formalizations of analogue computing, and of quantum computing,[9] have also been shown to satisfy the Church-Turing thesis.

请注意，这是一个推测性的论题，而不是能够证明的理论。我们不知道可能有多少种不同的等价系统。不过，它是可证伪的：某个新系统仍可能被证明具有不同于当前已知系统的性质，因而无法相互翻译。也就是说，丘奇—图灵论题可以作为经验性的常规科学的一部分接受实验研究。事实上，我们将在后文论证，它对于更广泛地理解现实具有核心意义。

Note that this is a speculative *thesis* not a provable *theory*. We do not know how many different equivalent systems there might be. Nonetheless, it is a falsifiable *thesis*, as it may yet be proved that some new system has different properties to those that are currently known, and so there is no mutual translation. That is, the Church-Turing *thesis* is subject to experimental investigation as part of empirical normal science. Indeed, we will argue subsequently that it is central to a wider understanding of reality.

从根本上说，所有这些系统都能够实现计算。这在图灵机中十分明确：机器处理纸带上的数据。对于组合子和λ演算，指令与数据没有明显的区分，规则作用于公式，直到公式无法再简化。

At heart, all these systems enable computation. This is explicit in TMs, where the machine manipulates the data on the tape. For combinators and λ calculus, where there is no obvious separation of instructions and data, rules are applied to formula until they cannot be simplified any further.

图灵机与其他形式体系的根本区别就在这里。一组图灵机指令是建造实际机器的蓝图；这种机器能像数字计算机一样，自主地处理其数据。相反，其他形式系统需要人或解释装置来执行规则。我们认为，这是支持图灵机在阐述唯物主义时具有首要地位的有力论据，后文还会讨论。

Herein lies the fundamental difference between TMs and other formalisms. A set of TM instructions is a blueprint for building an actual machine which, like a digital computer, will run autonomously on its data. In contrast, other formal systems require a human being, or an interpreting device, to carry out their rules. We think that this is a strong argument for the primacy of TMs in elaborating a materialism, and will return to this subsequently.

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英国辩证唯物主义者的回应

British dialectical materialist responses

对辩证唯物主义者来说，哥德尔和图灵的成果进一步证明了没有辩证法的逻辑所具有的局限。1938年，[10]共产主义数学家阿利斯特·沃森（1908—1982）[11]撰文认为，一切悖论

For the dialectical materialists, Gödel’s and Turing’s results were further demonstrations of the limitations of logic without dialectics. Writing in 1938,[10] the Communist mathematician Alister Watson (1908–82)[11] thought that all the paradoxes

不过以这样或那样的方式表达了那种众所周知的困难：当我们试图把一个无限过程看作已经完成时，就会出现这种困难。（第 450 页）

merely express in one way or another the well-known difficulties which arise when we attempt to treat an infinite process as completed. (p450)

沃森指的是亚里士多德对潜无限与实无限的区分。我们同意沃森：假定实无限存在，会使康托尔对角线法之类的方法成问题。不过，如果沃森把图灵机的纸带认作已完成的无限，那就错了。无论如何，图灵关于停机问题不可判定的结果，可以不依靠对角线法来确立。

Watson is referring to Aristotle’s distinction between potential and actualized infinity. We agree with Watson that the assumption of actualized infinities makes, for example, Cantor diagonalization problematic. However, Watson would be wrong to identify TM tapes as completed infinities. In any case, Turing’s result of the undecidability of the halting problem may be established without diagonalization.

沃森对元数学的怀疑十分清楚：

Watson’s scepticism about meta-mathematics is clear:

在数学基础这一领域，试图为抽象看待的数学论证辩护或加以否定的种种尝试，造成了大量混乱；人们付出极其巨大的劳动，才摆脱这些混乱。（第 451 页）

The attempts which have been made in the subject of the Foundations of Mathematics to justify or condemn mathematical arguments taken in the abstract, have given rise to a host of confusions, from which it has taken the most immense labour to escape. (p451)

他没有直接运用辩证唯物主义对形式逻辑的批判。不过，他在这段话的脚注中说，自己所反对的是戴德金的主张：数学基础不需要提及“可测量的量”。这令人想到马克思主义者对不从已知量导出的变量所持的反对意见。

He does not directly deploy the dialectical materialist critique of formal logic. However, in a footnote to this passage, he says that he was writing in opposition to Dedekind’s claim that the foundations of mathematics did not require any mention of ‘measurable quantities’. This is reminiscent of the Marxist objection to variables that are not derived from known quantities.

1938 年，进化生物学家 J. B. S. 霍尔丹（1892—1964）在《马克思主义哲学与科学》[12]中指出：

In 1938, in *The Marxist Philosophy and the Sciences*,[12] the evolutionary biologist J. B. S. Haldane (1892–1964) observed:

总体上，我们可以认为，马克思主义者对那些抱负更大的逻辑理论颇为怀疑。例如，只要能够作出足够清晰的分类，罗素和怀特海在《数学原理》中的体系无疑就是真的，或大体是真的。

On the whole we may take it that Marxists are rather sceptical of the more ambitious logical theories. For example, the system of Russell and Whitehead, in the *Principia Mathematica* is doubtless true, or largely true, if sufficiently sharp classification is possible.

p. 192–193 也就是说，这一逻辑系统的真假取决于能否建立按类排列的具体“存在物”“关系”和“命题”，从中再抽象出更多的类。不过，霍尔丹怀疑是否可能建立这样的本体论：

That is, the truth or falsity of this logical system depends on the ability to elaborate concrete ‘existents’, ‘relations’ and ‘propositions’ arranged in classes, such that further classes may be abstracted. However, Haldane doubts the possibility of setting up such ontologies:

p. 193 很可能，人们过分强调了某些逻辑系统；这些系统只适用于具有某些高度抽象性质的材料，而现实世界所体现的这些性质，远比逻辑学家愿意相信的更少见、更不完整。

it is probable that too great an emphasis has been attached to logical systems which will only work for material that has certain highly abstract properties, which are rather less frequently and much less completely exemplified in the real world than logicians would like to think.

1939 年，霍尔丹在《自然辩证法》英译本序言[13]中，谈及数学家声称已消除了恩格斯六十年前指出的数学矛盾时写道：

In 1939, in the Preface to the English translation of *Dialectics of Nature*,[13] Haldane wrote of the mathematicians’ claims to have removed contradictions in mathematics, noted by Engels sixty years earlier:

实际上，他们只是把矛盾推到了后台，矛盾仍然存在于数理逻辑领域。不仅从一组公理及其应用规则推导全部数学的每一次努力都失败了，而且哥德尔还证明，这些努力必定失败。

Actually they have only pushed the contradictions into the background, where they remain in the field of mathematical logic. Not only has every effort to deduce all mathematics from a set of axioms, and rules for applying them, failed, but Gödel has proved that they must.

这忽略了关键一点：哥德尔的结果虽然限定了能够证明什么，却没有告诉我们如何确立单个公式的性质。事实上，基于公理系统的自动化技术，正越来越多地适用于现实世界中的重要问题，例如数字计算机设计。

This is missing the key point that, while Gödel’s results circumscribe what can be proved, they tell us nothing about establishing properties of individual formula. Indeed, automated techniques based on axiomatic systems are proving increasingly applicable to substantial real-world problems, like digital computer design.

与沃森一样，霍尔丹没有直接使用辩证唯物主义的语言。不过，与他评论哥德尔结果时一样，他暗示逻辑不足以把握具体现实，而这需要辩证法。

Like Watson, Haldane does not use the language of dialectical materialism directly. Nonetheless, his implication, as for his comments on Gödel’s results, is that logic is not adequate for concrete reality, for which dialectics is required.

孟什维克化的唯心主义之后的苏联逻辑学

Soviet logic after Menshevizing Idealism

20 世纪 30 年代以来苏联数学的命运已有广泛记述，尽管并非总是冷静客观的，例如武西尼奇、[14]洛伦兹、[15]塞内塔[16]和库塔特拉泽[17]的著作。马蒂亚斯那篇标题耸动的《逻辑与恐怖》[18]讨论了 1950 年以前的苏联逻辑学。阿内利斯[19]批评马蒂亚斯依赖菲利

The fortunes of Soviet mathematics, from the 1930s onwards, have been widely documented, though not always dispassionately, for example, in Vucinich,[14] Lorentz,[15] Seneta[16] and Kutateldze.[17] Mathias’s egregiously titled ‘Logic and Terror’ [18] discusses Soviet logic to 1950.

波夫[20]的“论战性且有偏见的叙述”。不过，马蒂亚斯仍引用了当时出版物中一些颇能说明问题的文字，虽然出处标注不充分：

形式逻辑始终是占统治地位的剥削阶级手中最可靠的武器。是宗教和蒙昧主义的堡垒。（引自 1934 年一部关于辩证唯物主义和历史唯物主义的著作）

形式逻辑的规律与辩证逻辑的规律相对立。形式逻辑是空洞、贫乏、抽象的，因为它所建立的规律和范畴不符合客观现实。（《简明哲学词典》，1940）（第 7 页）

其中还包括 1936 年的《苏联大百科全书》（1936）：

形式逻辑是一种形而上学的思维形式……是人类知识发展的最低阶段，被作为最高思维形式的辩证法所取代。

正如我们所见，形式逻辑并非包含在辩证法中，而是被辩证法取代、驳斥和克服。

……全联盟共产党（VKP(b)）内的反列宁主义偏向。形式逻辑思维是孟什维主义的一个特征，列宁曾经多次指出；他对孟什维克的三段论和诡辩的形式逻辑推演，进行了毁灭性的辩证批判。（第 7 页）

虽然逻辑学的广泛研究受到限制，柯尔莫哥洛夫、亚诺夫斯卡娅等“红色专家”仍能研究和教授逻辑，并接触西方研究。例如，亚诺夫斯卡娅于 1936 年开始在莫斯科国立大学教授数理逻辑，1943 年获任数理逻辑研讨班主任。[21]

第二次世界大战后，逻辑学在苏联的地位得到恢复，尽管围绕它与辩证法的关系仍有持续的激烈争论，下文将予讨论。

Anellis[19] criticizes Mathias for relying on the ‘polemical and prejudiced account’ in Philipov.[20] Nonetheless, Mathias contains telling, if poorly referenced, quotes from period publications:

Formal logic is always a most trustworthy weapon in the hands of the predominant exploiting classes. a bastion of religion and obscurantism. (from a 1934 work on Dialectical and Historical Materialism)

the laws of formal logic are opposed to the law of dialectical logic. formal logic is empty, poor, abstract, for the laws and categories which it sets up do not correspond to objective reality. (Concise philosophical Dictionary, 1940) (p7)

including the 1936 *Large Soviet Encyclopedia* (1936):

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formal logic is a metaphysical form of thinking . . . the lowest stage in the development of human knowledge, replaced by dialectic as the highest form of thinking.

Formal logic, as we have seen, is not included in dialectic, but is displaced refuted and overcome by it.

. . . the anti-Leninist deviations in the All-Union Communist Party (VKP(b)). Formal logic thinking is a characteristic trait of Menshevism frequently noted by Lenin who levelled devastating dialectical criticisms at the Menshevik formal-logical deductions of syllogisms and sophisms. (p7)

While the broad study of logic was curtailed, ‘Red experts’ like Kolmogorov and Yanovskaya were still able to explore and teach logic, and had access to western research. For example, Yanovskaya started to teach mathematical logic in 1936 at Moscow State University and, in 1943, was appointed Director of the Seminar in Mathematical Logic.[21]

The fortunes of logic in the Soviet Union were restored after World War Two, though not without continued fierce dispute over its relationship to dialectics, as we shall discuss below.

数字计算机

Digital computers

数字计算机的发展很可能被证明是 20 世纪的关键进展。可以肯定，没有它们，21 世纪初的生活几乎难以想象。计算机史同样已有详尽记述。这里将非常简要地关注其中几个突出方面。

第二次世界大战造成空前的大规模残酷杀戮与贫困，也加速了现代计算机的发展。[22]哈佛 Mark 1 由霍华德 • 艾肯（1900—1973）于 1937 年设计，IBM 以自动顺序控制计算器（ASCC）的名称建造，于 1944 年首次运行。它是机电式的，能存储数据，却不能存储程序。[23]指令编码在穿孔纸带上，控制线性排列的操作序列。循环程序靠重复纸带实现，分支靠更换纸带实现。约翰 • 冯 • 诺伊曼（1903—1957）曾用 Mark 1 为研制最初原子弹的曼哈顿计划进行计算。

The key development of the twentieth century may well prove to have been that of digital computers. Certainly, early twenty-first-century life would be pretty well unthinkable without them. The history of computers is again thoroughly documented. Here, we will focus on salient aspects, albeit very briefly.

The Second World War, unprecedented in mass cruelty and immiseration, hastened the development of modern computers.[22] The Harvard Mark 1, built by IBM as the Automatic Sequence Controlled Calculator (ASCC), was designed by Howard Aitken (1900–73) in 1937, and first ran in 1944. It was electro-mechanical and could store data, but lacked the capability to store programs.[23] Instructions were encoded on punched paper tape to control a linear sequence of operations. Looping programs were accomplished by repeating the tape, and branching by changing tapes. The Mark 1 was used

by John von Neumann (1903–57) to perform calculations for the Manhattan Project, developing the first atomic bombs.

电子数字积分计算机（ENIAC）是最早的全电子通用计算机之一，于 1945 年完成。它比哈佛 Mark 1 快得多，通过将部件按适当配置插接起来进行编程。ENIAC 的用途包括为最初的氢弹进行计算。

The Electronic Numerical Integrator and Computer (ENIAC), among the first all electronic general-purpose computers, was completed in 1945. Much faster than the Harvard Mark 1, it was programmed by plugging components together in appropriate configurations. ENIAC was used, among other things, in the calculations for the first hydrogen bombs.

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冯・诺伊曼在 1925 年的博士研究中从事集合论，并于 20 世纪 30 年代初与哥德尔交流不完备性结果。他深受图灵关于通用图灵机的思想影响：数据与指令共用同一存储器。他把这一现在称为冯・诺伊曼架构的设计，纳入影响深远的《EDVAC 报告初稿》。[2425]这份报告自由流传，影响了世界各地早期数字计算机的发展。到世纪中叶，已有许多公司开发和销售冯・诺伊曼架构的计算机，尤其是在美国和英国。

von Neumann had worked on set theory for his 1925 PhD, and had communicated with Gödel in the early 1930s about the incompleteness results. Strongly influenced by Turing’s ideas about the UTM, where data and instructions share the same memory, von Neumann included this design, now known as the von Neumann architecture, in the highly influential *First Draft of a Report on the EDVAC*.[2425] The report was circulated freely, influencing early digital computer development worldwide. Mid-century, there were numerous firms developing and selling von Neumann architecture computers, particularly in the USA and the UK.

因此，尽管辩证唯物主义者和西欧哲学家都把希尔伯特纲领的失败视为对数学能力范围的限制，这一纲领仍产生了深远而持久的影响。此外，数理逻辑学家在 20 世纪 30 年代发展的形式体系，尤其是递归函数论和 λ 演算，长期以来一直是实用编程语言语义和设计的基础。

Thus, while dialectical materialists, and Western European philosophers alike, saw the failure of Hilbert’s programme as limiting the reach of mathematics, nonetheless it had profound and very long-lasting influences. Furthermore, the 1930s formalisms developed by mathematical logicians, especially recursive function theory and λ calculus, have long been the basis of the semantics, and the design, of practical programming languages.

控制论

Cybernetics

控制论研究“动物和机器中的控制与通信”，是二战后与计算机发展并行的一个重要研究领域。诺伯特・维纳（1894—1964）影响深远的著作[26]阐述了其关键原理，借助信息论、统计力学和巴甫洛夫的行为心理学，说明机器如何通过反馈来学习，并根据观察到的结果优化活动。

Cybernetics, the study of ‘control and communication in the animal and the machine’, was an area of major activity after WW2, alongside the development of computers. Norbert Wiener’s (1894 – 1964) highly influential book[26] set out the key principles, drawing on information theory, statistical mechanics and Pavlovian behavioural psychology to elaborate on how machines might learn through feedback to optimize activities against observed outcomes.

与图灵一样，维纳的工作以对人类行为的抽象为基础，但侧重神经系统层面，而非高级认知。他观察到，在需要不断预测未来行为的活动中，例如操纵航行器或追踪移动目标，反馈对调节活动十分重要，而且这种调节是自主进行的（第 6—8 页）。维纳也承认数理逻辑对控制论的强大影响。他指出，无论对形式主义者还是直觉主义者而言：

Like Turing’s, Wiener’s work was based on abstracting from human behaviour, but at the level of the nervous system rather than higher cognition. Wiener observed the importance of feedback in governing activity that involved continuously predicting future behaviour, for example, in steering a craft, or following a moving target, and that this was carried out autonomously (pp6–8). Wiener also acknowledged the strong influence of mathematical logic on cybernetics. He observes that, for both formalists and intuitionists:

> the development of a mathematico-logical theory is subject to the same sort of restrictions as those that limit the perfor-

数理逻辑理论的发展，受到的约束与限制计算机器性能的约束属于同一种类。正如下文将看到的，甚至康托尔悖论和罗素悖论也可以这样解释。（第 13 页）

mance of a computing machine. As we shall see later, it is even possible to interpret in this way the paradoxes of Cantor and of Russell. (p13)

也就是说，维纳认为，这些悖论限制的是一般的人类推理，而不只是数学，因为大脑就是机器。他引用图灵写道：

That is, Wiener saw the paradoxes as limiting human reasoning in general, not just mathematics, because brains are machines. Citing Turing, he wrote:

逻辑研究必须归结为逻辑机器的研究，无论它是神经性的还是机械性的，都必须包括它所有不可消除的局限和缺陷。（第 125 页）

the study of logic must reduce to the study of the logical machine, whether nervous or mechanical, with all its non-removable limitations and imperfections. (p125)

然而，维纳认为信息既不是物质，也不是能量。因此，我们强烈反对他的如下主张：

Nonetheless, Wiener saw information as neither matter nor energy. Hence, we strongly contest his claim that: p. 196

任何不承认这一点的唯物主义，在今天都不可能生存。（第 132 页）

No materialism which does not admit this can survive at the present day. (p132)

因为信息必须具有物质载体。

because information must be embodied.

维纳充分意识到控制论和计算的社会影响。他认为：

Wiener was well aware of the social implications of cybernetics and computing. He thought that:

现代工业革命……必定使人脑贬值，至少在那些较简单、较例行的决策方面如此。（第 27 页）

The modern industrial revolution is . . . bound to devalue the human brain, at least in its simpler and more routine decisions. (p27)

然而，正如格罗维奇[27]所述，意识形态上对控制论的反对，成为苏联计算机发展的重大障碍。要说明这一点，我们需要作一个看似偏离主题的转折，考察苏联关于语言学的思想。

However, ideological objections to cybernetics proved a major barrier to the development of Soviet computers, as Gerovitch[27] recounts. To clarify this, we need to make an apparent segue sideways, and consider Soviet thinking about linguistics

语言学

Linguistics

在 19 世纪的大部分时间里，语文学主导语言学研究。它通过识别不同语言词语中的共同词根，来追溯语言的起源。这些研究中的许多，因着意确立当代民族群体在历史上的先在地位而受到扭曲。

For much of the nineteenth century, philology dominated linguistics. This sought to trace languages back to their origins by identifying common roots in words from different languages. Much of this work was distorted by concerns with establishing the historical primacy of contemporary national groupings.

现代语言学由费迪南·德·索绪尔（1857—1913）奠基。他身后出版的《普通语言学教程》[28]于 1916 年问世。索绪尔区分语言（langue）与言语（parole）：

Modern linguistics was founded by Ferdinand de Saussure (1857–1913), whose posthumous *Cours de linguistique générale*[28] was published in 1916. Saussure distinguished between language (*langue*) and speaking (*parole*):

把（1）社会性的与个人性的分开；（2）本质的与附属的、多少带有偶然性的分开。（第 14 页）

separating: (1) what is social from what is individual; and (2) what is essential from what is accessory and more or less accidental. (p14)

索绪尔进一步区分了两种情况：同时代的人使用相对不变的语言；以及随着使用者和用法改变，语言如何在时间中发生变化：

Saussure further distinguished the use of a relatively unchanging language by contemporary speakers from how language changes in time as users and usages changes:

凡与我们这门科学的静态方面有关的，都是共时的；凡与演变有关的，都是历时的。同样，共时性和历时性分别指语言的某种状态和某个演变阶段。（第 81 页）

Everything that relates to the static side of our science is synchronic; everything that has to do with evolution is diachronic. Similarly, *synchrony* and *diachrony* designate respectively a language-state and an evolutionary phase. (p81)

与语文学家不同，索绪尔认为，历时研究对于理解语言用处不大。他把语言学视为范围更广的符号学的一个组成部分：

Unlike the philologists, Saussure thought diachrony as of little use for understanding language. Rather, he saw linguistics as a component of a wider semiology:

可以设想一门研究符号在社会中生活的科学；它应是社会心理学的一部分，因而也是普通心理学的一部分。我将称之为符号学（semiology，来自希腊语 sēmeîon，意为“符号”）。（第 16 页）

A science that studies the life of signs within society is conceivable; it would be a part of social psychology and consequently of general psychology; I shall call it *semiology* (from Greek *sēmeîon* ‘sign’). (p16)

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索绪尔特别关心符号如何将能指——进行指示的东西——与所指——被指示的东西，即概念或观念——结合起来。他倾向于用“符号”（sign）而非“象征”（symbol）这个词，因为符号是任意的，象征则根据其所暗示的内容来选定（第 68 页）。请注意，索绪尔主要关心系统中符号之间的关系，而不是语义本身；他认为意义的改变涉及“所指与能指之间关系的转移”（第 75 页）。

Sussure was particularly concerned with how *signs* combine a *signifier*, that which points, and the *signified*, that which is pointed at, i.e., a concept or idea. Saussure prefered the term ‘sign’ to ‘symbol’, as *signs* are arbitrary, where symbols are chosen for what they suggest (p68). Note that Saussure was primarily concerned with relations between *signs* in systems, not with semantics in itself, seeing changes in meaning as involving ‘*a shift in the relationship between the signified and the signifier*’ (p75)

最后，索绪尔暗示了语言活动与计算之间的关系，第 10 章将予讨论：

Finally, Saussure hinted at the relationship between linguistic activity and computation, discussed in Chapter 10:

语言的机制由接续各项的相互作用构成，类似机器的运转：其中的部件尽管排列在一个维度上，却具有彼此作用的功能。（第 128 页）

The mechanism of language, which consists of the interplay of successive terms, resembles the operation of a machine in which the parts have a reciprocating function even though they are arranged in a single dimension. (p128)

苏联语言学家瓦连京·沃洛希诺夫（1895—1936）试图在 1929 年的《马克思主义与语言哲学》[29]中，发展一种研究语言的辩证方法。沃洛希诺夫把洪堡学派[30]的个人主义主观主义与索绪尔的抽象客观主义相对置（第 48 页）。他拒斥索绪尔的抽象，把语言认定为完全社会性的东西，即由相互交往的说话者产生的（第 98 页）。因此，沃洛希诺夫将主题，即语义，视为理解的核心（第 99 页及以下）。

The Soviet linguist Valentin Vološinov (1895–1936) sought to develop a dialectical approach to language in *Marxism and the Philosophy of Language*[29] from 1929. Vološinov counterpoised the *individualistic subjectivism* of the Humboldt school,[30] to the *abstract objectivism* of Saussure (p48). Vološinov rejected Saussure’s abstraction, identifying language as entirely sociological, that is produced by interacting speakers (p98). Thus, Vološinov saw *themes*, that is semantics, as central to understanding (p99ff).

对沃洛希诺夫而言，语言是厘清马克思主义中基础与上层建筑关系问题的关键；基础即物质条件，上层建筑即社会形式（第 18 页及以下）。意识形态由基础决定，而词语在基础与上层建筑之间起中介作用。

For Vološinov, language was key to disentangling the Marxist problematic of the relationship between base, that is, material conditions, and superstructure, that is, social forms (pp18ff). Ideology is determined by the base, with the word mediating between base and superstructure.

沃洛希诺夫认为，物化于言语中的语言具有首要地位。主题承载意识形态，符号具有“社会多重重音性”（第 23 页）。词语的评价性重音由表达性语调决定（第 103 页）。语调及其解释方式也反映说话者与听者的阶级取向。因此，“符号成为阶级斗争的场所”（第 23 页），在这里：

Vološinov saw language materialized in speech as primary. Themes bear ideology, and signs have ‘social *multiaccentuality*’ (p23). Words have *evaluative accent* determined by *expressive intonation* (p103). The intonation, and how it is interpreted, also reflect the class orientations of speakers and listeners. Thus, ‘Sign becomes an arena of class struggle’ (p23), where:

统治阶级力图赋予意识形态符号一种超阶级的、永恒的性质，消灭符号内部社会价值判断之间的斗争，或者把这种斗争压入内部，使符号只有单一重音。（第 23 页）

The ruling class strives to impart a supraclass, eternal character to the ideological sign, to extinguish or drive inward the struggle between social value judgements which occurs in it, to make the sign uniaccentual. (p23)

沃洛希诺夫赞同地引用同时代的尼古拉·马尔（1864—1934），断言“语言学是语文学的产物”。二人都批评传统语文学把话语当作脱离动态言语交往的独白来研究（第 72 页），但都认为可以在物质文化的背景中追溯语言的起源。

Vološinov approvingly cites his contemporary Nikolai Marr (1864–1934) in asserting that ‘linguistics is the child of philology’. Both criticized traditional philology for focusing on utterances as monologues separated from dynamic verbal interactions (p72), but both thought it possible to derive the origins of languages in the contexts of material cultures.

不过，与沃洛希诺夫不同，马尔认为不存在无阶级的语言。马泰卡[31]将沃洛希诺夫失势归因于这一分歧（第 173 页）。

However, unlike Vološinov, Marr thought that no language could be classless. Matejka[31] attributes Vološinov’s fall from favour to this difference (p173).

马尔也拒斥形式逻辑，认为它以阶级为基础：

Marr also rejected formal logic as class based:

形式逻辑是阶级思维的产物，它连同创造它的阶级，被无产阶级的辩证唯物主义思维一扫而去；在这种思维中，思想取得了凌驾于语言之上的地位。（转引自马蒂亚斯[32]）

Formal logic, a product of class thinking, together with the class that created it, is swept away by the dialectical materialist thinking of the proletariat in which thought gains ascendancy over language. (cited in Mathias[32])

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直到 1950 年，马尔一直是苏联最受推崇的语言学家。

Marr was the preferred Soviet linguistic until 1950.

苏联逻辑学的复兴

The revival of Soviet logic

1945 年美国引爆三枚原子弹，从根本上改变了苏联科学工作的优先次序，削弱了意识形态，尤其是反形式主义，在政策中的支配地位。

The detonation of three atomic bombs by the USA in 1945 radically changed Soviet priorities in science and weakened the dominance of ideology in policy, in particular anti-formalism.

1945 年 8 月，苏联成立两个机构来管理原子弹研发。[33]（第 131 页）苏联第一个计算机项目于 1946 年在基辅启动，由谢尔盖 • 列别杰夫（1902—1974）领导，MESM 于 1950 年投入运行。然而，维纳的著作[34]出版后，一场反控制论运动展开了，理由是控制论属于资本主义创新，旨在削弱工人阶级组织，最终完全取代工人，使他们陷于赤贫。[35]（第 128 页）这表明，这些意识形态家对马克思主义经济学，以及活劳动在资本主义剩余价值生产中的核心作用，了解甚少。在马克思的体系中，利润来自人的活动，而非机器。而且，正如乌斯丁[36]所述，苏联军方迅速采用控制论，并“将意识形态考虑放到一边”（第 312 页）。

Two Soviet agencies were set up in August 1945 to manage Soviet atomic bomb development[33] (p131). The first Soviet computer project began in Kiev in 1946, directed by Sergey Lebedev (1902–74), and the MESM became operational in 1950. However, after the publication of Wiener’s book,[34] a campaign against cybernetics was mounted on the grounds that it was a capitalist innovation to weaken working-class organization and, ultimately, entirely supplant workers who would be left destitute[35] (p128). This suggests that the ideologues had little grasp of Marxist economics, and the central role of living labour in the production of surplus value under capitalism. In Marx’s scheme, profits derive from human activity, not machines. And, as Usdin[36] notes, the Soviet military was quick to deploy cybernetics and ‘push ideological considerations aside’ (p312).

尽管如此，计算机仍不得不被称为“数学机器”，以避开针对控制论的审查。[37]人们明确区分两者：用控制论模拟人类行为不可接受，而用计算机进行计算和自动化则可以。把计算机类比于人脑被视为“荒谬”。[38]（第 142—143 页）对唯物主义来说，这一立场似乎完全是一种倒退。如果人不只是机器，那么人的额外特性就必然来自非物质的属性。

Nonetheless, computers had to be presented as ‘mathematical machines’ to evade cybernetic scrutiny.[37] A clear distinction was made between the unacceptable use of cybernetics to model human behaviour and the use of computers for calculations and automation. Analogies between computers and human brains were deemed ‘absurd’[38] (p142–3). This stance seems entirely retrogressive for materialism. If humans are more than machines, then their additional qualities must derive from non-corporeal properties.

回到逻辑学，苏联教育中的变化来得很快。1946 年苏共（布）中央委员会的一项指令，[39]由坎贝尔[40]引述，指出

Returning to logic, change came quickly in Soviet education. A 1946 CPSU(B) Central Committee directive,[39] cited by Campbell,[40] noted that it was:

中学不教授逻辑学和心理学，是很不恰当的。（第 343 页）

quite improper that logic and psychology are not taught in secondary schools. (p343)

并制定了开设这些课程的计划，配以资源。马蒂亚斯[41]因此叙述，切尔帕诺夫 1918 年版的《逻辑学教科书》于 1946 年重印，随后出版了斯特罗戈维奇的《逻辑学》、阿斯穆斯的一部新教科书，以及斯特罗戈维奇著作的又一版（第 8 页）。

and set out plans, with resources, for their introduction. Thus, Mathias[41] recounts how a 1918 edition of Chelapanov’s *Textbook of Logic* was republished in 1946, followed by Strogovich’s *Logic*, a new textbook by Asmus, and a further edition of Strogovich (p8).

洛伦兹[42]指出，继李森科在遗传学上取得所谓成

p. 198–199

就之后，1948 年曾有人推动苏联更广泛的科学工作系统地服从“科学具有阶级性”这一观念。然而，数学和物理学方面的计划被原子导弹项目负责人贝利亚叫停，因为“有影响力的物理学家向他解释，这可能损害这些项目”（第 217—218 页）。

Lorentz[42] observes that, following Lysenko’s alleged achievements in genetics, there were moves in 1948 to systematically align wider Soviet science with the notion that science had a class character. However, the plans for mathematics and physics were halted by Beria, the chief of Atomic Missile Projects, after ‘influential physicists explained to him that this may damage these projects’ (p217–8).

p. 199

1952 年，坎贝尔[43]指出，苏联逻辑教育面临着如何调和“资产阶级”逻辑与辩证逻辑的问题。他概述奥斯马科夫[44],[45]1950 年的一篇文章，认为其主张是：

In 1952, Campbell[43] noted the problems for Soviet logic education of squaring ‘bourgoise’ logic with dialectical logic. He summarizes a 1950 article by Osmakov[44],[45] as arguing that:

> 与世界观不同，思维逻辑是一种无阶级的现象。（第 281 页）

> Unlike a world outlook, the logic of thinking is a *classless* phenomenon. (p281)

此外，还有一句令人强烈联想到亚里士多德的表述：

Further, in a formulation strongly reminiscent of Aristotle:

> 思维逻辑据以进行的概念，对客观现实反映得好或不好，取决于思考者的意识形态或世界观。（第 282 页）

> The concepts on the basis of which the logic of thinking proceeds reflect objective reality well or badly according to the ideology or world view of the thinker. (p282)

巴扎诺夫[46]和基拉科斯[47]叙述了亚诺夫斯卡娅如何成为苏联形式逻辑复兴的核心人物。1947 年，她翻译了形式主义者希尔伯特和阿克曼的《理论逻辑纲要》，1948 年又翻译了塔尔斯基的《逻辑与演绎科学方法论导论》。[48]1947 年，亚诺夫斯卡娅主张，应当把“数理逻辑的方法论形式主义”与其背后的唯心主义哲学区分开来，因为[49]数理逻辑

Bazhanov[46] and Kilakos[47] recount how Yanovskya was central to the revival of Soviet formal logic. In 1947, she had translated *Grundzüge der theoretische Logik* by the formalists Hilbert and Ackermann and, in 1948, Tarski’s *Introduction to logic and the methodology of deductive sciences.*[48] In 1947, Yanovskya was arguing that ‘methodological formalism of mathematical logic’ should be distinguished from the idealist philosophy underlying it, because[49] mathematical logic

> 不仅可以被视为数学的逻辑，也可以被视为逻辑的数学，因为它在很大程度上是将数学方法应用于逻辑问题的结果。[50]

> can be considered not only as logic of mathematics but also as mathematics of logic, for it is in large part the result of the application of mathematical methods to the problems of logic.[50]

这一论证回到布尔对逻辑的数学化，巧妙地倒转了元数学所宣称的目标。

This argument returns to Boole’s mathematicization of logic, neatly inverting the stated objective of metamathematics.

亚诺夫斯卡娅在此后的一生中始终推动形式逻辑的发展。她享有很高声望，从 1959 年起担任莫斯科国立大学数理逻辑教席，直至 1966 年去世。[51]

Yanovskaya promoted formal logic throughout the rest of her life. Held in high esteem, she was the Chair of Mathematical Logic at Moscow State University from 1959 until her death in 1966.[51]

斯大林论语言学

Stalin on linguistics

1950 年，斯大林对马尔的批驳，预示着反形式主义立场公开逆转。正如洛伦兹[52]所述，斯大林密切关注精密科学的发展，虽支持李森科，却怀疑科学普遍具有阶级性：[53]

In 1950, the public reversal of anti-formalism was heralded by Stalin’s repudiation of Marr. As Lorentz[52] observes, Stalin paid close attention to developments in exact science and, while endorsing Lysenko, expressed scepticism about science’s general class character:[53]:

> 哈哈哈。那数学呢？达尔文主义呢？（第 217 页）

> Ha, ha, ha. And what about mathematics? And about Darwinism? (p217)

p. 199–200

《马克思主义和语言学问题》[54]于 1950 年发表在《真理报》，随后又有四次进一步澄清的交流。在最初的文章中，斯大林彻底拒斥马尔语言学的两项核心主张，认为语言既不是基础，也不是上层建筑，并且不具有阶级标记。这些批评也适用于沃洛希诺夫。

Marxism and Problems of Linguistics[54] was published in *Pravda* in 1950, followed by four further clarificatory exchanges. In the original article, Stalin roundly rejects two key tenets of Marr’s linguistics, arguing that lan-

在后来答复克拉舍宁尼科娃时，斯大林承认语义学对语言学的重要性，同时批评马尔过分强调语义学。从上下文看，斯大林似乎再次批评马尔按说话者的阶级，赋予词语和表达不同的意义。他认为，这种差异很少，存在于个别词语中，而不在所有说话者共有的语法中。

guage is neither base nor superstructure, and is not class marked. These criticisms also apply to Vološinov.

In a subsequent reply to Krasheninnikova, Stalin chided Marr for overemphasizing semantics while acknowledging its importance for linguistics. From the context, Stalin appears to again be criticizing Marr for attributing different meanings to words and expressions, depending on a speaker’s class. He suggests that such differences are very few, and lie in individual words, not in grammar, which is common to all speakers.

p. 200

最重要的是，斯大林表明，以形式主义之名作为决定性批评的做法已经结束：

N. Y. 马尔认为，语法是空洞的“形式”，把语法体系视为语言基础的人是形式主义者。这完全是愚蠢的。

我认为，“形式主义”是“新学说”的作者们发明的，目的是便利他们与语言学对手的斗争。

Most significantly, Stalin signalled the end of the attribution of formalism as a decisive critique:

N. Y. Marr considered that grammar is an empty ‘formality’, and that people who regard the grammatical system as the foundation of language are formalists. This is altogether foolish.

I think that ‘formalism’ was invented by the authors of the ‘new doctrine’ to facilitate their struggle against their opponents in linguistics.

此后，苏联逻辑学蓬勃发展，尽管辩证法与逻辑关系的争论仍在继续。[55]十七年后，即 1967 年，科普宁[56]仍在援引恩格斯[57]来捍卫黑格尔的真理。

Thereafter, Soviet logic flourished, though debate about the relationship between dialectics and logic continued.[55] Seventeen years later, in 1967, Kopnine[56] was still citing Engels[57] in defending the Hegelian verities.

不过，1959 年一份访苏美国计算机科学家的报告[58]表明，苏联逻辑与计算领域的进展给他们留下了深刻印象。他们指出：

Nonetheless, a 1959 report[58] by US computer scientists visiting the Soviet Union showed that they were very impressed by progress in logic and computing. They note that:

莫斯科、基辅和列宁格勒大学里，直接或间接与计算机有关的逻辑学家和数学家，显然比西欧或美国任何大学的都更多、更优秀。（第 17 页）

There are obviously more and better logicians and mathematicians connected directly or indirectly with computers at Moscow, Kiev, and Leningrad Universities than at any universities in Western Europe or the United States. (p17)

他们还指出，除 MIT 以外，美国没有能够与莫斯科国立大学相比的“面向计算机”的逻辑教学项目（第 17 页），并警告说，苏联培养的

They further note that, outside of MIT, the USA lacked programs in ‘computer oriented’ logic comparable to that at Moscow State University (p17), and warned that the production of:

合格的、面向计算机的数学家，而不只是计算机程序员，其人数可能很快超过美国。（第 17 页）

qualified computer oriented mathematicians, not just computer programmers - may, soon surpass that of the United States. (p17)

构造主义

Constructivism

苏联构造逻辑学派由小安德烈·马尔可夫[59]（1903—1979）创立。[60,61]构造主义根植于直觉主义，却像柯尔莫哥洛夫的进路一样，具有唯物主义取向。

The Soviet school of *constructive* logic was founded by Andrey Markov Jr.[59] (1903 – 79).[60,61] Constructivism had its roots in intuitionism, but with a materialist orientation, as in Kolmogorov’s approach.

直觉主义以唯心主义为基础，因此苏联逻辑学家接受它，似乎令人意外。不过，马尔可夫的合作者纳戈尔尼认为，在“解冻”之前，集合论尽管具有柏拉图主义根源，却被视为唯物主义而非唯心主义。[62]此外，范杜拉基斯[63]指出，一旦将布劳威尔直觉主义的逻辑框架与其哲学分开，前者就被认为与辩证唯物主义相容。正如沙宁在 1962 年所写：[64]

Given the idealism underlying intuitionism, its embrace by Soviet logicians may seem surprising. However, Markov’s collaborator Ngorny suggests that, before the ‘thaw’, set theory was seen as materialist rather than idealist, despite its Platonic roots.[62] Further, Vandoulakis[63] notes that Brouwer’s intuitionism was seen as consistent with dialectical materialism, once its log-

p. 200–201

ical framework was separated from his philosophy. As Ŝanin wrote in 1962:[64]

p. 201 越过直觉主义哲学的表层，人们在许多情况下能够发现：对于数学抽象形成过程和数学逻辑基础所涉及的根本问题，存在非常有价值的具体观察和深刻的具体分析。（第 7 页）

outside of the surface of intuitionistic philosophy one finds in many cases very valuable concrete observations and profound concrete analysis of the fundamental problems relating to the processes of forming mathematical abstractions and logical foundations of mathematics. (p7)

马尔可夫的学生库什纳指出其构造主义的四个核心特征。[65]首先，研究对象是有限的，由有限的构造过程，从有限字母表出发，依照明确的算法形成规则生成。沙宁正确地把这一点视为构造主义取得成功的关键：

Markov’s student Kushner identified four central characteristics of Markov’s constructivism.[65] First of all, the objects under investigation are finite and generated by finite constructive processes from a finite alphabet according to definite rules of algorithm formation. Ŝanin rightly saw this as key to the success of constructivism:

以精确的算术算法概念丰富数学，为许多作者沿新方向开展富有成果的研究提供了起点。（第 6 页）

The enrichment of mathematics with the precise concept of arithmetic algorithm served as a starting point for fruitful investigation in a new direction by many authors. (p6)

其次，库什纳指出，构造主义与直觉主义一样，拒斥排中律和双重否定，存在性证明必须以构造为基础。第三，它接受潜无限而不接受实无限。最后，可计算性与算法相联系；它接受经经验验证的丘奇—图灵论题，即所有可计算性模型的等价性。

Secondly, Kushner notes that, as with intuitionism, both the law of the excluded middle and double negation are rejected, and proofs of existence must be based on construction. Third, potential infinities are accepted, but not actualized infinities. And, finally, computability is associated with algorithms, and the Church-Turing thesis of the equivalence of all models of computability, demonstrated empirically, is accepted.

与图灵一样，马尔可夫从作为纸笔物质过程的数学出发。例如，他考虑在构成书稿的纸上画出一排竖线。在讨论这些线条随后如何经印刷复制时，他指出：

Like Turing, Markov started from mathematics as a material process using pen and paper. As an example, he considers drawing a row of vertical pen strokes on the paper forming his book manuscript. Discussing how this is subsequently reproduced by printing, he observes that:

这个构造对象［即原始图画］是由纸张和干墨水组成的物体，上面的图画则是这一构造对象的副本，由纸张和干的印刷油墨组成。它也是构造对象，因为制作副本可以被视为一种构造活动。[66]

The constructive object [ie the original drawing] is a material body consisting of paper and dried ink, and the drawing given above is a copy of this constructive object, consisting of paper and dried typographic paint. It, too, is a constructive object, since the preparation of a copy may be considered a constructive act.[66]

马尔可夫明确承认物质现实的限制：

Markov explicitly acknowledges the limits of material reality:

p. 201–202 在执行构造过程时，我们常常碰到时间、空间和材料不足所造成的障碍。通常总能设法绕过这些障碍。然而，我们的构造能力确实有限，也没有理由假定，由这种限制引起的障碍总能消除。恰恰相反，现代物理学和宇宙学似乎表明，这些障碍原则上无法克服。[67]

Carrying out constructive processes, we often come up against obstacles connected with a lack of time, space and material. One usually succeeds in somehow by-passing these obstacles. However, our constructive possibilities really are limited, and there are no grounds for supposing that the obstacles caused by their restrictedness can always be obviated. Rather to the contrary, it seems that modern physics and cosmology testify to the impossibility in principle of surmounting such obstacles.[67]

p. 202 不过，用亚里士多德式的表述来说，潜在可行性这一抽象

使我们能够考虑任意长的构造过程和任意大的构造对象。它们的可行性是潜在的：如果有足够的空间、时间和材料，它们在实践中就是可行的。[68]

Nonetheless, in an Aristotelian formulation, the *abstraction of potential feasibility*:

allows us to consider arbitrarily long constructive processes and arbitrarily large constructive objects. Their feasibility is potential: they would be feasible in practice, had we available sufficient space, time and material.[68]

马尔可夫明确把这种进路与基于集合论的经典数学相对照。后者允许实无限的抽象，[69]并以否定不存在来确立存在，其中“甚至可能不知道如何构造所需对象”。[70]

马尔可夫的记号体系以重写符号序列的规则为基础，类似乔姆斯基后来对形式文法类别的刻画。一套这样的规则称为算法；马尔可夫将其特征概括为：

a) 规定精确，不留任意选择的余地，而且具有普遍可理解性——算法的确定性；

b) 可以从在给定范围内变化的初始数据出发——算法的通用性；

c) 算法以获得某种预期结果为目标，使用适当初始数据时，最终确实能够得到该结果——算法的得出结果性。[71]

使用这一记号体系，马尔可夫通过归纳法系统地重构了皮亚诺算术的相当大一部分。

马尔可夫认为，他的进路满足他所谓的丘奇论题，即算法模型的等价性。他还认为，图灵机“极具说服力”，因为

在本质方面，图灵机的运行充分模拟了从事计算的数学家的行为，[72]

即从一种思维状态转到另一种思维状态。

这套记号旨在用于形式系统的实际实验。例如，“元算法”编程语言 Refal[73]把马尔可夫规则纳入一个更常规的框架，用以辅助探索形式语义。然而，库什纳[74]指出，虽然马尔可夫的工作在西方已为人所知，记号的复杂性却妨碍了更广泛的采用。

与直觉主义一样，接受构造主义对无限推理的限制，会抽掉许多重要的经典数学的基础。因此，人们也像直觉主义者那样，开展大量研究，试图在有限性基础上重建这些数学。

具体说来，构造主义接受整数和有理数，却不接受必然涉及无限的实数。因此，把实数重新表述为构造函数，是至关重要的一步。构造主义的进路借鉴图灵与外尔的工作，用算法来定义实数：一个算法生成精度不断提高的有理数，而精度由另一个算法确定。正如沙宁所说：

Markov explicitly contrasts this approach with classical mathematics based on set theory, which allows abstractions of actual infinity[69] and for which existence is a consequence of refuting non-existence, where ‘a method for constructing the desired object may even be unknown’.[70]

Markov’s notation was based on rules for rewriting symbol sequences, similar to Chomsky’s subsequent characterization of classes of formal grammars. A *scheme* of such rules is called an *algorithm*, which Markov characterized as:

a) the precision of the prescription, leaving no place to arbitrariness, and its universal comprehensibility –the definiteness of the algorithm;

b) the possibility of starting out with initial data, which may vary within given limits –the generality of the algorithm;

c) the orientation of the algorithm towards obtaining some desired result, which is indeed obtained in the end with proper initial data, the conclusivenes of the algorithm.[71]

Using this notation, Markov systematically reconstructs a considerable portion of Peano arithmetic using induction.

Markov suggests that his approach satisfies what he calls Church’s Thesis, the equivalence of models of algorithms. He further argues that TMs are ‘extremely convincing’ as

in the essentials, a Turing machine’s performance adequately simulates the behaviour of a computing mathematician[72]

going from one state of mind to another.

The notation was intended for practical experimentation with formal systems. For example, the ‘meta-algorithmic’ programming language Refal[73] incorporated Markov rules within a more conventional framework as an aid to exploring formal semantics. However, Kushner[74] observes that, while Markov’s work was known in the west, wider take-up was hampered by notational complexity.

As with intuitionism, accepting constructivist limitations on infinitary reasoning removed the foundations of much essential classical mathematics. Thus, as with intuitionism, considerable research was undertaken to refound such mathematics on a finitary base.

In particular, constructivism accepted integers and rational numbers, but not real numbers, which are necessarily infinitary. Hence, a vital step was the reformulation of real numbers as constructive functions. In the constructivist approach, drawing on work by both Turing and Weyl, real numbers are defined by an algorithm that generates rational numbers of increasing precision, determined by another algorithm. As Ŝanin put it:

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the actual use of concrete real numbers in the natural sciences and engineering is essentially based on the possibility of

自然科学和工程中对具体实数的实际使用，本质上基于这样一种可能性：从一个实数的个别表示中，提取出给出其一系列有理近似值的算法。[75]

这明确承认，在实际应用中，由于测量有限，操纵物理现实时使用的是有理数。

尽管马尔可夫采取唯物主义进路，他仍赞同这样一种观点：不可判定性结果限制的是数学，而不是人：

因此，数学中（以及其他任何学问分支中）带有意向的研究事业，永远不会转移给机器；机器只能辅助人，而不能取代人。[76]

extracting from an individual representation of a real number an algorithm giving a sequence of rational approximating values for it.[75]

This recognized explicitly that, in practical applications, rational numbers are used in manipulating physical reality because measurement is bounded.

Despite Markov’s materialist approach, he shared the view that the undecidability results were limitations on mathematics but not human beings:

Therefore the conative, research enterprise in mathematics (as well as any other branch of learning) will never be transferred to machines, capable only of assisting man but not replacing him.[76]

第 9 章原注 / Chapter 9 Notes

↑1. K. Gödel：“论《数学原理》及相关系统中的形式不可判定命题”，载 J. van Heijenoort 编《从弗雷格到哥德尔：数理逻辑文献集，1879—1931》（Harvard University Press，1967），第 596—616 页。

1 K. Gödel, ‘On Formally Undecideable Propositions of *Principia Mathematica and Related Systems*’, in From Frege to Gödel: A Source Book in Mathematical Logic, 1879-1931, ed. J. van Heijenoort (Harvard University Press, 1967), 596–616. p. 258

↑2. Kleene：《元数学导论》。

2 Kleene, *Introduction to Meta-Mathematics*. p. 258

↑3. A. M. Turing：“论可计算数及其在判定问题上的应用”，Proceedings of the London Mathematical Society s2–42，第 1 期（1937）：230—265。

3 A. M. Turing, ‘On Computable Numbers, with an Aapplication to the Eentscheidungsproblem’, *Proceedings of the London Mathematical Society* s2–42, no. 1 (1937): 230–65. p. 258

↑4. S. C. Kleene：“自然数的一般递归函数”，Mathematische Annalen 112（1936）：727—742。

4 S. C. Kleene, ‘General Recursive Functions of Natural Numbers’, *Mathematische Annalen* 112 (1936): 727–42. p. 258

↑5. M. Schönfinkel：“论数理逻辑的构造块”，载 J. van Heijenoort 编《从弗雷格到哥德尔：数理逻辑文献集，1879—1931》（Harvard University Press，1967），第 357—366 页。

5 M. Schönfinkel, ‘On the Building Blocks of Mathematical Logic’, in *From Frege to Gödel: A Source Book in Mathematical Logic, 1879-1931*, ed. J. van Heijenoort (Harvard University Press, 1967), 357–66. p. 258

↑6. H. Curry：《组合逻辑基础》（Grundlagen Der Kombinatorischen Logik），F. Kamaraddine、J. Seldin 英译（College Publications，2017）。

6 H. Curry, *Foundations of Combinatory Logic: (Grundlagen Der Kombinatorischen Logik*), trans. F. Kamaraddine and J. Seldin (College Publications, 2017). p. 258

↑7. A. Church：“逻辑基础的一组公设”，Annals of Mathematics，第 2 系列，33，第 2 期（1932）：346—366。

7 A. Church, ‘A Set of Postulates for the Foundation of Logic’, *Annals of Mathematics* Series 2. 33, no. 2 (1932): 346–66. p. 258

↑8. S. J. Kleene［姓名首字母按底本］：“λ 可定义性与递归性”，Duke Mathematical Journal 2，第 2 期（1936）：340—353；A. M. Turing：“可计算性与 λ 可定义性”，Journal of Symbolic Logic 2（1937）：153—164。

8 S. J. Kleene, ‘λ-definability and Recursiveness’, *Duke Mathematical Journal* 2, no. 2 (1936): 340–53; A. M. Turing, ‘Computability and λ-Definability’, *Journal of Symbolic Logic* 2 (1937): 153–64. p. 258

↑9. D. Deutch［底本拼法］:“量子理论、丘奇—图灵原理与通用量子计算机”,Proceedings of the Royal Society: A, Mathematical, Physiscal and Engineering Sciences 400，第 1818 期（1985）：97—117。

9 D. Deutch, ‘Quantum Theory, the Church–Turing Principle and the Universal Quantum Computer’, *Proceedings of the Royal Society: A, Mathematical, Physiscal and Engineering Sciences* 400, no. 1818 (1985): 97–117. p. 258

↑10. A. G. D. Watson：“数学及其基础”，Mind 47，第 188 期（1938）：440—451。

10 A. G. D. Watson, ‘Mathematics and its Foundations’, in *Mind*, 47 no. 188 (1938): 440–51. p. 258

↑11. 沃森曾把图灵介绍给路德维希·维特根斯坦，并曾被指称为苏联间谍：`https://en.wikipedia.org/wiki/Alister_Watson`。

11 Watson introduced Turing to Ludwig Wittgenstein, and was alleged to have been a Soviet spy: `https://en.wikipedia.org/wiki/Alister_Watson`. p. 258

↑12. J. B. S. Haldane：《马克思主义哲学与科学》（George Allen and Unwin Ltd，1938）。

12 J. B. S. Haldane, *The Marxist Philosophy and the Sciences* (George Allen and Unwin Ltd, 1938). p. 258

↑13. J. B. S. Haldane：“序言”，载《自然辩证法》，苏联，1939，`https://www.marxists.org/archive/marx/works/1883/don/preface.htm`。

13 J. B. S. Haldane, ‘Preface’, in *Dialectics of Nature*, USSR, 1939, `https://www.marxists.org/archive/marx/works/1883/don/preface.htm`. p. 258

↑14. Vucinich：“苏联的数学与辩证法：斯大林以前的时期”。

14 Vucinich, ‘Mathematics and Dialectics in the Soviet Union: The Pre-Stalin Period’. p. 258

↑15. Lorentz：“1928 至 1953 年苏联的数学与政治”。

15 Lorentz, ‘Mathematics and Politics in the Soviet Union from 1928 to 1953’. p. 258

↑16. E. Seneta：“20 世纪 30 年代苏联的数学、宗教与马克思主义”，Historia Mathematica 31（2004）：337—367。

16 E. Seneta, ‘Mathematics, Religion, and Marxism in the Soviet Union in the 1930s’, *Historia Mathematica* 31 (2004): 337–67. p. 258

↑17. S. S. Kutateladze：“俄国数学的悲剧”，Siberian

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22 G. Dyson, *Turing’s Cathedral. The Origins of the Digital Universe* (Penguin, 2012).

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32 Mathias, ‘Logic and Terror’, 7.

p. 259 ↑33. Gerovitch:《从新话到控制论话语：苏联控制论史》。

33 Gerovitch, *From Newspeak to Cyberspeak: A History of Soviet Cybernetics*.

p. 259 ↑34. Wiener:《控制论，或动物和机器中的控制与通信》。

34 Wiener, *Cybernetics, or Control and Communication in the Animal and the Machine*.

↑35. Gerovitch：《从新话到控制论话语：苏联控制论史》。

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↑43. Campbell：“苏联的逻辑概念”。

43 Campbell, ‘The Soviet Concept of Logic’. p. 259

↑44. 苏联高等教育部社会科学教学管理机构的哲学部门负责人。

44 Head of the Philosophy Department in the Directorate for the Teaching of Social Sciences, USSR Ministry for Higher Education. p. 259

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↑48. Bazhanov：“复兴：S. A. 亚诺夫斯卡娅的逻辑学道路”，第 132 页。

48 Bazhanov, ‘Restoration: S. A. Yanovskya’s Path in Logic’, 132. p. 259

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51 Bazhanov, ‘Restoration: S. A. Yanovskya’s Path in Logic’, 132. p. 260

↑52. Lorentz：“1928 至 1953 年苏联的数学与政治”，第 217 页。

52 Lorentz, ‘Mathematics and Politics in the Soviet Union from 1928 to 1953’, 217. p. 260

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60 The son of Andrey Markov (1856–1922), known for foundational work on stochastic processes.

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p. 260 ↑63. I. M. Vandoulakis：“论 A. A. 马尔可夫对布劳威尔直觉主义的态度”，Philosophia Scientiæ 2015/1，第 19-1 期（2015）：第 46 段。

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p. 260 ↑66. A. A. Markov Jr、N. M. Nagorny:《算法理论》，M. Greendlinger 英译（Kluwer Academic，1988），第 1 页。

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67 Markov Jr and Nagorny, *The Theory of Algorithms*, 10.

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p. 260 ↑69. Markov Jr、Nagorny:《算法理论》，第 10 页。

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p. 260 ↑70. Markov Jr、Nagorny:《算法理论》，第 12 页。

70 Markov Jr and Nagorny, *The Theory of Algorithms*, 12.

p. 260 ↑71. Markov Jr:《算法理论》，第 1 页。

71 Markov Jr, *The Theory of Algorithms*, 1.

p. 260 ↑72. Markov Jr、Nagorny:《算法理论》，第 109 页。

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p. 260 ↑73. V. F. Turchin:“一种元算法语言”，Kibernetika 4，第 4 期（1967）：45—54。

73 V. F. Turchin,‘A Meta-algorithmic Language’, *Kibernetika* 4, no. 4 (1967): 45–54.

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74 B. A. Kushner,‘The Constructive Mathematics of A. A. Markov’, *The American Mathematical Monthly* 113, no. 6 (2006): 562.

p. 260 ↑75. Sanin：“构造实数与构造数空间”，第 13 页。

75 Sanin,‘Constructive Real Numbers and Constructive Number Spaces’, 13.

p. 260 ↑76. Markov Jr:《算法理论》，第 441 页。

76 Markov Jr, *The Theory of Algorithms*, 441.

第 10 章语言、自动机与意义

Chapter 10: LANGUAGE, AUTOMATA AND MEANING

语言与意义

Language and meaning

长期以来，人们一直区分陈述的意义的两个方面：它的值，以及它所谈论的内容。这些区别可以称为指称与涵义、指谓与含蕴，以及外延与内涵。

There is a longstanding distinction between the meaning of a statement as its value compared with what it is about. These may be termed reference and sense, denotation and connotation and extension and intension. p. 205

考虑对一个数的描述：

Consider the descriptions of a number:

比二多一

one more than two

它的外延意义来自将它求得一个值：三。另一种说法是，外延意义就是所有能在下列陈述中替换“那个数”、

The extensional meaning results from evaluating it to a value: three. Another way of expressing this is that the extensional meaning is all the things that can replace *the number that* in

那个数比二多一

the number that is one more than two

并保持陈述为真的东西。也就是说，我们把“那个数”当作变量，其唯一有效的值是“三”。

and retain the statement’s truth. That is, we treat *the number that* as a variable whose sole valid value is ‘three’.

现在考虑对一个数的描述：

Now consider the description of a number:

六的一半

half of six

它的外延意义也是三。但这两个陈述的内涵意义很不一样：前者涉及计数，后者涉及除法。我们可以[1]用皮亚诺算术将它们形式化，并证明二者等价。于是，它们具有可替换同义性。也就是说，在关于数的形式表达式中，可以互换使用二者，而不改变包含它们的表达式的意义。

which also has the extensional meaning of three. These statements’ intensional meanings are quite different. The first involves counting, and the second dividing. We could[1] use Peano arithmetic to formalize them, and prove that they are equivalent. They then have the property of substitutive synonymy. That is, they can be used interchangeably in formal expressions about numbers without changing the meaning of the expression that uses them.

现在假设克里斯会数数，却不会做除法。那么：

Now, suppose that Chris can count but can’t do division. Then:

克里斯知道三比二多一。

Chris knows that three is one more than two.

这是真的。但我们不能将“比二多一”替换为“六的一半”：

is true. But we cannot replace *one more than two* with ‘*half of six*’: p. 206

克里斯知道三是六的一半。

Chris knows that three is half of six.

同时还保持陈述为真。

and retain truth.

伯特兰·罗素试图分析并重新表述内涵性构造，使它们成为外延性的。[2]按照罗素的进路，我们可以写成：

Bertrand Russell sought to analyse and reformulate intensional constructs to make them extensional.[2] In Russell’s approach, we might write:

克里斯知道“有一个数比二多一”。

Chris knows that ‘there is a number that is one more than two’.

以明确“知道”的语境。如果将引号中的短语换成一个表面上等价的表述：

to make the context of ‘knows’ clear. If we replace the quoted phrase with an apparent equivalent:

克里斯知道“有一个数是六的一半”。

Chris knows that ‘there is a number that is half of six’.

那么，就不能把“有一个数”替换为“三”而仍保持陈述为真。

then we cannot replace *there is a number that* with *three* and retain truth.

鲁道夫·卡尔纳普（1891—1970）是一位逻辑实证主义者，试图从实证主义科学中寻找事实，以供

Rudolf Carnap (1891–1970) was a logical positivist who sought fact in positivist science for manipulation by

逻辑操作。起初，他认为一切陈述要么是外延性的，要么没有意义。在 1928 年的著作《世界的逻辑构造：哲学中的伪问题》[3]中，他提出一种全面的进路，否认任何不能以事实为基础的语言构造有意义。他尤其否认实在论（即唯物主义）或唯心主义有意义。也就是说，卡尔纳普认为知识建立在经验主义的验证上。尽管如此，他仍自称“物理主义者”，认为物理学示范了如何确立事实。

logic. Originally, Carnap considered all statements to be either extensional or meaningless. In his 1928 book *The logical structure of the world; pseudoproblems in philosophy*,[3] he enunciated a totalizing approach to denying meaning to any linguistic constructs that could not be grounded in fact. In particular, he disputed that either realism (i.e. materialism) or idealism was meaningful. That is, Carnap saw knowledge as grounded in verification via empiricism. Nonetheless, Carnap termed himself a ‘physicalist’ who saw physics as exemplifying how to establish facts.

威拉德 • 奎因（1908—2000）在 1953 年的《从逻辑的观点看》[4]中批评逻辑实证主义。在《经验论的两个教条》（第 20—46 页）中，他首先否认分析真理与综合真理之间存在深刻区别：前者基于意义而不考虑事实，后者基于事实（第 20 页）。例如，可以把形式化算术内部数“三”的性质，与某人对数“三”的认识相比较；后者是一个被报告的事实。其次，奎因拒绝那种还原论，即只承认能重构为由事实组成的逻辑陈述的陈述有意义（第 20 页）。最终留下的是一种明确的实用主义。

Willard Quine (1908–2000), critiqued logical positivism in his 1953 *From a Logical Point of View.*[4] In the essay *Two Dogmas of Empiricism* (pp20–46), he first disputes that there is a deep distinction between analytic truths, based on meanings without considering facts, and synthetic truths, based on facts (p20). For example, we might contrast properties of the number three, which are internal to formalized arithmetic, with what someone knows about the number three, which is a reported fact. Secondly, Quine rejects the reductionism that only accepts as meaningful those statements that may be reconstituted as logical statements constructed from facts (p20). He is left with an explicit pragmatism.

奎因特别关注能否用可替换性作为意义同义性的标准，探问替换在什么语境中正当，并指出替换必然改变陈述的形式（第 56 页）。他的结论同样是实用主义的：

Quine was particularly concerned with substitutivity as a criterion of synonymy of meaning, asking in what contexts substitution is legitimate and pointing out that substitutions necessarily change the form of statements (p56). His conclusion is again pragmatic:

> 真正重要的是在相关方面相似……这是经验科学的一个典型问题。（第 60 页）

> What matters rather is likeness in *relevant respects* . . . a problem typical of empirical science. (p60)

对奎因来说，形式系统以变量标记抽象的位置，因此：

For Quine, given that formal systems are constructed with variables marking points of abstraction:

> 被假定为一个实体，简而言之，就是被看作变量的值。（第 13 页）

p. 207

> To be assumed as an entity is, purely and simply, to be reckoned as the value of a variable. (p13)

于是：

Then:

> 一个理论所承诺的，正是且仅是这样的实体：为使该理论中的断言为真，理论的约束变量必须能够指称它们。（第 13—14 页）

> a theory is committed to those and only those entities to which the bound variables of the theory must be capable of referring in order that the affirmations made in the theory be made true. (pp13–14)

这是一种令人欣慰的宽容立场。它不仅容许在逻辑系统中操作事实，也容许操作系统能够正当操作的任何东西。

This is a pleasingly generous position. It admits not just of manipulating facts in logical systems, but of anything a system can legitimately manipulate.

模型论

Model theory

在 1921 年的《逻辑哲学论》[5]中，维特根斯坦主张，应当用事态来理解世界，即事物——他称之为对象——的配置。他断言：

In the 1921 *Tractatus Logico-Philosophicus,*[5] Wittgenstein suggested that the world should be understood in terms of *states of affairs*, that is, configurations of things, which he termed objects. He asserts that:

> 4.1 命题表现事态的存在与不存在。

> 4.1 Propositions represent the existence and non-existence of states of affairs.

4.11 真命题的总和就是整个自然科学（或自然科学的全部体系）。（第 25 页）

维特根斯坦还把命题的涵义界定为：

4.2 ……与事态存在和不存在的可能性相符合或不相符合。（第 30 页）

他随后提议用真值表来考察这些可能性（第 30 页），并已论证：

4.25 如果一个基本命题为真，则该事态存在；如果一个基本命题为假，则该事态不存在。

卡尔纳普深受维特根斯坦事态概念的影响，在 1947 年的《意义与必然性：语义学与模态逻辑研究》[6]中，通过状态描述这一观念，系统地形式化了外延意义概念：

有且仅有一个状态描述，描述宇宙的实际状态；它包含所有真的原子句，以及那些假的原子句的否定。因此，它只包含真句子，我们便称之为真状态描述。任何形式的句子为真，当且仅当它在真状态描述中成立。（第 10 页）

请注意，维特根斯坦谈的是确定某一事态是否存在，这相对于已知内容而言；卡尔纳普却断言，可以描述“宇宙的实际状态”。

卡尔纳普随后系统地构建一种形式对象语言，通过代入状态描述中的值来表达外延意义。内涵性陈述则用一种受约束的自然元语言[7]作出，以便译成对象语言，再对照状态描述加以验证。

到 1961 年，卡尔纳普已不再认为

关于事物的一切陈述，都能翻译为关于感觉材料的陈述。[8]

此外，他现在对内涵性陈述能够多严格地转化为外延性陈述，也持谨慎态度：

因此，我后来提出一个较弱的版本，主张每个非外延性陈述都能翻译为外延性语言中逻辑等价的陈述。这一论题似乎适用于迄今已知的一切非外延性陈述的例子，但尚未得到证明；我们只能把它作为一个猜想提出。（第 ix 页）

不过，维特根斯坦和卡尔纳普的工作构成了模型论的基础，而模型论如今支撑着编程语言的语义。卡尔纳普的工作也继续得到发展，例如马里安·普热莱茨基的《经验理论的逻辑》。[9]

对普热莱茨基而言，模型是用集合论形式化的事态或状态描述。他把模型限定于他所称的具有谓词性质的物理对象。他认为，这不足以成为物理学这类科学的基础，需要加入实数及关系、函数等高阶逻辑构造，尽管他本人并未作这种扩展（第 104 页）。由于科学不断发展，他还提出，可以通过截取一个

4.11 The totality of true propositions is the whole of natural science (or the whole corpus of the natural sciences). (p25)

Wittgenstein further identifies the *sense* of a proposition with:

4.2 . . . agreement and disagreement with the possibilities of existence and non-existence of states of affairs. (p30)

He then proposed the use of truth tables to explore these possibilities (p30), having argued that:

4.25 If an elementary proposition is true, then the state of affairs exists: if an elementary proposition is false, the state of affairs does not exist.

In his 1947 book *Meaning and Necessity: A Study in Semantics and Modal Logic,*[6] Carnap, strongly influenced by Wittgenstein’s states of affairs, systematically formalized the concept of extensional meaning through the idea of a *state description*:

There is one and only one state-description which describes the actual state of the universe; it is that which contains all true atomic sentences and the negations of those which are false. Hence it contains only true sentences; therefore, we call it the true state-description. A sentence of any form is true if and only if it holds in the true state-description. (p10)

Note that, where Wittgenstein talks about determining the existence of a state of affairs, which is relative to what is known, Carnap asserts the possibility of describing ‘the actual state of the universe’. p. 208

Carnap then systematically elaborated a formal *object language* for expressing extensional meanings through substitution of values from state descriptions. Intensional statements were made in a constrained natural *meta-language*[7] for translation into the *object language* for verification against state-descriptions.

By 1961, Carnap no longer thought that

all statements about things can be translated into statements about sense data.[8]

Further, he was now cautious about how rigorously intensional statements might be converted to extensional:

Hence I have later proposed a weaker version which claims that every nonextensional statement can be translated into a logically equivalent statement of an extensional language. It seems that this thesis holds for all hitherto known examples of nonextensional statements, but this has not yet been demonstrated; we can propose it only as a conjecture. (pix)

Nonetheless, the work of Wittgenstein and Carnap formed the basis of *model theory*, which now underpins the semantics of programming languages. And the development of Carnap’s work continued, for example, in Marian Przelecki’s *Logic of Empirical Theories*.[9]

For Przelecki, a *model* is state of affairs or a state description, formalized in set theory. Przelecki restricts models to what he calls *physical objects* with predicative properties. He suggests that this is insufficient as a basis for a science like physics, and it needs to be extended, though he doesn’t do so, with real numbers,

“横截面”来刻画一门学科的现状，从而确定其中确定的分析知识与不确定的综合知识之间不断变化的平衡（第 105—106 页）。

and higher order logic constructs like relations and functions (p104). Given that science continuously develops, Przelecki further suggests that the current state of a discipline might be characterized by taking a ‘cross section’ to determine the changing balance between determinate analytic and indeterminate synthetic knowledge (p105–6).

巴迪欧、模型论与唯物主义

Badiou, model theory and materialism

阿兰 • 巴迪欧的《模型的概念》[10]以 1968 年的讲课为基础，试图调和形式逻辑与马克思主义唯物主义传统。他批评卡尔纳普和奎因，认为两种观点都区分经验“事实”与逻辑形式（第 7 页），并主张这实际上将形式科学与经验科学联结起来；也就是说，二者之间存在

Alain Badiou’s *The Concept of Model*,[10] based on lectures given in 1968, sought to reconcile formal logic with the Marxist materialist tradition. Criticizing both Carnap and Quine, he sees the distinction between empirical ‘fact’ and logical form as common to both perspectives (p7), and claims that this actually serves to bind together formal and empirical science: that is, there is a

> 逻辑新实证主义与模型论之间的辩证共谋。（第 19 页）

> dialectical complicity between logical neo-positivism and model theory. (p19)

p. 209

巴迪欧分析的一大长处在于，他把数学刻画为统一的物质实践，将形式规则系统与实际计算联系起来：

The great strength of Badiou’s analysis is that he characterizes mathematics as a unified material practice, linking formal systems of rules to practical calculation:

> 有效程序这一哲学范畴——即可以通过一系列无歧义的书写操作明确计算出来的东西——真正处于一切数学认识论的中心。（第 26 页）

> The philosophical category of effective procedure –of that which is explicitly calculable by a series of unambiguous scriptural manipulations –is truly at the centre of every epistemology of mathematics. (p26)

巴迪欧强调，虽然模型是为探索形式系统而构造的，

Badiou emphasizes that while models are made to explore formal systems:

> 模型是关于逻辑—数学系统区分能力的、可用数学构造的概念。[11]（第 40 页）

> *a model is the mathematically constructable concept of the differentiating power of a logico-mathematical system.*[11] (p40)

反过来，模型也要接受实验研究：

In turn, models are subject to experimental investigation:

> 正因为形式装置本身是一种物质化的理论、一个数学结果，它才能进入数学知识的生产过程；在这一过程中，模型概念所指的不是一个等待形式化的外部，而是有待检验的数学材料。（第 47 页）

> It is because it is itself a materialised theory, a mathematical result, that the formal apparatus can enter into the process of production of mathematical knowledge; and in this process, the concept of model does not designate an outside to be formalised, but a mathematical material to be tested. (p47)

巴迪欧进一步认为，形式系统实实在在是一台“数学生产机器”，并指出

Badiou further argues that a formal system is quite literally a ‘machine for mathematical production’, and notes that the

> 这些系统的理论与自动机或计算机器理论之间日益明显的亲缘关系，鲜明地说明了形式主义的实验使命。（第 43 页）

> increasingly evident kinship between the theory of these systems and the theory of automata, or of calculating machines, strikingly illustrates the experimental vocation of formalism. (p43)

如下文将讨论的，我们不只把自动机本身视为实验装置，还认为它为语义提供一种动态的物理基础，超越数学构造，无论这些构造如何物质化。

As discussed below, we see automata themselves as not just experimental apparatuses but as providing a dynamic, physical basis for semantics that goes beyond mathematical constructs, no matter how materialized.

巴迪欧在构造模型时优先考虑集合论和整数，却没有提到丘奇—图灵论题。这个论题意味着，所有图灵完备（TC）系统都可以相互充当形式模型：证明一个新系统图灵完备，恰恰要求能够在它与某个已知图灵完备系统之间进行翻译。既然计算机是物

Badiou prioritizes set theory and integers for model making but does not mention the Church-Turing thesis. This implies that all Turing complete (TC) systems may serve as formal models for each other: demonstrating that a new system is TC requires precisely the ability to

理的图灵完备系统，它们便使形式模型立足于实际现实。

translate between it and some known TC system. Given that computers are physical TC systems, they ground formal models in actual reality.

自动机

Automata

第 9 章介绍了图灵机，以及它有界但可以任意扩展的纸带。我们还看到，根据丘奇—图灵论题，图灵机是一切可能计算的模型。

In Chapter 9, we met the Turing machine and saw how it has a bounded but arbitrarily extensible tape. We also saw that, according to the Church-Turing thesis, the TM is a model for all possible computations.

现在将探讨，限制图灵机纸带的性质如何改变它的计算性质。霍普克罗夫特和乌尔曼[12]对此有简明的论述。

We will now explore how restricting TMs’ tape properties changes their computational properties. Hopcroft and Ullman[12] provide a succinct account.

可以将图灵机纸带视为输入与输出的结合，同时也是存储器。初始纸带是输入，最终纸带是输出，图灵机能够任意检查、修改和扩展纸带。下面考虑通过改变纸带的性质，对图灵机行为施加的三类限制。

We might consider a TM tape as a combined input and output, as well as a memory. The initial tape is the input, and the final tape is the output, and the TM can arbitrarily inspect, modify and extend the tape. We’ll now consider three broad restrictions that we might place on TM behaviour by changing properties of the tape.

p. 210

首先，可以规定纸带不得扩展，即输出必须与输入一样大。这称为线性有界自动机（LBA），大致对应实际计算机。当然，计算机可以接收任意长度的流式输入，但任何时刻能保存在内存中的量都是有限的。实践中，我们仍把计算机当作图灵机，内存耗尽时就会抱怨。

First of all, we might insist that the tape cannot be extended, that is, the output must be the same size as the input. This is called a *linear bounded automaton (LBA)*, and broadly corresponds to actual computers. Of course, computers may be given streamed input of arbitrary length, but only a finite amount can be held in memory at any moment. In practice, though, we treat computers as if they were Turing machines, and get grumpy when they run out of memory.

其次，可以给有限状态自动机（FSA）配一条能沿一个方向扩展的纸带。它像栈一样工作：可以在顶部添加（压入）或移除（弹出）项，但只能检查栈顶项。这称为下推自动机（PDA）。下推自动机并不是实际机器的常见形式，却是计算机程序初步句法分析软件的核心。

Secondly, we might give a FSA a tape which is extensible in one direction. The tape behaves like a *stack*, where items may be added to (*pushed*) or removed from (*popped*) the top, but only the top item can be inspected. This is known as a *push down automaton (PDA)*. PDAs are not a common form of practical machine but are central to software for initial syntax analysis of computer programmes.

第三，可以把对有限纸带的访问限定为只从头到尾检查，不得修改纸带或改变方向。因此，机器只能一路处理纸带，从一个状态转到另一个状态。如果不止一个旧状态都能进入同一个新状态，机器便不记得自身的计算历史。这类机器称为有限状态自动机（FSA）或摩尔机。

Third, we might restrict access to a finite tape to only inspecting it from start to finish without changing the tape or reversing the direction. Thus, the machine can only chew through the tape, changing from state to state. If it is possible to enter a new state from more than one old state, then the machine has no memory of its computation history. This sort of machine is called a *finite state automaton (FSA)* or *Moore machine*.

还能够产生输出的有限状态自动机称为米利机。这相当于给摩尔机加上一条只有一个单元格的可写纸带。米利机广泛用于控制按固定动作序列运行的过程，例如一组交通信号灯或洗衣机的不同洗涤程序。

A FSA, which can also emit outputs, is called a *Mealey machine*. This is equivalent to adding a single-cell writable tape to a Moore machine. Mealey machines are very widely used for controlling processes that go through fixed sequences of actions, for example, sets of traffic lights or different washing machine cycles.

语言游戏

Language games

我们略去了两者的区别：检查符号序列是否具有所要求的结构，以及更广泛的计算。它们的联系在于，给定某种类型的文法，就有算法能够生成相应的检查机器。不过，确认结构与符号序列的意义如何相关，还不清楚。

We have elided the distinction between checking whether or not a symbol sequence has a required structure and wider computing. The connection is that, given a grammar of some type, there is an algorithm to generate its checking machine. However, it’s not yet clear how affirming structure relates to the meanings of symbol sequences.

关键在于，机器检查序列时还做了什么。一旦把意义定位于物质化系统的状态变化中，就能以语言交流来阐明更广泛的互动。

The key lies in what else the machine does as it checks the sequence. Once we locate meaning in state changes in materialized systems, we can clarify wider interaction in terms of linguistic exchanges.

与卡尔纳普一样，维特根斯坦也退离了系统地阐述事态的总体性构想。在《逻辑哲学论》之后，他生前发表很少，后期工作通过身后出版的文集为人所知。

Just like Carnap, Wittgenstein retreated from his totalizing vision of systematically elaborating states of affairs. He published little in his lifetime after *Tractatus*, and his later work is known through posthumous collections.

p. 210–211

在 1953 年出版的《哲学研究》[13]中，维特根斯坦探究语言游戏这一观念，以考察语言如何发挥作用。与其他游戏一样，语言游戏包含参与者依规则操作事物。游戏通过参与者遵循共同规则，轮流以说话和行动来互动而进行。

In *Philosophical Investigations*,[13] published in 1953, Wittgenstein explores the idea of a *language game* as a way of exploring how language functions. Like other games, language games involve players manipulating things according to rules. They proceed by players taking it in turn to interact through talking and doing things, according to rules they share.

p. 211

例如：

“我们来下一盘国际象棋吧”这个表达的涵义，与游戏的全部规则之间，是在哪里建立联系的？——在游戏的规则表中，在游戏的教学中，在日常下棋的实践中。（第 80 页）

For example:

Where is the connection effected between the sense of the expression ‘Let’s play a game of chess’ and all the rules of the game? – Well, in the list of rules of the game, in the teaching of it, in the day-to-day practice of playing. (p80)

维特根斯坦将此与奥古斯丁的实指意义观相对置：意义基于名称与事物的联系，因此儿童通过他人指着事物并说出名称来学习语言。维特根斯坦则认为，儿童通过训练学习语言，这种训练包含说话，也就是互动（第 2—4 页）。

Wittgenstein counterposes this to the Augustinian idea of ostensive meanings based on the association of names with things, so children learn language by having things named as they’re pointed at. Rather, Wittgenstein suggests, children learn language through training, which involves talking, that is, interacting (pp2–4).

他引入一个建筑工人与助手的语言游戏：工人喊出一个构件的名称，助手便递上该构件（第 3 页）。然后，他考虑工人询问助手有多少块石板的情形。助手回答“五块石板”，与工人说“五块石板”、期望助手递来五块石板，很不相同（第 10 页）。也就是说，即使广义理解命名，问题也不在给事物命名：同样的语言构造，在不同语境中有不同意义。

He introduces a language game of a builder and assistant, where the builder calls out the name of a component and the assistant provides it (p3). He then considers the builder, asking the assistant how many slabs there are. Then, the assistant saying ‘five slabs’ in response is quite different from the builder saying ‘five slabs’ and expecting the assistant to hand over five slabs (p10). That is, rather than naming things, even broadly understood, the same linguistic constructs have different meanings in different contexts.

对维特根斯坦而言，我们在各种具体情境中参与众多小型语言游戏：

For Wittgenstein, we take part in a multiplicity of small language games in diverse concrete circumstances:

我们没有意识到日常语言游戏的惊人多样性，因为语言的外衣使一切显得相同。（第 224 页）

We remain unconscious of the prodigious diversity of all the everyday language games because the clothing of our language makes everything alike. (p224)

不过，语言游戏并非固定不变，而是在互动中发展；如果参与者行动不当，或者行动产生了未预见

However, language games are not fixed, but develop through interaction and may break down if players don’t act appropriately or if actions have consequences that

的后果，游戏便可能失灵。由于我们浸润于语言使用之中，这种情况可以得到解决：

weren’t foreseen. This may be resolved because we are immersed in language use:

我们规定规则，规定一种游戏的技法，可是当我们遵循规则时，事情并不像我们假定的那样发展。于是，我们仿佛纠缠在自己的规则之中了。（第 50 页）

we lay down rules, a technique for a game, and then when we follow the rules, things do not turn out as we had assumed. That we are therefore as it were entangled in our own rule. (p50)

文法与自动机

Grammars and automata

诺姆·乔姆斯基[14]考察了不同类别的合式符号序列，而不直接关注其意义。他区分了四类文法，它们已被证明与上述自动机相对应。[15]即图灵机可以检查 0 型文法，线性有界自动机检查 1 型文法，下推自动机检查 2 型文法，有限状态自动机检查 3 型文法。这些类型因此构成一个层级，每一型所能定义的范围都比下一型更广，正如相应机器按计算能力排列一样。

Noam Chomsky[14] explored different classes of well-formed symbol sequences without direct concern for their meanings. He identified four types of grammar that have been shown to correspond to the automata we discussed above.[15] That is, a Turing machine can check Type 0 grammars, an LBA Type 1 grammars, a PDA Type 2 grammars and a FSA Type 3. Thus, these types form a *hierarchy*, where each type can define more than the next, just as the corresponding machines are ordered by computational power.

能力最强的 0 型文法能够涵盖任何图灵完备计算。不过，它们不用于实际计算，因为通过界定符号序列的上下文来思考问题实在很困难；这些上下文可以任意庞大而复杂。人们转而使用编程语言：它们在严谨性上像形式系统，在表达力上像自然语言。

The most powerful grammar, Type 0, can capture any TC computation. However, they are not used for practical computing because it is really hard to think through problems in terms of bounding symbol sequence contexts, which may be arbitrarily large and complex. Rather, programming languages are used. These are like formal systems in their rigour, and natural languages in their expressiveness.

p. 212

使用编程语言，就要依据它所支持的抽象来思考，尤其要考虑变量能够表示哪类东西。在心理层面，这略去了计算的物理根基：计算在物理计算机中进行，操作符号编码的物理实例。因此，最终决定什么能够计算的，是现实的物理性质。对于涉及任意小的构造（如实数）或任意大的构造（如可无止境增长的数据库）的计算，这一点尤其重要。

Using a programming language involves thinking in terms of the abstractions that it supports, in particular what sorts of things variables may represent. Psychologically, this elides the physical grounding of computations in physical computers, manipulating physical instances of symbol encodings. Thus, what can be computed is determined, ultimately, by the physical nature of reality. This is particularly significant for computations that involve arbitrarily small constructs, like real numbers, or arbitrarily large ones, like databases that may grow open-endedly.

如果现实有限，我们最终就会耗尽可用于计算的物质。我们或许以为自己在一种拥有潜在无限资源的 0 型语言中编程，但实际上内存有界。而且，即使现实无限，谈论在有限时间内动用无限资源的有限计算，也没有意义。[16]

If reality is finite, then eventually we run out of stuff with which to do computations. We may think we’re programming in a Type 0 language with potentially infinite resources, but we actually have bounded memory. And, even if reality is infinite, it makes no sense to talk about finite computations that deploy infinite resources in a finite time.[16]

结论

Conclusion

我们对形式逻辑发展的叙述支撑着我们的唯物主义。与图灵一样，我们把数学视为植根于物理机器的机械活动；与巴迪欧一样，我们把建模与逻辑视为相互补充的实验活动；与维特根斯坦一样，我们认为意义产生于改变现实的互动；与马尔可夫一样，我们用建模为自动机的算法过程来刻画意义。

Our account of the development of formal logic underpins our materialism. With Turing, we see mathematics as a mechanical activity grounded in physical machines. With Badiou, we see model-making and logic as complementary experimental activities. With Wittgenstein, we see meanings created in interactions that change re-

依据丘奇—图灵论题，我们的唯物主义是还原论的。如果可计算性是理解现实的关键，而各种说明都能被证明具有等价的解释力，那么如何选择便是实用问题，而非原则问题。具体说来，我们拒绝柏拉图主义赋予纯数学的首要地位，却承认它在抽象方面的力量。

我们的唯物主义根本上是科学的。我们拒绝黑格尔和马克思主义传统赋予辩证法的首要地位，但承认它在探究现实过程中揭示和解决矛盾的力量。

我们的唯物主义也是有限主义的。与构造主义者一样，我们拒绝以实无限为基础的论证。

ality. And, with Markov, we see meaning characterized by algorithmic processes modelled as automata.

From the Church-Turing thesis, our materialism is reductionist. If computability is key to understanding reality and all accounts may be demonstrated to have equivalent explanatory power, then their choice is a matter of pragmatism, not principle. In particular, we reject the Platonist primacy of pure mathematics but recognize its power in abstraction.

Our materialism is fundamentally scientific. We reject the primacy given to dialectics in the Hegelian and Marxist traditions, but recognize its power in exposing and resolving contradictions in interrogating reality.

And our materialism is finitist. With the constructivists, we reject arguments based on actualized infinities.

第 10 章原注 / Chapter 10 Notes

↑1. 但这里不这样做。

1 But won’t.

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↑2. B. Russell:“论指称”,Mind 14,第 56 期(1905):479—493。

2 B. Russell, ‘On Denoting’, *Mind* 14, no. 56 (1905): 479–93.

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↑3. R. Carnap:《世界的逻辑构造:哲学中的伪问题》,R. A. George 英译(University of California Press,1967)。

3 R. Carnap, *The Logical Structure of the World; Pseudoproblems in Philosophy*, trans. R. A. George (University of California Press, 1967).

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↑4. W. V.-O. Quine:《从逻辑的观点看:逻辑哲学论文集》(Harper Torchbook,1963)。

4 W. V.-O. Quine, *From A Logical Point of View: Logico-philosophical Essays* (Harper Torchbook, 1963).

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↑5. Wittgenstein:《逻辑哲学论》。

5 Wittgenstein, *Tractatus Logico-Philosophicus*.

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↑6. R. Carnap:《意义与必然性:语义学与模态逻辑研究》(University of Chicago Press,1947)。

6 R. Carnap, *Meaning and Necessity: A Study in Semantics and Modal Logic* (University of Chicago Press, 1947).

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↑7. 是英语,而非黑格尔偏爱的德语。

7 English rather than Hegel’s preferred German.

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↑8. Carnap:《世界的逻辑构造:哲学中的伪问题》,第 viii 页。

8 Carnap, *The Logical Structure of the World; Pseudoproblems in Philosophy*, viii.

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↑9. M. Przelecki:《经验理论的逻辑》(Routledge and Kegan Paul,1969)。

9 M. Przelecki, *The Logic of Empirical Theories* (Routledge and Kegan Paul, 1969).

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↑10. A. Badiou:《模型的概念:数学唯物主义认识论导论》,Z. L. Fraser、T. Tao 英译(re.press,2007)。

10 A. Badiou, *The Concept of Model: An Introduction to the Materialist Epistemology of Mathematics*, trans. Z. L. Fraser and T. Tao (re.press, 2007).

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↑11. 原文为斜体。

11 Italics in original.

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↑12. J. E. Hopcroft、J. E. Ullman:《形式语言及其与自动机的关系》(Addison-Wesley,1969)。

12 J. E. Hopcroft and J. E. Ullman, *Formal Languages and Their Relation to Automata* (Addison-Wesley, 1969).

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↑13. L. Wittgenstein:《哲学研究》,G. E. M. Anscombe 英译(Blackwell,1958)。

13 L. Wittgenstein, *Philosophical Investigations*, trans. G. E. M. Anscombe (Blackwell, 1958).

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↑14. N. Chomsky:“论文法的某些形式性质”,Information and Control 2(1959):137—167。

14 N. Chomsky, ‘On Certain Formal Properties of Grammars’, *Information and Control* 2 (1959): 137–67.

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↑15. Hopcroft、Ullman:《形式语言及其与自动机的关系》。

15 Hopcroft and Ullman, *Formal Languages and Their Relation to Automata*.

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↑16. Cockshott、Mackenzie、Michaelson:《计算及其限度》。

16 Cockshott, Mackenzie, and Michaelson, *Computation and Its Limits*.

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第 11 章辩证唯物主义与机械论唯物主义

Chapter 11: DIALECTICAL VERSUS MECHANIST MATERIALISMS

信息论与唯物主义

Information theory and materialism

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20 世纪中叶以来，信息技术改变了我们与世界的互动方式，信息论也使科学与哲学发生根本转变。这些进展鲜明地表明，认识中最根本的变化，往往可以源于实践要求。19 世纪，革新了我们对时间、秩序和生命的理解的热力学，来自蒸汽机工程师的实际需要。我们已经提到，20 世纪 30 年代末，克劳德·香农认识到布尔代数为理解开关电路提供了形式体系。当时，他在美国电报电话公司贝尔工作。他使用布尔代数，是为了使自动电话交换机的设计合理化。随后，布尔代数逐渐成为支撑数字电路的框架；这些电路用于 20 世纪 40 年代和 50 年代新兴的计算机技术。

From the mid-twentieth century, our interaction with the world has been transformed by information *technology*, while information *theory* produced a radical shift in sciences and philosophy. These advances are striking illustrations of how the most radical changes in understanding can often arise from the demands of practice. In the nineteenth century, thermodynamics, which revolutionized our understanding of time, order and life, arose from the practical requirements of engineers working with steam engines. We have already mentioned how, in the late 1930s, Claude Shannon realized that Boolean algebra provided a formalism for the understanding of switching circuits. He was working for the US telegraph and telephone company Bell. His use of Boolean algebra stemmed from the need to rationalize the design of automatic telephone exchanges. From there, it spread out to become an enabling framework for digital circuitry employed by the nascent computer *technology* of the 1940s and 1950s.

香农的贡献并未止于此。他接着创立了现代信息论。如今已很平常的观念——信息可以用比特客观计量——来自他为贝尔写的一篇技术论文。[1]电报工程师在实践中必须估算不同信息传输方式的容量：普通电报线、使用频分复用的电缆、无线电链路等。为此，需要共同的计量单位。这类似于约一个半世纪前瓦特面临的问题：如何量化不同动力形式所提供的功。当时的办法是定义一匹标准马的功率。对 20 世纪中叶已经习惯于取值为 [0,1] 的布尔逻辑的工程师而言，信息的自然单位就是比特。

Shannon’s contribution did not stop there. He went on to found modern information theory. The notion that information is something that can be objectively measured in bits, something that is commonplace now, stems from a technical paper that he wrote for Bell.[1] Telegraph engineers had, as a practical matter, to estimate the carrying capacity of different ways of transmitting information: simple telegraph wires, cables with frequency division multiplexing, radio links etc. To do this, you needed a common unit of measure. This is analogous to the problem that Watt had faced about a century and a half earlier of quantifying the work provided by different forms of motive power. The solution then was to define the power of a standard horse. For a mid-twentieth-century engineer already used dealing with Boolean logic, which took on the values [0,1] the obvious unit of information was the bit.

如果他仅仅务实地采用二进制数字，即同时代工程师已熟悉的东西，其工作的影响就会有限。它之所以具有革命性，是因为他没有止步于布尔，而是进一步用概率论来定义比特。[2]他不只说比特是 0 或 1，而将其定义为区分两个等概率结果所用的信息量。

Had he just stopped at a pragmatic adoption of the binary digits, something with which his engineering contemporaries were already familiar, his work would have had limited impact. What made it revolutionary was that he did not stop at Boole. He went beyond Boole to define bits in terms of probability theory.[2] Instead of just saying that a bit was either a 0 or a 1, he defined it as the quantity of information used to discriminate between two equally probable outcomes.

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假设纽约的爱丽丝抛一枚公平硬币三次。波士顿的鲍勃知道，每次正面和反面出现的概率相等，却不知道爱丽丝哪次抛出了正面，哪次抛出了反面。如果爱丽丝用电传机向鲍勃传送结果序列，那么无论她采用什么编码，无论是

Suppose Alice in New York tosses a fair coin three times. Bob in Boston knows that on each toss, HEADS or TAILS are equally likely, but he cannot tell which falls or Alice’s coin were HEADS and which were TAILS. If Alice transmits her sequence of Bob on a Telex machine,

then whatever code she uses to transmit the falls, whether it be

HEADS（正面）

HEADS（正面）

TAILS（反面）

还是 HHT 或 110，按香农的理论，她都只向鲍勃发送了 3 比特信息。如果采用 20 世纪 40 年代使用的 5 位二进制博多码，第一种逐词拼出的方式就需要 105 个二进制位。[3]若用 HHT，则发送 15 个二进制位；若在线路上发送原始二进制数据，就只用 3 个二进制位。

HEADS

HEADS

TAILS

or HHT or 110 she has, according to Shannon, sent to Bob only 3 bits of information. If the information was sent using the 5-binary-digit Baudot Code in use in the 1940s, then the first alternative, spelling it out word at a time would have requires 105 binary digits.[3] If as HHT then 15 binary digits would have been sent. If raw binary data had been sent on a line then only 3 binary digits would have been used.

因此，对香农而言，信息与数据不同。数据编码信息的效率可能很低。第一个英语文本例子的效率，仅为最紧凑编码的三十五分之一。

So for Shannon information and data were something distinct. Data might be very inefficient at encoding information. The first exampl of English text was thirty-five times less efficient than the most compact encoding.

目前我们只考虑了典型的公平硬币。如果硬币很不公平呢？

如果你的随机数发生器有偏呢？

如果硬币有四分之三的概率正面朝上呢？

So far, we have been considering only the paradigmatic fair coin. What if we have a very unfair coin?

What if your random number generator is biased?

What if the coin falls HEADS three quarters of the time?

那么显然，每次传送一条“正面”消息，发送的信息都必定少于公平硬币的情形。看到一长串抛掷结果后，鲍勃能估计正面的先验概率。如果要猜下一条消息，他每次都会猜正面，这样猜对的比例就会超过 50%。因此，现在每条“正面”消息携带的信息都比以前少，不到 1 比特。

Then it is clear that each transmission of a HEAD message must send less information than in the case of a fair coin. Having seen a long sequence of tosses, Bob can estimate the a priori probability of a HEAD. Were he to guess what the next message was going to be, he would guess HEADS each time. When he did so, he would be correct more than 50 per cent of the time. Consequently, each HEADS message now conveys less information than before. It conveys less than one bit.

香农提出了一个公式，计算这种有偏符号流的信息量。假设长数据流有 n 种可能的不同符号，第 i 种符号出现的概率为 p_i，则每个数据符号的平均信息量为

Shannon came up with a formula for the information content of such biased streams of symbols. Suppose we have a long data stream with n possible distinct symbols, such that the ith symbol occurs in our stream with probability p_i, then the average information content per data symbol is given by

$$H = -\sum_{i=1}^{n} p_i \log_2(p_i)$$

负号保证得到正的结果，因为这些概率都小于 1，因此 $\log_2(p_i) < 0$。

The minus sign is there to ensure we get a positive result since the probabilities are all <1.and hence $\log_2(p_i) < 0$.

所有不同符号等概率出现时，这一公式达到最大值。但人类语言不是这样，书面文本中字母的出现频率很不均匀。按香农的说法，这意味着书面文本即使在最基本的字符层面，也有效率不足或冗余。

This formula will be at a maximum when all the distinct symbols occur with equal probability. But in human languages, this is not the case. The frequency of occurrence of letters in written text is very non-uniform. In Shannon’s terms, this means that written text is inefficient or redundant, even at the most basic character level.

香农将符号的信息量与其原始编码区分开来，这在密码学[4]和数据压缩中有实际用途。每当观看数字

His distinction between the information content of symbols and their raw codes has practical applications in cryptography[4] and data compression. Whenever you

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电视或收听 MP3 音轨时，你都在不知不觉中依赖香农的信息论。

但从唯物主义的角度看，这个公式最引人注目之处，在于它的信息公式在函数形式上与玻尔兹曼的熵公式相同。[5]

这表明，在深层意义上，信息与熵是同一回事。从哲学角度看，这令人震惊。

view a digital TV signal or listen to an MP3 track, you are implicitly relying on Shannon's information theory.

But the most striking feature of his formula, from the standpoint of materialism, is that his formula for information is functionally the same[5] as Boltzmann's formula for entropy.

It indicated that, at a deep level, information and entropy were the same thing. This was, from a philosophical standpoint, startling.

物理意义

Physical meanings

此前，哲学并未费心定义信息，但这个词的通常用法与人所能理解的信息紧密相连。在这种背景下，认为信息是物理的，而且与物理学的原子层面相联系，认为可以完全撇开信息的意义、纯粹进行统计分析，都是违反直觉的。

人们可能天真地以为，诸如印刷文本之类的数据，只因读者能够主观地赋予其意义，才包含信息。反过来，也可能认为，这种意义及其信息源自原作者的主观意图。

对于当代文本，这似乎说得通。但难以理解的信息呢？

在发现罗塞塔石碑以前，象形文字文本中的信息又如何呢？

罗塞塔石碑发现前的数百年间，这些埃及文本对所有人都没有意义。发现石碑后，它们便能够被阅读。不可回避的结论是：数百年来，信息一直客观存在于文本中，存在于纸莎草纸上墨迹的空间配置中。1799 年后，人们能够理解埃及文本，是因为文本客观地包含信息，即使其信息编码曾经无法破译。香农坚称，信息必须与其编码表示区分开来。

Philosophy had not bothered to define information up to this point, but the normal usage of the word was so tied up with information understandable to humans that information was something physical and, moreover, something connected with the atomic level of physics, the idea that information could be analysed purely statistically, completely ignoring its meaning, was counterintuitive.

One might naively assume that data, for example, in the form of printed text only contained information because a reader was subjectively able to attach meaning to it. Conversely, this meaning, and thus the information, could be assumed to derive from the subjective intention of the original author.

That seems plausible when dealing with contemporary texts. But what about arcane information?

What about information in hieroglyphic texts prior to the discovery of the Rosetta stone?

Over centuries before the Rosetta's discovery, these Egyptian texts were meaningless to everyone. After it was found, they could be read. The inescapable conclusion is that the information objectively existed for centuries in the texts in the spatial configurations of ink on papyrus. The Egyptian texts could be understood after 1799 because they had objectively contained information, even though that information's code remained indecipherable. Shannon was adamant that information had to be considered distinct from its encoded representation.

要定义实际信息这一概念，请考虑如下情形。假设一个信息源在产生譬如英文文本。文本可以翻译或编码为许多其他形式，例如摩尔斯电码，并且能够解码恢复原文。就大多数通信目的而言，这些形式同样有效，可以认为包含相同信息。给定任意一种编码形式，都能得到其他任何形式，尽管这当然可能需要复杂的计算。因此，我们将一个随机过程的实际信息，定义为所有能够通过可逆编码操作从原过程得到的随机过程所共有的东西。[6]

To define a concept of actual information, consider the following situation. Suppose a source is producing, say, English text. This may be translated or encoded into many other forms (eg Morse code) in such a way that it is possible to decode and recover the original. For most purposes of communication, any of these forms is equally good and may be considered to contain the same information. Given any particular encoded form, any of the others may be obtained, (although of course it may require an involved computation to do so). Thus we are led to define the actual information of a stochastic process as that which is common to all stochastic processes which may be obtained from the original by reversible encoding operations.[6]

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编码信息完全独立于任何人类的主观参与，无论这种参与来自作者还是读者。20 世纪 50 年代遗传

That encoded information exists quite independently of any human subjective involvement, either as author

密码的发现，使这一点更加清楚。在这里，信息存在于 DNA 中。随着 DNA 测序仪的发展，这种分子密码可以打印成人能够阅读的形式：由 ATGC 四个字母构成的序列。香农的观点是，无论信息以分子形式高效地表示，还是以墨水和纸张上的文本这种冗余得多的形式表示，其信息量都相同。分子形式的信息既可以由现代测序仪“读取”，也可以由核糖体“读取”。受过训练的人能够阅读打印结果，并在一定程度上加以解释，这一事实与 DNA 中的信息无关。

or reader, became even clearer in the 1950s with the discovery of the genetic code. Here, information exists in DNA. With the development of DNA sequencing machines, this molecular code can be printed out in a form people can read: a sequence made up of the letters ATGC. Shannon’s point is that the information content is the same whether it is efficiently represented in molecular form or much more redundantly as an ink-and-paper text. The information in molecular form can be ‘read’ either by a modern sequencing machine or by a ribosome. The fact that a trained person can read and, to some degree, interpret the printout is irrelevant to the information in the DNA.

它并不是因为人能阅读才成为信息。人之所以能够阅读它，是因为它是信息。更一般地说，信息量不依赖于人赋予的意义。关系恰好相反：我们处理意义的能力依赖于客观存在的信息，包括神经系统外部和内部的信息。

It is not information because a person can read it. A person can read it because it is information. More generally, information content does not depend on human meanings. The opposite holds. Our ability to handle meaning depends on objectively existing information: both external to and internal to our nervous system.

生物体会对环境作出反应。即使单细胞生物也能以有目的的方式作出反应。单细胞水生光合物种衣藻的个体能够向光源游动。[7]这种有目的的行为具有明显的进化优势：获得光能。但要做到这一点，细胞必须能够储存一种与外部环境状态相关联的内部状态。这里的环境状态就是光照方向；从几何学上说，它是三维空间中的一个向量。

Living organisms respond to their environment. Even single-celled organisms can respond in a purposeful way. Individuals of the unicellular aquatic photosynthetic species *Chlamydomonas* can swim towards a light source.[7] This purposeful behaviour has an obvious evolutionary advantage: access to light energy. But to achieve it, the cell must be able to store an internal state that is associated with the external state of its environment. The direction of illumination, which in geometrical terms is a vector in three-space.

让我们考虑这种向量的一种可能的内部编码。只需为相互垂直的 x, y, z 轴各配上三个状态 $(-1, 0, +1)$，表示入射光向量背离该轴、与该轴成直角或朝向该轴。采用一种简单的编码方式，细胞就需要能够记录 $3 \times \log_2(3) = 4.755$ 比特的信息。[8]

Let us consider a possible internal encoding of such a vector. It would suffice to associate three states $(-1, 0, +1)$ to each of the mutually perpendicular x, y, z axes, indicating whether the incident light vector is away from, at right angles to, or towards that axis. With a naive encoding, this would require the cell to be able to register $3 \times \log_2(3) = 4.755$ bits of information.[8]

这 4.755 比特的“意义”是什么？显然，从某种意义上说，它意味着光源位于某个确定的大致方向。这种方向规定随后就能引导细胞运动。从信息论角度看，要使传感器具有意义，环境状态（光照方向）与趋光细胞的内部状态之间就必须存在互信息。[9]因此，这里的意义就是互信息。

What is the ‘meaning’ of these 4.755 bits? Obviously, in one sense, it means that the light source is in a specified general direction. This specification can then steer the cell’s motion. In terms of information theory, for the sensors to have meaning, there must be mutual information[9] between the states of the environment (directions of illumination) and internal states of the phototactic cell. The meaning then is the mutual information.

本节考察了信息论的一个在哲学上令人震惊的推论：信息是客观的、物理的东西，完全独立于人类意识。无需强调，这具有深刻的唯物主义意蕴。互信息的形式定义，为文献[10]所捍卫的知识表征模型提供了科学基础。

In this section, we have been examining one philosophically startling implication of information theory: that information is something objective and physical, something quite independent of human consciousness. It need hardly be emphasized that this had profound materialist implications. The formal definition of mutual information provides the scientific foundation for the representational model of knowledge defended in.[10]

随机性

Randomness

本节将考察香农所发表成果的第二个令人惊讶的推论。

In this section, we look at a second surprising implication of what Shannon published.

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香农表明，信息的公式在函数形式上与熵的公式完全相同。事实上，此后熵与信息这两个术语常被认为可以互换。

Shannon showed that the formula for information is functionally identical to that of entropy. Indeed, the terms entropy and information have since then often been considered interchangeable.

但把信息等同于熵，似乎与人们通常把熵理解为无序或随机性的看法相矛盾。自然过程产生随机结果。风化岩面看起来粗糙而杂乱。经人手加工、刻上铭文的石头则立即可与自然物区分开来，恰恰因为它总体上相对平滑而有序。因此，记录人类信息似乎伴随着有序程度的增加[11]和随机性的减少。

But this equation of information with entropy appears to contradict popular understanding of entropy as disorder or randomness. Natural processes produce random results. A weathered rock face has a rough disordered appearance. A rock worked by human hand to bear an inscription is immediately distinguishable from something natural precisely because it is, in the main, relatively smooth and orderly. It would thus seem that recording human information goes hand in hand with an increase in order[11] and a reduction in randomness.

我们如何把这一点与香农将信息等同于熵、从而等同于无序的观点协调起来？

How are we to reconcile this with Shannon’s identification of information with entropy, and thus with disorder?

当我们忽略传统人类记录的冗余性时，就会出现这一悖论。碑刻石匠希望文字清晰可读、醒目突出。为此，他们做了两件事：

This paradox arises when we neglect the redundancy of traditional human records. Monumental masons wanted their text to be clearly readable to stand out. To ensure this, they did two things:

1. 他们造出平整表面。这些表面不包含人类信息，却具有低熵和高度有序性。

1. They created flat surfaces. These contain no anthropic information but are low entropy and highly orderly.

2. 随后，他们刻上字母，字母较大，刻痕较深，以抵御风化。这些字母确实包含信息，但表示信息的方式具有很高的冗余度。

2. They then incised letters, which were large and relatively deeply cut so that they would withstand weathering. These do contain information, but represent it very redundantly.

先创造低熵基底，再以冗余方式在其上记录信息：从印刷到微芯片制造，许多生产过程[12]都具有这种特征。

This process of first creating a low-entropy substrate on which information is then redundantly recorded is characteristic of lots of production processes[12] from printing to micro-chip manufacture.

前面已经看到，抛硬币的信息可以用文本冗余地记录，也可以用二进制数字高效地记录。文本即便存在错误，仍容易阅读。HEADS 一词发生一个比特错误，可能变成譬如 HECDS，但在上下文中依然可以理解。相反，在用来编码正反面的纯二进制流中，一个比特错误会不被察觉。

We saw earlier that information about coin tosses could be recorded redundantly as text or efficiently as binary digits. The text is easily readable in the presence of errors. A one-bit error in the word HEADS could transform it to, for example, HECDS, which, in context, would still be understandable. A one-bit error in a pure binary stream encoding heads and tails, on the other hand, would pass unnoticed.

实用而稳健的信息存储方式利用冗余和有序性，因而具有低熵。由此可见，在 0 和 1 出现频率相等的完全随机比特流中，能够编码最多的信息。这一点对图灵和香农研究的密码学具有实际意义。破译密码依赖于利用明文中的非随机性，这种非随机性以某种方式泄露为密文中的非随机性。与此相关，文件压缩技术（如.zip 文件）会产生近似随机的比特串。

Practical and resilient ways of storing information use redundancy, order and as such are low entropy. A consequence of this is that the greatest possible information is encoded in a completely random bit stream in which 0 and 1 occur with equal frequency. This fact was of practical relevance in the cryptography on which Turing and Shannon had worked. Breaking codes was reliant on exploiting non-randomness in the clear-text, which somehow leaked through to non-randomness in the cipher text. Related to this, file compression techniques,

like .zip files, produce bit strings which approximate to randomness.

但这引出一个新问题：我们所说的随机性是什么意思？

But this raises a new question. What do we mean by randomness?

幸运眷顾勇敢者（Audaces fortuna iuvat）

Audaces fortuna iuvat

“Alea jacta est”（骰子已经掷出），凯撒如是说；骰子一旦掷出，他便只能等待，看它如何落下。结果不可预知：凯旋、胜利，或是遇刺。

Alea jacta est said Caesar; the die once thrown, he must wait and see how it lands. The result was unknowable: triumphs, victories or assassination.

Caesar forte regebatur。凯撒受命运支配。那么，命运与规则有何不同？

Caesar forte regebatur. He was ruled by fortune. But how then does fortune differ from rules?

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我们怎样通过观察，辨别某件事只是随机发生，还是遵循某种对我们而言可能并不明显的模式？

How can we, by observation, tell if something is just random or is following some pattern, which pattern may not be obvious to us?

以骰子为喻。给定一系列掷骰结果，我们如何判断骰子是公平的，还是存在某种可以据以稳妥下注的偏向？

Take the metaphor of a die. How can we tell, given a sequence of die casts, if the die is fair or if there is a bias on which we can soundly bet?

假定有一枚标着 0 至 9 的十面骰子，它给出如下序列：

Suppose we have a ten-sided die labelled 0.9, and this gives us the following sequence:

4, 1, 5, 9, 2, 6, 5, 3, 5, 8, 9, 7, 9, 3, 2, 3, 8, 4, 6, 2, 6, 4, 3, 3, 8, 3, 2, 7, 9, 5, 0, 2, 8, 8, 4, 1, 9, 7, 1, 6, 9, 3, 9, 9, 3, 7, 5, 1, 0, 5, 8, 2, 0, 9, 7, 4, 9, 4, 4, 5, 9, 2,

4, 1, 5, 9, 2, 6, 5, 3, 5, 8, 9, 7, 9, 3, 2, 3, 8, 4, 6, 2, 6, 4, 3, 3, 8, 3, 2, 7, 9, 5, 0, 2, 8, 8, 4, 1, 9, 7, 1, 6, 9, 3, 9, 9, 3, 7, 5, 1, 0, 5, 8, 2, 0, 9, 7, 4, 9, 4, 4, 5, 9, 2,

我们如何知道它是不是随机的？

How do we know if this was random?

我们可以查看每个数字出现多少次。

Well, we can look at how often each digit occurs.

数字	出现次数
0	3
1	4
2	7
3	8
4	7
5	7
6	4
7	5
8	6
9	11

Digit	Occurences
0	3
1	4
2	7
3	8
4	7
5	7
6	4
7	5
8	6
9	11

到这里，我们的骰子看起来似乎不容易掷出 0，而容易掷出 9。似乎只要押注不会出现 0，长期来看就能赢。让我们试着生成更多数字，看这一点是否成立。

At this point, it looks as if our die is biased against zeros and towards nines. It would seem that were we to bet against zeros, we would, over the long run, win. Let’s try generating more digits and see if that holds.

图 11.1 表明，随着产生的数字越来越多，数列中 0 所占的百分比趋近 10%，这正是我们对无偏骰子的预期。

如果要更严格地检验随机性，我们可以考察成对的数字。也许 0 的出现频率是 10%，但它仍可能在 6 之后比在 7 之后更容易出现。如果在足够大的样本中，我们发现每个数字的出现都不受前一个数字影响，就可以进一步考察三个数字、四个数字组成的组，以此类推。

随着所考察的数字组长度增加，样本所需的数字数量会呈指数增长。如果需要 44,000 个数字才能使 0 所占的比例收敛到 10%，那么可以预期，需要考察超过 400,000 个数字，这些 0 之前的数字为 6 的比例才会收敛到 10%。对于三个数字的组合，就需要检查 400 万个数字，以此类推。

我们试图探测的模式越长，证明随机性就越困难。可能存在的模式越微妙，就越不可能通过穷尽搜索将其找出。

Figure 11.1 shows that as we get more and more digits, the percentage of zeros in the stream approaches 10 per cent, which is what we expect from an unbiased die.

If we wanted a more stringent test on randomness, we could look at pairs of digits. Perhaps zeros occur 10 per cent of the time, but they are still slightly more likely to occur after six than after a seven. If we find that, over a suitably large sample, the occurrence of each digit is unaffected by the preceding digit, then we could look at groups of three digits, groups of four digits, etc.

As we increase the length of groups we are examining, then the number of digits required in our sample will grow exponentially. If it took 44,000 digits for the share of zeros to converge to 10 per cent, then we could expect to look at over 400,000 digits before the percentage of sixes preceding these zeros would converge to 10%. For triples, we would need to check four million digits etc.

Proving randomness becomes increasingly difficult the longer are the patterns we are trying to detect. The subtler the pattern that might exist, the more impossible it is to find by exhaustive searching.

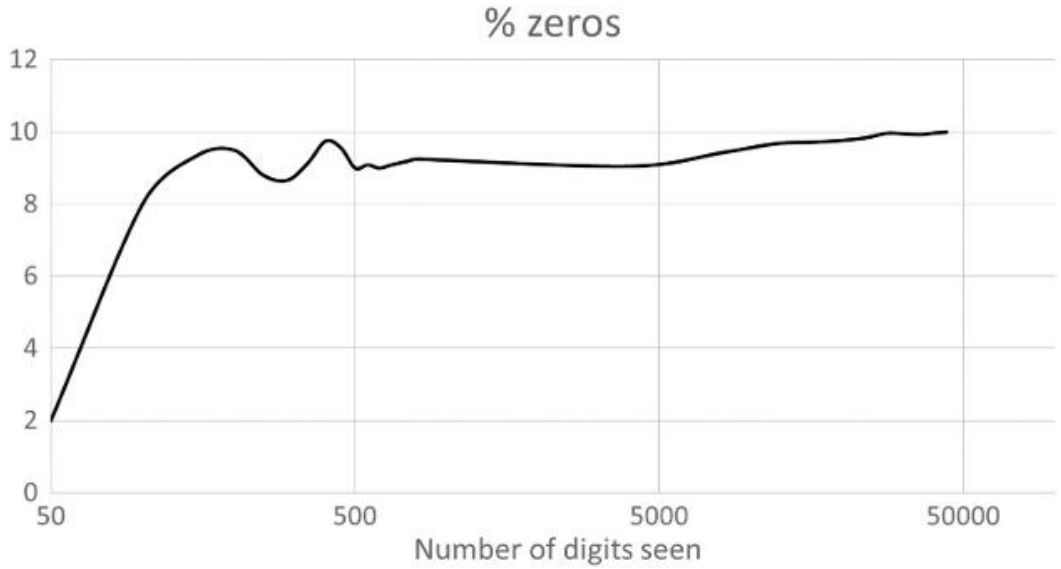


图 11.1 随着产生的数字越来越多，最初 0 出现得偏少的现象逐渐消失。作者绘图。

Figure 11.1 The initial bias against zero vanishes as we generate more and more digits. Figure by authors.
% zeros：0 所占的百分比；Number of digits seen：已观察的数字数量。

因此，对先前问题的回答是否定的。一般而言，我们无法通过观察证明一个长数字序列，或一长串对自然的观测结果，是随机的。

显然，迄今使用的数字序列实际上并不是通过投掷十面骰子产生的。掷 44,000 次骰子并不现实。

我们本可以使用某种电子硬件产生真正的随机数。[13]实际上，我们确实使用了计算机生成的序列，但它并非真正随机。我们采用了 π 的连续数字作为

So the answer to our earlier question is no. We cannot in general prove by observation that a long sequence of numbers, or a long set of observations of nature, are random.

Obviously the digit sequence we have been using up until now was not actually generated by throwing a ten-sided die. Throwing it 44,000 times would have been impractical.

We could have used some electronic hardware to generate true random numbers.[13] In fact, we did use a computer-generated sequence, but it was not a truly random one. We used successive digits of π as the source.[14]

来源。[14]然而，π 的连续数字似乎完美地模仿了随机性。

Yet the successive digits of π seem to perfectly mimic randomness.

这便是现实的一个深刻悖论：circulus regitur forte——圆受偶然支配。圆是自然秩序最典型的范例，却由看似随机的东西支配。如今，规则与偶然的区别又在哪里？

Here is a deep paradox of reality. *circulus regitur forte* The circle, the most paradigmatic of nature’s orders, is ruled by what appears to be randomness. Where now lies the difference between rule and fortune?

难道我们在作出重大决定以前，也应该请教占卜者、阅读《易经》吗？

Should we too consult the augurs and read the book of changes before our great decisions?

我们的科学定律大厦难道只是掩盖混沌的幻象？元数学家蔡廷[15]给出了一个答案：他基于计算机程序提出随机性的新定义。按照他的定义，随机数字序列就是不能由比它更短的计算机程序输出的序列。

Is our edifice of scientific laws a mere illusion hiding chaos? An answer to this was provided by the metamathematician Chaitin,[15] who gave a new definition of randomness based on computer programmes. According to him, a random sequence of digits is one which cannot be printed out by a shorter computer programme.

按照这一定义，π 不是随机的。下面是 Dik T. Winter 编写的一段占 160 个字符的 C 程序，它能输出 π 的 800 个字符：

By this definition, π is not random. Here is a C programme by Dik T. Winter occupying 160 characters that will print out 800 characters of it:

```
int a=10000,b,c=2800,d,e,f[2801],g;main(){for(;b-c;)
f[b++]=a/5;
for(;d=0,g=c*2;c-=14,printf("%.4d",e+d/a),e=d%a)fo
r(b=c;d+=f[b]*a,
f[b]=d%--g,d/=g--,--b;d*=b);}
```

还有其他短程序，能够输出 π 的数百万位数字。

Other short programmes exist that will print π to millions of digits.

苏联辩证唯物主义

Soviet diamat

辩证唯物主义由狄慈根创立，他在思想上影响了德国社会民主党；这一哲学后来成为共产主义的官方哲学，由德国社会民主党传承而来。然而，这一传统与物质世界的随机性基础是否相容，是值得怀疑的。19 世纪的唯物主义者，如玻尔兹曼，曾经苦苦思索：在由偶然性和热力学定律支配的世界中，如何理解有方向性的变化？我们又该如何在历史语境中理解同一个悖论？

Dialectical Materialism was invented by Dietzgen, an intellectual influence on German Social Democracy, and became the official communist philosophy inherited from the German Social Democrats. However, the compatibility of this tradition with the stochastic basis of the material world is questionable. Nineteenth-century materialists like Boltzmann had previously struggled with the problem of directional change in a world governed by chance and the laws of thermodynamics. How should we understand the same paradox in the context of history?

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苏联的“辩证唯物主义”（Diamat）学说，在多大程度上与本书迄今所阐述的唯物主义相容？

To what extent is the Soviet doctrine of ‘Diamat’ compatible with the account of materialism we have been giving in this book?

长期被奉为正统权威的斯大林提出了[16]唯物辩证法的以下四项原则：

Stalin, for long the orthodox authority, gave[16] the following four principles of materialist dialectics:

1. 自然界是相互联系和相互决定的

1. Nature is connected and determined

2. 自然界处于持续运动和变化的状态

2. Nature is a state of continuous motion and change

3. 自然界的量变导致质变

3. Natural quantitative change leads to qualitative change

4. 矛盾是自然界所固有的

4. Contradictions are inherent in nature

我们将依次考察这些原则。

We will consider these in turn.

自然界是相互联系和相互决定的

Nature is connected and determined

与形而上学相反，辩证法并不把自然界看作事物和现象的偶然聚集，仿佛它们彼此毫无联系、相互孤立、相互独立；而把自然界看作一个相互联系的完整整体，其中事物和现象彼此有机联系、相互依存、相互决定。[17]

Contrary to metaphysics, dialectics does not regard nature as an accidental agglomeration of things, of phenomena, unconnected with, isolated from, and independent of, each other, but as a connected and integral whole, in which things, phenomena ,are organically connected with, dependent on, and determined by, each other.[17]

这一论述立即引人注意的一点，是它完全属于正统力学。在两次世界大战之间，苏联的“机械论者”与德博林派曾展开激烈争论。[18]但斯大林在 20 世纪 30 年代对辩证唯物主义的阐述所说的、自然界通过超距作用力体现的决定关系和联系，正是经典力学和量子力学的共同基础。在量子力学中，情况也相同，正如我们在 [sec:qcrypt] 节对量子密码学的论述所示。物理学家玻姆[19]将其称为实在的“隐含序”。因此，断言自然界具有决定性并相互联系，并非辩证法所特有的东西。辩证唯物主义若持这一观点，那是因为它忠实地追随了力学。

〔校注：原书此处的交叉引用未解析，印作“Section [sec:qcrypt]”；本稿保留该标记。〕

What strikes one immediately about this is that it is absolutely orthodox mechanics. Within the USSR in the interwar period, there had been a vigourous debate between the ‘mechanists’ and the Deborinists.[18] But Stalin’s 1930s exposition on dialectical materialism states that nature’s determination and connection via action at a distance forces are fundamental to both classical and quantum mechanics. In quantum mechanics, as our account of quantum cryptography in Section [sec:qcrypt] shows, the same applies. The physicist Bohm[19] called this the ‘implicate order’ of reality. So an assertion that nature is deterministic and connected is not something specifically dialectical. If dialectical materialism held this, it was because it was a faithful follower of mechanics.

自然界处于持续运动和变化的状态

Nature is a state of continuous motion and change

与形而上学相反，辩证法认为，自然界并非处于静止不动、停滞不变的状态，而是处于持续运动和变化、持续更新和发展的状态；其中总有某些事物在产生和发展，也总有某些事物在瓦解和消亡。[20]

Contrary to metaphysics, dialectics holds that nature is not a state of rest and immobility, stagnation and immutability, but a state of continuous movement and change, of continuous renewal and development, where something is always arising and developing, and something always disintegrating and dying away.[20]

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运动是任何力学动力学理论的基础。一切统计力学也建立在这样一幅图景之上。通过斯密、恩格斯和达尔文，这幅图景被扩展到社会和生物学领域。确实，自古代原子论者以来，所有唯物主义者都主张自然界中存在持续的运动和变化。

Motion underlies any mechanical theory of dynamics. All statistical mechanics, too, are based on such a picture. With Smith, Engels and Darwin it gets extended to society and biology. Continuous motion and change in nature have indeed been asserted by all materialists since the ancient atomists.

太阳运行许多年之后，手指上的戒指因佩戴而在内侧变薄，檐下滴水在石头上凿出凹穴，弯曲的铁犁铧在田间不知不觉地磨损；我们看见石铺街道被众人的脚步磨平，城门旁的青铜雕像也显出右手的消瘦，那是无数过路人向它致意时触摸的结果。

after the revolution of many of the sun’s years a ring on the finger is thinned on the under side by wearing, the dripping from the eaves hollows a stone, the bent plowshare of iron imperceptibly decreases in the fields, and we behold the stone-paved streets worn down by the feet of the multitude; the brass statues too at the gates show their right hands to be wasted by the touch of the numerous passers by who greet them.

因此，我们看到这些东西因这样磨损而变小；但在任何给定时刻究竟有哪些物体离去，视觉的本性却严严实实地阻止我们看见。

These things then we see are lessened, since they have been thus worn down; but what bodies depart at any given time the nature of vision has jealously shut out our seeing.

最后，时间和自然一点一点地给事物添加物体，迫使它们按适当尺度生长，无论怎样用力观看，眼睛也看不见这些物体；同样，无论事物因年岁而衰老腐败，还是悬于海上的岩石被啃啮般的咸水浪沫侵蚀，你都看不见它们在任何给定时刻失去了什么。因此，自然是通过看不见的物体来运作的。[21]

Lastly the bodies which time and nature add to things by little and little, constraining them to grow in due measure, no exertion of the eyesight can behold; and so too wherever things grow old by age and decay, and when rocks hanging over the sea are eaten away by the gnawing salt spray, you cannot see

what they lose at any given moment Nature therefore works by unseen bodies.[21]

自然界的量变导致质变

Natural quantitative change leads to qualitative change

量变导致质变这一观念，似乎因恩格斯的《自然辩证法》而广为流传。该书认为，在物理学中，我们研究物体的分子状态以及使分子活动起来的运动形式。每一种变化都是量向质的转化，[22]源自物体所固有的或传递给物体的运动的量变。物理常数标示着节点，在这些节点上，运动的增加或减少改变物体的状态，从而把量转化为质。

The idea that quantitative change leads to qualitative change seems to have gained popularity from Engels' Dialectics of Nature, which argued that in physics, we work with bodies' molecular states and motion forms, which bring molecules into play. Every change is a transformation of quantity to quality,[22] resulting from a quantitative change of motion inherent in the body or communicated to it. Physical constants are designations of nodal points where addition/subtraction of motion alters the state of the body, transforming quantity into quality.

恩格斯在这里描述的，基本上就是我们现在所说的相变现象。他所举的液态与气态之间的变化，确实是相变的例子。不过，与物理学的术语相比，恩格斯谈论量与质时使用的术语含义模糊。物理学家会把相变定义为熵随温度的变化变得不连续的情形：在达到 100 摄氏度以前，每升高 1 摄氏度需要 1,000 卡热量；而从 99 摄氏度的水变成 100 摄氏度的蒸汽，则需要 540,000 卡，增幅巨大。水从表面蒸发时，每克消耗 540 卡热量。在没有外界热量输入时，水会冷却 0.5°C。这只有在密闭容器中才是现实的，而沸点以下的蒸发速率与开尔文温度成正比。对于炉灶上的敞口锅，把 1 升水升高 1 摄氏度所需的热量输入，在达到 100°C 之前是非线性的；到了 100°C，仍会发生阶跃式跃变。

What Engels was describing here is basically what we now call phase-change phenomena. His examples of changes between liquid and gaseous states are valid phase changes. The terminology Engels used though, writing of quantity and quality is ambiguous compared to that used in physics. A physicist would define a phase change as occurring when the change in entropy becomes discontinuous for temperature, you require 1,000 calories per degree Celsius of heating until you reach 100 Celsius, then to go between water at 99 Celsius and steam at 100 Celsius you need 540,000 calories, a huge increase. Water evaporates from the surface using 540 calories of heat per gram. In the absence of external heat, water cools down by 0.5°C. This is realistic only for a closed container, and the evaporation rate below the boiling point is proportional to the temperature in Kelvin. For an open pan on the stove, the heat input required to raise a litre by 1 Celsius will be non-linear up to 100°C, at which point you still get a step jump.

辩证唯物主义话语的问题在于，“量”和“质”这两个术语含糊不清，定义不明确。在熔化和沸腾的例子中，凭常识我们知道固、液、气三种相态之间存在质的区别。但若采用超临界锅炉，即在高于 3,200 psi 的压力下运行的锅炉，那么液态水与蒸汽就成为无法区分的相。人们可以用质的话语来包装这一现象，说在高于 3,200 psi 时，水与蒸汽之间没有质的区别；也可以更精确地说，在高于 3,200 psi 时，加热水不会造成熵的阶跃式跃变。

The problem with the diamat language is that the terms quantity and quality are vague and ill-defined. In the melting/boiling example, we have a commonsense understanding of there being a qualitative difference between solid, liquid and gas phases. But if we have a super-critical boiler, one that operates above 3,200 psi, then liquid water and steam become indistinguishable as phases. One may dress this up in the language of quality by saying that above 3,200 psi, there is no qualitative difference between water and steam. Or one can be more precise and say that above 3,200 psi, there is no step jump in entropy as we heat the water up.

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恩格斯语言的含糊，使他的论述出现了明显的错误。白炽灯这个例子完全错了。

The vagueness of Engels' s language leads him to make what are clear errors in his account. The example of the incandescent lamp is completely wrong.

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改变灯的供电电压，灯丝温度便逐渐升高。随着温度上升，不同波长的发光强度逐渐变化，光色从红、橙、黄趋向近乎白色。凡是使用过白炽灯调光器的人，都见过这种现象。这里没有明确的相变，也没有鲜明的阈值电压。发光二极管则有明确的阈值电压，低于它时不发光；超过该阈值之后，发出的光色也不改变。

If you alter the supply voltage to a lamp you get a gradual rise in the temperature of the filament. As the temperature rises, the intensity of light given off at different wavelengths gradually changes from red, orange and yellow to almost white. Anyone who has used a dimmer control for an incandescent bulb has seen this. There is no distinct phase change and no sharp threshold voltage.

With a Light Emitting Diode, there is a distinct threshold voltage below which no light is admitted, and the colour of light emitted does not change above that threshold.

只有某些系统会发生相变，其他系统则连续变化。这个混乱的灯泡例子，似乎正是恩格斯术语含糊造成的。

Only some systems undergo phase changes, others change continuously. This confused light bulb example seems to be a consequence of Engels's ambiguous terminology.

什么算作质变？什么算作量变？所谓量，他指的是整数，还是连续统？在他关于沸腾相关温度变化的正确例子中，量变沿一个连续尺度发生。他关于白炽灯泡的错误例子也一样。[23]电压是一种模拟变量，也就是连续变量。

What counts as a qualitative change? What counts as a quantitative change? For quantity, is he referring to whole numbers or to a continuum? In the case of his correct example of temperature change associated with boiling, then the quantitative change is along a continuous scale. The same applies to his faulty example of the incandescent bulb.[23] Voltage is an analogue or continuous variable.

但在其他地方，恩格斯又希望把量理解为整数。

But in other places Engels wants quantity to be interpreted in terms of integer numbers.

然而，黑格尔所发现的自然规律取得最重要胜利的领域，是化学。化学可以称为研究物体因数量组成改变而发生质变的科学。黑格尔本人已经知道这一点。以氧为例：如果三个原子而非通常的两个原子结合成一个分子，就得到臭氧，一种在气味和反应上都与普通氧大不相同的物体。又如，氧可以按不同的比例与氮或硫结合，每一种比例都产生在质上不同于其他任何一种的物质！笑气（一氧化氮，N_2O）与硝酸酐（五氧化二氮，N_2O_5）多么不同！前者是气体，后者在常温下是固态结晶物质。然而，二者在组成上的全部区别，仅在于后者含氧量是前者的五倍；二者之间还有另外三种氮的氧化物（N0、N_2O_3、NO_2），每一种在质上都与前两种及彼此不同。[24]

〔校注：原书将笑气 N_2O 称作“nitrogen monoxide”（一氧化氮）；N_2O 的通常中文名称为一氧化二氮。后列第一式原书印作“N0”（末位为数字零），依上下文应为 NO。正文保留原书写法。〕

The sphere, however, in which the law of nature discovered by Hegel celebrates its most important triumphs is that of chemistry. Chemistry can be termed the science of the qualitative changes of bodies as a result of changed quantitative composition. That was already known to Hegel himself. As in the case of oxygen: if three atoms unite into a molecule, instead of the usual two, we get ozone, a body which is very considerably different from ordinary oxygen in its odour and reactions. Again, one can take the various proportions in which oxygen combines with nitrogen or sulphur, each of which produces a substance qualitatively different from any of the others! How different laughing gas (nitrogen monoxide N_2O) is from nitric anhydride (nitrogen pentoxide, N_2O_5) ! The first is a gas, the second at ordinary temperatures a solid crystalline substance. And yet the whole difference in composition is that the second contains five times as much oxygen as the first, and between the two of them are three more oxides of nitrogen (N0, N_2O_3, NO_2), each of which is qualitatively different from the first two and from each other.[24]

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于是，我们竟有了一条由黑格尔发现的“自然规律”！

So, we have a ‘law of nature’ discovered by Hegel!

得了吧，弗雷德。你要找的名字是道尔顿，约翰·道尔顿，而不是黑格尔。被引为这条“规律”出处的黑格尔《逻辑学》出版于 1812 年；道尔顿的《化学哲学新体系》早在四年前，即 1808 年，就已问世。这充其量是黑格尔事后把他人的发现说成自己的发现，就像 1801 年他声称自己对牛顿万有引力定律有更高明的证明一样。我们在第 3.4 节已经指出，黑格尔这些关于独创性的主张是不连贯的胡言乱语。这里也是如此。化合物中各组成元素的原子必须以自然数之比存在，这是道尔顿原子理论不可分割的一部分。恩格斯援引黑格尔，在水沸腾时的相变与这些数值比例之间作类比，其不连贯程度与黑格尔关于引力的论证如出一辙。

Give us a break Fred. The name you are looking for is Dalton, John Dalton, not Hegel. Hegel's *Science of Logic*, cited as the source for this ‘law’ is dated 1812. Dalton's *A New System of Chemical Philosophy* came out four years earlier in 1808. At best we have a retrospective claim by Hegel to other's discoveries, as was the case when in 1801 Hegel claimed to have a superior proof of Newton's law of gravitation. We showed in Section 3.4 that those claims to originality by Hegel were incoherent nonsense. The same applies here. That chemical compounds must contain natural number ratios of atoms of their constituent elements was an integral part of Dalton's atomic theory. The analogy that Engels, with reference to Hegel, makes between phase change in boiling water and these numeric ratios is as incoherent as Hegel's arguments about gravity.

水沸腾的情形中，有一个连续的模拟尺度，即温度；在这一尺度的某些点上发生相变，其标志是另一个模拟尺度，即熵，出现不连续。在化学的情形

In the case of water boiling, you have one continuous analogue scale: temperature at certain points on which you get a phase change, marked by a discontinuity in

中，并不存在连续的模拟尺度。逐渐增加氮，并不会使化合物突然改变。在 NO 与 N_2O 之间，不可能有 $N_{1.275}O$ 这样的化合物。每种原子的数量都必须是整数。这是原子在化学上不可分的结果。因此，这里并非渐进的量变导致突发的质变，而是存在一组离散的化合物，每种都含有离散数量的原子。

another analogue scale: entropy. In the chemical case, there is no continuous analogue scale. In the chemical case, you do not get a sudden change in compound as you gradually add more Nitrogen. You cannot have a compound $N_{1.275}O$ between NO and N_2O. The number of each type of atom has to be an integer. This is a consequence of the (chemical) indivisibility of the atom. So instead of a gradual quantitative change leading to a sudden qualitative change, you have a discrete set of compounds, each containing discrete numbers of atoms.

“量转化为质”这条“规律”具有强烈的政治吸引力。关于量变导致质变的说法，可以被理解为一种隐喻：对工人阶级的剥削在量上加重，最终导致突然爆发的暴力革命。把革命比作紧张局势沸腾后溢出的结果，早在 19 世纪就已有这种用法。[25]

The political appeal of the quantity into quality ‘law’ was strong. Statements about quantitative change leading to qualitative change could be taken as a metaphor for the increased quantitative exploitation of the working class leading to a sudden violent revolution. The metaphor of revolutions being the result of tensions boiling over was already in use by the nineteenth century.[25]

这条所谓量转化为质的普遍规律，与许多唯心主义哲学一样，建立在肤浅的类比和双关之上。作为普遍规律，它对预测毫无用处。事实上，即使对于革命，这种类比是否说得通也值得怀疑。在苏联的革命图像中，群众团结为有组织的队列，向红色未来行进，或聚集在正在演讲的列宁面前。革命发生于熵减之时，也就是群众获得秩序和方向、不再漫无目的地运动之时。

This alleged general law of quantity into quality is based, like much idealist philosophy, on superficial analogy and *double entendre*. As a general law, it is useless for making predictions. Indeed, even in the case of revolutions it is questionable whether the analogy makes sense. In Soviet revolutionary imagery, the masses unite into organized columns, marching towards the red future or gather in front of Lenin as he speaks. Revolution happens when entropy decreases, that is, when the masses gain order and direction instead of moving aimlessly.

有人可能反驳，相变这个概念有时也颇具隐喻性。如果我们嘲笑黑格尔式的量变质变观念，又为什么允许科学家谈论其他相变？

It might be objected that the notion of phase change is also sometimes used in a rather metaphorical way. If we ridicule the Hegelian quantity in quality notion, why should we allow scientists to speak of other phase transitions?

这种反驳不能成立，因为在科学文献中使用相变一词，就应当证明：当一个变量连续变化时，另一个类似于熵的变量发生了不连续变化。

No, because, in the scientific literature, when one uses the term phase change, it is expected that one will demonstrate that there is a discontinuity in a variable analogous to entropy while another variable undergoes continuous change.

让我们举一个使用相变这一术语的非物理学例子，说明这一术语的应用必须满足严格的理论约束。

Let’s give a non-physics example where the term *phase transition* is used and show how the application of the term has to meet tight theoretical constraints.

在符号逻辑中，有一个所谓的可满足性问题（SAT）。给定一个定义在 n 个布尔变量 $v_1, \ldots, v_n$ 上的布尔代数公式 F，问题要求找出一种变量真值赋值，使 F 为真。

In symbolic logic, there is something called the satisfaction problem (SAT). The problem takes some formula F in Boolean algebra defined over n Boolean variables $v_1, \ldots, v_n$ and seeks to find an assignment of truth values to the variables such that F is true.

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F 通常以合取范式给出，例如：

F is typically given in conjunctive normal form for example:

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$$F(v_1, v_2, v_3, v_4) = (v_1 + v_2 + \neg v_3) \cdot (v_3 + \neg v_1 + v_4)$$

人们观察到，找到这样的变量赋值往往很容易，但有时又十分困难；在极端情况下，需要尝试给 n 个变量赋予真或假的全部 2^n 种不同方案。

It has been observed that it is often very easy to find such an assignment of variables, but at other times it becomes very hard, requiring in the limiting case that one tries all 2^n possible distinct assignments of true or false to the n variables.

对于另一些 F，却很容易证明不存在任何能使 F 为真的变量赋值。

For other examples of *F*, it turns out to be very easy to prove that there is no possible assignment of variables that will make *F* true.

进一步研究表明，从容易满足到难以满足、再到容易证明不可满足的变化，发生在一个被称为相变的临界点。[26]这一领域的实验工作为相变这一用语提供了依据：它表明，F 存在一个约束程度的度量[27]m，随着 m 增大，问题难度先急剧上升，随后下降。为什么这类似于相变？

Further investigation has shown that the change from easy to satisfy, to hard to satisfy, to easy to prove unsatisfiable occurs at a critical point, which has been termed a phase transition.[26] The experimental work in this field justifies the term phase transition by showing that there is a constrainedness measure[27] *m* on *F* such that as *m* increases the hardness of the problem rises sharply and then falls. Why is it like a phase transition?

它与水沸腾有什么共同之处？二者相似，是因为约束水平 m 相当于温度 T，而难度（用计算运行时间衡量）相当于 $\frac{dH}{dT}$，即熵对温度的导数。显然，在水转化为蒸汽的相变中，$\frac{dH}{dT}$ 出现尖峰，类似于 F 的难度随着 m 增大而出现尖峰。

What does it have in common with water boiling? They are similar because the constraint level m acts as an analogue for temperature T, and the hardness (measured as computational run time) is analogous to $\frac{dH}{dT}$ the derivative of entropy with respect to temperature. Obviously, in the water-to-steam phase, transition $\frac{dH}{dT}$ spikes analogous to the way that the hardness of F spikes as m increases.

矛盾是自然界所固有的

Contradictions are inherent in nature

辩证唯物主义的支持者说，他们所说的“矛盾”存在于现实世界中，而不只存在于逻辑陈述中。显然，他们使用这个词的含义，与它通常的含义有所不同。形式逻辑的矛盾——非（非 x 且 x）；x 或非 x——取决于真值的性质。但斯大林或毛泽东所说的矛盾并非这个意思。斯大林这样写道：

Supporters of dialectical materialism say that what they call ‘contradictions’ exist in the real world, not just in logical statements. It is clear that they are using the word in a somewhat different sense to what it usually means. Formal logical contradictions –not (not x and x); x or not x –depend on properties of truth values. But this is not what Stalin or Mao meant by contradictions. So we have Stalin writing:

〔校注：原书把所列两式称作“contradictions”。按通常的经典命题逻辑，两式分别表达非矛盾律与排中律，均非矛盾式；此处保留作者表述。〕

> 与形而上学相反，辩证法认为，内在矛盾是自然界一切事物和现象所固有的，因为它们都有否定和肯定的方面，有过去和未来，有正在消亡的东西和正在发展的东西；而这些对立面之间的斗争，即新与旧、正在消亡的与正在诞生的、正在消失的与正在发展的东西之间的斗争，构成了发展过程的内部内容，构成了量变转化为质变的内部内容。[28]

> Contrary to metaphysics, dialectics holds that internal contradictions are inherent in all things and phenomena of nature, for they all have their negative and positive sides, a past and a future, something dying away and something developing; and that the struggle between these opposites, the struggle between the old and the new, between that which is dying away and that which is being born, between that which is disappearing and that which is developing, constitutes the internal content of the process of development, the internal content of the transformation of quantitative changes into qualitative changes.[28]

所以，斯大林说矛盾，是以隐喻的方式谈论世界中的变化过程。但为什么承认变化就被认为是辩证法所特有的？

So, by contradiction, Stalin is talking metaphorically about the process of change in the world. But why is the recognition of change supposed to be something specific to dialectics?

事物有过去和未来，事物发展又消亡，这是人人都承认的。用“矛盾”这个术语来指称这一点，究竟增加了什么特殊成分？

That things have a past and a future, that things develop and die, is recognized by everyone. What special ingredient is added by using the term ‘contradiction’ to refer to this?

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斯大林详细解释其含义时，诉诸的不是辩证法，而是马克思所谓的唯物主义历史理论。斯大林在这里称之为“研究社会现象的历史方法”。

When Stalin elaborates on what he means, he has recourse not to dialectics but to what Marx had called the materialist theory of history. Stalin refers to it here as ‘the historical approach to social phenomena’.

在现代条件下，奴隶制会是无意义、愚蠢而不自然的。但在原始公社制度解体的条件下，奴隶制却是完全可以理

The slave system would be senseless, stupid and unnatural under modern conditions. But under the conditions of a dis-

解而且自然的现象，因为它代表着相对于原始公社制度的进步。

……

一切都取决于条件、时间和地点。

显然，没有这种研究社会现象的历史方法，历史科学就不可能存在和发展；因为只有这种方法，才能使历史科学免于沦为偶然事件的杂乱集合和荒谬错误的堆积。[29]

正如我们在第 5.1 节所述，亚当 • 斯密已经清楚地认识到，社会形式、生产关系及其相应的政治上层建筑是相继更替的。斯密并不觉得需要借助矛盾或辩证法来解释这一点。马克思和恩格斯在首次阐述历史唯物主义的《德意志意识形态》中，虽然从未提到辩证法，却确实经常使用矛盾这个术语。以下只是《德意志意识形态》中的几个例子，马克思和恩格斯在其中以后来成为正统“辩证唯物主义”的方式使用矛盾一词：

分工意味着单个个人或单个家庭的利益与一切彼此交往的个人的共同利益之间的矛盾。

……

各个市民的生活条件，由于与现存关系以及由这些关系决定的劳动方式相矛盾，变成了他们所有人共有的、独立于每个个人的条件。市民在摆脱封建束缚的程度上创造了这些条件，同时又在受自身与既存封建制度的对立所决定的程度上，被这些条件所塑造。当各个城市开始联合时，这些共同条件便发展成阶级条件。相同的条件、相同的矛盾、相同的利益，必然在各地大体上产生相同的习俗。

……

生产力与交往形式之间的这种矛盾，正如我们所见，在过去的历史中曾多次发生，虽然并未危及其基础，却必然每次都以革命的形式爆发，同时呈现出各种附属形式，如全面的冲突、不同阶级之间的冲突、意识的矛盾、思想斗争等等，以及政治冲突等等。[30]

1932 年，《德意志意识形态》出版，引入了“矛盾”这一术语。斯大林在 1938 年撰写自己的文本时，想必已熟悉这个术语。“阶级矛盾”是较晚才进入正统马克思主义的说法，首次出现在布哈林 1925 年著作《历史唯物主义》的英文版中。在毛泽东的辩证法中，这种意义上的矛盾成为核心。

integrating primitive communal system, the slave system is a quite understandable and natural phenomenon, since it represents an advance on the primitive communal system

. . .

Everything depends on the conditions, time and place.

It is clear that without such a historical approach to social phenomena, the existence and development of the science of history is impossible; for only such an approach saves the science of history from becoming a jumble of accidents and an agglomeration of most absurd mistakes.[29]

That there are successive social forms, successive relations of production and associated political superstructures was already evident to Adam Smith, as we have recounted in Section 5.1. Smith felt no need to invoke contradiction or dialectics to explain this. In their own first account of historical materialism, *The German Ideology*, Marx and Engels did often use the term contradiction even though they never referred to dialectics. Here are just a few examples from *The German Ideology* where Marx and Engels use the term contradiction in what later became the orthodox ‘diamat’ style:

the division of labour implies the *contradiction* between the interest of the separate individual or the individual family and the communal interest of all individuals who have intercourse with one another.

. . .

The conditions of life of the individual burghers became, on account of their *contradiction* to the existing relationships and of the mode of labour determined by these, conditions which were common to them all and independent of each individual. The burghers had created the conditions insofar as they had torn themselves free from feudal ties, and were created by them insofar as they were determined by their antagonism to the feudal system which they found in existence. When the individual towns began to enter into associations, these common conditions developed into class conditions. The same conditions, the same *contradiction*, the same interests necessarily called forth on the whole similar customs everywhere.

. . .

This *contradiction* between the productive forces and the form of intercourse, which, as we saw, has occurred several times in past history, without, however, endangering the basis, necessarily on each occasion burst out in a revolution, taking on at the same time various subsidiary forms, such as all-embracing collisions, collisions of various classes, *contradiction* of consciousness, battle of ideas etc, political conflict etc.[30]

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In 1932, *The German Ideology* was published and introduced the term ‘contradiction’. Stalin would have been familiar with this term when he wrote his own text in 1938. The phrase ‘class contradiction’ is a relatively late addition to orthodox Marxism, first appearing in the English version of Bukharin’s 1925 book *Historical Materialism*. In Maoist dialectics, this sense of contradiction became central.

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毛泽东对辩证法的阐述

Mao’s account of dialectics

毛泽东的《矛盾论》[31]是迄今出版的最具影响力的辩证法著作。它的影响有两个方面。首先，中国的庞大规模使它得以大量印行，拥有数千万读者。但这之所以可能，正是因为以其作者为领导的中国共产党人，把以《矛盾论》为基础的政治战略付诸实践，取得胜利并建立了中华人民共和国。

Mao’s *On Contradiction*[31] was the most influential work of dialectics ever published. Its influence was twofold. Firstly, the great size of China means that it was printed in huge numbers and read by tens of millions. But that was only possible because the Chinese Communists, led by its author, put into practice a political strategy based on *On Contradiction* and emerged victorious to establish the People’s Republic of China.

这篇文章比斯大林的《论辩证唯物主义和历史唯物主义》早一年问世；阅读它，必须兼顾当时苏联正在进行的争论和中国革命战争的背景。

Coming out a year before Stalin’s *Dialectical and Historical Materialism*, it has to be read in the context of both the ongoing debates in the Soviet Union and the Chinese Revolutionary War.

前言将这篇文章定位为对受黑格尔影响的德博林学派的批判：

The preface positions the text as an attack on Hegel-influenced Deborin school:

近年来，苏联哲学界对德博林学派唯心主义的批判，引起了我们的极大兴趣。德博林的唯心主义在中国共产党内产生了很坏的影响，不能说我们党内的教条主义思想与该学派的研究方法毫无关系。因此，我们当前研究哲学，应当以根除教条主义思想为主要目标。

The criticism to which the idealism of the Deborin school has been subjected in Soviet philosophical circles in recent years has aroused great interest among us. Deborin’s idealism has exerted a very bad influence in the Chinese Communist Party, and it cannot be said that the dogmatist thinking in our Party is unrelated to the approach of that school. Our present study of philosophy should therefore have the eradication of dogmatist thinking as its main objective.

毛泽东与斯大林一样，认为辩证唯物主义世界观同对时间与变化的解释相联系。他把自己所称的形而上学或庸俗进化论的时间观，与马克思主义观点相对照。他说，庸俗进化论者承认变化，却只承认三种形式：增加、减少和循环重复。从上下文看，他所说的庸俗进化论显然不是达尔文理论，而是对社会演化的简单化观念：

Mao, like Stalin, sees the dialectical materialist world view as being tied to accounts of time and change. He contrasts what he calls metaphysical or vulgar evolutionist accounts of time with the view of Marxism. The vulgar evolutionists, he says, recognize change, but only in three forms: increase, decrease and cyclical repetition. From the context, it is clear that by vulgar evolutionism he does not mean the Darwinian theory, but simplified notions of social evolution:

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形而上学者认为，宇宙中各种事物及其一切特征，自产生之初就一直如此。此后的一切变化，都只是量的增加或减少。他们主张，一种事物只能作为同一种事物不断重复自身，不能变成别的事物。在他们看来，资本主义剥削、资本主义竞争、资本主义社会的个人主义意识形态等等，都可以在古代奴隶社会，甚至在原始社会中找到，而且会永远不变地存在下去。他们把社会发展的原因归于社会外部的因素，例如地理和气候。

Metaphysicians hold that all the different kinds of things in the universe and all their characteristics have been the same ever since they first came into being. All subsequent changes have simply been increases or decreases in quantity. They contend that a thing can only keep on repeating itself as the same kind of thing and cannot change into anything different. In their opinion, capitalist exploitation, capitalist competition, the individualist ideology of capitalist society, and so on, can all be found in ancient slave society, or even in primitive society, and will exist forever unchanged. They ascribe the causes of social development to factors external to society, such as geography and climate.

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当今这种庸俗进化论观点的例子，或许可以在图尔钦的著作中找到。[32]他的著作虽然包含丰富的历史内容，并承认阶级冲突，却确实用数千年来基本不变的动力机制来阐述问题。图尔钦理论的关键遗漏，是马克思所说的生产力的发展，以及这种发展如何制约社会的生产关系。

An example of such a vulgar evolutionist view today might be Turchin’s work,[32] which, despite its historical richness and recognition of class conflict, does formulate things in terms of what are basically unchanging dynamics over thousands of years. The key omission in the Turchin theory is the development of what Marx called the productive forces, and the way such developments condition social relations of production.

毛泽东认为，社会通过内部冲突发展，而非由外部原因推动。他认为，中国的发展将取决于国内的

Mao believed that societies develop through internal conflicts, not external causes. He argued that China’s

阶级斗争，即工人、农民和爱国资本家同地主、买办资本家之间的斗争。从印度和中国此后的经济与社会发展轨迹来看，这一观点被证明是正确的。

development would be determined by the domestic class struggle between workers, peasants and patriotic capitalists versus landlords and comprador capitalists. This perspective proved correct in light of the subsequent economic and social trajectories of India and China.

他讨论了矛盾的普遍性和特殊性。前者概括恩格斯与列宁的概念，后者则是这篇小册子的重点，分析中国革命中的具体矛盾。他批评教条主义，并突出列举了一些矛盾普遍性的例子；这些例子并不比恩格斯和斯大林的更有说服力。不过，他确实提出了一些值得更多关注的例子。他引用了列宁列出的一组例子，用来说明矛盾的普遍性。

He discusses the universality and particularity of contradiction. The former is a summary of concepts from Engels and Lenin, while the latter is the main focus of the pamphlet, analyzing specific contradictions in the Chinese Revolution. He criticizes dogmatism and highlights some examples of the universality of contradiction, which are no more convincing than those of Engels and Stalin. He did raise some examples that are worth greater attention. He quotes a list by Lenin of examples that are intended to illustrate the universal nature of contradictions.

辩证法内容的这一方面是否正确，必须由科学史来检验。辩证法的这一方面（例如在普列汉诺夫那里）通常没有得到足够注意：对立面的同一性被当作例子的总和［“例如，一粒种子”，“例如，原始共产主义”。恩格斯那里也是如此。但这是“为了普及……”］，而不是当作认识的规律（以及客观世界的规律）。

The correctness of this aspect of the content of dialectics must be tested by the history of science. This aspect of dialectics (eg in Plekhanov) usually receives inadequate attention: the identity of opposites is taken as the sum-total of examples [’for example, a seed,’ ‘for example, primitive communism.’ The same is true of Engels. But it is ‘in the interests of popularisation . . .’] and not as a law of cognition (and as a law of the objective world).

1. 在数学中：＋与－。微分与积分。

1. In mathematics: + and −. Differential and integral.

2. 在力学中：作用与反作用。

2. In mechanics: action and reaction.

3. 在物理学中：正电与负电。

3. In physics: positive and negative electricity.

4. 在化学中：原子的结合与解离。

4. In chemistry: the combination and dissociation of atoms.

5. 在社会科学中：阶级斗争。

5. In social science: the class struggle.

对立面的同一性（说它们的“统一性”也许更准确，尽管同一性与统一性这两个术语的区别在这里并不特别重要；在某种意义上，二者都正确），就是承认（发现）自然界（包括思维和社会）的一切现象和过程中存在着相互矛盾、相互排斥、相互对立的趋势。要认识世界的一切过程的“自己运动”、自发发展和实际生活，条件就是把这些过程作为对立面的统一来认识。[3334]

The identity of opposites (it would be more correct, perhaps, to say their ‘unity’, —although the difference between the terms identity and unity is not particularly important here. In a certain sense both are correct) is the recognition (discovery) of the contradictory, mutually exclusive, opposite tendencies in all phenomena and processes of nature (including mind and society). The condition for the knowledge of all processes of the world in their ‘self-movement’, in their spontaneous development, in their real life, is the knowledge of them as a unity of opposites.[3334]

我们认为，列宁对这些例子的解释，以及毛泽东对这一解释的默许，具有误导性。为说明这一点，先把例子分组：（1）属于算术，即人类的一种技艺或技术；（2、3、4）属于自然科学；（5）属于社会科学。

We think that the gloss given by Lenin to these examples, implicitly accepted by Mao, is misleading. To show this, let’s group the examples: (1) applies to arithmetic, a human art or technology; (2,3,4) apply to the natural sciences; (5) applies to social science.

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加法与减法、乘法与除法、微分与积分等，都是数学中互逆运算的例子。如果 α, β 是互逆运算符，那么 $((A\alpha B)\beta B) = A$；把 α, β 称作相反的运算，是完全合理的。

Addition and subtraction, multiplication and division, differentiation and integration etc., are examples of inverse operators in maths. If α, β are inverse operators then $((A\alpha B)\beta B) = A$, and it is quite reasonable to call α, β *opposite* operations.

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算术是一门技术性技艺。它涉及实物，如筹码、算珠和账簿中的符号。在 20 世纪 30 年代，俄国或中国的商业算术主要使用算盘。加 2 时，操作者把 2 颗算珠从右移到左；减 2 时，把 2 颗算珠从左移到右。先加后减之后，算珠就回到起初的状态。正是

Arithmetic is a technical art. It involves physical objects such as tokens, beads and signs in ledgers. In the 1930s, commercial arithmetic in Russia or China was largely performed on an abacus. When adding 2, the operator slid 2 beads from right to left. When subtracting

这种回到起始状态的性质，使加法与减法成为作用于状态的互逆运算。

2, the operator moved 2 beads left to right. An addition followed by a subtraction then left the beads in the same state as they started out. It is this return to the starting state that qualifies addition and subtraction as inverse operations on the state.

但若要声称对立统一具有普遍性，就会遇到问题。首先，很容易找出并非彼此互逆的运算符对。在逻辑中，与（∧）和或（∨）作为两个基本运算符成对出现，但它们并不互逆：$((A \wedge B) \vee B) \neq A$。那么，为什么要声称相反的运算符是根本的呢？

But if one wants to claim that the unity of opposites is universal, then there are problems. For a start, it is easy to come up with pairs of operators that are not inverse of one another. In logic, AND (∧) is paired with OR (∨) as the two basic operators, but these are not inverses: $((A \wedge B) \vee B) \neq A$. So why claim that opposite operators are fundamental?

这难道不是挑选有利例子来支持论点吗？是的，声称对立统一是数学的根本，是武断的。的确，数学有互逆运算符，但也有其他重要的运算符。

Is this not cherry-picking examples to support an argument? Yes, it is arbitrary to claim that the unity of opposites is fundamental to mathematics. Yes, mathematics does have inverse operators, but it has other important ones as well.

如果回到加法和减法使算盘或其他计算器恢复起始状态这一想法，情况会有所不同吗？

If we go back to the idea that addition and subtraction return an abacus or other calculator to its start state, does this help?

使某物恢复起始状态，总是需要一对互逆运算吗？

Does returning something to its start state always require a pair of inverse operations?

不。考虑把一个轮子旋转半圈的操作。执行两次，轮子便回到起始状态。算术中的类似操作是乘以－1。对一个数执行两次，就回到原来的数。因此，回到起点这一观念，同样不能证明对立统一的必然性。

No. Consider the operation of rotating a wheel by a half circle. If we perform this operation twice, the wheel returns to its starting state. An arithmetic analogue would be multiplication by −1. Apply that twice to a number, and you get back to your starting number. So the idea of a return to a starting point does not establish the necessity of a unity of opposites either.

接下来三个例子——作用与反作用、正负电荷以及原子的结合与解离——确实有共同之处，但诉诸对立统一，只能部分说明这种共同之处，遗漏了最重要的东西。

The next three examples of action and reaction, positive and negative charge and combination/dissociation of atoms do have something in common, but reference to a unity of opposites gives only a partial account of what they have in common, missing out the most important thing.

它们都是底层守恒定律的表达方式，或是这些定律的推论。

All of these are ways of expressing, or consequences of, underlying conservation laws.

大小相等、方向相反的力源于动量守恒。例如，地球与月球彼此施加大小相等、方向相反的引力，使这一系统所受的合力为零。如果没有太阳从外部施加的引力，地月系统就会沿直线运动，保持动量守恒。

Equal and opposite forces result from the conservation of momentum. For instance, the Earth and the Moon exert equal and opposite gravitational forces on each other, resulting in no net force on the system. Without the Sun’s external attraction, the Earth/Moon system would move in a straight line, conserving momentum.

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正电与负电涉及另一条基本守恒定律，即电荷守恒。带有＋6电荷的碳14原子核发生β衰变时，会形成一个带有＋7电荷的氮14原子核，同时产生一个β粒子，也就是一个带负电的电子。净电荷仍为＋6。电荷守恒意味着，在所有这类核转变中，总净电荷保持不变。

Positive and negative electricity relate to another basic conservation law: the conservation of charge. When a carbon 14 nucleus with a charge of +6 undergoes β decay, it forms a nitrogen 14 nucleus with a charge of +7 along with a β particle, that is to say, a negatively charged electron. The net charge is still +6. The conservation of charge means that in all such nuclear transitions, the total net charge remains unchanged.

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化学反应中原子的结合与解离，也表达了一种守恒关系：在化学过程中，每种元素的原子总数保持不变。

Association and disassociation of atoms in the chemical reactions also express a conservation relation: the conservation of the total number of atoms of each element in chemical processes.

列宁的这三个例子都是守恒过程，并不能证明对立统一这一辩证法原理。它们的共同性质源于卢克莱修的格言：“无中不能生有。”尽管苏联辩证唯物主义批评力学，却正是力学世界观表达了列宁这些例子背后的守恒定律。

Lenin’s three examples are conservative processes and do not demonstrate the dialectical principle of the unity of opposites. The source of their shared property lies in Lucretius’maxim, ‘nothing comes from nothing’. Despite Soviet diamat’s criticism of mechanics, it is the mechanical world view that expresses the conservation laws behind Lenin’s examples.

最后一个例子是阶级冲突。但它也涉及守恒约束，因为在任何社会中，有限的劳动力和有限的工作时间，都意味着剩余劳动增加会使必要劳动减少。这在工人与雇主之间造成不可调和的阶级冲突：资本家阶级占有的劳动每增加一分，可用于满足工人阶级需要的劳动就相应减少一分。

The last example given is class conflict. But this too relates to a conservation constraint since in every society, a finite labour force and limited working hours mean that an increase in surplus labour leads to a decrease in necessary labour. This creates irreconcilable class conflict between workers and employers, as any increase in labour appropriated by the capitalist class results in a corresponding decrease in the labour available to satisfy the needs of the working class.

守恒定律恰恰来源于那备受鄙视的力学。源自力学的思维方式，运用守恒原理解释世界如何运作。在第 5.3 节，我们论证过，马克思揭示资本主义社会“运动规律”的志向，也属于这一思想脉络。

It is from the despised mechanics that conservation laws originate. Forms of thought deriving from mechanics use conservation principles to explain how the world works. In Section 5.3, we argued that Marx’s ambition to uncover the ‘laws of motion’ of capitalist society belonged within this line of thought.

毛泽东在《矛盾论》中引用恩格斯和列宁时，只是在向正统敷衍地表示认同。他的创新来自对矛盾特殊性的处理。在这里，他运用了一套全新的概念术语及其含义，把辩证法的语言转化为军事和政治战略的工具。

When Mao was quoting of Engels and Lenin in *On Contradiction*, he was making perfunctory gestures towards orthodoxy. His innovation came in his treatment of the particularity of contradiction. Here, he deployed a set of entirely new conceptual terms and meanings, turning the language of dialectics into a tool of military and political strategy.

毛泽东主张，不同科学的区别在于它们所研究的矛盾的特殊性。他对科学方法作了如下描述：

Mao claimed that it was the particularity of the contradictions that they studied that distinguished the different sciences. His description of the scientific method was:

这就是认识的两个过程：一个是从特殊到一般，另一个是从一般到特殊。因此，认识总是循环往复地进行，而且只要严格遵循科学方法，每一个循环都会把人类知识提高一步，使之日益深刻。[35]

These are the two processes of cognition: one, from the particular to the general, and the other, from the general to the particular. Thus cognition always moves in cycles and (so long as scientific method is strictly adhered to) each cycle advances human knowledge a step higher and so makes it more and more profound.[35]

只要把术语从力换成矛盾，这一循环过程就与牛顿在《自然哲学的数学原理》导言中所说的完全一致：“哲学的一切困难似乎都在于此——从运动现象探究自然界的力，然后由这些力论证其他现象。”[36]毛泽东是在用矛盾的术语描述牛顿的科学方法。

Allowing for a shift in vocabulary from forces to contradictions, this cyclical process exactly matches what Newton said in the introduction to the Principia: ‘all the difficulty of philosophy seems to consist in this — from the phænomena of motions to investigate the forces of nature, and then from these forces to demonstrate the other phænomena’.[36] Mao was describing, in his terminology of contradictions, the Newtonian scientific method.

毛泽东把科学方法与受德博林影响的教条主义者的方法相对照：

Mao contrasted the scientific approach to that of the Deborin influenced dogmatists:

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我们的教条主义者是一群懒汉。他们拒绝对具体事物作任何艰苦的研究，把一般真理看作凭空产生的东西，把它们变成纯粹抽象、难以捉摸的公式，从而完全否定并颠倒人类认识真理的正常顺序。他们也不懂认识的两个过程——从特殊到一般，再从一般到特殊——之间的相互联系。他们根本不懂马克思主义的认识论。[37]

Our dogmatists are lazy-bones. They refuse to undertake any painstaking study of concrete things, they regard general truths as emerging out of the void, they turn them into purely abstract unfathomable formulas, and thereby completely deny and reverse the normal sequence by which man comes to know truth. Nor do they understand the interconnection of the two processes in cognition –from the particular to the general and

then from the general to the particular. They understand nothing of the Marxist theory of knowledge.[37]

要把更深刻的理解建立在对真实历史条件的具体分析之上，这在本书作者看来很有道理。以马克思的利润率下降规律为例。根据我们的经验，只有对英国从 19 世纪 50 年代到 20 世纪末的积累过程进行具体的统计分析，[38]才能深化对其机制的理解，并精确地用数学表述其动态行为。[39]如果我们只根据《资本论》的论述作推演，而没有实际历史统计数据的指引，就不大可能得出正确的动力学方程。没有那个方程、那条运动规律，就无法理解 20 世纪最后十年西方盈利能力部分恢复的有限性与矛盾性，也无法理解 2000 年以来中国与非洲资本回报率的不同趋势。

This need to base ones deeper understanding on concrete analysis of real historical conditions seems very plausible to the authors. Take the example of Marx’s law of the falling rate of profit. It was our experience that only by means of a concrete statistical analysis of the accumulation process in the UK from the 1850s to the late twentieth century[38] were we able to deepen our understanding of its mechanism and come up with a precise mathematical formulation of its dynamical behaviour.[39] Had we just deduced things from the account given in Capital, without the guidance of actual historical statistics, it is unlikely we would have come up with the correct dynamic equations. Without that eq uation, that law of motion, you cannot grasp the limited and contradictory nature of the partial recovery in Western profitability that occurred in the last decade of the twentieth century, nor the contrasting trends of capital returns in China and Africa since 2000.

毛泽东写道：

任何重大事物的发展过程中，都有许多矛盾。例如，在条件极其复杂的中国资产阶级民主革命过程中，存在着中国社会一切被压迫阶级与帝国主义之间的矛盾，广大人民群众与封建主义之间的矛盾，无产阶级与资产阶级之间的矛盾，农民和城市小资产阶级与资产阶级之间的矛盾，各反动统治集团之间的矛盾，等等。[40]

Mao wrote:

There are many contradictions in the course of development of any major thing. For instance, in the course of China’s bourgeois-democratic revolution, where the conditions are exceedingly complex, there exist the contradiction between all the oppressed classes in Chinese society and imperialism, the contradiction between the great masses of the people and feudalism, the contradiction between the proletariat and the bourgeoisie, the contradiction between the peasantry and the urban petty bourgeoisie on the one hand and the bourgeoisie on the other, the contradiction between the various reactionary ruling groups, and so on.[40]

这些矛盾具有鲜明的中国特殊性。但同样的多重矛盾话语，也提供了考察西方资本主义危机的一种方式；在这里，需要考虑：

These are very specific to China. But the same language of *many contradictions* provides a way of looking at the crisis of capitalism in the West where one has to consider:

• 劳动与资本之间的“矛盾”（阶级冲突），它体现为剩余价值率。

• The‘contradiction’(class conflict) between labour and capital expressed in the rate of surplus value.

• 生产性劳动与非生产性劳动之间的“矛盾”。用于支付非生产性劳动工资的剩余价值份额越高，能够作为新增生产资料积累的剩余价值流量就越少。

• The‘contradiction’between productive and unproductive labour. The higher the portion of surplus value used to pay wages for unproductive labour, the lower the flow surplus value that can be accumulated as a new means of production.

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• 资本与土地所有权之间的“矛盾”，它体现为剩余价值中用作地租的份额上升。

• The ‘contradiction’between capital and landed property expressed as a rising share of surplus value going as rent.

• 积累本身的矛盾效应：它通过提高资本与劳动之比，趋向于压低利润率。

• The contradictory effect of accumulation itself, which, by raising the capital/ labour ratio, tends to depress the profit rate.

毛泽东关于社会发展过程中多重矛盾的话语，为思考这些过程提供了框架。或许可以用数学方式表述这些过程，使“矛盾”成为数学项：$s' = \frac{s}{v}$，$p = s - u - r$（利润＝剩余价值－非生产性劳动工资－地租）。

Mao’s language of multiple contradictions in the process of the development of a society provides a framework for thinking through these processes. One can perhaps formulate these in mathematical terms so that the‘contradictions’become mathematical terms $s' = \frac{s}{v}$, $p = s - u - r$ (profit = surplus value - unproductive wages - rent).

在其他情况下，可以用较为熟悉的动力学和微积分语言表达矛盾关系。考虑积累与利润率之间的矛盾关系。

利润率，用 π 表示，由 $\pi = \frac{p}{K}$ 给出，其中 p 的定义同前，K 是资本存量。

资本存量相对于时间的变化率写作 $\left(\frac{dK}{dt}\right)$，而它又由 $\frac{dK}{dt} = \frac{A}{K}$ 给出，其中 A 为年度积累。将利润中用于净积累的份额记作 $\alpha = \frac{A}{p}$。现在考察利润率相对于时间的变化率 $\pi' = \frac{d\pi}{dt}$，也就是说，利润率随时间趋于上升还是下降。由前面的方程显然可见，$\frac{d\pi'}{d\alpha} < 0$，即利润变化率对积累率的偏导数为负。高积累率导致利润率下降。这是一种负反馈或矛盾关系，在数学中用牛顿与莱布尼茨的微积分描述，在其他领域则用不同术语描述。注意，只要稍作深入分析，就会发现上面最初列表中的所有矛盾都涉及负的（小于 0 的）偏导数。剩余价值对工资的导数为负，利润对地租的导数为负，等等。

〔校注：原书此处写作 dK/dt=A/K。按文中 K 为资本存量、A 为年度积累量的定义，A/K 对应相对增长率，而 dK/dt 对应存量的变化率；此处可能存在记号或排印问题，本稿不作无声更正。原书以 d 表示导数而在文字中称其为偏导数，亦照录。〕

事实上，如果用数字电路使逻辑机械化，逻辑否定与负偏导数之间的这种关系就会显现。

在计算机设计中，用导线分别表示逻辑信号本身及其反相信号，是完全标准的做法。如果像恩格斯和列宁谈论唯物辩证法时那样作肤浅的类比，那么同时包含 A 与非 A 信号的电路，可以说包含了“矛盾”或“对立面的同一性”；而这些矛盾项确实通过负偏导数联系在一起。在图 11.2 中，非 A 信号的电压对 A 信号电压的偏导数为负。在物理系统中，负偏导数用来表示否定。逻辑是用来描述这些系统的一种抽象，无论系统是电路，还是使用记事本的人。因此，对于任何受负导数支配的物质过程，我们都可以隐喻地使用“否定”这一术语。

In other cases, one can express contradictory relationships in the more familiar language of dynamics and calculus. Consider the contradictory relationship between accumulation and the rate of profit.

The rate of profit, let us use π, is given by $\pi = \frac{p}{K}$ where p as defined earlier, and K is capital stock.

The rate of change of the capital stock with respect to time would be written $\left(\frac{dK}{dt}\right)$ and this in turn is given by $\frac{dK}{dt} = \frac{A}{K}$ where A is annual accumulation. If we denote the share of profits going as net accumulation as $\alpha = \frac{A}{p}$. Now consider the rate of change of the rate of profit with respect to time $\pi' = \frac{d\pi}{dt}$ that is to say whether the rate of profit tends to rise or fall over time. It is clear from the previous equations that $\frac{d\pi'}{d\alpha} < 0$ that is to say the partial derivative of the rate of change of profit with respect to the rate of accumulation is negative. A high rate of accumulation leads to a decrease in the rate of profit. This is a negative feedback or contradictory relationship, described using the calculus of Newton and Leibniz in mathematics or different terminologies in other fields. Note that all of the contradictions in the initial list above, when analysed slightly more closely, involve negative (<0) partial derivatives. The derivative of surplus value with respect to wages is negative. The derivative of profit with respect to rent is negative, etc.

Indeed, if you mechanize logic with digital circuits, this relationship between logical negation and negative partial derivatives emerges.

It is absolutely standard practice in computer design to have wires representing both the true and the complement of a logic signal. If we make the kind of superficial analogies that Engels and Lenin use about materialist dialectics, then a circuit that contains the signals A and not A might be said to contain ‘contradiction’ or ‘identity of opposites’, and it turns out that these contradictory terms are related by negative partial derivatives. In Figure 11.2, the partial derivative of the voltage on the not A signal with respect to the voltage on the A signal will be negative. Negative partial derivatives are used to represent negation in physical systems. Logic is an abstraction used to describe these systems, whether they are electric circuits or people using notepads. As such, we can use the term ”negation” metaphorically for any material process governed by negative derivatives.

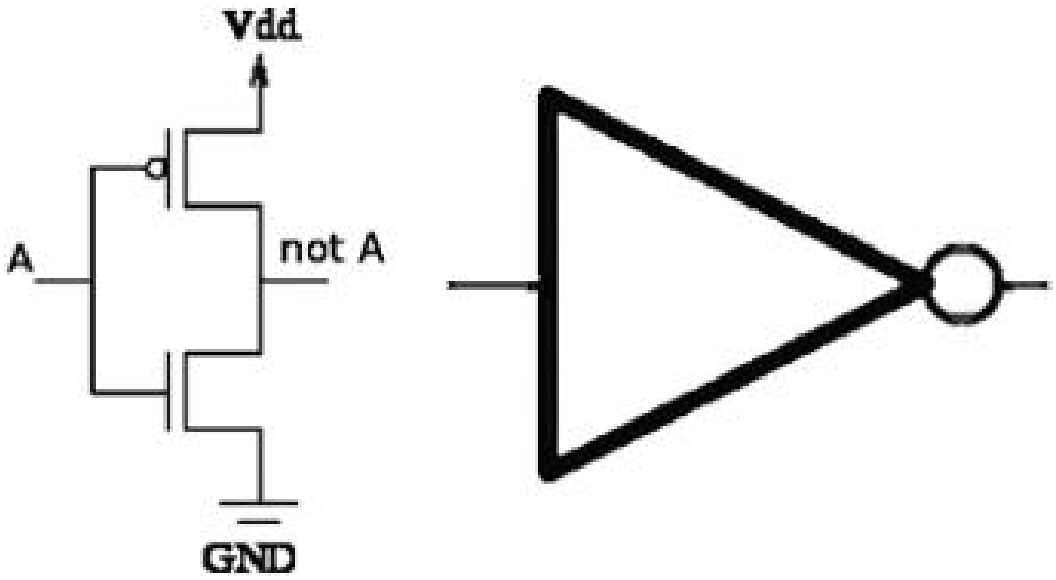


图 11.2 左：计算机芯片中的一种标准元件——CMOS 反相器。向输入端 A 施加正电压，会使上方晶体管关断、下方晶体管导通，从而把非 A 输出端的电压拉低至地电位。右：反相器的逻辑符号。逻辑符号使我们能够撇开实现逻辑的具体电路。作者绘图。

Figure 11.2 Left a standard component in computer chips is the CMOS inverter. Applying a positive voltage to input A switches the top transistor off and the bottom one on, resulting in the output voltage of not A being driven down to ground. Right the logic symbol for an inverter. A logic symbol allows you to abstract from the specific electrical circuit used to implement the logic. Figure by authors.

A：输入信号；not A：非 A 输出；Vdd：电源；GND：地。

把同时包含一个信号及其反相信号的电路描述为“矛盾”，或许是相当自由的隐喻；因为虽然 A 与非 A 信号可以同时存在于逻辑电路中，但其中只有一个会被置为有效。

Describing a circuit containing both true and complement of a signal as a ‘contradiction’ is perhaps a rather free metaphor since, while signals A and not A can exist in a logic circuit, only one of them will be asserted.

毛泽东强调，不同矛盾有其特定的“解决”形式。

Mao emphsisizes that different contradictions have their own specific forms of ‘resolution’.

性质不同的矛盾，只有用性质不同的方法才能解决。例如，无产阶级与资产阶级的矛盾，用社会主义革命的方法解决；广大人民群众与封建制度的矛盾，用民主革命的方法解决；殖民地与帝国主义的矛盾，用民族革命战争的方法解决；社会主义社会中工人阶级与农民阶级之间的矛盾，用农业集体化和机械化的方法解决；[41]

Qualitatively different contradictions can only be resolved by qualitatively different methods. For instance, the contradiction between the proletariat and the bourgeoisie is resolved by the method of socialist revolution; the contradiction between the great masses of the people and the feudal system is resolved by the method of democratic revolution; the contradiction between the colonies and imperialism is resolved by the method of national revolutionary war; the contradiction between the working class and the peasant class in socialist society is resolved by the method of collectivisation and mechanisation in agriculture;[41]

他还引入了“主要矛盾”的概念，大致是指在决定社会发展进程方面最重要的矛盾。主要矛盾这一概念本质上与政治相联系。他说，在资本主义中，“处于矛盾之中的两个力量——无产阶级和资产阶级——构成主要矛盾”。这是因为，这些相互对立的阶级塑造了政治结构。他将它与所谓基本矛盾或根本矛盾区分开来；后者作用于生产关系层面，直到社会发展的某个阶段完成，才会消失。[42]

He also introduces the notion of a ‘principal contradiction’, by which he means, roughly, the contradiction that is most important in determining the course of social development. This notion of a *principle contradiction* was essentially related to politics. He says that for capitalism, ‘the two forces in contradiction, the proletariat and the bourgeoisie, form the principal contradiction’. That is because these opposing classes shape the structure of politics. He distinguishes this from what he calls a *basic* or *fundamental* contradiction, which operates at the level of production relations and will not disappear until a given stage in the development of society is completed.[42]

马克思和恩格斯把事物中的矛盾规律应用于社会历史

p. 232–233

过程的研究时，发现了生产力与生产关系之间的矛盾，发现了剥削阶级与被剥削阶级之间的矛盾，以及由此产生的经济基础与其上层建筑（政治、意识形态等）之间的矛盾，并发现这些矛盾如何在不同类型的阶级社会中，必然导致不同类型的社会革命。

When Marx and Engels applied the law of contradiction in things to the study of the socio-historical process, they discovered the contradiction between the productive forces and the relations of production, they discovered the contradiction between the exploiting and exploited classes and also the resultant contradiction between the economic base and its superstructure (politics, ideology etc), and they discovered how these contradictions inevitably lead to different kinds of social revolution in different kinds of class society.

马克思把这一规律应用于资本主义社会经济结构的研究时，发现这一社会的基本矛盾，是生产的社会性与所有制的私人性质之间的矛盾。这种矛盾表现为个别企业内部生产的有组织性与整个社会生产的无政府性之间的矛盾。在阶级关系上，它表现为资产阶级与无产阶级之间的矛盾。[43]

When Marx applied this law to the study of the economic structure of capitalist society, he discovered that the basic contradiction of this society is the contradiction between the social character of production and the private character of ownership. This contradiction manifests itself in the contradiction between the organized character of production in individual enterprises and the anarchic character of production in society as a whole. In terms of class relations, it manifests itself in the contradiction between the bourgeoisie and the proletariat.[43]

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将这一点应用于现代资本主义，生产力与生产关系之间的矛盾，就以利润率下降趋势的形式产生一种基本经济矛盾。在阶级层面，它表现为工人与雇主围绕工资、工作日及工作条件展开的经济阶级斗争。正是这一阶级矛盾居于主要地位，压倒了资本家与土地所有权之间的阶级冲突。理解这些矛盾之间的关系，就能说明危机如何在经济和政治层面展开。

Applying this to modern capitalism, the contradiction between the forces and relations of production produces a basic economic contradiction in the tendency of the rate of profit to fall. At the class level, it is expressed in the economic class struggle between workers and employers over wages, the working day and conditions of work. It is this class contradiction that is the principle one, overruling the class conflict between capitalists and landed property. Understanding the relationship between the contradictions explains how the economic and political working out of a crisis can proceed.

只要政治层面上工人阶级与资本家之间的主要矛盾还没有通过社会主义革命解决，基本经济矛盾就依然存在。利润率下降的趋势并未消除！

So long as the principle contradiction at the political level between the working class and the capitalists is not resolved by a socialist revolution, the basic economic contradiction persists. The tendency of the rate of profit to fall is not removed!

作为问题解决方法的辩证法

Dialectics as problem solving

现在，让我们根据上述讨论，重新审视辩证法的三条经典规律。我们认为，最善意而有成效的理解，是把它们看作分析方法，而不是解释方法。

Let us now revisit the classic three laws of the dialectic in light of the foregoing discussion. We think that these are most charitably and fruitfully considered as methods of analysis rather than of explanation.

首先，量向质的转化，如今通常被称为涌现。观察性质在可测量的边界上发生变化，并试图解释其原因，是科学的核心内容。然而，声称某事物通过量向质的转化而改变，是在诉诸魔法，而不只是承认某些可观察的现象尚无法用科学术语解释。诉诸涌现的说法，最常见于人工智能（AI）的讨论，尤其是奇点这一概念，即人工智能将获得超过人类的能力的时刻。人工智能日益取得成功的说法，掩盖了奇点究竟如何到来的问题；据说这些成功会以某种方式加速，并导致质变。我们当然不排除通用人工智能的可能性，但当前证据表明，在特定领域取得的小规模而且确实令人印象深刻的成果，并不能很好地扩展规模、推广到其他领域或与其他成果整合。

First of all, the transformation of quantity into quality is now popularly referred to as emergence. Observing that properties change on measureable boundaries and seeking to explain why is a core part of science. However, claiming that something changes through the transformation of quantity into quality is a way of invoking magic, rather than simply acknowledging that some observable phenomenum are not yet explicable in scientific terms. Appeals to emergence are most commonly found in discussions of artificial intelligence (AI), in particular in the notion of the singularity, when AIs will achieve competences greater than humans. How the singularity will come about is elided by appeal to increasing AI success, which somehow will gather pace and lead to qualitative change. While we are certainly open to the possi-

bility of general AI, current evidence suggests that small, and indeed impressive, domain specific achievements do not scale, generalize or integrate well with other achievements.

p. 234 同样，否定之否定常被表述为正题、反题、合题三元结构的抽象。与量向质的转化一样，它通常被用作事后解释。从纯粹逻辑的角度说，对一个否定再作否定，就会回到起点。

Similarly, the negation of the negation is often presented as the abstraction of the trinity of thesis/antithesis/synthesis. Like the transformation of quantity into quality, it is typically invoked as a post-hoc explanation. In purely logical terms, the negation of a negation returns to the starting point.

最后，对立面的相互渗透，表达的是过程往往包含相互制衡、彼此作用的子过程。就其本身而言，寻找相互作用的“对立面”，是通过分解来解决问题的诸多进路之一。因此，在当代计算思维中，对立面的相互渗透是一种分而治之的形式，[44]其中把问题分解为两个子问题，是最简单的情况。在解决任意问题时，这些子问题通常相互补充，而不是相互矛盾。尽管如此，通过寻找社会阶级之间的矛盾来理解社会过程，仍是历史唯物主义的关键洞见。马克思、恩格斯、列宁和毛泽东娴熟地运用了这一洞见，为了改变世界而解释世界。

Finally, the interpenetration of opposites is a way of expressing that processes often involve countervailing sub-processes, which act on each other. In itself, seeking interacting ‘opposites’ is one of many approaches to problem solving through decomposition. Thus, in contemporary computational thinking, the interpenetration of opposites is a form of divide and conquer,[44] where problem decomposition into two sub-problems is the simplest case. In solving arbitrary problems, such sub-problems are usually complementary, rather than contradictory. Nonetheless, understanding social processes by seeking contradictions between social classes is the key insight of historical materialism, so ably deployed by Marx, Engels, Lenin and Mao, to interpret the world in order to change it.

第 11 章原注 / Chapter 11 Notes

↑1. Shannon：“通信的数学理论”。

1 Shannon, ‘A Mathematical Theory of Communication’. p. 261

↑2. 费希尔在此前十年已经尝试用概率定义信息：Ronald Aylmer Fisher，“不确定推理逻辑中的概率、似然与信息量”，Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character 146，第 856 期（1934）：1—8；但香农的处理更为一般，影响也更持久。

2 Attempts to define information in probability terms had been started by Fisher in the previous decade Ronald Aylmer Fisher, ‘Probability Likelihood and Quantity of Information in the Logic of Uncertain Inference’, *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character* 146, no. 856 (1934): 1–8, but Shannon’s treatment was much more general and of more lasting influence. p. 261

↑3. 每个单词 5 个字符，加上行末的回车与换行字符，每词便有 $(5+2)\times 5 = 35$ 个二进制位，再乘以 3 个单词。

3 Five characters per word, plus carriage return and line feed characters at the end of lines so $(5+2)\times 5 = 35$ binary digits per word, × three words. p. 261

↑4. 德国“Fish”编码系统用来加密博多码的随机数并非完全均匀，再加上德语字母频率不等，使第一台数字计算机 Colossus 能够破解这些编码消息：B. Jack Copeland，《Colossus：布莱切利园破译计算机的秘密》（Oxford University Press，2010）。香农在战时曾与图灵合作研究密码学：Andrew Hodges，《艾伦·图灵：谜》（Touchstone，1983）；图灵十分熟悉 Colossus 的进路。

4 The existence of slight deviations from uniformity in the random numbers used to encrypt Baudot code in the German ‘Fish’ encoding system, along with the unequal frequency of letters in the German language was what enabled the first digital computer Colossus to break the encoded messages B Jack Copeland, *Colossus: The secrets of Bletchley Park’s Code-breaking Computers* (Oxford University Press, 2010). It is worth noting that Shannon had cooperated with Turing on cryptography during the war Andrew Hodges, *Alan Turing the Enigma* (Touchstone, 1983) and Turing was closely familiar with the Colossus approach. p. 261

↑5. 区别仅在于使用离散空间的求和，而非连续空间的积分。

5 Apart from the use of summation over a discrete space rather than integration over a continuous one. p. 261

↑6. Claude Shannon：“信息的格论”，Transactions of the IRE Professional Group on Information Theory 1，第 1 期（1953）：105—107。

6 Claude Shannon, ‘The Lattice Theory of Information’, *Transactions of the IRE Professional Group on Information Theory* 1, no. 1 (1953): 105–7. p. 261

↑7. Rachel R. Bennett 与 Ramin Golestanian：“衣藻趋光性的转向机制”，Journal of The Royal Society Interface 12，第 104 期（2015）：20141164。

7 Rachel R. Bennett and Ramin Golestanian, ‘A Steering Mechanism for Phototaxis in Chlamydomonas’, *Journal of The Royal Society Interface* 12, no. 104 (2015): 20141164. p. 261

↑8. 实际上，正如 Bennett 和 Golestanian 所示，衣藻具有更高效的编码方式。它只有一个眼点，沿螺旋路径游动，同时绕自身轴线旋转。这使一个眼点能够以约 60 毫秒的采样间隔，对三个空间轴进行时分复用扫描。假定在所有可能朝向中，有 $\frac{1}{6}$ 会激活眼点，那么根据香农公式，一个样本的平均信息量为 0.65 比特，因此眼点每秒提供约 10 比特的信息。

8 In fact, as Bennett and Golestanian show, *Chlamydomonas* has a more efficient encoding. It has a single eye spot and swims in a spiral path spinning on its axis as it does so. This enables the one eye spot to perform time-multiplexed scanning of the three spatial axes with a sampling interval of about 60 ms. Assuming that the eye spot activates for $\frac{1}{6}$ of possible orientations and using Shannon’s formula the mean information content of a sample is 0.65 bits so the eye spot provides about 10 bits per second of information. p. 261–262

↑9. 两个随机变量 X,Y 之间的互信息记为 $I(X;Y) = H(X) - H(X \mid Y)$，其中 $H(X)$ 为 X 的信息熵，$H(X \mid Y) = \sum_{x,y} p(x,y) \log_2 \frac{1}{p(x|y)}$ 为给定 Y 时 X 的条件熵。

9 Mutual information between two random variables X,Y is denoted by $I(X;Y) = H(X) - H(X \mid Y)$ where $H(X)$ is the information entropy of X and $H(X \mid Y) = \sum_{x,y} p(x,y) \log_2 \frac{1}{p(x|y)}$ is the conditional entropy of X given Y. p. 262

↑10. 列宁，《唯物主义和经验批判主义》。

10 Lenin, *Materialism and Empirio-criticism*. p. 262

↑11. 这种直觉或许解释了为什么某些通俗论述将信息称为负熵（negentropy）。这一术语用法源于对 Léon Brillouin 论文“信息的负熵原理”的误

11 This intuition is perhaps why some popular accounts refer to information as negative entropy (negentropy). This usage of the term stems from a misreading p. 262

读，见 Journal of Applied Physics 24，第 9 期（1953）：1152—1163。Brillouin 实际做的是引入一个新变量 $N = -S$，作为推广热力学第二定律的一部分：

“我们可以定义量 N（负熵）：$N = -S, \Delta(N+I) \leq 0$（7b）。负熵与信息之和必定始终减小。负熵对应于开尔文‘能量退化’原理中的能量‘品位’。”（Brillouin）

由此，他得以就获取信息的观测所产生的熵效应作出结论：

“凡是从一个物理系统获取信息的实验，平均而言都对应于该系统或其周围环境中熵的增加。这种平均增量始终大于（或等于）所获得的信息量。”（Brillouin）

他能够利用这一关系证明麦克斯韦妖是不可行的。

of Leon Brillouin, ‘The Negentropy Principle of Information’, *Journal of Applied Physics* 24, no. 9 (1953): 1152–63. What Brillouin did was introduce a new variable $N = -S$ as part of a generalization of the second law of thermodynamics:

We may define the quantity N (negentropy) $N = -S, \Delta(N + I) \leq 0$ (7b) The sum of negentropy plus information must always decrease. Negentropy corresponds to the ‘grade’ of energy in Kelvin’s principle of ‘degradation of energy’, Brillouin

From this he was able to draw conclusions about the entropic effect of observations that obtain information:

Any experiment by which an information is obtained about a physical system corresponds in average to an increase of entropy in the system or in its surroundings. This average increase is always larger than (or equal to) the amount of information obtained. Brillouin

He was able to use this relation to demonstrate that Maxwell’s daemon was unfeasible.

p. 262 ↑12. 关于资本主义工业中熵减少与信息增加交替过程的详细分析，见 Cockshott 等，《经典经济物理学：古典政治经济学、热力学与信息论论集》。

12 For an extended analysis of this alternation between entropy reduction and information addition in capitalist industry see Cockshott et al., *Classical Econophysics: Essays on Classical Political Economy, Thermodynamics and Information Theory*.

p. 262 ↑13. 自 2018 年以来，英特尔处理器支持 RDRAND 指令，该指令提供据称是真正的随机数。Intel，《Intel64 和 IA-32 架构软件开发者手册》，第 2B 卷（Intel，2021）。

13 Since 2018 Intel processors have supported the RDRAND instruction which provides what are claimed to be true random numbers Intel, *Intel64 and IA-32 Architectures Software Developer’s Manual*, vol. 2B (Intel, 2021).

p. 262 ↑14. 我们从第三个数字开始，以免来源一目了然。

14 We started at the third digit to prevent our source being obvious.

p. 262 ↑15. Gregory J. Chaitin：“随机性与数学证明”，Scientific American 232，第 5 期（1975）：47—53。

15 Gregory J. Chaitin, ‘Randomness and Mathematical Proof’, *Scientific American* 232, no. 5 (1975): 47–53.

p. 262 ↑16. 约瑟夫·维萨里奥诺维奇·斯大林，《论辩证唯物主义和历史唯物主义》（Lawrence & Wishart，1943）。

16 Josif Vissarionovič Stalin, *Dialectical and Historical Materialism* (Lawrence & Wishart, 1943).

p. 262 ↑17. 斯大林，《论辩证唯物主义和历史唯物主义》，第 4 页。

17 Stalin, *Dialectical and Historical Materialism*, 4.

p. 262 ↑18. 参见 Emery：“辩证法与机械论：关于科学方法的一场共产主义论争”；本书前面的章节曾提及此文。

18 See Emery, ‘Dialectics versus Mechanics. A Communist Debate on Scientific Method’, to whom we have refered in an earlier chapter.

p. 262 ↑19. 玻姆，《整体性与隐含序》。

19 Bohm, *Wholeness and the Implicate Order*.

p. 262 ↑20. 斯大林，《论辩证唯物主义和历史唯物主义》，第 5 页。

20 Stalin, *Dialectical and Historical Materialism*, 5.

p. 262 ↑21. Titus Lucretius Carus，《卢克莱修全集》（Delphi Classics，2015），H. A. J. Munro 译。

21 Titus Lucretius Carus, *The Complete Works of Lucretius* (Delphi Classics, 2015), H. A. J. Munro translation.

p. 263 ↑22. 弗里德里希·恩格斯，《自然辩证法》（Wellred Books，1960），第 2 章。

22 Friedrich Engels, *Dialectics of Nature* (Wellred Books, 1960), Chap 2.

p. 263 ↑23. 正如本书前文所述，灯泡等物体发出的黑体辐射所涉及的连续能级与离散能级问题，是爱因斯坦针对马赫学派确证原子论的关键。

23 The issue of continuous versus discrete energy levels associated with black body radiation like that from a bulb, was, as we have recounted earlier, fundamental to

Einstein’s validation of atomism against the Machian school.

†24. 恩格斯，《自然辩证法》，第 2 章。

24 Engels, *Dialectics of Nature*, Chap 2.

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†25. 例如，Charles Merivale 的《帝国时期罗马人史》（Longmans，1851）中写道：“但在这个时代，实施这种不义行径不可能不受惩罚。战争的因素正在整个半岛酝酿；成千上万破产的人突然被抛入那动荡不定、心怀不满的人群中，这群人只缺少这一成分，就会在新的革命中沸腾爆发。”（第 223 页）

25 For instance in Charles Merivale, *A History of the Romans under the Empire* (Longmans, 1851) we have: But this was not a time in which such injustice could be perpetrated with impunity. The elements of war were fermenting throughout the peninsula; thousands of broken fortunes were suddenly thrown into the floating mass of discontent, which wanted only this ingredient to boil over in a new revolution. (223)

p. 263

†26. Ian P. Gent 等：“CSP 相变中的尺度效应”，载 International Conference on Principles and Practice of Constraint Programming（Springer，1995），70—87；P. Prosser：“二元约束满足问题中相变的实证研究”，Artificial Intelligence 81，第 1—2 期（1996）：81—109。

26 Ian P Gent et al.,‘Scaling Effects in the CSP Phase Transition’, in *International Conference on Principles and Practice of Constraint Programming* (Springer, 1995), 70–87; P. Prosser,‘An Empirical Study of Phase Transitions in Binary Constraint Satisfaction Problems’, *Artificial Intelligence* 81, no. 1–2 (1996): 81–109.

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†27. 假设 F 的每个求和项包含 3 个变量（3SAT），那么约束程度的一个度量，可以是 F 中求和项的数量与变量数 n 的比值。

27 Suppose each sum term in F contains 3 variables (3SAT) then a measure of the constrainedness might be the ratio of the number of sum terms in F to the number of variable *n*.

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†28. 斯大林，《论辩证唯物主义和历史唯物主义》。

28 Stalin, *Dialectical and Historical Materialism*.

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†29. 斯大林，《论辩证唯物主义和历史唯物主义》。

29 Stalin, *Dialectical and Historical Materialism*.

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†30. 卡尔·马克思、弗里德里希·恩格斯，《德意志意识形态》（Progress Publishers，1976）。

30 Karl Marx and Frederick Engels, *The German Ideology* (Progress Publishers, 1976).

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†31. 毛泽东，《矛盾论》，载《毛泽东选集》第 1 卷（1937）。

31 Tse-Tung Mao, *On Contradiction, in Selected Works of Mao Tse-Tung*, vol. 1 (1937).

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†32. Peter Turchin、Sergey A. Nefedov：“长期周期”，载《长期周期》（Princeton University Press，2009）。

32 Peter Turchin and Sergey A Nefedov, ‘*Secular Cycles*’, in *Secular Cycles* (Princeton University Press, 2009).

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†33. 毛泽东只引用了第 1 至第 5 项。我们把前后段落也列出，以提供上下文，因为毛泽东在文中其他地方似乎隐含地依赖这一段。第 1 至第 5 项的明确编号是我们添加的，列宁原文中它们只是一个未编号的列表。

33 Mao only quotes items 1 to 5, we include the surrounding paragraphs for context, since Mao appears elsewhere in the text to implicitly rely on this paragraph. The explicit numbering of items 1 to 5 is by us, in the Lenin text they are rendered simply as list.

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†34. V. I. Lenin：“谈谈辩证法问题”，载《列宁全集》第 38 卷（Progress Publishers，1915）。

34 V. I. Lenin, ‘On the Question of Dialectics’, in *Lenin Collected Works*, by Lenin, vol. 38 (Progress Publishers, 1915).

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†35. 毛泽东，《矛盾论》，第 321 页。

35 Mao, *On Contradiction*, 321.

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†36. “Omnis enim Philosophiæ difficultas in eo versari videtur, ut a Phænomenis motuum investigemus vires Naturæ, deinde ab his viribus demonstremus phænomena reliqua”。〔译：哲学的一切困难似乎都在于此：从运动现象探究自然界的力，然后由这些力论证其他现象。〕

36‘Omnis enim Philosophiæ difficultas in eo versari videtur, ut a Phænomenis motuum investigemus vires Naturæ, deinde ab his viribus demonstremus phænomena reliqua’.

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†37. 毛泽东，《矛盾论》，第 321 页。

37 Mao, *On Contradiction*, 321.

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†38. Paul Cockshott、Allin Cottrell、Greg Michaelson：“检验马克思：英国数据的一些新结果”，Capital & Class 19，第 1 期（1995）：103—130。

38 Paul Cockshott, Allin Cottrell, and Greg Michaelson,‘Testing Marx: Some New Results from UK Data’, *Capital & Class* 19, no. 1 (1995): 103–30.

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†39. Cockshott 等，《经典经济物理学：古典政治经济学、热力学与信息论论集》；Allin Cottrell、Paul Cockshott：“人口与利润率下降”，Indian Development Review 4，第 1 期（2006）：39—59。

39 Cockshott et al., *Classical Econophysics: Essays on Classical Political Economy, thermodynamics and Information Theory*; Allin Cottrell and Paul Cockshott,

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‘Demography and the Falling Rate of Profit’, *Indian Development Review* 4, no. 1 (2006): 39–59 .

p. 263 ↑40. 毛泽东，《矛盾论》，第 322 页。

40 Mao, *On Contradiction*, 322.

p. 263 ↑41. 毛泽东，《矛盾论》，第 321 页。

41 Mao, *On Contradiction*, 321.

p. 263 ↑42. 毛泽东，《矛盾论》，第 325 页。

42 Mao, *On Contradiction*, 325.

p. 263 ↑43. 毛泽东，《矛盾论》，第 328—329 页。

43 Mao, *On Contradiction*, 328–9.

p. 263 ↑44. G. Michaelson：“编程范式、图灵完备性与计算思维”，The Art, Science, and Engineering of Programming 4，第 3 期（2020）：第 4 篇，`https://arxiv.org/abs/2002.06178`。

44 G. Michaelson, ‘Programming Paradigms, Turing Completeness and Computational Thinking’, *The Art, Science, and Engineering of Programming* 4, no. 3 (2020): Article 4, `https://arxiv.org/abs/2002.06178`.

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